A First Course in Abstract Algebra

Sections 0–56 and the matrix appendix: collected solutions

Detailed worked solutions

Mathjock

Text and numbering. John B. Fraleigh, *A First Course in Abstract Algebra*, seventh edition (2003), following the scanned textbook linked on Mathjock. A label such as 12.3 means Exercise 3 in Section 12; A.3 means Exercise 3 in the matrix appendix. Section 0 is included. These are independently written solutions; brief descriptions identify the exercises, and the textbook supplies their original statements.

How to read the solutions. Identify the assumptions and the conclusion before beginning a proof. Keep track of the underlying set, its operations, and the hypotheses of each theorem. For a calculation, follow the intermediate steps and check the result in the original definition. A counterexample must satisfy all assumptions and fail the proposed conclusion.

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0 Sets and Relations

Starting point. Read the defining condition before calculating. An equality of sets requires both inclusions; an equivalence relation requires all three properties.

Exercise 0.1: Real square roots

The equation $x^2 = 3$ has two real solutions, $x = \sqrt{3}$ and $x = -\sqrt{3}$. Both square to 3, and factoring $x^2 - 3 = (x - \sqrt{3})(x + \sqrt{3})$ shows there are no others. The set is $\{-\sqrt{3}, \sqrt{3}\}$.

Exercise 0.2: Integer square roots

An integer whose square is 3 would have absolute value strictly between 1 and 2, which is impossible. Thus the set is $\varnothing$. Notice that changing the domain from $\mathbb{R}$ to $\mathbb{Z}$ changes the answer.

Exercise 0.3: Divisors of sixty

The condition means that m divides 60. Since $60 = 2^2 \cdot 3 \cdot 5$, its positive divisors are $1, 2, 3, 4, 5, 6, 10, 12, 15, 20, 30, 60$. Every negative of such a number is also a divisor, using a negative quotient. Thus the set is

$$\{\pm 1, \pm 2, \pm 3, \pm 4, \pm 5, \pm 6, \pm 10, \pm 12, \pm 15, \pm 20, \pm 30, \pm 60\}.$$

Zero is excluded because $0n$ cannot equal 60.

Exercise 0.4: An integer inequality

Solve $m^2 - m < 115$ by checking where the upward-opening quadratic crosses zero. Its roots are $(1 \pm \sqrt{461})/2$, approximately -10.235 and 11.235. Hence the integers strictly between them are

$$\{-10, -9, -8, -7, -6, -5, -4, -3, -2, -1, 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11\}.$$

At the endpoints $m = -10, 11$ the value is 110; at the next integers $-11, 12$ it is 132.

Exercise 0.5: The word large

This is not well defined: no cutoff for "large" is specified, so membership cannot be decided unambiguously. For example, $\{n \in \mathbb{Z}^+ : n > 1000\}$ would be a well-defined replacement, but its cutoff is additional information, not part of the question.

Exercise 0.6: Negative integer squares

Every integer square is nonnegative. Thus no integer satisfies $n^2 < 0$, and the specified object is the well-defined empty set $\varnothing$.

Exercise 0.7: An interval for a cube

Since $3^3 = 27$, $4^3 = 64$, and cubing is strictly increasing, there is no integer with $39 < n^3 < 57$. The set is well defined and equals $\varnothing$.

Exercise 0.8: Almost an integer

"Almost" gives no numerical tolerance. Therefore this description does not define a set. A precise replacement could be $\{q \in \mathbb{Q} : |q - m| < 1/100 \text{ for some } m \in \mathbb{Z}\}$; other tolerances give other sets.

Exercise 0.9: A large denominator

This is a well-defined description of all of $\mathbb{Q}$. If $q = a/b$ with $b > 0$, then $q = (ka)/(kb)$ for every positive integer k. Choose k so that $kb > 100$. The question does not require the fraction to be in lowest terms.

Exercise 0.10: A small positive denominator

The allowed denominators are $1, 2, 3$. Consequently the set is

$$\{a/b : a \in \mathbb{Z},\ b \in \{1, 2, 3\}\} = \mathbb{Z} \cup \tfrac{1}{2}\mathbb{Z} \cup \tfrac{1}{3}\mathbb{Z}.$$

This is well defined. For example $1/2$ is included and $1/4$ is not: $1/4 = a/b$ would force $b = 4a$, impossible for $b = 1, 2, 3$.

Exercise 0.11: A Cartesian product

Choose a first coordinate from $\{a, b, c\}$ and independently a second from $\{1, 2, c\}$. The result is

$$\{(a, 1), (a, 2), (a, c), (b, 1), (b, 2), (b, c), (c, 1), (c, 2), (c, c)\}.$$

There are $3 \cdot 3 = 9$ ordered pairs; an ordered pair is not an unordered set.

Exercise 0.12: Functions, injections, and surjections

A function must assign exactly one value to each of $1, 2, 3$; it is injective when those values are distinct, and onto when all of $2, 4, 6$ occur.

(a) A function, but not injective since $1, 2$ both map to 4, and not onto since 2 is absent.

(b) A function, but not injective since $1, 3$ both map to 4, and not onto since 2 is absent.

(c) Not a function: 1 has three different outputs, while $2, 3$ have none.

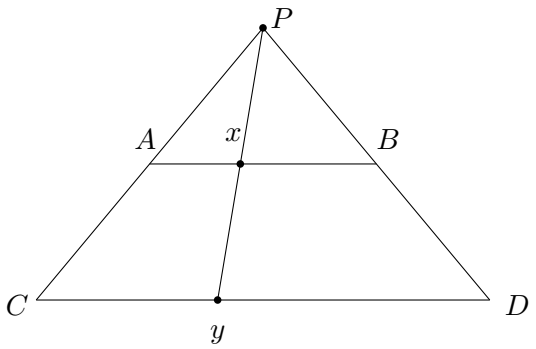
(d) A function and a bijection: $1 \mapsto 6$, $2 \mapsto 2$, $3 \mapsto 4$ uses each target exactly once.

(e) A function, but not injective since all inputs map to 6, and not onto since $2, 4$ are absent.

(f) Not a function: 2 has outputs 6 and 4, and 3 has no output.

Exercise 0.13: Pairing two segments

Draw the line through P and x, and take $y = Px \cap CD$. Projection from P sends A to C, B to D, and each intermediate point to exactly one intermediate point. Conversely the line from P through any point on CD meets AB once. Thus this pairing is a bijection.



Exercise 0.14: Bijections of closed intervals

(a) $f(x) = 2x$ maps $[0, 1]$ onto $[0, 2]$; its inverse is $y/2$.

(b) $f(x) = 10x - 5$ maps 1 to 5 and 3 to 25; its inverse is $(y + 5)/10$.

(c) For $a < b$ and $c < d$, put $f(x) = c + (d - c)(x - a)/(b - a)$. The positive slope makes it injective; for any $y \in [c, d]$, the inverse $x = a + (b - a)(y - c)/(d - c)$ lies in $[a, b]$. Thus it is onto as well.

Exercise 0.15: An open interval and the real line

Use $f(x) = \tan(\pi(x - 1/2))$ for $0 < x < 1$. Its argument ranges through $(-\pi/2, \pi/2)$, where tangent is strictly increasing and has limits $-\infty, +\infty$. Explicitly $f^{-1}(y) = 1/2 + (\arctan y)/\pi$ lies in $(0, 1)$ for every real y. This proves both injectivity and surjectivity.

Exercise 0.16: Four power sets

The empty subset must be included, even when the original set is empty.

(a) $\mathcal{P}(\varnothing) = \{\varnothing\}$ has 1 element.

(b) $\mathcal{P}(\{a\}) = \{\varnothing, \{a\}\}$ has 2 elements.

(c) $\mathcal{P}(\{a, b\}) = \{\varnothing, \{a\}, \{b\}, \{a, b\}\}$ has 4 elements.

(d) $\mathcal{P}(\{a, b, c\}) = \{\varnothing, \{a\}, \{b\}, \{c\}, \{a, b\}, \{a, c\}, \{b, c\}, \{a, b, c\}\}$ has 8 elements.

Exercise 0.17: Counting subsets

If $|A| = s$, then $|\mathcal{P}(A)| = 2^s$. Label the elements $a_1, \ldots, a_s$. To specify a subset, decide independently for each a_i whether to include it. There are two choices at each of s positions, hence 2^s subsets. This

also covers $s = 0$, when there is just the empty subset.

Exercise 0.18: Subsets and binary-valued functions

For $f : A \to \{0, 1\}$ define $T(f) = \{a \in A : f(a) = 1\}$. If $T(f) = T(g)$, then for every a either both functions equal 1 or both equal 0, so $f = g$. Given a subset $U \subseteq A$, its characteristic function χ_U, equal to 1 on U and 0 elsewhere, satisfies $T(\chi_U) = U$. Thus T is a bijection, for finite or infinite A.

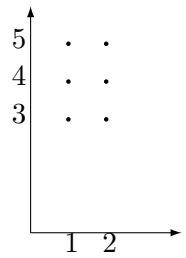
Exercise 0.19: Cantor diagonalization

Suppose $f : A \to \mathcal{P}(A)$ were onto. Define $D = \{a \in A : a \notin f(a)\}$. If $D = f(d)$, then $d \in D$ holds exactly when $d \notin f(d) = D$, a contradiction. Hence no such surjection exists, while $a \mapsto \{a\}$ is injective. Repeated power sets therefore give strictly increasing infinite cardinalities starting with $\mathbb{Z}$. A set U containing everything would contain every subset of itself, so $\mathcal{P}(U) \subseteq U$; this contradicts the same diagonal argument (or forms the contradictory set $\{x \in U : x \notin x\}$). Thus a universal set is not acceptable in ordinary set theory.

Exercise 0.20: Cardinal sums and products

(a) Since A, B are disjoint, their union has the five elements $1, 2, 3, 4, 5$. Similarly adjoining three new objects to $\mathbb{Z}^+$ still permits one infinite list, so $3 + \aleph_0 = \aleph_0$. Two disjoint copies of $\mathbb{Z}^+$ interleave, for example $(n, 0) \mapsto 2n - 1$, $(n, 1) \mapsto 2n$; hence $\aleph_0 + \aleph_0 = \aleph_0$.

(b) $A \times B$ has the six plotted positions $(1, 3), (1, 4), (1, 5), (2, 3), (2, 4), (2, 5)$. An infinite positive-integer grid can be listed diagonal by diagonal, so $\aleph_0\aleph_0 = \aleph_0$.



Exercise 0.21: Infinite strings of digits

There are $10^2 = 100$ two-place strings and $10^5 = 100000$ five-place strings. Each fixed-length string gives a distinct number in $[0, 1)$, allowing leading and trailing zeros. Infinite decimal strings give $10^{\aleph_0} = |\mathbb{R}|$: every point of $[0, 1)$ has a decimal expansion, and the ambiguity of terminating decimals does not change this cardinality. More explicitly $2^{\aleph_0} \le 10^{\aleph_0} \le (2^4)^{\aleph_0} = 2^{\aleph_0}$, and binary strings inject into $[0, 1]$ as ternary expansions using only digits $0, 2$. Since $2 \le 12 \le 2^4$, the same argument gives $12^{\aleph_0} = 2^{\aleph_0} = |\mathbb{R}|$.

Exercise 0.22: An increasing cardinal sequence

Cantor's theorem says $|X| < 2^{|X|}$. Since $|\mathbb{R}| = 2^{\aleph_0}$, the requested continuation is

$$\aleph_0, \quad |\mathbb{R}|, \quad 2^{|\mathbb{R}|}, \quad 2^{2^{|\mathbb{R}|}}, \quad 2^{2^{2^{|\mathbb{R}|}}}.$$

Each next term is the cardinality of the power set of a set having the preceding cardinality.

Exercise 0.23: Counting partitions of 1 element

The number is $B_1 = 1$, where B_n denotes the number of partitions of an n-element set. The sole partition is $\{\{a\}\}$.

Exercise 0.24: Counting partitions of 2 elements

The number is $B_2 = 2$, where B_n denotes the number of partitions of an n-element set. They are $\{\{a, b\}\}$ and $\{\{a\}, \{b\}\}$.

Exercise 0.25: Counting partitions of 3 elements

The number is $B_3 = 5$, where B_n denotes the number of partitions of an n-element set. The cell-size patterns are 3 (one partition), $2 + 1$ (three choices for the singleton), and $1 + 1 + 1$ (one partition), giving $1 + 3 + 1 = 5$.

Exercise 0.26: Counting partitions of 4 elements

The number is $B_4 = 15$, where B_n denotes the number of partitions of an n-element set. The patterns $4, 3 + 1, 2 + 2, 2 + 1 + 1, 1 + 1 + 1 + 1$ contribute respectively $1, 4, 3, 6, 1$: for two pairs divide the six choices of a first pair by two. Their sum is 15.

Exercise 0.27: Counting partitions of 5 elements

The number is $B_5 = 52$, where B_n denotes the number of partitions of an n-element set. Choose the cell containing a distinguished element. If exactly k of the other four elements lie outside that cell, choose them in $\binom{4}{k}$ ways and partition them in B_k ways. With $B_0 = 1$, this gives $B_5 = \sum_{k=0}^{4} \binom{4}{k} B_k = 1 + 4 + 12 + 20 + 15 = 52$.

Exercise 0.28: A partition gives an equivalence relation

Every element x belongs to the same cell as itself, proving reflexivity. For transitivity, suppose x, y share a cell C and y, z share a cell D. A partition puts y into exactly one cell, so $C = D$. Therefore x, z share a cell. Together with the symmetry already discussed in the text, all three properties hold.

Exercise 0.29: A strictly positive product

This relation is not reflexive on $\mathbb{Z}$: $0 \cdot 0 = 0$ is not strictly positive. Thus it is not an equivalence relation on the stated domain. On $\mathbb{Z} \setminus \{0\}$ it would have the two classes of positive and negative integers, but removing zero changes the question.

Exercise 0.30: The order relation

The relation $x \geq y$ is reflexive and transitive, but is not symmetric: $1 \geq 0$ while $0 \not\geq 1$. Hence it is not an equivalence relation, and no equivalence-class partition is associated to it.

Exercise 0.31: Equal absolute values

Equality of absolute values is reflexive, symmetric, and transitive because equality itself has those properties. The class of x is $\{x, -x\}$, with the special singleton class $\{0\}$. These classes partition $\mathbb{R}$.

Exercise 0.32: Distance at most three

Reflexivity holds since $|x - x| = 0$, and symmetry holds since $|x - y| = |y - x|$. Transitivity fails: $|0 - 3| = 3$ and $|3 - 6| = 3$, but $|0 - 6| = 6 > 3$. Thus this is not an equivalence relation.

Exercise 0.33: Equal numbers of decimal digits

The number of decimal digits is a function from $\mathbb{Z}^+$ to $\mathbb{Z}^+$. Equality of its values is an equivalence relation. The classes are

$$\{1, \ldots, 9\}, \quad \{10, \ldots, 99\}, \quad \{100, \ldots, 999\}, \quad \ldots,$$

or $\{10^{k-1}, \ldots, 10^k - 1\}$ for $k \geq 1$.

Exercise 0.34: Equal final digits

Two positive integers have the same final digit exactly when they have the same remainder modulo 10. Equality of remainders is reflexive, symmetric, and transitive. The ten classes are $\{10, 20, 30, \ldots\}$ and $\{r, r + 10, r + 20, \ldots\}$ for $r = 1, \ldots, 9$.

Exercise 0.35: Positive residue classes

(a) For $n = 2$: $\{2, 4, 6, \ldots\}$ and $\{1, 3, 5, \ldots\}$.

(b) For $n = 3$: $\{3, 6, 9, \ldots\}$, $\{1, 4, 7, \ldots\}$, $\{2, 5, 8, \ldots\}$.

(c) For $n = 5$: $\{5, 10, 15, \ldots\}$, $\{1, 6, 11, \ldots\}$, $\{2, 7, 12, \ldots\}$, $\{3, 8, 13, \ldots\}$, $\{4, 9, 14, \ldots\}$.

The class beginning with n is the zero-remainder class, even though zero is absent from $\mathbb{Z}^+$.

Exercise 0.36: Congruence on all integers

(a) $r - r = n \cdot 0$ proves reflexivity. If $r - s = nq$, then $s - r = n(-q)$ proves symmetry. If $r - s = nq$ and $s - t = np$, then $r - t = n(q + p)$ proves transitivity.

(b) Write $r = nq + a$, $s = np + b$ with $0 \leq a, b < n$. Their difference is divisible by n exactly when $a - b$ is; since $|a - b| < n$, this happens exactly when $a = b$. Thus the restriction is equality of remainders.

(c) For $n = 2$ the classes are $\{\ldots, -4, -2, 0, 2, 4, \ldots\}$ and $\{\ldots, -3, -1, 1, 3, \ldots\}$. For $n = 3$ they are $\{\ldots, -6, -3, 0, 3, 6, \ldots\}$, $\{\ldots, -5, -2, 1, 4, 7, \ldots\}$, $\{\ldots, -4, -1, 2, 5, 8, \ldots\}$. For $n = 5$ they are $\{\ldots, -10, -5, 0, 5, 10, \ldots\}$, $\{\ldots, -9, -4, 1, 6, 11, \ldots\}$, $\{\ldots, -8, -3, 2, 7, 12, \ldots\}$, $\{\ldots, -7, -2, 3, 8, 13, \ldots\}$, and $\{\ldots, -6, -1, 4, 9, 14, \ldots\}$.

Exercise 0.37: What injectivity actually says

Every function already sends one input to one output. Injectivity adds that *two different inputs must produce two different outputs*: $a \neq b \Rightarrow f(a) \neq f(b)$. Calling this "two-to-two" would emphasize comparing two distinct inputs rather than describing the image of just one input. The logically equivalent test is $f(a) = f(b) \Rightarrow a = b$; no assumption of being onto is involved.

1 Introduction and Examples

Exercise 1.1: A cube of i

$i^3 = i^2 i = -i$, so the answer is $0 - i$.

Exercise 1.2: A fourth power of i

$i^4 = (i^2)^2 = (-1)^2 = 1$, or $1 + 0i$.

Exercise 1.3: A high power of i

Since $23 = 4 \cdot 5 + 3$ and $i^4 = 1$, $i^{23} = (i^4)^5 i^3 = -i$.

Exercise 1.4: A power of minus i

$(-i)^{35} = (-1)^{35} i^{35} = -i^3 = i$, because $35 \equiv 3 \pmod 4$.

Exercise 1.5: Multiplying two binomials

Distribute and use $i^2 = -1$: $(4 - i)(5 + 3i) = 20 + 12i - 5i - 3i^2 = 23 + 7i$.

Exercise 1.6: Another binomial product

$(8 + 2i)(3 - i) = 24 - 8i + 6i - 2i^2 = 26 - 2i$. The real contribution from $-2i^2$ is $+2$.

Exercise 1.7: A product plus a complex number

$(2 - 3i)(4 + i) = 8 + 2i - 12i - 3i^2 = 11 - 10i$. Adding $6 - 5i$ yields $17 - 15i$.

Exercise 1.8: Cubing one plus i

First $(1 + i)^2 = 1 + 2i + i^2 = 2i$. Thus $(1 + i)^3 = 2i(1 + i) = -2 + 2i$.

Exercise 1.9: A fifth power by the binomial theorem

Expand $(1 - i)^5 = 1 - 5i + 10i^2 - 10i^3 + 5i^4 - i^5$. Replacing $i^2 = -1, i^3 = -i, i^4 = 1, i^5 = i$ gives $(1 - 10 + 5) + (-5 + 10 - 1)i = -4 + 4i$.

Exercise 1.10: The modulus of three minus four i

For $a + bi$ the modulus is $\sqrt{a^2 + b^2}$. Hence $|3 - 4i| = \sqrt{9 + 16} = 5$.

Exercise 1.11: The modulus of six plus four i

$|6 + 4i| = \sqrt{36 + 16} = \sqrt{52} = 2\sqrt{13}$. The modulus is nonnegative.

Exercise 1.12: Polar form: Three minus four i

Compute $r = |3 - 4i| = 5$ and divide the real and imaginary coordinates by r. This gives

$$3 - 4i = 5\left(3/5 + (-4/5)\, i\right).$$

The two displayed normalized coordinates have squared sum 1, so the parenthesized complex number lies on the unit circle. This is the requested form $r(p + qi)$.

Exercise 1.13: Polar form: Minus one plus i

Compute $r = |-1 + i| = \sqrt{2}$ and divide the real and imaginary coordinates by r. This gives

$$-1 + i = \sqrt{2}\left(-1/\sqrt{2} + \left(1/\sqrt{2}\right) i\right).$$

The two displayed normalized coordinates have squared sum 1, so the parenthesized complex number lies on the unit circle. This is the requested form $r(p + qi)$.

Exercise 1.14: Polar form: Twelve plus five i

Compute $r = |12 + 5i| = 13$ and divide the real and imaginary coordinates by r. This gives

$$12 + 5i = 13\left(12/13 + (5/13)\, i\right).$$

The two displayed normalized coordinates have squared sum 1, so the parenthesized complex number lies on the unit circle. This is the requested form $r(p + qi)$.

Exercise 1.15: Polar form: Minus three plus five i

Compute $r = |-3 + 5i| = \sqrt{34}$ and divide the real and imaginary coordinates by r. This gives

$$-3 + 5i = \sqrt{34}\left(-3/\sqrt{34} + \left(5/\sqrt{34}\right) i\right).$$

The two displayed normalized coordinates have squared sum 1, so the parenthesized complex number lies on the unit circle. This is the requested form $r(p + qi)$.

Exercise 1.16: Fourth roots of one

Write $z = re^{i\theta}$. Then $r^4 = 1$, so $r = 1$, and $4\theta = 2\pi k$. Distinct roots occur for $k = 0, 1, 2, 3$:

$$z = e^{i\pi k/2} \in \{1, i, -1, -i\}.$$

Larger integers repeat these roots; a fourth-degree polynomial has no more than four distinct roots.

Exercise 1.17: Fourth roots of minus one

Since $-1 = e^{i(\pi+2\pi k)}$, the roots are

$$z = e^{i(\pi/4+k\pi/2)}, \quad k = 0, 1, 2, 3.$$

In Cartesian form these are $(1+i)/\sqrt{2}$, $(-1+i)/\sqrt{2}$, $(-1-i)/\sqrt{2}$, and $(1-i)/\sqrt{2}$. Each has modulus 1 and fourth power -1.

Exercise 1.18: Cube roots of minus eight

Here $-8 = 8e^{i(\pi+2\pi k)}$. Thus $r = 2$ and $\theta = \pi/3 + 2\pi k/3$. For $k = 0, 1, 2$ the roots are $1 + i\sqrt{3}$, -2, and $1 - i\sqrt{3}$. Distinct arguments modulo 2π give all three solutions.

Exercise 1.19: Cube roots of minus twenty-seven i

Use $-27i = 27e^{i(-\pi/2+2\pi k)}$. The roots are $3e^{i(-\pi/6+2\pi k/3)}$ for $k = 0, 1, 2$, namely

$$\frac{3\sqrt{3}}{2} - \frac{3}{2}i, \qquad 3i, \qquad -\frac{3\sqrt{3}}{2} - \frac{3}{2}i.$$

Cubing triples each argument and cubes the modulus, giving $-27i$ in every case.

Exercise 1.20: Sixth roots of one

The six roots are $e^{i\pi k/3}$, $k = 0, \ldots, 5$. Explicitly:

$$1, \quad \tfrac{1}{2} + \tfrac{\sqrt{3}}{2}i, \quad -\tfrac{1}{2} + \tfrac{\sqrt{3}}{2}i, \quad -1, \quad -\tfrac{1}{2} - \tfrac{\sqrt{3}}{2}i, \quad \tfrac{1}{2} - \tfrac{\sqrt{3}}{2}i.$$

They follow from $r^6 = 1$ and $6\theta \in 2\pi\mathbb{Z}$.

Exercise 1.21: Sixth roots of minus sixty-four

Here $r^6 = 64$ gives $r = 2$ and $6\theta = \pi + 2\pi k$. Thus $z = 2e^{i(\pi/6+k\pi/3)}$, $k = 0, \ldots, 5$, or

$$\sqrt{3} + i, \quad 2i, \quad -\sqrt{3} + i, \quad -\sqrt{3} - i, \quad -2i, \quad \sqrt{3} - i.$$

These have six distinct arguments modulo 2π.

Exercise 1.22: Addition modulo seventeen

$10 + 16 = 26 = 17 + 9$, so $10 +_{17} 16 = 9$. In modular addition, choose the unique representative between zero inclusive and the modulus exclusive.

Exercise 1.23: Addition modulo ten

$8 + 6 = 14 = 10 + 4$, so $8 +_{10} 6 = 4$. In modular addition, choose the unique representative between zero inclusive and the modulus exclusive.

Exercise 1.24: Real modular addition

$20.5 + 19.3 = 39.8 = 25 + 14.8$, so the representative in $[0, 25)$ is 14.8. In modular addition, choose the unique representative between zero inclusive and the modulus exclusive.

Exercise 1.25: Addition modulo one

$1/2 + 7/8 = 11/8 = 1 + 3/8$, hence $1/2 +_1 7/8 = 3/8$. In modular addition, choose the unique representative between zero inclusive and the modulus exclusive.

Exercise 1.26: Angles modulo two pi

$3\pi/4 + 3\pi/2 = 9\pi/4 = 2\pi + \pi/4$, so the result is $\pi/4$. In modular addition, choose the unique representative between zero inclusive and the modulus exclusive.

Exercise 1.27: An irrational modulus

$2\sqrt{2} + 3\sqrt{2} = 5\sqrt{2}$, and $\sqrt{32} = 4\sqrt{2}$. Subtracting that modulus once leaves $\sqrt{2} \in [0, 4\sqrt{2})$, the answer. In modular addition, choose the unique representative between zero inclusive and the modulus exclusive.

Exercise 1.28: Inputs outside the underlying set

The operation $+_6$ on $\mathbb{R}_6 = [0, 6)$ accepts only ordered pairs of elements of that set. The number 8 is not an element of $\mathbb{R}_6$, so $5 +_6 8$ is not an operation on the stated domain. One could first replace 8 by 2, but that is an additional reduction, not the given expression as written.

Exercise 1.29: A linear equation modulo fifteen

The congruence is $x + 7 \equiv 3 \pmod{15}$, so $x \equiv -4 \equiv 11 \pmod{15}$. The only representative in $\mathbb{Z}_{15}$ is 11, and $11 +_{15} 7 = 3$.

Exercise 1.30: An angular equation

Subtract $3\pi/2$ from $3\pi/4$: $x \equiv -3\pi/4 \pmod{2\pi}$. Adding 2π gives $x = 5\pi/4 \in [0, 2\pi)$. Its sum with $3\pi/2$ is $11\pi/4$, whose remainder is $3\pi/4$.

Exercise 1.31: Twice a residue modulo seven

We need $2x \equiv 3 \pmod{7}$. Since $2 \cdot 4 \equiv 1$, multiplication by 4 gives $x \equiv 12 \equiv 5$. Checking, $5 +_7 5 = 3$. The inverse of 2 makes the answer unique.

Exercise 1.32: Three times a residue modulo seven

The equation is $3x \equiv 5 \pmod 7$. Since $3 \cdot 5 \equiv 1$, we get $x \equiv 25 \equiv 4$. Indeed $4 + 4 + 4 = 12 \equiv 5 \pmod 7$.

Exercise 1.33: Twice a residue modulo twelve

The equation $2x \equiv 2 \pmod{12}$ means $12 \mid 2(x-1)$, equivalently $6 \mid x - 1$. Representatives in $\mathbb{Z}_{12}$ are therefore $x = 1$ and $x = 7$. Both give remainder 2 when doubled. Dividing a congruence here also divides its modulus.

Exercise 1.34: Four times a residue modulo four

Every residue $x \in \{0, 1, 2, 3\}$ satisfies $4x \equiv 0 \pmod 4$. Hence all four residues are solutions; the coefficient has no multiplicative inverse modulo 4.

Exercise 1.35: An isomorphism from eighth roots

Multiplication of powers corresponds to addition, so the image of ζ^m must be $5m \pmod 8$. For the six requested exponents:

m	0	3	4	5	6	7
$5m \bmod 8$	0	7	4	1	6	3

This also gives $m = 2 \mapsto 2$, as required in the question. Since $\gcd(5, 8) = 1$, all eight images are distinct.

Exercise 1.36: An isomorphism from seventh roots

The image of ζ^m is $4m \pmod 7$ because powers become repeated sums. Thus

m	0	2	3	4	5	6
$4m \bmod 7$	0	1	5	2	6	3

Multiplication by 4 is a permutation of $\mathbb{Z}_7$, since 4 is relatively prime to 7.

Exercise 1.37: Why the proposed image is impossible

The element $\zeta = e^{i\pi/3}$ generates all six roots, so its image under an isomorphism must generate $\mathbb{Z}_6$. But 4 generates only $\{0, 4, 2\}$, since $3 \cdot 4 \equiv 0 \pmod 6$. Alternatively, $\zeta^3 = -1 \neq 1$ would map to 0, violating injectivity. Thus no isomorphism can have this image.

Exercise 1.38: Angle addition from Euler formula

Compute the product in two ways:

$$e^{ia}e^{ib} = e^{i(a+b)} = \cos(a + b) + i \sin(a + b),$$

whereas direct multiplication gives

$$(\cos a + i\sin a)(\cos b + i\sin b) = (\cos a\cos b - \sin a\sin b) + i(\sin a\cos b + \cos a\sin b).$$

Equality of real parts proves the cosine identity; equality of imaginary parts proves the sine identity. This argument explains why multiplying unit complex numbers adds their angles.

Exercise 1.39: Multiplying polar forms

Put $r_j = |z_j|$. Distributing the factors gives

$$z_1 z_2 = r_1 r_2\{(\cos\theta_1\cos\theta_2 - \sin\theta_1\sin\theta_2) + i(\sin\theta_1\cos\theta_2 + \cos\theta_1\sin\theta_2)\}.$$

The two addition identities replace the braces by $\cos(\theta_1+\theta_2) + i\sin(\theta_1+\theta_2)$. Thus the moduli multiply and the angles add. If one number is zero, the same product formula is zero regardless of the chosen angle.

Exercise 1.40: The triple-angle identity

(a) Cubing Euler's formula gives $e^{3i\theta} = (\cos\theta + i\sin\theta)^3$. Its real part is $\cos^3\theta - 3\cos\theta\sin^2\theta$, since the terms with odd powers of i are imaginary and $i^2 = -1$. Hence $\cos 3\theta = \cos^3\theta - 3\cos\theta\sin^2\theta$.

(b) Substitute $\sin^2\theta = 1 - \cos^2\theta$ to obtain $\cos 3\theta = \cos^3\theta - 3\cos\theta(1-\cos^2\theta) = 4\cos^3\theta - 3\cos\theta$.

Exercise 1.41: Euler's formula from power series

Start with $e^z = \sum_{n=0}^{\infty} z^n/n!$ and substitute $z = i\theta$. Separate even and odd powers using $i^{2k} = (-1)^k$ and $i^{2k+1} = i(-1)^k$:

$$\begin{aligned} e^{i\theta} &= \sum_{k=0}^{\infty} \frac{(-1)^k\theta^{2k}}{(2k)!} + i\sum_{k=0}^{\infty}\frac{(-1)^k\theta^{2k+1}}{(2k+1)!} \\ &= \cos\theta + i\sin\theta. \end{aligned}$$

The two real series are exactly the given cosine and sine expansions. This is the requested formal derivation; their absolute convergence for every real θ also justifies separating the two subseries analytically.

2 Binary Operations

Exercise 2.1: Reading the operation table

Use the first factor's row and the second factor's column. Table 2.26 gives $b * d = e$ and $c * c = b$. For the nested expression, work from the innermost parentheses: $a * c = c$, then $c * e = a$, then $a * a = a$. Thus $[(a * c) * e] * a = a$.

Exercise 2.2: A test that happens to agree

The table gives $(a * b) * c = b * c = a$ and $a * (b * c) = a * a = a$. These values agree, but one successful triple does not establish associativity: that law must hold for every ordered triple. This computation alone therefore cannot decide whether the operation is associative.

Exercise 2.3: A failed associativity test

Since $b * d = e$, $(b * d) * c = e * c = a$. On the other hand $d * c = b$, so $b * (d * c) = b * b = c$. The elements a, c differ, so this one triple disproves associativity.

Exercise 2.4: Testing commutativity

The operation is not commutative: $a * e = d$ whereas $e * a = b$. A commutative table must be symmetric across its main diagonal; these two off-diagonal entries already violate that requirement.

Exercise 2.5: A symmetric completion

Reflect each known entry across the diagonal, leaving the given diagonal entries unchanged. This forces

$*$	a	b	c	d
a	a	b	c	d
b	b	d	a	c
c	c	a	d	b
d	d	c	b	a

For instance $c * b = a$ forces $b * c = a$, and $b * d = c$ forces $d * b = c$. Every resulting entry is in the specified set and the table is symmetric, as required.

Exercise 2.6: An associative completion

The known entry $c * b = d$ lets us compute every missing product: associativity requires $d * x = (c * b) * x = c * (b * x)$. For $x = a, b, c, d$, the given row of b is b, a, c, d; applying the given row of c gives d, c, c, d. Thus

$*$	a	b	c	d
a	a	b	c	d
b	b	a	c	d
c	c	d	c	d
d	d	c	c	d

The question permits assuming an associative completion exists. The preceding equations show it must be this one; direct checking of its 4^3 triples confirms it is associative.

Exercise 2.7: Subtraction on the integers

This is neither commutative nor associative. For commutativity, $1 * 2 = -1$ while $2 * 1 = 1$. For associativity, $(1 * 2) * 3 = -1 - 3 = -4$, whereas $1 * (2 * 3) = 1 - (-1) = 2$. Every output is still an integer, so it is a binary operation despite these failed laws.

Exercise 2.8: Multiplication followed by adding one

Commutativity holds because $ab + 1 = ba + 1$. However

$$(a * b) * c = (ab + 1)c + 1 = abc + c + 1, \qquad a * (b * c) = a(bc + 1) + 1 = abc + a + 1.$$

Taking $a = 0, b = 0, c = 1$ gives 2 versus 1, so associativity fails.

Exercise 2.9: A scaled product

Commutativity follows from $ab/2 = ba/2$. For every rational a, b, c,

$$(a * b) * c = ((ab)/2)c/2 = abc/4 = a((bc)/2)/2 = a * (b * c).$$

Thus the operation is both commutative and associative. Products and division by 2 keep all outputs rational.

Exercise 2.10: Two raised to a product

The operation is commutative since $ab = ba$. It is not associative: with $(a, b, c) = (1, 1, 3)$,

$$(1 * 1) * 3 = 2 * 3 = 2^6 = 64, \qquad 1 * (1 * 3) = 1 * 8 = 2^8 = 256.$$

All outputs are positive integers, so closure is not an issue.

Exercise 2.11: Exponentiation

This operation is not commutative: $2^3 = 8 \neq 9 = 3^2$. It is not associative either:

$$(2 * 2) * 3 = 4^3 = 64, \qquad 2 * (2 * 3) = 2^8 = 256.$$

The distinction is between $(a^b)^c = a^{bc}$ and a^{b^c}, not a failure of closure.

Exercise 2.12: Counting all binary operations

For a set of size n, there are n^2 ordered input pairs and independently n choices for each output. Thus there are n^{n^2} operations. For one, two, and three elements this gives respectively 1, $2^4 = 16$, and $3^9 = 19683$. The table is an arbitrary function from $S \times S$ into S.

Exercise 2.13: Counting commutative operations

Only the diagonal and one half of the off-diagonal entries can be chosen freely. There are $n + n(n-1)/2 = n(n+1)/2$ such positions, each with n possible outputs, so the answer is $n^{n(n+1)/2}$. It is $2^3 = 8$ for two elements and $3^6 = 729$ for three elements. The remaining entries are forced by symmetry.

Exercise 2.14: A precise commutativity definition

A binary operation $*$ on a set S is commutative if and only if $a * b = b * a$ *for all* $a, b \in S$. The universal quantifier and the underlying set must be specified: agreement for just one pair does not suffice.

Exercise 2.15: An acceptable associativity definition

The given definition is correct: $(b * c) * a = b * (c * a)$ for all $a, b, c \in S$ is the associative law with the dummy variables renamed. For the more usual arrangement, one may write $(x * y) * z = x * (y * z)$ for all $x, y, z \in S$. No correction of its mathematical content is needed.

Exercise 2.16: Closure of a subset

A subset $H \subseteq S$ is closed under $*$ if and only if $a * b \in H$ for all $a, b \in H$. The proposed version incorrectly quantifies over all of S. Closure only asks about two inputs already in H.

Exercise 2.17: A subtraction rule on positive integers

The numerical output $a - b$ is uniquely determined for every input, so Condition 1 holds. Condition 2, closure, fails: $1 * 1 = 0 \notin \mathbb{Z}^+$ (and $1 * 2 = -1$ is another failure). Thus this does not define a binary operation on $\mathbb{Z}^+$.

Exercise 2.18: An exponentiation rule on positive integers

For each positive integer pair a, b, the integer a^b exists uniquely and is positive. Both uniqueness/everywhere-definition (Condition 1) and closure (Condition 2) hold. Hence this is a binary operation on the given set.

Exercise 2.19: A subtraction rule on all real numbers

For any two real numbers a, b, the difference $a - b$ is one uniquely determined real number. Therefore Conditions 1 and 2 both hold; subtraction is a binary operation on $\mathbb{R}$. Associativity is not required in the definition of binary operation.

Exercise 2.20: The least integer above two inputs

For positive integers a, b, the specified integer is exactly $\max(a, b) + 1$. It exists, is unique, and is positive. Thus both conditions hold and the rule defines a binary operation.

Exercise 2.21: An ambiguous lower bound

The condition permits every integer $c \geq a + b + 5$. For example, when $a = b = 1$, both 7 and 8 are allowed. Thus Condition 1 fails because no unique output is specified. All allowed outputs are positive, so there is no violation of Condition 2. The rule is not a binary operation.

Exercise 2.22: The largest integer below a product

Since ab is an integer, the largest integer strictly below it is uniquely $ab - 1$. Thus Condition 1 holds. But $1 * 1 = 0$ is not positive, so Condition 2 fails and the rule is not a binary operation on $\mathbb{Z}^+$.

Exercise 2.23: A matrix-shaped subset

Write $M(a, b) = \begin{pmatrix} a & -b \\ b & a \end{pmatrix}$. For arbitrary real a, b, c, d,

$$M(a, b) + M(c, d) = M(a + c, b + d),$$

so **(a)** it is closed under addition. Also

$$M(a, b)M(c, d) = \begin{pmatrix} ac - bd & -(ad + bc) \\ ad + bc & ac - bd \end{pmatrix} = M(ac - bd, ad + bc),$$

so **(b)** it is closed under multiplication. Exhibiting the same form proves closure for all inputs.

Exercise 2.24: Ten statements about operations

(a) False: addition on $\mathbb{Z}$ has $1 + 1 \neq 1$.

(b) True: treat $b * c \in S$ as one element and apply commutativity to it and a.

(c) False: matrix multiplication is associative but generally noncommutative; specifically take $a = E_{12}$, $b = E_{21}$, $c = I_2$. Then $a(bc) = E_{11}$ but $(bc)a = E_{22}$.

(d) False: function composition, set union, and matrix multiplication are important operations on nonnumeric objects.

(e) False: the equation must hold for every pair. Subtraction has $0-0 = 0-0$ but is not commutative.

(f) True: on $\{s\}$ every product must be s, so both laws hold.

(g) True: exactly one output in particular means at least one.

(h) True: exactly one output in particular means at most one.

(i) True: this is the defining requirement, together with the specified codomain.

(j) False: multiple allowed outputs contradict being a function.

Exercise 2.25: Two operations on a nonnumeric set

Take S to be the set of all finite words in the alphabet $\{A, B\}$, including the empty word. Define $u * v$ to be concatenation and $u *' v = u$. Each ordered pair has exactly one output in S. They are different: for $u = A, v = B$, concatenation gives AB, while the second operation gives A. The alphabet and finiteness requirement make the set unambiguous.

Exercise 2.26: Rearranging four factors

Use only the stated laws, one step at a time:

$$\begin{aligned}(a * b) * (c * d) &= (c * d) * (a * b) && \text{(commutativity)}\\ &= ((c * d) * a) * b && \text{(associativity on } c * d, a, b)\\ &= ((d * c) * a) * b && \text{(commutativity on } c, d).\end{aligned}$$

Each parenthesized product is an element of S by closure, so applying the laws to such intermediate elements is legitimate.

Exercise 2.27: The singleton case

Let $S = \{s\}$. The only possible output of $s * s$ is s. Hence $s * s = s * s$ proves commutativity, and $(s * s) * s = s * s = s = s * (s * s)$ proves associativity. These exhaust all pairs and triples because the set has one element.

Exercise 2.28: Commutativity need not imply associativity

On $\{0, 1\}$ define $a * b = 1 - ab$ (the values are again 0 or 1). It is commutative because $ab = ba$. But $(0 * 0) * 1 = 1 * 1 = 0$, while $0 * (0 * 1) = 0 * 1 = 1$. Thus this commutative two-element operation is not associative.

Exercise 2.29: Associativity of function addition

For every real x, $((f + g) + h)(x) = (f(x) + g(x)) + h(x) = f(x) + (g(x) + h(x)) = (f + (g + h))(x)$. The middle equality is associativity of real addition. Since the two functions agree at every input, they are equal, so function addition is associative.

Exercise 2.30: Commutativity of function subtraction

False. Let $f(x) = 1$ and $g(x) = 0$ for every real x. Then $(f - g)(x) = 1$ while $(g - f)(x) = -1$, so the resulting functions differ. Constant functions are allowed elements of F.

Exercise 2.31: Associativity of function subtraction

False. Let f, g, h all be the constant function 1. Then $(f - g) - h$ is the constant -1, whereas $f - (g - h)$ is the constant 1. This counterexample uses functions with exactly the required domain and codomain.

Exercise 2.32: Commutativity of function multiplication

For every real x, $(fg)(x) = f(x)g(x) = g(x)f(x) = (gf)(x)$ by commutativity of real multiplication. Pointwise agreement proves $fg = gf$ as functions. Thus the statement is true.

Exercise 2.33: Associativity of function multiplication

For every x, $((fg)h)(x) = (f(x)g(x))h(x) = f(x)(g(x)h(x)) = (f(gh))(x)$. Associativity of real multiplication gives the central equality. Thus the two functions agree on all of $\mathbb{R}$, proving the assertion.

Exercise 2.34: Commutativity of composition

False. Set $f(x) = x + 1$ and $g(x) = 2x$. Then $(f \circ g)(x) = 2x + 1$ but $(g \circ f)(x) = 2x + 2$. The order of applying functions matters, even though composition is associative.

Exercise 2.35: An arbitrary distributive law

False: choose both operations to be ordinary addition on $\mathbb{Z}$. With $a = 1, b = c = 0$, the left side $a + (b + c)$ is 1, whereas $(a + b) + (a + c)$ is 2. Distributivity is an extra relation between operations; it does not follow from their merely being binary operations.

Exercise 2.36: Closure of elements commuting with everything

Let $a, b \in H$ and $x \in S$. Reassociate and use that a, b commute with x:

$$(a * b) * x = a * (b * x) = a * (x * b) = (a * x) * b = (x * a) * b = x * (a * b).$$

Since this holds for every $x \in S$, the element $a * b$ belongs to H. No commutativity of the whole set is assumed; we only used the defining property of these two elements.

Exercise 2.37: Closure of idempotents

Take $a, b \in H$, so $a * a = a$ and $b * b = b$. Associativity and commutativity allow regrouping as

$$(a * b) * (a * b) = (a * a) * (b * b) = a * b.$$

For justification, first write $a*(b*(a*b))$, commute b past a inside, then reassociate into $(a*a)*(b*b)$. Thus $a * b$ is idempotent and lies in H.

3 Isomorphic Binary Structures

Exercise 3.1: The three isomorphism checks

Check that the map is (1) one-to-one, (2) onto the specified target set, and (3) operation-preserving: $\phi(a * b) = \phi(a) *' \phi(b)$ for every $a, b \in S$. The first two make a bijection; the third ensures that it respects the algebraic structure. Also confirm initially that the formula actually takes values in the stated codomain.

Exercise 3.2: Negation on the integers

This is an isomorphism. If $-m = -n$ then $m = n$, proving injectivity. Each integer y is the image of $-y$, proving surjectivity. Finally $\phi(m+n) = -(m+n) = -m-n = \phi(m)+\phi(n)$, proving preservation of addition.

Exercise 3.3: Doubling integers

This map preserves addition since $2(m + n) = 2m + 2n$ and is injective because $2m = 2n$ implies $m = n$. However its image consists only of even integers; 1 has no preimage in $\mathbb{Z}$. Thus it is not onto and is not an isomorphism onto the specified target.

Exercise 3.4: Translating integers

The map $n \mapsto n + 1$ is bijective, with inverse $y \mapsto y - 1$, but does not preserve addition. Indeed $\phi(0 + 0) = 1$, while $\phi(0) + \phi(0) = 2$. A bijection alone is not an isomorphism.

Exercise 3.5: Halving rational numbers

If $x/2 = y/2$, then $x = y$, so the map is injective. A rational target q has rational preimage $2q$, so it is onto. Also $(x + y)/2 = x/2 + y/2$. All three requirements hold, making this an additive isomorphism.

Exercise 3.6: Squaring rational numbers

Squaring preserves multiplication: $(xy)^2 = x^2y^2$. It is not injective, since 1 and -1 have the same image, and is not onto $\mathbb{Q}$, since no rational square equals -1. Hence it is not an isomorphism.

Exercise 3.7: Cubing real numbers

The cubic function is strictly increasing, so it is injective, and every real y has real preimage $\sqrt[3]{y}$, so it is onto. Since $(xy)^3 = x^3y^3$, it preserves multiplication. Thus it is an isomorphism of the indicated multiplicative binary structures, even though the structures include zero and are not groups.

Exercise 3.8: Determinants of two by two matrices

Determinants preserve multiplication: $\det(AB) = \det(A)\det(B)$. The map is onto since $\det \operatorname{diag}(r, 1) = r$. It is not injective: the distinct matrices I_2 and $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ both have determinant 1. Thus it is not an isomorphism.

Exercise 3.9: Determinants of one by one matrices

A one by one matrix $[r]$ has determinant r. The map $[r] \mapsto r$ is bijective, with inverse $r \mapsto [r]$. Also $[r][s] = [rs]$, whose determinant is rs. Consequently this determinant map is an isomorphism.

Exercise 3.10: A decreasing exponential

The codomain here is $\mathbb{R}^+$, the positive reals. The function $\phi(r) = (1/2)^r$ is strictly decreasing and has inverse $y \mapsto \log_{1/2} y$ for $y > 0$, so it is bijective. The exponent law gives $\phi(r+s) = (1/2)^{r+s} = \phi(r)\phi(s)$. Hence it is an isomorphism from addition to multiplication.

Exercise 3.11: Differentiating smooth functions

Differentiation preserves addition: $(f + g)' = f' + g'$. It is onto the smooth functions: if h is smooth, $f(x) = \int_0^x h(t)\,dt$ is smooth and has derivative h. But it is not injective, since the constant functions 0 and 1 both map to zero. Thus it is not an isomorphism.

Exercise 3.12: Evaluating a derivative at zero

The map is additive because $(f + g)'(0) = f'(0) + g'(0)$. It is onto $\mathbb{R}$, since $f(x) = cx$ has $f'(0) = c$ for any real c. It is not injective: distinct constant functions have the same image 0. Therefore it is not an isomorphism.

Exercise 3.13: Integration from zero

Integration is additive, and if $\int_0^x f(t)\,dt = \int_0^x g(t)\,dt$ for all x, differentiation gives $f = g$, proving injectivity. However each image function is zero at $x = 0$, so the constant function 1 is not in the image. Thus it is not onto F and is not an isomorphism.

Exercise 3.14: Differentiating the integral

The fundamental theorem of calculus gives $\phi(f)(x) = \frac{d}{dx}\int_0^x f(t)\,dt = f(x)$, since every smooth function is continuous. Thus the map is the identity on F, so is bijective and preserves addition. It is an isomorphism.

Exercise 3.15: Multiplication by the independent variable

The map is not onto: every image $xf(x)$ vanishes at zero, whereas the constant function 1 does not. It also fails to preserve multiplication: taking $f = g = 1$ gives $\phi(fg)(x) = x$ but $(\phi(f)\phi(g))(x) = x^2$, different at $x = 2$. Thus it is not an isomorphism for either reason.

Exercise 3.16: Transporting addition by a translation

The inverse of $\phi(n) = n + 1$ is $\phi^{-1}(u) = u - 1$.

(a) To send ordinary addition to $*$, define $u * v = \phi((u - 1) + (v - 1)) = u + v - 1$. Then $\phi(m) * \phi(n) = (m + 1) + (n + 1) - 1 = m + n + 1 = \phi(m + n)$. Its identity is $\phi(0) = 1$.

(b) To send $*$ to ordinary addition, require $\phi(m*n) = \phi(m)+\phi(n)$. This gives $m*n+1 = m+n+2$, or $m * n = m + n + 1$. Its identity is $-1 = \phi^{-1}(0)$.

Exercise 3.17: Transporting multiplication by a translation

Use $\phi(n) = n + 1$ and its inverse $u - 1$.

(a) Define $u * v = \phi((u - 1)(v - 1)) = (u - 1)(v - 1) + 1 = uv - u - v + 2$. Then $\phi(m) * \phi(n) = mn + 1 = \phi(mn)$. The identity is $\phi(1) = 2$.

(b) Require $\phi(m * n) = \phi(m)\phi(n)$, giving $m * n = (m + 1)(n + 1) - 1 = mn + m + n$. Its identity is $0 = \phi^{-1}(1)$.

These are binary operations on integers; the construction does not assert that every integer has a multiplicative inverse.

Exercise 3.18: Transporting rational addition

Here $\phi(x) = 3x - 1$ and $\phi^{-1}(u) = (u + 1)/3$.

(a) Define $u*v = \phi((u+1)/3+(v+1)/3) = u+v+1$. This satisfies $\phi(x)*\phi(y) = 3x+3y-1 = \phi(x+y)$. Its identity is $-1 = \phi(0)$.

(b) Define $x * y = \phi^{-1}(\phi(x) + \phi(y)) = (3x + 3y - 2 + 1)/3 = x + y - 1/3$. Its identity is $1/3 = \phi^{-1}(0)$.

All displayed formulas preserve rationality, so their domains and codomains are correct.

Exercise 3.19: Transporting rational multiplication

Again $\phi^{-1}(u) = (u + 1)/3$.

(a) Define $u * v = \phi(\phi^{-1}(u)\phi^{-1}(v)) = (u + 1)(v + 1)/3 - 1$. Then $\phi(x) * \phi(y) = 3xy - 1 = \phi(xy)$. Its identity is $\phi(1) = 2$.

(b) Define $x * y = \phi^{-1}(\phi(x)\phi(y)) = ((3x - 1)(3y - 1) + 1)/3 = 3xy - x - y + 2/3$. Its identity is $\phi^{-1}(1) = 2/3$.

In each direction the equation used to define the operation is exactly the required preservation condition, and the original map is already bijective.

Exercise 3.20: Commuting with the operations

There are two routes from an input pair (a, b) to an element of S'. One first computes $a * b$ in S and then applies ϕ. The other first applies ϕ to both inputs and then computes $\phi(a) *' \phi(b)$ in S'. Saying that ϕ commutes with these operations means precisely that the two routes give the same result. It does not mean ϕ itself is a binary operation.

Exercise 3.21: Completing the isomorphism definition

Let $*$ and $*'$ be binary operations on S and S'. An isomorphism $\phi : (S, *) \to (S', *')$ is a bijection such that $\phi(a * b) = \phi(a) *' \phi(b)$ for every $a, b \in S$. The original version omitted bijectivity and the universal quantifier.

Exercise 3.22: The identity definition

The stated definition is correct: an element $e \in S$ is an identity for $*$ when $s * e = s = e * s$ for every $s \in S$. The middle equality to s says multiplication by e leaves every element unchanged, and the two sides specify both a left and a right identity. Merely requiring $s * e = e * s$, without equality to s, would only require e to commute with every element and would not suffice.

Exercise 3.23: Synopsis of identity uniqueness

Compute the product of two proposed identities: using the first as a left identity and the second as a right identity makes that same product equal to each of them, forcing them to coincide.

Exercise 3.24: Left and right identities

(a) A left identity $e_L \in S$ satisfies $e_L * s = s$ for every $s \in S$. It need not be unique: on $S = \{a, b\}$ define $x * y = y$. Then both elements are left identities. In the two-sided uniqueness proof, the step $e_L * e'_L = e_L$ is unavailable: it would require e'_L to be a right identity.

(b) A right identity $e_R \in S$ satisfies $s * e_R = s$ for every $s \in S$. On $S = \{a, b\}$ define $x * y = x$. Then both elements are right identities. The step $e_R * e'_R = e'_R$ fails because it requires e_R to be a left identity.

Exercise 3.25: A left identity and a right identity together

They cannot differ. If l is a left identity and r a right identity, then $l * r = r$ by the left identity property and $l * r = l$ by the right identity property. Thus $l = r$, and this element is consequently a two-sided identity. No associativity assumption is needed.

Exercise 3.26: The inverse of an isomorphism

The inverse ϕ^{-1} is a bijection because ϕ is. Given $u, v \in S'$, write $a = \phi^{-1}(u)$ and $b = \phi^{-1}(v)$. Preservation by ϕ gives $\phi(a * b) = \phi(a) *' \phi(b) = u *' v$. Applying ϕ^{-1} yields

$$\phi^{-1}(u *' v) = a * b = \phi^{-1}(u) * \phi^{-1}(v).$$

Thus the inverse also preserves the operations, completing all requirements.

Exercise 3.27: Composing isomorphisms

A composite of two bijections is a bijection (its inverse is $\phi^{-1} \circ \psi^{-1}$). Also

$$(\psi \circ \phi)(a * b) = \psi(\phi(a) *' \phi(b)) = \psi(\phi(a)) *'' \psi(\phi(b)).$$

The first preservation step uses ϕ and the second uses ψ. This is precisely preservation by $\psi \circ \phi$, so the composite is an isomorphism.

Exercise 3.28: Isomorphism is an equivalence relation

Reflexivity: the identity map on any binary structure is bijective and operation-preserving. Symmetry: if one structure is isomorphic to another, the inverse isomorphism from Exercise 3.26 gives the reverse direction. Transitivity: compose the two isomorphisms using Exercise 3.27. Thus all three equivalence-relation conditions hold on any specified set of binary structures.

Exercise 3.29: Commutativity is structural

Suppose $\phi : S \to S'$ is an isomorphism and $*$ is commutative. For arbitrary $u, v \in S'$, surjectivity gives $u = \phi(a)$ and $v = \phi(b)$. Then

$$u *' v = \phi(a * b) = \phi(b * a) = v *' u.$$

Hence $*'$ is commutative. Applying the same argument to ϕ^{-1} proves the converse. Therefore commutativity depends on structure, not on the names of elements.

Exercise 3.30: Associativity is structural

For arbitrary $u = \phi(a), v = \phi(b), w = \phi(c)$ in S', preservation gives

$$(u *' v) *' w = \phi((a * b) * c) = \phi(a * (b * c)) = u *' (v *' w).$$

Surjectivity ensures every target triple has these preimages; associativity in S gives the middle equality. Repeat with the inverse to obtain the reverse implication. Thus associativity is invariant under isomorphism.

Exercise 3.31: Existence of square roots is structural

Assume each $c \in S$ has some x with $x * x = c$. Given $d \in S'$, choose $c = \phi^{-1}(d)$ and a square root x of c. Then $y = \phi(x)$ satisfies $y *' y = \phi(x * x) = \phi(c) = d$. The same argument using ϕ^{-1} proves the converse. We require existence, not uniqueness, of these roots.

Exercise 3.32: Existence of an idempotent is structural

If $b * b = b$ in S, then $\phi(b) *' \phi(b) = \phi(b * b) = \phi(b)$, so $\phi(b)$ is idempotent in S'. Conversely any target idempotent pulls back under ϕ^{-1} to an idempotent in S. Thus the existence of such an element is a structural property.

Exercise 3.33: Complex numbers as real matrices

Define $\phi(a+bi) = M(a,b) = \begin{pmatrix} a & -b \\ b & a \end{pmatrix}$. Equality of matrices recovers both a and b, proving injectivity, and every matrix in H has this form, proving surjectivity.

(a) $M(a,b) + M(c,d) = M(a+c, b+d) = \phi((a+bi)+(c+di))$, so the map preserves addition.

(b) $M(a,b)M(c,d) = M(ac-bd, ad+bc) = \phi((a+bi)(c+di))$, so it preserves multiplication.

The same bijection therefore gives both requested isomorphisms.

Exercise 3.34: Classifying two-element binary structures

Encode a table on $\{a,b\}$ by the four entries $(a*a, a*b, b*a, b*b)$. The only relabelings are the identity and the swap of a, b. Here are ten representative tables, specified by their four entries in that order:

	$a*a$	$a*b$	$b*a$	$b*b$
1	a	a	a	a
2	a	a	a	b
3	a	a	b	a
4	a	a	b	b
5	a	b	a	a
6	a	b	a	b
7	a	b	b	a
8	b	a	a	a
9	b	a	b	a
10	b	b	a	a

To see completeness without guessing, encode a, b as $0, 1$. A swap sends (t_1, t_2, t_3, t_4) to $(1-t_4, 1-t_3, 1-t_2, 1-t_1)$. Exactly four tables are fixed: choose t_1, t_2 freely and then $t_4 = 1-t_1, t_3 = 1-t_2$. The other twelve tables form six pairs. Hence there are $4 + 12/2 = 10$ classes; the ten displayed representatives lie in distinct swap classes and exhaust them.

4 Groups

Exercise 4.1: Integer multiplication

Integer multiplication is associative ($\mathcal{G}_1$) and has identity 1 ($\mathcal{G}_2$). But not every integer has a multiplicative inverse in $\mathbb{Z}$: $0x = 1$ is impossible, and $2x = 1$ has no integer solution. Thus the first failing axiom is $\mathcal{G}_3$, so this is not a group.

Exercise 4.2: Even integers under addition

The sum $2m + 2n = 2(m + n)$ is even, so closure holds. Addition is associative, $0 = 2 \cdot 0$ is the identity, and the inverse of $2m$ is $-2m$, again even. All three group axioms hold, so this is an abelian group.

Exercise 4.3: The geometric-mean operation

Associativity already fails, so $\mathcal{G}_1$ is the first failing axiom. For example, on $\mathbb{R}^+$ let $a = 1, b = 1, c = 16$. Then $(a * b) * c = 1 * 16 = 4$, whereas $a * (b * c) = 1 * 4 = 2$. The positive square roots make this a binary operation, but not a group.

Exercise 4.4: Rational multiplication including zero

Multiplication is associative and has identity 1. Zero has no inverse, since $0q = 0$ for every rational q. Thus $\mathcal{G}_3$ is the first failure and the set $\mathbb{Q}$, as opposed to $\mathbb{Q} \setminus \{0\}$, is not a multiplicative group.

Exercise 4.5: Division as an operation

The operation is closed on nonzero reals, but is not associative. With $a = 1, b = 1, c = 2$, $(a * b) * c = (1/1)/2 = 1/2$, whereas $a * (b * c) = 1/(1/2) = 2$. Therefore the first failing axiom is $\mathcal{G}_1$.

Exercise 4.6: The absolute value of a product

Associativity holds: $(a * b) * c = \big||ab|c\big| = |a||b||c| = \big|a|bc|\big| = a * (b * c)$. However no identity exists in $\mathbb{C}$. Any product $a * e = |ae|$ is a nonnegative real number and so cannot equal, for example, $a = i$. Thus $\mathcal{G}_2$ is the first failure.

Exercise 4.7: An abelian group of order one thousand

Take $\mathbb{Z}_{1000}$ under addition modulo 1000. It has exactly 1000 elements, addition of residues is associative and commutative, zero is the identity, and k has inverse the representative of $-k$ modulo 1000. Hence this example meets every requirement.

Exercise 4.8: Units modulo eight

Multiply each row label by each column label and reduce modulo 8:

$\cdot_8$	1	3	5	7
1	1	3	5	7
3	3	1	7	5
5	5	7	1	3
7	7	5	3	1

The table shows closure, identity 1, and that each element is its own inverse. Associativity is inherited from integer multiplication modulo 8. Thus this is an abelian group (the four-element noncyclic group).

Exercise 4.9: Distinguishing the circle group

In U the equation $z^3 = 1$ has three solutions, the three cube roots of unity. In the additive group $\mathbb{R}$, the corresponding equation $3x = 0$ has only $x = 0$. In the multiplicative group $\mathbb{R}^*$, $x^3 = 1$ has only $x = 1$. An isomorphism sends the identity to the identity and preserves products and powers, so would biject these solution sets. Their different sizes rule out both proposed isomorphisms.

Exercise 4.10: Multiples of a positive integer

(a) Sums of multiples of n are multiples of n, addition is associative, $0 = n \cdot 0$ belongs, and the negative of nm is $n(-m)$. Thus $(n\mathbb{Z}, +)$ is a group.

(b) Define $\phi : n\mathbb{Z} \to \mathbb{Z}$ by $\phi(nm) = m$. This is well defined since $n \neq 0$ makes the representation unique; it is bijective with inverse $m \mapsto nm$. Finally $\phi(nm + nk) = m + k = \phi(nm) + \phi(nk)$, so it is an isomorphism.

Exercise 4.11: Diagonal matrices under addition

This is a group. Adding diagonal matrices adds their diagonal entries and leaves off-diagonal entries zero. Associativity follows from matrix addition; the zero matrix is the identity; and negating every diagonal entry gives an additive inverse still in the set. The group is abelian.

Exercise 4.12: All diagonal matrices under multiplication

The set is closed, multiplication is associative, and I_n is its identity, but the zero matrix has no multiplicative inverse. Therefore it is not a group. Requiring nonzero diagonal entries would remove this obstruction.

Exercise 4.13: Invertible diagonal matrices

The product of $\operatorname{diag}(a_1, \ldots, a_n)$ and $\operatorname{diag}(b_1, \ldots, b_n)$ is $\operatorname{diag}(a_1b_1, \ldots, a_nb_n)$ with nonzero diagonal. The identity is I_n, the inverse is $\operatorname{diag}(1/a_1, \ldots, 1/a_n)$, and multiplication is associative. Thus these matrices form an abelian group.

Exercise 4.14: Diagonal sign matrices

A product of two signs is a sign, so these diagonal matrices are closed. The identity has all diagonal entries 1, and each matrix is its own inverse since $(\pm 1)^2 = 1$. Associativity is inherited from matrix multiplication. Hence this is a group of 2^n elements.

Exercise 4.15: All upper-triangular matrices multiplicatively

Products remain upper triangular and I_n belongs, but singular upper-triangular matrices are also included. In particular the zero matrix cannot have a multiplicative inverse. Thus the given set is not a group under multiplication.

Exercise 4.16: Upper-triangular matrices additively

Sums and negatives remain upper triangular. Matrix addition is associative, the zero matrix is an additive identity in this set, and $-A$ is the additive inverse of A. Therefore these matrices form an abelian group under addition.

Exercise 4.17: Upper-triangular matrices of determinant one

If A, B are upper triangular with determinant 1, so is AB, since $\det(AB) = 1$. The identity belongs. Each A is invertible, and A^{-1} is upper triangular: solving $Ax = e_j$ by back substitution gives zeros below the jth position in column j of the inverse. Also $\det(A^{-1}) = 1$. Associativity is inherited, so this is a group.

Exercise 4.18: Matrices of determinant plus or minus one

The determinant of a product is the product of determinants, so closure holds in $\{1, -1\}$. Every such matrix is invertible, I_n belongs, and $\det(A^{-1}) = 1/\det A \in \{1, -1\}$. Associativity is matrix associativity, so the indicated set is a group.

Exercise 4.19: A shifted multiplicative group

Use the identity $a * b + 1 = (a + 1)(b + 1)$.

(a) If $a, b \neq -1$, their shifted values are nonzero, so $(a * b) + 1 \neq 0$ and $a * b \neq -1$. The formula specifies one real output, proving it is a binary operation on S.

(b) The bijection $\phi : S \to \mathbb{R}^*$, $\phi(a) = a + 1$, satisfies $\phi(a * b) = \phi(a)\phi(b)$, so it transports the group laws from $\mathbb{R}^*$. Explicitly, both bracketings have shifted value $(a + 1)(b + 1)(c + 1)$, proving associativity; the identity is 0, and the inverse of a is $-a/(a+1)$, whose shifted value is $1/(a+1)$.

(c) Apply ϕ to $2 * x * 3 = 7$: $3(x + 1)4 = 8$. Thus $x + 1 = 2/3$ and $x = -1/3 \in S$. Checking directly, $2 * (-1/3) = 1$ and $1 * 3 = 7$.

Exercise 4.20: The groups of order four

No repeated entry can occur in a row or column of a group table, by cancellation. Filling accordingly gives the following three tables in the order e, a, b, c:

T_1	e	a	b	c
e	e	a	b	c
a	a	e	c	b
b	b	c	e	a
c	c	b	a	e

T_2	e	a	b	c
e	e	a	b	c
a	a	e	c	b
b	b	c	a	e
c	c	b	e	a

T_3	e	a	b	c
e	e	a	b	c
a	a	b	c	e
b	b	c	e	a
c	c	e	a	b

The first two have $a^2 = e$; the third has $a^2 = b$. In T_2, b has successive powers e, b, a, c, whereas in T_3, a has successive powers e, a, b, c. Therefore $T_2 \cong T_3$ via $e \mapsto e, a \mapsto b, b \mapsto a, c \mapsto c$. In T_1 every nonidentity element squares to e, so it cannot be isomorphic to either cyclic table.

(a) Yes: all three tables are symmetric, so every group of order four is abelian.

(b) T_3 is U_4 via $e \mapsto 1, a \mapsto i, b \mapsto -1, c \mapsto -i$; T_2 is also U_4 via its preceding relabeling.

(c) T_1 is the diagonal-sign group for $n = 2$: send $a \mapsto \operatorname{diag}(-1, 1)$, $b \mapsto \operatorname{diag}(1, -1)$, $c \mapsto -I_2$. Multiplying these matrices verifies the table and therefore its associativity.

Exercise 4.21: Counting labeled small group operations

On a fixed two-element set there are two group operations: either element can be the identity, and that choice forces the other element to square to it. On a fixed three-element set there are three operations, one for each choice of identity; after that choice the two other elements must be inverses and their squares are forced. Swapping those two nonidentity labels leaves the same cyclic group table, so does not create an additional operation. Thus the answers are 2 and 3.

Exercise 4.22: Ordering the group axioms

The inverse axiom $\mathcal{G}_3$ refers to the identity e introduced in $\mathcal{G}_2$. Thus $\mathcal{G}_2$ must precede $\mathcal{G}_3$. The unacceptable orders are $\mathcal{G}_1\mathcal{G}_3\mathcal{G}_2$, $\mathcal{G}_3\mathcal{G}_1\mathcal{G}_2$, and $\mathcal{G}_3\mathcal{G}_2\mathcal{G}_1$. The three orders with $\mathcal{G}_2$ first relative to $\mathcal{G}_3$ are acceptable; associativity has no such dependence on the other two.

Exercise 4.23: Critiquing four proposed definitions

A correct definition specifies a binary operation on G, associativity, an element $e \in G$ satisfying $ex = xe = x$ for every x, and for each $a \in G$ an $a^{-1} \in G$ satisfying $aa^{-1} = a^{-1}a = e$.

(a) The equality ending "x = identity" is incorrect: x is arbitrary, not necessarily the identity. The identity equation needs "for every $x \in G$," the inverse must belong to G, and the same operation symbol must be used consistently.

(b) It refers to "the operation" without specifying a binary operation on G. It should define what the identity and inverse equations mean and explicitly put each inverse in G.

(c) Associativity is missing. "An inverse exists" must mean an inverse for every element, relative to the specified identity; merely defining a binary operation is already part of the setup.

(d) "Associative under addition" confuses the operation with another operation. The notation $\{e\}$ denotes a set, not the element e; the identity equation must end in a, not e. The inverse must be in G, and the universal quantifiers must govern the appropriate equations.

Exercise 4.24: Identity and inverses without associativity

Consider

$*$	e	a	b
e	e	a	b
a	a	e	e
b	b	e	e

The first row and column show e is a two-sided identity. Both a and b are their own two-sided inverses, so $\mathcal{G}_2, \mathcal{G}_3$ hold. But $(a * a) * b = e * b = b$, whereas $a * (a * b) = a * e = a$. Thus associativity fails, as requested.

Exercise 4.25: Ten statements about groups

(a) False: two identities e, f satisfy $e = ef = f$.

(b) True: the cancellation rule forces the unique three-element cyclic table, up to relabeling.

(c) True for the linear equations in this section: $ax = b$ and $ya = b$ have the solutions $a^{-1}b$ and ba^{-1}.

(d) False: understanding and stating the exact content matters; identical wording is unnecessary.

(e) False: one implication could allow a definition that excludes some actual groups. Equivalence requires both directions.

(f) True: equivalence of the two definitions is precisely what is needed.

(g) True: the one-, two-, and three-element group tables are all commutative.

(h) True: $axb = c$ has the unique solution $x = a^{-1}cb^{-1}$, obtained by multiplying on the indicated sides.

(i) False: a group must contain its identity, so it is nonempty.

(j) True: a group is a set with a binary operation satisfying additional axioms.

Exercise 4.26: Synopsis of left cancellation

Multiply an equality with the same left factor on the left by that factor's inverse, then reassociate and use the identity to remove that factor.

Exercise 4.27: Synopsis of solving a group equation

Multiplying the desired right-hand side on the left by the inverse of the coefficient produces a solution. Left cancellation shows that any two solutions must agree.

Exercise 4.28: Isomorphisms preserve inverses

Let a^{-1} be the inverse of a and let e' be the target identity. Since $\phi(e) = e'$,

$$\phi(a)\phi(a^{-1}) = \phi(aa^{-1}) = e', \qquad \phi(a^{-1})\phi(a) = \phi(a^{-1}a) = e'.$$

Thus $\phi(a^{-1})$ is a two-sided inverse of $\phi(a)$. The inverse of a group element is unique, so $\phi(a^{-1}) = \phi(a)^{-1}$.

Exercise 4.29: An involution in a group of even order

Pair each element x with its inverse. Elements with $x \neq x^{-1}$ occur in disjoint pairs and hence contribute an even number of elements. If e were the only self-inverse element, the total size would therefore be even plus one, which is odd. Since the group has even size, there must be some $a \neq e$ with $a = a^{-1}$, equivalently $a^2 = e$.

Exercise 4.30: Mixed one-sided axioms

On $\mathbb{R}^*$ put $a * b = |a|b$.

(a) Outputs are nonzero, and $(a * b) * c = ||a|b|c = |a||b|c = a * (b * c)$, proving associativity.

(b) The element 1 is a left identity, since $1 * b = b$. Each a has a right inverse relative to 1, namely $b = 1/|a|$, since $a * b = 1$.

(c) This is not a group. A right identity r would need $1 * r = 1$, hence $r = 1$, but then $(-1) * r = 1 \neq -1$.

(d) A left identity combined with right inverses does not suffice for the group axioms. The valid one-sided alternatives require the identity and inverse axioms to have matching sides (both left, or both right). This example even has two left identities, 1 and -1.

Exercise 4.31: Idempotents in a group

The identity is idempotent since $e^2 = e$. If $x^2 = x$, multiply on the left by x^{-1} to obtain $x = e$. Thus every idempotent equals the identity, and there is exactly one.

Exercise 4.32: A group of exponent two is abelian

Assume $x^2 = e$ for every $x \in G$. In particular $(ab)^2 = e$, so $abab = e$. Multiplying this equality on the left by a and on the right by b gives $ba = ab$, because $a^2 = b^2 = e$. Since a, b were arbitrary, the group is abelian. No commutation was used before establishing the conclusion.

Exercise 4.33: Powers of commuting products

Fix a, b in an abelian group. For $n = 1$, $(ab)^1 = a^1b^1$. If $(ab)^n = a^nb^n$, then

$$(ab)^{n+1} = (ab)^n ab = a^nb^nab = a^nab^nb = a^{n+1}b^{n+1}.$$

Commutativity permits moving a past b^n, and associativity permits regrouping. Thus mathematical induction proves the identity for every positive integer n.

Exercise 4.34: A finite-order power

Let $|G| = m$. The $m + 1$ powers $e, a, a^2, \ldots, a^m$ cannot all be distinct, so $a^i = a^j$ for some $0 \leq i < j \leq m$. Multiply by a^{-i} to get $a^{j-i} = e$. Since $j - i > 0$, this supplies the required positive exponent.

Exercise 4.35: A squared-product identity forces commutation

The hypothesis is $abab = aabb$. Multiply on the left by a^{-1} to get $bab = abb$, then on the right by b^{-1} to get $ba = ab$. The sides of multiplication matter; cancellation is valid without assuming the group is abelian.

Exercise 4.36: When inverses keep their order

In any group $(ab)^{-1} = b^{-1}a^{-1}$, since this element gives e on multiplication on either side. If $ab = ba$, then their inverses also commute, so this equals $a^{-1}b^{-1}$. Conversely, if $(ab)^{-1} = a^{-1}b^{-1}$, comparison with the general formula gives $b^{-1}a^{-1} = a^{-1}b^{-1}$. Taking inverses reverses both products and yields $ab = ba$. This proves both implications.

Exercise 4.37: Cyclically rotating a product equal to the identity

From $abc = e$, multiply on the left by a^{-1} to get $bc = a^{-1}$. Multiplying on the right by a then gives $bca = a^{-1}a = e$. This is a cyclic rotation, not an arbitrary interchange of factors.

Exercise 4.38: Matching left axioms imply the group axioms

Assume associativity, a left identity e, and a left inverse a' for every a.

First obtain left cancellation.

If $ax = ay$, multiply on the left by a'. Then $(a'a)x = (a'a)y$ gives $ex = ey$, hence $x = y$.

Prove the identity is also right-sided.

Since $a'(ae) = (a'a)e = ee = e = a'a$, left cancellation of a' gives $ae = a$.

Prove each left inverse is also right-sided.

Now $a'(aa') = (a'a)a' = ea' = a' = a'e$. Cancel a' on the left to get $aa' = e$. Thus e is a two-sided identity and each a' is a two-sided inverse. All ordinary group axioms hold.

Exercise 4.39: Solvability of all linear equations

Choose $a \in G$, possible since G is nonempty. By the solvability hypothesis choose e with $ea = a$. For any $g \in G$, choose z with $az = g$. Associativity gives $eg = e(az) = (ea)z = az = g$, so e is a left

identity for every element. For any $c \in G$, solvability of $yc = e$ gives a left inverse c' with $c'c = e$. Thus the matching left axioms of Exercise 4.38 hold, and its proof shows that G is a group.

Exercise 4.40: The reversed operation

Write the original multiplication as juxtaposition and set $a * b = ba$. Then $(a * b) * c = cba = a * (b * c)$, the original identity remains an identity, and the original a^{-1} is also the $*$-inverse. Thus $(G, *)$ is a group. Define $\phi(a) = a^{-1}$. It is bijective because applying it twice gives a, and

$$\phi(a * b) = \phi(ba) = (ba)^{-1} = a^{-1}b^{-1} = \phi(a)\phi(b).$$

Hence it is an isomorphism from the reversed group to the original group.

Exercise 4.41: Conjugation is an automorphism

For fixed $g \in G$ define $i_g(x) = gxg^{-1}$. Its inverse map is $i_{g^{-1}}$, since $g^{-1}(gxg^{-1})g = x$ and the reverse composition also equals the identity. Thus i_g is bijective. Moreover

$$i_g(x)i_g(y) = (gxg^{-1})(gyg^{-1}) = gx(g^{-1}g)yg^{-1} = gxyg^{-1} = i_g(xy).$$

Therefore conjugation preserves multiplication and is an isomorphism of G with itself.

5 Subgroups

Exercise 5.1: Real numbers in the additive complex group

Yes. The real numbers contain 0, and for any reals a, b both $a + b$ and $-a$ are real. Associativity is inherited from complex addition. These facts verify the subgroup conditions.

Exercise 5.2: Positive rationals additively

No. The additive identity 0 is not positive, so it is not in $\mathbb{Q}^+$. Also a positive rational has a negative additive inverse, outside the set. Closure under addition alone is insufficient.

Exercise 5.3: Multiples of seven

Yes. The set $7\mathbb{Z}$ contains zero. If $7m, 7n \in 7\mathbb{Z}$, then $7m + 7n = 7(m + n)$ and $-(7m) = 7(-m)$ remain in the set. Thus it is an additive subgroup of $\mathbb{C}$.

Exercise 5.4: Pure imaginary numbers

Yes. The identity is $0 = i \cdot 0$. For real a, b, $ia + ib = i(a + b)$ and $-(ia) = i(-a)$ are pure imaginary. Therefore closure and inverse closure hold.

Exercise 5.5: Rational multiples of pi

Yes. The set is $\pi\mathbb{Q} = \{\pi q : q \in \mathbb{Q}\}$. It contains zero, sums have the form $\pi(q + r)$, and negatives have the form $\pi(-q)$. Since $q + r, -q$ are rational, all subgroup conditions hold under addition.

Exercise 5.6: Integral powers of pi additively

No. Every π^n is positive, so zero is absent and no negative additive inverse is included. Thus $\{\pi^n : n \in \mathbb{Z}\}$ is not an additive subgroup, regardless of whether particular sums happen to lie in it.

Exercise 5.7: The same six sets multiplicatively

The only multiplicative subgroups of $\mathbb{C}^*$ among the six are $\mathbb{Q}^+$ and $\{\pi^n : n \in \mathbb{Z}\}$. The positive rationals contain 1, are closed under products, and have positive rational reciprocals. Integral powers of π contain $\pi^0 = 1$, multiply by adding exponents, and invert by negating exponents. The other four sets $(\mathbb{R}, 7\mathbb{Z}, i\mathbb{R}, \pi\mathbb{Q})$ contain 0, so are not even subsets of the stated group $\mathbb{C}^*$.

Exercise 5.8: Matrices of determinant two

No. The identity matrix has determinant 1, so it is absent. In addition, the product of two matrices of determinant 2 has determinant 4, violating closure. Such matrices exist, for example $\mathrm{diag}(2, 1, \ldots, 1)$.

Exercise 5.9: Invertible diagonal matrices

Yes. The identity is diagonal, products multiply corresponding nonzero diagonal entries, and reciprocating those entries produces a diagonal inverse. Thus this nonempty subset is closed under products and inverses, hence is a subgroup of $GL(n, \mathbb{R})$.

Exercise 5.10: Invertible upper-triangular matrices

Yes. Products remain upper triangular and their diagonal entries are the corresponding products of nonzero diagonal entries. The identity belongs. Back substitution in $Ax = e_j$ shows A^{-1} is upper triangular, and its diagonal entries are $1/a_{jj}$, hence nonzero. Thus this is a subgroup.

Exercise 5.11: Matrices of determinant minus one

No. The identity has determinant 1, and the product of two determinant--1 matrices has determinant 1, outside the set. Hence both identity membership and product closure fail.

Exercise 5.12: Matrices of determinant one or minus one

Yes. The identity belongs, products have determinant in $\{1, -1\}$, and the inverse has determinant $1/\det A$, again in $\{1, -1\}$. All matrices in the set are invertible, so the subgroup criterion applies.

Exercise 5.13: Orthogonal matrices

The identity satisfies $I_n^T I_n = I_n$. If $A^T A = B^T B = I_n$, then $(AB)^T(AB) = B^T A^T AB = B^T B = I_n$, so products remain orthogonal. The equation $A^T A = I_n$ gives $A^{-1} = A^T$, and then $(A^{-1})^T A^{-1} = AA^T = I_n$. Thus inverses also belong, and this is a subgroup.

Exercise 5.14: Nowhere-zero functions

(a) Not an additive subgroup: the zero function is absent; also $1 + (-1) = 0$ shows failure of closure. **(b)** Yes, this is the entire multiplicative group $\widetilde{F}$. Products are nowhere zero, the constant 1 is the identity, and the inverse of f is the function $x \mapsto 1/f(x)$, defined everywhere because f never vanishes.

Exercise 5.15: Functions vanishing at one

(a) Yes. The zero function belongs, and if $f(1) = g(1) = 0$, then $(f + g)(1) = 0$ and $(-f)(1) = 0$. **(b)** No. Every function in the specified set vanishes at 1, so none belongs to $\widetilde{F}$. It is not a subset of the proposed ambient multiplicative group.

Exercise 5.16: Nowhere-zero functions equal to one at one

(a) No: the zero function is absent (and adding two members gives value 2 at 1). **(b)** Yes. The constant function 1 belongs, products have value $1 \cdot 1 = 1$ at 1, and reciprocals have value $1/1 = 1$

there. Nowhere-zero values are preserved by products and reciprocals.

Exercise 5.17: Nowhere-zero functions equal to one at zero

(a) No, since the additive identity is absent and adding two such functions gives value 2 at zero. **(b)** Yes: the constant 1 belongs, and both $(fg)(0) = f(0)g(0) = 1$ and $(1/f)(0) = 1$ hold. All resulting functions remain nowhere zero.

Exercise 5.18: Nowhere-zero functions equal to minus one at zero

(a) No: the zero function does not satisfy the defining condition. **(b)** No: the multiplicative identity has value 1, not -1, at zero. Furthermore multiplying two members gives value $(-1)(-1) = 1$ at zero, so product closure fails.

Exercise 5.19: Nonzero constant functions

(a) No: the zero constant is not in this subset of $\widetilde{F}$, and the constants $1, -1$ add to it. **(b)** Yes: nonzero constants contain 1, multiply to nonzero constants, and have nonzero constant reciprocals. Thus they form a multiplicative subgroup.

Exercise 5.20: All subgroup relations among nine groups

Include the nine reflexive relations $G_i \leq G_i$. All strict relations are the transitive consequences of

$$G_2 < G_8 < G_7 < G_1 < G_4, \qquad G_9 < G_3 < G_5, \qquad G_6 < G_5.$$

Explicitly the additional pairs (i, j) with $G_i < G_j$ are

$$(2,8),(2,7),(2,1),(2,4),(8,7),(8,1),(8,4),$$
$$(7,1),(7,4),(1,4),(9,3),(9,5),(3,5),(6,5).$$

The additive chain is $12\mathbb{Z} \subset 6\mathbb{Z} \subset 3\mathbb{Z} \subset \mathbb{Z} \subset \mathbb{R}$. Multiplicatively, powers of 6 are positive rationals, and powers of π are positive reals. The latter are not all rational, since π is irrational; the two cyclic power groups do not contain each other, since π is not a power of 6 and 6 is not an integral power of π (use $\pi < 6 < \pi^2$). A subgroup must use the ambient operation, so no cross-relations between the additive and multiplicative lists are allowed.

Exercise 5.21: Five elements of three cyclic groups

(a) In $25\mathbb{Z}$: $-50, -25, 0, 25, 50$, obtained using integer multiples $-2, -1, 0, 1, 2$ of 25.

(b) In $\{(1/2)^n : n \in \mathbb{Z}\}$: $4, 2, 1, 1/2, 1/4$, from those same exponents.

(c) In $\{\pi^n : n \in \mathbb{Z}\}$: $\pi^{-2}, \pi^{-1}, 1, \pi, \pi^2$.

Zero exponents in a multiplicative cyclic group give its identity 1; they never give the number 0.

Exercise 5.22: A signed interchange matrix

Let $A = \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}$. Direct multiplication gives $A^2 = I_2$, while $A \neq I_2$. Thus every even integral power is I_2 and every odd integral power is A (also for negative exponents, since $A^{-1} = A$). The subgroup is precisely $\{I_2, A\}$.

Exercise 5.23: A shear matrix

For $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$, multiplication shows

$$\begin{pmatrix} 1 & m \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & m+n \\ 0 & 1 \end{pmatrix}.$$

Thus induction gives $A^n = \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}$ for positive n, and the inverse $A^{-1} = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}$ extends it to every integer n. These, and only these, are the elements of $\langle A \rangle$.

Exercise 5.24: A diagonal generator

If $A = \operatorname{diag}(3, 2)$, then $A^n = \operatorname{diag}(3^n, 2^n)$ for every $n \in \mathbb{Z}$; negative powers use reciprocals of the diagonal entries. Therefore

$$\langle A \rangle = \{\operatorname{diag}(3^n, 2^n) : n \in \mathbb{Z}\}.$$

The same exponent must occur in both positions; the subgroup does not allow them to be chosen independently.

Exercise 5.25: A scaled signed interchange

Write $B = \begin{pmatrix} 0 & -2 \\ -2 & 0 \end{pmatrix} = 2A$ with $A^2 = I_2$ from Exercise 5.22. Since scalar matrices commute with A, $B^n = 2^n A^n$. Thus its elements are

$$\{2^{2k} I_2 : k \in \mathbb{Z}\} \cup \left\{ 2^{2k+1} \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix} : k \in \mathbb{Z} \right\}.$$

This includes negative powers; $B^2 = 4I_2$ shows that the group is infinite.

Exercise 5.26: Which six groups are cyclic?

G_1 : Cyclic, generated exactly by $1, -1$: the multiples of an integer a fill $\mathbb{Z}$ only if $|a| = 1$.

G_2 : Not cyclic. If $q \neq 0$ were an additive generator, $q/2$ could not be an integral multiple of q; $q = 0$ plainly cannot generate $\mathbb{Q}$.

G_3 : Not cyclic. If a positive rational generator differs from 1, replace it by its reciprocal if necessary so it is $q > 1$. Then the rational $(1 + q)/2$ lies strictly between 1 and q, while no integral power of q does. The generator 1 also fails.

G_4 : Cyclic, with precisely the generators $6, -6$, since division by 6 identifies this group with $\mathbb{Z}$.

G_5 : Cyclic, with precisely $6, 1/6$ as generators. If 6^k generates, there must be an integer m with $6^{km} = 6$, so $km = 1$ and $k = \pm 1$.

G_6 : Not cyclic. If $a + b\sqrt{2}$ generated it, expressing $1 = k(a + b\sqrt{2})$ with integers forces $kb = 0$ and $ka = 1$, by irrationality of $\sqrt{2}$. Thus $b = 0$, $a = \pm 1$, and its multiples cannot include $\sqrt{2}$.

Exercise 5.27: The subgroup generated by three modulo four

Successive multiples of 3 modulo 4 are $0, 3, 2, 1$, then 0 again. These four residues are distinct, so the generated subgroup has order 4.

Exercise 5.28: A nonidentity element of the Klein group

Table 5.11 gives $c^2 = e$ and $c \neq e$. Therefore the subgroup generated by c is $\{e, c\}$ and has order 2.

Exercise 5.29: An angle of two pi over three

The generator is $e^{2\pi i/3}$. Its kth power is 1 exactly when $2\pi k/3$ is an integral multiple of 2π, or $3 \mid k$. The least positive such k is 3, so the subgroup has order 3, even though the ambient group has six elements.

Exercise 5.30: An angle of four pi over five

The generator is $e^{4\pi i/5}$. Its kth power is 1 if and only if $5 \mid 2k$. Since 2 and 5 are relatively prime, this is equivalent to $5 \mid k$. Thus the order is 5.

Exercise 5.31: An angle of three pi over two

The generator is $e^{3\pi i/2} = -i$. Its powers are $1, -i, -1, i$, then 1 again. Consequently its cyclic subgroup has order 4.

Exercise 5.32: An angle of five pi over four

The generator is $e^{5\pi i/4} = e^{2\pi i(5/8)}$. Its kth power is 1 exactly when $8 \mid 5k$. Because $\gcd(5, 8) = 1$, the least positive such k is 8. Thus this element generates all of U_8.

Exercise 5.33: A permutation matrix of order two

On the standard column vectors the matrix interchanges e_1, e_3 and interchanges e_2, e_4. Applying it twice fixes all four basis vectors, so $A^2 = I_4$. Since one application does not fix e_1, $A \neq I_4$. Therefore

its cyclic subgroup has order 2.

Exercise 5.34: A permutation matrix of order four

Track the first standard basis vector: $e_1 \mapsto e_3 \mapsto e_2 \mapsto e_4 \mapsto e_1$. This is a cycle containing all four basis vectors. Four applications fix each vector, and fewer than four do not fix e_1. Thus $A^4 = I_4$ with no smaller positive identity power, giving order 4.

Exercise 5.35: A permutation matrix of order three

This matrix sends $e_1 \mapsto e_4 \mapsto e_2 \mapsto e_1$ and fixes e_3. Thus three applications fix every standard basis vector. Neither one nor two applications fix e_1, so the least positive identity power is 3, and the generated subgroup has order 3.

Exercise 5.36: All subgroups generated in Z six

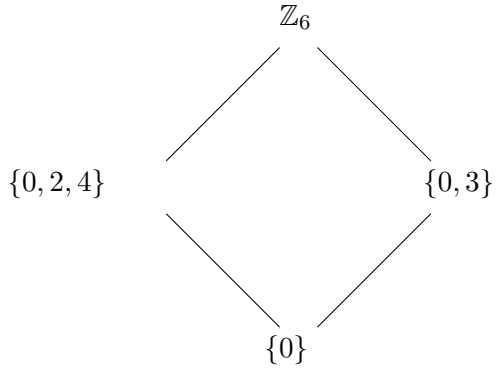
(a) Add row and column labels modulo 6:

$+$	0	1	2	3	4	5
0	0	1	2	3	4	5
1	1	2	3	4	5	0
2	2	3	4	5	0	1
3	3	4	5	0	1	2
4	4	5	0	1	2	3
5	5	0	1	2	3	4

(b) $\langle 0 \rangle = \{0\}$, $\langle 1 \rangle = \langle 5 \rangle = \mathbb{Z}_6$, $\langle 2 \rangle = \langle 4 \rangle = \{0, 2, 4\}$, $\langle 3 \rangle = \{0, 3\}$. These follow by repeatedly adding the generator until zero reappears.

(c) Exactly 1 and 5 generate all six residues.

(d) The inclusion diagram is the diamond below. Neither middle subgroup contains the other, since $2 \notin \{0, 3\}$ and $3 \notin \{0, 2, 4\}$.



Exercise 5.37: A complete subgroup definition

A subgroup H of a group G is a subset that is itself a group under the operation inherited from G. Equivalently, it contains e, is closed under products, and contains the inverse of each of its elements. The proposed definition omits product closure. For instance $\{0, 1, -1\}$ in the additive group $\mathbb{Z}$ contains the identity and all negatives but is not closed under addition.

Exercise 5.38: The definition of cyclic

The stated definition is correct: there exists $a \in G$ such that every element of G is a^n for some integer n. Both negative powers and $n = 0$ are allowed. In additive notation the same condition is $G = \{na : n \in \mathbb{Z}\}$.

Exercise 5.39: Ten subgroup statements

(a) True: associativity is one of the group axioms.

(b) False: multiplication by inverses proves both cancellation laws in every group.

(c) True: the inherited operation makes G a subgroup of itself.

(d) False: the text calls only G itself the improper subgroup; $\{e\}$ is the trivial subgroup, usually proper when $G \neq \{e\}$.

(e) False: in $\mathbb{Z}_4$, the element 2 generates only $\{0, 2\}$.

(f) False: $1, -1$ are distinct generators of the infinite cyclic group $\mathbb{Z}$.

(g) False: $\mathbb{Z}$ is a group under addition but zero has no multiplicative inverse.

(h) False as a proposed complete definition: a subgroup is a subset satisfying additional operation and group conditions.

(i) True: 1 generates $\mathbb{Z}_4$ by addition modulo 4.

(j) False: $\{1\}$ is a subset of the additive group $\mathbb{Z}$ but is not a subgroup, since it lacks zero.

Exercise 5.40: More than two square roots of the identity

In the Klein four-group $V = \{e, a, b, c\}$ every element satisfies $x^2 = e$. Thus there are four solutions, exceeding two. The familiar degree bound for polynomial roots over a field cannot be applied to an arbitrary group equation.

Exercise 5.41: An isomorphism carries subgroups to subgroups

Since H contains e, its image contains $\phi(e) = e'$. If $u = \phi(h)$ and $v = \phi(k)$ lie in $\phi[H]$, then $u *' v = \phi(h * k)$ belongs to $\phi[H]$ because $h * k \in H$. Also $u^{-1} = \phi(h^{-1})$ belongs because $h^{-1} \in H$. Thus the image contains the identity and is closed under products and inverses, so is a subgroup of G'.

Exercise 5.42: An isomorphism carries a cyclic group to a cyclic group

Let $G = \langle a \rangle$. Every $g' \in G'$ is $\phi(g)$ for some $g \in G$, by surjectivity. Write $g = a^n$. A homomorphism preserves positive powers by repeated multiplication, zero powers because it preserves the identity, and negative powers because it preserves inverses. Hence $g' = \phi(a)^n$. Thus $G' = \langle \phi(a) \rangle$ is cyclic.

Exercise 5.43: Products of two subgroups in an abelian group

Let $HK = \{hk : h \in H, k \in K\}$. It contains $ee = e$. If $h_1k_1, h_2k_2 \in HK$, then by commutativity

$$(h_1k_1)(h_2k_2) = (h_1h_2)(k_1k_2) \in HK.$$

Also $(hk)^{-1} = k^{-1}h^{-1} = h^{-1}k^{-1} \in HK$. The factors remain in their respective subgroups. Therefore HK is a subgroup; commutativity was essential in regrouping the factors.

Exercise 5.44: The hidden nonemptiness assumption

The argument starts "let $a \in H$," which is not possible when $H = \varnothing$. The empty subset is vacuously closed under products and inverses, so Conditions 1 and 3 alone do not force identity membership. If one separately assumes H nonempty, the argument becomes valid. The flaw is therefore not the algebraic calculation, but the unjustified choice of an element.

Exercise 5.45: The one-step subgroup criterion

If H is a subgroup, then $b^{-1} \in H$ and hence $ab^{-1} \in H$ for $a, b \in H$. Conversely assume $H \neq \varnothing$ and that all such products belong. Choose $h \in H$; then $hh^{-1} = e \in H$. For $b \in H$ use $a = e$ to get $b^{-1} \in H$. Finally for $a, b \in H$ the element b^{-1} also belongs, so $a(b^{-1})^{-1} = ab \in H$. Thus identity, inverse closure, and product closure all follow.

Exercise 5.46: Only one generator

If $G = \langle a \rangle$, then a^{-1} is also a generator because $(a^{-1})^{-n} = a^n$. With only one generator we must have $a = a^{-1}$, so $a^2 = e$. Every power of a is then either e or a, giving $|G| \leq 2$. This includes the trivial group, whose sole generator is its identity.

Exercise 5.47: The solutions of x squared equals identity

Let $H = \{x : x^2 = e\}$ in an abelian group. The identity belongs. If $x, y \in H$, commutativity gives $(xy)^2 = x^2y^2 = e$. Also $(x^{-1})^2 = (x^2)^{-1} = e$. Thus H is closed under products and inverses and is a subgroup. Without commutativity, the regrouping of $(xy)^2$ would not be justified.

Exercise 5.48: The solutions of a fixed power equation

For fixed $n \geq 1$, put $H = \{x : x^n = e\}$. The identity is in H. If $x, y \in H$, the commuting-product power law gives $(xy)^n = x^ny^n = e$, and $(x^{-1})^n = (x^n)^{-1} = e$. Hence the subgroup criterion holds. For $n = 1$ this is simply the trivial subgroup.

Exercise 5.49: A positive identity power in a finite group

The powers $e, a, a^2, \ldots, a^{|G|}$ form $|G|+1$ elements counted with repetitions in a set of size $|G|$. Thus $a^i = a^j$ for some $i < j$. Multiplying by a^{-i} gives $a^{j-i} = e$ with positive exponent, as required.

Exercise 5.50: A finite nonempty product-closed subset

Choose $h \in H$. By closure all positive powers of h are in H, and finiteness forces $h^i = h^j$ for some positive $i < j$. In the ambient group cancel to get $h^{j-i} = e \in H$. For any $a \in H$, the same finite-power argument produces $a^m = e$ with $m \geq 1$. If $m = 1$, then $a = e$ and its inverse is in H; otherwise $a^{-1} = a^{m-1} \in H$. Thus H contains the identity and all inverses, and the given product closure completes the subgroup conditions.

Exercise 5.51: The centralizer of one element

The identity commutes with a. If $xa = ax$ and $ya = ay$, then $(xy)a = x(ya) = x(ay) = (xa)y = a(xy)$. To obtain inverse closure, multiply $xa = ax$ on the left and right by x^{-1} to get $ax^{-1} = x^{-1}a$. Thus H_a contains e and is closed under products and inverses, making it a subgroup.

Exercise 5.52: The centralizer of a set and the center

(a) The identity commutes with every $s \in S$. If x, y do so, then for every such s, $(xy)s = x(ys) = x(sy) = (xs)y = s(xy)$. Similarly $xs = sx$ implies $x^{-1}s = sx^{-1}$. Hence H_S is a subgroup. If S is empty the condition is vacuous and $H_S = G$, still a subgroup.

(b) For $S = G$, take $x, y \in H_G$. Since x commutes with *every* element of G, it commutes in particular with y. Thus the subgroup H_G is abelian.

Exercise 5.53: An equivalence relation from a subgroup

Reflexivity follows from $aa^{-1} = e \in H$. If $ab^{-1} \in H$, its inverse ba^{-1} belongs to H, proving symmetry. If $ab^{-1}, bc^{-1} \in H$, their product is $ab^{-1}bc^{-1} = ac^{-1} \in H$, proving transitivity. These are all three required conditions; no commutativity or normality is assumed.

Exercise 5.54: Intersecting two subgroups

Both H and K contain e, so their intersection is nonempty. If $x, y \in H \cap K$, their product lies in H and lies in K, so lies in the intersection. The same applies to x^{-1} because both subgroups are inverse-closed. Associativity is inherited from G, and the subgroup criterion therefore proves $H \cap K \leq G$.

Exercise 5.55: A cyclic group is abelian

Write $G = \langle a \rangle$. Given $x, y \in G$, there are integers r, s with $x = a^r, y = a^s$. Then $xy = a^{r+s} = a^{s+r} = yx$, using commutativity of integer addition, not a preexisting commutativity assumption about G. Thus all elements of G commute.

Exercise 5.56: The set of all nth powers

A sufficient hypothesis is that G is abelian (for any fixed positive integer n). Then $e = e^n$ lies in G_n, the product $a^n b^n = (ab)^n$ is again an nth power, and $(a^n)^{-1} = (a^{-1})^n$ is again an nth power. Thus $G_n \leq G$. The hypothesis is sufficient rather than necessary; when $n = 1$, for example, $G_n = G$ for every group.

Exercise 5.57: No proper nontrivial subgroup forces cyclicity

If $G = \{e\}$ it is cyclic. Otherwise choose $a \neq e$. Its cyclic subgroup $\langle a \rangle$ is nontrivial because it contains a. By hypothesis it cannot be a proper subgroup, so $\langle a \rangle = G$. Thus G is cyclic. Choosing a nonidentity element is the step that excludes the trivial subgroup from this argument.

6 Cyclic Groups

Exercise 6.1: Quotient and nonnegative remainder

The division algorithm requires $n = mq + r$ with $0 \leq r < m$. Here

$$42 = 9(4) + 6, \qquad 0 \leq 6 < 9.$$

Thus the quotient is 4 and the remainder is 6. The nonnegative-remainder condition determines the quotient even when the dividend is negative.

Exercise 6.2: Quotient and nonnegative remainder

The division algorithm requires $n = mq + r$ with $0 \leq r < m$. Here

$$-42 = 9(-5) + 3, \qquad 0 \leq 3 < 9.$$

Thus the quotient is -5 and the remainder is 3. The nonnegative-remainder condition determines the quotient even when the dividend is negative.

Exercise 6.3: Quotient and nonnegative remainder

The division algorithm requires $n = mq + r$ with $0 \leq r < m$. Here

$$-50 = 8(-7) + 6, \qquad 0 \leq 6 < 8.$$

Thus the quotient is -7 and the remainder is 6. The nonnegative-remainder condition determines the quotient even when the dividend is negative.

Exercise 6.4: Quotient and nonnegative remainder

The division algorithm requires $n = mq + r$ with $0 \leq r < m$. Here

$$50 = 8(6) + 2, \qquad 0 \leq 2 < 8.$$

Thus the quotient is 6 and the remainder is 2. The nonnegative-remainder condition determines the quotient even when the dividend is negative.

Exercise 6.5: Greatest common divisor: Thirty-two and twenty-four

$32 = 1 \cdot 24 + 8$ and $24 = 3 \cdot 8 + 0$, so $\gcd(32, 24) = 8$. At each Euclidean step, the common divisors of the two current numbers are exactly those of the divisor and remainder; hence the final nonzero remainder is the greatest common divisor.

Exercise 6.6: Greatest common divisor: Forty-eight and eighty-eight

$88 = 1 \cdot 48 + 40$, $48 = 1 \cdot 40 + 8$, and $40 = 5 \cdot 8 + 0$. Thus $\gcd(48, 88) = 8$. At each Euclidean step, the common divisors of the two current numbers are exactly those of the divisor and remainder; hence the final nonzero remainder is the greatest common divisor.

Exercise 6.7: Greatest common divisor: Three hundred sixty and four hundred twenty

$420 = 1 \cdot 360 + 60$ and $360 = 6 \cdot 60 + 0$, so $\gcd(360, 420) = 60$. At each Euclidean step, the common divisors of the two current numbers are exactly those of the divisor and remainder; hence the final nonzero remainder is the greatest common divisor.

Exercise 6.8: Generators at order 5

If a generates a cyclic group of order 5, then a^k is a generator exactly when $\gcd(k, 5) = 1$. Indeed its order is $5/\gcd(k, 5)$. The eligible residues are $\{1, 2, 3, 4\}$, so the number of generators is $\varphi(5) = 4$.

Exercise 6.9: Generators at order 8

If a generates a cyclic group of order 8, then a^k is a generator exactly when $\gcd(k, 8) = 1$. Indeed its order is $8/\gcd(k, 8)$. The eligible residues are $\{1, 3, 5, 7\}$, so the number of generators is $\varphi(8) = 4$.

Exercise 6.10: Generators at order 12

If a generates a cyclic group of order 12, then a^k is a generator exactly when $\gcd(k, 12) = 1$. Indeed its order is $12/\gcd(k, 12)$. The eligible residues are $\{1, 5, 7, 11\}$, so the number of generators is $\varphi(12) = 4$.

Exercise 6.11: Generators at order 60

If a generates a cyclic group of order 60, then a^k is a generator exactly when $\gcd(k, 60) = 1$. Indeed its order is $60/\gcd(k, 60)$. The eligible residues are $\{1, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 49, 53, 59\}$, so the number of generators is $\varphi(60) = 16$.

Exercise 6.12: Automorphisms of Z 2

An additive map from $\mathbb{Z}_2$ is determined by the image k of 1: it must send x to $kx \pmod 2$. It is an automorphism exactly when k generates $\mathbb{Z}_2$, equivalently $\gcd(k, 2) = 1$. The choices are $k \in \{1\}$, so there are 1 automorphisms.

Exercise 6.13: Automorphisms of Z 6

An additive map from $\mathbb{Z}_6$ is determined by the image k of 1: it must send x to $kx \pmod 6$. It is an automorphism exactly when k generates $\mathbb{Z}_6$, equivalently $\gcd(k, 6) = 1$. The choices are $k \in \{1, 5\}$, so there are 2 automorphisms.

Exercise 6.14: Automorphisms of Z 8

An additive map from $\mathbb{Z}_8$ is determined by the image k of 1: it must send x to $kx \pmod 8$. It is an automorphism exactly when k generates $\mathbb{Z}_8$, equivalently $\gcd(k, 8) = 1$. The choices are $k \in \{1, 3, 5, 7\}$, so there are 4 automorphisms.

Exercise 6.15: Automorphisms of the integers

An additive map is determined by $\phi(1)$, since $\phi(n) = n\phi(1)$ for every integer, including negative integers. To be onto, this image must generate $\mathbb{Z}$, so it is 1 or -1. Both maps $n \mapsto n$ and $n \mapsto -n$ are automorphisms. Hence there are exactly 2.

Exercise 6.16: Automorphisms of Z twelve

The image of 1 can be any generator, and determines the whole map by $x \mapsto kx \pmod{12}$. The generators are $1, 5, 7, 11$, the residues relatively prime to 12. Each gives an invertible map (indeed each squares to 1 modulo 12), so there are 4 automorphisms.

Exercise 6.17: Twenty-five modulo thirty

The order of a residue k in the additive group $\mathbb{Z}_n$ is $n/\gcd(n, k)$. Here it is $30/5 = 6$. Directly, the generated subgroup is $\{0, 25, 20, 15, 10, 5\}$; the next multiple is zero and no previous positive multiple is zero.

Exercise 6.18: Thirty modulo forty-two

The subgroup has order $42/\gcd(42, 30) = 42/6 = 7$. Its elements, listed in increasing order, are $\{0, 6, 12, 18, 24, 30, 36\}$. They arise because integer multiples of 30 modulo 42 are exactly the multiples of 6 modulo 42.

Exercise 6.19: The powers of i

The powers are $1, i, -1, -i$, then 1 again. Thus the subgroup has exactly 4 elements. In particular $i^4 = 1$ and no smaller positive power equals 1.

Exercise 6.20: A normalized complex number

The generator $(1 + i)/\sqrt{2} = e^{i\pi/4}$ lies on the unit circle. Its kth power is 1 exactly when $k\pi/4 \in 2\pi\mathbb{Z}$, or $8 \mid k$. Hence its subgroup has 8 elements.

Exercise 6.21: The powers of one plus i

The modulus of $1 + i$ is $\sqrt{2}$. If $(1 + i)^m = (1 + i)^n$, taking moduli gives $(\sqrt{2})^m = (\sqrt{2})^n$, which forces $m = n$. Thus all integral powers are distinct and the generated subgroup is infinite (countably infinite). The modulus need not be 1 in the ambient group $\mathbb{C}^*$.

Exercise 6.22: All subgroups of Z 12

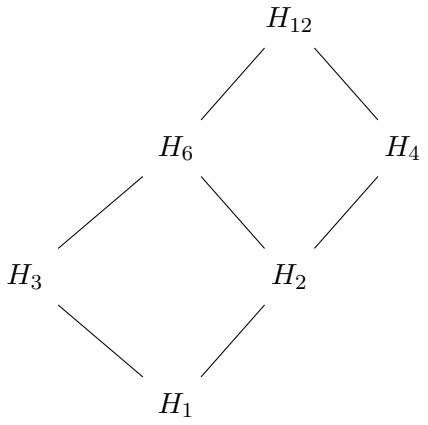
Every subgroup of a cyclic group is cyclic, and there is exactly one subgroup of each order d dividing 12. Label it $H_d = \langle 12/d \rangle$. Its complete element description is

$$H_d = \{0, 12/d, 2(12/d), \ldots, (d-1)(12/d)\}, \qquad |H_d| = d.$$

The complete list is specified by the following table:

d	1	2	3	4	6	12
generator n/d	12	6	4	3	2	1

For $d = 1$, the generator 12 represents 0 modulo 12, so $H_1 = \{0\}$; $H_{12} = \mathbb{Z}_{12}$. Inclusion satisfies $H_d \subseteq H_e$ exactly when $d \mid e$. The diagram joins only immediate inclusions (larger groups above):



Exercise 6.23: All subgroups of Z 36

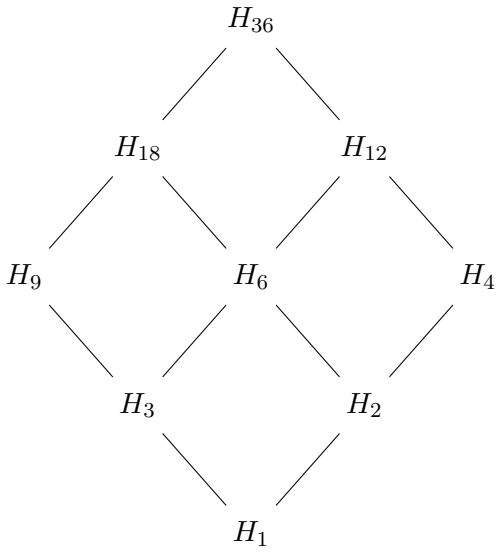
Every subgroup of a cyclic group is cyclic, and there is exactly one subgroup of each order d dividing 36. Label it $H_d = \langle 36/d \rangle$. Its complete element description is

$$H_d = \{0, 36/d, 2(36/d), \ldots, (d-1)(36/d)\}, \qquad |H_d| = d.$$

The complete list is specified by the following table:

d	1	2	3	4	6	9	12	18	36
generator n/d	36	18	12	9	6	4	3	2	1

For $d = 1$, the generator 36 represents 0 modulo 36, so $H_1 = \{0\}$; $H_{36} = \mathbb{Z}_{36}$. Inclusion satisfies $H_d \subseteq H_e$ exactly when $d \mid e$. The diagram joins only immediate inclusions (larger groups above):



Exercise 6.24: All subgroups of Z 8

Every subgroup of a cyclic group is cyclic, and there is exactly one subgroup of each order d dividing 8. Label it $H_d = \langle 8/d \rangle$. Its complete element description is

$$H_d = \{0, 8/d, 2(8/d), \ldots, (d-1)(8/d)\}, \qquad |H_d| = d.$$

The complete list is specified by the following table:

d	1	2	4	8
generator n/d	8	4	2	1

For $d = 1$, the generator 8 represents 0 modulo 8, so $H_1 = \{0\}$; $H_8 = \mathbb{Z}_8$. Inclusion satisfies $H_d \subseteq H_e$ exactly when $d \mid e$. The diagram joins only immediate inclusions (larger groups above):



Exercise 6.25: Subgroup orders in Z 6

The possible orders are exactly the positive divisors of 6:

$$\{1, 2, 3, 6\}.$$

A subgroup of order d actually exists for each listed divisor: it is generated by the residue $6/d$. Conversely the classification of subgroups of a cyclic group excludes all other orders.

Exercise 6.26: Subgroup orders in Z 8

The possible orders are exactly the positive divisors of 8:

$$\{1, 2, 4, 8\}.$$

A subgroup of order d actually exists for each listed divisor: it is generated by the residue $8/d$. Conversely the classification of subgroups of a cyclic group excludes all other orders.

Exercise 6.27: Subgroup orders in Z 12

The possible orders are exactly the positive divisors of 12:

$$\{1, 2, 3, 4, 6, 12\}.$$

A subgroup of order d actually exists for each listed divisor: it is generated by the residue $12/d$. Conversely the classification of subgroups of a cyclic group excludes all other orders.

Exercise 6.28: Subgroup orders in Z 20

The possible orders are exactly the positive divisors of 20:

$$\{1, 2, 4, 5, 10, 20\}.$$

A subgroup of order d actually exists for each listed divisor: it is generated by the residue $20/d$. Conversely the classification of subgroups of a cyclic group excludes all other orders.

Exercise 6.29: Subgroup orders in Z 17

The possible orders are exactly the positive divisors of 17:

$$\{1, 17\}.$$

A subgroup of order d actually exists for each listed divisor: it is generated by the residue $17/d$. Conversely the classification of subgroups of a cyclic group excludes all other orders.

Exercise 6.30: The definition of element order

An element a has order $n \in \mathbb{Z}^+$ if and only if n is the *least positive* integer such that $a^n = e$. Merely requiring $a^n = e$ is not sufficient: if a has order 2, it also satisfies $a^4 = e$, but its order is not 4. If no such positive exponent exists, the order is infinite.

Exercise 6.31: The definition of greatest common divisor

The definition is correct: the greatest common divisor of two positive integers is their largest positive common divisor. Existence is assured because 1 is a common divisor and all positive common divisors are bounded above by the smaller integer. A useful stronger property, proved via the Euclidean algorithm, is that every common divisor divides this greatest common divisor.

Exercise 6.32: Ten statements about cyclic groups

(a) True: powers of a common generator commute.

(b) False: the Klein four-group is abelian but not cyclic.

(c) False: if $q \neq 0$ generated $\mathbb{Q}$ additively, $q/2$ would not be an integer multiple of q.

(d) False: zero in a nontrivial additive cyclic group generates only its trivial subgroup.

(e) True: $\mathbb{Z}_n$ is abelian of order n for every positive n.

(f) False: the Klein four-group has order 4 and is not cyclic.

(g) False: 1 and 9 both generate $\mathbb{Z}_{20}$ but are not prime numbers; relative primality to 20 is the criterion.

(h) False in general: two groups may even have disjoint underlying sets, so their intersection is empty and cannot be a group. An operation shared in an ambient group is not specified here.

(i) True: the intersection contains the shared identity and is closed under the shared operation and inverses.

(j) True: if a generates a cyclic group of order greater than 2, then a^{-1} is a distinct generator; equality would imply $a^2 = e$ and order at most 2. The same reasoning covers infinite order.

Exercise 6.33: A finite noncyclic group

Use the Klein four-group $V = \{e, a, b, c\}$. Each nonidentity element has order 2, so every cyclic subgroup has at most two elements. None can equal the four-element group, proving that it is not cyclic.

Exercise 6.34: An infinite noncyclic group

The additive group $\mathbb{Q}$ is infinite and not cyclic. The zero element cannot generate it. If a nonzero rational q were a generator, then $q/2$ would have to be nq for some integer n; dividing by q gives $n = 1/2$, impossible.

Exercise 6.35: A cyclic group with one generator

The group $\mathbb{Z}_2$ works: the element 1 generates both residues, whereas 0 generates just itself. Hence it has exactly one generator. The trivial group is another possible example, since its identity generates it.

Exercise 6.36: An infinite cyclic group with four generators

No such group exists. Every infinite cyclic group is isomorphic to $\mathbb{Z}$. The only generators of $\mathbb{Z}$ are $1, -1$, since $k\mathbb{Z} = \mathbb{Z}$ forces $|k| = 1$. Isomorphisms preserve generators, so every infinite cyclic group has exactly two.

Exercise 6.37: A finite cyclic group with four generators

Use $\mathbb{Z}_5$. Its generators are $1, 2, 3, 4$, since each is relatively prime to 5 and therefore has order 5. The identity 0 is not a generator, so there are exactly four.

Exercise 6.38: Primitive roots for n equals 4

The roots of unity are $e^{2\pi ik/4}$, and such a root is primitive exactly when $\gcd(k,4)=1$, since its order is $4/\gcd(k,4)$. Therefore the primitive roots are

$$\{e^{2\pi ik/4} : k \in \{1,3\}\}.$$

These are $i, -i$.

Exercise 6.39: Primitive roots for n equals 6

The roots of unity are $e^{2\pi ik/6}$, and such a root is primitive exactly when $\gcd(k,6)=1$, since its order is $6/\gcd(k,6)$. Therefore the primitive roots are

$$\{e^{2\pi ik/6} : k \in \{1,5\}\}.$$

These are $1/2 + i\sqrt{3}/2$ and $1/2 - i\sqrt{3}/2$.

Exercise 6.40: Primitive roots for n equals 8

The roots of unity are $e^{2\pi ik/8}$, and such a root is primitive exactly when $\gcd(k,8)=1$, since its order is $8/\gcd(k,8)$. Therefore the primitive roots are

$$\{e^{2\pi ik/8} : k \in \{1,3,5,7\}\}.$$

These are $(1+i)/\sqrt{2}$, $(-1+i)/\sqrt{2}$, $(-1-i)/\sqrt{2}$, and $(1-i)/\sqrt{2}$.

Exercise 6.41: Primitive roots for n equals 12

The roots of unity are $e^{2\pi ik/12}$, and such a root is primitive exactly when $\gcd(k,12)=1$, since its order is $12/\gcd(k,12)$. Therefore the primitive roots are

$$\{e^{2\pi ik/12} : k \in \{1,5,7,11\}\}.$$

Their arguments are $\pi/6, 5\pi/6, 7\pi/6, 11\pi/6$; equivalently the roots are $(\pm\sqrt{3} \pm i)/2$ with independent signs.

Exercise 6.42: Synopsis of cyclic implies abelian

Express any two elements as powers of one generator and use addition of their integer exponents to see that their products in the two orders agree.

Exercise 6.43: Synopsis of the cyclic-subgroup theorem

For a nontrivial subgroup of a cyclic group, choose the least positive exponent whose generator power belongs to the subgroup. Divide the exponent of any other subgroup element by this least exponent; closure puts the remainder power in the subgroup, and minimality forces that remainder to be zero. Thus every element of the subgroup is a power of the chosen one.

Exercise 6.44: A homomorphism from a cyclic group is determined by one value

Write an arbitrary $x \in G$ as a^m for an integer m. Preservation of products and inverses gives $\phi(x) = \phi(a^m) = \phi(a)^m$. If $\psi(a) = \phi(a)$, then $\psi(x) = \psi(a)^m = \phi(a)^m = \phi(x)$. This works for all x, including the identity and negative powers. Thus the maps agree everywhere once they agree on a generator.

Exercise 6.45: Integer linear combinations form a subgroup

Let $H = \{nr + ms : n, m \in \mathbb{Z}\}$. It contains $0 = 0r + 0s$. The sum of $nr + ms$ and $n'r + m's$ is $(n+n')r + (m+m')s$, again in H. The negative is $(-n)r + (-m)s$, also in H. Hence H is a subgroup of the additive group $\mathbb{Z}$ (indeed it is $\gcd(r, s)\mathbb{Z}$).

Exercise 6.46: The orders of ab and ba

The elements are conjugate: $ba = a^{-1}(ab)a$. Therefore for every positive integer k,

$$(ba)^k = a^{-1}(ab)^k a,$$

proved by cancellation of the adjacent aa^{-1} factors. Consequently $(ba)^k = e$ if and only if $(ab)^k = e$. Their least positive identity exponents are therefore the same, so ba also has order n. This proves equality, rather than merely showing its order divides n.

Exercise 6.47: Least common multiples via subgroups

(a) The intersection $r\mathbb{Z} \cap s\mathbb{Z}$ is a subgroup of $\mathbb{Z}$ containing $rs > 0$. Its unique positive generator is $\ell = \operatorname{lcm}(r, s)$: its positive members are exactly the positive common multiples, and its least positive member generates it.

(b) $\ell = rs$ exactly when $\gcd(r, s) = 1$.

(c) Put $d = \gcd(r, s)$ and write $r = du, s = dv$ with $\gcd(u, v) = 1$. If M is a common multiple, write $M = duk$. Since $dv \mid M$, we have $v \mid uk$; relatively prime u, v imply $v \mid k$ (multiply a Bezout identity for u, v by k). Thus $duv \mid M$. Conversely duv is a common multiple. Therefore $\ell = duv$, and $d\ell = d^2uv = rs$. Since r, s are positive, this formula also proves the equivalence in part (b).

Exercise 6.48: Finitely many subgroups force a finite group

First, no element can have infinite order. If a did, its infinite cyclic subgroup would contain the infinitely many distinct subgroups $\langle a^m \rangle$, $m = 1, 2, \ldots$, all also subgroups of G. They are distinct because in an infinite cyclic group the exponent sets $m\mathbb{Z}$ are distinct. Thus every cyclic subgroup of G is finite. There are only finitely many subgroups by hypothesis, hence only finitely many cyclic subgroups. Every element lies in the cyclic subgroup it generates, so G is a finite union of finite cyclic subgroups and is finite.

Exercise 6.49: All proper subgroups cyclic does not suffice

The Klein four-group is not cyclic, since every element has order at most 2. Its proper subgroups are $\{e\}$, $\{e, a\}$, $\{e, b\}$, and $\{e, c\}$, all cyclic. Indeed any subgroup containing two different nonidentity elements must also contain their product, the third nonidentity element, and hence must be all of V. Thus this is the required counterexample.

Exercise 6.50: A unique element of order two is central

For arbitrary $x \in G$, compute $(xax^{-1})^2 = xa^2x^{-1} = e$. Also $xax^{-1} \neq e$, because otherwise multiplying by x^{-1} on the left and x on the right would give $a = e$. Therefore this conjugate also has order 2, and uniqueness forces $xax^{-1} = a$. Multiplying on the right by x yields $xa = ax$. Since x was arbitrary, a commutes with every element.

Exercise 6.51: Generators at order pq

A residue modulo pq generates the cyclic group exactly when it is not divisible by p or by q. Among the pq residues $0, \ldots, pq-1$, there are q multiples of p, p multiples of q, and one residue (zero) counted in both lists. Inclusion-exclusion gives

$$pq - q - p + 1 = (p-1)(q-1)$$

generators. Distinctness and primality of p, q ensure that these are exactly the forbidden prime divisors.

Exercise 6.52: Generators at a prime-power order

A residue modulo p^r generates the additive group precisely when it is relatively prime to p^r, equivalently not divisible by p. Among $0, 1, \ldots, p^r - 1$, the multiples of p are $0, p, 2p, \ldots, (p^{r-1} - 1)p$, exactly p^{r-1} residues. Therefore the number of generators is

$$p^r - p^{r-1} = p^{r-1}(p-1).$$

This includes $r = 1$, giving $p - 1$ nonzero generators in $\mathbb{Z}_p$.

Exercise 6.53: A power equation when the exponent divides the order

Let $G = \langle a \rangle$ have order n, and write $n = mq$. Every element has a unique representation a^k with $0 \leq k < n$. The equation $(a^k)^m = e$ holds if and only if $n \mid km$, which, since $n = mq$, is equivalent to $q \mid k$. The eligible exponents are therefore

$$0, q, 2q, \ldots, (m-1)q,$$

exactly m distinct residues. The corresponding m distinct group elements are all the solutions. The argument includes the cases $m = 1$ and $m = n$.

Exercise 6.54: A power equation for a general exponent

Put $d = \gcd(n, m)$ and write $n = dn'$, $m = dm'$ with $\gcd(n', m') = 1$. For $x = a^k$, the equation $x^m = e$ is equivalent to $n \mid mk$, hence to $n' \mid m'k$. Relative primality makes this equivalent to $n' \mid k$. There are exactly d such residues modulo n, namely $k = 0, n', \ldots, (d-1)n'$. Thus there are $\gcd(n, m)$ solutions. In the question's case $1 < m < n$ and $m \nmid n$, this number is strictly less than m (and equals 1 if n, m are relatively prime).

Exercise 6.55: Subgroups at prime order

Every subgroup of $\mathbb{Z}_p$ has order dividing p, by the classification of subgroups of a cyclic group. Since p is prime, the only divisors are 1 and p, yielding respectively the trivial subgroup and the whole group. Hence there is no proper nontrivial subgroup. Equivalently, any nonzero residue k is relatively prime to p and itself generates all of $\mathbb{Z}_p$.

Exercise 6.56: Finding a cyclic subgroup of a least-common-multiple order

Write $H = \langle a \rangle$, $K = \langle b \rangle$, with a, b of orders r, s.

(a) Assume $\gcd(r, s) = 1$. Any element of $H \cap K$ has order dividing both r and s, hence is the identity. Since the group is abelian, $(ab)^{rs} = e$. Conversely if $(ab)^t = e$, then $a^t = b^{-t}$ lies in $H \cap K$, so $a^t = b^t = e$. Thus $r \mid t$ and $s \mid t$, giving $rs \mid t$. Therefore ab has order exactly rs, and $\langle ab \rangle$ is the required subgroup.

(b) Factor r, s into prime powers. For each prime p appearing, take the larger of its two prime-power factors, say p^{γ_p}. If that factor comes from r, choose $c_p = a^{r/p^{\gamma_p}}$; its order is exactly p^{γ_p}, by the order formula for a power. If it comes from s, choose $c_p = b^{s/p^{\gamma_p}}$ instead (ties can use either). All these chosen elements commute because G is abelian. Their orders are pairwise relatively prime, so applying part (a) repeatedly shows that the product $c = \prod_p c_p$ has order $\prod_p p^{\gamma_p} = \operatorname{lcm}(r, s)$. Thus $\langle c \rangle$ is the required cyclic subgroup. If $r = s = 1$, simply take $c = e$.

The product ab itself need not have the least-common-multiple order when r, s overlap; the prime-power selection prevents cancellation of common factors.

7 Generating Sets and Cayley Digraphs

Exercise 7.1: Two and three modulo twelve

The generated subgroup contains $3-2=1$, because subgroups are closed under addition and negatives. Once it contains 1, it contains every residue. Therefore it is $\mathbb{Z}_{12}=\{0,1,2,3,4,5,6,7,8,9,10,11\}$.

Exercise 7.2: Four and six modulo twelve

Every integer combination of 4 and 6 is even, so the subgroup lies inside the even residues. It contains $6-4=2$, and hence every even residue. Thus it is exactly $\{0,2,4,6,8,10\}$.

Exercise 7.3: Eight and ten modulo eighteen

Every combination of $8,10$ and the modulus 18 is even, so no odd residue can occur. The difference $10-8=2$ belongs, and its multiples give all even residues. Thus the subgroup is $\{0,2,4,6,8,10,12,14,16\}$.

Exercise 7.4: Twelve and thirty modulo thirty-six

The difference $30-2\cdot 12=6$ belongs. Every input and the modulus are divisible by 6, so only multiples of 6 occur, and all of them occur by adding 6. The subgroup is therefore $\{0,6,12,18,24,30\}$.

Exercise 7.5: Twelve and forty-two in the integers

All generated elements are combinations $12m+42n$, hence multiples of 6. Conversely $42-3\cdot 12=6$ belongs, so every integer multiple of 6 belongs. The subgroup is $6\mathbb{Z}=\{\ldots,-12,-6,0,6,12,\ldots\}$.

Exercise 7.6: Three integer generators

All three inputs $18,24,39$ are divisible by 3. The generated subgroup contains $24-18=6$ and then $39-6\cdot 6=3$. It therefore contains every multiple of 3, and contains no other integer. The answer is $3\mathbb{Z}=\{\ldots,-6,-3,0,3,6,\ldots\}$.

Exercise 7.7: Products in the eight-element digraph

The outer solid cycle gives $a^4=e$. Dashed edges give $b^2=e$, and the reversed inner solid cycle gives $ba=a^{-1}b$. Thus $ba^k=a^{-k}b$ for every integer k.

(a) $(a^2b)a^3=a^2a^{-3}b=a^{-1}b=a^3b$.

(b) $(ab)(a^3b)=aa^{-3}b^2=a^{-2}=a^2$.

(c) $b(a^2b)=a^{-2}b^2=a^{-2}=a^2$.

Each result can also be checked by starting at the first factor's vertex and following the right factor's directed edge sequence.

Exercise 7.8: The square digraph table

Both indicated generator types are involutions, and traversing one of each takes e to c in either order. Thus $a^2 = b^2 = e$, $ab = ba = c$, giving

	e	a	b	c
e	e	a	b	c
a	a	e	c	b
b	b	c	e	a
c	c	b	a	e

For an order-two generator an undirected edge stands for the two opposite directed edges. The resulting group is the Klein four-group.

Exercise 7.9: The alternating-edge hexagon table

Read the two edge types as involutions a, b. The paths give $c = ab$, $d = ba$, and $f = aba = bab$. One concrete model is $a = (12)$ and $b = (23)$, so $c = (123)$, $d = (132)$, and $f = (13)$. Multiplying these permutations, with the rightmost one applied first, yields the table in the requested alphabetical order:

	e	a	b	c	d	f
e	e	a	b	c	d	f
a	a	e	c	b	f	d
b	b	d	e	f	a	c
c	c	f	a	d	e	b
d	d	b	f	e	c	a
f	f	c	d	a	b	e

For example $ab = c$ but $ba = d$, so this six-element group is nonabelian. The model also verifies the whole table's associativity.

Exercise 7.10: The two-triangle digraph table

The outside arrows give $a^3 = e$ and $c = a^2$. The dashed edges give $b^2 = e$, $ab = f$, $a^2b = d$. Reading the inside arrows yields $ba = a^{-1}b$, so this is the triangular dihedral group. Using $a = (123)$, $b = (12)$ realizes these labels concretely and gives

	e	a	b	c	d	f
e	e	a	b	c	d	f
a	a	c	f	e	b	d
b	b	d	e	f	a	c
c	c	e	d	a	f	b
d	d	f	c	b	e	a
f	f	b	a	d	c	e

For example $ab = f$ and $ba = d$, consistent with the distinct paths in the figure. This diagram and the preceding one describe isomorphic groups, but their element names represent different permutations.

Exercise 7.11: Recognizing commutativity from paths

Compare the endpoints of paths using two generator types in opposite orders. If some s-then-t path ends differently from the corresponding t-then-s path at the same starting point, the group is noncommutative. Conversely, if every pair of generators commutes, their inverses also commute; any two words in generators and inverses can then be reordered past each other. Since such words are all group elements, the group is abelian. Thus this two-step-path comparison is a complete criterion.

Exercise 7.12: The eight-element example is nonabelian

Starting at e, traversing an a edge and then a b edge ends at ab, whereas traversing b then a ends at a^3b. These are different labeled vertices in Fig. 7.11(b). Therefore $ab \neq ba$ and the group is noncommutative.

Exercise 7.13: Recognizing cyclicity

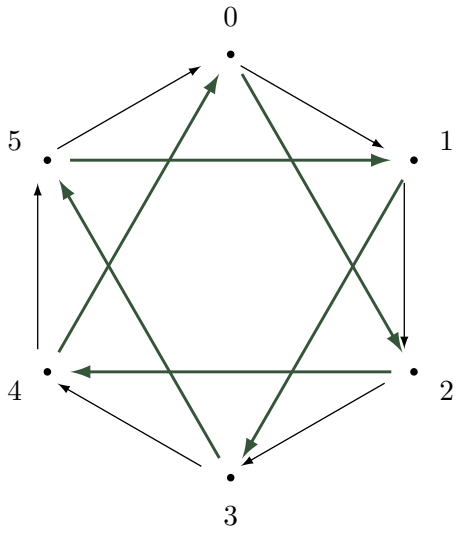
A single generator type forming a directed cycle through every vertex proves cyclicity, but its absence does not disprove cyclicity. A diagram might use a redundant generating set without displaying any individual generator of the whole group. For example $\mathbb{Z}_6$ generated by $\{2, 3\}$ is cyclic, while the 2-edges form triangles and the 3-edges form pairs. Thus cyclicity need not be visually apparent from one edge type; one may need a repeated word in several edge types.

Exercise 7.14: The inner triangle is not a subgroup

The smaller triangle in Fig. 7.9(b) has the vertices $\{1, 3, 5\}$. It does not contain the additive identity 0, so it cannot be a subgroup of $\mathbb{Z}_6$. It is also not closed: $1 + 1 = 2$ is outside it. It is a translated copy of the subgroup $\{0, 2, 4\}$, which explains the identical triangle shape without making it a subgroup.

Exercise 7.15: A Cayley digraph with generators one and two

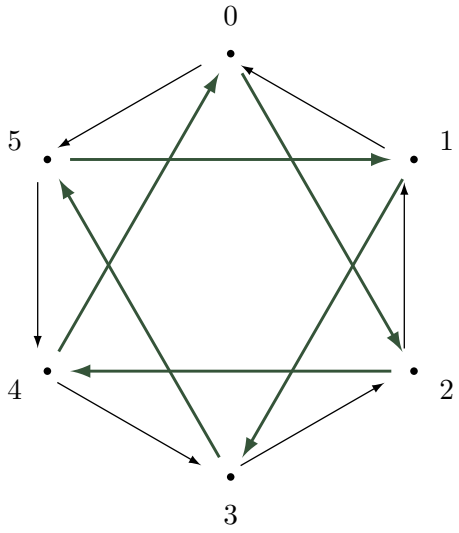
Place the six residues around a hexagon in the order $0, 1, 2, 3, 4, 5$ clockwise. Draw a thin black arrow $x \to x + 1$ and a thick green arrow $x \to x + 2$, reducing modulo 6:



The black arrows form $0 \to 1 \to 2 \to 3 \to 4 \to 5 \to 0$. The green arrows form $0 \to 2 \to 4 \to 0$ and $1 \to 3 \to 5 \to 1$. Thus each vertex has exactly one incoming and one outgoing arrow of each generator type, as required.

Exercise 7.16: A Cayley digraph with generators two and five

Use the same labeled hexagon. Draw thick green arrows $x \to x+2$ and thin black arrows $x \to x+5 = x - 1$ modulo 6:



The green triangles are $0 \to 2 \to 4 \to 0$ and $1 \to 3 \to 5 \to 1$. The black outside cycle is $0 \to 5 \to 4 \to 3 \to 2 \to 1 \to 0$. Since 5 alone is a generator, the combined diagram is connected and represents all of $\mathbb{Z}_6$.

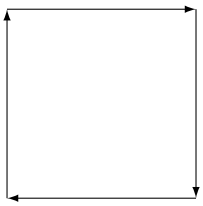
Exercise 7.17: Relations read from closed paths

(a) Begin at the identity and follow a closed path. Write down the generator for each forward edge and its inverse for each backward edge. The resulting word is a product equal to e, hence a relation. Conversely any such relation returns the identity to itself when its word is followed.

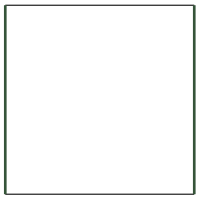
(b) In Fig. 7.11(b), the outside four-cycle gives $a^4 = e$, a dashed round trip gives $b^2 = e$, and the path expressing $ba = a^{-1}b$ gives $baba = e$. Thus three relations are $a^4 = e$, $b^2 = e$, and $(ba)^2 = e$. The last can also be written $bab = a^{-1}$.

Exercise 7.18: Minimal digraphs at order four

For the cyclic group use one generator: a directed four-cycle. For the Klein group one generator never suffices, since every element has order at most 2; use two involutions and a square with different edge types in the two directions. Here a double-headed edge is represented by an undirected line, as usual for an involution.



$\mathbb{Z}_4$: one generator



V: two generators

Each square has four vertices at its corners. In the right-hand diagram horizontal edges have one generator type and vertical edges the other; both types are their own reverses.

Exercise 7.19: Two involutions generating a nonabelian group of order twice n

Take symmetries of a regular n-gon, $n \geq 3$. Let r rotate through $2\pi/n$ and s reflect in an axis through its center. The symmetries are exactly r^k and $r^k s$ for $0 \leq k < n$: these $2n$ transformations are distinct because the first preserve orientation and the second reverse it, and the rotations themselves are distinct. They are closed under composition because $sr = r^{-1}s$.

Choose two order-two generators.

The elements s and rs are reflections. They satisfy $s^2 = e$ and $(rs)^2 = r(sr)s = rr^{-1}s^2 = e$, and neither is the identity. Their product $(rs)s = r$ gives the basic rotation, so they generate every $r^k, r^k s$.

Check noncommutativity.

The products rs and $sr = r^{-1}s$ differ: equality would give $r = r^{-1}$, contradicting that r has order $n \geq 3$. Thus the group is nonabelian, has exactly $2n$ elements, and is generated by two elements of order 2.

8 Groups of Permutations

Conventions. Permutations compose from right to left. The identity is e. A group generated by an element a is denoted $\langle a \rangle$.

Exercise 8.1: Product $\tau\sigma$

Plan. Apply the rightmost permutation first. The intermediate images are

x	1	2	3	4	5	6
$\sigma(x)$	3	1	4	5	6	2
$\tau(\sigma(x))$	1	2	3	6	5	4

The final row is the image of $1, \ldots, 6$ under the required product. Thus

$$\tau\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 1 & 2 & 3 & 6 & 5 & 4 \end{pmatrix}.$$

Exercise 8.2: Product $\tau^2\sigma$

Plan. Apply the rightmost permutation first. The intermediate images are

x	1	2	3	4	5	6
$\sigma(x)$	3	1	4	5	6	2
$\tau(\sigma(x))$	1	2	3	6	5	4
$\tau(\tau(\sigma(x)))$	2	4	1	5	6	3

The final row is the image of $1, \ldots, 6$ under the required product. Thus

$$\tau^2\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 2 & 4 & 1 & 5 & 6 & 3 \end{pmatrix}.$$

Exercise 8.3: Product $\mu\sigma^2$

Plan. Apply the rightmost permutation first. The intermediate images are

x	1	2	3	4	5	6
$\sigma(x)$	3	1	4	5	6	2
$\sigma(\sigma(x))$	4	3	5	6	2	1
$\mu(\sigma(\sigma(x)))$	3	4	1	6	2	5

The final row is the image of $1, \ldots, 6$ under the required product. Thus

$$\mu\sigma^2 = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 3 & 4 & 1 & 6 & 2 & 5 \end{pmatrix}.$$

Exercise 8.4: Product $\sigma^{-2}\tau$

Plan. Apply the rightmost permutation first. The intermediate images are

x	1	2	3	4	5	6
$\tau(x)$	2	4	1	3	6	5
$\sigma^{-1}(\tau(x))$	6	3	2	1	5	4
$\sigma^{-1}(\sigma^{-1}(\tau(x)))$	5	1	6	2	4	3

The final row is the image of $1, \ldots, 6$ under the required product. Thus

$$\sigma^{-2}\tau = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 5 & 1 & 6 & 2 & 4 & 3 \end{pmatrix}.$$

Exercise 8.5: Product $\sigma^{-1}\tau\sigma$

Plan. Apply the rightmost permutation first. The intermediate images are

x	1	2	3	4	5	6
$\sigma(x)$	3	1	4	5	6	2
$\tau(\sigma(x))$	1	2	3	6	5	4
$\sigma^{-1}(\tau(\sigma(x)))$	2	6	1	5	4	3

The final row is the image of $1, \ldots, 6$ under the required product. Thus

$$\sigma^{-1}\tau\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 2 & 6 & 1 & 5 & 4 & 3 \end{pmatrix}.$$

Exercise 8.6: Order of the subgroup generated by σ

Following 1 gives $1 \mapsto 3 \mapsto 4 \mapsto 5 \mapsto 6 \mapsto 2 \mapsto 1$. Thus $\sigma = (1, 3, 4, 5, 6, 2)$. It takes exactly six applications to return 1, and six applications fix every point. Therefore σ has order 6, so $|\langle\sigma\rangle| = 6$.

Exercise 8.7: Order of the subgroup generated by au^2

The cycles of τ are $(1, 2, 4, 3)(5, 6)$. Squaring advances two steps in each cycle, giving $\tau^2 = (1, 4)(2, 3)$, with $5, 6$ fixed. This permutation is not the identity, but its square is. Consequently $\langle\tau^2\rangle = \{e, \tau^2\}$ and the requested order is 2.

Exercise 8.8: The power σ^{100}

Since $\sigma^6 = e$ and $100 = 16 \cdot 6 + 4$, $\sigma^{100} = \sigma^4$. Advance four steps around $(1, 3, 4, 5, 6, 2)$ to obtain

$$\sigma^{100} = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 6 & 5 & 2 & 1 & 3 & 4 \end{pmatrix} = (1, 6, 4)(2, 5, 3).$$

For example $1 \mapsto 3 \mapsto 4 \mapsto 5 \mapsto 6$, checking the first image.

Exercise 8.9: The power μ^{100}

Here $\mu = (1,5)(3,4)$, so $\mu^2 = e$. Since 100 is even, $\mu^{100} = (\mu^2)^{50} = e$. The points $2, 6$ are fixed throughout.

Exercise 8.10: Partition into isomorphism classes

Plan. First separate finite groups by order, then use commutativity, torsion, and divisibility to distinguish infinite groups. The complete partition is

$$\{(\mathbb{Z},+),(17\mathbb{Z},+),(3\mathbb{Z},+),\langle\pi\rangle\},$$
$$\{\mathbb{Z}_6, G\},\quad \{\mathbb{Z}_2, S_2\},\quad \{S_6\},$$
$$\{(\mathbb{Q},+)\},\quad \{(\mathbb{R},+),(\mathbb{R}^+,\cdot)\},$$
$$\{(\mathbb{R}^*,\cdot)\},\quad \{(\mathbb{Q}^*,\cdot)\},\quad \{(\mathbb{C}^*,\cdot)\}.$$

The maps $n \mapsto 17n$, $n \mapsto 3n$, and $n \mapsto \pi^n$ give the first class. The generator specified for G is $(1,3,4)(2,5)$, of order 6, so $G \cong \mathbb{Z}_6$. Both S_2 and $\mathbb{Z}_2$ are cyclic of order 2. The group S_6 has 720 elements and is nonabelian. The exponential $x \mapsto e^x$ identifies $(\mathbb{R},+)$ with $(\mathbb{R}^+,\cdot)$. The additive group $\mathbb{Q}$ is countable and divisible (every equation $nx = q$, $n > 0$, has a solution), whereas $\mathbb{Z}$ is not divisible. The group $\mathbb{Q}^*$ is countable but has the element -1 of order 2, distinguishing it from the countable torsion-free groups. The uncountable group $\mathbb{R}^*$ has exactly two torsion elements, $1, -1$, and is not divisible because $x^2 = -1$ has no real solution. The group $\mathbb{C}^*$ has roots of unity of every positive order. These observations also exclude all remaining possible identifications.

Exercise 8.11: Orbit under σ

Repeatedly applying the permutation gives $1 \mapsto 3 \mapsto 4 \mapsto 5 \mapsto 6 \mapsto 2 \mapsto 1$. Negative powers traverse the same cycle backwards, so the orbit of 1 is exactly $\{1,2,3,4,5,6\}$.

Exercise 8.12: Orbit under au

Repeatedly applying the permutation gives $1 \mapsto 2 \mapsto 4 \mapsto 3 \mapsto 1$. Negative powers traverse the same cycle backwards, so the orbit of 1 is exactly $\{1,2,3,4\}$.

Exercise 8.13: Orbit under μ

Repeatedly applying the permutation gives $1 \mapsto 5 \mapsto 1$. Negative powers traverse the same cycle backwards, so the orbit of 1 is exactly $\{1,5\}$.

Exercise 8.14: Six geometric labels for S_3

Let r be a 120° rotation of the equilateral triangle and s its specified reflection. The rotations are e, r, r^2. Applying s first and then each rotation gives the three reflections s, rs, r^2s, whose reflection axes pass through the three different vertices. They are distinct, and no reflection is a rotation because reflections reverse orientation. Hence e, r, r^2, s, rs, r^2s are precisely the six elements of S_3.

Exercise 8.15: Eight geometric labels for D_4

Choose the quarter-turn $r = \rho_1$ and any one reflection s, for example $s = \mu_1$. The eight elements are

$$e, r, r^2, r^3, s, rs, r^2s, r^3s.$$

The first four preserve orientation and give the four rotational placements of a square. The last four reverse orientation and send a fixed vertex to four distinct vertices, so they are all different reflections. Thus they give every element of D_4.

Exercise 8.16: Permutations fixing 3

Fixing 3 in a permutation of $\{1, 2, 3, 4\}$ leaves a free permutation of $\{1, 2, 4\}$. There are $3 \cdot 2 \cdot 1 = 6$ such choices, and each extends uniquely by $3 \mapsto 3$. The answer is 6.

Exercise 8.17: Permutations sending 2 to 5

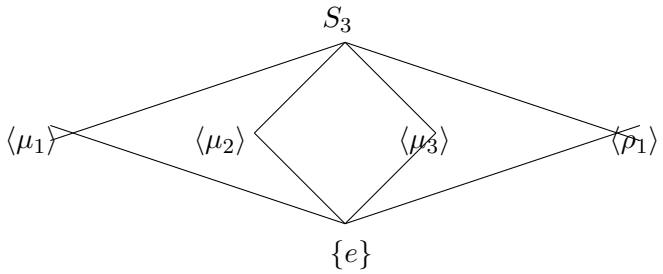
The image of 2 is already fixed. The other four arguments must map bijectively to the four unused targets $1, 2, 3, 4$. There are $4! = 24$ ways to do this. Notice that fixing an image different from the argument does not change the count.

Exercise 8.18: Subgroups of S_3

(a) Both nonidentity rotations generate $\{e, \rho_1, \rho_2\}$, since $\rho_1^2 = \rho_2$ and $\rho_1^3 = e$. Since $\mu_1^2 = e$, $\langle \mu_1 \rangle = \{e, \mu_1\}$. **(b)** The entire subgroup list is

$$\{e\}, \quad \langle \mu_1 \rangle, \langle \mu_2 \rangle, \langle \mu_3 \rangle, \quad \langle \rho_1 \rangle, \quad S_3.$$

An element is either a reflection or a rotation. Two distinct reflections have a nontrivial rotation as product, and a reflection together with a nontrivial rotation generates all six elements. Thus there are no additional subgroups. In the subgroup diagram $\{e\}$ lies below each of the four proper nontrivial subgroups, and each of those lies below S_3; there are no inclusions between those four.



Exercise 8.19: All subgroups of D_4

Write $r = \rho_1$ and $s = \mu_1$, so $r^4 = s^2 = e$ and $sr = r^{-1}s$. The cyclic subgroups generated by single elements are

$$\{e\}, \quad \langle r^2 \rangle, \quad \langle r \rangle, \quad \langle s \rangle, \langle rs \rangle, \langle r^2 s \rangle, \langle r^3 s \rangle.$$

Here $\langle r\rangle$ has order 4, while the five other nontrivial cyclic subgroups have order 2. If a subgroup contains a reflection and r or r^3, it is all of D_4. If it contains two reflections $r^i s, r^j s$, their product is r^{i-j}. For odd $i-j$ this again generates D_4; for even nonzero $i-j$, it produces one of the two Klein groups

$$K_0 = \{e, r^2, s, r^2 s\}, \qquad K_1 = \{e, r^2, rs, r^3 s\}.$$

These are all possibilities for subgroups with two or more generators. Thus the diagram has top D_4; the next level $\langle r\rangle, K_0, K_1$; the next level the five order-2 groups; and bottom $\{e\}$. The group $\langle r^2\rangle$ connects to all three groups of order 4, $\langle s\rangle, \langle r^2 s\rangle$ connect to K_0, and $\langle rs\rangle, \langle r^3 s\rangle$ connect to K_1. These are exactly the edges in the supplied diagram, verifying it exhaustively.

Exercise 8.20: The cyclic subgroup generated by ho

The permutation is $\rho = (1,2,4)(3,5)$, so its order is $\operatorname{lcm}(3,2) = 6$. Indeed, $\rho^k = e$ precisely when both $3 \mid k$ and $2 \mid k$. In the following table $e = \rho^0 = \rho^6$:

	e	ρ	ρ^2	ρ^3	ρ^4	ρ^5
e	e	ρ^1	ρ^2	ρ^3	ρ^4	ρ^5
ρ^1	ρ^1	ρ^2	ρ^3	ρ^4	ρ^5	e
ρ^2	ρ^2	ρ^3	ρ^4	ρ^5	e	ρ^1
ρ^3	ρ^3	ρ^4	ρ^5	e	ρ^1	ρ^2
ρ^4	ρ^4	ρ^5	e	ρ^1	ρ^2	ρ^3
ρ^5	ρ^5	e	ρ^1	ρ^2	ρ^3	ρ^4

The rule is $\rho^i \rho^j = \rho^{i+j \bmod 6}$. This group is abelian and cyclic, so it is not isomorphic to the nonabelian group S_3, despite having the same order.

Exercise 8.21: Six permutation matrices

(a) Each displayed matrix has exactly one 1 in each row and column, and all other entries zero. Such a matrix permutes the standard basis vectors. If $P_\sigma e_j = e_{\sigma(j)}$, then

$$P_\sigma P_\tau e_j = P_\sigma e_{\tau(j)} = e_{\sigma(\tau(j))},$$

so $P_\sigma P_\tau = P_{\sigma\tau}$. Thus the six matrices are closed under multiplication, contain I, and have inverses $P_{\sigma^{-1}}$; associativity comes from matrix multiplication. **(b)** There are exactly $3! = 6$ permutations of three basis vectors, and these are the six displayed matrices. The map $\sigma \mapsto P_\sigma$ is a bijective homomorphism, so the group is isomorphic to S_3.

Exercise 8.22: Matrices representing D_4

Use the rotation and reflection matrices

$$R = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad S = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

The requested eight matrices are $I, R, R^2, R^3, S, RS, R^2S, R^3S$, explicitly

$$\begin{pmatrix}1&0\\0&1\end{pmatrix}, \begin{pmatrix}0&-1\\1&0\end{pmatrix}, \begin{pmatrix}-1&0\\0&-1\end{pmatrix}, \begin{pmatrix}0&1\\-1&0\end{pmatrix},$$
$$\begin{pmatrix}1&0\\0&-1\end{pmatrix}, \begin{pmatrix}0&1\\1&0\end{pmatrix}, \begin{pmatrix}-1&0\\0&1\end{pmatrix}, \begin{pmatrix}0&-1\\-1&0\end{pmatrix}.$$

They act on the square with vertices $(\pm1, \pm1)$ as exactly its eight symmetries. The equations $R^4 = S^2 = I$ and $SR = R^{-1}S$ prove closure and the same multiplication rule as D_4, giving the desired isomorphism.

Exercise 8.23: Symmetries of the three-armed figure

The three equal arms meet at 120°. Every symmetry fixes their common junction and permutes the three endpoints. Rotations through $0°, 120°, 240°$ and the three reflections through an arm realize all six permutations of those endpoints. The action is faithful because an isometry fixing three noncollinear endpoints is the identity. Thus the group is $D_3 \cong S_3$.

Exercise 8.24: Symmetries of the divided circle

The distinguished diameter must map to itself. The possibilities are the identity, the half-turn, reflection in that diameter, and reflection in its perpendicular diameter. Each nonidentity element squares to the identity, and the product of the two reflections is the half-turn. Hence the symmetry group is the Klein four-group $V \cong \mathbb{Z}_2 \times \mathbb{Z}_2$.

Exercise 8.25: Symmetries of the cross

The four equal arms have endpoints forming a square. A symmetry of the cross gives a symmetry of those endpoints, and every symmetry of the square preserves the cross. The four rotations and four reflections therefore give exactly D_4.

Exercise 8.26: Symmetries of the infinite decorated line

The downward vertical teeth and upward slanting teeth distinguish the two sides of the horizontal line and also distinguish left from right. Thus neither a reflection nor a half-turn preserves the figure. The allowed translations are exactly integral multiples of the repeating horizontal spacing a. Their composition adds the integers: $T_{ma}T_{na} = T_{(m+n)a}$. Hence the symmetry group is the infinite cyclic group $\mathbb{Z}$.

Exercise 8.27: Regular representations

Left regular representation of $\mathbb{Z}_4$. For $k \in \mathbb{Z}_4$, put $L_k(x) = k + x \pmod 4$. The four permutations are

$$L_0 = e, \quad L_1 = (0,1,2,3), \quad L_2 = (0,2)(1,3), \quad L_3 = (0,3,2,1),$$

with table

	L_0	L_1	L_2	L_3
L_0	L_0	L_1	L_2	L_3
L_1	L_1	L_2	L_3	L_0
L_2	L_2	L_3	L_0	L_1
L_3	L_3	L_0	L_1	L_2

Right regular representation of S_3. The book's homomorphic convention is $R_a(x) = xa^{-1}$, not xa. With columns in the order $\rho_0, \rho_1, \rho_2, \mu_1, \mu_2, \mu_3$, the images are

	ρ_0	ρ_1	ρ_2	μ_1	μ_2	μ_3
R_{ρ_0}	ρ_0	ρ_1	ρ_2	μ_1	μ_2	μ_3
R_{ρ_1}	ρ_2	ρ_0	ρ_1	μ_3	μ_1	μ_2
R_{ρ_2}	ρ_1	ρ_2	ρ_0	μ_2	μ_3	μ_1
R_{μ_1}	μ_1	μ_3	μ_2	ρ_0	ρ_2	ρ_1
R_{μ_2}	μ_2	μ_1	μ_3	ρ_1	ρ_0	ρ_2
R_{μ_3}	μ_3	μ_2	μ_1	ρ_2	ρ_1	ρ_0

For example $R_{\rho_1}(\rho_0) = \rho_2$. Since $R_a R_b(x) = (xb^{-1})a^{-1} = x(ab)^{-1}$, the multiplication table of the representation is

	R_{ρ_0}	R_{ρ_1}	R_{ρ_2}	R_{μ_1}	R_{μ_2}	R_{μ_3}
R_{ρ_0}	R_{ρ_0}	R_{ρ_1}	R_{ρ_2}	R_{μ_1}	R_{μ_2}	R_{μ_3}
R_{ρ_1}	R_{ρ_1}	R_{ρ_2}	R_{ρ_0}	R_{μ_3}	R_{μ_1}	R_{μ_2}
R_{ρ_2}	R_{ρ_2}	R_{ρ_0}	R_{ρ_1}	R_{μ_2}	R_{μ_3}	R_{μ_1}
R_{μ_1}	R_{μ_1}	R_{μ_2}	R_{μ_3}	R_{ρ_0}	R_{ρ_1}	R_{ρ_2}
R_{μ_2}	R_{μ_2}	R_{μ_3}	R_{μ_1}	R_{ρ_2}	R_{ρ_0}	R_{ρ_1}
R_{μ_3}	R_{μ_3}	R_{μ_1}	R_{μ_2}	R_{ρ_1}	R_{ρ_2}	R_{ρ_0}

In each case evaluation at the identity distinguishes the permutations, so the maps from the original group are injective.

Exercise 8.28: Definition of a permutation

A permutation of S is a *bijection* $f : S \to S$: it must be both one-to-one and onto. The original wording omits surjectivity. For instance $n \mapsto n + 1$ on $\{0, 1, 2, \ldots\}$ is one-to-one but misses 0, and therefore is not a permutation.

Exercise 8.29: Definition of the left regular representation

The definition is correct: $g \mapsto L_g$, with $L_g(x) = gx$, maps G into S_G. To check the terms are justified, $L_g^{-1} = L_{g^{-1}}$, so each L_g is a permutation, and $L_g L_h = L_{gh}$. Finally $L_g(e) = g$, so this representation is one-to-one.

Exercise 8.30: Translation as a permutation

Yes. If $x + 1 = y + 1$, subtracting 1 gives $x = y$, so the map is one-to-one. Given a real target u, choosing $x = u - 1$ gives $f(x) = u$, so it is onto. Its inverse is $u \mapsto u - 1$.

Exercise 8.31: Squaring as a permutation

No. The two inputs 1 and -1 have the same image 1, so the map is not one-to-one. It also misses every negative real number because $x^2 \geq 0$. Either failure excludes it from $S_{\mathbb{R}}$.

Exercise 8.32: The map $x \mapsto -x^3$

Yes. If $-x^3 = -y^3$, then $x^3 = y^3$ and hence $x = y$, since the real cube function is strictly increasing. Given $u \in \mathbb{R}$, choose $x = \sqrt[3]{-u}$. Then $-x^3 = u$, so the map is onto. Its inverse is $u \mapsto \sqrt[3]{-u}$.

Exercise 8.33: The exponential map

No. Although e^x is strictly increasing and therefore one-to-one, its range is $(0, \infty)$, not all of $\mathbb{R}$. In particular there is no real input with image 0. It would be a bijection from $\mathbb{R}$ to $\mathbb{R}^+$, but a permutation here must map $\mathbb{R}$ onto itself.

Exercise 8.34: A cubic polynomial

The polynomial is $f(x) = x^3 - x^2 - 2x = x(x-2)(x+1)$. Thus $f(-1) = f(0) = f(2) = 0$ at three distinct inputs, so it is not one-to-one and is not a permutation. (It is onto $\mathbb{R}$ by continuity and its opposite infinite end limits, but that does not repair injectivity.)

Exercise 8.35: True or false

(a) True: a permutation is a bijection and hence injective.

(b) False: injectivity does not ensure surjectivity or even equal domain and codomain; the shift on nonnegative integers is a counterexample.

(c) True: if a finite set has n elements, a surjection onto all n targets cannot repeat an image, or at most $n - 1$ targets would occur.

(d) True: Cayley's theorem uses $g \mapsto L_g$ to embed G into S_G.

(e) True: if $ab = ba$ for all elements in a group, the same equality holds when both elements are restricted to any subgroup.

(f) True: $\langle a \rangle = \{a^k : k \in \mathbb{Z}\}$ is closed under products and inverses and contains e.

(g) False: S_{10} has 10!, not 10, elements.

(h) False: S_3 is nonabelian, whereas every cyclic group is abelian.

(i) False: S_1 is trivial and cyclic, and S_2 is cyclic of order 2.

(j) True: this is the permutation-group form of Cayley's theorem.

Exercise 8.36: A nonabelian group with abelian proper subgroups

Take S_3. Its proper subgroups are $\{e\}$, the three order-2 groups generated by transpositions, and the cyclic order-3 group A_3. All are cyclic and therefore abelian. Nevertheless $(1,2)(2,3) = (1,2,3)$ and $(2,3)(1,2) = (1,3,2)$ differ, so S_3 itself is nonabelian.

Exercise 8.37: All self-maps under composition

Function composition of maps $A \to A$ is an associative binary operation and has identity id_A. Thus the set of all self-maps is a monoid (and in particular a semigroup). It is generally not a group: if A has at least two elements, a constant map is not injective and cannot have an inverse. If A is a singleton, there is only the identity map and it is the trivial group.

Exercise 8.38: A Cayley digraph for D_n

Let r be a rotation of order n and s a reflection, so $sr = r^{-1}s$. Label the two rows (or two concentric n-gons) by r^i and $r^i s$, with i read modulo n. For right-multiplication arcs, draw

$$r^i \xrightarrow{\ r\ } r^{i+1}, \qquad r^i s \xrightarrow{\ r\ } r^{i-1}s, \qquad r^i \text{—}_s\ r^i s.$$

Thus the two n-cycles have opposite directions and a reflection edge joins corresponding vertices. This fully specifies the digraph, with $2n$ vertices and the correct outgoing arc of each type at every vertex. Schematically:

$$\begin{array}{ccccccccccc} e & \longrightarrow & r & \longrightarrow & r^2 & \longrightarrow \cdots \longrightarrow & r^{n-1} & \longrightarrow & e \\ \updownarrow s & & \updownarrow s & & \updownarrow s & & \updownarrow s & & \\ s & \longleftarrow & rs & \longleftarrow & r^2 s & \longleftarrow \cdots \longleftarrow & r^{n-1}s & \longleftarrow & s \end{array}$$

The reflection arcs are undirected because $s^2 = e$.

Exercise 8.39: Synopsis of Cayley's theorem

Each $g \in G$ determines a permutation $L_g : x \mapsto gx$ of G, and $L_g L_h = L_{gh}$, so $g \mapsto L_g$ is a homomorphism into S_G. It is injective because $L_g(e) = g$, hence identifies G with the subgroup $\{L_g : g \in G\}$.

Exercise 8.40: The stabilizer of one point

Yes. The identity fixes b. If $\sigma(b) = b$ and $\tau(b) = b$, then $(\sigma\tau)(b) = \sigma(b) = b$. Also $\sigma(b) = b$ implies $\sigma^{-1}(b) = b$. Thus the set contains the identity and is closed under composition and inverses, and is a subgroup of S_A.

Exercise 8.41: Permutations sending a point into a subset

Not necessarily. Let $A = \{1,2,3\}$, $B = \{1,2\}$, and $b = 1$. The permutations $\sigma = (2,3)$ and $\tau = (1,2)$ both send 1 into B, but $(\sigma\tau)(1) = \sigma(2) = 3 \notin B$. Thus the proposed set can fail closure under products.

Exercise 8.42: Permutations mapping a subset into itself

Not necessarily, because B may be infinite. Let $A = \mathbb{Z}$ and $B = \{0, 1, 2, \ldots\}$. The shift $\sigma(n) = n + 1$ satisfies $\sigma[B] \subseteq B$, but its inverse sends 0 to $-1 \notin B$. Hence the set can fail closure under inverses. If B is finite, inclusion does force equality, but no finiteness is assumed here.

Exercise 8.43: The setwise stabilizer of a subset

Yes. The identity maps B onto itself. If $\sigma[B] = B$ and $\tau[B] = B$, then $(\sigma\tau)[B] = \sigma[\tau[B]] = \sigma[B] = B$. Applying σ^{-1} to $\sigma[B] = B$ gives $\sigma^{-1}[B] = B$. These three facts prove that the set is a subgroup.

Exercise 8.44: Order and rotation subgroup of D_n

Choose the image of one vertex: n possibilities. Its adjacent vertex must go to one of the two adjacent vertices of the chosen image: 2 possibilities. These choices determine the entire polygon placement, and all occur through rotations or reflections. Therefore $|D_n| = 2n$. The rotations through multiples of $2\pi/n$ form the cyclic subgroup $\langle r \rangle = \{e, r, \ldots, r^{n-1}\}$ of order n, exactly half of $2n$.

Exercise 8.45: Rotations of a cube

Choose which of the six faces is on the bottom and then which of its four neighboring faces points forward. These choices determine a cube rotation, so the rotation group has $6 \cdot 4 = 24$ elements. For each of the three axes through centers of opposite faces, the quarter-turns form a distinct cyclic subgroup $\{e, r, r^2, r^3\}$ of order 4. For each of the four long diagonals joining opposite vertices, the 120° rotations form a distinct cyclic subgroup of order 3. Different axes give different nonidentity rotations, establishing the requested lower bounds on numbers of subgroups.

Exercise 8.46: Noncommutativity of S_n

For $n \geq 3$, let $a = (1, 2)$ and $b = (2, 3)$, fixing all other symbols. Then $(ab)(1) = a(1) = 2$, whereas $(ba)(1) = b(2) = 3$. Thus $ab \neq ba$ and S_n is nonabelian. Comparing the images of one point is enough to distinguish two permutations.

Exercise 8.47: Only the identity commutes with all permutations

Plan. Show a permutation moving a point fails to commute with a suitable transposition. Suppose $\sigma(a) = b \neq a$. Choose c different from a, b, possible because $n \geq 3$, and let $\tau = (b, c)$. Then $\tau(a) = a$, so $(\sigma\tau)(a) = b$, while $(\tau\sigma)(a) = \tau(b) = c$. These images differ. Hence no nonidentity σ commutes with every permutation. The identity does commute with every permutation, proving the claim.

Exercise 8.48: Intersecting orbits coincide

Suppose $c = \sigma^m(a) = \sigma^n(b)$ lies in both orbits. Applying σ^{-n} gives $b = \sigma^{m-n}(a)$. Therefore any $\sigma^k(b) = \sigma^{k+m-n}(a)$ lies in the orbit of a, giving $O_{b,\sigma} \subseteq O_{a,\sigma}$. Interchanging a, b gives the reverse inclusion, so the two orbits are equal.

Exercise 8.49: A finite cyclic transitive permutation group

Write the nonempty finite set as $A = \{a_1, \ldots, a_m\}$. Define $\sigma(a_i) = a_{i+1}$ for $i < m$ and $\sigma(a_m) = a_1$. The cycle σ has order m, so $H = \langle\sigma\rangle$ has $|H| = m = |A|$. Given a_i, a_j, the power σ^{j-i} carries a_i to a_j (with exponents modulo m). Thus H is transitive. For $m = 1$ the same construction gives the identity group.

Exercise 8.50: Transitivity and a single orbit

If $\langle\sigma\rangle$ is transitive and $a \in A$, then for every $b \in A$ some power σ^k sends a to b. Hence $O_{a,\sigma} = A$. Conversely, if this orbit is all of A, write any $b = \sigma^r(a)$ and $c = \sigma^s(a)$. Then $\sigma^{s-r}(b) = c$, so $\langle\sigma\rangle$ is transitive. The asserted existence of a presupposes $A \neq \varnothing$; for the empty set transitivity is vacuous and this existential formulation needs that qualification.

Exercise 8.51: The opposite group

(a) Viewing the written equations through glass reverses left-to-right multiplication order: $a * b = c$ reads as $b *' a = c$. The associativity equation is reversed on both sides, so it is still associativity for the reversed operation. The identity and inverse equations are likewise preserved with the factors interchanged. **(b)** Define $x *' y = y * x$. Closure follows from closure of $*$. Compute

$$(x *' y) *' z = z * (y * x) = (z * y) * x = x *' (y *' z),$$

using associativity of $*$. The old identity satisfies $e *' x = x * e = x$ and $x *' e = e * x = x$. The old inverse satisfies $x *' x^{-1} = x^{-1} * x = e$ and $x^{-1} *' x = x * x^{-1} = e$. Thus the reversed operation gives a group, called the opposite group.

Exercise 8.52: The group of right translations

For $a \in G$, define $\rho_a(x) = xa$. Each is a permutation with inverse $\rho_{a^{-1}}$. Composition reverses the subscripts:

$$\rho_a\rho_b(x) = (xb)a = x(ba), \qquad \rho_a\rho_b = \rho_{ba}.$$

Thus these permutations contain the identity, products, and inverses and form a group. The map $a \mapsto \rho_a$ need not be a homomorphism. Instead use $\Phi(a) = \rho_{a^{-1}}$; then $\Phi(a)\Phi(b) = \rho_{b^{-1}a^{-1}} = \rho_{(ab)^{-1}} = \Phi(ab)$. It is onto the set of right translations, and equality of two such translations, evaluated at e, gives $a^{-1} = b^{-1}$ and then $a = b$. Hence Φ is an isomorphism.

Exercise 8.53: Permutation matrices and the Klein group

(a) List the elements of a finite group G in some order and let L_g be left multiplication by g. Define $P_g e_x = e_{gx}$ on the standard basis indexed by G. Each P_g is a permutation matrix. On every basis vector,

$$P_g P_h e_x = e_{ghx} = P_{gh} e_x,$$

so $g \mapsto P_g$ is a homomorphism. If $P_g = P_h$, applying both sides to e_e gives $g = h$, so it is an isomorphism onto its image. **(b)** Order V as e, a, b, c, with $a^2 = b^2 = c^2 = e$ and $ab = c$. Then

$$P_e = I_4, \quad P_a = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix},$$

$$P_b = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}, \quad P_c = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix}.$$

For example $P_a e_e = e_a$ and $P_a e_b = e_c$, matching multiplication by a. The construction in part (a) already verifies every product, including $P_a P_b = P_c$.

9 Orbits, Cycles, and the Alternating Groups

Conventions. Permutations compose from right to left. The identity is e. A group generated by an element a is denoted $\langle a \rangle$.

Exercise 9.1: Orbits of a finite permutation

Plan. Start at an unused symbol, repeatedly apply the permutation until it returns, and repeat with a symbol not yet used. The nontrivial cycles are $(1,5,2)(4,6)$. Each fixed point forms a singleton orbit, so the complete orbit list is

$$\{1,2,5\}, \quad \{4,6\}, \quad \{3\}.$$

Every symbol occurs in exactly one listed set; negative powers move backwards within the same set and produce no further points.

Exercise 9.2: Orbits of a finite permutation

Plan. Start at an unused symbol, repeatedly apply the permutation until it returns, and repeat with a symbol not yet used. The nontrivial cycles are $(1,5,8,7)(2,6,3)$. Each fixed point forms a singleton orbit, so the complete orbit list is

$$\{1,5,7,8\}, \quad \{2,3,6\}, \quad \{4\}.$$

Every symbol occurs in exactly one listed set; negative powers move backwards within the same set and produce no further points.

Exercise 9.3: Orbits of a finite permutation

Plan. Start at an unused symbol, repeatedly apply the permutation until it returns, and repeat with a symbol not yet used. The nontrivial cycles are $(1,2,3,5,4)(7,8)$. Each fixed point forms a singleton orbit, so the complete orbit list is

$$\{1,2,3,4,5\}, \quad \{7,8\}, \quad \{6\}.$$

Every symbol occurs in exactly one listed set; negative powers move backwards within the same set and produce no further points.

Exercise 9.4: Orbits of the integer shift by 1

The rth power sends an integer a to $a + 1r$, where r ranges over $\mathbb{Z}$. Hence the orbit of a is $a + 1\mathbb{Z}$, its residue class modulo 1. The distinct orbits are

$$\mathbb{Z}.$$

The sign of the shift only changes the direction in which these same points are traversed.

Exercise 9.5: Orbits of the integer shift by 2

The rth power sends an integer a to $a + 2r$, where r ranges over $\mathbb{Z}$. Hence the orbit of a is $a + 2\mathbb{Z}$, its residue class modulo 2. The distinct orbits are

$$0 + 2\mathbb{Z}, \quad 1 + 2\mathbb{Z}.$$

The sign of the shift only changes the direction in which these same points are traversed.

Exercise 9.6: Orbits of the integer shift by -3

The rth power sends an integer a to $a + -3r$, where r ranges over $\mathbb{Z}$. Hence the orbit of a is $a + 3\mathbb{Z}$, its residue class modulo 3. The distinct orbits are

$$0 + 3\mathbb{Z}, \quad 1 + 3\mathbb{Z}, \quad 2 + 3\mathbb{Z}.$$

The sign of the shift only changes the direction in which these same points are traversed.

Exercise 9.7: A product of cycles

Plan. Read the factors from right to left, leaving a symbol unchanged whenever it is absent from a factor. The images of the eight arguments are

$$\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 4 & 1 & 3 & 5 & 8 & 6 & 2 & 7 \end{pmatrix}.$$

Following each chain of these images yields the disjoint-cycle answer

$$(1, 4, 5)(7, 8)(2, 5, 7) = (1, 4, 5, 8, 7, 2).$$

Any symbol omitted from the final cycles is fixed. The two-row form provides a check of every image, not just of one cycle.

Exercise 9.8: A product of cycles

Plan. Read the factors from right to left, leaving a symbol unchanged whenever it is absent from a factor. The images of the eight arguments are

$$\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 3 & 7 & 2 & 8 & 5 & 4 & 1 & 6 \end{pmatrix}.$$

Following each chain of these images yields the disjoint-cycle answer

$$(1, 3, 2, 7)(4, 8, 6) = (1, 3, 2, 7)(4, 8, 6).$$

Any symbol omitted from the final cycles is fixed. The two-row form provides a check of every image, not just of one cycle.

Exercise 9.9: A product of cycles

Plan. Read the factors from right to left, leaving a symbol unchanged whenever it is absent from a factor. The images of the eight arguments are

$$\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 5 & 4 & 3 & 7 & 8 & 6 & 2 & 1 \end{pmatrix}.$$

Following each chain of these images yields the disjoint-cycle answer

$$(1,2)(4,7,8)(2,1)(7,2,8,1,5) = (1,5,8)(2,4,7).$$

Any symbol omitted from the final cycles is fixed. The two-row form provides a check of every image, not just of one cycle.

Exercise 9.10: Disjoint cycles and transpositions

1. Follow the images.

The nontrivial chains are

$$1 \mapsto 8 \mapsto 1; \quad 3 \mapsto 6 \mapsto 4 \mapsto 3; \quad 5 \mapsto 7 \mapsto 5.$$

Thus the disjoint-cycle expression is $(1,8)(3,6,4)(5,7)$; any missing point is fixed.

2. Split each cycle.

Use $(a_1, a_2, \ldots, a_k) = (a_1, a_k) \cdots (a_1, a_3)(a_1, a_2)$, verified by following each a_i. This gives

$$(1,8)(3,6,4)(5,7) = (1,8)(3,4)(3,6)(5,7).$$

Exercise 9.11: Disjoint cycles and transpositions

1. Follow the images.

The nontrivial chains are

$$1 \mapsto 3 \mapsto 4 \mapsto 1; \quad 2 \mapsto 6 \mapsto 2; \quad 5 \mapsto 8 \mapsto 7 \mapsto 5.$$

Thus the disjoint-cycle expression is $(1,3,4)(2,6)(5,8,7)$; any missing point is fixed.

2. Split each cycle.

Use $(a_1, a_2, \ldots, a_k) = (a_1, a_k) \cdots (a_1, a_3)(a_1, a_2)$, verified by following each a_i. This gives

$$(1,3,4)(2,6)(5,8,7) = (1,4)(1,3)(2,6)(5,7)(5,8).$$

Exercise 9.12: Disjoint cycles and transpositions

1. Follow the images.

The nontrivial chains are

$$1 \mapsto 3 \mapsto 4 \mapsto 7 \mapsto 8 \mapsto 6 \mapsto 5 \mapsto 2 \mapsto 1.$$

Thus the disjoint-cycle expression is $(1, 3, 4, 7, 8, 6, 5, 2)$; any missing point is fixed.

2. Split each cycle.

Use $(a_1, a_2, \ldots, a_k) = (a_1, a_k) \cdots (a_1, a_3)(a_1, a_2)$, verified by following each a_i. This gives

$$(1, 3, 4, 7, 8, 6, 5, 2) = (1, 2)(1, 5)(1, 6)(1, 8)(1, 7)(1, 4)(1, 3).$$

Exercise 9.13: Orders of cycles and permutations

(a) $(1, 4, 5, 7)$ has order 4: four applications return each listed point, and no smaller positive number returns 1.

(b) A cycle of length k has order k. Each application advances one step, so a power fixes the cycle exactly when the exponent is divisible by k.

(c) The disjoint cycles of $\sigma = (4, 5)(2, 3, 7)$ have lengths $2, 3$, giving order 6. Those of $\tau = (1, 4)(3, 5, 7, 8)$ have lengths $2, 4$, giving order 4.

(d) Exercises 10, 11, 12 have nontrivial cycle lengths $(2, 3, 2)$, $(3, 2, 3)$, and (8), respectively. Their orders are $6, 6, 8$.

(e) The order of a permutation is the least common multiple of its disjoint-cycle lengths: a power is the identity precisely when it is a multiple of every cycle length.

Exercise 9.14: Maximum element order in S_5

Plan. Disjoint-cycle lengths are positive integers summing to 5, and the order is their least common multiple. Thus check partitions, rather than individual permutations. Grouping the partitions by their largest part L, the largest possible least common multiple in each group is

L	1	2	3	4	5
maximum	1	2	6	4	5

For example, these values can be reproduced by listing the remaining nondecreasing parts, each at most L, with sum $5 - L$, and taking the least common multiple with L. The overall maximum is 6, attained by disjoint cycles of lengths $2, 3$. This gives both an example and an exhaustive upper bound.

Exercise 9.15: Maximum element order in S_6

Plan. Disjoint-cycle lengths are positive integers summing to 6, and the order is their least common multiple. Thus check partitions, rather than individual permutations. Grouping the partitions by

their largest part L, the largest possible least common multiple in each group is

L	1	2	3	4	5	6
maximum	1	2	6	4	5	6

For example, these values can be reproduced by listing the remaining nondecreasing parts, each at most L, with sum $6 - L$, and taking the least common multiple with L. The overall maximum is 6, attained by disjoint cycles of lengths $1, 2, 3$. This gives both an example and an exhaustive upper bound.

Exercise 9.16: Maximum element order in S_7

Plan. Disjoint-cycle lengths are positive integers summing to 7, and the order is their least common multiple. Thus check partitions, rather than individual permutations. Grouping the partitions by their largest part L, the largest possible least common multiple in each group is

L	1	2	3	4	5	6	7
maximum	1	2	6	12	10	6	7

For example, these values can be reproduced by listing the remaining nondecreasing parts, each at most L, with sum $7 - L$, and taking the least common multiple with L. The overall maximum is 12, attained by disjoint cycles of lengths $3, 4$. This gives both an example and an exhaustive upper bound.

Exercise 9.17: Maximum element order in S_{10}

Plan. Disjoint-cycle lengths are positive integers summing to 10, and the order is their least common multiple. Thus check partitions, rather than individual permutations. Grouping the partitions by their largest part L, the largest possible least common multiple in each group is

L	1	2	3	4	5	6	7	8	9	10
maximum	1	2	6	12	30	12	21	8	9	10

For example, these values can be reproduced by listing the remaining nondecreasing parts, each at most L, with sum $10 - L$, and taking the least common multiple with L. The overall maximum is 30, attained by disjoint cycles of lengths $2, 3, 5$. This gives both an example and an exhaustive upper bound.

Exercise 9.18: Maximum element order in S_{15}

Plan. Disjoint-cycle lengths are positive integers summing to 15, and the order is their least common multiple. Thus check partitions, rather than individual permutations. Grouping the partitions by their largest part L, the largest possible least common multiple in each group is

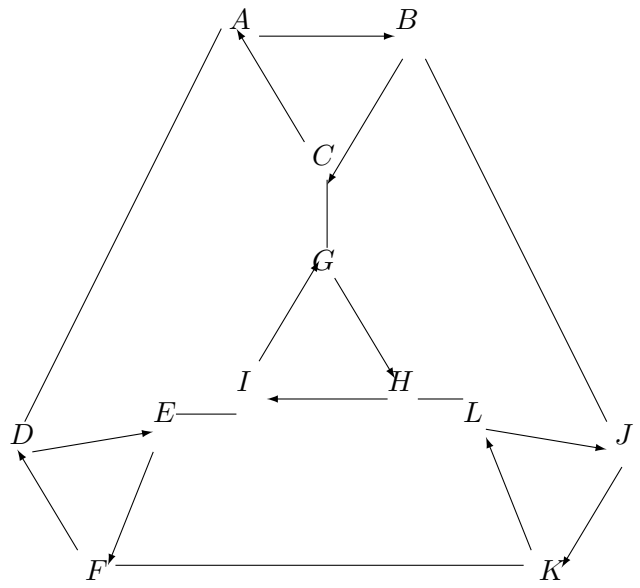
L	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
maximum	1	2	6	12	60	60	105	56	45	30	44	12	26	14	15

For example, these values can be reproduced by listing the remaining nondecreasing parts, each at most L, with sum $15 - L$, and taking the least common multiple with L. The overall maximum is 105, attained by disjoint cycles of lengths $3, 5, 7$. This gives both an example and an exhaustive upper bound.

Exercise 9.19: Labeling the Cayley digraph of A_4

Put $a = (1,2,3)$ and $b = (1,2)(3,4)$. Follow each solid arrow by right multiplying by a, and each dashed edge by right multiplying by b. The following table labels every position in the supplied diagram; the letters also appear in the schematic below.

vertex	position	permutation
A	$topleft$	e
B	$topright$	$(1,2,3)$
C	$belowtoppair$	$(1,3,2)$
D	$leftoutside$	$(1,2)(3,4)$
E	$leftinner$	$(2,4,3)$
F	$bottomleft$	$(1,4,3)$
G	$middletop$	$(2,3,4)$
H	$middleright$	$(1,3)(2,4)$
I	$middleleft$	$(1,4,2)$
J	$rightoutside$	$(1,3,4)$
K	$bottomright$	$(1,2,4)$
L	$rightinner$	$(1,4)(2,3)$



The six edges without arrowheads represent the dashed b edges; both directions give the same operation since $b^2 = e$. For instance $D = b$, $E = ba$, and $F = ba^2$. The twelve entries are distinct and exhaust A_4, checking that no vertex has been missed.

Exercise 9.20: Definition of an orbit

The proposed definition is correct when "minimal" means minimal under inclusion among nonempty invariant subsets: an orbit is a nonempty subset $O \subseteq A$ with $\sigma[O] = O$ and no proper nonempty

subset having that property. Indeed, $O_a = \{\sigma^k(a) : k \in \mathbb{Z}\}$ is invariant, and any nonempty invariant subset containing a point of O_a contains its entire two-sided orbit. Conversely, pick a in a minimal such set; its orbit lies inside the set and hence must equal it.

Exercise 9.21: Definition of a cycle

A cycle in S_n is a permutation with *at most one orbit having more than one element*. Its length is the size of its largest orbit. Fixed points outside the moving cycle produce additional singleton orbits, so "only one orbit" is too restrictive. The book includes the identity as a cycle of length 1.

Exercise 9.22: Definition of the alternating group

Specify the common finite underlying set: the alternating group A_n is the subgroup of S_n consisting of all even permutations of $\{1, \ldots, n\}$. Without specifying this set or n, "all even permutations" does not describe a single group. Closure follows because products and inverses of even permutations are even.

Exercise 9.23: True or false

(a) False: $(1, 2)(3, 4)$ has two nontrivial orbits.

(b) True: a cycle is a special kind of permutation.

(c) False: first one must prove that the parity of the number of transpositions is independent of the chosen expression; otherwise the definition could assign both parities to one permutation.

(d) False: $\langle(1, 2, 3, 4)\rangle$ has an odd permutation but its elements are e, a 4-cycle, a double transposition, and the inverse 4-cycle; none is a transposition.

(e) False: $|A_5| = 5!/2 = 60$.

(f) False: $S_2 = \{e, (1, 2)\}$ is cyclic.

(g) True: A_3 is cyclic of order 3.

(h) True: extend each permutation of $\{1, \ldots, 7\}$ by fixing 8.

(i) True: relabel the seven symbols other than 5, and extend by fixing 5; the particular omitted symbol is immaterial.

(j) False: the identity is even and so is absent, and the product of two odd permutations is even.

Exercise 9.24: The table of A_3

The even permutations are $\rho_0 = e$, $\rho_1 = (1, 2, 3)$, and $\rho_2 = (1, 3, 2)$. Each 3-cycle is a product of two transpositions, while μ_1, μ_2, μ_3 are single transpositions and are odd. Thus

	ρ_0	ρ_1	ρ_2
ρ_0	ρ_0	ρ_1	ρ_2
ρ_1	ρ_1	ρ_2	ρ_0
ρ_2	ρ_2	ρ_0	ρ_1

The entries follow from $\rho_1^3 = e$ and $\rho_2 = \rho_1^2$.

Exercise 9.25: Synopsis of the determinant proof

A permutation matrix obtained from the identity by k row transpositions has determinant $(-1)^k$, so two transposition expressions for the same permutation must have the same parity because one matrix cannot have determinants 1 and -1 simultaneously.

Exercise 9.26: Synopsis of the orbit-counting proof

Multiplying a permutation by one transposition merges two distinct orbits or splits one orbit in two, changing its number of orbits by exactly 1. Starting from the identity's n orbits, the parity of the number of transpositions is therefore determined by the final number of orbits and cannot depend on the expression.

Exercise 9.27: Bounds on numbers of transpositions

Let the disjoint-cycle lengths, including fixed points, be $k_1, \ldots, k_t$, with sum n. A k_i-cycle uses $k_i - 1$ transpositions, so the entire permutation uses

$$(k_1 - 1) + \cdots + (k_t - 1) = n - t.$$

(a) Since $t \geq 1$, this is at most $n - 1$. The identity can be represented by zero transpositions, the empty product. **(b)** A permutation which is not a cycle has at least two nontrivial orbits, hence $t \geq 2$, giving at most $n - 2$ transpositions. **(c)** If the permutation is odd, the displayed number $m = n - t$ is odd. Since $m \leq n - 1 < 2n + 3$ and $2n + 3 - m$ is even, insert $(2n + 3 - m)/2$ pairs $(1, 2)(1, 2)$ into that expression to obtain exactly $2n + 3$ factors. For an even permutation m is even and the same insertion of $(2n + 8 - m)/2$ pairs gives exactly $2n + 8$ factors. Each pair is the identity and so does not change the permutation.

Exercise 9.28: Merging orbits when a point is fixed

(a) Write the orbit containing j as $(j, x_1, \ldots, x_k)$ and suppose i is a different fixed point. Then direct calculation gives

$$(i, j)(j, x_1, \ldots, x_k) = (i, j, x_1, \ldots, x_k).$$

Thus the two old orbit diagrams and the new diagram can be drawn as

$$\underbrace{i \longrightarrow i}_{\text{one orbit}} \qquad \underbrace{j \longrightarrow x_1 \longrightarrow \cdots \longrightarrow x_k \longrightarrow j}_{\text{one orbit}} \quad \rightsquigarrow \quad i \longrightarrow j \longrightarrow x_1 \longrightarrow \cdots \longrightarrow x_k \longrightarrow i.$$

The other orbits are unchanged, so the orbit count decreases by one. **(b)** If j is fixed too, the two singleton loops $i \to i$ and $j \to j$ become the single 2-cycle $i \to j \to i$ under $(i, j)\sigma$. Again exactly two orbits have become one.

Exercise 9.29: Even permutations in a subgroup

If H has no odd permutations, every element is even. Otherwise fix an odd $\tau \in H$. Left multiplication by τ maps the even elements of H to odd elements of H, because signs multiply. Its inverse is left multiplication by τ^{-1}, which is also odd. Thus it is a bijection between the even and odd elements of H. As H is finite, exactly half its elements are even.

Exercise 9.30: How many points a cycle moves

A cycle of length $n > 1$ moves exactly its n listed points: each goes to the next distinct listed point, and all other points are fixed. Under the book's convention, a cycle of length 1 is the identity and moves 0 points. Thus the usual answer n requires the nontrivial-cycle condition $n > 1$.

Exercise 9.31: Finitely supported permutations form a subgroup

For a permutation σ, write $\operatorname{supp}(\sigma) = \{a : \sigma(a) \neq a\}$. The identity has empty support. If both σ and τ fix a point a, their product fixes a, so

$$\operatorname{supp}(\sigma\tau) \subseteq \operatorname{supp}(\sigma) \cup \operatorname{supp}(\tau).$$

A union of two finite sets is finite, proving product closure. A point is fixed by σ if and only if it is fixed by σ^{-1}, so the two supports are equal. Hence inverses also have finite support, and the given set is a subgroup.

Exercise 9.32: A uniform bound on support size

No. Choose 100 distinct points of the infinite set A, and let σ be a 50-cycle on the first 50, while τ is a 50-cycle on the other 50. Extend both by the identity elsewhere. Each moves exactly 50 points, so both belong to K. Their disjoint supports mean $\sigma\tau$ moves all 100 selected points, so it is not in K. Thus product closure fails.

Exercise 9.33: The odd permutations form a coset

Let σ be the fixed odd permutation and let π be any odd permutation. Set $\alpha = \sigma^{-1}\pi$. Both factors are odd, so their product is even and $\alpha \in A_n$. Then $\pi = \sigma\alpha$. Conversely, the product of σ with any even permutation is odd. Thus the set of odd permutations is exactly σA_n.

Exercise 9.34: Squaring an odd-length cycle

Let $\sigma = (a_0, a_1, \ldots, a_{n-1})$ with n odd. Its square moves a_i to $a_{i+2 \bmod n}$. If $2k \equiv 0 \pmod{n}$, oddness implies $n \mid k$, so the first positive return occurs after n steps. Hence all n symbols lie in one cycle of σ^2, and every point outside the original cycle stays fixed. For $n = 1$, both permutations are the identity, also a cycle by convention.

Exercise 9.35: When a power of a cycle is a cycle

Plan. Advancing r steps around an n-cycle partitions its n points into $d = \gcd(n, r)$ cycles, each of length n/d. Indeed, $\sigma^{rk}(a) = a$ precisely when $n \mid rk$, whose least positive solution is $k = n/d$. If $n/d > 1$, all d orbits are nontrivial, so the power is a cycle exactly when $d = 1$. There is an important convention in this text: the identity is also a cycle. If $n \mid r$, then $\sigma^r = e$ and it is therefore a cycle as well. With that convention the exact condition is

$$\boxed{\gcd(n, r) = 1 \quad \text{or} \quad n \mid r.}$$

If "cycle" is intended to mean a nonidentity n-cycle, the familiar condition is just $\gcd(n, r) = 1$ (for $n > 1$). The extra identity case must not be silently omitted under Definition 9.6.

Exercise 9.36: Left multiplication is a permutation

The inverse of $\lambda_a : g \mapsto ag$ is $\lambda_{a^{-1}}$, because $a^{-1}(ag) = g$ and $a(a^{-1}g) = g$. Equivalently, cancellation gives injectivity, and a desired target h is reached by the input $a^{-1}h$, giving surjectivity. Thus $\lambda_a \in S_G$.

Exercise 9.37: The subgroup of left translations

By the preceding exercise every λ_a is a permutation. The identity is λ_e. Associativity in G gives

$$(\lambda_a \lambda_b)(g) = a(bg) = (ab)g = \lambda_{ab}(g),$$

so the set is closed under composition. Finally $\lambda_a^{-1} = \lambda_{a^{-1}}$ also lies in it. These facts prove it is a subgroup of S_G.

Exercise 9.38: Left translations act transitively

Given $x, y \in G$, solve $ax = y$ by taking $a = yx^{-1}$. The left translation λ_a then sends x to y. Since this works for every pair of points in the set G, the left-translation subgroup is transitive on G.

Exercise 9.39: Two generators for the symmetric group

Put $c = (1, 2, \ldots, n)$ and $t = (1, 2)$. Conjugating a transposition relabels its entries, so

$$c^k t c^{-k} = (c^k(1), c^k(2)).$$

For $k = 0, \ldots, n-2$, these include all adjacent transpositions $s_i = (i, i+1)$, $1 \le i < n$. For $i < j$ one has

$$(i, j) = s_i s_{i+1} \cdots s_{j-2} s_{j-1} s_{j-2} \cdots s_{i+1} s_i.$$

To verify it, the product before the middle transposition conjugates $(j-1, j)$ to (i, j); alternatively follow i, j and each intermediate point. Thus every transposition lies in $\langle c, t \rangle$. Every permutation is a product of transpositions, so $\langle c, t \rangle = S_n$.

10 Cosets and the Theorem of Lagrange

Conventions. Permutations compose from right to left. The identity is e. A group generated by an element a is denoted $\langle a \rangle$.

Exercise 10.1: Cosets of $4\mathbb{Z}$ in $\mathbb{Z}$

In additive notation, a coset is $a + 4\mathbb{Z} = \{a + 4k : k \in \mathbb{Z}\}$. Two representatives give the same coset precisely when their difference is divisible by 4. The distinct cosets in $\mathbb{Z}$ are

$$0 + 4\mathbb{Z}, \quad 1 + 4\mathbb{Z}, \quad 2 + 4\mathbb{Z}, \quad 3 + 4\mathbb{Z}.$$

The division algorithm shows every allowed representative belongs to one of these, and their different residues show the listed cosets are distinct.

Exercise 10.2: Cosets of $4\mathbb{Z}$ in $2\mathbb{Z}$

In additive notation, a coset is $a + 4\mathbb{Z} = \{a + 4k : k \in \mathbb{Z}\}$. Two representatives give the same coset precisely when their difference is divisible by 4. The distinct cosets in $2\mathbb{Z}$ are

$$0 + 4\mathbb{Z}, \quad 2 + 4\mathbb{Z}.$$

The division algorithm shows every allowed representative belongs to one of these, and their different residues show the listed cosets are distinct.

Exercise 10.3: Cosets of $\langle 2 \rangle$ in $\mathbb{Z}_{12}$

First list the subgroup generated by 2 by repeatedly adding it modulo 12:

$$H = \langle 2 \rangle = \{0, 2, 4, 6, 8, 10\}.$$

Adding a representative to every element of H gives all distinct cosets:

$$0 + H = \{0, 2, 4, 6, 8, 10\},$$
$$1 + H = \{1, 3, 5, 7, 9, 11\}.$$

These sets are disjoint and together contain all 12 residues, verifying that the list is complete.

Exercise 10.4: Cosets of $\langle 4 \rangle$ in $\mathbb{Z}_{12}$

First list the subgroup generated by 4 by repeatedly adding it modulo 12:

$$H = \langle 4 \rangle = \{0, 4, 8\}.$$

Adding a representative to every element of H gives all distinct cosets:

$$0 + H = \{0, 4, 8\},$$
$$1 + H = \{1, 5, 9\},$$
$$2 + H = \{2, 6, 10\},$$
$$3 + H = \{3, 7, 11\}.$$

These sets are disjoint and together contain all 12 residues, verifying that the list is complete.

Exercise 10.5: Cosets of $\langle 18\rangle$ in $\mathbb{Z}_{36}$

First list the subgroup generated by 18 by repeatedly adding it modulo 36:

$$H = \langle 18\rangle = \{0, 18\}.$$

Adding a representative to every element of H gives all distinct cosets:

$$\begin{aligned}
0 + H &= \{0, 18\},\\
1 + H &= \{1, 19\},\\
2 + H &= \{2, 20\},\\
3 + H &= \{3, 21\},\\
4 + H &= \{4, 22\},\\
5 + H &= \{5, 23\},\\
6 + H &= \{6, 24\},\\
7 + H &= \{7, 25\},\\
8 + H &= \{8, 26\},\\
9 + H &= \{9, 27\},\\
10 + H &= \{10, 28\},\\
11 + H &= \{11, 29\},\\
12 + H &= \{12, 30\},\\
13 + H &= \{13, 31\},\\
14 + H &= \{14, 32\},\\
15 + H &= \{15, 33\},\\
16 + H &= \{16, 34\},\\
17 + H &= \{17, 35\}.
\end{aligned}$$

These sets are disjoint and together contain all 36 residues, verifying that the list is complete.

Exercise 10.6: Left cosets of a reflection subgroup in D_4

Let $H = \{\rho_0, \mu_2\}$. For $a \in D_4$, the left coset is $aH = \{a, a\mu_2\}$. Reading the indicated products from Table 8.12 gives

$$\begin{aligned}
\rho_0 H &= \{\rho_0, \mu_2\},\\
\rho_1 H &= \{\rho_1, \delta_2\},\\
\rho_2 H &= \{\rho_2, \mu_1\},\\
\rho_3 H &= \{\rho_3, \delta_1\}.
\end{aligned}$$

The four pairs are disjoint and include all eight group elements, so these are all the left cosets.

Exercise 10.7: Right cosets of the reflection subgroup

A right coset is $H\rho_i = \{\rho_i, \mu_2\rho_i\}$, so the list is

$$\begin{aligned} H\rho_0 &= \{\rho_0, \mu_2\}, \\ H\rho_1 &= \{\rho_1, \delta_1\}, \\ H\rho_2 &= \{\rho_2, \mu_1\}, \\ H\rho_3 &= \{\rho_3, \delta_2\}. \end{aligned}$$

These do not give the same partition as the left cosets: for example $\rho_1 H = \{\rho_1, \delta_2\}$ while $H\rho_1 = \{\rho_1, \delta_1\}$. Thus H is not normal.

Exercise 10.8: Reordered table for the reflection subgroup

Order the rows and columns in consecutive left cosets from Exercise 6. The resulting table is

	ρ_0	μ_2	ρ_1	δ_2	ρ_2	μ_1	ρ_3	δ_1
ρ_0	ρ_0	μ_2	ρ_1	δ_2	ρ_2	μ_1	ρ_3	δ_1
μ_2	μ_2	ρ_0	δ_1	ρ_3	μ_1	ρ_2	δ_2	ρ_1
ρ_1	ρ_1	δ_2	ρ_2	μ_1	ρ_3	δ_1	ρ_0	μ_2
δ_2	δ_2	ρ_1	μ_2	ρ_0	δ_1	ρ_3	μ_1	ρ_2
ρ_2	ρ_2	μ_1	ρ_3	δ_1	ρ_0	μ_2	ρ_1	δ_2
μ_1	μ_1	ρ_2	δ_2	ρ_1	μ_2	ρ_0	δ_1	ρ_3
ρ_3	ρ_3	δ_1	ρ_0	μ_2	ρ_1	δ_2	ρ_2	μ_1
δ_1	δ_1	ρ_3	μ_1	ρ_2	δ_2	ρ_1	μ_2	ρ_0

The block $(\rho_0 H)(\rho_1 H)$ contains elements from different left cosets: $\rho_0\rho_1 = \rho_1 \in \rho_1 H$, but $\mu_2\rho_1 = \delta_1 \in \rho_3 H$. Both ρ_0 and μ_2 represent the same coset H, so this demonstrates that changing a representative changes the product coset. Thus coset multiplication is not well defined. There is no coset group here, so it is isomorphic to neither $\mathbb{Z}_4$ nor V.

Exercise 10.9: Left cosets of the half-turn subgroup

Set $K = \{\rho_0, \rho_2\}$. The distinct left cosets are

$$\begin{aligned} K = \{\rho_0, \rho_2\}, \quad \rho_1 K = \{\rho_1, \rho_3\}, \\ \mu_1 K = \{\mu_1, \mu_2\}, \quad \delta_1 K = \{\delta_1, \delta_2\}. \end{aligned}$$

Each pair comes from multiplying the representative by ρ_0 and by ρ_2, and the four pairs partition the eight elements.

Exercise 10.10: Right cosets of the half-turn subgroup

The half-turn ρ_2 commutes with every square symmetry: algebraically, $r^2 r^i = r^i r^2$ and $r^2(r^i s) = (r^i s)r^2$, using $sr^2 = r^{-2}s = r^2 s$. Hence $aK = Ka$ for each $a \in D_4$. The right cosets are therefore the same four pairs listed in Exercise 9; K is normal.

Exercise 10.11: Reordered table and the Klein quotient

Using the order $\rho_0, \rho_2, \rho_1, \rho_3, \mu_1, \mu_2, \delta_1, \delta_2$, the table is

	ρ_0	ρ_2	ρ_1	ρ_3	μ_1	μ_2	δ_1	δ_2
ρ_0	ρ_0	ρ_2	ρ_1	ρ_3	μ_1	μ_2	δ_1	δ_2
ρ_2	ρ_2	ρ_0	ρ_3	ρ_1	μ_2	μ_1	δ_2	δ_1
ρ_1	ρ_1	ρ_3	ρ_2	ρ_0	δ_1	δ_2	μ_2	μ_1
ρ_3	ρ_3	ρ_1	ρ_0	ρ_2	δ_2	δ_1	μ_1	μ_2
μ_1	μ_1	μ_2	δ_2	δ_1	ρ_0	ρ_2	ρ_3	ρ_1
μ_2	μ_2	μ_1	δ_1	δ_2	ρ_2	ρ_0	ρ_1	ρ_3
δ_1	δ_1	δ_2	μ_1	μ_2	ρ_1	ρ_3	ρ_0	ρ_2
δ_2	δ_2	δ_1	μ_2	μ_1	ρ_3	ρ_1	ρ_2	ρ_0

Every 2×2 block lies entirely in a single coset, so multiplication descends to the four cosets. Label them $K, A = \rho_1 K, B = \mu_1 K, C = \delta_1 K$. The induced table is

	K	A	B	C
K	K	A	B	C
A	A	K	C	B
B	B	C	K	A
C	C	B	A	K

Every nonidentity element has order 2. Thus the quotient is the Klein four-group V, not $\mathbb{Z}_4$.

Exercise 10.12: Index of $\langle 3 \rangle$ in $\mathbb{Z}_{24}$

The multiples of 3 modulo 24 are $0, 3, 6, 9, 12, 15, 18, 21$, so the subgroup has order 8. Lagrange's counting formula gives $(\mathbb{Z}_{24} : \langle 3 \rangle) = 24/8 = 3$. Equivalently the cosets are the three residue classes modulo 3.

Exercise 10.13: Index of a reflection subgroup in S_3

A reflection μ_1 has order 2, so $\langle \mu_1 \rangle = \{e, \mu_1\}$. Since $|S_3| = 6$, its index is $6/2 = 3$.

Exercise 10.14: Index of a reflection subgroup in D_4

The element μ_2 is a reflection and has order 2. Its subgroup has two elements, whereas D_4 has eight. The number of cosets is therefore $8/2 = 4$.

Exercise 10.15: Index of the subgroup generated by σ

The factors in $\sigma = (1, 2, 5, 4)(2, 3)$ overlap, so their orders cannot be combined by a least-common-multiple rule. Follow their images from right to left:

$$1 \mapsto 2 \mapsto 3 \mapsto 5 \mapsto 4 \mapsto 1.$$

Thus $\sigma = (1, 2, 3, 5, 4)$ has order 5. Therefore $|\langle \sigma \rangle| = 5$ and $(S_5 : \langle \sigma \rangle) = 5!/5 = 24$.

Exercise 10.16: Index of the subgroup generated by μ

The cycles $(1,2,4,5)$ and $(3,6)$ are disjoint and have lengths $4, 2$. Hence μ has order $\operatorname{lcm}(4,2) = 4$. Its cyclic subgroup has four elements, and the index in S_6 is $6!/4 = 720/4 = 180$.

Exercise 10.17: Definition of a left coset

Require that H be a *subgroup* of G, not merely a subset, and that $a \in G$. Then the left coset of H containing a is $aH = \{ah : h \in H\}$. Because $e \in H$, it indeed contains $a = ae$. For an arbitrary subset, the proposed phrase "containing a" need not be true.

Exercise 10.18: Definition of index

The definition using the number of right cosets is correct. The number of left cosets is equal to it, so either convention defines the same index $(G : H)$. For finite G this number is $|G|/|H|$; the definition itself does not require finiteness.

Exercise 10.19: True or false

(a) True: aH is defined for every subgroup $H \leq G$ and every $a \in G$.

(b) True: $|G| = (G : H)|H|$, so the number of cosets divides $|G|$.

(c) True: a group of prime order is cyclic, and cyclic groups are abelian.

(d) False: the finite subgroup $\{0\}$ of $\mathbb{Z}$ has singleton cosets $n + \{0\}$.

(e) True: $H = eH$.

(f) False: for example $\mathbb{Z} \leq \mathbb{R}$ has cosets $x + \mathbb{Z}$.

(g) True: for $n > 1$, half the permutations are even, so $[S_n : A_n] = 2$.

(h) This is a statement of taste rather than a mathematical proposition; one may agree that the theorem is nice, but there is no objectively defined truth value to prove.

(i) False: V has order 4 but no element of order 4.

(j) True: if $G = \langle g \rangle$ has order m and $d \mid m$, then $g^{m/d}$ has order d.

Exercise 10.20: Different left and right partitions in an abelian group

Impossible. For every $a \in G$ and $h \in H$, commutativity gives $ah = ha$, so $aH = Ha$. Thus the left and right coset partitions must agree for every subgroup of an abelian group.

Exercise 10.21: A single coset

Take $H = G$, for example $H = G = \mathbb{Z}_6$. Every $aH = aG = G$, so there is exactly one cell in the partition. Conversely an index-1 subgroup must be all of G, since its identity coset H is the only cell.

Exercise 10.22: Six cosets in a group of order six

Take $G = \mathbb{Z}_6$ and the trivial subgroup $H = \{0\}$. Its six cosets are the six singletons, so the partition has six cells. The formula $[G : H] = 6/1$ confirms the count.

Exercise 10.23: Twelve cosets in a group of order six

Impossible. Every coset is nonempty, so a partition of a six-element set has at most six cells. Equivalently, the index would force $|H| = 6/12 = 1/2$, which is not a possible subgroup order.

Exercise 10.24: Four cosets in a group of order six

Impossible. Lagrange's formula would give $6 = 4|H|$, so $|H| = 3/2$, not an integer. Thus the index of a subgroup of an order-6 group cannot be 4.

Exercise 10.25: Synopsis of Lagrange's theorem

The left cosets of H partition finite G, and each has $|H|$ elements by the bijection $h \mapsto ah$, so $|G| = [G : H]|H|$ and $|H|$ divides $|G|$.

Exercise 10.26: The right-coset equivalence relation

The relation is $a \sim_R b$ if and only if $ab^{-1} \in H$. It is reflexive because $aa^{-1} = e \in H$. If $ab^{-1} \in H$, inverse closure gives $(ab^{-1})^{-1} = ba^{-1} \in H$, so it is symmetric. If $ab^{-1} \in H$ and $bc^{-1} \in H$, their product is $ac^{-1} \in H$, proving transitivity. Hence it is an equivalence relation; its class containing a is Ha.

Exercise 10.27: A bijection with a right coset

Define $f : H \to Hg$ by $f(h) = hg$. Its values lie in Hg by definition. If $f(h_1) = f(h_2)$, then $h_1 g = h_2 g$; multiplying on the right by g^{-1} gives $h_1 = h_2$, proving injectivity. Every $x \in Hg$ has the form hg for some $h \in H$, so $x = f(h)$ and the map is surjective. Its inverse is $x \mapsto xg^{-1}$.

Exercise 10.28: Conjugation condition implies equal cosets

Suppose $g^{-1}hg \in H$ for all $g \in G, h \in H$. If $hg \in Hg$, write $hg = g(g^{-1}hg) \in gH$, so $Hg \subseteq gH$. For the other inclusion replace g by g^{-1} in the hypothesis: $ghg^{-1} \in H$. Then $gh = (ghg^{-1})g \in Hg$, so $gH \subseteq Hg$. The two inclusions give equality.

Exercise 10.29: Equal coset partitions imply the conjugation condition

Assume the partitions agree. The left coset gH and the right coset Hg both contain g, so they must be the same cell of that common partition. For $h \in H$, the element hg is in $Hg = gH$, so $hg = gh'$ for some $h' \in H$. Multiplying on the left by g^{-1} gives $g^{-1}hg = h' \in H$, as required.

Exercise 10.30: Do equal left cosets imply equal right cosets?

No. In S_3, take $H = \{e, (1,2)\}$, $a = (1,3)$, and $b = a(1,2) = (1,2,3)$. Since $b = ah$ for $h \in H$, $aH = bH$. If $Ha = Hb$, the right-coset criterion would give $ba^{-1} \in H$. But

$$ba^{-1} = a(1,2)a^{-1} = (3,2) \notin H.$$

Thus $Ha \neq Hb$, giving a counterexample satisfying the assumed left-coset equality.

Exercise 10.31: Membership from equal right cosets

True. Since $e \in H$, the element $b = eb$ belongs to Hb. The assumption $Ha = Hb$ then gives $b \in Ha$. This conclusion is a membership statement and does not require commutativity or normality.

Exercise 10.32: Inverting equal cosets

True. If $aH = bH$, take the inverses of every element of both sets. Since inversion is a bijection and $H^{-1} = H$,

$$(aH)^{-1} = H^{-1}a^{-1} = Ha^{-1}, \qquad (bH)^{-1} = Hb^{-1}.$$

Equal sets have equal inverse sets, so $Ha^{-1} = Hb^{-1}$.

Exercise 10.33: Squaring representatives of equal cosets

False. Use $H = \{e, (1,2)\}$, $a = (1,3)$, and $b = (1,2,3) = a(1,2)$ as in Exercise 30. Then $aH = bH$, but $a^2 = e$, so $a^2H = H$. Meanwhile $b^2 = (1,3,2) \notin H$, so $b^2H \neq H$. Thus squaring a representative need not preserve left-coset equality when the subgroup is not normal.

Exercise 10.34: Proper subgroups of a group of order pq

By Lagrange, the order of a subgroup divides pq. If the subgroup is proper, its order is $1, p$, or q (when $p = q$, the list reduces to $1, p$). The trivial group is cyclic. Any group of prime order is cyclic: a nonidentity element generates a subgroup whose order is a divisor greater than 1 of that prime, and so generates the whole group. Hence every proper subgroup is cyclic.

Exercise 10.35: A bijection between left and right cosets

Define F on left cosets by $F(aH) = Ha^{-1}$. It is well defined because $aH = bH$ implies $Ha^{-1} = Hb^{-1}$, as proved in Exercise 32. Conversely $Ha^{-1} = Hb^{-1}$ implies $aH = bH$ after taking inverse sets, so F is injective. Every right coset Hg equals $F(g^{-1}H)$, proving surjectivity. Thus the collections have the same cardinality even for infinite groups.

Exercise 10.36: A unique involution in an abelian group of order $2n$

Existence follows by pairing nonidentity elements with their inverses in a group of even order: at least one is its own inverse. For uniqueness, suppose distinct involutions a, b exist. Since the group

is abelian, the four elements e, a, b, ab form a subgroup: all square to e, and they are distinct, since $ab = e, a, b$ would force $a = b$, $b = e$, or $a = e$. Thus 4 divides $2n$ by Lagrange, contradicting oddness of n. Hence there is exactly one element of order 2.

Exercise 10.37: No proper nontrivial subgroups

Choose $a \neq e$. Its subgroup $\langle a \rangle$ is nontrivial, so by hypothesis it equals G and G is cyclic. It cannot be infinite cyclic, because then $\langle a^2 \rangle$ is a proper nontrivial subgroup: it contains $a^2 \neq e$ but not a. Thus G has a finite order $m > 1$. If m were composite, choose a proper divisor d with $1 < d < m$; the element $a^{m/d}$ generates a subgroup of order d, again a contradiction. Therefore m is prime.

Exercise 10.38: Multiplication of subgroup indices

Let the distinct left cosets of H in G be $a_i H$ for $1 \leq i \leq r$, and those of K in H be $b_j K$ for $1 \leq j \leq s$.

1. Cover G.

Given $g \in G$, write $g = a_i h$ for some i and $h \in H$. Then $h = b_j k$ for some $j, k \in K$, so $g \in a_i b_j K$.

2. Show the rs cosets are distinct.

If $a_i b_j K = a_{i'} b_{j'} K$, representatives differ by a factor in $K \subseteq H$, so $a_i H = a_{i'} H$ and $i = i'$. Cancelling a_i then gives $b_j K = b_{j'} K$, hence $j = j'$. Therefore these are exactly rs distinct left cosets of K in G, proving $[G : K] = [G : H][H : K]$. The argument does not require G itself to be finite.

Exercise 10.39: A subgroup of index two is normal

If $g \in H$, both gH and Hg equal H. If $g \notin H$, the two left cosets are H and gH, so $gH = G \setminus H$. The right cosets likewise give $Hg = G \setminus H$. Hence $gH = Hg$ for every g, proving normality. In fact the same complement argument works without the given finiteness assumption.

Exercise 10.40: Every element satisfies $a^n = e$

Let the order of a be d. By Lagrange applied to $\langle a \rangle$, d divides $n = |G|$, so write $n = dk$. Then $a^n = (a^d)^k = e^k = e$. The order d is finite because G is finite.

Exercise 10.41: A unique representative in $[0, 1)$

Let a coset be $a + \mathbb{Z}$. Set $x = a - \lfloor a \rfloor$. Since $\lfloor a \rfloor \leq a < \lfloor a \rfloor + 1$, we have $0 \leq x < 1$; subtracting the integer $\lfloor a \rfloor$ also shows $x \in a + \mathbb{Z}$. For uniqueness, if x, y both lie in this coset and in $[0, 1)$, their difference is an integer with $-1 < x - y < 1$. The only such integer is 0, so $x = y$.

Exercise 10.42: Sine is constant on a coset

The subgroup generated by 2π in additive $\mathbb{R}$ is $2\pi\mathbb{Z}$. If y belongs to the coset $x + 2\pi\mathbb{Z}$, then $y = x + 2\pi k$ for some integer k. By the 2π-periodicity of sine, $\sin y = \sin x$. Thus assigning $x + 2\pi\mathbb{Z} \mapsto \sin x$ is independent of the representative chosen and defines a function on the coset set.

Exercise 10.43: Double cosets

(a) Reflexivity: $a = eae$ with $e \in H, K$. Symmetry: if $a = hbk$, then $b = h^{-1}ak^{-1}$, with the inverse factors still in their subgroups. Transitivity: if $a = h_1bk_1$ and $b = h_2ck_2$, then $a = (h_1h_2)c(k_2k_1)$, with $h_1h_2 \in H$ and $k_2k_1 \in K$. Thus the relation is an equivalence relation. **(b)** Its class containing a consists exactly of all hak, with $h \in H$ and $k \in K$, namely the double coset HaK. The order of the two factors on the right in the transitivity calculation matters because the group need not be abelian.

Exercise 10.44: A point stabilizer and its transport coset

(a) Let $H = S_{c,c}$. The identity fixes c; products of maps fixing c fix c; and inverses fix c too. Hence $H \leq S_A$. **(b)** If $d \neq c$, the identity does not send c to d, so it is absent from $S_{c,d}$. Therefore this set is not a subgroup. **(c)** Choose a permutation τ sending c to d, for example the transposition (c, d). Then $S_{c,d} = \tau H$. Indeed $(\tau h)(c) = \tau(c) = d$ for $h \in H$. Conversely, if $\sigma(c) = d$, then $(\tau^{-1}\sigma)(c) = c$, so $\tau^{-1}\sigma \in H$ and $\sigma \in \tau H$. Thus it is a left coset of the stabilizer.

Exercise 10.45: The subgroups of a finite cyclic group

Let $G = \langle g \rangle$ have order n. For a divisor $d \mid n$, the element $g^{n/d}$ has order d, so $H_d = \langle g^{n/d} \rangle$ is a subgroup of order d. Conversely, every subgroup of a cyclic group is cyclic. If $H = \langle g^k \rangle$ has order d, then $d = n/\gcd(n, k)$, so $\gcd(n, k) = n/d$. The subgroup generated by g^k is exactly the subgroup generated by $g^{\gcd(n,k)}$: the exponent gcd is an integer combination of n, k, and $g^n = e$. Thus $H = \langle g^{n/d} \rangle = H_d$. This proves existence and uniqueness for each divisor, and Lagrange excludes other subgroup orders.

Exercise 10.46: The divisor sum of the Euler function

Partition the n elements of $G = \langle g \rangle$ according to their exact orders. For each divisor $d \mid n$, there is one subgroup of order d, and the elements of exact order d are precisely its generators. A cyclic order-d group has $\varphi(d)$ generators, because its generator powers h^k generate exactly when $\gcd(k, d) = 1$. Therefore counting all elements gives

$$\boxed{\sum_{d|n} \varphi(d) = n.}$$

The divisor $d = 1$ contributes the identity, since $\varphi(1) = 1$.

Exercise 10.47: A bound on roots forces cyclicity

Put $n = |G|$, and for each divisor d of n let c_d count elements of exact order d. If $c_d > 0$, choose such an element a. Its cyclic subgroup has d elements, all satisfying $x^d = e$. By the hypothesis there are at most d solutions in all of G, so these are all the solutions. Thus every element of exact order d belongs to $\langle a \rangle$, and $c_d = \varphi(d)$. If no such element exists, $c_d = 0$. In either case $c_d \leq \varphi(d)$. Lagrange says every element order divides n, so

$$n = \sum_{d|n} c_d \leq \sum_{d|n} \varphi(d) = n.$$

Equality of the two sums of nonnegative terms forces equality in every term: otherwise at least one strictly smaller term and no larger terms would make the sum smaller. In particular $c_n = \varphi(n) > 0$. An element of order n generates all n elements of G, so G is cyclic.

11 Direct Products and Finitely Generated Abelian Groups

Conventions. Permutations compose from right to left. The identity is e. A group generated by an element a is denoted $\langle a \rangle$.

Exercise 11.1: Elements and orders in $\mathbb{Z}_2 \times \mathbb{Z}_4$

The elements are all pairs (a, b) with $0 \le a < 2$ and $0 \le b < 4$. Addition is componentwise. A multiple $k(a, b)$ is zero exactly when both $ka \equiv 0 \pmod 2$ and $kb \equiv 0 \pmod 4$. Hence its order is the least common multiple of the two component orders. The table lists the order of each pair, with first coordinate indexing rows and second coordinate indexing columns:

$a\backslash b$	0	1	2	3
0	1	4	2	4
1	2	4	2	4

No element has order 8, so the order-8 group is not cyclic.

Exercise 11.2: Elements and orders in $\mathbb{Z}_3 \times \mathbb{Z}_4$

The elements are all pairs (a, b) with $0 \le a < 3$ and $0 \le b < 4$. Addition is componentwise. A multiple $k(a, b)$ is zero exactly when both $ka \equiv 0 \pmod 3$ and $kb \equiv 0 \pmod 4$. Hence its order is the least common multiple of the two component orders. The table lists the order of each pair, with first coordinate indexing rows and second coordinate indexing columns:

$a\backslash b$	0	1	2	3
0	1	4	2	4
1	3	12	6	12
2	3	12	6	12

The element $(1, 1)$ has order $\operatorname{lcm}(3, 4) = 12$, equal to the group order, so it generates the whole group; the group is cyclic.

Exercise 11.3: Order of $(2, 6)$

In $\mathbb{Z}_m$, the order of a is $m/\gcd(m, a)$. In the given group $\mathbb{Z}_4 \times \mathbb{Z}_{12}$, the component orders are

$$\frac{4}{\gcd(4, 2)} = 2, \quad \frac{12}{\gcd(12, 6)} = 2.$$

All components return to zero simultaneously at the least common multiple of these orders, so

$$\operatorname{ord}(2, 6) = \operatorname{lcm}(2, 2) = 2.$$

Exercise 11.4: Order of $(2, 3)$

In $\mathbb{Z}_m$, the order of a is $m/\gcd(m, a)$. In the given group $\mathbb{Z}_6 \times \mathbb{Z}_{15}$, the component orders are

$$\frac{6}{\gcd(6, 2)} = 3, \quad \frac{15}{\gcd(15, 3)} = 5.$$

All components return to zero simultaneously at the least common multiple of these orders, so

$$\operatorname{ord}(2,3) = \operatorname{lcm}(3,5) = 15.$$

Exercise 11.5: Order of $(8, 10)$

In $\mathbb{Z}_m$, the order of a is $m/\gcd(m,a)$. In the given group $\mathbb{Z}_{12} \times \mathbb{Z}_{18}$, the component orders are

$$\frac{12}{\gcd(12,8)} = 3, \quad \frac{18}{\gcd(18,10)} = 9.$$

All components return to zero simultaneously at the least common multiple of these orders, so

$$\operatorname{ord}(8,10) = \operatorname{lcm}(3,9) = 9.$$

Exercise 11.6: Order of $(3, 10, 9)$

In $\mathbb{Z}_m$, the order of a is $m/\gcd(m,a)$. In the given group $\mathbb{Z}_4 \times \mathbb{Z}_{12} \times \mathbb{Z}_{15}$, the component orders are

$$\frac{4}{\gcd(4,3)} = 4, \quad \frac{12}{\gcd(12,10)} = 6, \quad \frac{15}{\gcd(15,9)} = 5.$$

All components return to zero simultaneously at the least common multiple of these orders, so

$$\operatorname{ord}(3,10,9) = \operatorname{lcm}(4,6,5) = 60.$$

Exercise 11.7: Order of $(3, 6, 12, 16)$

In $\mathbb{Z}_m$, the order of a is $m/\gcd(m,a)$. In the given group $\mathbb{Z}_4 \times \mathbb{Z}_{12} \times \mathbb{Z}_{20} \times \mathbb{Z}_{24}$, the component orders are

$$\frac{4}{\gcd(4,3)} = 4, \quad \frac{12}{\gcd(12,6)} = 2, \quad \frac{20}{\gcd(20,12)} = 5, \quad \frac{24}{\gcd(24,16)} = 3.$$

All components return to zero simultaneously at the least common multiple of these orders, so

$$\operatorname{ord}(3,6,12,16) = \operatorname{lcm}(4,2,5,3) = 60.$$

Exercise 11.8: Largest orders of cyclic subgroups

In $\mathbb{Z}_m \times \mathbb{Z}_n$, every element order divides $\operatorname{lcm}(m,n)$, since that many additions kill both coordinates. The element $(1,1)$ attains this order, so it is the largest cyclic-subgroup order. Thus the answers are

$$\operatorname{lcm}(6,8) = 24, \qquad \operatorname{lcm}(12,15) = 60.$$

These need not equal the group orders 48 and 180, respectively.

Exercise 11.9: Proper nontrivial subgroups of $\mathbb{Z}_2^2$

All three nonzero elements have order 2. They generate the three subgroups

$$\{(0,0),(1,0)\}, \quad \{(0,0),(0,1)\}, \quad \{(0,0),(1,1)\}.$$

Any subgroup containing two different nonzero vectors contains their sum, the third nonzero vector, and so is the whole group. Thus the three listed subgroups are exactly the proper nontrivial ones.

Exercise 11.10: Proper nontrivial subgroups of $\mathbb{Z}_2^3$

Regard the group as the three-dimensional vector space over $\mathbb{Z}_2$; subgroup closure is exactly linear-subspace closure, because the only scalars are $0, 1$. The seven subgroups of order 2 are $\{0, v\}$ for

$$v = (1,0,0), (0,1,0), (0,0,1), (1,1,0), (1,0,1), (0,1,1), (1,1,1).$$

The seven subgroups of order 4 are the planes described, respectively, by

$$x = 0, \quad y = 0, \quad z = 0, \quad x + y = 0, \quad x + z = 0, \quad y + z = 0, \quad x + y + z = 0,$$

where arithmetic is modulo 2. Each displayed equation has four solutions. Every two-dimensional subspace is the kernel of one nonzero linear functional, and over $\mathbb{Z}_2$ no two distinct nonzero functionals are scalar multiples. Consequently this lists all seven planes. Proper nontrivial subgroups have dimension 1 or 2, so these fourteen are all of them.

Exercise 11.11: Order-four subgroups of $\mathbb{Z}_2 imes \mathbb{Z}_4$

An element has order 4 precisely when its second coordinate is odd. There are four such elements, and each cyclic order-4 subgroup has two generators. The two distinct cyclic subgroups are

$$\langle (0,1) \rangle = \{(0,0), (0,1), (0,2), (0,3)\},$$

$$\langle (1,1) \rangle = \{(0,0), (1,1), (0,2), (1,3)\}.$$

Any noncyclic group of order 4 is a Klein group and all its elements are killed by 2. The elements killed by 2 here are exactly

$$\{(0,0), (1,0), (0,2), (1,2)\},$$

which form one more subgroup. This exhausts both possible types of order-4 subgroup.

Exercise 11.12: Klein subgroups of $\mathbb{Z}_2 imes \mathbb{Z}_4 imes \mathbb{Z}_4$

Every element of a Klein subgroup is killed by 2, so it must lie in

$$T = \{(a, 2b, 2c) : a, b, c \in \mathbb{Z}_2\} \cong \mathbb{Z}_2^3.$$

A Klein subgroup is a two-dimensional subspace of this three-dimensional space. Thus all seven are given by $(a, 2b, 2c)$ subject to one of the conditions

$$a = 0,\ b = 0,\ c = 0,\ a + b = 0,\ a + c = 0,\ b + c = 0,\ a + b + c = 0$$

in $\mathbb{Z}_2$. For example the first is $\{(0,0,0), (0,2,0), (0,0,2), (0,2,2)\}$. Each condition gives four elements, all of exponent 2, hence a Klein group. The seven-plane classification from the preceding exercise proves completeness.

Exercise 11.13: Direct product descriptions of $\mathbb{Z}_{60}$

A finite direct product of cyclic groups is cyclic precisely when their orders are pairwise relatively prime. Since $60 = 4 \cdot 3 \cdot 5$, the prime-power blocks $4, 3, 5$ must remain intact but can be grouped. Ignoring order of factors, all descriptions using at least two nontrivial factors are

$$\mathbb{Z}_4 \times \mathbb{Z}_{15}, \quad \mathbb{Z}_3 \times \mathbb{Z}_{20}, \quad \mathbb{Z}_5 \times \mathbb{Z}_{12}, \quad \mathbb{Z}_4 \times \mathbb{Z}_3 \times \mathbb{Z}_5.$$

These exhaust the partitions of the three prime-power blocks into at least two groups. If factors $\mathbb{Z}_1$ are permitted, one can insert arbitrarily many trivial factors, giving infinitely many formally different descriptions without changing the group; the finite list uses the usual convention of omitting trivial factors.

Exercise 11.14: Fill in the numerical blanks

(a) $\operatorname{ord}_{\mathbb{Z}_{24}}(18) = 24/\gcd(24, 18) = 24/6 = 4$.

(b) $|\mathbb{Z}_3 \times \mathbb{Z}_4| = 3 \cdot 4 = 12$.

(c) The orders of $4 \in \mathbb{Z}_{12}$ and $2 \in \mathbb{Z}_8$ are 3 and 4, so $(4, 2)$ has order 12.

(d) $V \cong \mathbb{Z}_2 \times \mathbb{Z}_2$; both blanks are 2.

(e) A finite-order element of $\mathbb{Z}_2 \times \mathbb{Z} \times \mathbb{Z}_4$ must have zero middle coordinate. There are $2 \cdot 1 \cdot 4 = 8$ such elements, all indeed of finite order.

Exercise 11.15: Maximum order in $\mathbb{Z}_4 times \mathbb{Z}_6$

Every component is killed by its modulus, so $12 = \operatorname{lcm}(4, 6)$ kills every pair. The pair $(1, 1)$ has component orders $4, 6$ and hence order 12. Thus 12 is the maximum possible order.

Exercise 11.16: Comparing two groups of order 24

Yes. Splitting relatively prime cyclic factors gives

$$\mathbb{Z}_2 \times \mathbb{Z}_{12} \cong \mathbb{Z}_2 \times \mathbb{Z}_4 \times \mathbb{Z}_3, \qquad \mathbb{Z}_4 \times \mathbb{Z}_6 \cong \mathbb{Z}_4 \times \mathbb{Z}_2 \times \mathbb{Z}_3.$$

The right-hand products differ only in order of coordinates, so they are isomorphic. An explicit map from the second group to the first is $(a, b) \mapsto (b \bmod 2, c)$, where c is the unique residue modulo 12 satisfying $c \equiv a \pmod 4$, $c \equiv b \pmod 3$; existence and uniqueness follow from coprimality of $4, 3$.

Exercise 11.17: Maximum order in a three-factor product

For $\mathbb{Z}_8 \times \mathbb{Z}_{10} \times \mathbb{Z}_{24}$ the maximum is $\operatorname{lcm}(8, 10, 24) = 2^3 \cdot 3 \cdot 5 = 120$. Every element is killed by 120, and $(1, 1, 1)$ has exactly that order, proving both the upper bound and its attainment.

Exercise 11.18: Comparing the groups in Exercise 18

The groups are not isomorphic, although both have the same order and the same maximum element order 120. Count elements killed by 4. In $\mathbb{Z}_m$ the congruence $4x = 0$ has $\gcd(4, m)$ solutions. Therefore the counts are

$$\gcd(4, 8)\gcd(4, 10)\gcd(4, 24) = 4 \cdot 2 \cdot 4 = 32,$$

$$\gcd(4, 4)\gcd(4, 12)\gcd(4, 40) = 4 \cdot 4 \cdot 4 = 64.$$

An isomorphism preserves the equation $4x = 0$ and hence its number of solutions, so the different counts rule out an isomorphism.

Exercise 11.19: Maximum order in $\mathbb{Z}_4 imes \mathbb{Z}_{18} imes \mathbb{Z}_{15}$

The least common multiple of the factor orders is $\operatorname{lcm}(4, 18, 15) = 2^2 \cdot 3^2 \cdot 5 = 180$. It kills every element, and $(1, 1, 1)$ has that order. Therefore the maximum is 180.

Exercise 11.20: Comparing the groups in Exercise 20

Yes. Decompose the factors into prime powers:

$$\mathbb{Z}_4 \times \mathbb{Z}_{18} \times \mathbb{Z}_{15} \cong \mathbb{Z}_4 \times \mathbb{Z}_2 \times \mathbb{Z}_9 \times \mathbb{Z}_3 \times \mathbb{Z}_5,$$
$$\mathbb{Z}_3 \times \mathbb{Z}_{36} \times \mathbb{Z}_{10} \cong \mathbb{Z}_3 \times \mathbb{Z}_4 \times \mathbb{Z}_9 \times \mathbb{Z}_2 \times \mathbb{Z}_5.$$

The same five prime-power cyclic factors occur, so reordering coordinates gives an isomorphism. Splitting was legitimate because each original factor was split into relatively prime factors.

Exercise 11.21: Abelian groups of order 8

The classification theorem assigns one group to each partition of 3, using one cyclic factor $\mathbb{Z}_{2^k}$ for each part k. Thus the complete list is

$$\mathbb{Z}_8,$$
$$\mathbb{Z}_4 \times \mathbb{Z}_2,$$
$$\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2.$$

The partitions are distinct, so the uniqueness in the classification theorem shows the listed groups are pairwise nonisomorphic. There are 3 groups in all.

Exercise 11.22: Abelian groups of order 16

The classification theorem assigns one group to each partition of 4, using one cyclic factor $\mathbb{Z}_{2^k}$ for each part k. Thus the complete list is

$$\mathbb{Z}_{16},$$
$$\mathbb{Z}_8 \times \mathbb{Z}_2,$$
$$\mathbb{Z}_4 \times \mathbb{Z}_4,$$
$$\mathbb{Z}_4 \times \mathbb{Z}_2 \times \mathbb{Z}_2,$$
$$\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2.$$

The partitions are distinct, so the uniqueness in the classification theorem shows the listed groups are pairwise nonisomorphic. There are 5 groups in all.

Exercise 11.23: Abelian groups of order 32

The classification theorem assigns one group to each partition of 5, using one cyclic factor $\mathbb{Z}_{2^k}$ for each part k. Thus the complete list is

$$\begin{gathered}\mathbb{Z}_{32},\\ \mathbb{Z}_{16}\times\mathbb{Z}_2,\\ \mathbb{Z}_8\times\mathbb{Z}_4,\\ \mathbb{Z}_8\times\mathbb{Z}_2\times\mathbb{Z}_2,\\ \mathbb{Z}_4\times\mathbb{Z}_4\times\mathbb{Z}_2,\\ \mathbb{Z}_4\times\mathbb{Z}_2\times\mathbb{Z}_2\times\mathbb{Z}_2,\\ \mathbb{Z}_2\times\mathbb{Z}_2\times\mathbb{Z}_2\times\mathbb{Z}_2\times\mathbb{Z}_2.\end{gathered}$$

The partitions are distinct, so the uniqueness in the classification theorem shows the listed groups are pairwise nonisomorphic. There are 7 groups in all.

Exercise 11.24: Abelian groups of order 720

Factor $720 = 2^4 \cdot 3^2 \cdot 5$. The possible 2-primary factors are

$$T \in \{\mathbb{Z}_{16},\ \mathbb{Z}_8\times\mathbb{Z}_2,\ \mathbb{Z}_4\times\mathbb{Z}_4,\ \mathbb{Z}_4\times\mathbb{Z}_2^2,\ \mathbb{Z}_2^4\},$$

and the possible 3-primary factors are $U \in \{\mathbb{Z}_9, \mathbb{Z}_3^2\}$. The 5-primary factor must be $\mathbb{Z}_5$. All groups, with none omitted or repeated up to isomorphism, are therefore

$$\boxed{T\times U\times\mathbb{Z}_5 \quad \text{for every one of the five } T \text{ and two } U.}$$

This specifies $5\cdot 2 = 10$ groups explicitly by independent choices; uniqueness of the prime-power cyclic decomposition distinguishes them.

Exercise 11.25: Abelian groups of order 1089

Since $1089 = 33^2 = 3^2\cdot 11^2$, each primary component has two possibilities. The four groups are

$$\begin{aligned}&\mathbb{Z}_9\times\mathbb{Z}_{121}, \qquad &&\mathbb{Z}_9\times\mathbb{Z}_{11}^2,\\ &\mathbb{Z}_3^2\times\mathbb{Z}_{121}, \qquad &&\mathbb{Z}_3^2\times\mathbb{Z}_{11}^2.\end{aligned}$$

The exponent-2 partition choices are 2 and $1+1$ for each prime. They produce all primary decompositions and hence all isomorphism types.

Exercise 11.26: Counts for orders 24, 25, and 600

The printed numbers are 24 and 25, not 2^4 and 2^5. Since $24 = 2^3\cdot 3$, the three partitions of 3 give three groups:

$$\mathbb{Z}_8\times\mathbb{Z}_3, \quad \mathbb{Z}_4\times\mathbb{Z}_2\times\mathbb{Z}_3, \quad \mathbb{Z}_2^3\times\mathbb{Z}_3.$$

For order $25 = 5^2$ the two possibilities are $\mathbb{Z}_{25}$ and $\mathbb{Z}_5^2$. Because 24 and 25 are coprime, their primary decompositions can be chosen independently. Hence for $(24)(25) = 600$ there are $3 \cdot 2 = 6$ abelian groups up to isomorphism.

Exercise 11.27: Multiplicativity for coprime group orders

Suppose $\gcd(m, n) = 1$. The prime factors of m and n are disjoint. By the finite abelian classification theorem, any abelian group of order mn is uniquely determined by its primary factors. Collecting the factors at primes dividing m gives a group of order m; collecting those at primes dividing n gives one of order n. Conversely, the direct product of any one of the r types of order m and any one of the s types of order n gives a group of order mn. Uniqueness of primary decompositions means different pairs give different isomorphism types. Thus there are exactly rs.

Exercise 11.28: Groups of order 10^5

Factor $10^5 = 2^5 \cdot 5^5$. There are seven partitions of 5, hence seven abelian 2-groups of order 2^5 and seven abelian 5-groups of order 5^5. These choices are independent by coprimality, so the answer is $7 \cdot 7 = 49$.

Exercise 11.29: Partition counts for prime-power orders

(a) An abelian group of order p^n corresponds to a partition of n: parts $\lambda_1, \ldots, \lambda_k$ give $\mathbb{Z}_{p^{\lambda_1}} \times \cdots \times \mathbb{Z}_{p^{\lambda_k}}$. The required table is

n	2	3	4	5	6	7	8
number	2	3	5	7	11	15	22

One reproducible way to obtain it is the recurrence $P(n, k) = P(n, k-1) + P(n-k, k)$, splitting partitions according to whether a part k occurs, with $P(0, k) = 1$ and $P(n, 0) = 0$ for $n > 0$; the desired count is $P(n, n)$. **(b)** Distinct primes permit independent choices. Thus

$$\begin{aligned} \text{(i) } & p^3q^4r^7 : 3 \cdot 5 \cdot 15 = 225, \\ \text{(ii) } & (qr)^7 = q^7r^7 : 15 \cdot 15 = 225, \\ \text{(iii) } & q^5r^4q^3 = q^8r^4 : 22 \cdot 5 = 110. \end{aligned}$$

In (iii) the two powers of the same prime q must first be combined; they are not independent primary factors.

Exercise 11.30: A Cayley digraph on a toroidal grid

Place a vertex (i, j) in an m-by-n grid, with $0 \le i < m$, $0 \le j < n$. Draw one solid arrow

$$(i, j) \longrightarrow (i + 1 \bmod m, j)$$

for the generator $(1, 0)$, and one dashed arrow

$$(i, j) \dashrightarrow (i, j + 1 \bmod n)$$

for $(0,1)$. Thus each row wraps into an m-cycle and each column into an n-cycle. A corner of the schematic is

$$\begin{array}{ccccc} (0,2) & \longrightarrow & (1,2) & \longrightarrow & \cdots \\ \uparrow & & \uparrow & & \\ (0,1) & \longrightarrow & (1,1) & \longrightarrow & \cdots \\ \uparrow & & \uparrow & & \\ (0,0) & \longrightarrow & (1,0) & \longrightarrow & \cdots \end{array}$$

Vertical arrows represent the second arc type; include the wrap arrows from $i = m-1$ to $i = 0$ and from $j = n-1$ to $j = 0$. These rules specify every labeled vertex and arc, including the boundary identifications.

Exercise 11.31: Two concentric polygon digraphs

Let r denote the solid-arrow generator, of order n, and s the dashed-edge generator, of order 2. Label one polygon $e, r, \ldots, r^{n-1}$ and the matching vertices in the other $s, rs, \ldots, r^{n-1}s$. **(a)** The group is abelian exactly when the arrows on the two polygons run in the same direction. In that case traversing a dashed edge before or after a solid edge reaches the same vertex, so $rs = sr$; because r, s generate the group, this proves commutativity. Opposite directions instead give $sr = r^{-1}s \neq rs$ for $n \geq 3$. **(b)** In the abelian case all $2n$ elements are uniquely $r^i s^j$ with $0 \leq i < n$, $j = 0, 1$, and multiplication adds both exponents. Thus $G \cong \mathbb{Z}_n \times \mathbb{Z}_2$. **(c)** This product is cyclic exactly when $\gcd(n, 2) = 1$, that is, n is odd. **(d)** Opposite arrow directions give exactly the usual dihedral relations, and hence the $2n$-element group is D_n.

Exercise 11.32: True or false: direct products

(a) True: $(g_1, g_2) \mapsto (g_2, g_1)$ is an isomorphism.

(b) Componentwise computation is precisely the multiplication rule; the description "easy" is informal, but the intended mathematical content is true.

(c) False: $\mathbb{Z} \times \mathbb{Z}$ is a direct product of infinite groups.

(d) True: the orders of two proper nontrivial finite factors would both exceed 1 and multiply to the prime group order, which is impossible.

(e) False: every element of $\mathbb{Z}_2 \times \mathbb{Z}_4$ has order dividing 4, unlike a generator of $\mathbb{Z}_8$.

(f) False: the product has 8 elements, while S_8 has 8!; it is also abelian.

(g) False: both groups have 24 elements, but $\mathbb{Z}_3 \times \mathbb{Z}_8$ is abelian and S_4 is not.

(h) False: the identity has order 1; other orders can also be 2 or 4.

(i) False: $|\mathbb{Z}_{12} \times \mathbb{Z}_{15}| = 12 \cdot 15 = 180$.

(j) True: there are independently m choices in the first coordinate and n in the second.

Exercise 11.33: An indecomposable abelian group

Take $G = \mathbb{Z}_p$ for a prime p. Its only subgroups are $\{0\}$ and itself, so there are no proper nontrivial subgroups to use as internal factors. Alternatively, an internal product of two nontrivial finite groups would have composite order, contrary to $|G| = p$.

Exercise 11.34: Subgroups isomorphic to the full product

(a) Exactly one. The finite group $\mathbb{Z}_5 \times \mathbb{Z}_6$ has 30 elements, and a subgroup isomorphic to it must also have 30 elements, hence equal the whole group. **(b)** Infinitely many: for each positive integer k, $k\mathbb{Z} \times \mathbb{Z}$ is a subgroup of $\mathbb{Z}^2$ isomorphic to $\mathbb{Z}^2$ via $(a, b) \mapsto (ka, b)$. These subgroups are distinct. More precisely there are countably infinitely many, since any subgroup isomorphic to $\mathbb{Z}^2$ has two generators in the countable set $\mathbb{Z}^2$, and there are only countably many ordered pairs of possible generators.

Exercise 11.35: A composite-order group with no nontrivial direct decomposition

Take $\mathbb{Z}_4$. Its only proper nontrivial subgroup is $\{0, 2\}$. Any two proper nontrivial internal factors would therefore coincide and have nontrivial intersection, so cannot give an internal direct product. Equivalently the only possible nontrivial factor orders are $2, 2$, whose external product is $\mathbb{Z}_2^2$, with no element of order 4, and hence is not $\mathbb{Z}_4$.

Exercise 11.36: True or false: classification of abelian groups

(a) True: every group of prime order is cyclic.

(b) False: $\mathbb{Z}_2^2$ has prime-power order 4 and is not cyclic.

(c) False: every sum of 4's and 6's modulo 8 is even, so these generate only $\{0, 2, 4, 6\}$.

(d) True: $5 - 4 = 1$, so $\{4, 5, 6\}$ generates 1 and hence $\mathbb{Z}_8$.

(e) True: a finite group is finitely generated, so the theorem applies.

(f) False: $\mathbb{Z}$ and $\mathbb{Z} \times \mathbb{Z}_2$ both have Betti number 1, but only the latter has nontrivial torsion.

(g) True: the primary decomposition contains a 5-power cyclic factor, which has an element of order 5.

(h) False: $\mathbb{Z}_2^2$ has order divisible by 4 but no element of order 4.

(i) True: the primary decomposition contains elements of orders 2 and 3 in different primary factors; their product has order 6.

(j) True: a nonzero free $\mathbb{Z}$ factor would make the group infinite, so a finite group's Betti number is 0.

Exercise 11.37: Changing the prime in a prime-power order

The numbers are equal. For either prime, the isomorphism types are parametrized by the same partitions of the exponent r. Replacing every factor $\mathbb{Z}_{p^k}$ in a decomposition by $\mathbb{Z}_{q^k}$ gives a bijection

of the two lists of isomorphism types. Thus the count depends on r alone, not on the prime.

Exercise 11.38: Order-eight and order-four subgroups of an order-72 group

Write $G = P \times Q$ with $|P| = 8$ and $|Q| = 9$, using the primary decomposition. **(a)** There is exactly one subgroup of order 8, namely $P \times \{0\}$. An element of a group of order 8 has order a power of 2, so its projection into the order-9 group Q is zero. Thus any order-8 subgroup lies in $P \times \{0\}$ and must equal it. **(b)** The number of order-4 subgroups is not determined by $|G|$ alone. If $P = \mathbb{Z}_8$, it has exactly one order-4 subgroup. If $P = \mathbb{Z}_2^3$, it has seven planes of order 4. Taking, for example, $Q = \mathbb{Z}_9$ gives two groups of order 72 with different answers.

Exercise 11.39: The torsion subgroup

Let T consist of elements of finite order. The identity is in T. If $a^r = e$ and $b^s = e$, commutativity gives $(ab)^{rs} = a^{rs}b^{rs} = e$, so ab has finite order and belongs to T. If $a^r = e$, then $(a^{-1})^r = (a^r)^{-1} = e$, so $a^{-1} \in T$. Thus T is a subgroup. The commutativity assumption is used in expanding the power of ab; it cannot simply be omitted.

Exercise 11.40: Orders of two torsion subgroups

In a product with a $\mathbb{Z}$ coordinate, finite order forces that coordinate to be 0, because a nonzero integer has infinite additive order. All elements in the remaining finite factors are torsion. Hence the torsion subgroups are

$$\mathbb{Z}_4 \times \{0\} \times \mathbb{Z}_3 \quad \text{of order } 12,$$

$$\mathbb{Z}_{12} \times \{0\} \times \mathbb{Z}_{12} \quad \text{of order } 144.$$

Exercise 11.41: Torsion in the nonzero real numbers

If $x^n = 1$ for a real nonzero x, then $|x|^n = 1$, so $|x| = 1$ and $x = \pm 1$. Conversely 1 has order 1 and -1 has order 2. Thus the torsion subgroup is $\{1, -1\} \cong \mathbb{Z}_2$.

Exercise 11.42: Torsion in the nonzero complex numbers

A complex number has finite multiplicative order exactly when it is a root of unity. Indeed $z^n = 1$ implies $z = e^{2\pi ik/n}$ for some integer k, and any such number has a finite power equal to 1. Therefore

$$T = \{e^{2\pi iq} : q \in \mathbb{Q}\},$$

the group of all roots of unity. It is isomorphic to $\mathbb{Q}/\mathbb{Z}$ under the map $q + \mathbb{Z} \mapsto e^{2\pi iq}$.

Exercise 11.43: Torsion and free parts of a finitely generated abelian group

By the classification theorem choose an isomorphism $G \cong F \times \mathbb{Z}^r$, where F is a finite product of finite cyclic prime-power groups. An element has finite order precisely when all its $\mathbb{Z}$ coordinates are zero, so the torsion subgroup corresponds to $F \times \{0\}$. The subgroup corresponding to $\{e\} \times \mathbb{Z}^r$ is torsion

free. These subgroups intersect trivially, commute elementwise, and every element is their product; transporting them back through the isomorphism gives an internal direct-product decomposition of G. This includes the case of a trivial finite or free part.

Exercise 11.44: Torsion coefficients

(a) Since $\gcd(4, 9) = 1$, $\mathbb{Z}_4 \times \mathbb{Z}_9 \cong \mathbb{Z}_{36}$. The only torsion coefficient is 36. **(b)** The group is $\mathbb{Z}_6 \times \mathbb{Z}_{12} \times \mathbb{Z}_{20}$. Split into prime-power factors and sort the exponents at each prime, padding with zero exponents on the left:

p	first	second	third
2	1	2	2
3	0	1	1
5	0	0	1

Combining each column gives 2, $2^2 \cdot 3 = 12$, and $2^2 \cdot 3 \cdot 5 = 60$. Hence the torsion coefficients are $2, 12, 60$, and $2 \mid 12 \mid 60$. **(c)** For an arbitrary finite product, factor all moduli into prime powers. For each prime put the exponents in nondecreasing order and pad shorter lists with zero exponents at their left ends to obtain equal lengths. Multiply the prime powers column by column. The resulting column products divide the next ones, because every prime exponent is nondecreasing. Reordering primary factors and combining relatively prime factors proves this process preserves the isomorphism type. Discard any leading coefficient 1.

Exercise 11.45: Synopsis of the coprime-product criterion

The element $(1, 1) \in \mathbb{Z}_m \times \mathbb{Z}_n$ has order $\operatorname{lcm}(m, n)$, so when $\gcd(m, n) = 1$ it generates the entire order-mn group. If the gcd is $d > 1$, every element is killed by $mn/d < mn$, so no element can generate the whole group and the product is not cyclic.

Exercise 11.46: A direct product of abelian groups is abelian

Let $a = (a_1, \ldots, a_r)$ and $b = (b_1, \ldots, b_r)$ belong to the product. Since each factor is abelian, $a_i b_i = b_i a_i$ for every i. Therefore

$$ab = (a_1 b_1, \ldots, a_r b_r) = (b_1 a_1, \ldots, b_r a_r) = ba.$$

This is exactly commutativity of the product operation.

Exercise 11.47: Elements of order at most two

The given set is $H = \{g \in G : g^2 = e\}$. It contains e. If $a, b \in H$, commutativity gives $(ab)^2 = a^2 b^2 = e$, so $ab \in H$. Also $(a^{-1})^2 = (a^2)^{-1} = e$. Thus H is a subgroup. Any nonidentity element killed by 2 has order exactly 2, so this is precisely the set described in the question.

Exercise 11.48: Identity together with all elements of an exact order

For order 3, the answer is yes: the set is $\{g : g^3 = e\}$, and commutativity shows it is closed under products and inverses. For order 4, the answer is no. In $\mathbb{Z}_4$, 1 has order 4 but $1 + 1 = 2$ has order 2, so the set $\{0, 1, 3\}$ is not closed. In general the universal property holds exactly for $n = 1$ and for prime

n. For prime p, every nonidentity solution of $g^p = e$ has order p, so the same kernel-style closure argument works. If n is composite, choose a prime $p \mid n$ with $p < n$ and work in $\mathbb{Z}_n \times \mathbb{Z}_n$. Both

$$a = (1, 0), \qquad b = (-1, n/p)$$

have order n because their first coordinates do. But $a + b = (0, n/p)$ has order p, which is neither 1 nor n. Thus the prescribed set fails closure. This also explains the distinction between exact order n and order dividing n.

Exercise 11.49: Why commutativity is needed for involutions

In the nonabelian group S_3, the identity and three transpositions are exactly the elements of order 1 or 2. But $(1, 2)(2, 3) = (1, 2, 3)$ has order 3 and does not lie in that set. Hence it is not a subgroup, providing a counterexample when the abelian hypothesis is removed.

Exercise 11.50: The coordinate subgroups in an external product

Write the two coordinate subgroups as $\widetilde{H} = H \times \{e_K\}$ and $\widetilde{K} = \{e_H\} \times K$. **(a)** Every (h, k) equals $(h, e_K)(e_H, k)$. **(b)** Componentwise multiplication gives

$$(h, e_K)(e_H, k) = (h, k) = (e_H, k)(h, e_K),$$

so the two subgroups commute elementwise even if the individual groups are nonabelian. **(c)** An element in both coordinate subgroups has first coordinate e_H and second coordinate e_K, so the intersection is exactly $\{(e_H, e_K)\}$.

Exercise 11.51: Recognizing an internal direct product

By the first assumed condition, every $g \in G$ is hk. If $hk = h'k'$, rearrange to obtain

$$h'^{-1}h = k'k^{-1} \in H \cap K = \{e\}.$$

Thus $h = h'$ and $k = k'$, proving uniqueness. Define $\Phi : H \times K \to G$ by $\Phi(h, k) = hk$. Existence and uniqueness say it is bijective. Since every element of H commutes with every element of K,

$$\Phi(h, k)\Phi(h', k') = hkh'k' = hh'kk' = \Phi(hh', kk').$$

It is therefore a homomorphism and an isomorphism, making precise the claimed structural identity.

Exercise 11.52: Detecting a noncyclic finite abelian group

Suppose first G contains $\mathbb{Z}_p^2$. This subgroup is not cyclic: every nonidentity element has order p, while the group has order p^2. Every subgroup of a cyclic group is cyclic, so G cannot be cyclic. Conversely, decompose finite abelian G into its prime-power cyclic factors. If each prime occurred in at most one factor, these factor orders would be pairwise coprime, and their product would be cyclic. Since G is not cyclic, some prime p occurs in at least two factors, say $\mathbb{Z}_{p^a}$ and $\mathbb{Z}_{p^b}$. Their order-p subgroups generated by p^{a-1} and p^{b-1} have direct product $\mathbb{Z}_p^2$, which embeds in G. This proves the equivalence.

Exercise 11.53: Element orders in a prime-power group

If $|G| = p^r$, then for any $g \in G$, Lagrange's theorem says $\operatorname{ord}(g)$ divides p^r. The positive divisors are exactly p^s for $0 \le s \le r$, so every element order is a power of p (including $p^0 = 1$). This proof did not use commutativity; therefore the abelian hypothesis can be dropped.

Exercise 11.54: Cancellation for finitely generated abelian groups

By classification, a finitely generated abelian group is specified up to isomorphism by its free rank and, for every prime p and exponent $j \ge 1$, the multiplicity of the factor $\mathbb{Z}_{p^j}$. In a direct product these ranks and multiplicities add. The isomorphism $G \times K \cong H \times K$ therefore gives

$$r_G + r_K = r_H + r_K, \qquad m_G(p,j) + m_K(p,j) = m_H(p,j) + m_K(p,j).$$

All quantities are finite nonnegative integers, since the groups are finitely generated. Cancelling the common terms yields equal free ranks and all equal prime-power multiplicities for G, H. The uniqueness part of the theorem then gives $G \cong H$.

12 Plane Isometries

Conventions. Permutations compose from right to left. The identity is e. A group generated by an element a is denoted $\langle a \rangle$.

Exercise 12.1: Symmetries depend on ambient dimension

(a) For a point $c \in \mathbb{R}$, the isometries fixing it are $x \mapsto x$ and $x \mapsto 2c - x$, the reflection through c. An isometry of the line has form $x \mapsto x + t$ or $x \mapsto -x + t$, and the condition of fixing c determines t in each case.

(b) For a point P in the plane, all rotations about P and all reflections across lines through P fix it, together with the identity. There are infinitely many, unlike in the line. Translations and genuine glide reflections fix no points.

(c) A nondegenerate segment in $\mathbb{R}$ has only the identity and reflection through its midpoint: its two endpoints must either both stay fixed or be interchanged.

(d) In $\mathbb{R}^2$ the symmetries are the identity, reflection in the line containing the segment, reflection in its perpendicular bisector, and the half-turn about its midpoint. They form a Klein four-group. Fixing both endpoints allows reflection in the supporting line, and interchanging them allows the other reflection or the half-turn.

(e) In $\mathbb{R}^3$ examples include every rotation about the supporting line, reflection in any plane containing that line, and reflection in the perpendicular midpoint plane. More fully, with midpoint at the origin and segment on the x-axis, all maps $(x, y, z) \mapsto (\varepsilon x, Q(y, z))$ with $\varepsilon = \pm 1$ and Q any orthogonal 2×2 matrix preserve the segment.

Exercise 12.2: Orientation under composition

Assign sign $+1$ to an orientation-preserving map P and -1 to an orientation-reversing map R. Under composition these signs multiply, as can be seen from the determinants of the orthogonal linear parts of the isometries. Thus

	P	R
P	P	R
R	R	P

In particular two reversals restore orientation, while one reversal changes it.

Exercise 12.3: Types of products of plane isometries

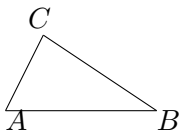
Use t for a nonzero translation, p for a nonidentity rotation, μ for a reflection, and γ for a genuine glide reflection. With identity outcomes omitted, every possible type is recorded by

	t	p	μ	γ
t	t	p	μ, γ	μ, γ
p	p	t, p	μ, γ	μ, γ
μ	μ, γ	μ, γ	t, p	t, p
γ	μ, γ	μ, γ	t, p	t, p

Here is why the lists are exhaustive. Orientation rules eliminate t, p from a product with exactly one reversing factor, and eliminate μ, γ from a product with zero or two reversing factors. Translations compose by adding vectors. A translation times a rotation keeps the same nonzero rotational angle, hence is another rotation. Two rotations can add to a nonzero angle (rotation) or cancel their angles about different centers (translation). Two reflections across intersecting lines give a rotation; across distinct parallel lines they give a translation. Adding a translation to an orientation-reversing map can introduce or cancel displacement along its axis, giving either a reflection or a glide reflection. Choosing intersecting versus parallel axes for products involving reversing factors gives respectively the rotation and translation alternatives. These constructions realize every entry; inverse choices may give the omitted identity.

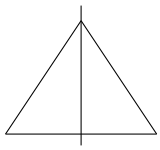
Exercise 12.4: A figure with only the identity symmetry

Take a scalene triangle, for example vertices $A = (0, 0)$, $B = (4, 0)$, $C = (1, 2)$. Its three side lengths are $4, \sqrt{5}, \sqrt{13}$, all distinct. Any symmetry must preserve side lengths, so it fixes all three vertices and is the identity. A sketch is



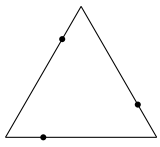
Exercise 12.5: A figure with two symmetries

Use an isosceles triangle that is not equilateral, such as vertices $(-2, 0), (2, 0), (0, 3)$. The unique vertex between the two equal sides must be fixed; the base endpoints can either stay fixed or interchange. Thus the only symmetries are the identity and reflection across the vertical altitude, forming $\mathbb{Z}_2$.



Exercise 12.6: A figure with three symmetries

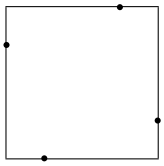
Draw an equilateral triangle and mark a heavy dot one quarter of the way along each edge, always proceeding counterclockwise around the boundary. The dots are part of the figure.



The rotations through $0°, 120°, 240°$ preserve the marks. A reflection reverses the boundary direction and would put the marks three quarters of the way along each counterclockwise edge, where there are no marks. Since the convex hull is the equilateral triangle, every possible symmetry was already among its rotations and reflections. Thus exactly the three rotations remain, giving $\mathbb{Z}_3$.

Exercise 12.7: A figure with cyclic order-four symmetry

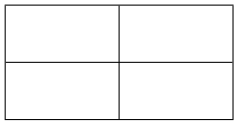
Use a square with one heavy dot one quarter of the way along each edge, in a consistent counterclockwise direction:



All four quarter-turn rotations preserve the figure. Every square reflection reverses the direction along its boundary and moves each quarter-edge marker to a missing three-quarter-edge marker. Thus none is a symmetry. The convex hull forces every symmetry to be a square symmetry, so the group consists of the four rotations and is $\mathbb{Z}_4$.

Exercise 12.8: A figure with Klein-four symmetry

A rectangle with unequal side lengths has the identity, reflection in each of its two midlines, and the half-turn about its center. A quarter-turn would interchange its unequal dimensions, so is not allowed. Each of its three nonidentity symmetries has order 2, and the two reflections multiply to the half-turn, giving the Klein group V.



Exercise 12.9: Orders of the four isometry types

A nonzero translation has infinite order, because its kth power translates by k times its nonzero vector. A rotation through angle θ has finite order $q \geq 2$ when $\theta/(2\pi) = p/q$ in lowest terms, and infinite order when this ratio is irrational. Every reflection has order 2. A genuine glide reflection has infinite order: if its nonzero displacement along the axis is v, its square is the nonzero translation by $2v$. An element whose square has infinite order cannot have finite order.

Exercise 12.10: Which types have fixed points?

A nonidentity rotation fixes its center. A reflection fixes every point of its axis. A nonzero translation has no fixed point because $P + v = P$ would force $v = 0$. A genuine glide reflection also has none: its component parallel to the axis moves every point by the same nonzero amount, which cannot vanish. Thus exactly rotations and reflections can have fixed points.

Exercise 12.11: Exactly one fixed point

Only a nonidentity rotation. Its center is fixed, and any other point moves around a circle by a nonzero angle. A reflection has an entire line of fixed points, while translations and genuine glide reflections have none.

Exercise 12.12: Exactly two fixed points

None of the four nonidentity types. A rotation has one, a reflection has infinitely many, and translations and glide reflections have none. The identity fixes the whole plane and also does not have exactly two.

Exercise 12.13: Infinitely many fixed points

Among the four nonidentity types, only reflections: their fixed-point set is their entire axis. If the identity is also allowed, it fixes every point of the plane and therefore is another isometry with infinitely many fixed points.

Exercise 12.14: Three noncollinear fixed points force the identity

Let A, B, C be the fixed noncollinear points and let X be any point. An isometry preserves its distances to all three: the image Y must satisfy $YA = XA$, $YB = XB$, and $YC = XC$. The first two circles meet in at most two points, which, when distinct, are mirror images across AB. Since C is off AB, those two mirror points have different distances to C; hence the third distance selects X uniquely. Thus $Y = X$ for every X and the isometry is the identity. This is the geometric reason three noncollinear distance references determine a plane point.

Exercise 12.15: Isometries agreeing on three points

Suppose $\phi(P_i) = \psi(P_i)$ at the three given noncollinear points. Form the isometry $\eta = \psi^{-1}\phi$. Then $\eta(P_i) = \psi^{-1}(\psi(P_i)) = P_i$ for each i. By Exercise 14, η is the identity. Multiplying $\psi^{-1}\phi = \mathrm{id}$ on the left by ψ yields $\phi = \psi$.

Exercise 12.16: All rotations together with the identity

They do not form a subgroup. Let $r_A(X) = 2A - X$ and $r_B(X) = 2B - X$ be half-turns about two distinct centers. Then

$$(r_B r_A)(X) = 2B - (2A - X) = X + 2(B - A),$$

a nonzero translation. This product is neither a rotation nor the identity, so closure fails.

Exercise 12.17: Translations together with the identity

Yes. Write $T_v(X) = X + v$, allowing $v = 0$ for the identity. Then $T_v T_w = T_{v+w}$ and $T_v^{-1} = T_{-v}$. The sums and negatives of plane vectors are plane vectors, so the set contains the identity and is closed

under products and inverses.

Exercise 12.18: Rotations about a common point

Yes. After translating coordinates so that the fixed center is the origin, composition of rotations adds angles, and inversion negates the angle. Thus $R_\alpha R_\beta = R_{\alpha+\beta}$ and $R_\alpha^{-1} = R_{-\alpha}$, with angles read modulo 2π. Angle 0 gives the identity, so these maps form a subgroup.

Exercise 12.19: One reflection together with the identity

Yes. If s is reflection across the specified line, then $s^2 = e$. Hence $\{e, s\}$ is closed under multiplication, and each element is its own inverse. This is a subgroup isomorphic to $\mathbb{Z}_2$.

Exercise 12.20: All glide reflections together with the identity

No. A genuine glide reflection g has nonzero displacement v along its axis, and g^2 is translation by $2v$. This is neither a genuine glide reflection nor the identity. Thus the product of an allowed element with itself leaves the proposed set.

Exercise 12.21: Isometries in finite subgroups

Every element of a finite group has finite order. Nonzero translations and genuine glide reflections have infinite order and cannot occur. Reflections can occur with order 2, and rotations can occur when their angle is a rational multiple of 2π. Thus the possible nonidentity types are precisely reflections and finite-order rotations.

Exercise 12.22: The rotation subgroup of a finite plane group

Restrict the orientation sign map to G: its value is $+1$ on preserving elements and -1 on reversing elements, and signs multiply. Its $+1$ set H is therefore closed under products and inverses and contains the identity. In a finite group, the preceding exercise excludes nonzero translations, so H consists exactly of the rotations and identity described in the question. If there is no reversing element, $H = G$. Otherwise fix a reversing $s \in G$; multiplication by s is a bijection from H to the reversing elements, because it changes the sign and has inverse multiplication by s^{-1}. Hence $|G| = 2|H|$.

Exercise 12.23: A finite group of rotations about a point is cyclic

If $G = \{e\}$, it is already cyclic. Otherwise choose the smallest positive angle $\theta \in (0, 2\pi)$ occurring among its finitely many rotations, and let r be that rotation. For any rotation angle $\alpha \in [0, 2\pi)$, division by θ writes $\alpha = q\theta + \delta$ with $0 \le \delta < \theta$. The rotation $r^{-q}R_\alpha$ lies in G and has angle δ. Minimality of θ forces $\delta = 0$, so every element is a power of r. This proves $G = \langle r \rangle$. Since its order is finite, r also has finite order and its multiples close up exactly around the circle.

Exercise 12.24: Repeated F figures

Convention. Here "nontrivial glide" means one not expressible as a product of a translation and reflection that are themselves symmetries, as defined on printed page 116.

(a) No.

(b) No.

(c) No.

(d) No.

(e) The group is isomorphic to $\mathbb{Z}$.

The upper and lower parts of F differ, and its left and right sides differ. Only integral translations by the spacing preserve the repeated upright figures.

Exercise 12.25: Repeated T figures

Convention. Here "nontrivial glide" means one not expressible as a product of a translation and reflection that are themselves symmetries, as defined on printed page 116.

(a) No.

(b) No.

(c) Yes.

(d) No.

(e) The group is isomorphic to D_∞.

Each T has a vertical mirror, but no horizontal mirror or half-turn symmetry. A unit translation t and a vertical reflection s generate the group, with $s^2 = e$ and $sts = t^{-1}$.

Exercise 12.26: Repeated E figures

Convention. Here "nontrivial glide" means one not expressible as a product of a translation and reflection that are themselves symmetries, as defined on printed page 116.

(a) No.

(b) Yes.

(c) No.

(d) No.

(e) The group is isomorphic to $\mathbb{Z} \times \mathbb{Z}_2$.

Horizontal reflection through the middle bar preserves every E; vertical reflection and half-turn do not. This reflection commutes with the unit translation. Glides obtained by combining these two symmetries are trivial in the book's sense.

Exercise 12.27: Repeated Z figures

Convention. Here "nontrivial glide" means one not expressible as a product of a translation and reflection that are themselves symmetries, as defined on printed page 116.

(a) Yes.

(b) No.

(c) No.

(d) No.

(e) The group is isomorphic to D_∞.

A half-turn about the center of a Z preserves it, but neither horizontal nor vertical reflection does. The half-turn r reverses translation: $rtr = t^{-1}$, yielding the infinite dihedral group.

Exercise 12.28: Repeated H figures

Convention. Here "nontrivial glide" means one not expressible as a product of a translation and reflection that are themselves symmetries, as defined on printed page 116.

(a) Yes.

(b) Yes.

(c) Yes.

(d) No.

(e) The group is isomorphic to $D_\infty \times \mathbb{Z}_2$.

The H has both horizontal and vertical mirrors and a half-turn. Translation and a vertical mirror generate D_∞, while the horizontal mirror is an independent commuting involution. Every glide is the product of an existing translation and a horizontal reflection.

Exercise 12.29: Alternating bent strokes

Convention. Here "nontrivial glide" means one not expressible as a product of a translation and reflection that are themselves symmetries, as defined on printed page 116.

(a) No.

(b) No.

(c) No.

(d) Yes.

(e) The group is isomorphic to $\mathbb{Z}$.

The upper and lower bent strokes are horizontal mirror images displaced by half a repeat. A glide g through that half spacing interchanges the two rows, and g^2 is the full translation. No pure reflection or half-turn matches the direction of the short horizontal stubs. All symmetries are powers of g.

Exercise 12.30: Alternating cup and cap figures

Convention. Here "nontrivial glide" means one not expressible as a product of a translation and reflection that are themselves symmetries, as defined on printed page 116.

(a) Yes.

(b) No.

(c) Yes.

(d) Yes.

(e) The group is isomorphic to D_∞.

Vertical lines through cup or cap centers are mirrors. A half-period glide g in the horizontal baseline interchanges cups and caps; its reflection component alone is not a symmetry. Together with a vertical reflection s, it gives $sgs = g^{-1}$, so the group is infinite dihedral. Their product also yields half-turns.

Exercise 12.31: The square grid

(a) Rotations through $90°$ and $180°$ occur, about square centers or appropriate grid vertices; no smaller positive angle is possible because the two perpendicular edge directions must be preserved as a set. **(b)** Yes: horizontal, vertical, and diagonal mirror lines preserve the grid. **(c)** No nontrivial glides in the book's sense. A symmetry has the form $X \mapsto AX + v$ with v a lattice vector and $A \in D_4$ a symmetry fixing a grid vertex. If it is a glide, A is a reflection, so it is already a product of a translation and a reflection in the symmetry group. Ordinary glides do exist; the word "nontrivial" is essential here.

Exercise 12.32: The staggered square or brick pattern

(a) A half-turn is a symmetry, but a quarter-turn is not: vertical grid lines continue indefinitely, whereas horizontal edges terminate at the staggered joins. Thus the only angle in $(0, 180°]$ is $180°$. **(b)** Yes. Take unit square tiles in column k shifted vertically by $k/2$. There are horizontal mirrors at integer or half-integer heights and vertical mirrors through column centers. **(c)** No nontrivial glides under the book's product definition. The group is generated by its centered rectangular translation lattice and the two perpendicular reflections through a column center on a horizontal mirror. Every glide has one of those reflection linear parts and differs from such a reflection by a translation already in the group. The translation/reflection factors need not be the canonical parallel-axis factors; this is why specifying the book's definition matters.

Exercise 12.33: Square tiles marked L

(a) No nonidentity rotation preserves the identically upright L markings. **(b)** No mirror preserves these markings. **(c)** No nontrivial glide occurs: an orientation reversal would turn each L into a reflected letter, which no translation restores. Thus only the square-lattice translations remain. As usual for a letter-decorated pattern, the intended L is the ordinary nonsymmetric letterform in the source, not a specially designed equal-arm right-angle symbol with an extra diagonal mirror.

Exercise 12.34: Square tiles marked E

(a) No nonidentity rotation preserves upright E. **(b)** Yes: reflection in a horizontal line through a row of letter centers (and suitable midway lines) preserves the pattern. Vertical or diagonal reflections do not. **(c)** No nontrivial glide in the source's sense: every glide has horizontal reflection linear part, and composing it with a horizontal mirror symmetry yields a translation symmetry. Hence it is a product of existing translation and reflection symmetries.

Exercise 12.35: Square tiles marked H

(a) The angle $180°$ occurs, but $90°$ turns each H sideways and is not allowed. **(b)** Yes: horizontal and vertical mirror lines through suitable rows and columns preserve H. **(c)** No nontrivial glide: both possible reflection directions already occur as mirrors. Composing any glide with a parallel mirror symmetry gives a translation symmetry, expressing the glide as a product of symmetries of these two types.

Exercise 12.36: The regular hexagonal tiling

(a) The possible angles in $(0, 180°]$ are $60°, 120°, 180°$. Rotations about hexagon centers by multiples of $60°$ preserve the tiling, and a symmetry must permute its three unoriented edge directions, excluding other angles. **(b)** Yes: lines through opposite vertices or opposite side midpoints of a hexagon are mirrors of the entire tiling. **(c)** No nontrivial glide in the book's sense. Taking a hexagon center as origin, each symmetry is a lattice translation followed by an element of the hexagon's dihedral point group; when reversing orientation, the latter is an existing reflection. Thus any glide already has the required factorization.

Exercise 12.37: Hexagons marked with upright triangles

(a) The angle $120°$ occurs; $60°$ and $180°$ turn the centered upward triangle to a downward one, so they fail. **(b)** Yes: the three reflection axes of each upward equilateral triangle also preserve the hexagonal tiling. **(c)** No nontrivial glides. The remaining point group is D_3, and every orientation-reversing linear part is one of those existing mirrors; composing a glide with that mirror leaves a translation in the same symmetry group. This gives precisely the product excluded by the word "nontrivial."

Exercise 12.38: Horsemen tessellation

Ignore color shading as the instructions require, but preserve the outlines and the internal horse-and-rider markings. **(a)** Half-turns ($180°$) occur; quarter-turns and smaller rotations do not, since

a horizontal horse would be turned into an absent vertical orientation. **(b)** There are no reflection axes: a mirror reverses the handed rider/horse outline and its offset adjoining shapes. **(c)** There are nontrivial glides in both horizontal and vertical directions. Reflecting and shifting by half the appropriate period matches adjacent oppositely oriented rows or columns; the reflection alone does not preserve the pattern. **(d)** Translation generators may be chosen horizontal and vertical, with directions $(1, 0)$ and $(0, 1)$. Choose the lengths as the smallest full repeats of a marked upright horse. The half-repeat displacements attached to the glides are not pure translations. These observations describe the usual *pgg* pattern.

Exercise 12.39: Imaginary human figures tessellation

(a) Rotations through $90°$ and $180°$ occur. Quarter-turn centers are the junctions where four corresponding feet form a pinwheel; a quarter-turn carries one figure orientation to the next. **(b)** Yes: horizontal and vertical lines along the bilateral body axes are mirrors. There are no diagonal mirrors. Candidate axes must preserve the repeating junction pattern. In one full repeat, diagonal candidates through a four-foot junction reverse its chiral pinwheel, while shifted diagonal candidates between junction rows fail to match the neighboring head and body orientations. Horizontal and vertical mirrors can exchange different junctions, so this exclusion does not assert that every pinwheel has the same handedness. **(c)** Yes, there are nontrivial diagonal glide reflections. To see why, compose a quarter-turn symmetry with a horizontal mirror symmetry. The result reverses orientation and has the linear reflection part for a diagonal axis. Since no diagonal reflection is a symmetry, this composition must be a genuine glide reflection. It cannot be a product of a translation symmetry and a reflection symmetry: a translation does not change the linear part, so such a factorization would require an absent diagonal mirror. This distinguishes the pattern $(p4g)$ from an ordinary square grid. **(d)** Translation generators have diagonal directions $(-1, 1)$ and $(1, 1)$, with lengths taken to the first repeat of a figure in the same orientation. Following these diagonal chains, rather than moving to a differently rotated neighboring figure, identifies the translation periods. Color differences are disregarded. The printed odd-answer entry for part (c) says "No"; the diagonal glides just constructed give "Yes" under the section's own definition.

Exercise 12.40: Reptile tessellation

(a) Ignoring shading, rotations through $60°, 120°, 180°$ occur. The six reptile orientations cycle around the sixfold junctions; rotations through $120°$ also occur at the threefold junctions. **(b)** No: a reflection reverses the handed arrangement of the curled bodies, feet, and tails, and that reversed arrangement is absent. **(c)** No glide reflections occur, since a glide also reverses orientation and cannot restore the handed reptile markings. **(d)** Choose two primitive translation directions along rows of identically oriented reptiles in the triangular lattice, for example $(1, 0)$ and $(1, \sqrt{3})$, with suitable lengths to reach the next full repeat. These directions have angle $60°$; the directions, rather than a scale read from the printed image, are requested. The uncolored pattern has point group C_6 (wallpaper type $p6$).

Exercise 12.41: Human figures tessellation

(a) The possible nonzero angle at most $180°$ is $120°$, rotating the three figure orientations around their threefold junctions. A $60°$ turn would require an orientation not present. **(b)** Yes: the bilateral human figures align along mirror axes in the pattern. **(c)** No nontrivial glides in the book's sense; an existing reflection with the same linear part removes the reversing component and leaves a translation

symmetry. **(d)** Two translation-generator directions are $(0, 1)$ and $(1, \sqrt{3})$; choose their lengths to carry a figure to the next identically oriented one. These are directions only, as requested, so the lengths need not be inferred from the reproduction. Shading distinctions are not retained.

Exercise 12.42: Cube rotations and S_4

Plan. Let a cube rotation act on the four long (unoriented) diagonals through its center. This produces a homomorphism Φ into S_4, because performing two rotations in succession composes their diagonal permutations. We show its kernel is trivial. If a rotation preserves all four diagonal lines, choose unit vectors $v_1, \ldots, v_4$ along them directed to the vertices of one inscribed regular tetrahedron; then $v_i \cdot v_j = -1/3$ for $i \neq j$ and $v_1 + \cdots + v_4 = 0$. The rotation sends v_i to $\varepsilon_i v_i$. Preservation of the nonzero pairwise dot products gives $\varepsilon_i \varepsilon_j = 1$ for every pair, so all signs agree. If all are $+1$, three independent diagonals are fixed and the rotation is the identity. If all are -1, the linear map is $-I_3$, which has determinant -1 and is not a rotation. Thus Φ is injective. There are 24 cube rotations: choose a bottom face in six ways and a front face in four ways. Since $|S_4| = 24$, the injective map is also surjective and gives an isomorphism. It is important to use lines rather than just one endpoint of each diagonal, and to rule out central inversion as above.

13 Homomorphisms

Conventions. Permutations compose from right to left. The identity is e. A group generated by an element a is denoted $\langle a \rangle$.

Exercise 13.1: Inclusion of the integers in the reals

Yes. Both groups use addition, so the required identity is $\phi(m+n) = \phi(m) + \phi(n)$. The left side is $m+n$ and the right side is also $m+n$. Thus the inclusion preserves the operation and is a homomorphism.

Exercise 13.2: The greatest-integer map

No. Choose $x = y = 3/5$. Then $\phi(x) = \phi(y) = 0$, but $\phi(x+y) = \lfloor 6/5 \rfloor = 1$. Hence $\phi(x+y) \neq \phi(x) + \phi(y)$, so the additive homomorphism law fails.

Exercise 13.3: Absolute value under multiplication

Yes. Absolute value sends nonzero reals to nonzero reals, so the map has the stated codomain. For every $x, y \in \mathbb{R}^*$,

$$\phi(xy) = |xy| = |x||y| = \phi(x)\phi(y).$$

The kernel is $\{1, -1\}$, showing that a homomorphism need not be injective.

Exercise 13.4: Reduction from $\mathbb{Z}_6$ to $\mathbb{Z}_2$

Yes. Reduction of an integer modulo 2 is unchanged if a multiple of 6 is added, because 6 is even. Thus reduction descends to $\mathbb{Z}_6$, and reducing the sum of two representatives modulo 2 gives the sum of their reductions. Formally $\phi([a+b]_6) = [a+b]_2 = [a]_2 + [b]_2$, proving the homomorphism law.

Exercise 13.5: Remainder parity on $\mathbb{Z}_9$

No. Using the canonical representatives, take 8 and 1. In $\mathbb{Z}_9$ their sum is 0, so $\phi(8+1) = \phi(0) = 0$. But $\phi(8) + \phi(1) = 0 + 1 = 1$ in $\mathbb{Z}_2$. Subtracting the modulus 9 changes parity, causing the homomorphism law to fail.

Exercise 13.6: Exponentiation from addition to multiplication

Yes. The values 2^x are positive and hence nonzero reals. The domain operation is addition and the target operation is multiplication, so the correct identity is

$$\phi(x+y) = 2^{x+y} = 2^x 2^y = \phi(x)\phi(y).$$

The change in operations is part of the homomorphism definition and must be kept explicit.

Exercise 13.7: A coordinate inclusion into a direct product

Yes. The map puts g_i in its own coordinate and identity elements in all others. Multiplying $\phi_i(a)$ and $\phi_i(b)$ multiplies only the ith coordinates nontrivially, giving ab there. Consequently $\phi_i(a)\phi_i(b) = \phi_i(ab)$. It is also injective because equality of the product vectors forces equality of their ith coordinates.

Exercise 13.8: The inverse map

For an arbitrary group, the answer depends on commutativity. One has

$$\phi(ab) = (ab)^{-1} = b^{-1}a^{-1}, \qquad \phi(a)\phi(b) = a^{-1}b^{-1}.$$

Thus it is a homomorphism if the group is abelian. Conversely, if it is a homomorphism, these expressions are equal for every a, b; taking inverses gives $ab = ba$, so the group must be abelian. For example it fails in S_3 using two noncommuting transpositions. The exact answer is: if and only if G is abelian.

Exercise 13.9: Taking a second derivative

Yes. The second derivative of a function having derivatives of all orders again has derivatives of all orders, so the map stays in F. Differentiation is additive, hence

$$\phi(f + g) = (f + g)'' = f'' + g'' = \phi(f) + \phi(g).$$

This is the homomorphism law for the additive function groups.

Exercise 13.10: Integration as an additive map

Yes. A continuous function has a finite Riemann integral on $[0, 4]$. Linearity of the integral gives

$$\phi(f + g) = \int_0^4 (f(x) + g(x))\,dx = \int_0^4 f(x)\,dx + \int_0^4 g(x)\,dx = \phi(f) + \phi(g).$$

Thus the map preserves the additive operations.

Exercise 13.11: Multiplying a function by three

Yes. For pointwise addition,

$$\phi(f + g) = 3(f + g) = 3f + 3g = \phi(f) + \phi(g).$$

The equality holds at every real input, so it is an equality of functions. There is no requirement here that the functions be continuous or differentiable.

Exercise 13.12: Determinant on an additive matrix group

For $n \geq 2$, no. The domain operation is matrix addition, so a homomorphism would require $\det(A + B) = \det A + \det B$. Taking $A = B = I_n$ gives $\det(2I_n) = 2^n$, while $\det I_n + \det I_n = 2$; these

are unequal for $n \geq 2$. The familiar multiplicative determinant identity concerns matrix multiplication and does not apply to this domain operation. For the exceptional case $n = 1$, the determinant is just the single entry, so the map is additive and is a homomorphism. Thus a blanket negative answer implicitly assumes $n > 1$.

Exercise 13.13: Trace on an additive matrix group

Yes. The diagonal entries of $A + B$ are $a_{ii} + b_{ii}$, so

$$\operatorname{tr}(A + B) = \sum_i (a_{ii} + b_{ii}) = \sum_i a_{ii} + \sum_i b_{ii} = \operatorname{tr}(A) + \operatorname{tr}(B).$$

This proves the additive homomorphism law for every positive matrix size.

Exercise 13.14: Trace on the multiplicative general linear group

No. A homomorphism from a multiplicative group to additive $\mathbb{R}$ must send the multiplicative identity to 0. But $\operatorname{tr}(I_n) = n \neq 0$ for $n \geq 1$. Equivalently, $I_n I_n = I_n$ would require $n = n + n$, which is false. The same formula that works on the additive matrix group fails for this different operation.

Exercise 13.15: Integration on a multiplicative function group

The map is well defined: a continuous nonvanishing function on $\mathbb{R}$ has constant sign, so its integral on $[0, 1]$ is nonzero. But it is not multiplicative. Take

$$f(x) = g(x) = 2 + \sin(2\pi x),$$

which is everywhere positive. Then $\phi(f) = \phi(g) = 2$, while

$$\phi(fg) = \int_0^1 \left(4 + 4\sin(2\pi x) + \sin^2(2\pi x)\right) dx = 4 + 0 + \tfrac{1}{2} = \tfrac{9}{2}.$$

This differs from $\phi(f)\phi(g) = 4$, so the homomorphism law fails.

Exercise 13.16: Kernel of the parity homomorphism

The map of Example 13.3 sends even permutations to $0 \in \mathbb{Z}_2$ and odd permutations to 1. The target identity is 0, so its inverse image is exactly the even permutations:

$$\operatorname{Ker}\phi = A_3 = \{e, (1,2,3), (1,3,2)\} = \{\rho_0, \rho_1, \rho_2\}.$$

Exercise 13.17: An integer homomorphism into $\mathbb{Z}_7$

Since 1 generates $\mathbb{Z}$ and $\phi(1) = 4$, additivity forces $\phi(n) = 4n \pmod 7$ for every integer n. Therefore $\phi(25) = 100 \equiv 2 \pmod 7$. The kernel consists of integers with $7 \mid 4n$. Since $\gcd(4, 7) = 1$, this is equivalent to $7 \mid n$, so $\operatorname{Ker}\phi = 7\mathbb{Z}$.

Exercise 13.18: An integer homomorphism into $\mathbb{Z}_{10}$

The image of n is $6n \pmod{10}$. Thus $\phi(18) = 108 \equiv 8 \pmod{10}$. Also $10 \mid 6n$ if and only if $5 \mid 3n$, which is equivalent to $5 \mid n$. Consequently $\operatorname{Ker}\phi = 5\mathbb{Z}$.

Exercise 13.19: An integer homomorphism into S_8

Let $g = (1,4,2,6)(2,5,7)$. Its factors overlap, so first compute their composition:

$$1 \mapsto 4 \mapsto 2 \mapsto 5 \mapsto 7 \mapsto 6 \mapsto 1, \qquad g = (1,4,2,5,7,6).$$

It has order 6, and $\phi(n) = g^n$, so $\operatorname{Ker}\phi = 6\mathbb{Z}$. Since $20 \equiv 2 \pmod{6}$,

$$\phi(20) = g^2 = (1,2,7)(4,5,6),$$

with the remaining symbols fixed.

Exercise 13.20: A homomorphism from $\mathbb{Z}_{10}$ to $\mathbb{Z}_{20}$

The formula is $\phi(n) = 8n \pmod{20}$, and it is consistent with the domain modulus because $8 \cdot 10 = 80 \equiv 0 \pmod{20}$. The kernel condition $20 \mid 8n$ reduces to $5 \mid n$, so in $\mathbb{Z}_{10}$ the kernel is $\{0,5\}$. Finally $\phi(3) = 24 \equiv 4 \pmod{20}$.

Exercise 13.21: A cyclic homomorphism into S_8

Put $g = (2,5)(1,4,6,7)$. The disjoint cycles have lengths $2, 4$, so g has order 4. Hence $\phi(n) = g^n$ is well defined on $\mathbb{Z}_{24}$, since $4 \mid 24$. Its kernel consists of multiples of 4:

$$\operatorname{Ker}\phi = \{0,4,8,12,16,20\}.$$

Because $14 \equiv 2 \pmod{4}$, squaring the cycles gives

$$\phi(14) = (1,6)(4,7).$$

Exercise 13.22: A map from two integer coordinates to one

By additivity,

$$\phi(m,n) = m\phi(1,0) + n\phi(0,1) = 3m - 5n.$$

Thus $\phi(-3,2) = -9 - 10 = -19$. For the kernel, solve $3m = 5n$. Coprimality forces $m = 5k$ and then $n = 3k$, with $k \in \mathbb{Z}$. Therefore

$$\operatorname{Ker}\phi = \{(5k,3k) : k \in \mathbb{Z}\} = \langle (5,3) \rangle.$$

Exercise 13.23: A map between integer lattices

The generator images give

$$\phi(m,n) = m(2,-3) + n(-1,5) = (2m - n, -3m + 5n).$$

Therefore $\phi(4,6) = (8-6, -12+30) = (2,18)$. If the image is $(0,0)$, the first coordinate gives $n = 2m$, and the second then gives $-3m + 10m = 7m = 0$. Hence $m = n = 0$, so the kernel is $\{(0,0)\}$.

Exercise 13.24: A two-generator homomorphism to S_{10}

Write $a = (3,5)(2,4)$ and $b = (1,7)(6,10,8,9)$. Their supports are disjoint, so they commute, making the formula $\phi(m,n) = a^m b^n$ consistent with the abelian domain. The orders are 2 and 4. If $a^m b^n = e$, restriction to the support of a gives $2 \mid m$, and restriction to that of b gives $4 \mid n$. Conversely those divisibilities clearly suffice. Thus

$$\operatorname{Ker}\phi = 2\mathbb{Z} \times 4\mathbb{Z}.$$

For the specified point, $a^3 = a$ and $b^{10} = b^2 = (6,8)(10,9)$, so

$$\phi(3,10) = (3,5)(2,4)(6,8)(9,10).$$

Exercise 13.25: Homomorphisms of $\mathbb{Z}$ onto itself

Every homomorphism has the form $\phi(n) = kn$, where $k = \phi(1) \in \mathbb{Z}$. Its image is $k\mathbb{Z}$. This is all of $\mathbb{Z}$ precisely when $k = 1$ or $k = -1$. Thus there are exactly two surjective homomorphisms, $n \mapsto n$ and $n \mapsto -n$.

Exercise 13.26: Homomorphisms of $\mathbb{Z}$ into itself

For each integer k, the formula $\phi_k(n) = kn$ is additive, so is a homomorphism. Conversely additivity and the generator 1 force this formula for every homomorphism, including negative inputs by $\phi(-n) = -\phi(n)$. Therefore the homomorphisms are parametrized by $k \in \mathbb{Z}$: countably infinitely many.

Exercise 13.27: Homomorphisms of $\mathbb{Z}$ into $\mathbb{Z}_2$

The image of 1 is either 0 or 1, and that choice determines the map on every integer. Both choices work: $n \mapsto 0$ and $n \mapsto n \bmod 2$ preserve addition. Thus there are exactly two homomorphisms.

Exercise 13.28: When is left multiplication a homomorphism?

A homomorphism must send identity to identity. Here $\phi_g(e) = ge = g$, so this requires $g = e$. Conversely, when $g = e$, ϕ_g is the identity map and is a homomorphism. Thus precisely the identity element qualifies. Being a permutation of G is much weaker than being a group homomorphism.

Exercise 13.29: When is conjugation a homomorphism?

For every $g \in G$. Indeed

$$\phi_g(xy) = gxyg^{-1} = (gxg^{-1})(gyg^{-1}) = \phi_g(x)\phi_g(y),$$

where the middle $g^{-1}g$ cancels. Its inverse map is conjugation by g^{-1}, so it is even an automorphism for every g.

Exercise 13.30: Definition of a group homomorphism

A complete definition specifies the algebraic structures and the quantifier: a homomorphism from a group G to a group G' is a function $\phi : G \to G'$ satisfying $\phi(xy) = \phi(x)\phi(y)$ for all $x, y \in G$. The first product is in G and the second is in G'. Merely saying "a map" without domains, operations, and a universal condition leaves the original statement incomplete.

Exercise 13.31: Definition of the kernel

The definition is correct. For a group homomorphism $\phi : G \to G'$ with target identity e', the kernel is the inverse image of that identity:

$$\operatorname{Ker} \phi = \{x \in G : \phi(x) = e'\}.$$

It always contains the identity of G and is a normal subgroup; the identity used in the test is the target identity, which need not have the same notation or value as the domain identity.

Exercise 13.32: True or false

(a) True: A_n is the kernel of the sign homomorphism (with the trivial small cases included).

(b) True: the constant map $g \mapsto e'$ is always a homomorphism.

(c) False: a nontrivial group mapping constantly to the target identity has many points in its kernel.

(d) True: $\phi(x) = \phi(y)$ is equivalent to $xy^{-1} \in \operatorname{Ker} \phi$, giving injectivity exactly for a trivial kernel.

(e) False: an image of a six-element group has order dividing 6, so cannot have order 4.

(f) False: a function on six points has at most six image points, so cannot have an image of size 12.

(g) True: $\mathbb{Z}_6 \to \mathbb{Z}_{12}$, $[k]_6 \mapsto [2k]_{12}$, is a homomorphism, even injective.

(h) True: at least the trivial homomorphism exists; a nontrivial example is $\mathbb{Z}_6 \to \mathbb{Z}_{10}$, $[k] \mapsto [5k]$.

(i) False: the domain identity always maps to the target identity, so the kernel is never empty.

(j) False: $\mathbb{Z}_2 \to \mathbb{R}^*$, $[k] \mapsto (-1)^k$, is a nontrivial homomorphism into an infinite group.

Exercise 13.33: A nontrivial map from $\mathbb{Z}_{12}$ to $\mathbb{Z}_5$?

No such map exists. Its image order must divide both the domain order 12 and the target order 5. Their gcd is 1, so the image has order 1 and the map is trivial. Equivalently, if $a = \phi(1)$, then $12a = 0$ in $\mathbb{Z}_5$ forces $a = 0$.

Exercise 13.34: A nontrivial map from $\mathbb{Z}_{12}$ to $\mathbb{Z}_4$

Use reduction modulo 4: $\phi([k]_{12}) = [k]_4$. It is well defined because changing k by 12 does not change its residue modulo 4, and it preserves addition. It is nontrivial because $\phi(1) = 1 \neq 0$, and is in fact surjective.

Exercise 13.35: A nontrivial map between two direct products

For the source $\mathbb{Z}_2 \times \mathbb{Z}_4$ and target $\mathbb{Z}_2 \times \mathbb{Z}_5$, define $\phi(a, b) = (a, 0)$. Addition is coordinatewise, so

$$\phi((a, b) + (c, d)) = (a + c, 0) = (a, 0) + (c, 0).$$

Thus it is a homomorphism. The value $\phi(1, 0) = (1, 0)$ is nonidentity, so it is nontrivial.

Exercise 13.36: A nontrivial map from $\mathbb{Z}_3$ to $\mathbb{Z}$?

Impossible. If $k = \phi(1)$, then $3k = \phi(3 \cdot 1) = \phi(0) = 0$ in the integers. This forces $k = 0$, so every element maps to zero. The obstruction is that a finite-order element cannot have a nonzero image in the torsion-free group $\mathbb{Z}$.

Exercise 13.37: A nontrivial map from $\mathbb{Z}_3$ to S_3

Let $r = (1, 2, 3)$ and define $\phi([k]_3) = r^k$. Since $r^3 = e$, this is independent of the representative. The exponent law gives $\phi(a + b) = r^{a+b} = r^a r^b = \phi(a)\phi(b)$. Its image is A_3, and $\phi(1) = r \neq e$, so it is nontrivial.

Exercise 13.38: A nontrivial map from $\mathbb{Z}$ to S_3

Take $\phi(k) = (1, 2)^k$. Integer powers satisfy $\phi(k + \ell) = \phi(k)\phi(\ell)$, including negative exponents. The map is nontrivial because $\phi(1) = (1, 2) \neq e$. Its image is the order-2 subgroup $\{e, (1, 2)\}$.

Exercise 13.39: A nontrivial map from $\mathbb{Z}^2$ to $2\mathbb{Z}$

Define $\phi(m, n) = 2m$. Its values lie in the even integers, and

$$\phi((m, n) + (r, s)) = 2(m + r) = 2m + 2r = \phi(m, n) + \phi(r, s).$$

It is nontrivial since $\phi(1, 0) = 2$, and is even onto $2\mathbb{Z}$.

Exercise 13.40: A nontrivial map from $2\mathbb{Z}$ to $\mathbb{Z}^2$

For an even integer $2k$, define $\phi(2k) = (k, 0)$. The integer k is unique, so the map is well defined. Then $\phi(2k + 2\ell) = (k + \ell, 0) = (k, 0) + (\ell, 0)$. It is nontrivial because $\phi(2) = (1, 0)$.

Exercise 13.41: A nontrivial map from D_4 to S_3

View D_4 as permutations of its four vertices. Define $\phi(g) = e$ when g is even and $\phi(g) = (1, 2)$ when g is odd. Parity adds modulo 2, and $(1, 2)^2 = e$, so this is a homomorphism into S_3. A quarter-turn is a 4-cycle and is odd, so it has nonidentity image, proving the map is nontrivial.

Exercise 13.42: A nontrivial map from S_3 to S_4

Extend a permutation of $\{1, 2, 3\}$ by fixing 4. The extension of a composite equals the composite of the extensions, because this is true on $1, 2, 3$ and all maps fix 4. Thus the extension is an injective homomorphism and is nontrivial.

Exercise 13.43: A nontrivial map from S_4 to S_3

Send each even permutation to $e \in S_3$ and each odd permutation to $(1, 2)$. This is the sign homomorphism with its two values relabeled by the order-2 subgroup $\{e, (1, 2)\}$ of S_3. Products preserve parity, so it preserves the group operation. It is nontrivial because a transposition has image $(1, 2)$. Surjectivity onto S_3 is not required by the exercise.

Exercise 13.44: Image order divides a finite domain order

Let $K = \operatorname{Ker}\phi$. For an image element $\phi(a)$, its full fiber is aK: indeed $\phi(x) = \phi(a)$ if and only if $\phi(a^{-1}x) = e'$, or $a^{-1}x \in K$. Each fiber therefore has $|K|$ elements by $k \mapsto ak$, and the fibers partition finite G. Thus

$$|G| = |\phi[G]|\,|K|.$$

The image is finite and its order divides $|G|$. This proof obtains the count directly without presupposing the isomorphism theorem of the next section.

Exercise 13.45: Image order divides a finite target order

The image $\phi[G]$ is a subgroup of G': it contains $e' = \phi(e)$, products satisfy $\phi(a)\phi(b) = \phi(ab)$, and inverses satisfy $\phi(a)^{-1} = \phi(a^{-1})$. Since G' is finite, its subgroup is finite and Lagrange gives $|\phi[G]| \mid |G'|$. The domain need not be finite for this argument.

Exercise 13.46: Agreement on generators determines a homomorphism

Every element of the group generated by the a_i is a finite word $a_{i_1}^{\varepsilon_1} \cdots a_{i_k}^{\varepsilon_k}$ with $\varepsilon_j = \pm 1$. A homomorphism carries this word to the product of the corresponding image powers, because it preserves products and inverses. Since $\phi(a_i) = \mu(a_i)$ for every generator, the two maps give equal products on every such word. They also agree at the identity. Thus they agree on every element of G, meaning $\phi = \mu$. No assumption that the generators are independent is needed.

Exercise 13.47: A homomorphism with prime-order domain

If $|G| = p$ is prime, its kernel is a subgroup and must have order 1 or p by Lagrange. In the first case the kernel is trivial, so the homomorphism is injective: equal images imply the quotient of the inputs lies in that kernel. In the second case the kernel is all of G, so every input maps to the target identity and the homomorphism is trivial. These are the only possibilities.

Exercise 13.48: Kernel of the sign map

By definition $\operatorname{sgn}(\sigma) = +1$ exactly for even permutations. The identity of the multiplicative target $\{1, -1\}$ is $+1$, so

$$\operatorname{Ker}(\operatorname{sgn}_n) = A_n.$$

For $n \geq 2$, a transposition has sign -1, so the map is onto both target values. For $n = 1$ it is still a homomorphism with kernel $S_1 = A_1$, but it is not onto the two-element target. This map is the general version of the parity map $S_3 \to \mathbb{Z}_2$.

Exercise 13.49: Composition of homomorphisms

For $a, b \in G$, apply first the homomorphism law for ϕ, then the one for ψ:

$$(\psi\phi)(ab) = \psi(\phi(ab)) = \psi(\phi(a)\phi(b)) = \psi(\phi(a))\psi(\phi(b)).$$

The result is $(\psi\phi)(a)(\psi\phi)(b)$, which is the required law for the composite $G \to G''$.

Exercise 13.50: An abelian image and commutators

For any $x, y \in G$,

$$\phi(xyx^{-1}y^{-1}) = \phi(x)\phi(y)\phi(x)^{-1}\phi(y)^{-1}.$$

If the image is abelian, the right side cancels to the identity, so every displayed commutator lies in the kernel. Conversely, if it is always in the kernel, the right side equals the identity. Multiplying that equality on the right by $\phi(y)\phi(x)$ gives $\phi(x)\phi(y) = \phi(y)\phi(x)$. Every image element is some $\phi(x)$, so the image is abelian.

Exercise 13.51: The universal power map from the integers

For all integers m, n, $a^{m+n} = a^m a^n$, so $\phi(n) = a^n$ is a homomorphism. Its image is exactly $\langle a \rangle$. If a has infinite order, only the exponent 0 gives e, so the kernel is $\{0\}$. If a has finite order r, then $a^n = e$ exactly when $r \mid n$, so the kernel is $r\mathbb{Z}$. This includes $r = 1$, when $a = e$ and the kernel is all of $\mathbb{Z}$.

Exercise 13.52: A fiber is a kernel coset

Let $H = \operatorname{Ker}\phi$. For $x \in G$,

$$\phi(x) = \phi(a) \iff \phi(xa^{-1}) = e' \iff xa^{-1} \in H \iff x \in Ha.$$

The first equivalence uses preservation of inverses and cancellation in the target; the last is the definition of a right coset. Thus the two described sets are equal. Since a kernel is normal, this fiber is also aH.

Exercise 13.53: A two-generator power formula

The necessary and sufficient condition is $hk = kh$. If $\phi(m, n) = h^m k^n$ is a homomorphism, the commuting domain elements $(1, 0)$ and $(0, 1)$ must have commuting images. Their images are h, k,

proving necessity. Conversely suppose h, k commute. Their integer powers then commute, including negative powers, and for all integers m, n, r, s,

$$\phi(m,n)\phi(r,s) = h^m k^n h^r k^s = h^{m+r} k^{n+s} = \phi(m+r, n+s).$$

This is the additive-domain homomorphism law, proving sufficiency.

Exercise 13.54: When all two-generator power formulas work

The condition is that G be abelian. If every pair h, k gives a homomorphism in Exercise 53, its necessary condition says every pair commutes, so G is abelian. Conversely, if G is abelian, every pair commutes and the sufficient calculation in that exercise proves every such formula is a homomorphism.

Exercise 13.55: Power maps from a finite cyclic group

The exact condition is

$$\boxed{h^n = e,}$$

equivalently, h has finite order dividing n. Necessity follows because $n[1] = [0]$ in $\mathbb{Z}_n$, so a homomorphism must satisfy $h^n = \phi(n[1]) = \phi([0]) = e$. For sufficiency, if integers i, j differ by a multiple of n, then $h^i = h^j$ because $h^n = e$. Thus $\phi([i]) = h^i$ is well defined on residue classes, and

$$\phi([i] + [j]) = h^{i+j} = h^i h^j = \phi([i])\phi([j]).$$

The source writes $0 \leq i \leq n$, duplicating $[0] = [n]$; the same condition resolves that endpoint consistency. It requires $h^n = e$, not $h = e$: for $n = 2$ and $G = \mathbb{Z}_2$, sending 1 to the nonzero element gives the identity homomorphism and has $h \neq e$.

14 Factor Groups

Conventions. Permutations compose from right to left. The identity is e. A group generated by an element a is denoted $\langle a \rangle$.

Exercise 14.1: Order of $\mathbb{Z}_6/\langle 3 \rangle$

The subgroup is $\langle 3 \rangle = \{0, 3\}$, of order 2. The quotient order is the number of cosets, $6/2 = 3$. Explicitly the cosets are $\{0, 3\}$, $\{1, 4\}$, $\{2, 5\}$.

Exercise 14.2: Quotient by a product subgroup

In $\mathbb{Z}_4$, $\langle 2 \rangle$ has order 2; in $\mathbb{Z}_{12}$, $\langle 2 \rangle$ has order 6. Their product therefore has order $2 \cdot 6 = 12$. The ambient product has order $4 \cdot 12 = 48$, so the quotient order is $48/12 = 4$. All groups here are abelian, so the subgroup is normal and the quotient is defined.

Exercise 14.3: Quotient by $\langle (2, 1) \rangle$

In $\mathbb{Z}_4 \times \mathbb{Z}_2$, the coordinates 2 and 1 each have order 2, so $(2, 1)$ has order 2 and generates $\{(0, 0), (2, 1)\}$. The ambient order is $4 \cdot 2 = 8$. Thus the quotient has order $8/2 = 4$.

Exercise 14.4: Quotient by the second coordinate factor

The subgroup $\{0\} \times \mathbb{Z}_5$ has five elements in the fifteen-element group $\mathbb{Z}_3 \times \mathbb{Z}_5$, so the quotient has order 3. Equivalently the projection $(a, b) \mapsto a$ has this subgroup as kernel and image $\mathbb{Z}_3$.

Exercise 14.5: Quotient of $\mathbb{Z}_2 times \mathbb{Z}_4$ by a diagonal cycle

The element $(1, 1)$ has order $\operatorname{lcm}(2, 4) = 4$, so its generated subgroup has four elements. The product has 8 elements, giving a quotient of order $8/4 = 2$.

Exercise 14.6: Quotient of $\mathbb{Z}_{12} times \mathbb{Z}_{18}$

The order of $4 \in \mathbb{Z}_{12}$ is $12/\gcd(12, 4) = 3$, and that of $3 \in \mathbb{Z}_{18}$ is $18/\gcd(18, 3) = 6$. Thus $(4, 3)$ has order $\operatorname{lcm}(3, 6) = 6$. Dividing the ambient order $12 \cdot 18 = 216$ by this subgroup order gives quotient order $216/6 = 36$.

Exercise 14.7: A quotient of $\mathbb{Z}_2 times S_3$

The element $(1, \rho_1)$ has order $\operatorname{lcm}(2, 3) = 6$. Its subgroup equals $\mathbb{Z}_2 \times A_3$: both have six elements, and all its powers lie there. This is normal because $A_3 \triangleleft S_3$ and the first coordinate is unrestricted. The ambient order is $2 \cdot 6 = 12$, so the quotient has order $12/6 = 2$. Normality matters here because the ambient group is not abelian.

Exercise 14.8: A diagonal subgroup of a coprime product

The element $(1,1)$ in $\mathbb{Z}_{11} \times \mathbb{Z}_{15}$ has order $\operatorname{lcm}(11,15) = 165$, equal to the ambient order. It therefore generates the whole group, so the quotient has just one element. Its order is 1.

Exercise 14.9: Order of $5 + \langle 4 \rangle$ in a cyclic quotient

The subgroup of $\mathbb{Z}_{12}$ is $H = \{0, 4, 8\}$. The order of $5 + H$ is the least positive k with $5k \in H$. Testing multiples modulo 12 gives $5, 10, 3, 8$ at $k = 1, 2, 3, 4$, respectively; the first three are outside H and 8 is inside. Thus the order is 4. Equivalently reduce to $5 \equiv 1$ modulo 4.

Exercise 14.10: Order of $26 + \langle 12 \rangle$

The quotient $\mathbb{Z}_{60}/\langle 12 \rangle$ is identified with $\mathbb{Z}_{12}$ by reduction modulo 12, because the kernel consists exactly of the multiples of 12 modulo 60. The coset of 26 corresponds to $2 \in \mathbb{Z}_{12}$, whose order is $12/\gcd(12,2) = 6$. Thus the requested order is 6.

Exercise 14.11: Order of a coset in $\mathbb{Z}_3 times \mathbb{Z}_6$

Let $H = \langle (1,1) \rangle$. Its elements are $(t \bmod 3, t \bmod 6)$. A multiple $k(2,1)$ lies in H if and only if an integer t satisfies

$$t \equiv 2k \pmod 3, \qquad t \equiv k \pmod 6.$$

The second congruence implies $t \equiv k \pmod 3$, so consistency is exactly $3 \mid k$. This condition is also sufficient by taking $t = k$. Therefore the least positive such k is 3, giving order 3.

Exercise 14.12: Order of a coset in $\mathbb{Z}_4^2$

The homomorphism $(x, y) \mapsto x - y \pmod 4$ has kernel $\langle (1,1) \rangle$, so the coset of $(3,1)$ corresponds to $3 - 1 = 2 \in \mathbb{Z}_4$. This element has order 2. Directly, $(3,1)$ is not on the diagonal, but $2(3,1) = (2,2)$ is, giving the same least positive return.

Exercise 14.13: Order modulo $\langle (0,2) \rangle$

In $\mathbb{Z}_4 \times \mathbb{Z}_8$, the subgroup consists of $(0,0), (0,2), (0,4), (0,6)$. For $k(3,1)$ to lie in it, the first coordinate must vanish, requiring $4 \mid 3k$ or $4 \mid k$, and the second coordinate must be even, requiring $2 \mid k$. The combined least positive solution is $k = 4$. Hence the coset has order 4.

Exercise 14.14: Order modulo $\langle (1,2) \rangle$

In $\mathbb{Z}_4 \times \mathbb{Z}_8$, use $f(x,y) = y - 2x \pmod 8$. This is well defined: changing x by 4 changes $-2x$ by -8. Its kernel is exactly $\{(t, 2t) : t \in \mathbb{Z}_4\} = \langle (1,2) \rangle$. Thus the coset of $(3,3)$ has the same order as $f(3,3) = 3 - 6 = -3 \equiv 5 \pmod 8$. Since $\gcd(5,8) = 1$, its order is 8. This also verifies that no smaller positive multiple enters the subgroup.

Exercise 14.15: A coset already equal to the identity

In $\mathbb{Z}_6 \times \mathbb{Z}_8$ one has $2(4,4) = (8,8) = (2,0)$. Hence $(2,0)$ already belongs to $H = \langle (4,4) \rangle$. The coset $(2,0) + H$ is therefore H, the identity of the quotient, and its order is 1.

Exercise 14.16: Conjugating a reflection subgroup of S_3

The source convention is $i_g(x) = gxg^{-1}$. Here $\rho_1 = (1,2,3)$ and $\mu_1 = (2,3)$. Conjugation relabels a transposition by applying the conjugating permutation to its entries:

$$\rho_1(2,3)\rho_1^{-1} = (\rho_1(2), \rho_1(3)) = (3,1) = \mu_2.$$

The identity remains the identity, so $i_{\rho_1}[H] = \{\rho_0, \mu_2\}$.

Exercise 14.17: Correcting the first normal-subgroup definition

The definition must read: a subgroup $H \leq G$ is normal if $gH = Hg$ for every $g \in G$. The original condition $hG = Gh$ for $h \in H$ says nothing about normality, because for any $h \in G$ both hG and Gh are just the whole group G. The subgroup being translated must be H, and the quantifier must range over all of G.

Exercise 14.18: The conjugation definition of normality

The stated definition is correct, provided H is a subgroup: $g^{-1}hg \in H$ for every $h \in H$ and $g \in G$ characterizes normality. Although this initially states an inclusion $g^{-1}Hg \subseteq H$, replacing g by g^{-1} gives the reverse inclusion after conjugation, so equality follows. Thus it is equivalent to invariance under every inner automorphism.

Exercise 14.19: Definition of an automorphism

An automorphism of G is an *isomorphism* from G onto itself, meaning a bijective homomorphism $G \to G$. A homomorphism merely mapping G into itself is an endomorphism and need not be invertible. For example, the constant map to e is an endomorphism of every group but is not an automorphism unless the group is trivial.

Exercise 14.20: Why normal subgroups matter

Normality is exactly the condition making the rule $(aH)(bH) = (ab)H$ independent of the chosen representatives, so that the cosets form the factor group G/H. Every normal subgroup is the kernel of the canonical homomorphism $g \mapsto gH$, and conversely every kernel is normal. Thus normal subgroups allow us to collapse specified elements to the identity while retaining a valid group operation, and describe the information lost by homomorphisms.

Exercise 14.21: Writing a proof about elements of a quotient

(a) The instructor suspects that the writer is confusing elements of G with elements of G/H, which are cosets. Naming quotient elements a, b is formally permissible, but then one cannot invoke commutativity of G on those objects without choosing representatives. The difficulty is not the letter names themselves; it is failure to track which group they belong to. **(b)** A clear beginning is: "Let $aH, bH \in G/H$, where $a, b \in G$. We must show $(aH)(bH) = (bH)(aH)$." **(c)** Since G is abelian, $ab = ba$. Therefore

$$(aH)(bH) = (ab)H = (ba)H = (bH)(aH).$$

The quotient multiplication is valid because H is normal. This proves G/H is abelian.

Exercise 14.22: A quotient of a torsion group

(a) As in Exercise 21, the likely error is treating a coset as though it were an element of the original group. The sentence "let $x \in G/H$" is not itself false, but the torsion hypothesis is about representatives in G, so the proof must display one. **(b)** Write: "Let $aH \in G/H$ with $a \in G$. Since G is torsion, a has finite order r." **(c)** Then $(aH)^r = a^r H = eH = H$, the quotient identity. Thus aH has finite order dividing r. Since the coset was arbitrary, the quotient is torsion. Its order need not equal the representative's order; it can decrease.

Exercise 14.23: True or false

(a) True: the standard coset product defines a group exactly when $N \triangleleft G$.

(b) True: $ghg^{-1} = h$ in an abelian group.

(c) True: conjugating x by g gives $gxg^{-1} = x$ when the group is abelian.

(d) True: its order is $|G|/|H|$.

(e) True: if $a^r = e$, then $(aH)^r = H$.

(f) False: $\mathbb{Z}$ is torsion free but $\mathbb{Z}/2\mathbb{Z}$ has a nonidentity element of order 2.

(g) True: $(aH)(bH) = abH = baH = (bH)(aH)$.

(h) False: $S_3/A_3 \cong \mathbb{Z}_2$ is abelian, although S_3 is not.

(i) True for positive integer n: the class of 1 generates all n residue classes.

(j) False in general: for nonzero integer n, $n\mathbb{R} = \mathbb{R}$, since $x = n(x/n)$, so the quotient is trivial, not of order n unless $n = 1$.

Exercise 14.24: The alternating subgroup and its quotient

For $n \geq 2$, the sign map $\operatorname{sgn} : S_n \to \{1, -1\}$ is a surjective homomorphism with kernel A_n. A kernel is normal, so $A_n \triangleleft S_n$, and the homomorphism theorem gives

$$S_n/A_n \cong \{1, -1\} \cong \mathbb{Z}_2.$$

Equivalently there are two cosets, the even and the odd permutations; odd times odd is even. For $n = 1$, $S_1 = A_1$ and the quotient is trivial, which is the small-case exception to the order-2 conclusion.

Exercise 14.25: Well-defined coset multiplication forces an inclusion

Let $h \in H$. Since $hH = H = eH$, well-definedness of multiplication gives

$$(ha)H = (hH)(aH) = (eH)(aH) = aH.$$

The element ha lies in its own left coset $(ha)H$, so $ha \in aH$. Since this holds for every $h \in H$, it proves $Ha \subseteq aH$. This is the desired missing direction and uses equality of cosets rather than an unproved normality assumption.

Exercise 14.26: Quotienting out all torsion

The torsion elements form a subgroup T by the earlier proof for abelian groups, and every subgroup of an abelian group is normal. Suppose gT has finite order in G/T. Then $(gT)^n = T$ for some positive n, so $g^n \in T$. Because that element is torsion, $(g^n)^m = e$ for some positive m. Thus $g^{nm} = e$, so $g \in T$ and $gT = T$. The quotient has no nonidentity torsion elements, which means it is torsion free.

Exercise 14.27: Conjugacy is an equivalence relation on subgroups

Reflexivity follows from $H = eHe^{-1}$. For symmetry, if $K = gHg^{-1}$ then $H = g^{-1}Kg$. For transitivity, if $K = gHg^{-1}$ and $L = hKh^{-1}$, then

$$L = hgHg^{-1}h^{-1} = (hg)H(hg)^{-1}.$$

Thus conjugacy of subgroups is reflexive, symmetric, and transitive. Each conjugation maps subgroups to subgroups, since it is an automorphism.

Exercise 14.28: Normal subgroups as conjugacy classes

A subgroup H is normal exactly when its conjugacy class among subgroups is the singleton $\{H\}$. Indeed, normality means $gHg^{-1} = H$ for every g, so there are no other conjugate subgroups. Conversely a singleton class means every conjugate equals H, which is the conjugation criterion for normality. This is a class of subgroups, not a conjugacy class of individual group elements.

Exercise 14.29: Conjugate reflection subgroups in S_3

The subgroup $\{\rho_0, \mu_2\}$ is generated by the transposition $(1, 3)$. Conjugating a transposition gives $(g(1), g(3))$, and as g ranges over S_3 this realizes every transposition. Hence the conjugate subgroups are exactly

$$\{\rho_0, \mu_1\}, \quad \{\rho_0, \mu_2\}, \quad \{\rho_0, \mu_3\}.$$

There are no others, since any conjugate subgroup has two elements and its nonidentity element is a transposition.

Exercise 14.30: Powers lying in a normal subgroup of finite index

Interpret $m = [G : H]$ as finite, as required for the integer exponent in the statement. The factor group has order m, so Lagrange gives $(aH)^m = H$ for each coset. Since $(aH)^m = a^m H$, this is $a^m H = H$, equivalent to $a^m \in H$. The ambient group itself need not be finite; finite index is enough. If the index is infinite, the displayed power a^m is not an ordinary group power and the statement needs the finite-index qualification.

Exercise 14.31: Intersections of normal subgroups

Let $N = \bigcap_{i \in I} N_i$ with each $N_i \triangleleft G$. It is a subgroup: all identities belong, and if $a, b \in N$, then ab^{-1} belongs to every N_i, hence to N. For $g \in G$ and $x \in N$, normality of every N_i gives $gxg^{-1} \in N_i$ for all i, so $gxg^{-1} \in N$. Thus N is normal. For an empty family the intersection is understood to be G, which is also normal.

Exercise 14.32: The normal closure of a subset

Let $\mathcal{N}$ be the collection of normal subgroups containing S. It is nonempty because G itself belongs to it. Set

$$N_S = \bigcap_{N \in \mathcal{N}} N.$$

By Exercise 31 this is normal. It contains S, since every intersected subgroup does. If M is any normal subgroup containing S, then $M \in \mathcal{N}$ and $N_S \subseteq M$ by the definition of intersection. Thus it is the unique smallest normal subgroup containing S, called the normal closure.

Exercise 14.33: The commutator quotient is abelian

By definition every commutator $aba^{-1}b^{-1}$ lies in the normal subgroup C. Therefore in G/C,

$$(aC)(bC)(aC)^{-1}(bC)^{-1} = (aba^{-1}b^{-1})C = C.$$

Multiplying by $(bC)(aC)$ on the right gives $(aC)(bC) = (bC)(aC)$. All quotient elements are such cosets, so the quotient is abelian. This proof uses only the inclusion of all commutators, not that each element of C must itself be a single commutator.

Exercise 14.34: A unique subgroup of a given order is normal

For any $g \in G$, conjugation is an automorphism, so gHg^{-1} is a subgroup of the same order as H. If H is the only subgroup of that order, $gHg^{-1} = H$ for every g. This is exactly normality. Finiteness ensures that "order" in the premise is the usual integer cardinality, although the uniqueness argument itself works for arbitrary cardinality.

Exercise 14.35: Intersecting with a normal subgroup

For $h \in H$ and $x \in H \cap N$, the conjugate hxh^{-1} lies in H because H is a subgroup, and in N because $N \triangleleft G$. Thus it lies in $H \cap N$, proving $H \cap N \triangleleft H$. It need not be normal in G: take $G = N = S_3$ and

$H = \{e, (1,2)\}$. Then N is normal in G and $H \cap N = H$. But conjugating $(1,2)$ by $(1,2,3)$ gives $(2,3) \notin H$, so this intersection is not normal in G.

Exercise 14.36: Intersection of all subgroups of a fixed finite order

Let K be the intersection of all subgroups of order s; the family is nonempty by hypothesis. Conjugation by a fixed $g \in G$ permutes that family, since $H \mapsto gHg^{-1}$ preserves subgroup order and has inverse conjugation by g^{-1}. Thus

$$gKg^{-1} = g\left(\bigcap_{|H|=s} H\right)g^{-1} = \bigcap_{|H|=s} gHg^{-1} = \bigcap_{|H|=s} H = K.$$

Conjugation commutes with intersection because it is a bijection. Since an intersection of subgroups is a subgroup, the equality proves $K \triangleleft G$ even though the individual subgroups need not be normal.

Exercise 14.37: Automorphisms and inner automorphisms

(a) The identity is an automorphism. A composite of two bijective homomorphisms is again a bijective homomorphism. The inverse of a bijective homomorphism is a homomorphism: writing $x = \phi(a)$, $y = \phi(b)$ gives $\phi^{-1}(xy) = \phi^{-1}(\phi(ab)) = ab = \phi^{-1}(x)\phi^{-1}(y)$. Composition is associative because function composition is. Thus $\mathrm{Aut}(G)$ is a group. **(b)** First prove the subgroup requirement: $i_g i_h = i_{gh}$, i_e is the identity, and $(i_g)^{-1} = i_{g^{-1}}$, so $\mathrm{Inn}(G) = \{i_g : g \in G\}$ is a subgroup. For any $\phi \in \mathrm{Aut}(G)$ and $x \in G$,

$$(\phi i_g \phi^{-1})(x) = \phi(g\phi^{-1}(x)g^{-1}) = \phi(g)x\phi(g)^{-1} = i_{\phi(g)}(x).$$

Thus conjugation by every automorphism sends inner automorphisms to inner automorphisms, proving $\mathrm{Inn}(G) \triangleleft \mathrm{Aut}(G)$.

Exercise 14.38: Elements inducing the identity inner automorphism

The condition $i_g = \mathrm{id}$ means $gxg^{-1} = x$ for every $x \in G$, equivalent to $gx = xg$ for every x. Thus the set is the center $Z(G)$. It is normal because it is the kernel of the homomorphism

$$G \to \mathrm{Aut}(G), \qquad g \mapsto i_g,$$

whose homomorphism law is $i_g i_h = i_{gh}$. Alternatively, conjugating a central element leaves it unchanged, directly giving normality.

Exercise 14.39: An induced map on factor groups

Define $\phi_* : G/H \to G'/H'$ by $\phi_*(gH) = \phi(g)H'$.

1. Check representatives.

If $gH = kH$, then $k^{-1}g \in H$. By the assumption $\phi[H] \subseteq H'$, its image $\phi(k)^{-1}\phi(g)$ is in H'. Thus $\phi(g)H' = \phi(k)H'$ and the proposed map is well defined.

2. Check products.

Normality makes both quotient products valid, and

$$\phi_*((gH)(kH)) = \phi(gk)H' = \phi(g)\phi(k)H' = (\phi(g)H')(\phi(k)H').$$

Therefore ϕ_* is a homomorphism. It is natural because it sends a coset to the coset of the original image without any arbitrary choices.

Exercise 14.40: Normal subgroups defined by determinants

(a) The determinant is a homomorphism $\det : GL(n, \mathbb{R}) \to \mathbb{R}^*$ because $\det(AB) = \det A \det B$ and invertibility gives nonzero determinant. The determinant-1 matrices form its kernel, hence a normal subgroup. Explicitly products and inverses keep determinant 1, and $\det(BAB^{-1}) = \det A$. **(b)** The map $A \mapsto |\det A|$ is a homomorphism to $\mathbb{R}^+$, because absolute value is multiplicative. Its kernel is exactly the matrices with determinant $+1$ or -1. Therefore this set is also a normal subgroup. The identity belongs to both sets because $\det I_n = 1$.

Exercise 14.41: Multiplication of subsets and the quotient group

(a) The set product is associative: $(AB)C$ and $A(BC)$ both consist of all abc with $a \in A, b \in B, c \in C$, since multiplication in G is associative. The identity subset is $\{e\}$, because $\{e\}A = A\{e\} = A$. But the powerset is not a group: the empty set has product $\varnothing$ with every subset and cannot have product $\{e\}$ with an inverse. **(b)** If $N \triangleleft G$, then for $a, b \in G$,

$$(aN)(bN) = \{an_1bn_2 : n_1, n_2 \in N\} = abN.$$

For inclusion in abN, rewrite $an_1bn_2 = ab(b^{-1}n_1b)n_2$, whose final factor lies in N. Conversely every abn is the product of $a \in aN$ and $bn \in bN$, so equality holds. Thus cosets are closed under the subset product, and the rule agrees with representative multiplication. **(c)** Associativity is inherited from subset multiplication. The subset N is an identity on cosets, since $N(aN) = aN = (aN)N$ by the displayed equality and normality. Each aN has inverse $a^{-1}N$, because their products in both orders equal N. Hence the cosets form a group, proved directly from subsets. Its identity is N, whereas the identity of the whole powerset monoid is $\{e\}$; they coincide exactly when $N = \{e\}$.

15 Factor-Group Computations and Simple Groups

Exercise 15.1: Removing an entire direct factor

The projection $(a, b) \mapsto a$ from $\mathbb{Z}_2 \times \mathbb{Z}_4$ onto $\mathbb{Z}_2$ has kernel $\{0\} \times \mathbb{Z}_4 = \langle (0, 1) \rangle$. The first isomorphism theorem therefore identifies the quotient with $\mathbb{Z}_2$.

Exercise 15.2: Reducing the second coordinate modulo two

The map $(a, b) \mapsto (a, b \bmod 2)$ is a surjective homomorphism onto $\mathbb{Z}_2 \times \mathbb{Z}_2$. Its kernel is $\{(0, 0), (0, 2)\} = \langle (0, 2) \rangle$. Hence this is the quotient's classification.

Exercise 15.3: A diagonal relation changes the quotient

Define $\phi : \mathbb{Z}_2 \times \mathbb{Z}_4 \to \mathbb{Z}_4$ by $\phi(a, b) = 2a + b \pmod 4$. It is well defined because changing a by 2 changes the result by 4, and it is onto since $\phi(0, 1) = 1$. Its kernel is exactly $(0, 0), (1, 2)$. Thus the quotient by $\langle (1, 2) \rangle$ is $\mathbb{Z}_4$, rather than $\mathbb{Z}_2 \times \mathbb{Z}_2$.

Exercise 15.4: A cyclic quotient of order eight

The surjective map $(a, b) \mapsto b - 2a \pmod 8$ from $\mathbb{Z}_4 \times \mathbb{Z}_8$ to $\mathbb{Z}_8$ is well defined. Its kernel consists exactly of $(a, 2a)$, the subgroup generated by $(1, 2)$. Therefore the quotient is $\mathbb{Z}_8$. Its order is also $32/4 = 8$, checking the computation.

Exercise 15.5: Eliminating one coordinate in a triple product

Map $\mathbb{Z}_4 \times \mathbb{Z}_4 \times \mathbb{Z}_8$ onto $\mathbb{Z}_4 \times \mathbb{Z}_8$ by

$$\phi(a, b, c) = (b - 2a \pmod 4,\ c - 4a \pmod 8).$$

Changing the representative for a by 4 adds multiples of 4 and 8 to the outputs, so the formula is well defined. It is onto using $a = 0$. Its kernel is $(a, 2a, 4a)$, precisely $\langle (1, 2, 4) \rangle$. Thus the quotient is $\mathbb{Z}_4 \times \mathbb{Z}_8$, of order $128/4 = 32$.

Exercise 15.6: An infinite coordinate projection

The map $\mathbb{Z}^2 \to \mathbb{Z}$, $(a, b) \mapsto a$, is onto with kernel $\langle (0, 1) \rangle$. Hence the quotient is the infinite cyclic group $\mathbb{Z}$.

Exercise 15.7: A primitive diagonal in the integer lattice

Use $\phi(a, b) = b - 2a$. This homomorphism onto $\mathbb{Z}$ has kernel the pairs $(a, 2a)$, exactly $\langle (1, 2) \rangle$. Thus the quotient is $\mathbb{Z}$, with no finite torsion factor.

Exercise 15.8: A primitive diagonal in three dimensions

The map $\phi : \mathbb{Z}^3 \to \mathbb{Z}^2$ given by $(a, b, c) \mapsto (b - a, c - a)$ is onto and has kernel the triples (a, a, a). Therefore the quotient by $\langle (1, 1, 1) \rangle$ is $\mathbb{Z} \times \mathbb{Z}$.

Exercise 15.9: A finite relation on an infinite coordinate

Reducing the first integer coordinate modulo 3 gives the quotient $\mathbb{Z}_3 \times \mathbb{Z} \times \mathbb{Z}_4$. Since 3 and 4 are coprime, $\mathbb{Z}_3 \times \mathbb{Z}_4 \cong \mathbb{Z}_{12}$ by the Chinese remainder theorem. In invariant-factor form the answer is $\mathbb{Z} \times \mathbb{Z}_{12}$. The middle free coordinate remains infinite.

Exercise 15.10: A fourth-multiple relation

The relation only reduces the second integer coordinate modulo 4. The map $(a, b, c) \mapsto (a, b \bmod 4, c)$ is onto $\mathbb{Z} \times \mathbb{Z}_4 \times \mathbb{Z}_8$ with kernel $\langle (0, 4, 0) \rangle$. Thus that direct product is the classification, and $4 \mid 8$ puts the finite factors in invariant-factor order.

Exercise 15.11: A nonprimitive diagonal

Map $\mathbb{Z}^2$ onto $\mathbb{Z} \times \mathbb{Z}_2$ by $\phi(a, b) = (a - b, b \bmod 2)$. Given $(u, \bar{v})$, choose an integer representative v and use $(u + v, v)$, proving surjectivity. The kernel requires $a = b$ and b even, hence is $\langle (2, 2) \rangle$. Therefore the quotient is $\mathbb{Z} \times \mathbb{Z}_2$. The factor of 2 in the diagonal produces the torsion.

Exercise 15.12: A triple diagonal with torsion

The map $(a, b, c) \mapsto (b - a, c - a, a \bmod 3)$ is onto $\mathbb{Z}^2 \times \mathbb{Z}_3$. Its kernel consists of $a = b = c$ with $3 \mid a$, exactly $\langle (3, 3, 3) \rangle$. Thus the quotient is $\mathbb{Z} \times \mathbb{Z} \times \mathbb{Z}_3$.

Exercise 15.13: Center and commutators of the square group

Write $D_4 = \langle r, s \rangle$ with $r^4 = s^2 = e$ and $srs = r^{-1}$. A rotation r^k commutes with s exactly when $r^k = r^{-k}$, meaning k even; no reflection commutes with r, since that would give $r = r^{-1}$. Thus $Z(D_4) = \{e, r^2\}$. The commutator $rsr^{-1}s^{-1} = r^2$ lies in C. On the other hand the quotient by $\langle r^2 \rangle$ has $r^{-1} = r$, so its defining generators commute and it is abelian. The commutator subgroup is therefore contained in $\langle r^2 \rangle$ as well, proving $C = \{e, r^2\}$.

Exercise 15.14: A direct product with a symmetric group

For any two groups, the center of their product is the product of their centers: commuting is checked in each coordinate. Likewise $[G \times H, G \times H] = [G, G] \times [H, H]$, since commutators are coordinatewise and commutators in either individual coordinate are obtained by setting the other coordinate to the identity. As $\mathbb{Z}_3$ is abelian, $Z(S_3) = \{e\}$, and $[S_3, S_3] = A_3$, the answers are

$$Z(\mathbb{Z}_3 \times S_3) = \mathbb{Z}_3 \times \{e\}, \qquad C = \{0\} \times A_3 \cong \mathbb{Z}_3.$$

For the last standard fact, S_3/A_3 is abelian, while the commutator of two different transpositions is a nonidentity three-cycle, generating A_3.

Exercise 15.15: Two nonabelian direct factors

Apply the coordinatewise center and commutator rules proved in Exercise 15.14. Since $Z(S_3) = \{e\}$, $Z(D_4) = \{e, r^2\}$, $[S_3, S_3] = A_3$, and $[D_4, D_4] = \langle r^2 \rangle$,

$$Z(S_3 \times D_4) = \{e\} \times \langle r^2 \rangle \cong \mathbb{Z}_2, \qquad C = A_3 \times \langle r^2 \rangle \cong \mathbb{Z}_6.$$

The last isomorphism uses the coprime orders 3 and 2.

Exercise 15.16: Every small subgroup of a sixteen-element group

Let $G = \mathbb{Z}_4 \times \mathbb{Z}_4$. Its complete list of subgroups of order at most 4, with quotients, is

subgroup	order	G/H
$\{(0,0)\}$	1	$\mathbb{Z}_4 \times \mathbb{Z}_4$
$\langle (2,0) \rangle$	2	$\mathbb{Z}_2 \times \mathbb{Z}_4$
$\langle (0,2) \rangle$	2	$\mathbb{Z}_2 \times \mathbb{Z}_4$
$\langle (2,2) \rangle$	2	$\mathbb{Z}_2 \times \mathbb{Z}_4$
$\langle (1,0) \rangle,\ \langle (0,1) \rangle$	4	$\mathbb{Z}_4$ (each)
$\langle (1,1) \rangle,\ \langle (1,2) \rangle$	4	$\mathbb{Z}_4$ (each)
$\langle (1,3) \rangle,\ \langle (2,1) \rangle$	4	$\mathbb{Z}_4$ (each)
$\{(0,0), (2,0), (0,2), (2,2)\}$	4	$\mathbb{Z}_2 \times \mathbb{Z}_2$

Here each bracket denotes all multiples of its displayed generator, so the actual subgroup elements are explicit from that rule. To verify completeness, exactly three nonzero elements have order 2. The remaining twelve nonzero elements have order 4, with two generators per cyclic subgroup, giving exactly six cyclic subgroups as listed. A noncyclic order-four subgroup must consist of the identity and all three order-two elements, so there is exactly one more. For an order-two subgroup, the quotient has order 8, exponent at most 4, and an element of order 4: use a coordinate not killed by that subgroup, or $(1,0) + H$ for $H = \langle (2,2) \rangle$. Its classification is therefore $\mathbb{Z}_2 \times \mathbb{Z}_4$. Each cyclic order-four generator has an odd coordinate; elementary invertible changes of coordinates send it to $(1,0)$, so its quotient is $\mathbb{Z}_4$. The remaining subgroup is $2G$, and reducing both coordinates modulo 2 gives its quotient $\mathbb{Z}_2^2$.

Exercise 15.17: The center must contain exactly the commuting elements

Replace "contains all" by "is the set of all." Precisely,

$$Z(G) = \{z \in G : zg = gz \text{ for every } g \in G\}.$$

Saying only that it contains these elements would allow extraneous elements and would not determine a subgroup uniquely.

Exercise 15.18: The commutator subgroup requires generation

The commutator subgroup is the subgroup *generated* by all commutators, not merely their set. It is

$$C = \langle a^{-1}b^{-1}ab : a, b \in G \rangle.$$

This convention gives the same subgroup as $aba^{-1}b^{-1}$, by replacing a, b with their inverses. Individual commutators need not themselves be closed under multiplication, so generation is essential.

Exercise 15.19: Ten statements about factor and simple groups

(a) True. If a generates G, then aN generates every quotient G/N.

(b) False. $\mathbb{Z}_2^2$ is not cyclic, but quotienting by one coordinate gives $\mathbb{Z}_2$.

(c) False. $1/2 + \mathbb{Z}$ has order 2.

(d) True. $1/n + \mathbb{Z}$ has order n, including the identity when $n = 1$.

(e) False. An element killed by 4 must have representative $k/4$. Modulo integers there are only four such classes, and exactly two have order 4: $1/4 + \mathbb{Z}, 3/4 + \mathbb{Z}$.

(f) True. If every commutator is identity, $aba^{-1}b^{-1} = e$ gives $ab = ba$.

(g) False. The inclusion is $C \subseteq H$, because all commutators become identity in an abelian quotient.

(h) False. A simple cyclic group of prime order is abelian and has trivial commutator subgroup.

(i) True. C is normal; it is nontrivial in a nonabelian group, so simplicity forces $C = G$.

(j) False. A_5 is simple and has order 60, which is not prime.

Exercise 15.20: Functions modulo constants

Let $H = \{f : \mathbb{R} \to \mathbb{R} : f(0) = 0\}$ under addition. Define $T : F \to H$ by $T(f)(x) = f(x) - f(0)$. This is an additive homomorphism and is onto because it fixes every function in H. Its kernel is exactly the constant functions K. Thus $F/K \cong H$. Choosing a basepoint turns each coset into its unique representative vanishing there.

Exercise 15.21: Nonvanishing functions modulo constant factors

Let $H^* = \{f : \mathbb{R} \to \mathbb{R} \setminus \{0\} : f(0) = 1\}$ under multiplication. The map $T(f)(x) = f(x)/f(0)$ is well defined and multiplicative, and fixes every function in H^*, so is onto. Its kernel consists exactly of the nonzero constant functions. Therefore $F^*/K^* \cong H^*$.

Exercise 15.22: No order-two additive coset modulo continuous functions

No. If $f + K$ were killed by 2, then $2f$ would be continuous. Multiplying it by the constant $1/2$ would make f continuous as well, so $f \in K$ and its coset is identity, not an element of order 2. The same argument shows the additive quotient has no nonidentity element of any finite order.

Exercise 15.23: An order-two multiplicative coset

Yes. Define $f(x) = 1$ for $x \geq 0$ and $f(x) = -1$ for $x < 0$. It never vanishes but is discontinuous at 0, so $f \notin K^*$. Yet f^2 is the constant function 1 and lies in K^*. Thus fK^* is nonidentity and its square is identity, giving order exactly 2. Unlike scalar multiplication, squaring loses the discontinuous sign information.

Exercise 15.24: A translated full circle

Since $z_0 \in U$, multiplication by z_0 maps U onto U: for any $w \in U$, $w = z_0(z_0^{-1}w)$ and $z_0^{-1}w \in U$. Hence $z_0U = U$, which is certainly a subgroup. Its quotient $U/z_0U = U/U$ is the trivial group.

Exercise 15.25: The circle modulo a half-turn

The homomorphism $U \to U$, $z \mapsto z^2$, is onto because every unit complex number has a unit square root. Its kernel is $\{1, -1\} = \langle -1 \rangle$. Thus $U/\langle -1 \rangle \cong U$ by the first isomorphism theorem.

Exercise 15.26: The circle modulo the nth roots

The map $U \to U$, $z \mapsto z^n$, is a surjective homomorphism. Its kernel is the cyclic subgroup of all nth roots of unity, generated by ζ_n. Consequently $U/\langle \zeta_n \rangle \cong U$. When $n = 1$ this is the identity-map case.

Exercise 15.27: Real numbers modulo integers

The map $\phi : \mathbb{R} \to U$ given by $\phi(t) = \cos(2\pi t) + i \sin(2\pi t)$ is a homomorphism from addition to multiplication, by the angle-addition formulas. It is onto, and $\phi(t) = 1$ exactly when $t \in \mathbb{Z}$. Thus $\mathbb{R}/\mathbb{Z} \cong U$.

Exercise 15.28: Torsion created by quotienting

Take the torsion-free additive group $G = \mathbb{Z}$ and $H = 2\mathbb{Z}$. In G, $na = 0$ with $n > 0$ forces $a = 0$, so no nonzero element has finite order. But $G/H \cong \mathbb{Z}_2$, and all its elements have finite order. Alternatively $\mathbb{Q}/\mathbb{Z}$ gives an infinite quotient with the same property.

Exercise 15.29: Isomorphic kernels with different quotients

Let $G = \mathbb{Z} \times \mathbb{Z}_2$, $H = \mathbb{Z} \times \{0\}$, and $K = 2\mathbb{Z} \times \{0\}$. They are normal because G is abelian, and both are infinite cyclic, hence isomorphic. However $G/H \cong \mathbb{Z}_2$, while $G/K \cong \mathbb{Z}_2 \times \mathbb{Z}_2$. Their different orders show the quotients are not isomorphic.

Exercise 15.30: Centers of simple groups

(a) If a simple group is abelian, every element commutes with every other, so $Z(G) = G$. A nontrivial simple abelian group is cyclic of prime order.

(b) If it is nonabelian, its normal center cannot be G. Simplicity leaves only $Z(G) = \{e\}$.

Exercise 15.31: Commutators of simple groups

(a) In an abelian group, every commutator is e, so $C = \{e\}$.

(b) In a nonabelian group at least one commutator is nonidentity. Thus C is a nontrivial normal subgroup, and simplicity forces $C = G$.

Exercise 15.32: Synopsis of cyclic quotients

If $G = \langle a \rangle$, then every coset is $a^k N = (aN)^k$, so aN generates G/N. This one observation proves the cyclic-quotient theorem.

Exercise 15.33: Synopsis of maximal normal subgroups

Under the quotient map $G \to G/M$, normal subgroups properly between M and G correspond exactly to nontrivial proper normal subgroups of G/M. Thus absence of the former, meaning M is maximal normal, is equivalent to simplicity of G/M.

Exercise 15.34: Index two prevents simplicity

A subgroup H of index 2 is normal: for $g \notin H$, both gH and Hg are the complement of H, while for $g \in H$ both equal H. The subgroup is proper because its index is 2 and nontrivial by hypothesis. It is therefore a nontrivial proper normal subgroup of G, so G is not simple.

Exercise 15.35: The image of a normal subgroup

The image $\phi[N]$ is a subgroup of $\phi[G]$. For $\phi(g) \in \phi[G]$ and $\phi(n) \in \phi[N]$,

$$\phi(g)\phi(n)\phi(g)^{-1} = \phi(gng^{-1}) \in \phi[N],$$

since N is normal in G. Thus the image is normal in $\phi[G]$. It need not be normal in all of G' when ϕ is not onto.

Exercise 15.36: The inverse image of a normal subgroup

Set $H = \phi^{-1}[N']$. It is a subgroup because identity, products and inverses map into the subgroup N'. For $g \in G, h \in H$,

$$\phi(ghg^{-1}) = \phi(g)\phi(h)\phi(g)^{-1} \in N',$$

by normality of N' in G'. Hence $ghg^{-1} \in H$, proving normality in G. No surjectivity assumption is needed.

Exercise 15.37: A cyclic central quotient forces commutativity

Prove the contrapositive. Suppose $G/Z(G)$ is generated by $aZ(G)$. Each $x \in G$ then has form $a^m z$ with $z \in Z(G)$, and similarly $y = a^n w$. Since the central factors commute with all elements,

$$xy = a^{m+n} zw = a^{n+m} wz = yx.$$

Thus G is abelian. Therefore a nonabelian group cannot have a cyclic quotient by its center.

Exercise 15.38: A nonabelian group of order a product of primes

The center is a subgroup, so its order divides pq. If it were the whole group, G would be abelian. If it were nontrivial and proper, its order would be p or q (also when $p = q$). The quotient would then have prime order and hence be cyclic. Exercise 15.37 would make G abelian, again a contradiction. Thus $Z(G) = \{e\}$.

Exercise 15.39: A complete proof of simplicity of the alternating group

We compose permutations from right to left. All letters appearing distinctly in a displayed identity are assumed distinct. Put $c_i = (r, s, i)$ for fixed distinct r, s and $i \notin \{r, s\}$. The source's description of "n special cycles" means these $n - 2$ actual three-cycles; repeated entries $i = r, s$ are not cycles.

(a) A three-cycle is a product of two transpositions, for example $(a, b, c) = (a, c)(a, b)$, so is even and lies in A_n.

(b) Every even permutation is a product of an even number of transpositions. Group them into adjacent pairs. Equal transpositions cancel. A pair sharing one letter is a three-cycle (or its inverse). For a disjoint pair, direct computation gives $(a, b)(c, d) = (a, c, b)(a, c, d)$. Thus every such pair is a product of three-cycles, and these generate A_n.

(c) To show the special three-cycles suffice, compute

$$c_i^2 = (r, i, s), \qquad c_j c_i^2 = (r, i, j),$$
$$c_j^2 c_i = (s, i, j), \qquad c_i^2 c_k c_j^2 c_i = (i, j, k).$$

Their inverses give the opposite orientations. These cover three-cycles containing both r, s, only r, only s, or neither. Thus every three-cycle is a product of the special ones, and (b) proves generation. If $n = 3$, only c_i, c_i^{-1} are needed.

(d) Suppose $N \trianglelefteq A_n$ contains a three-cycle, relabeled as c_i. Then $c_i^2 \in N$. For any other $j \notin \{r, s, i\}$, the double transposition $h = (r, s)(i, j)$ is even, and conjugation gives $hc_i^2 h^{-1} = (r, s, j) = c_j \in N$. Thus N contains all special cycles (and c_i itself). Part (c) gives $N = A_n$. For $n = 3$ the contained three-cycle already generates A_3.

(e) Take a nonidentity $\sigma \in N$ and use its disjoint-cycle decomposition. We check every case in the source. For any $t \in A_n$, normality gives $\sigma^{-1} t\sigma t^{-1} \in N$: the first factor σ^{-1} and the conjugate $t\sigma t^{-1}$ both belong to N.

Case I: N contains a three-cycle as an element.

Part (d) applies immediately. Merely having a three-cycle among the disjoint factors of a larger permutation does not suffice; those cases are treated below.

Case II: there is a cycle of length $\ell > 3$.

Write $\sigma = \mu(a_1, a_2, \ldots, a_\ell)$, with μ disjoint from that cycle. Set $t = (a_1, a_2, a_3)$. Conjugation replaces the entries by their preimages under σ, so

$$\sigma^{-1}t\sigma t^{-1} = (a_\ell, a_1, a_2)(a_1, a_3, a_2) = (a_1, a_3, a_\ell).$$

This is a three-cycle in N, so (d) applies.

Case III: at least two three-cycles occur.

Write $\sigma = \mu(a_4, a_5, a_6)(a_1, a_2, a_3)$, all factors disjoint, and choose $t = (a_1, a_2, a_4)$. Then

$$\sigma^{-1}t\sigma t^{-1} = (a_3, a_1, a_6)(a_1, a_4, a_2) = (a_1, a_4, a_2, a_6, a_3).$$

This five-cycle lies in N, reducing to Case II, hence to Case I.

Case IV: precisely one three-cycle and some two-cycles occur.

Here $\sigma = \mu(a_1, a_2, a_3)$ with μ a disjoint product of transpositions. Since $\mu^2 = e$ and disjoint factors commute, $\sigma^2 = (a_1, a_3, a_2) \in N$, a three-cycle.

Case V: all nontrivial cycles are transpositions.

Since σ is even, there are an even number of them, at least two. Write $\sigma = \mu(a_3, a_4)(a_1, a_2)$ with μ disjoint. Taking $t = (a_1, a_2, a_3)$ gives

$$\sigma^{-1}t\sigma t^{-1} = (a_2, a_1, a_4)(a_1, a_3, a_2) = (a_1, a_3)(a_2, a_4) = \alpha \in N.$$

Because $n \geq 5$, choose a letter j different from these four and let $\beta = (a_1, a_3, j) \in A_n$. Another conjugation calculation gives

$$\beta^{-1}\alpha\beta\alpha = (j, a_1)(a_2, a_4)(a_1, a_3)(a_2, a_4) = (a_1, a_3, j) \in N.$$

This is again a three-cycle. These cases exhaust all possible nonidentity even permutations: a longer cycle gives II; otherwise the number of three-cycles is at least two, exactly one, or zero, giving III, IV, or V. Every nontrivial normal subgroup is therefore A_n, proving simplicity for $n \geq 5$.

Exercise 15.40: The product of a subgroup and a normal subgroup

The set HN contains both subgroups. For $h_1n_1, h_2n_2 \in HN$, normality gives

$$(h_1n_1)(h_2n_2) = h_1h_2(h_2^{-1}n_1h_2)n_2 \in HN.$$

Also $(hn)^{-1} = n^{-1}h^{-1} = h^{-1}(hn^{-1}h^{-1}) \in HN$. Together with the identity this proves HN is a subgroup. Any subgroup containing both H and N contains every product hn, so contains HN. Thus HN is the smallest such subgroup.

Exercise 15.41: A product of two normal subgroups

By Exercise 15.40, NM is a subgroup. If $g \in G$ and $nm \in NM$, then $g(nm)g^{-1} = (gng^{-1})(gmg^{-1})$. The two factors lie in N and M, respectively, by their normality, so the product lies in NM. Therefore NM is normal in G.

Exercise 15.42: Disjoint normal subgroups commute elementwise

For $h \in H, k \in K$, let $c = hkh^{-1}k^{-1}$. It belongs to K, because $hkh^{-1} \in K$ and $k^{-1} \in K$. It also belongs to H, because $kh^{-1}k^{-1} \in H$ and $c = h(kh^{-1}k^{-1})$. The intersection hypothesis gives $c = e$, so $hk = kh$. Thus all pairs of elements from the two subgroups commute.

16 Group Action on a Set

For a group action, gx denotes the result of applying g to x; it need not be multiplication in a group containing x. The axioms are $ex = x$ and $(gh)x = g(hx)$. The orbit Gx consists of all possible images of x, the stabilizer G_x consists of the group elements fixing x, and X_g is the set of points fixed by one specified g. Keep these three sets distinct. In the classification exercises, a transitive set is understood to be nonempty, as in the book's examples. If the empty set is admitted, the quantified transitivity condition holds vacuously there and adds one empty case.

Exercise 16.1: All fixed sets in the square action

Plan. In each row of Table 16.10, retain exactly the column labels that the row leaves unchanged. Write X for the full seventeen-element set.

$$X_{\rho_0} = X, \qquad X_{\rho_1} = \{C\}, \qquad X_{\rho_3} = \{C\},$$
$$X_{\rho_2} = \{m_1, m_2, d_1, d_2, C\},$$
$$X_{\mu_1} = \{s_1, s_3, m_1, m_2, C, P_1, P_3\},$$
$$X_{\mu_2} = \{s_2, s_4, m_1, m_2, C, P_2, P_4\},$$
$$X_{\delta_1} = \{2, 4, d_1, d_2, C\},$$
$$X_{\delta_2} = \{1, 3, d_1, d_2, C\}.$$

For example, a half-turn exchanges opposite vertices, sides, and midpoints, but leaves each axis and each diagonal as a set. An object in X can be a whole line or side, so "fixed" does not require each point of that object to be fixed individually.

Exercise 16.2: Every stabilizer in the square action

Plan. Read down a column and collect the row labels that reproduce its heading. Let $e = \rho_0$. The answers can be grouped by identical stabilizers:

x	G_x
$1, 3$	$\{e, \delta_2\}$
$2, 4$	$\{e, \delta_1\}$
s_1, s_3, P_1, P_3	$\{e, \mu_1\}$
s_2, s_4, P_2, P_4	$\{e, \mu_2\}$
m_1, m_2	$\{e, \rho_2, \mu_1, \mu_2\}$
d_1, d_2	$\{e, \rho_2, \delta_1, \delta_2\}$
C	D_4

For example, vertex 1 is fixed by the identity and by reflection in its own diagonal d_2, and by no other square symmetry. The orbit–stabilizer check gives orbit size $8/2 = 4$, as expected for the vertices.

Exercise 16.3: The six orbits of the square action

Starting from a vertex reaches every vertex and no other kind of object. The same observation for each kind of object gives

$$\{1, 2, 3, 4\}, \quad \{s_1, s_2, s_3, s_4\}, \quad \{m_1, m_2\}, \quad \{d_1, d_2\}, \quad \{C\}, \quad \{P_1, P_2, P_3, P_4\}.$$

They are disjoint, their sizes sum to 17, and each is closed under all square symmetries. This verifies that the list exhausts X.

Exercise 16.4: Correcting faithful action

The quantifier "for every x" is essential. An action is faithful if

$$(gx = x \text{ for every } x \in X) \implies g = e.$$

Equivalently, the homomorphism $G \to S_X$ determined by the action is injective. Requiring this implication for a single fixed point would mean that all point stabilizers are trivial, which is the stronger condition called a free action. A reflection of a square fixes some vertices without acting as the identity on all vertices.

Exercise 16.5: Correcting transitive action

For every $x, y \in X$ there must exist $g \in G$ such that $gx = y$. The group element may depend on both points. One fixed g cannot send a single x to several different outputs. For a nonempty X, an equivalent formulation is: the orbit of any one chosen point is all of X.

Exercise 16.6: Subsets preserved by an action

Conclusion. The sub-G-sets are precisely the unions of whole G-orbits.

If S is a sub-G-set and $s \in S$, closure gives $Gs \subseteq S$. Since $s = es$, every s is in its own orbit, so $S = \bigcup_{s \in S} Gs$. Conversely, if S is a union of orbits and $s \in Gx \subseteq S$, then $s = hx$ for some h. For any g, $gs = (gh)x \in Gx \subseteq S$. The empty union also yields the empty sub-G-set.

Exercise 16.7: Transitivity in terms of orbits

For nonempty X, transitivity is equivalent to having exactly one orbit. If every y equals gx for some g, then $Gx = X$. Conversely, when $Gx = X$, any $y = hx$ and $z = kx$ satisfy $(kh^{-1})y = z$. If empty sets are allowed under the literal quantified definition, the corresponding characterization is "at most one orbit."

Exercise 16.8: True or false: action properties

(a) False. A group can act on any set; a three-element set with the trivial $\mathbb{Z}_2$ action need not come equipped with a group operation.

(b) True. This is the axiom $ex = x$.

(c) False. In the trivial action every g fixes every point; faithfulness is an additional condition.

(d) True. Apply g^{-1} to $gx_1 = gx_2$ and use the action axioms to get $x_1 = x_2$.

(e) False. In the trivial action of a nontrivial group all group elements send a chosen x to the same point.

(f) True. It is preserved by G, and if hx, kx are two points in the orbit then kh^{-1} sends the first to the second.

(g) True. Restrict the action map to $H \times X$; the same two axioms hold.

(h) False. Take $H = \{e\}$ inside a group that acts transitively on more than one point. Its orbits are singletons, while the original orbit is larger.

(i) True. The inverse of the map $x \mapsto gx$ is $x \mapsto g^{-1}x$.

(j) True. Orbit–stabilizer gives $|Gx| = [G : G_x] = |G|/|G_x|$ when G is finite.

Exercise 16.9: Isomorphic orbits and other subactions

(a) The side orbit and midpoint orbit are isomorphic by $s_i \mapsto P_i$. Every square symmetry sends the midpoint of a side to the midpoint of its image side, so this bijection commutes with the action.

(b) The reflection μ_1 fixes two sides, s_1, s_3, but no vertices. An equivariant bijection must carry the fixed points of any g bijectively onto the fixed points of the same g. Thus no such bijection can identify the vertex and side orbits.

(c) No. The question permits sub-D_4-sets that are unions of orbits. For instance $\{s_1, s_2, s_3, s_4, C\}$ and $\{P_1, P_2, P_3, P_4, C\}$ are different isomorphic sub-D_4-sets: extend the bijection in part (a) by $C \mapsto C$. Among individual orbits, the only isomorphic pair is indeed the side and midpoint pair: the other four-point orbit is excluded by part (b), while μ_1 fixes both axes and exchanges the diagonals, distinguishing the two two-point orbits.

Exercise 16.10: Faithfulness of the square action

(a) Yes. A symmetry fixing all seventeen objects fixes all four vertices. Only the identity symmetry does that.

(b) The vertex, side, and midpoint orbits each give faithful actions: inspection of their columns in Table 16.10 shows that no nonidentity row fixes the entire orbit. The axis orbit and diagonal orbit are not faithful, since the nonidentity half-turn ρ_2 fixes both members in either case. The center orbit is not faithful, since every symmetry fixes C.

Exercise 16.11: Faithfulness and distinct permutations

What we must show. Prove that faithfulness says precisely that different group elements induce different maps on X. Suppose first that the action is faithful and $gx = hx$ for every x. Applying h^{-1} gives $(h^{-1}g)x = x$ for all x, so $h^{-1}g = e$ and $g = h$. Conversely, if distinct group elements have distinct actions and $gx = x$ for every x, then g and e induce the same map. Hence $g = e$, proving faithfulness. Equality of actions here means equality on *every* point, not merely agreement at one point.

Exercise 16.12: Pointwise stabilizers of subsets

Let $G_Y = \{g : gy = y \text{ for every } y \in Y\}$. The identity belongs to G_Y. If $g, h \in G_Y$, then for every $y \in Y$, $(gh)y = g(hy) = gy = y$, so $gh \in G_Y$. If $g \in G_Y$, applying g^{-1} to $gy = y$ gives $g^{-1}y = y$, so $g^{-1} \in G_Y$. Thus G_Y is a subgroup. Equivalently $G_Y = \bigcap_{y \in Y} G_y$. When $Y = \varnothing$, all conditions are vacuous and $G_Y = G$.

Exercise 16.13: The real rotation action

(a) For $\theta \in (\mathbb{R}, +)$ use $R_\theta = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$. We have $R_0 = I$ and the angle-addition identities give $R_{\theta+\phi} = R_\theta R_\phi$. These are the two action axioms.

(b) Rotation preserves distance to the origin. Conversely every point at distance $|P| > 0$ has an angle differing from that of P by some real θ. Hence the orbit is the entire circle centered at the origin through P.

(c) A nonzero point returns to itself exactly after an integer multiple of a full turn, so $G_P = 2\pi\mathbb{Z}$ as a subgroup of $(\mathbb{R}, +)$. In particular, this real-number action is not faithful: 2π acts trivially.

Exercise 16.14: Unions of disjoint actions

(a) If the X_i are disjoint, every x in their union belongs to exactly one X_i. Define gx using that component's given action. This is well defined, stays in X_i, and satisfies $ex = x$ and $(gh)x = g(hx)$ because the component action does.

(b) Every $x \in X$ belongs to Gx. Two orbits meeting at a point are equal: if $gx = hy$, then $x = g^{-1}hy$, so every image of x is an image of y, and conversely. Thus the distinct orbits partition X, and their union, with the component actions, is exactly the original action.

Exercise 16.15: A transitive action is a coset action

Put $H = G_{x_0}$ and define $\phi(gx_0) = gH$; transitivity provides such a g for each point.

Prove the definition does not depend on the choice of g.

If $gx_0 = hx_0$, then $(h^{-1}g)x_0 = x_0$, so $h^{-1}g \in H$ and $gH = hH$. The same equivalences in reverse show that $gH = hH$ implies $gx_0 = hx_0$. Thus ϕ is both well defined and injective.

Check surjectivity and the action.

Every coset gH is the image of gx_0, so ϕ is onto. For $k \in G$,

$$\phi(k(gx_0)) = \phi((kg)x_0) = (kg)H = k\phi(gx_0).$$

This proves the required G-set isomorphism.

Exercise 16.16: Decomposing an arbitrary action

First partition X into its distinct orbits O_i. Choose a point x_i in each nonempty orbit and let $H_i = G_{x_i}$. Exercise 16.15 supplies an isomorphism $O_i \to G/H_i$, where G/H_i here denotes the *set of left cosets*; H_i need not be normal. Tag the coset set by i and let

$$g \cdot (kH_i, i) = (gkH_i, i).$$

The map taking $x \in O_i$ to $(\phi_i(x), i)$ is bijective, because the orbits partition X and each ϕ_i is bijective. It is equivariant in each component and hence globally. The tags prevent cosets from different components from being confused. For an arbitrary infinite family, choosing all the representatives uses the usual choice convention; finite actions need no such qualification.

Exercise 16.17: When two coset actions are isomorphic

(a) Let $H = G_{x_0}$. An element k fixes g_0x_0 exactly when $(g_0^{-1}kg_0)x_0 = x_0$, equivalently $g_0^{-1}kg_0 \in H$. Consequently $G_{g_0x_0} = g_0Hg_0^{-1}$.

(b) The conjecture is that G/H and G/K are isomorphic as G-sets exactly when H and K are conjugate subgroups.

(c) If $K = tHt^{-1}$, define $\phi(gH) = gt^{-1}K$. Replacing g by gh for $h \in H$ does not change the answer, because $ght^{-1}K = gt^{-1}(tht^{-1})K = gt^{-1}K$. Its inverse sends uK to utH, and left multiplication commutes with ϕ, proving sufficiency. Conversely, an equivariant bijection sends H to some aK and preserves stabilizers. The stabilizer of H is H, while that of aK is aKa^{-1}. Thus $H = aKa^{-1}$, proving necessity.

Exercise 16.18: All transitive actions of the group of order four

Plan. A transitive action is a coset action, and isomorphism types correspond to conjugacy classes of subgroups. Since $\mathbb{Z}_4$ is abelian, conjugation does not identify different subgroups. Its subgroups are $\mathbb{Z}_4$, $\{0, 2\}$, and $\{0\}$, giving exactly three types, with $1, 2, 4$ points respectively. Their action tables are

	a
0	a
1	a
2	a
3	a

	a	b
0	a	b
1	b	a
2	a	b
3	b	a

	a	b	c	d
0	a	b	c	d
1	b	c	d	a
2	c	d	a	b
3	d	a	b	c

.

In the four-point table, a, b, c, d represent $0, 1, 2, 3$; in the two-point table they represent the two cosets of $\{0, 2\}$.

Exercise 16.19: All transitive actions of the group of order six

There is one subgroup of each order $1, 2, 3, 6$ in $\mathbb{Z}_6$ and no others, since every subgroup of a cyclic group is cyclic. Conjugation is trivial. The four isomorphism types have sizes $6, 3, 2, 1$. For the first

three smaller sizes the tables are

	a
0	a
1	a
2	a
3	a
4	a
5	a

	a	b
0	a	b
1	b	a
2	a	b
3	b	a
4	a	b
5	b	a

	a	b	c
0	a	b	c
1	b	c	a
2	c	a	b
3	a	b	c
4	b	c	a
5	c	a	b

.

The six-point regular action is

	a	b	c	d	e	f
0	a	b	c	d	e	f
1	b	c	d	e	f	a
2	c	d	e	f	a	b
3	d	e	f	a	b	c
4	e	f	a	b	c	d
5	f	a	b	c	d	e

.

Each row adds its group element to the residue represented by the column, modulo the number of points. Since that number divides 6, this defines an action of $\mathbb{Z}_6$ in every case.

Exercise 16.20: All transitive actions of the symmetric group on three letters

Write $r = (1,2,3)$, $r^2 = (1,3,2)$, $s = (2,3)$, $t = (1,3)$, and $u = (1,2)$. The subgroups are S_3, A_3, the three conjugate order-two subgroups, and the identity subgroup. Hence there are four transitive isomorphism types, with $1, 2, 3, 6$ points. The one-point action is trivial; the two-point action records even or odd parity; the three-point action is the natural permutation action; and the six-point action is left multiplication on S_3. Here are all their action tables, with the requested row order:

	a
e	a
r	a
r^2	a
s	a
t	a
u	a

	a	b
e	a	b
r	a	b
r^2	a	b
s	b	a
t	b	a
u	b	a

	a	b	c
e	a	b	c
r	b	c	a
r^2	c	a	b
s	a	c	b
t	c	b	a
u	b	a	c

,

	a	b	c	d	e	f
e	a	b	c	d	e	f
r	b	c	a	f	d	e
r^2	c	a	b	e	f	d
s	d	e	f	a	b	c
t	e	f	d	c	a	b
u	f	d	e	b	c	a

.

In the last table the point labels a, b, c, d, e, f represent the group elements e, r, r^2, s, t, u respectively. For example, $rs = u$, so row r, column d has entry f. The point label e in that table is distinct from the row label for the group identity.

17 Applications of G-Sets to Counting

Counting method. All these exercises request Burnside's formula:

$$\#\text{orbits} = \frac{1}{|G|}\sum_{g\in G}|X_g|.$$

Choose the labeled objects X and the allowed symmetry group G first. A coloring fixed by a permutation must be constant on each cycle of that permutation. With q available colors and c cycles, there are q^c fixed colorings when repetition is allowed. For three-dimensional solids we use rigid rotations; a square piece of cardboard can also be turned over, which induces the plane reflections of its edges.

Exercise 17.1: A cyclic action on eight points

The generator $(1,3,5,6)$ has order four, so the group contains its zeroth, first, second, and third powers. The identity fixes all 8 points. Each of the other three powers moves all four points $1,3,5,6$ and fixes $2,4,7,8$, so each has 4 fixed points. Burnside gives

$$\#\text{orbits} = \frac{8+4+4+4}{4} = \boxed{5}.$$

As a check, the orbits are $\{1,3,5,6\}$ and the four singleton fixed points.

Exercise 17.2: Two generators with disjoint supports

Let $a=(1,3)$ and $b=(2,4,7)$. They commute, have orders 2 and 3, and generate the six distinct elements a^ib^j with $i=0,1$ and $j=0,1,2$. For a point in $\{1,3\}$ to be fixed we need $i=0$; this contributes two fixed points for each such element. A point in $\{2,4,7\}$ is fixed exactly when $j=0$, contributing three. The points $5,6,8$ are always fixed. Thus, in the order e,a,b,b^2,ab,ab^2, the fixed counts are $8,6,5,5,3,3$. Consequently

$$\#\text{orbits} = \frac{8+6+5+5+3+3}{6} = \boxed{5}.$$

Exercise 17.3: Tetrahedral dice with four different labels

The rotation group of a regular tetrahedron has 12 elements: one can choose the face on the bottom in four ways and rotate it into three orientations. There are $4! = 24$ labeled placements of the four different face labels. The identity fixes all 24. Every nonidentity rotation permutes at least two faces, so it cannot fix a placement with distinct labels. Burnside therefore gives

$$\frac{24+11\cdot 0}{12} = \boxed{2}$$

distinguishable dice. Reflections are not rigid rotations of a solid die.

Exercise 17.4: Cube faces with six different colors

There are $8 \cdot 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 = 20160$ assignments of six distinct colors to six specified faces. The rotation group has order 24. Only the identity can fix a coloring with all faces different, because any other rotation moves some face to another face of a different color. Burnside gives $20160/24 = \boxed{840}$ distinguishable blocks.

Exercise 17.5: Cube faces with arbitrary color repetition

Plan. Divide the 24 rotations into geometric types and count their cycles on the six faces. A fixed coloring is constant on each face cycle.

Rotation type	Number	Face cycle lengths	Cycles
Identity	1	$1,1,1,1,1,1$	6
Quarter-turn about face centers	6	$1,1,4$	3
Half-turn about face centers	3	$1,1,2,2$	4
Third-turn about opposite vertices	8	$3,3$	2
Half-turn about opposite edges	6	$2,2,2$	3

For instance a face-axis quarter-turn fixes the two axial faces and cycles the other four, giving three freely chosen colors. The entries in the second column sum to 24. With $q = 8$, Burnside gives

$$\frac{8^6 + 6 \cdot 8^3 + 3 \cdot 8^4 + 8 \cdot 8^2 + 6 \cdot 8^3}{24} = \frac{281088}{24} = \boxed{11712}.$$

Repeated colors are allowed independently on different cycles.

Exercise 17.6: Coloring the eight vertices of a cube

The same rotations now act on vertices, with different cycle counts:

Rotation type	Number	Vertex cycle lengths	Cycles
Identity	1	$1,1,1,1,1,1,1,1$	8
Face-axis quarter-turn	6	$4,4$	2
Face-axis half-turn	3	$2,2,2,2$	4
Vertex-axis third-turn	8	$1,1,3,3$	4
Edge-axis half-turn	6	$2,2,2,2$	4

For example, a third-turn fixes its two axial vertices and cycles the other six in two triples. With four available colors, Burnside gives

$$\frac{4^8 + 6 \cdot 4^2 + (3+8+6)4^4}{24} = \frac{69984}{24} = \boxed{2916}.$$

Here a color need not appear; if it does, it may appear on any number of vertices. If one additionally requires all four colors to appear, which is a different restriction, replace each 4^c by the number of onto maps from its c cycles to four colors; the result would be 1718.

Exercise 17.7: Painting the edges of a square

The allowed group is D_4, of order 8. Turning over the cardboard makes reflections physically available as well as in-plane rotations.

(a) With no repeated color there are $6 \cdot 5 \cdot 4 \cdot 3 = 360$ labeled paintings. Only the identity fixes any of them. Hence Burnside gives $360/8 = \boxed{45}$.

(b) On the four edges, the identity has four cycles; two quarter-turns have one cycle each; the half-turn has two cycles; two reflections have two cycles; and two reflections have three cycles. Therefore

$$\frac{6^4 + 2 \cdot 6 + (1+2)6^2 + 2 \cdot 6^3}{8} = \frac{1848}{8} = \boxed{231}.$$

A reflection through opposite side midpoints fixes those two edges and swaps the other two, accounting for the three-cycle type.

Exercise 17.8: Resistors on the edges of a tetrahedron

Let X be the 2^6 assignments of two resistor types to the six edges. Use the 12 rigid rotations of the tetrahedral framework. The identity has six edge cycles. Each of the eight third-turns around a vertex and the opposite face center divides the edges into two triples, hence has two cycles. Each of the three half-turns around opposite edge midpoints fixes those two edges as sets and swaps the other four in two pairs, hence has four cycles. Burnside gives

$$\frac{2^6 + 8 \cdot 2^2 + 3 \cdot 2^4}{12} = \frac{144}{12} = \boxed{12}.$$

The objects are actual frameworks in space, so mirror reflection is not an additional rigid rotation. If mirror images were expressly identified, the larger symmetry group would instead give 11 classes.

Exercise 17.9: Coloring a prism with two square ends

Identify the rotation group.

The unique long axis must map to itself, so arbitrary cube rotations cannot be used. There are four rotations about the long axis, through $0, 90, 180, 270$ degrees, and four half-turns about transverse axes. Thus $|G| = 8$. On the six faces the cycle counts are as follows:

Rotation type	Number	Face cycle lengths	Cycles
Identity	1	$1,1,1,1,1,1$	6
Long-axis quarter-turn	2	$1,1,4$	3
Long-axis half-turn	1	$1,1,2,2$	4
Transverse half-turn through side centers	2	$1,1,2,2$	4
Transverse half-turn through side-edge midpoints	2	$2,2,2$	3

The transverse rotations interchange the two square ends. In the first transverse type, two lateral faces remain individually fixed; in the second type the four lateral faces are interchanged in pairs.

(a) With all six colors distinct there are $6! = 720$ labeled colorings, and only the identity fixes any. Burnside gives $720/8 = \boxed{90}$.

(b) With repetition, each cycle can have any of six colors. Hence

$$\frac{6^6 + 3 \cdot 6^4 + 4 \cdot 6^3}{8} = \frac{46656 + 3888 + 864}{8} = \frac{51408}{8} = \boxed{6426}.$$

Answer-key discrepancy. The scanned book's odd-answer key prints 6246 for part (b). The rotation table above gives 6426. Independently, enumerating all 46656 labeled face colorings and identifying them under the eight signed-coordinate rotation matrices preserving the long axis also gives 6426 orbits. Thus the printed answer has transposed digits.

18 Rings and Fields

Exercise 18.1: A product modulo 24

Multiply the integer representatives first: $12 \cdot 16 = 192$. Since $192 = 8 \cdot 24$, its remainder modulo 24 is 0. Thus the product in $\mathbb{Z}_{24}$ is $\boxed{0}$.

Exercise 18.2: A product modulo 32

The integer product is $16 \cdot 3 = 48 = 32 + 16$. Reduction modulo 32 therefore gives $\boxed{16}$.

Exercise 18.3: A negative product modulo 15

Compute $11(-4) = -44$. Adding the multiple 45 of 15 gives $-44 \equiv 1 \pmod{15}$. The product is $\boxed{1}$.

Exercise 18.4: A negative product modulo 26

We have $20(-8) = -160 = -7 \cdot 26 + 22$. Therefore the representative in $\{0, \ldots, 25\}$ is $\boxed{22}$.

Exercise 18.5: Multiplication in a product ring

Operations in $\mathbb{Z}_5 \times \mathbb{Z}_9$ are performed separately in each coordinate. Hence

$$(2, 3)(3, 5) = (6 \bmod 5, 15 \bmod 9) = \boxed{(1, 6)}.$$

Exercise 18.6: Two modular coordinates

In $\mathbb{Z}_4 \times \mathbb{Z}_{11}$ the product is

$$(-3, 5)(2, -4) = (-6, -20) = (2, 2),$$

because $-6 \equiv 2 \pmod{4}$ and $-20 \equiv 2 \pmod{11}$.

Exercise 18.7: The multiples of an integer

For $n \geq 1$, sums, negatives, and products of multiples of n remain multiples of n: $na - nb = n(a - b)$ and $(na)(nb) = n(nab)$. The associative and distributive laws are inherited from $\mathbb{Z}$. Thus $n\mathbb{Z}$ is a commutative ring. If it has unity u, applying $u \cdot n = n$ in the integers gives $u = 1$. Consequently it has unity exactly when $n = 1$, and it is never a field: when $n = 1$, the nonzero integer 2 has no inverse in $\mathbb{Z}$. If $n = 0$ is allowed, $0\mathbb{Z} = \{0\}$ is the zero ring, whose identity equals zero and which is not a field.

Exercise 18.8: Positive integers

Both addition and multiplication are closed on $\mathbb{Z}^+$. Nevertheless addition is not a group operation: 0 is absent and no positive integer has its additive inverse in the set. Thus these operations do not form a ring.

Exercise 18.9: The product of two integer rings

Addition and multiplication are componentwise, so each ring axiom follows in each coordinate from its counterpart in $\mathbb{Z}$. The result is a commutative ring with unity $(1,1)$. It is not a field: $(1,0)$ is nonzero but cannot have an inverse, because its product with any (a,b) has second coordinate zero.

Exercise 18.10: An even-integer coordinate

The subset $2\mathbb{Z}\times\mathbb{Z}$ is closed under subtraction and multiplication, so it is a commutative subring of $\mathbb{Z}\times\mathbb{Z}$. If (u,v) were its unity, multiplying $(2,0)$ would require $2u=2$, hence $u=1$, contrary to $u\in 2\mathbb{Z}$. It has no unity and is not a field.

Exercise 18.11: Integral coefficients with a square root

For integers a,b,c,d,

$$(a+b\sqrt{2})-(c+d\sqrt{2})=(a-c)+(b-d)\sqrt{2},\quad (a+b\sqrt{2})(c+d\sqrt{2})=(ac+2bd)+(ad+bc)\sqrt{2}.$$

Thus $\mathbb{Z}[\sqrt{2}]$ is a commutative subring of $\mathbb{R}$, with unity 1. It is not a field: an inverse of 2 would be $1/2$, which is not $a+b\sqrt{2}$ with integers a,b. Indeed $b\neq 0$ would make $\sqrt{2}$ rational, and $b=0$ would require $a=1/2$.

Exercise 18.12: Rational coefficients with a square root

The subtraction and multiplication formulas in Exercise 18.11 also hold for rational coefficients, proving that this set is a commutative ring with unity. For $a+b\sqrt{2}\neq 0$,

$$(a+b\sqrt{2})^{-1}=\frac{a-b\sqrt{2}}{a^2-2b^2}.$$

The denominator is nonzero: otherwise either $a=b=0$ or $\sqrt{2}=a/b$ would be rational. The displayed inverse again has rational coefficients. Therefore $\mathbb{Q}(\sqrt{2})$ is a field.

Exercise 18.13: Pure imaginary numbers

Addition is closed, but multiplication is not: $i\cdot i=-1$ is not purely imaginary. Since a ring requires a binary multiplication on its underlying set, this is not a ring under the indicated operations.

Exercise 18.14: Units in the integers

A unit $u\in\mathbb{Z}$ must satisfy $uv=1$ for an integer v. The only integer factors of 1 are 1 and -1. Thus the units are $\boxed{\{1,-1\}}$, each its own inverse.

Exercise 18.15: Units in a product of integer rings

The equation $(a,b)(c,d)=(1,1)$ is equivalent to $ac=1$ and $bd=1$. Therefore each of a,b must be 1 or -1, independently. The four units are $\boxed{(1,1),(1,-1),(-1,1),(-1,-1)}$.

Exercise 18.16: Units modulo five

Since 5 is prime, every nonzero residue is a unit. Explicitly, $1^{-1} = 1$, $2^{-1} = 3$, $3^{-1} = 2$, and $4^{-1} = 4$, as multiplication verifies modulo 5. The units are $\boxed{1, 2, 3, 4}$.

Exercise 18.17: Units in the rational field

Every nonzero rational a/b has rational inverse b/a. Zero cannot multiply any rational to give 1. Thus the unit group is $\boxed{\mathbb{Q} \setminus \{0\}}$.

Exercise 18.18: Units in a three-factor ring

An inverse must work in all three coordinates. The first and last integer coordinates must therefore be ± 1, and the rational coordinate must be nonzero. All units are

$$\boxed{\{(\varepsilon, q, \delta) : \varepsilon, \delta \in \{1, -1\},\ q \in \mathbb{Q} \setminus \{0\}\}}.$$

The inverse is $(\varepsilon, q^{-1}, \delta)$.

Exercise 18.19: Units modulo four

The invertible residues are precisely those relatively prime to 4, namely $1, 3$. Directly, $1 \cdot 1 = 1$ and $3 \cdot 3 = 9 \equiv 1 \pmod 4$, whereas products with 0 or 2 cannot be odd. Thus the units are $\boxed{1, 3}$.

Exercise 18.20: The matrix ring over two elements

(a) Each of the four entries has two choices, giving $2^4 = \boxed{16}$ matrices. **(b)** A matrix over the field $\mathbb{Z}_2$ is invertible exactly when its determinant is nonzero. Its first column can be any of the three nonzero vectors; its second must be outside the span of the first, leaving two choices. The six units are

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}, \quad \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}, \quad \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}.$$

Every listed determinant is 1 in $\mathbb{Z}_2$, and the counting argument shows that none are missing.

Exercise 18.21: A homomorphism that does not preserve unity

Use $\phi : \mathbb{Z} \longrightarrow \mathbb{Z} \times \mathbb{Z}$, $\phi(n) = (n, 0)$. Both addition and multiplication are preserved componentwise. However, $\phi(1) = (1, 0)$ is neither the zero $(0, 0)$ nor the target unity $(1, 1)$. The book's ring-homomorphism convention does not require unity preservation.

Exercise 18.22: The determinant map

For $n \geq 2$, determinants preserve multiplication but not addition. Take $A = \operatorname{diag}(1, 0, 0, \ldots)$ and $B = \operatorname{diag}(0, 1, 1, \ldots)$. Both determinants are zero, whereas $A + B = I_n$ has determinant 1. Hence $\det(A + B) \neq \det A + \det B$, and the map is not a ring homomorphism. For the exceptional case $n = 1$, the determinant is simply the entry, so it is an isomorphism $M_1(\mathbb{R}) \to \mathbb{R}$.

Exercise 18.23: Homomorphisms from the integers to themselves

Additivity forces $\phi(n) = n\phi(1)$ for every integer n, including negative integers. Set $k = \phi(1)$. Multiplicativity applied to $1^2 = 1$ gives $k^2 = k$, so $k = 0$ or 1. Both choices work. Thus the only maps are the zero map and the identity map.

Exercise 18.24: Homomorphisms into a product

Write $\phi(1) = (u, v)$. Additivity gives $\phi(n) = (nu, nv)$, and multiplicativity requires $(u^2, v^2) = (u, v)$. In $\mathbb{Z}$, an idempotent is 0 or 1. Consequently there are four homomorphisms:

$$n \mapsto (0,0), \quad n \mapsto (n,0), \quad n \mapsto (0,n), \quad n \mapsto (n,n).$$

Each preserves the two ring operations by direct substitution.

Exercise 18.25: Homomorphisms out of a product

Let $e_1 = (1,0)$ and $e_2 = (0,1)$. Since they are idempotent, their images u, v in $\mathbb{Z}$ lie in $\{0,1\}$. Since $e_1 e_2 = 0$, we must have $uv = 0$. Additivity then gives $\phi(a,b) = au + bv$. The possible pairs (u,v) are $(0,0), (1,0), (0,1)$, yielding exactly the zero map and the two coordinate projections.

Exercise 18.26: Three source coordinates

The three coordinate idempotents must map to 0 or 1 in $\mathbb{Z}$, and the product of any two different images must be zero. Therefore at most one image is 1. There are four choices: all zero, or one of the three coordinate positions equal to 1. These give the zero map and the three projections, so the answer is $\boxed{4}$.

Exercise 18.27: A false zero-product argument for matrices

The factorization is valid because I_3 commutes with X. The invalid step is inferring that a factor must be zero: matrix rings have zero divisors. For example,

$$X = \operatorname{diag}(1, 1, -1)$$

satisfies $X^2 = I_3$ while $X \neq I_3$ and $X \neq -I_3$. Here both $X - I_3$ and $X + I_3$ are nonzero, although their product is zero.

Exercise 18.28: A quadratic congruence modulo fourteen

Factor the polynomial as $(x+3)(x-2)$. A product is zero modulo 14 exactly when it is zero modulo both 2 and 7. Modulo 2 the original polynomial is $x(x+1)$, zero for both residues. Modulo 7, the two roots are 2 and $-3 = 4$. Lifting these residues to $\mathbb{Z}_{14}$ gives

$$\boxed{x = 2, 4, 9, 11}.$$

For example, at $x = 4$, neither $x + 3 = 7$ nor $x - 2 = 2$ is zero modulo 14, but their product is. This is precisely the zero-divisor issue in Exercise 18.27.

Exercise 18.29: Correcting the definition of a field

The missing requirement is commutativity of multiplication. A field is a *commutative* ring with unity $1 \neq 0$ such that every nonzero element has a multiplicative inverse; equivalently its nonzero elements form an abelian group under multiplication. The printed definition, without commutativity, describes a division ring.

Exercise 18.30: Correcting the definition of a unit

Magnitude is irrelevant in a general ring. In a ring with unity 1, an element u is a unit if there exists v in the ring with $uv = vu = 1$. Both equations are required in a possibly noncommutative ring.

Exercise 18.31: A product of nonzero elements that vanishes

In $\mathbb{Z}_6$, the residues 2 and 3 are nonzero, but $2 \cdot 3 = 6 = 0$. This supplies the requested example of nonzero zero divisors.

Exercise 18.32: A subring with its own unity

In $R = \mathbb{Z} \times \mathbb{Z}$, let $S = \mathbb{Z} \times \{0\}$. It is closed under subtraction and multiplication, so it is a subring. Its unity is $(1, 0)$, which is nonzero and differs from the unity $(1, 1)$ of R. Indeed $(1, 0)(a, 0) = (a, 0) = (a, 0)(1, 0)$.

Exercise 18.33: True or false: ring and field axioms

(a) True: a field is a ring with additional properties.

(b) False: $2\mathbb{Z}$ has no multiplicative identity.

(c) False: $\mathbb{Z}_2$ has just one unit, 1, since $-1 = 1$ there.

(d) False: $\mathbb{Z}_5$ has four units.

(e) True: $\mathbb{Z} \subset \mathbb{Q}$ is a subring but not a subfield, because $2^{-1} = 1/2 \notin \mathbb{Z}$.

(f) False: distributivity is an essential ring axiom, linking the two operations.

(g) True: commutative multiplication is part of the definition of a field.

(h) True: this is the nonzero multiplicative group in the field definition.

(i) True: the additive group of a ring is required to be abelian.

(j) True: the additive-group axiom provides an additive inverse for every element.

Exercise 18.34: Pointwise multiplication of functions

The operations are pointwise: $(fg)(x) = f(x)g(x)$ and $(f + g)(x) = f(x) + g(x)$. For every point x,

$$((fg)h)(x) = (f(x)g(x))h(x) = f(x)(g(x)h(x)) = (f(gh))(x).$$

Thus multiplication is associative. Likewise

$$(f(g+h))(x) = f(x)(g(x)+h(x)) = f(x)g(x) + f(x)h(x) = (fg+fh)(x),$$

and $((f+g)h)(x) = (fh+gh)(x)$. Equality at every input means equality of functions, proving the required associative and both distributive axioms.

Exercise 18.35: Evaluation preserves multiplication

For the evaluation map $\phi_a(f) = f(a)$, the definition of pointwise product gives

$$\phi_a(fg) = (fg)(a) = f(a)g(a) = \phi_a(f)\phi_a(g).$$

This verifies the multiplicative requirement, without assuming that a general function map preserves products.

Exercise 18.36: Ring isomorphism is an equivalence relation

Reflexivity: the identity map preserves both operations and is bijective. **Symmetry:** if $\phi : R \to S$ is an isomorphism, write $u = \phi(a)$, $v = \phi(b)$. Then $\phi^{-1}(u+v) = a+b$ and $\phi^{-1}(uv) = ab$, so its inverse is also an isomorphism. **Transitivity:** if $\psi : S \to T$ is an isomorphism as well, then

$$(\psi\phi)(a+b) = \psi(\phi(a)+\phi(b)) = (\psi\phi)(a) + (\psi\phi)(b),$$

and the same calculation with products proves multiplicativity. The composition is bijective. These are exactly the three equivalence-relation properties.

Exercise 18.37: The group of units

Let u, v be units. Their product is a unit because

$$(uv)(v^{-1}u^{-1}) = 1 = (v^{-1}u^{-1})(uv).$$

Thus multiplication is closed on the units; its associativity is inherited from the ring. The unity is a unit, and the inverse of a unit is itself a unit, with inverse u. Consequently all group axioms hold. The order reversal in $(uv)^{-1}$ matters when multiplication is not commutative.

Exercise 18.38: A difference-of-squares criterion

If R is commutative, expansion gives $(a+b)(a-b) = a^2 - ab + ba - b^2 = a^2 - b^2$. Conversely, suppose the stated identity holds for every a, b. Expanding its right side and canceling a^2 and $-b^2$ in the additive group gives $-ab + ba = 0$, hence $ab = ba$. Therefore multiplication is commutative.

Exercise 18.39: Zero multiplication

Define $ab = 0$ for every pair. This is a closed multiplication, and both $(ab)c$ and $a(bc)$ are zero. Furthermore $a(b+c) = 0 = 0+0 = ab+ac$ and $(a+b)c = 0 = ac+bc$. Together with the assumed abelian additive group, these facts prove all ring axioms.

Exercise 18.40: Two pairs of nonisomorphic rings

The rings $2\mathbb{Z}$ and $3\mathbb{Z}$. An additive isomorphism must carry the additive generator 2 to either 3 or -3. If $\phi(2) = 3\varepsilon$ with $\varepsilon = \pm 1$, additivity gives $\phi(4) = 6\varepsilon$, whereas multiplicativity gives $\phi(4) = \phi(2)^2 = 9$. These cannot agree.

The fields $\mathbb{R}$ and $\mathbb{C}$. Any field isomorphism carries unity to unity: its image of 1 acts as an identity on the whole target. Therefore a supposed map $\phi : \mathbb{C} \to \mathbb{R}$ would give $\phi(i)^2 = \phi(-1) = -1$. No real number has square -1. Hence these fields cannot be isomorphic.

Exercise 18.41: The prime-characteristic binomial identity

In a commutative ring the binomial expansion is valid. For $1 \le k \le p-1$, the binomial coefficient $\binom{p}{k}$ is divisible by the prime p: its denominator $k!(p-k)!$ has no factor p, whereas its numerator $p!$ has one. In $\mathbb{Z}_p$ all these intermediate coefficients vanish. Thus

$$(a+b)^p = a^p + \sum_{k=1}^{p-1} \binom{p}{k} a^{p-k}b^k + b^p = a^p + b^p.$$

Exercise 18.42: The unity of a subfield

Let e be the unity of a subfield $K \subset F$. It is nonzero, and $e^2 = e$ in F. Multiply this equation by the inverse of e in the field F to obtain $e = 1_F$. Cancellation in a field is why a subfield cannot exhibit the different-unity phenomenon of Exercise 18.32.

Exercise 18.43: Uniqueness of a multiplicative inverse

If b and c are both two-sided inverses of a, then

$$b = b1 = b(ac) = (ba)c = 1c = c.$$

Associativity and the two inverse equations justify every step. Thus a unit has exactly one two-sided inverse.

Exercise 18.44: Idempotents and a product ring

(a) If $a^2 = a$, $b^2 = b$, and multiplication is commutative, then $(ab)^2 = a^2b^2 = ab$. Hence the product is idempotent.

(b) An idempotent residue x satisfies $x(x-1) \equiv 0$. In $\mathbb{Z}_6$, testing the six residues gives $\{0, 1, 3, 4\}$. In $\mathbb{Z}_{12}$, the requirement is that $x \equiv 0$ or 1 modulo each of 4 and 3, giving $\{0, 1, 4, 9\}$. Idempotence is coordinatewise, so all sixteen idempotents are

$$\boxed{\{0,1,3,4\} \times \{0,1,4,9\}}.$$

Exercise 18.45: An idempotent projection matrix

Put $B = (A^TA)^{-1}$, so $P = ABA^T$. Associativity of matrix multiplication, with no illicit reordering, gives

$$P^2 = ABA^TABA^T = AB(A^TA)BA^T = ABI_nA^T = ABA^T = P.$$

The dimensions are consistent: A is $m \times n$, B is $n \times n$, and P is $m \times m$. Thus it is an idempotent in $M_m(\mathbb{R})$. The printed reference to $n \times n$ matrices is a dimension slip: the formula for P gives $m \times m$.

Exercise 18.46: Adding nilpotent elements

Choose positive integers r, s with $a^r = 0$ and $b^s = 0$. Because the ring is commutative, expand $(a+b)^{r+s-1}$. In each term $a^kb^{r+s-1-k}$, either $k \geq r$, or else $k \leq r-1$ and $r+s-1-k \geq s$. The term is zero in either case. The entire expansion vanishes, so $a+b$ is nilpotent.

Exercise 18.47: Testing nilpotence using squares

If no nonzero nilpotent exists, an element satisfying $x^2 = 0$ must be zero. Conversely, suppose every square-zero element is zero. If a were nonzero nilpotent, choose the least $n \geq 2$ with $a^n = 0$. Then $b = a^{n-1} \neq 0$ by minimality, but $b^2 = a^{2n-2} = 0$ because $2n-2 \geq n$. This contradicts the square-zero condition.

Exercise 18.48: The subring test

A subring necessarily contains zero and is closed under subtraction and multiplication. Conversely suppose the three listed conditions hold. Setting $a = 0$ shows $-b \in S$; then $a + b = a - (-b) \in S$. Thus S is an additive subgroup of the abelian group $(R, +)$. Closure supplies a multiplication on S, and its associativity and both distributive laws are inherited from R. Hence S is a ring with the induced operations. A shared multiplicative unity is not required by the book's subring convention.

Exercise 18.49: Intersections of subrings and subfields

(a) Every subring in the family contains 0. If a, b are in their intersection, then $a - b$ and ab lie in every member, hence in the intersection. Apply Exercise 18.48.

(b) For a nonempty family of subfields of F, their common unity is 1_F by Exercise 18.42. The intersection is a subring and contains this nonzero unity. If $a \neq 0$ belongs to it, its unique inverse in F lies in every subfield, hence in the intersection. It is therefore a field. With the convention that an empty intersection is the ambient object, the empty-family conclusions hold as well.

Exercise 18.50: A one-sided annihilator is a subring

For $I_a = \{x : ax = 0\}$, first $a0 = 0$, so $0 \in I_a$. If $x, y \in I_a$, distributivity gives $a(x-y) = ax - ay = 0$. Associativity gives $a(xy) = (ax)y = 0y = 0$. Thus the subring test applies. No commutativity of the ambient ring is needed.

Exercise 18.51: The subring generated by one element

Let H consist of all finite sums $\sum_{k=1}^{N} m_k a^k$ with integer coefficients. This is exactly the additive subgroup generated by the positive powers of a. It contains a and is closed under subtraction. Products remain in H, because distributivity expresses a product as a finite integer combination of powers a^{i+j}. Therefore H is a subring containing a, so $R_a \subseteq H$ by the definition of R_a as an intersection. Conversely every subring containing a contains all positive powers and their integer linear combinations, so $H \subseteq R_a$. The two sets are equal. A constant term is not included: subrings here need not contain the ambient unity.

Exercise 18.52: Two simultaneous congruences

For coprime r, s, the map $\mathbb{Z}_{rs} \to \mathbb{Z}_r \times \mathbb{Z}_s$, $[x]_{rs} \mapsto ([x]_r, [x]_s)$, is a ring isomorphism as in the text. In particular it is onto, so the pair $([m]_r, [n]_s)$ has a preimage. Any integer representative x of that preimage satisfies both required congruences. Concretely, choose integers u, v with $ur + vs = 1$; then $x = nur + mvs$ has residues m modulo r and n modulo s.

Exercise 18.53: The general Chinese remainder theorem

(a) For pairwise coprime positive moduli $b_1, \ldots, b_n$, put $B = \prod b_i$. The map

$$\Phi : \mathbb{Z}_B \longrightarrow \prod_{i=1}^{n} \mathbb{Z}_{b_i}, \qquad [x]_B \mapsto ([x]_{b_1}, \ldots, [x]_{b_n})$$

is a well-defined ring homomorphism. Its kernel is zero: if every b_i divides x, their pairwise coprimality implies $B \mid x$. Both finite sets have B elements, so injectivity implies bijectivity. This proves the direct-product isomorphism.

(b) Surjectivity applied to the tuple of residues a_i produces the desired integer solution. Adding a sufficiently large multiple of B makes it positive without changing any congruence. For an explicit formula, let $B_i = B/b_i$ and choose u_i with $u_i B_i \equiv 1 \pmod{b_i}$. Then $x = \sum_i a_i u_i B_i$ has residue a_j modulo b_j, because every other summand is divisible by b_j.

Exercise 18.54: Recovering commutative addition in a division ring

We must prove the missing additive commutativity. Distributivity and additive cancellation first show $a0 = 0 = 0a$, since, for example, $a0 = a(0 + 0) = a0 + a0$. The multiplicative identity 1 on the nonzero elements also satisfies $10 = 01 = 0$, so it acts as unity on all of S. Expand the same product in two ways:

$$(1 + 1)(a + b) = a + b + a + b = (1 + 1)a + (1 + 1)b = a + a + b + b.$$

Cancel the initial a and final b in the additive group to obtain $b + a = a + b$. Thus $(S, +)$ is abelian. Multiplication is associative when all factors are nonzero by the assumed group axiom; if any factor is zero both associated products are zero. The assumed distributive laws now give all ring axioms, with nonzero unity and inverses for every nonzero element. Hence S is a division ring.

Exercise 18.55: Boolean rings are commutative

For any a, idempotence of $-a$ gives $(-a)^2 = -a$. But $(-a)^2 = a^2 = a$, so $a = -a$, equivalently $2a = 0$. Next expand the identity $(a+b)^2 = a+b$:

$$a^2 + ab + ba + b^2 = a + b.$$

Since $a^2 = a$ and $b^2 = b$, cancellation yields $ab + ba = 0$. Therefore $ab = -ba = ba$, using the already proved fact that every element equals its negative. The ring is commutative.

Exercise 18.56: A Boolean ring of subsets

(a) Abbreviate $\varnothing, \{a\}, \{b\}, \{a,b\}$ by $0, A, B, S$. The addition is symmetric difference and multiplication is intersection:

$+$	0	A	B	S
0	0	A	B	S
A	A	0	S	B
B	B	S	0	A
S	S	B	A	0

$\cdot$	0	A	B	S
0	0	0	0	0
A	0	A	0	A
B	0	0	B	B
S	0	A	B	S

.

(b) Associate to each subset $T \subseteq S$ its indicator function $\chi_T : S \to \mathbb{Z}_2$. At each point, $\chi_{T \triangle U} = \chi_T + \chi_U$ and $\chi_{T \cap U} = \chi_T \chi_U$, with operations in $\mathbb{Z}_2$. The bijection between subsets and their indicator functions therefore transfers the pointwise ring axioms to these set operations. The additive identity is $\varnothing$, each subset is its own additive inverse, and the unity is S. Finally $T \cap T = T$, so every element is idempotent and this is a Boolean ring.

19 Integral Domains

Exercise 19.1: A cubic congruence

Factor $x^3 - 2x^2 - 3x = x(x-3)(x+1)$. Because $12 = 3 \cdot 4$, it vanishes modulo 12 exactly when it vanishes modulo both 3 and 4. Modulo 3 the admissible residues are $0, 2$; modulo 4 they are $0, 1, 3$. Pairing these conditions, or testing the twelve residues, gives

$$\boxed{x = 0, 3, 5, 8, 9, 11 \text{ in } \mathbb{Z}_{12}}.$$

For example $x = 8$ gives $8 \cdot 5 \cdot 9 = 360$, divisible by 12. Factoring alone does not permit setting one factor equal to zero in a ring with zero divisors.

Exercise 19.2: Solving linear equations in two prime fields

In $\mathbb{Z}_7$, $3^{-1} = 5$, since $3 \cdot 5 = 15 \equiv 1$; hence $x = 5 \cdot 2 = \boxed{3}$. In $\mathbb{Z}_{23}$, $3^{-1} = 8$, since $3 \cdot 8 = 24 \equiv 1$; hence $x = \boxed{16}$. Multiplication by a nonzero field element is bijective, so these solutions are unique.

Exercise 19.3: A quadratic modulo six

For $x = 0, 1, 2, 3, 4, 5$, the residues of $x^2 + 2x + 2$ modulo 6 are respectively $2, 5, 4, 5, 2, 1$. None is zero, so there are $\boxed{\text{no solutions}}$. This exhaustive check is valid because these are all six elements of the ring.

Exercise 19.4: A second quadratic modulo six

Evaluating $x^2 + 2x + 4$ at $x = 0, 1, 2, 3, 4, 5$ gives residues $4, 1, 0, 1, 4, 3$ modulo 6. Therefore the complete solution set is $\boxed{\{2\}}$.

Exercise 19.5: Characteristic of the even integers

The characteristic is 0. For every positive integer n, the sum of n copies of the element 2 is the nonzero integer $2n$. Thus no positive n annihilates every element.

Exercise 19.6: Characteristic of two integer coordinates

For every positive n, $n(1, 0) = (n, 0) \neq (0, 0)$. Hence there is no common positive additive annihilator, and $\boxed{\text{char}(\mathbb{Z} \times \mathbb{Z}) = 0}$.

Exercise 19.7: A finite and an infinite coordinate

Although the first factor $\mathbb{Z}_3$ has characteristic 3, the element $(0, 3)$ in $\mathbb{Z}_3 \times 3\mathbb{Z}$ has infinite additive order. Its n-fold sum is $(0, 3n) \neq 0$ for every positive n. The product therefore has characteristic $\boxed{0}$.

Exercise 19.8: Two characteristic-three factors

Three copies of any element of $\mathbb{Z}_3 \times \mathbb{Z}_3$ sum to zero. One or two copies of $(1,0)$ do not. The least common positive annihilator is consequently $\boxed{3}$.

Exercise 19.9: Relatively prime finite characteristics

A positive integer annihilates $\mathbb{Z}_3 \times \mathbb{Z}_4$ exactly when it is divisible by both 3 and 4. The least such integer is $\operatorname{lcm}(3,4) = \boxed{12}$.

Exercise 19.10: A least common multiple of characteristics

The characteristic of $\mathbb{Z}_6 \times \mathbb{Z}_{15}$ is $\operatorname{lcm}(6,15) = \boxed{30}$. Indeed $n(1,0) = 0$ requires $6 \mid n$, while $n(0,1) = 0$ requires $15 \mid n$; these two conditions are also sufficient to annihilate every element.

Exercise 19.11: A fourth power in characteristic four

Commutativity allows the binomial theorem:

$$(a+b)^4 = a^4 + 4a^3b + 6a^2b^2 + 4ab^3 + b^4.$$

In characteristic 4, coefficients are reduced modulo 4: $4 = 0$ and $6 = 2$. Thus the simplified expression is $\boxed{a^4 + 2a^2b^2 + b^4}$. The middle term need not vanish.

Exercise 19.12: A fifth power in characteristic three

The binomial coefficients for exponent 5 are $1, 5, 10, 10, 5, 1$, reducing modulo 3 to $1, 2, 1, 1, 2, 1$. Therefore

$$\boxed{(a+b)^5 = a^5 + 2a^4b + a^3b^2 + a^2b^3 + 2ab^4 + b^5}.$$

Each reduction is justified because $3r = 0$ for every element r of the ring.

Exercise 19.13: A sixth power in characteristic three

First $(a+b)^3 = a^3 + b^3$, since the intermediate binomial coefficients are 3. Squaring and using commutativity gives

$$\boxed{(a+b)^6 = (a^3+b^3)^2 = a^6 + 2a^3b^3 + b^6}.$$

This agrees with reducing the sixth-row binomial coefficients modulo 3.

Exercise 19.14: Exhibiting a matrix zero divisor

Let

$$A = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}, \qquad B = \begin{pmatrix} -2 & 0 \\ 1 & 0 \end{pmatrix}.$$

Both matrices are nonzero, but $AB = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$, as row-by-column multiplication verifies. Thus A is a divisor of zero in $M_2(\mathbb{Z})$.

Exercise 19.15: Correcting zero-divisor terminology

One must exclude zero factors. A nonzero element a is a divisor of zero if there is a nonzero element b with $ab = 0$ or $ba = 0$ (in a commutative ring these are the same condition). The mere equation $ab = 0$ does not suffice if, for instance, $b = 0$.

Exercise 19.16: Correcting characteristic

The characteristic is the *least positive integer* n such that $na = 0$ for every $a \in R$, if such a positive integer exists. If none exists, the characteristic is 0. The printed condition omits both minimality and the characteristic-zero case.

Exercise 19.17: True or false: integral domains

(a) False: $n\mathbb{Z}$, as a subring of $\mathbb{Z}$, never has nonzero zero divisors, regardless of primality.

(b) True: a field is commutative with nonzero unity, and $ab = 0$, $a \neq 0$ implies $b = a^{-1}0 = 0$.

(c) False: for positive n, $n\mathbb{Z}$ has characteristic 0, since $kn \neq 0$ for positive k.

(d) False: for $n > 1$, $n\mathbb{Z}$ has no unity, whereas $\mathbb{Z}$ has one; an onto isomorphism would transport unity. Equivalently compare the images of 1 and 1^2.

(e) True: an isomorphism transports a zero-product or a cancellation equation to the integral domain, where cancellation holds.

(f) True: the elements $1, 2 \cdot 1, 3 \cdot 1, \ldots$ are distinct in characteristic zero.

(g) False: in a product of two domains, $(1, 0)(0, 1) = 0$ with both factors nonzero.

(h) True: multiplying $ab = 0$ by the inverse of a would force $b = 0$.

(i) False in general: for $n > 1$, $n\mathbb{Z}$ lacks unity and is not an integral domain under the book's definition. For $n = 1$ it is a subdomain.

(j) False: $\mathbb{Z}$ is a subring of $\mathbb{Q}$, but it lacks the inverse of 2 and is not a field.

Exercise 19.18: Examples in the six ring regions

The regions of Figure 19.10 have the following examples.

1. $\mathbb{Q}$: a field.

2. $\mathbb{Z}$: an integral domain that is not a field, because 2 is not a unit.

3. $\mathbb{Z}_6$: a commutative ring with unity but with $2 \cdot 3 = 0$.

4. $2\mathbb{Z}$: a commutative ring without unity.

5. $M_2(\mathbb{R})$: a ring with unity but not commutative, since $E_{12}E_{21} = E_{11} \neq E_{22} = E_{21}E_{12}$.

6. $M_2(2\mathbb{Z})$: noncommutative by the same calculation using $2E_{12}, 2E_{21}$, and without unity. A unity would have to act as identity on $2E_{11}$ and $2E_{22}$, forcing it to be I_2, whose entries are not all even.

Exercise 19.19: Five matrix criteria for being a zero divisor

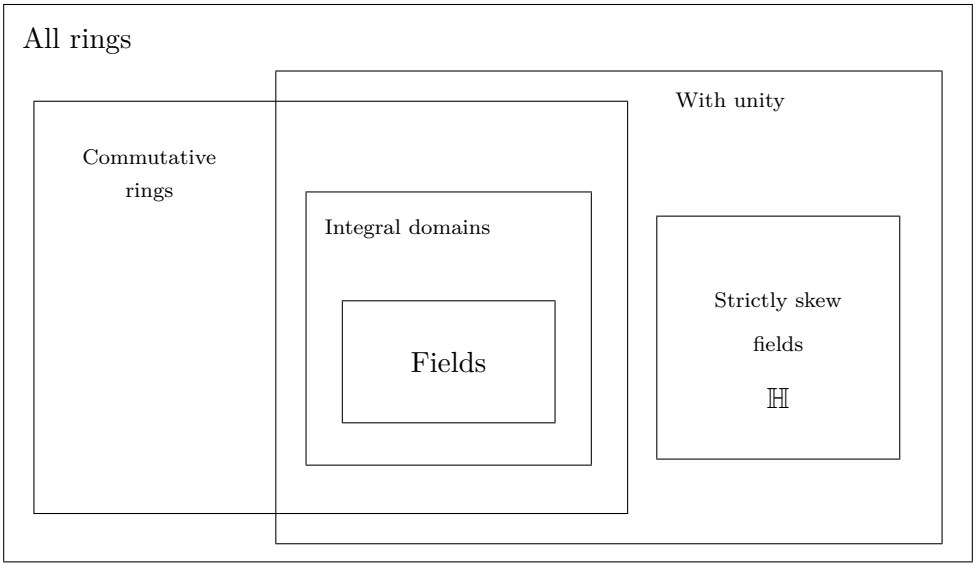
For a *nonzero* $A \in M_n(F)$, the following five equivalent tests characterize zero divisors:

$$\det A = 0; \quad \operatorname{rank} A < n; \quad \operatorname{Ker} A \neq \{0\}; \quad \text{columns of } A \text{ are dependent}; \quad A \text{ is not invertible.}$$

These are equivalent by the invertible-matrix theorem over a field. To connect them to the ring definition, if $Av = 0$ with $v \neq 0$, form B with first column v and all other columns zero. Then $B \neq 0$ and $AB = 0$. Conversely if $AB = 0$ with $B \neq 0$, some nonzero column v of B satisfies $Av = 0$. If instead $BA = 0$, invertibility would still be impossible. The exclusion $A \neq 0$ is essential because the book does not call zero itself a zero divisor.

Exercise 19.20: Locating strictly skew fields in the ring diagram

A strictly skew field is a division ring whose multiplication is not commutative. It therefore lies inside the rings-with-unity region and outside the commutative-rings region. Its set is disjoint from the integral domains and fields of the book's definitions. The quaternion division ring $\mathbb{H}$ supplies an example. The following inclusion diagram redraws the relevant regions using rectangles instead of circles.



Exercise 19.21: Synopsis of cancellation without zero divisors

If $ab = ac$ with $a \neq 0$, then $a(b - c) = 0$, so the absence of zero divisors forces $b - c = 0$ and hence $b = c$; the right-cancellation argument is identical with the factors reversed.

Exercise 19.22: Synopsis of the finite-domain theorem

For nonzero a in a finite integral domain, the map $x \mapsto ax$ is injective by cancellation and therefore surjective, so $ax = 1$ for some x, giving the required multiplicative inverse.

Exercise 19.23: Idempotents in a division ring

Both 0 and 1 are idempotent and are distinct. If $a^2 = a$ and $a \neq 0$, left multiplication by a^{-1} gives $a = 1$. Thus the only possibilities are exactly $\boxed{0, 1}$.

Exercise 19.24: Intersection of subdomains

A subdomain $D' \subseteq D$ must have the same unity as D: its nonzero unity e satisfies $e(e - 1_D) = 0$, hence $e = 1_D$ because D has no zero divisors. Thus an intersection of subdomains contains $1_D \neq 0$. It is a subring by the subring test, commutative by inheritance, and has no zero divisors because D has none. It is consequently a subdomain. For the empty family use the ambient domain as its intersection.

Exercise 19.25: A finite ring without zero divisors

Fix $a \neq 0$. The maps $x \mapsto ax$ and $x \mapsto xa$ are injective: equality of outputs gives a zero product after subtraction, so cancellation applies. Finiteness makes them onto. Hence there exist b, c with $ab = 1$ and $ca = 1$. They agree because $c = c(ab) = (ca)b = b$. Thus a has a two-sided inverse. This holds for every nonzero a, proving that R is a division ring; no commutativity was assumed.

Exercise 19.26: A unique generalized inverse forces a division ring

(a) Fix nonzero a and its unique b with $aba = a$. If either $ax = 0$ or $xa = 0$, then $axa = 0$, so $a(b + x)a = a$. Uniqueness implies $b + x = b$, hence $x = 0$. Thus there are no nonzero zero divisors.

(b) Associativity gives $a(bab)a = (aba)ba = aba = a$. Uniqueness of b gives $bab = b$.

(c) The element $e = ab$ is nonzero, since $aba = a \neq 0$, and $e^2 = (aba)b = ab = e$. For every x, $e(ex - x) = 0$ and $(xe - x)e = 0$. By (a), both differences are zero. Thus $ex = xe = x$, so e is a unity for the whole ring.

(d) Also ba is a nonzero idempotent and hence, by the same argument, is a unity. The uniqueness of unity gives $ba = ab = 1$. Every nonzero a therefore has a two-sided inverse, so R is a division ring.

Exercise 19.27: Characteristic of a subdomain

A subdomain $E \subseteq D$ has unity 1_D, as proved in Exercise 19.24. In any ring with unity the characteristic is the additive order of its unity, with infinite order recorded as characteristic zero. The sums $n1$ are identical whether computed in E or D, so their first positive zero, if any, occurs at the same n. Hence the two characteristics agree.

Exercise 19.28: The smallest subdomain

Let $P = \{n1_D : n \in \mathbb{Z}\}$. It contains 0 and 1_D, and

$$(m1_D) - (n1_D) = (m-n)1_D, \qquad (m1_D)(n1_D) = mn1_D.$$

It is therefore a commutative subring with nonzero unity, and has no zero divisors because it is contained in D. Thus it is a subdomain. Every subdomain contains the same unity 1_D and its integer sums and negatives, so every subdomain contains P.

Exercise 19.29: The characteristic of an integral domain

Suppose the positive characteristic is composite, say $n = rs$ with $1 < r, s < n$. Minimality gives $r1 \neq 0$ and $s1 \neq 0$, but

$$(r1)(s1) = rs1 = n1 = 0,$$

contradicting the absence of zero divisors. Also the characteristic cannot be 1, since $1 \neq 0$. Therefore the characteristic is either 0 or a prime.

Exercise 19.30: Adjoining a unity without changing characteristic

Write $K = \mathbb{Z}$ in characteristic zero, and $K = \mathbb{Z}_n$ in characteristic $n > 0$. On $S = R \times K$ use the stated operations. Integer scalar multiplication on R is well defined for residues modulo n because $nr = 0$.

(a) Addition is the direct-product abelian group. To check multiplication associativity, write three elements as $(r,k),(s,l),(t,m)$. The first coordinate of either associated product expands to

$$rst + k(st) + l(rt) + m(rs) + (kl)t + (km)s + (lm)r,$$

and the second coordinate is klm. The two distributive laws follow by expanding in the same way using distributivity in R and integer scalar multiplication. Thus this is a ring.

(b) Its unity is $(0,1)$, since

$$(0,1)(r,k) = (r,k) = (r,k)(0,1).$$

(c) If R has characteristic $n > 0$, both coordinates are killed by n, and the element $(0,1)$ has additive order exactly n. If R has characteristic zero, the integer coordinate of $(0,1)$ has infinite order. Therefore the characteristics agree. This includes the zero ring, with characteristic 1, if admitted.

(d) The map $r \mapsto (r,0)$ is injective and satisfies $(r,0)+(s,0) = (r+s,0)$ and $(r,0)(s,0) = (rs,0)$. It is consequently a ring isomorphism onto the subring $R \times \{0\}$.

20 Fermat's and Euler's Theorems

Exercise 20.1: A generator modulo 7

The nonzero residues form a multiplicative group of order 6. Successive powers of 3, with reduction modulo 7 after every multiplication, are

$$3^1, 3^2, \ldots, 3^6 \equiv 3, 2, 6, 4, 5, 1 \pmod 7.$$

Every nonzero residue appears once. Thus $\boxed{3}$ generates the entire multiplicative group.

Exercise 20.2: A generator modulo 11

The nonzero residues form a multiplicative group of order 10. Successive powers of 2, with reduction modulo 11 after every multiplication, are

$$2^1, 2^2, \ldots, 2^{10} \equiv 2, 4, 8, 5, 10, 9, 7, 3, 6, 1 \pmod{11}.$$

Every nonzero residue appears once. Thus $\boxed{2}$ generates the entire multiplicative group.

Exercise 20.3: A generator modulo 17

The nonzero residues form a multiplicative group of order 16. Successive powers of 3, with reduction modulo 17 after every multiplication, are

$$3^1, 3^2, \ldots, 3^{16} \equiv 3, 9, 10, 13, 5, 15, 11, 16, 14, 8, 7, 4, 12, 2, 6, 1 \pmod{17}.$$

Every nonzero residue appears once. Thus $\boxed{3}$ generates the entire multiplicative group.

Exercise 20.4: A large power modulo twenty-three

Fermat's theorem applies because 23 is prime and does not divide 3. It gives $3^{22} \equiv 1 \pmod{23}$. Since $47 = 2 \cdot 22 + 3$,

$$3^{47} \equiv 3^3 = 27 \equiv \boxed{4} \pmod{23}.$$

Exercise 20.5: A large power modulo seven

Reduce the base: $37 \equiv 2 \pmod 7$. Fermat gives $2^6 \equiv 1 \pmod 7$, and $49 = 8 \cdot 6 + 1$. Therefore $37^{49} \equiv 2^{49} \equiv \boxed{2} \pmod 7$.

Exercise 20.6: A power whose exponent is itself a power

Since 19 is prime, $2^{18} \equiv 1 \pmod{19}$, so the outer exponent may be reduced modulo 18. The powers 2^k modulo 18 for $k = 1, \ldots, 6$ are $2, 4, 8, 16, 14, 10$, and $2^{k+6} \equiv 2^k$ for $k \geq 1$, because $2(2^6 - 1) = 126$ is divisible by 18. Thus $2^{17} \equiv 2^5 \equiv 14 \pmod{18}$. Now

$$2^{14} = 2^{10}2^4 \equiv 17 \cdot 16 = 272 \equiv 6 \pmod{19},$$

where $2^{10} = 1024 \equiv 17$. Consequently $2^{(2^{17})} + 1 \equiv \boxed{7} \pmod{19}$.

Exercise 20.7: Euler phi values through thirty

The value $\varphi(n)$ counts integers from 1 through n relatively prime to n; in particular $\varphi(1) = 1$. For computation, if the distinct prime divisors of n are p, use $\varphi(n) = n\prod_{p\mid n}(1 - 1/p)$. The complete table is

n	1	2	3	4	5	6	7	8	9	10
$\varphi(n)$	1	1	2	2	4	2	6	4	6	4
n	11	12	13	14	15	16	17	18	19	20
$\varphi(n)$	10	4	12	6	8	8	16	6	18	8
n	21	22	23	24	25	26	27	28	29	30
$\varphi(n)$	12	10	22	8	20	12	18	12	28	8

Exercise 20.8: Euler phi for a prime square

Among $1, \ldots, p^2$, the integers not relatively prime to p^2 are exactly the p multiples $p, 2p, \ldots, p^2$. Removing them leaves $\boxed{\varphi(p^2) = p^2 - p = p(p-1)}$.

Exercise 20.9: Euler phi for a product of primes

If $p \neq q$, count the integers up to pq divisible by neither prime. There are q multiples of p, p multiples of q, and one multiple of both; inclusion–exclusion gives

$$\boxed{\varphi(pq) = pq - p - q + 1 = (p-1)(q-1)}.$$

If the primes are equal, the answer instead is $\varphi(p^2) = p(p-1)$ from Exercise 20.8. Distinctness cannot be silently assumed.

Exercise 20.10: Euler reduction modulo twenty-four

The base 7 is relatively prime to 24, and $\varphi(24) = 8$. Euler's theorem gives $7^8 \equiv 1 \pmod{24}$. Since $1000 = 125 \cdot 8$, the required remainder is $\boxed{1}$. One can also check directly that $7^2 = 49 \equiv 1 \pmod{24}$.

Exercise 20.11: A congruence with two residue solutions

The gcd is $\gcd(2, 4) = 2$, which divides 6. Divide the congruence by 2, including the modulus: $x \equiv 3 \pmod{2}$. Thus all integer solutions are $x = 1 + 2k$ with $k \in \mathbb{Z}$; the two incongruent solutions modulo 4 are $\boxed{1, 3}$.

Exercise 20.12: An invertible coefficient modulo fifteen

Reduce 22 modulo 15 to get $7x \equiv 5$. Since $7^{-1} = 13$ modulo 15, multiplication gives $x \equiv 65 \equiv 5 \pmod{15}$. Hence $\boxed{x = 5 + 15k, k \in \mathbb{Z}}$, one residue solution modulo 15.

Exercise 20.13: A congruence failing the gcd condition

Here $d = \gcd(36, 24) = 12$, but 12 does not divide 15. Every multiple $36x$ is divisible by 12, whereas $15 + 24k$ never is. Thus there are $\boxed{\text{no solutions}}$.

Exercise 20.14: Three residue solutions modulo twenty-four

Since $\gcd(45, 24) = 3$ divides 15, divide by 3 to obtain $15x \equiv 5 \pmod 8$, or $7x \equiv 5$. The inverse of 7 modulo 8 is 7, so $x \equiv 35 \equiv 3 \pmod 8$. Thus all integer solutions are $x = 3 + 8k$, and modulo 24 they are $\boxed{3, 11, 19}$.

Exercise 20.15: A second failure of the gcd condition

The coefficient 39 and modulus 9 have gcd 3. Since $3 \nmid 125$, the congruence has $\boxed{\text{no solutions}}$. Equivalently, its left side is zero modulo 3 but its right side is 2.

Exercise 20.16: One solution modulo nine

Reduce to $5x \equiv 8 \pmod 9$. Since $5 \cdot 2 \equiv 1$, multiply by 2: $x \equiv 16 \equiv 7 \pmod 9$. Thus $\boxed{x = 7 + 9k, k \in \mathbb{Z}}$.

Exercise 20.17: Five solutions modulo sixty-five

Here $\gcd(155, 65) = 5$, which divides 75. Divide by 5 to obtain $31x \equiv 15 \pmod{13}$, hence $5x \equiv 2$. Since $5^{-1} = 8$ modulo 13, $x \equiv 16 \equiv 3 \pmod{13}$. All integer solutions are $x = 3 + 13k$; the five residues modulo 65 are $\boxed{3, 16, 29, 42, 55}$.

Exercise 20.18: Thirteen solutions modulo one hundred thirty

The gcd is $\gcd(39, 130) = 13$, and $13 \mid 52$. Divide to obtain $3x \equiv 4 \pmod{10}$. Multiplying by $3^{-1} = 7$ gives $x \equiv 28 \equiv 8 \pmod{10}$. Hence all integer solutions are $\boxed{x = 8 + 10k}$, with $k \in \mathbb{Z}$; representatives modulo 130 are $8, 18, \ldots, 128$, thirteen in total.

Exercise 20.19: A factorial just below Wilson's factorial

Wilson's theorem gives $(p-1)! \equiv -1 \pmod p$. Since $(p-1)! = (p-1)(p-2)! \equiv -(p-2)!$, we obtain $\boxed{(p-2)! \equiv 1 \pmod p}$. For $p \geq 3$ the remainder is therefore 1.

Exercise 20.20: A factorial modulo thirty-seven

Write $36! = 36 \cdot 35 \cdot 34!$. Wilson's theorem and reduction of the last factors give $(-1)(-2)34! \equiv -1 \pmod{37}$, so $2 \cdot 34! \equiv -1$. The inverse of 2 is 19 modulo 37, and $-19 \equiv \boxed{18}$.

Exercise 20.21: A factorial modulo fifty-three

Wilson gives $52! \equiv -1 \pmod{53}$. Since $52! = 52 \cdot 51 \cdot 50 \cdot 49!$, we have $(-1)(-2)(-3)49! \equiv -1$. Thus $6 \cdot 49! \equiv 1$. The inverse of 6 modulo 53 is 9, because $6 \cdot 9 = 54 \equiv 1$. The remainder is $\boxed{9}$.

Exercise 20.22: A factorial modulo twenty-nine

Wilson gives $28! \equiv -1 \pmod{29}$. The four final factors are $28, 27, 26, 25 \equiv -1, -2, -3, -4$, whose product is 24. Hence $24 \cdot 24! \equiv -1$. Since $24 \cdot 23 = 552 \equiv 1$. Therefore $24! \equiv -23 \equiv \boxed{6} \pmod{29}$.

Exercise 20.23: True or false: Euler and congruences

(a) False: if $p \mid a$, then $a^{p-1} \equiv 0$, not 1.

(b) True: this is Fermat's theorem with its necessary nonzero-residue hypothesis.

(c) True: $\varphi(n)$ counts a subset of n integers, including $\varphi(1) = 1$.

(d) False as stated for all positive n: at $n = 1$, $\varphi(1) = 1 > 0 = n - 1$. It is true for $n \geq 2$, since n itself is then not coprime to n.

(e) True for $n \geq 2$, the usual nonzero-unity modular rings: the units are exactly residues with gcd 1 with n. If one admits $\mathbb{Z}_1$, the literal wording "positive integers less than n" omits its sole residue, so that case requires a separate convention.

(f) True: if u, v are units, $(uv)^{-1} = v^{-1}u^{-1}$.

(g) False: if ab is a unit in a commutative ring, $a[b(ab)^{-1}] = 1$ and similarly b is a unit; thus a product of two nonunits cannot be a unit.

(h) True by the same argument: a unit product would force the nonunit factor to be invertible.

(i) False: $0x \equiv 1 \pmod{p}$ has no solution.

(j) True: divide by d to obtain one solution modulo m/d, which lifts to exactly d distinct residue classes modulo m.

Exercise 20.24: The four units modulo twelve

The units are $1, 5, 7, 11$. Their table is

$\cdot$	1	5	7	11
1	1	5	7	11
5	5	1	11	7
7	7	11	1	5
11	11	7	5	1

.

Every nonidentity element has order 2, so this is not the cyclic group of order 4. It is the Klein four-group $\mathbb{Z}_2 \times \mathbb{Z}_2$, for example via $1 \mapsto (0,0)$, $5 \mapsto (1,0)$, $7 \mapsto (0,1)$, $11 \mapsto (1,1)$.

Exercise 20.25: Synopsis of Fermat's theorem

For nonzero $a \in \mathbb{Z}_p$, Lagrange's theorem in the multiplicative group of the $p-1$ nonzero residues gives $a^{p-1} = 1$, which is precisely the desired congruence.

Exercise 20.26: Synopsis of Euler's theorem

The unit represented by an integer coprime to n lies in the multiplicative unit group of $\mathbb{Z}_n$, whose order is $\varphi(n)$, so Lagrange's theorem gives $a^{\varphi(n)} \equiv 1 \pmod{n}$.

Exercise 20.27: Self-inverse nonzero residues

A self-inverse element satisfies $x^2 = 1$, hence $(x-1)(x+1) = 0$. Since $\mathbb{Z}_p$ is a field, one factor must vanish. Therefore $x = 1$ or $x = -1 = p-1$, and both do square to 1. For $p = 2$, these names describe the same element; for odd p, they are distinct.

Exercise 20.28: Wilson's congruence

For odd prime p, pair each nonzero residue other than 1 and -1 with its distinct multiplicative inverse. Every pair contributes product 1 to $(p-1)!$. By Exercise 20.27 the only unpaired factors are 1 and -1, giving $(p-1)! \equiv -1 \pmod{p}$. For $p = 2$, $1! = 1 \equiv -1 \pmod{2}$ directly.

For completeness, if an integer $n > 1$ satisfies this congruence, every prime divisor q of n must equal n: otherwise $q \leq n-1$ divides $(n-1)!$ and n, contradicting $(n-1)! \equiv -1 \pmod{q}$. Thus the converse also follows.

Exercise 20.29: A universal divisor of a polynomial value

Each prime p in $\{2, 3, 7, 13, 19, 37\}$ has $p - 1 \mid 36$. If $p \mid n$, then $n^{37} - n \equiv 0 \pmod{p}$. Otherwise Fermat gives $n^{36} \equiv 1 \pmod{p}$, again yielding $n^{37} - n = n(n^{36} - 1) \equiv 0$. These primes are distinct, so their product divides the difference:

$$2 \cdot 3 \cdot 7 \cdot 13 \cdot 19 \cdot 37 = \boxed{383838}.$$

Exercise 20.30: A larger universal divisor

Also $5 - 1 = 4$ divides 36, so the same Fermat argument shows $5 \mid n^{37} - n$ for every integer n. Since 5 is coprime to 383838, the larger number

$$\boxed{5 \cdot 383838 = 1919190}$$

divides every such difference.

21 The Field of Quotients of an Integral Domain

Exercise 21.1: Quotients of Gaussian integers

The field is $\boxed{\mathbb{Q}(i) = \{a + bi : a, b \in \mathbb{Q}\}}$. To see that every quotient lies here, rationalize its denominator:

$$\frac{m + ni}{r + si} = \frac{(mr + ns) + (nr - ms)i}{r^2 + s^2}, \qquad r + si \neq 0.$$

Both resulting coefficients are rational. Conversely, if $a, b \in \mathbb{Q}$, choose a common nonzero integer denominator d with $da, db \in \mathbb{Z}$. Then $a + bi = ((da) + (db)i)/d$ is a quotient of Gaussian integers. These two inclusions establish the description.

Exercise 21.2: Quotients with an irrational square root

The answer is $\boxed{\mathbb{Q}(\sqrt{2}) = \{a + b\sqrt{2} : a, b \in \mathbb{Q}\}}$. A quotient rationalizes as

$$\frac{m + n\sqrt{2}}{r + s\sqrt{2}} = \frac{(mr - 2ns) + (nr - ms)\sqrt{2}}{r^2 - 2s^2}.$$

The denominator is nonzero whenever $r + s\sqrt{2} \neq 0$: otherwise irrationality of $\sqrt{2}$ forces $r = s = 0$. Thus the quotient has rational coefficients. Conversely, clearing the two rational denominators expresses any $a + b\sqrt{2}$ as a quotient of two members of $\mathbb{Z}[\sqrt{2}]$.

Exercise 21.3: Correcting the field-of-quotients definition

The missing condition is that *every element of* F be a quotient of elements of the embedded copy of D. Precisely, a field of quotients of D is a field F with an injective ring map $i : D \to F$ such that every $x \in F$ is $i(a)i(b)^{-1}$ for some $a, b \in D$, $b \neq 0$. Merely making all nonzero elements of D invertible allows larger fields such as $\mathbb{R}$ for $D = \mathbb{Z}$, and does not characterize the field of quotients.

Exercise 21.4: True or false: fraction fields

(a) True: every rational number is a quotient of integers.

(b) False: an irrational real number cannot be a quotient of integers.

(c) True: every real r is the quotient $r/1$ of real numbers.

(d) False: a quotient of real numbers is real, so it cannot equal i.

(e) True: if D is already a field, all its fractions $a/b = ab^{-1}$ remain in D.

(f) True: nonzero denominators must have nonzero products, and cancellation is used to establish transitivity of the equivalence relation.

(g) False: zero is an element of D and has no multiplicative inverse in a field.

(h) True: the image of a nonzero b has inverse represented by the fraction $1/b$.

(i) True: in a fraction field of D, the fractions a/b with $a, b \in D'$ form a subfield isomorphic to the fraction field of D'. Equality is the same cross-product equality in both domains.

(j) True: uniqueness of the fraction field up to isomorphism applies to $\mathbb{Z}$, and $\mathbb{Q}$ is one such field.

Exercise 21.5: Proper subdomains with the same fraction field

Take $D' = \mathbb{Z}$ and $D = \mathbb{Z}[1/2] = \{m/2^k : m \in \mathbb{Z},\ k \geq 0\}$. These are integral domains and D' is proper because $1/2 \in D \setminus D'$. The fraction field of either is $\mathbb{Q}$: fractions of elements of D are rational, and every rational is already a quotient of integers. Thus $F' = \mathbb{Q}$ works for both.

Exercise 21.6: Associativity of fraction addition

Write $[a, b]$ for the class $[(a, b)]$, and let $b, d, f \neq 0$. The addition formula gives

$$([a, b] + [c, d]) + [e, f] = [(ad + bc)f + (bd)e, bdf],$$

whereas

$$[a, b] + ([c, d] + [e, f]) = [a(df) + b(cf + de), bdf].$$

The numerators are both $adf + bcf + bde$ by the distributive and associative laws in D. The resulting ordered pairs are identical, hence so are their classes. This proves Part 2 of Step 3.

Exercise 21.7: The additive identity in the fraction construction

Using the prescribed operations,

$$[a, b] + [0, 1] = [a \cdot 1 + b \cdot 0, b \cdot 1] = [a, b].$$

Commutativity of addition, established in Part 1, also gives $[0, 1] + [a, b] = [a, b]$. Thus $[0, 1]$ is an additive identity, proving Part 3.

Exercise 21.8: The additive inverse of a fraction

Since $b \neq 0$, its square is nonzero, and $[-a, b]$ is an admissible fraction. Then

$$[a, b] + [-a, b] = [ab + b(-a), b^2] = [0, b^2] = [0, 1].$$

The last equality uses the defining equivalence: $0 \cdot 1 = b^2 \cdot 0$. Commutative addition supplies the equation in the other order. Hence $[-a, b]$ is the additive inverse, proving Part 4.

Exercise 21.9: Associativity of fraction multiplication

For three fractions, the multiplication rule gives

$$([a, b][c, d])[e, f] = [(ac)e, (bd)f], \qquad [a, b]([c, d][e, f]) = [a(ce), b(df)].$$

Associativity in D makes both numerators and both denominators identical. Nonzero denominator products follow from the integral-domain hypothesis. Therefore the classes are equal, proving Part 5.

Exercise 21.10: Commutativity of fraction multiplication

The products are $[a,b][c,d] = [ac, bd]$ and $[c,d][a,b] = [ca, db]$. Both coordinates agree because D is commutative. Thus the products are equal, which proves Part 6.

Exercise 21.11: Distributivity in the fraction construction

Let $x = [a,b]$, $y = [c,d]$, $z = [e,f]$. The prescribed formulas give

$$x(y+z) = [a(cf+de), bdf], \qquad xy + xz = [acbf + abde, b^2df].$$

The latter numerator is $b\,a(cf+de)$, so the two fractions are equivalent: their cross products are both $a(cf+de)b^2df$. Thus $x(y+z) = xy + xz$. Multiplication has already been proved commutative in Part 6, so

$$(y+z)x = x(y+z) = xy + xz = yx + zx.$$

This proves both distributive laws, Part 7.

Exercise 21.12: Partial quotients and a unity

Use classes $[r,t]$ of pairs in $R \times T$, with $[r,t] = [s,u]$ when $ru = st$. Transitivity follows by canceling a denominator from T; cancellation is valid because its elements are nonzero and not zero divisors. Products of denominators remain in T. The ordinary fraction addition and multiplication formulas are well defined by cross multiplication, just as in the text. They give a commutative ring by the same coordinate expansions as Exercises 21.6–21.11.

To view R inside it, fix any $t \in T$ and set $i(r) = [rt, t]$. The class is independent of t, because $(rt)u = (ru)t$. The formulas show that i preserves sums and products. If $i(r) = i(s)$, then $(r-s)t^2 = 0$; two cancellations of the regular element t give $r = s$. Thus i is an embedding.

(a) The element $e = [t,t]$ is independent of $t \in T$. It is a unity, since $[t,t][r,u] = [tr, tu] = [r,u]$ by cross multiplication. No unity in R was needed.

(b) For $u \in T$ and $t \in T$, the element $i(u) = [ut, t]$ has inverse $[t, ut]$, whose denominator lies in T. Their product is $[ut^2, ut^2] = e$. Hence every embedded element of T is a unit.

Exercise 21.13: Adjoining unity using regular powers

For a *nonzero* element a that is not a zero divisor, let $T = \{a^n : n \geq 1\}$. It is nonempty and multiplicatively closed. None of its elements is zero or a zero divisor: an equation $a^n x = 0$ permits n successive cancellations of a and forces $x = 0$. Exercise 21.12 therefore embeds R in the commutative ring $Q(R,T)$ with unity.

The qualification $a \neq 0$ is necessary for this particular argument: under the book's convention zero itself is not called a zero divisor, but its powers cannot serve as denominators. If the printed hypothesis is read as allowing only $a = 0$, use the general unity-adjoining construction of Exercise 19.30 instead. That construction applies to every ring, whereas localization additionally makes the chosen regular denominators invertible.

Exercise 21.14: Partial quotients of the ring modulo four

The allowed denominators $1, 3$ are already units in $\mathbb{Z}_4$, with $3^{-1} = 3$. Send $[r, t]$ to $rt^{-1} \in \mathbb{Z}_4$. Cross multiplication shows this is well defined and injective, and every residue occurs as $[r, 1]$. It is a ring isomorphism. Thus the partial-quotient ring has $\boxed{4}$ elements.

Exercise 21.15: Inverting powers of two

Send a formal fraction $[m, 2^n]$ to the real rational $m/2^n$. Cross multiplication makes this a well-defined injective ring map. Its image is

$$\boxed{\mathbb{Z}[1/2] = \{m/2^n : m \in \mathbb{Z},\ n \geq 0\}}.$$

Allowing $n = 0$ does not enlarge the image: $m = (2m)/2$. These are the dyadic rationals; denominators may contain only the prime 2 after reduction.

Exercise 21.16: Quotients of multiples of three

The map $[3m, 6^n] \mapsto 3m/6^n$ identifies the required ring with a subring of $\mathbb{Q}$. Every such fraction has denominator involving only 2 and 3, so its image is contained in $\mathbb{Z}[1/6]$. Conversely any $k/6^j$, $j \geq 0$, equals $6k/6^{j+1}$; the numerator $6k$ lies in $3\mathbb{Z}$ and the denominator is an allowed positive power. Therefore the image is exactly

$$\boxed{\mathbb{Z}[1/6] = \{k/6^j : k \in \mathbb{Z},\ j \geq 0\}}.$$

Exercise 21.17: The failure when denominators are zero divisors

The cross-product relation need not be transitive. In $R = \mathbb{Z}_6$, with $T = \{1, 2, 4\}$, consider

$$(0, 1), \qquad (0, 2), \qquad (3, 1).$$

The first two are related because $0 \cdot 2 = 1 \cdot 0$. The last two are related because $0 \cdot 1 = 2 \cdot 3 = 0$ modulo 6. But the first and third are not related, because $0 \cdot 1 = 0 \neq 3 = 1 \cdot 3$. Thus Step 1 already fails to define an equivalence relation. The transitivity proof would have canceled 2, which is a zero divisor. More general localization can be defined by a different relation, but then elements annihilated by denominators collapse to zero and the original ring need not embed.

22 Rings of Polynomials

Exercise 22.1: Addition and multiplication modulo eight

Add like powers, then reduce coefficients modulo 8:

$$f + g = 2x^2 - 3 = \boxed{2x^2 + 5}.$$

For the product, distribute before reducing:

$$fg = 8x^3 - 26x^2 + 28x - 10 = \boxed{6x^2 + 4x + 6} \quad \text{in } \mathbb{Z}_8[x].$$

The degree drops because the leading coefficient 8 vanishes modulo 8.

Exercise 22.2: Addition and multiplication in characteristic two

Here $f = g = x+1$. Thus $f+g = 2x+2 = \boxed{0}$. Expanding the product gives $fg = x^2+2x+1 = \boxed{x^2+1}$, since $2 = 0$ in $\mathbb{Z}_2$.

Exercise 22.3: Polynomial arithmetic modulo six

Addition gives

$$f + g = 5x^2 + 5x + 7 = \boxed{5x^2 + 5x + 1}.$$

Distributive multiplication over the integers gives

$$fg = 6x^4 + 13x^3 + 24x^2 + 17x + 12.$$

Reducing each coefficient modulo 6 gives $\boxed{x^3 + 5x}$. The expected degree 4 disappears because $2 \cdot 3 = 0$ in the coefficient ring.

Exercise 22.4: Polynomial arithmetic modulo five

After collecting terms,

$$f + g = 3x^4 + 2x^3 + 4x^2 + 5x + 6 = \boxed{3x^4 + 2x^3 + 4x^2 + 1}.$$

The unreduced product is

$$6x^7 + 12x^6 + 9x^5 + 10x^4 + 16x^3 + 22x^2 + 16x + 8.$$

Thus in $\mathbb{Z}_5[x]$,

$$\boxed{fg = x^7 + 2x^6 + 4x^5 + x^3 + 2x^2 + x + 3}.$$

Exercise 22.5: Counting short binary polynomials

Every polynomial of degree at most 3, including zero, is uniquely $a_0 + a_1x + a_2x^2 + a_3x^3$. Each coefficient has two choices in $\mathbb{Z}_2$, independently. The count is $\boxed{2^4 = 16}$. Allowing leading coefficients zero includes all smaller degrees.

Exercise 22.6: Counting short polynomials modulo five

Write the polynomial uniquely as $a_0 + a_1x + a_2x^2$, allowing leading zeros. Each of the three coefficients has five choices. The answer is $\boxed{5^3 = 125}$, including the zero polynomial.

Exercise 22.7: Evaluation at a real point

Evaluation means substituting the indicated input. Therefore $\phi_2(x^2 + 3) = 2^2 + 3 = \boxed{7}$.

Exercise 22.8: Evaluation at the imaginary unit

Substitute i and use $i^2 = -1$, $i^3 = -i$:

$$\phi_i(2x^3 - x^2 + 3x + 2) = 2(-i) - (-1) + 3i + 2 = \boxed{3 + i}.$$

Exercise 22.9: Evaluation of a product modulo seven

Evaluation preserves products, so evaluate the two factors separately at 3. They give $3^4 + 2 \cdot 3 = 87 \equiv 3$ and $3^3 - 3 \cdot 3^2 + 3 = 3$. Their product is $9 \equiv \boxed{2} \pmod 7$.

Exercise 22.10: Three evaluated factors modulo seven

At $x = 5$ the factors reduce modulo 7 to

$$5^2 + 2 = 27 \equiv 6, \quad 4 \cdot 5^2 + 3 = 103 \equiv 5, \quad 5^7 + 3 \cdot 5^2 + 1 \equiv 5 + 5 + 1 \equiv 4.$$

Here $5^7 \equiv 5$ by Fermat. Multiply the residues: $6 \cdot 5 \cdot 4 = 120 \equiv \boxed{1} \pmod 7$.

Exercise 22.11: Large exponents evaluated modulo seven

Since $4 \neq 0$ in $\mathbb{Z}_7$, Fermat gives $4^6 = 1$. Reduce the exponents: $106 \equiv 4$, $99 \equiv 3$, $53 \equiv 5 \pmod 6$. The residues $4^3 = 1$, $4^4 = 4$, $4^5 = 2$ in $\mathbb{Z}_7$ give

$$\phi_4(3x^{106} + 5x^{99} + 2x^{53}) = 3 \cdot 4 + 5 \cdot 1 + 2 \cdot 2 = 21 = \boxed{0}.$$

Exercise 22.12: Zeros over two elements

There are only two inputs. At 0, $x^2 + 1 = 1$; at 1, it is $1 + 1 = 0$ in $\mathbb{Z}_2$. Thus the complete zero set is $\boxed{\{1\}}$.

Exercise 22.13: Zeros of a cubic over seven elements

Evaluating at $0, 1, 2, 3, 4, 5, 6$ gives, respectively, residues $2, 5, 0, 0, 4, 4, 6$. Hence the zeros are exactly $\boxed{2, 3}$. In particular, a cubic may have a repeated root rather than three distinct roots.

Exercise 22.14: Zeros of a fifth-degree polynomial over five elements

For each $a \in \mathbb{Z}_5$, Fermat gives $a^5 = a$, so the value becomes $3a^3 + a^2 + 3a$. At $a = 0, 1, 2, 3, 4$ the residues are $0, 2, 4, 4, 0$. Therefore the complete zero set is $\boxed{\{0, 4\}}$.

Exercise 22.15: Zeros of a product

Since $\mathbb{Z}_7$ is a field, $f(a)g(a) = 0$ if and only if at least one factor is zero. Evaluating $f = x^3 + 2x^2 + 5$ at $0, \ldots, 6$ gives $5, 1, 0, 1, 3, 5, 6$, so its only zero is 2. Factoring $g = 3x^2 + 2x = x(3x + 2)$ gives zeros 0 and 4, because $3 \cdot 4 + 2 = 14 = 0$. Taking the union, the product has zeros $\boxed{0, 2, 4}$.

Exercise 22.16: Evaluation with prime-field exponent reduction

For nonzero $3 \in \mathbb{Z}_5$, $3^4 = 1$. The exponents satisfy $231 \equiv 3$, $117 \equiv 1$, $53 \equiv 1 \pmod 4$. Hence

$$\phi_3(x^{231} + 3x^{117} - 2x^{53} + 1) = 3^3 + 3 \cdot 3 - 2 \cdot 3 + 1 = 31 \equiv \boxed{1} \pmod 5.$$

Exercise 22.17: All zeros of a large-degree polynomial

At $x = 0$ the polynomial is zero, since every exponent is positive. For $x \neq 0$, use $x^4 = 1$ in $\mathbb{Z}_5$ and reduce exponents modulo 4:

$$2x^{219} + 3x^{74} + 2x^{57} + 3x^{44} = 2x^3 + 3x^2 + 2x + 3.$$

This cubic factors as $(x^2 + 1)(2x + 3)$. Its zeros are $x = 2, 3$ from $x^2 = -1 = 4$, and $x = 1$ from $2x = -3 = 2$. Including the separately checked zero input, all zeros are $\boxed{0, 1, 2, 3}$. The exponent reduction was not applied blindly to $x = 0$.

Exercise 22.18: Correcting the definition of a polynomial

Add the condition that only finitely many coefficients are nonzero. A polynomial over R is a formal expression $\sum_{i \geq 0} a_i x^i$ with $a_i \in R$ and $a_i = 0$ for all sufficiently large i. Without that finiteness condition, the expression is a formal power series rather than necessarily a polynomial.

Exercise 22.19: The definition of a zero

The stated definition is correct: $a \in F$ is a zero of f when evaluation at a gives $\phi_a(f) = f(a) = 0$. More generally, if $F \subseteq E$, a zero in E is an element $a \in E$ satisfying the same equation under $\phi_a : F[x] \to E$. The domain of the evaluation map is the polynomial ring $F[x]$.

Exercise 22.20: Regrouping a polynomial in two variables

Collect the displayed expression by powers of x rather than powers of y. The x^4 terms contribute $(y+1)x^4$; the x^3 term is $3y^3x^3$; the x^2 terms give $(y^2-3)x^2$; and the linear terms give $(2y^3-6y^2-2y)x$. The remaining constant in x is $y^2 + 2$. Thus in $(\mathbb{Q}[y])[x]$ it is

$$\boxed{(y + 1)x^4 + 3y^3x^3 + (y^2 - 3)x^2 + (2y^3 - 6y^2 - 2y)x + (y^2 + 2)}.$$

Exercise 22.21: Six kernel elements for evaluation at five

The kernel consists of polynomials with rational coefficients vanishing at 5. Six distinct examples are

$$0, \quad x-5, \quad 2(x-5), \quad (x-5)^2, \quad x(x-5), \quad (x-5)^3.$$

Every displayed expression has value zero after substituting $x = 5$, which directly verifies membership. Their different coefficients or degrees show that the six polynomials are distinct.

Exercise 22.22: A positive-degree polynomial unit

In $\mathbb{Z}_4[x]$, take $u = 1 + 2x$. Its square is $u^2 = 1 + 4x + 4x^2 = 1$. Therefore u is a unit, with inverse itself, and has degree $1 > 0$. This phenomenon uses the zero-divisor coefficient 2.

Exercise 22.23: True or false: polynomial rings

(a) True: a formal polynomial is zero exactly when each coefficient is zero.

(b) True: commutative coefficient products allow the convolution formula for multiplication to be reversed.

(c) True: the product of nonzero leading coefficients in a domain remains nonzero, so nonzero polynomials cannot multiply to zero.

(d) True: regard a pair of nonzero zero-divisor coefficients as constant polynomials.

(e) False: multiplication cannot create a term of degree greater than $3 + 4 = 7$.

(f) False: in $\mathbb{Z}_4[x]$, $(2x^3)(2x^4) = 0$, so the degree need not be 7.

(g) True: evaluation is multiplicative, giving $(fg)(a) = f(a)g(a) = 0$.

(h) True: if $fg = 1$ over a field, degree additivity forces both degrees to be zero; the units are nonzero constants.

(i) True whenever x is interpreted as the polynomial indeterminate in a ring with unity: multiplication by x shifts each coefficient one place, so $xf = 0$ forces $f = 0$. The same coefficient-shift statement holds formally without a unity.

(j) False: $2x$ is a nonconstant zero divisor in $\mathbb{Z}_4[x]$, since $(2x)2 = 0$.

Exercise 22.24: A polynomial ring over a domain is a domain

The standard coefficient operations give a commutative ring $D[x]$ with unity the constant polynomial $1_D \neq 0$. Let f, g be nonzero, with leading terms $a_m x^m, b_n x^n$. The coefficient of x^{m+n} in fg is $a_m b_n$, which is nonzero because D has no zero divisors. Hence $fg \neq 0$. Thus $D[x]$ has no nonzero zero divisors and is an integral domain.

Exercise 22.25: Units in three polynomial rings

(a) If $fg = 1$ in $D[x]$, then $\deg f + \deg g = \deg 1 = 0$ by the leading-coefficient argument. Both degrees must be zero, so f is a constant whose value is a unit in D. Conversely every such constant has its constant inverse in $D[x]$.

(b) The units of $\mathbb{Z}$ are ± 1, so the units of $\mathbb{Z}[x]$ are precisely the two constant polynomials $1, -1$.

(c) Every nonzero element of the field $\mathbb{Z}_7$ is a unit. Thus the units of $\mathbb{Z}_7[x]$ are precisely the constant polynomials $1, 2, 3, 4, 5, 6$.

Exercise 22.26: Left distributivity from coefficient formulas

Write $f = \sum a_i x^i$, $g = \sum b_j x^j$, and $h = \sum c_j x^j$, extending coefficient lists by zeros. The coefficient of x^n in $f(g + h)$ is

$$\sum_{i+j=n} a_i(b_j + c_j) = \sum_{i+j=n} a_i b_j + \sum_{i+j=n} a_i c_j.$$

Distributivity in R gives the equality and its abelian addition allows regrouping the finite sum. The right side is exactly the coefficient of x^n in $fg + fh$. Equality for every coefficient proves $f(g + h) = fg + fh$.

Exercise 22.27: Formal differentiation as an additive map

(a) Coefficientwise differentiation gives $D(f+g) = Df+Dg$, so D is an additive-group homomorphism. It is not a ring homomorphism: $D(x^2) = 2x$ but $D(x)D(x) = 1$. These formal polynomials are distinct.

(b) If $Df = 0$, every coefficient ka_k with $k \geq 1$ vanishes. In characteristic zero the scalar k is nonzero and invertible in the field, so $a_k = 0$ for $k \geq 1$. Hence the kernel is exactly the constant polynomials F.

(c) For $g = \sum_{k=0}^{n} b_k x^k$, set $f = \sum_{k=0}^{n} (b_k/(k+1)) x^{k+1}$. All denominators are nonzero in the characteristic-zero field. Then $Df = g$, so the image is all of $F[x]$.

Exercise 22.28: Evaluation in several variables

(a) For $\boldsymbol{a} = (a_1, \ldots, a_n) \in E^n$, define

$$\phi_{\boldsymbol{a}}\left(\sum_{\boldsymbol{k}} c_{\boldsymbol{k}} x_1^{k_1} \cdots x_n^{k_n}\right) = \sum_{\boldsymbol{k}} c_{\boldsymbol{k}} a_1^{k_1} \cdots a_n^{k_n}.$$

All sums are finite. Distributing products in the commutative field E proves that this map preserves addition and multiplication. It fixes the coefficients in F and sends x_i to a_i.

(b) Substitution yields $(-3)^2 2^3 + 3(-3)^4 2 = 9 \cdot 8 + 3 \cdot 81 \cdot 2 = \boxed{558}$.

(c) A zero in E^n is a tuple $\boldsymbol{a}$ with $\phi_{\boldsymbol{a}}(f) = f(a_1, \ldots, a_n) = 0$. A tuple, rather than a single scalar, is required.

Exercise 22.29: Pointwise operations on all functions

The zero function is the additive identity, and $(-\phi)(r) = -\phi(r)$ provides additive inverses. Addition is commutative and associative because these identities hold at every value r in R. For multiplication,

$$((\phi\psi)\theta)(r) = (\phi(r)\psi(r))\theta(r) = \phi(r)(\psi(r)\theta(r)) = (\phi(\psi\theta))(r).$$

Likewise evaluating either distributive law at r gives the corresponding law in R, for example

$$(\phi(\psi + \theta))(r) = \phi(r)(\psi(r) + \theta(r)) = (\phi\psi + \phi\theta)(r).$$

Closure is automatic because the operations in R have values in R. Thus the function set is a ring. Its multiplication is pointwise multiplication, as stipulated, not composition.

Exercise 22.30: Polynomial functions form a smaller kind of ring

(a) The zero function is polynomial. If functions u, v are represented by polynomials f, g, then $u - v$ is represented by $f - g$, and uv by fg. The subring test proves that P_F is a subring of the full function ring.

(b) If F has q elements, there are only q^q functions $F \to F$, hence only finitely many polynomial functions. But the formal polynomials $1, x, x^2, \ldots$ are infinitely many distinct members of $F[x]$. A bijection, and thus a ring isomorphism, is impossible. Distinct formal polynomials can induce the same function; over $\mathbb{Z}_p$, for instance, x^p and x do so.

Exercise 22.31: All functions on a finite field are polynomial

(a) Each input can be assigned any output independently. There are $\boxed{2^2 = 4}$ functions $\mathbb{Z}_2 \to \mathbb{Z}_2$, and $\boxed{3^3 = 27}$ functions $\mathbb{Z}_3 \to \mathbb{Z}_3$.

(b) Recording the two, or three, function values is an additive-group isomorphism to a Cartesian power. Thus the two groups are $\boxed{\mathbb{Z}_2 \times \mathbb{Z}_2}$ and $\boxed{\mathbb{Z}_3 \times \mathbb{Z}_3 \times \mathbb{Z}_3}$ respectively. Addition in each coordinate is the field addition.

(c) List $F = \{a_1, \ldots, a_q\}$. Given any function $u : F \to F$, define

$$f(x) = \sum_{i=1}^{q} u(a_i) \prod_{j \neq i} \frac{x - a_j}{a_i - a_j}.$$

Every denominator is nonzero because the a_i are distinct. At $x = a_k$, all summands but the kth contain a zero factor; the kth product equals 1. Therefore $f(a_k) = u(a_k)$ for every k. This polynomial represents u, proving $F^F = P_F$.

23 Factorization of Polynomials over a Field

Exercise 23.1: Polynomial division 1

Work in $\mathbb{Z}_7[x]$, reducing coefficients after each subtraction. The inverse of the divisor's leading coefficient is 1. Repeatedly subtract the multiple that cancels the current highest power. The complete long-division steps are

term added to quotient	new remainder
x^4	$x^5 + 3x^4 + 4x^2 + 4x + 2$
x^3	$x^4 + 3x^3 + 4x^2 + 4x + 2$
x^2	$x^3 + 4x + 2$
x	$5x^2 + 2$
5	$4x + 3$

Thus $\boxed{q(x) = x^4 + x^3 + x^2 + x + 5}$ and $\boxed{r(x) = 4x + 3}$. The final remainder has degree below the divisor's degree; reversing the subtractions verifies $f = gq + r$.

Exercise 23.2: Polynomial division 2

Work in $\mathbb{Z}_7[x]$, reducing coefficients after each subtraction. The inverse of the divisor's leading coefficient is 5. Repeatedly subtract the multiple that cancels the current highest power. The complete long-division steps are

term added to quotient	new remainder
$5x^4$	$x^4 + 4x^2 + 4x + 2$
$5x^2$	$4x^3 + 5x^2 + 4x + 2$
$6x$	$x + 2$

Thus $\boxed{q(x) = 5x^4 + 5x^2 + 6x}$ and $\boxed{r(x) = x + 2}$. The final remainder has degree below the divisor's degree; reversing the subtractions verifies $f = gq + r$.

Exercise 23.3: Polynomial division 3

Work in $\mathbb{Z}_{11}[x]$, reducing coefficients after each subtraction. The inverse of the divisor's leading coefficient is 6. Repeatedly subtract the multiple that cancels the current highest power. The complete long-division steps are

term added to quotient	new remainder
$6x^4$	$3x^4 + 3x + 6$
$7x^3$	$4x^3 + 3x + 6$
$2x^2$	$9x^2 + 3x + 6$
$10x$	$4x + 6$
2	4

Thus $\boxed{q(x) = 6x^4 + 7x^3 + 2x^2 + 10x + 2}$ and $\boxed{r(x) = 4}$. The final remainder has degree below the divisor's degree; reversing the subtractions verifies $f = gq + r$.

Exercise 23.4: Polynomial division 4

Work in $\mathbb{Z}_{11}[x]$, reducing coefficients after each subtraction. The inverse of the divisor's leading coefficient is 9. Repeatedly subtract the multiple that cancels the current highest power. The complete long-division steps are

term added to quotient	new remainder
$9x^2$	$3x^3 + x^2$
$5x$	$6x^2 + x$
10	2

Thus $\boxed{q(x) = 9x^2 + 5x + 10}$ and $\boxed{r(x) = 2}$. The final remainder has degree below the divisor's degree; reversing the subtractions verifies $f = gq + r$.

Exercise 23.5: All multiplicative generators modulo 5

The group has order 4. The element 2 has this full order: it satisfies $2^4 = 1$, and the checks $2^2 \equiv 4 \pmod 5$ exclude every proper divisor of the order (check exponents $(4)/q$ for each prime q dividing 4). In a cyclic group of order 4, the generators are exactly g^k with $\gcd(k, 4) = 1$. Here those exponents are $1, 3$, yielding the complete generator set

$$\boxed{\{2, 3\}}.$$

Exercise 23.6: All multiplicative generators modulo 7

The group has order 6. The element 3 has this full order: it satisfies $3^6 = 1$, and the checks $3^3 \equiv 6, 3^2 \equiv 2 \pmod 7$ exclude every proper divisor of the order (check exponents $(6)/q$ for each prime q dividing 6). In a cyclic group of order 6, the generators are exactly g^k with $\gcd(k, 6) = 1$. Here those exponents are $1, 5$, yielding the complete generator set

$$\boxed{\{3, 5\}}.$$

Exercise 23.7: All multiplicative generators modulo 17

The group has order 16. The element 3 has this full order: it satisfies $3^{16} = 1$, and the checks $3^8 \equiv 16 \pmod{17}$ exclude every proper divisor of the order (check exponents $(16)/q$ for each prime q dividing 16). In a cyclic group of order 16, the generators are exactly g^k with $\gcd(k, 16) = 1$. Here those exponents are $1, 3, 5, 7, 9, 11, 13, 15$, yielding the complete generator set

$$\boxed{\{3, 5, 6, 7, 10, 11, 12, 14\}}.$$

Exercise 23.8: All multiplicative generators modulo 23

The group has order 22. The element 5 has this full order: it satisfies $5^{22} = 1$, and the checks $5^{11} \equiv 22, 5^2 \equiv 2 \pmod{23}$ exclude every proper divisor of the order (check exponents $(22)/q$ for each prime q dividing 22). In a cyclic group of order 22, the generators are exactly g^k with $\gcd(k, 22) = 1$. Here those exponents are $1, 3, 5, 7, 9, 13, 15, 17, 19, 21$, yielding the complete generator set

$$\boxed{\{5, 7, 10, 11, 14, 15, 17, 19, 20, 21\}}.$$

Exercise 23.9: Four linear factors modulo five

Since $4 = -1$ in $\mathbb{Z}_5$,

$$x^4 + 4 = x^4 - 1 = (x^2 - 1)(x^2 + 1).$$

Now $x^2 - 1 = (x - 1)(x + 1)$ and $x^2 + 1 = x^2 - 4 = (x - 2)(x + 2)$. Therefore the requested linear factorization is

$$\boxed{(x - 1)(x + 1)(x - 2)(x + 2)}.$$

Exercise 23.10: Three linear factors modulo seven

Substitution shows that $x = -1$ is a zero. Division gives

$$x^3 + 2x^2 + 2x + 1 = (x + 1)(x^2 + x + 1).$$

In $\mathbb{Z}_7$, $(x - 2)(x + 3) = x^2 + x - 6 = x^2 + x + 1$. Thus

$$\boxed{x^3 + 2x^2 + 2x + 1 = (x + 1)(x - 2)(x + 3)}.$$

Expansion checks all coefficients modulo 7.

Exercise 23.11: A nonmonic cubic modulo eleven

The residues $3, 4, -3$ are zeros, suggesting the three factors. Their product expands as

$$2(x - 4)(x - 3)(x + 3) = 2(x - 4)(x^2 - 9) = 2x^3 - 8x^2 - 18x + 72.$$

Reducing modulo 11 gives $2x^3+3x^2-7x-5$, as required. Hence this is the complete linear factorization, including its leading coefficient 2.

Exercise 23.12: A reducible cubic modulo five

The value at $x = 2$ is $8 + 4 + 3 = 15 = 0$ modulo 5, so the cubic is not irreducible. Division by $x - 2$ gives $x^2 + 2x + 1 = (x + 1)^2$. Therefore

$$\boxed{x^3 + 2x + 3 = (x - 2)(x + 1)^2} \quad \text{in } \mathbb{Z}_5[x].$$

All three linear factors are irreducible, with a repeated factor explicitly recorded.

Exercise 23.13: An irreducible cubic modulo five

A cubic over a field is reducible exactly when it has a zero in the field. The values of $2x^3 + x^2 + 2x + 2$ at $0, 1, 2, 3, 4$ are $2, 2, 1, 1, 4$ in $\mathbb{Z}_5$, none zero. It is therefore irreducible. If a monic irreducible factor is desired, write it as

$$\boxed{2(x^3 + 3x^2 + x + 1)}.$$

The scalar 2 is a unit, and the cubic factor is irreducible; there is no factorization into positive lower degrees.

Exercise 23.14: A quadratic over three fields

The discriminant of $x^2 + 8x - 2$ is $8^2 - 4(1)(-2) = 72$. Its roots are $-4 \pm 3\sqrt{2}$, neither rational. A quadratic over a field is reducible if and only if it has a root there, so it is irreducible over $\mathbb{Q}$. Both roots are real, however, giving

$$x^2 + 8x - 2 = (x + 4 - 3\sqrt{2})(x + 4 + 3\sqrt{2}).$$

Thus it is reducible over $\mathbb{R}$ and consequently over $\mathbb{C}$.

Exercise 23.15: A quadratic with negative discriminant

For $x^2 + 6x + 12$, the discriminant is $36 - 48 = -12$. There is no real root, hence no rational root. It is irreducible over both $\mathbb{Q}$ and $\mathbb{R}$. Over $\mathbb{C}$, its roots are $-3 \pm i\sqrt{3}$, so

$$x^2 + 6x + 12 = (x + 3 - i\sqrt{3})(x + 3 + i\sqrt{3})$$

is a factorization and it is reducible there.

Exercise 23.16: A cubic over the rationals

A reducible cubic over $\mathbb{Q}$ would have a rational root. The rational-root theorem for the monic polynomial $x^3 + 3x^2 - 8$ limits candidates to $\pm 1, \pm 2, \pm 4, \pm 8$. Their values, in that order, are

$$-4, -6, 12, -4, 104, -24, 696, -328,$$

all nonzero. Thus there is no rational root and the cubic is irreducible over $\mathbb{Q}$.

Exercise 23.17: A quartic with no quadratic factor

The only possible rational roots of the monic integer polynomial are ± 1, and both give -20, so no linear factor exists. By Gauss's lemma, any remaining rational factorization could be taken as

$$(x^2 + ax + b)(x^2 + cx + d), \qquad a, b, c, d \in \mathbb{Z}.$$

Comparing the x^3 and constant coefficients gives $c = -a$ and $bd = 1$, so $(b, d) = (1, 1)$ or $(-1, -1)$. The x^2 coefficient is $b + d - a^2 = -22$. In the first case this forces $a^2 = 24$; in the second it forces $a^2 = 20$. Neither is an integer square. Thus there is no quadratic factorization either, and $x^4 - 22x^2 + 1$ is irreducible over $\mathbb{Q}$.

Exercise 23.18: An Eisenstein quadratic

For $x^2 - 12$, use the prime $p = 3$. It does not divide the leading coefficient 1, divides all other coefficients (including the zero linear coefficient), and $3^2 = 9$ does not divide -12. Eisenstein's criterion therefore applies and proves irreducibility over $\mathbb{Q}$.

Exercise 23.19: An Eisenstein cubic

Use $p = 3$: it does not divide the leading coefficient 8, it divides the other coefficients $6, -9, 24$, and 9 does not divide the constant coefficient 24. Thus all Eisenstein hypotheses hold and the cubic is irreducible over $\mathbb{Q}$.

Exercise 23.20: An Eisenstein polynomial of degree ten

Use $p = 3$ for $4x^{10} - 9x^3 + 24x - 18$. Although $3 \nmid 4$ and 3 divides all lower coefficients, $3^2 = 9$ divides the constant coefficient -18. This prime fails the square condition. No other prime divides all the lower nonzero coefficients, since their gcd is 3. Thus the polynomial satisfies no Eisenstein criterion in its displayed form. Failure of the criterion alone does not decide reducibility.

Exercise 23.21: Another Eisenstein polynomial of degree ten

For $2x^{10} - 25x^3 + 10x^2 - 30$, the prime 5 divides every lower coefficient but not the leading coefficient 2. Its square 25 does not divide -30. Thus Eisenstein applies with $p = 5$, proving irreducibility over $\mathbb{Q}$.

Exercise 23.22: All rational zeros of a quartic

The rational-root theorem permits only fractions with numerator dividing 10 and denominator dividing 6. Testing candidates finds $x = -5/2$ and $x = 2/3$. Exact division, or multiplication, gives

$$6x^4 + 17x^3 + 7x^2 + x - 10 = (2x + 5)(3x - 2)(x^2 + x + 1).$$

The final quadratic has discriminant $1 - 4 = -3$, so it has no rational zero. Consequently the complete rational zero set is $\boxed{\{-5/2,\ 2/3\}}$. The displayed factorization checks the candidate search and excludes further rational roots.

Exercise 23.23: Correcting irreducibility

A nonconstant polynomial $f \in F[x]$ is irreducible over F if it cannot be written as $f = gh$ with *both* $g, h \in F[x]$ of positive degree. The printed definition forbids even $f = 1 \cdot f$, which every polynomial has. Trivial factorizations by nonzero constant units must be allowed.

Exercise 23.24: An equivalent irreducibility definition

The stated definition is correct. A nonconstant f is nonzero. In any factorization $f = gh$, both factors are nonzero, so a factor belonging to F is a nonzero constant and hence a unit in $F[x]$. Requiring such a factor in every factorization is equivalent to forbidding a product of two positive-degree polynomials.

Exercise 23.25: True or false: irreducibility and root bounds

(a) True: a degree-one polynomial cannot be a product of two positive-degree polynomials.

(b) True for the same reason; the nonzero scalar factor 3 is a unit in $\mathbb{Q}[x]$.

(c) True: $\sqrt{3}$ is irrational, so this quadratic has no rational root.

(d) False: in $\mathbb{Z}_7$, $2^2 + 3 = 7 = 0$, so $x^2 + 3$ is reducible.

(e) True: degree additivity forces any polynomial unit over a field to be a nonzero constant.

(f) True: the source repeats the same assertion as in (e), so the same reasoning applies.

(g) True: a polynomial of degree n is nonzero and has at most n distinct zeros over a field.

(h) True: regard it as the same nonzero degree-n polynomial over the extension field E and apply the root bound there.

(i) True: $ax + b$, $a \neq 0$, has zero $-b/a \in F$.

(j) False: over an infinite field such as $\mathbb{Q}$, the zero polynomial vanishes at infinitely many inputs. The assertion becomes true if one requires the polynomial to be nonzero.

Exercise 23.26: Primes for which a specified linear factor divides

The factor theorem says that $x + 2 = x - (-2)$ divides the polynomial modulo p exactly when its value at -2 is zero modulo p. This value is

$$16 - 8 + 4 + 2 + 1 = 15.$$

Thus p must divide 15. The prime possibilities are exactly $\boxed{3, 5}$, and both do satisfy the condition.

Exercise 23.27: Irreducible quadratics over two elements

Every monic quadratic has form $x^2 + ax + b$. It has no root at 0 only if $b = 1$; then its value at 1 is a, which is nonzero only if $a = 1$. Thus the only irreducible quadratic is $\boxed{x^2 + x + 1}$. There is no further nonzero scalar multiple to list because 1 is the only nonzero scalar in $\mathbb{Z}_2$.

Exercise 23.28: Irreducible cubics over two elements

A cubic is irreducible exactly when it has no field root. Write it $x^3 + ax^2 + bx + c$. Avoiding root 0 requires $c = 1$; avoiding root 1 requires $a + b \neq 0$, so $(a, b) = (1, 0)$ or $(0, 1)$. The complete list is

$$\boxed{x^3 + x^2 + 1, \qquad x^3 + x + 1}.$$

Exercise 23.29: Irreducible quadratics over three elements

Test the monic quadratics $x^2 + ax + b$ at $0, 1, 2$. The monic ones with no zero are exactly

$$x^2 + 1, \quad x^2 + x + 2, \quad x^2 + 2x + 2.$$

Their respective value triples are $(1, 2, 2)$, $(2, 1, 2)$, and $(2, 2, 1)$. All quadratic leading coefficients are 1 or 2, and multiplying by a nonzero constant preserves irreducibility. Thus the full answer is these three polynomials and twice each of them, six in total.

Exercise 23.30: Irreducible cubics over three elements

A cubic is irreducible exactly when none of $0, 1, 2$ is a zero. The monic list, together with values at these inputs, is

$h(x)$	$h(0)$	$h(1)$	$h(2)$
$x^3 + 2x^2 + 1$	1	1	2
$x^3 + 2x^2 + x + 1$	1	2	1
$x^3 + 2x + 1$	1	1	1
$x^3 + x^2 + 2x + 1$	1	2	2
$x^3 + x^2 + 2$	2	1	2
$x^3 + x^2 + x + 2$	2	2	1
$x^3 + 2x + 2$	2	2	2
$x^3 + 2x^2 + 2x + 2$	2	1	1

.

To see completeness, write a monic cubic as $x^3 + ax^2 + bx + c$. Necessarily $c = 1$ or 2; testing the nine pairs (a, b) for each c and discarding zeros at 1 or 2 gives precisely these eight rows. The full list of irreducible cubics consists of these eight polynomials and their multiples by 2, sixteen in all.

Exercise 23.31: Counting irreducible quadratics over a prime field

There are p^2 monic quadratics. A reducible monic quadratic is uniquely $(x-a)(x-b)$ for an unordered pair of field elements, allowing $a = b$. There are $\binom{p}{2} + p = p(p+1)/2$ such pairs. Therefore the number of monic irreducible quadratics is

$$p^2 - \frac{p(p+1)}{2} = \frac{p(p-1)}{2}.$$

Each has $p - 1$ distinct nonzero scalar multiples, and every nonmonic quadratic is a unique scalar times a monic one. Thus the requested count of *all* irreducible quadratics is $\boxed{p(p-1)^2/2}$.

Exercise 23.32: Synopsis of the bound on roots

Each distinct zero supplies a distinct linear factor by the factor theorem, and repeatedly removing these factors lowers the degree by one each time; a nonzero degree-n polynomial cannot supply more than n such factors.

Exercise 23.33: Synopsis of cyclicity of finite field subgroups

By the finite abelian group theorem, write the finite multiplicative subgroup with invariant factors $d_1 \mid \cdots \mid d_r$. Every element then satisfies $x^{d_r} = 1$. The polynomial root bound gives $|G| \le d_r$, while the product of invariant factors is $|G| \ge d_r$. Equality forces all earlier factors to be trivial, so the group is cyclic.

Exercise 23.34: A reducible prime-power polynomial

In $\mathbb{Z}_p$, Fermat gives $a^p = a$, and all intermediate binomial coefficients of exponent p vanish. Hence

$$(x + a)^p = x^p + a^p = x^p + a.$$

Since $p \geq 2$, this is a product of p linear factors, so it is not irreducible. This includes $a = 0$, when the polynomial is x^p.

Exercise 23.35: The reciprocal polynomial

Let $g(x) = a_n + a_{n-1}x + \cdots + a_0x^n$. Since $a \neq 0$, substitute a^{-1} and factor out a^{-n}:

$$g(a^{-1}) = a^{-n}(a_0 + a_1a + \cdots + a_na^n) = a^{-n}f(a) = 0.$$

Thus $1/a$ is a zero of the reciprocal polynomial, with the nonzero assumption justifying all inverse powers.

Exercise 23.36: The remainder theorem

Division by $x - a$ gives $f(x) = (x - a)q(x) + r(x)$, where $r = 0$ or $\deg r < 1$. Thus r is a constant. Evaluating at a yields $f(a) = 0 \cdot q(a) + r = r$. Hence the remainder is exactly the constant polynomial $\boxed{f(a)}$.

Exercise 23.37: Reduction of integer coefficients

(a) Reduce each coefficient modulo m. Sums are preserved coefficientwise; products are preserved because each coefficient of a product is a finite sum of coefficient products, and integer reduction preserves both operations. Every polynomial over $\mathbb{Z}_m$ can be lifted by choosing integer representatives for its finitely many coefficients, so the map is onto.

(b) Suppose instead that f is reducible over $\mathbb{Q}$. By Gauss's lemma it has a factorization $f = gh$ in $\mathbb{Z}[x]$ with $0 < \deg g, \deg h < n$. The leading coefficient of f remains nonzero modulo m by the degree hypothesis. Because it is the product of the leading coefficients of g, h, neither of those can become zero modulo m. Their reduced degrees are therefore unchanged. Reducing $f = gh$ then gives the forbidden factorization into two degrees below n, a contradiction. Thus f is irreducible over $\mathbb{Q}$.

(c) Modulo 5, the polynomial becomes $x^3 + 2x + 1$. Its values at $0, 1, 2, 3, 4$ are $1, 4, 3, 4, 3$, none zero. As a cubic over the field $\mathbb{Z}_5$, it is irreducible, and its degree remains 3. Part (b) proves that $x^3 + 17x + 36$ is irreducible over $\mathbb{Q}$.

24 Noncommutative Examples

Exercise 24.1: Addition in a three-element group algebra

Add the coefficients of each basis element, reducing in $\mathbb{Z}_5$:

$$(2e + 3a + 0b) + (4e + 2a + 3b) = 6e + 5a + 3b = \boxed{e + 0a + 3b}.$$

The group elements are formal basis labels; coefficients are combined only when their labels agree.

Exercise 24.2: Multiplication in a three-element group algebra

Since G is cyclic of order 3, its multiplication satisfies $a^2 = b$, $ab = ba = e$, and $b^2 = a$. Distribute in the given order:

$$(2e + 3a)(4e + 2a + 3b) = 8e + 4a + 6b + 12a + 6b + 9e.$$

Collecting coefficients gives $17e + 16a + 12b$. Reduction modulo 5 yields $\boxed{2e + a + 2b}$.

Exercise 24.3: A fourth power in the group algebra

Put $S = e + a + b$. For each fixed product label, exactly three of the nine terms in S^2 contribute to it, so $S^2 = 3S$. It follows that $S^4 = 9S^2 = 27S$. Therefore

$$(3S)^4 = 3^4 \cdot 27S = 2187S = 2S \quad \text{over } \mathbb{Z}_5.$$

The answer is $\boxed{2e + 2a + 2b}$.

Exercise 24.4: An ordered quaternion product

Use $i^2 = j^2 = k^2 = -1$, $ij = k$, $jk = i$, $ki = j$, and reverse these last three products with a minus sign. Then

$$(i + 3j)(4 + 2j - k) = 4i + 2k + j + 12j - 6 - 3i.$$

Collecting the four real coordinates gives $\boxed{-6 + i + 13j + 2k}$. Keeping the order of factors avoids the wrong signs in $ik = -j$ and $jk = i$.

Exercise 24.5: Quaternion powers and multiplication

Reduce the powers first: $i^2 = -1$, $j^3 = -j$, and $i^5 = i$. Thus

$$i^2 j^3 k j i^5 = (-1)(-j)kji = (jk)ji = iji = ki = \boxed{j}.$$

Each reassociation uses associativity, and no two noncommuting factors have been interchanged.

Exercise 24.6: The inverse of a pure quaternion

For $q = i + j$, its conjugate is $\overline{q} = -i - j$ and its squared norm is $1^2 + 1^2 = 2$. Hence

$$\boxed{(i+j)^{-1} = \frac{-i-j}{2}}.$$

For a direct check, $(i+j)^2 = i^2 + ij + ji + j^2 = -2$, so multiplication by the displayed inverse gives 1 on either side.

Exercise 24.7: The inverse of a quaternion product

Multiply the factors before inverting:

$$(1+3i)(4j+3k) = 4j + 3k + 12k + 9ik = -5j + 15k.$$

The squared norm is $25 + 225 = 250$, and the conjugate is $5j - 15k$. Therefore

$$\boxed{[(1+3i)(4j+3k)]^{-1} = \frac{j-3k}{50}}.$$

Indeed a pure quaternion $v = -5j + 15k$ has $v^2 = -250$, so $v(-v/250) = 1 = (-v/250)v$.

Exercise 24.8: A product in the group algebra of the symmetric group

Discard the zero coefficients. The factors are $A = \rho_1 + \mu_2 + \mu_3$ and $B = \rho_0 + \rho_1 + \mu_1 + \mu_3$. Using the source's right-to-left permutation composition, the required twelve products are

	ρ_0	ρ_1	μ_1	μ_3
ρ_1	ρ_1	ρ_2	μ_3	μ_2
μ_2	μ_2	μ_3	ρ_2	ρ_1
μ_3	μ_3	μ_1	ρ_1	ρ_0

.

The multiplicities of $\rho_0, \rho_1, \rho_2, \mu_1, \mu_2, \mu_3$ are $1, 3, 2, 1, 2, 3$. Coefficients are in $\mathbb{Z}_2$, so even multiplicities disappear and odd ones become 1. Thus

$$\boxed{AB = \rho_0 + \rho_1 + 0\rho_2 + \mu_1 + 0\mu_2 + \mu_3}.$$

Exercise 24.9: The center of the nonzero quaternion group

Every nonzero real scalar commutes with every quaternion, so $\mathbb{R} \setminus \{0\}$ is contained in the center. Conversely let $q = a + bi + cj + dk$ be central. Comparing

$$qi = -b + ai + dj - ck, \qquad iq = -b + ai - dj + ck$$

gives $c = d = 0$. Then comparing $(a+bi)j = aj + bk$ and $j(a+bi) = aj - bk$ gives $b = 0$. Therefore $q = a$ is real and nonzero. The center is exactly $\boxed{\mathbb{R} \setminus \{0\}}$ under multiplication.

Exercise 24.10: Two other complex planes inside the quaternions

Take $F_j = \{a + bj : a, b \in \mathbb{R}\}$ and $F_k = \{a + bk : a, b \in \mathbb{R}\}$. Because $j^2 = k^2 = -1$, their induced addition and multiplication obey exactly the complex-number formulas. The maps $a + bi \mapsto a + bj$ and $a + bi \mapsto a + bk$ are bijective ring homomorphisms from $\mathbb{C}$, hence field isomorphisms. These two subsets differ from each other and from the usual $\{a + bi\}$: for example j belongs only to F_j among these three planes.

Exercise 24.11: True or false: noncommutative examples

(a) False: $E_{11}E_{22} = 0$ in $M_2(F)$, with both factors nonzero.

(b) False: $\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$ is nonzero but has no inverse.

(c) False as stated for every abelian group: for the trivial group, the identity endomorphism equals the zero endomorphism. For every nontrivial A, the identity map is a nonzero unity.

(d) False: $A = \mathbb{Z}$ has a nonzero identity endomorphism as unity.

(e) False: for nontrivial A, the zero endomorphism is not an automorphism, so these automorphisms do not even contain the additive identity of a subring.

(f) False: take $R = \mathbb{Z}_2$. The group ring of the infinite cyclic group then has characteristic 2, whereas $\mathbb{Z}$ has characteristic zero, so the rings are not isomorphic.

(g) True: both the scalar coefficients and group labels commute, so every pair of formal sums commutes on expansion.

(h) False: $ij = k$ but $ji = -k$, so multiplication is not commutative. They form a division ring.

(i) True: quaternion multiplication is associative, 1 is the identity, and $q^{-1} = \overline{q}/\|q\|^2$ exists for every nonzero q.

(j) False: the real scalars, and also the usual complex plane, are subrings that are fields.

Exercise 24.12: Two failures of commutative-field conclusions

(a) In $\mathbb{H}$, the degree-two polynomial $x^2 + 1$ vanishes at $i, -i, j, -j, k, -k$, already six distinct zeros. More generally every $bi + cj + dk$ with $b^2 + c^2 + d^2 = 1$ has square -1. Thus the commutative-field root bound fails in a strictly skew field.

(b) The eight quaternions $\{\pm 1, \pm i, \pm j, \pm k\}$ form a finite multiplicative subgroup. Every element has order at most 4, so none generates all eight elements. This quaternion group is not cyclic.

Exercise 24.13: The other zero-divisor equation for an endomorphism

The source map is $\phi(m, n) = (m + n, 0)$. Define $\theta(m, n) = (m, -m)$; this is a nonzero additive endomorphism of $\mathbb{Z} \times \mathbb{Z}$. Then

$$(\phi\theta)(m, n) = \phi(m, -m) = (0, 0)$$

for every (m, n). Thus $\phi\theta = 0$ with $\theta \neq 0$, giving the required zero product on the side opposite to the source example's $\psi\phi = 0$.

Exercise 24.14: Six units over any field

The six matrices

$$\begin{pmatrix}1&0\\0&1\end{pmatrix}, \quad \begin{pmatrix}1&1\\0&1\end{pmatrix}, \quad \begin{pmatrix}0&1\\1&0\end{pmatrix}, \quad \begin{pmatrix}0&1\\1&1\end{pmatrix}, \quad \begin{pmatrix}1&1\\1&0\end{pmatrix}, \quad \begin{pmatrix}1&0\\1&1\end{pmatrix}$$

are distinct because $0 \neq 1$ in a field. Their determinants are $1, 1, -1, -1, -1, 1$, respectively, all nonzero even in characteristic 2 (where $-1 = 1$). The invertible-matrix criterion proves that all six are units of $M_2(F)$.

Exercise 24.15: Endomorphisms of cyclic additive groups

An additive endomorphism of $\mathbb{Z}$ is uniquely determined by the integer $k = \phi(1)$, and then $\phi(n) = kn$ for every integer n. The map $k \mapsto \phi_k$ is a bijection $\mathbb{Z} \to \mathrm{End}(\mathbb{Z})$, with

$$\phi_k + \phi_l = \phi_{k+l}, \qquad \phi_k \circ \phi_l = \phi_{kl}.$$

Thus it is a ring isomorphism.

For $\mathbb{Z}_n$, the same argument uses $k = \phi([1]) \in \mathbb{Z}_n$ and $\phi([r]) = [r]k$. Every k defines a well-defined endomorphism, since $nk = 0$ automatically. Addition and composition again correspond to $k + l$ and kl modulo n, proving $\mathrm{End}(\mathbb{Z}_n) \cong \mathbb{Z}_n$.

Exercise 24.16: Endomorphisms of a two-dimensional binary group

View $\mathbb{Z}_2 \times \mathbb{Z}_2$ as a two-dimensional vector space over $\mathbb{Z}_2$. Every additive endomorphism is linear and is determined by arbitrary images of its two basis vectors. Each image has four choices, so there are $4^2 = 16$ endomorphisms. The componentwise product ring $\mathbb{Z}_2 \times \mathbb{Z}_2$ has only four elements. Different cardinalities rule out a ring isomorphism. Equivalently the endomorphism ring is $M_2(\mathbb{Z}_2)$, which is also noncommutative.

Exercise 24.17: The differentiation–multiplication commutator

Here $Xf = xf$ and $Yf = f'$ are endomorphisms of the additive polynomial group. For every polynomial f,

$$(YX - XY)f = (xf)' - xf' = f + xf' - xf' = f.$$

The product rule can be checked on each monomial x^n: its two terms are $(n+1)x^n$ and nx^n. Thus $YX - XY$ acts as the identity on every polynomial and equals the unity endomorphism $\boxed{1}$.

Exercise 24.18: The group ring of a one-element group

If $G = \{e\}$, every formal sum has the form re. The map $re \mapsto r$ is bijective. Furthermore $(re) + (se) = (r + s)e$ and $(re)(se) = (rs)e$, since $e^2 = e$. The map preserves both operations, proving $RG \cong R$.

Exercise 24.19: A matrix realization of the quaternions

Put

$$E = I_2, \qquad I = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \qquad J = \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}.$$

The i in these matrix entries is the complex imaginary unit.

(a) Since quaternion $ij = k$, multiplicativity forces

$$\boxed{K = IJ = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}}.$$

(b) With $IJ = K$ used as this definition, the other eight multiplication-table equations to check are

$$I^2 = J^2 = K^2 = -E, \qquad JI = -K, \qquad JK = I, \qquad KJ = -I, \qquad KI = J, \qquad IK = -J.$$

Direct matrix multiplication verifies them: the three squares are $\mathrm{diag}(-1,-1)$, $JI = \mathrm{diag}(-i, i)$, $JK = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$, and the remaining reversed products have the stated signs. Identities involving E hold automatically. Real-linearity of the definition makes additivity automatic, and distributivity reduces multiplicativity for arbitrary quaternions to these basis products.

(c) We must also check injectivity, since surjectivity onto the image is automatic. The explicit matrix is

$$\phi(a + bi + cj + dk) = \begin{pmatrix} a + di & b + ci \\ -b + ci & a - di \end{pmatrix}.$$

If it is zero, real and imaginary parts of its entries give $a = d = b = c = 0$. Thus the kernel is zero, proving the required isomorphism onto $\phi[\mathbb{H}]$.

25 Ordered Rings and Fields

Exercise 25.1: The high-degree ordering of monomials

For P_{high}, the sign of a nonzero polynomial is the sign of its highest-degree coefficient. Thus $x - a$ is positive for every $a > 0$ in R, and $x^{n+1} - x^n$ is positive for every $n \geq 1$. The requested ordering is

$$\boxed{0 < a < x < x^2 < x^3 < \cdots < x^n < x^{n+1} < \cdots}.$$

The comparison $a < x$ concerns every positive constant, regardless of its size in the original ring.

Exercise 25.2: Laurent monomials in the low-degree ordering

The sign of a nonzero Laurent series is determined by the coefficient at its least exponent. For integers $i < j$, the difference $x^i - x^j$ has least term x^i with positive coefficient. Consequently $x^j < x^i$. All monomials are positive, and the full order is

$$\boxed{0 < \cdots < x^3 < x^2 < x < 1 < x^{-1} < x^{-2} < x^{-3} < \cdots}.$$

In particular x is positive and smaller than every positive constant of the coefficient field.

Exercise 25.3: Positive elements in the conjugate ordering

The ordering is transported by $\phi(m + n\sqrt{2}) = m - n\sqrt{2}$. Therefore an element is positive in this new ordering exactly when $m - n\sqrt{2} > 0$ in the usual real order. In integer-coefficient cases this means:

- if $n = 0$, then $m > 0$;
- if $n > 0$, then $m > 0$ and $m^2 > 2n^2$;
- if $n < 0$, then either $m \geq 0$, or $m < 0$ and $2n^2 > m^2$.

Squaring in these comparisons is valid only after the relevant sides have been made nonnegative. For example $-\sqrt{2}$ has $m = 0, n = -1$, so it is positive in this ordering.

Exercise 25.4: Ordering five polynomials, set 4

Compare two labeled polynomials by subtracting the earlier one from the later one. For P_{low} inspect the *lowest* nonzero term; for P_{high} inspect the *highest* nonzero term. A positive deciding coefficient means the difference is positive.

(i) The increasing order is $\boxed{a < c < d < e < b}$. The deciding terms of its four successive differences are

$$5, 2x, 2, 3.$$

All four coefficients are positive, verifying each adjacent comparison.

(ii) The increasing order is $\boxed{d < b < a < e < c}$. The deciding terms of its four successive differences are

$$7x^2, 6x, 4x^2, 3x^2.$$

All four coefficients are positive, verifying each adjacent comparison.

Exercise 25.5: Ordering five polynomials, set 5

Compare two labeled polynomials by subtracting the earlier one from the later one. For P_{low} inspect the *lowest* nonzero term; for P_{high} inspect the *highest* nonzero term. A positive deciding coefficient means the difference is positive.

(i) The increasing order is $\boxed{a < c < e < d < b}$. The deciding terms of its four successive differences are

$$1, 5x, 8x^2, 3x.$$

All four coefficients are positive, verifying each adjacent comparison.

(ii) The increasing order is $\boxed{e < c < b < a < d}$. The deciding terms of its four successive differences are

$$4x^5, 11x^4, 8x^3, x^5.$$

All four coefficients are positive, verifying each adjacent comparison.

Exercise 25.6: Ordering five polynomials, set 6

Compare two labeled polynomials by subtracting the earlier one from the later one. For P_{low} inspect the *lowest* nonzero term; for P_{high} inspect the *highest* nonzero term. A positive deciding coefficient means the difference is positive.

(i) The increasing order is $\boxed{c < a < b < e < d}$. The deciding terms of its four successive differences are

$$2, 3, 2x, 6x^3.$$

All four coefficients are positive, verifying each adjacent comparison.

(ii) The increasing order is $\boxed{e < c < a < b < d}$. The deciding terms of its four successive differences are

$$5x^5, 5x^2, x^3, 8x^4.$$

All four coefficients are positive, verifying each adjacent comparison.

Exercise 25.7: Ordering five polynomials, set 7

Compare two labeled polynomials by subtracting the earlier one from the later one. For P_{low} inspect the *lowest* nonzero term; for P_{high} inspect the *highest* nonzero term. A positive deciding coefficient means the difference is positive.

(i) The increasing order is $\boxed{d < a < b < c < e}$. The deciding terms of its four successive differences are

$$3x, 2x^2, 2x, x^2.$$

All four coefficients are positive, verifying each adjacent comparison.

(ii) The increasing order is $\boxed{d < c < e < a < b}$. The deciding terms of its four successive differences are

$$x^2, x^2, 5x^3, 4x^4.$$

All four coefficients are positive, verifying each adjacent comparison.

Exercise 25.8: Ordering five polynomials, set 8

Compare two labeled polynomials by subtracting the earlier one from the later one. For P_{low} inspect the *lowest* nonzero term; for P_{high} inspect the *highest* nonzero term. A positive deciding coefficient means the difference is positive.

(i) The increasing order is $\boxed{e < a < c < b < d}$. The deciding terms of its four successive differences are

$$x, 3x^2, 2x^2, x.$$

All four coefficients are positive, verifying each adjacent comparison.

(ii) The increasing order is $\boxed{c < d < a < e < b}$. The deciding terms of its four successive differences are

$$x, 6x^3, x^2, 4x^2.$$

All four coefficients are positive, verifying each adjacent comparison.

Exercise 25.9: Ordering five polynomials, set 9

Compare two labeled polynomials by subtracting the earlier one from the later one. For P_{low} inspect the *lowest* nonzero term; for P_{high} inspect the *highest* nonzero term. A positive deciding coefficient means the difference is positive.

(i) The increasing order is $\boxed{c < a < e < d < b}$. The deciding terms of its four successive differences are

$$x^3, 6x^2, 4x^3, 2.$$

All four coefficients are positive, verifying each adjacent comparison.

(ii) The increasing order is $\boxed{e < c < b < a < d}$. The deciding terms of its four successive differences are

$$8x^3, x^3, x, 4x^4.$$

All four coefficients are positive, verifying each adjacent comparison.

Exercise 25.10: Ordering Laurent monomials with coefficients

The least exponent controls the sign. The two negative terms come first: $-5x^{-2} < -3x^{-2}$. Among the positive terms, $4x < x^{-1} < 2x^{-1}$, because a positive x^{-1} term exceeds any positive multiple of x, and the coefficient 1 is less than 2 at exponent -1. Thus the increasing label order is $\boxed{b < d < e < a < c}$.

Exercise 25.11: Ordering five rational Laurent series

Use formal geometric-series division to expose each least term:

$$a = 1 + x + \cdots, \quad b = x^2 + \cdots, \quad c = x^{-1} + 1 + \cdots,$$
$$d = -x + \cdots, \qquad e = \tfrac{3}{4}x^{-1} - \tfrac{1}{2} + \cdots.$$

The negative series d comes first. The positive infinitesimal b is below a, whose constant term is 1. Both e and c have positive exponent--1 terms, so exceed a; their coefficients compare as $3/4 < 1$. Therefore $\boxed{d < b < a < e < c}$. The dots hide higher exponents, which cannot alter these deciding terms.

Exercise 25.12: Comparing equal constant terms in Laurent series

Expansion gives

$$a = \tfrac{5}{3}x^{-1} - \tfrac{26}{9} + \cdots, \qquad b = -\tfrac{1}{2} + \tfrac{5}{8}x + \cdots, \qquad c = \tfrac{7}{4} + \cdots,$$

$$d = \tfrac{9}{2} + \cdots, \qquad e = -\tfrac{1}{2} + \tfrac{2}{3}x + \cdots.$$

For b and e, the constant terms cancel; the coefficient of x in $e - b$ is $2/3 - 5/8 = 1/24 > 0$, proving $b < e$. Their negative constant terms put both below c and d. Since $7/4 < 9/2$, $c < d$, and the positive x^{-1} term puts a above them all. Thus $\boxed{b < e < c < d < a}$.

Exercise 25.13: A second comparison of equal constant terms

Formal expansions yield

$$a = 1 - 2x + \cdots, \quad b = 1 - \tfrac{10}{3}x + \cdots, \quad c = \tfrac{1}{4}x^{-1} - \tfrac{1}{16} + \cdots,$$

$$d = -\tfrac{1}{3}x^{-1} - \tfrac{1}{9} + \cdots, \qquad e = 4x + \cdots.$$

The negative x^{-1} term places d first, and the positive such term places c last. The positive infinitesimal e lies below the two series with constant term 1. Finally $a - b$ has least term $(4/3)x > 0$, so $b < a$. The increasing order is $\boxed{d < e < b < a < c}$.

Exercise 25.14: An ordered field containing nonreal numbers

Let $K = \mathbb{Q}(\sqrt[3]{2}) \subset \mathbb{R}$, with its induced real ordering, and let $L \subset \mathbb{C}$ be the isomorphic field in the question, containing $\sqrt[3]{2}(-1 + i\sqrt{3})/2$. If $\phi : K \to L$ is the stated isomorphism, declare $\phi(a) >_L 0$ exactly when $a > 0$ in K. The cone axioms transfer through the isomorphism, so this orders L, although L contains nonreal complex numbers. There is no contradiction: the order uses the algebraic field structure of L and is not a restriction of an order on all of $\mathbb{C}$. The obstruction $i^2 = -1$ rules out ordering $\mathbb{C}$ itself, not every nonreal subfield.

Exercise 25.15: True or false: orderings

(a) True: $1 = 1^2$ is positive; positive integer sums are then positive, and trichotomy forces the negatives to be negative.

(b) True: every usual positive real is a nonzero square, so must be positive in any field ordering. Trichotomy fixes all signs.

(c) False: $\mathbb{Q}(\sqrt{2})$ has its usual ordering and the ordering transported by conjugation $\sqrt{2} \mapsto -\sqrt{2}$, which disagree on $\sqrt{2}$.

(d) True: positive integers are positive, their inverses are positive, and hence all usual positive rationals are positive. Trichotomy leaves no alternative.

(e) True: either the low- or high-degree coefficient ordering extends the given constant ordering.

(f) False: without $a > 0$, take $a = 0, b = 1$; no positive n has $1 < n0$.

(g) True: this is equivalent to the Archimedean definition. For $b > 0$ it is that definition; for $b \leq 0$, choose $n = 1$ since $a > 0 \geq b$.

(h) False: take $a = -1$, so $-a = 1$ is positive.

(i) False: at $a = 0$, neither a nor $-a$ is positive. One is positive when $a \neq 0$.

(j) True: $1, 2 \cdot 1, 3 \cdot 1, \ldots$ are distinct because a positive sum of 1's cannot be zero.

Exercise 25.16: Making pi larger than every rational

Since π is transcendental, evaluation $\phi_\pi : \mathbb{Q}[x] \to \mathbb{Q}[\pi]$ is an isomorphism. Give $\mathbb{Q}[x]$ the high-degree ordering and transfer it through this map. Equivalently a nonzero $a_0 + a_1\pi + \cdots + a_n\pi^n$ is positive exactly when its highest nonzero rational coefficient is positive. Then for every $q \in \mathbb{Q}$, $\pi - q$ has highest coefficient 1, so $\boxed{q < \pi}$ in this ordering. Transcendence ensures a unique polynomial representation, so the sign rule is well defined.

Exercise 25.17: Conjugation of a quadratic ring

Representations $m + n\sqrt{2}$ are unique by irrationality of $\sqrt{2}$. For $u = m + n\sqrt{2}$, $v = r + s\sqrt{2}$, direct calculation gives

$$\phi(u + v) = (m + r) - (n + s)\sqrt{2} = \phi(u) + \phi(v).$$

Since $uv = (mr + 2ns) + (ms + nr)\sqrt{2}$,

$$\phi(uv) = (mr + 2ns) - (ms + nr)\sqrt{2} = (m - n\sqrt{2})(r - s\sqrt{2}).$$

Thus multiplication is preserved as well. In fact ϕ^2 is the identity, so the homomorphism is an automorphism.

Exercise 25.18: Cone membership means greater than zero

By the definition in Theorem 25.5, $0 < a$ means exactly $a - 0 \in P$. This equals $a \in P$, which is the hypothesis. Thus the conclusion follows directly from the definition of $<$.

Exercise 25.19: Equal positive multiples

First, an ordered ring has no nonzero zero divisors: for nonzero u, v, choose signs so that $\pm u, \pm v \in P$; closure makes one of $uv, -uv$ positive, so $uv \neq 0$. Hence if $c = 0$, then $bd = 0$ with $b \neq 0$ forces $d = 0$, and similarly with $d = 0$.

Suppose now that neither is zero. If $c \in P$, then $ac \in P$. If $-d \in P$, closure would also give $b(-d) = -bd = -ac \in P$, contradicting trichotomy. Thus $d \in P$, and $cd \in P$. If instead $-c \in P$, the same argument applied to $a(-c) = b(-d)$ gives $-d \in P$, whence $cd = (-c)(-d) \in P$. This proves exactly the stated alternatives using only cone closure and trichotomy.

Exercise 25.20: Reversing signs reverses an inequality

The hypothesis $a < b$ means $b - a \in P$. But

$$(-a) - (-b) = b - a \in P.$$

By the definition of $<$, this is precisely $-b < -a$.

Exercise 25.21: A negative times a positive

From $a < 0$ we obtain $-a \in P$, and from $0 < b$ we obtain $b \in P$. Closure under multiplication gives $(-a)b = -ab \in P$. Since $0 - ab = -ab$ is positive, the defining condition for $<$ gives $ab < 0$. No separate sign rule has been assumed.

Exercise 25.22: A positive quotient

Since $b > 0$, it is nonzero and b^{-1} exists. The square of any nonzero element is in P: trichotomy makes either u or $-u$ positive, and squaring that element gives $u^2 \in P$. Thus $(b^{-1})^2 \in P$. Multiplying by $b \in P$ gives $b^{-1} = (b^{-1})^2 b \in P$. Finally closure gives $a/b = ab^{-1} \in P$, proving positivity of the quotient.

Exercise 25.23: Inverting a positive element below one

The assumptions give $a \in P$ and $1 - a \in P$. By Exercise 25.22, $(1 - a)/a \in P$. But this quotient is $a^{-1} - 1$. Therefore $1 < a^{-1}$ by the defining relation $<$.

Exercise 25.24: Inverting a negative element above minus one

Let $b = -a$. The inequalities $-1 < a < 0$ imply $0 < b < 1$, using the difference definition of $<$ or Exercise 25.20. Exercise 25.23 gives $1 < b^{-1}$. Reversing signs yields $-b^{-1} < -1$. Since $a^{-1} = (-b)^{-1} = -b^{-1}$, the conclusion is $a^{-1} < -1$.

Exercise 25.25: Transporting an ordering across an isomorphism

Let $P' = \phi[P]$. It is nonempty because P is nonempty. If $u = \phi(a)$ and $v = \phi(b)$ with $a, b \in P$, then $u + v = \phi(a + b) \in P'$ and $uv = \phi(ab) \in P'$ by closure of P. For any $u \in R'$, take its unique preimage a. Exactly one of $a \in P$, $a = 0$, $-a \in P$ holds, and injectivity and surjectivity translate this into exactly one of $u \in P'$, $u = 0$, $-u \in P'$. Hence P' is an ordering cone. Finally

$$\phi(a) <' \phi(b) \iff \phi(b) - \phi(a) = \phi(b - a) \in \phi[P] \iff b - a \in P \iff a < b.$$

The middle reverse implication uses injectivity of ϕ. This proves Theorem 25.10 in full.

Exercise 25.26: The induced ordering on a subring

For a nonzero subring S, put $P_S = P \cap S$. It is nonempty: choose $s \in S \setminus \{0\}$, and trichotomy gives either s or $-s$ in P_S. Sums and products of elements of P_S lie in P by cone closure and in S by the

subring axioms. For any $s \in S$, exactly one of $s \in P$, $s = 0$, $-s \in P$ holds; since both $s, -s$ lie in S, this is exactly the cone trichotomy for P_S. Thus the induced order is the restriction of the original order.

The nonzero qualification matters: for $S = \{0\}$ the intersection is empty, contrary to the definition's nonempty-cone requirement. If one also imposes the section's standing requirement that every ordered ring have nonzero unity, restrict the assertion to subrings with nonzero unity. For nonunital nonzero subrings the same proof still supplies the asserted order and cone axioms.

Exercise 25.27: Recovering the cone from the order relation

Set $P = \{x : 0 < x\}$. Since the ring is nonzero, compare 0 with some $a \neq 0$. Trichotomy gives either $0 < a$ or $a < 0$; in the latter case additive isotonicity gives $0 < -a$. Thus P is nonempty. If $a, b \in P$, adding a to $0 < b$ gives $a < a + b$, and transitivity with $0 < a$ gives $0 < a + b$. Multiplicative isotonicity applied to $0 < b$ and $0 < a$ gives $0 = a0 < ab$. So P is closed under both operations.

Trichotomy of 0 and x, and additive isotonicity translating $x < 0$ to $0 < -x$, give exactly one of $x \in P$, $x = 0$, $-x \in P$. Thus this is a positive cone. Finally additive isotonicity, in both directions, gives

$$a < b \iff 0 < b - a \iff b - a \in P.$$

Therefore the relation induced by P is the original relation, proving the converse in Theorem 25.5.

Exercise 25.28: Injectivity of an odd power in an ordered domain

An odd power has the same sign as its nonzero base: a positive base gives a positive product; for a negative base, its negative is positive and $(-c)^{2n+1} = -c^{2n+1}$. Thus equal odd powers cannot come from bases of opposite signs, or from one zero and one nonzero base.

For positive $a < b$, factor

$$b^{2n+1} - a^{2n+1} = (b - a)\big(b^{2n} + b^{2n-1}a + \cdots + a^{2n}\big).$$

Every summand in the parenthesis is positive, so the difference is positive and equality of the powers is impossible. The case $0 < a = b$ already has the conclusion. If $a, b < 0$, their negatives are positive, and equality of odd powers gives equality of the odd powers of $-a, -b$, reducing to the positive case. Trichotomy therefore leaves only $\boxed{a = b}$.

Exercise 25.29: Eight different orderings in two variables

For each stage choose low (L) or high (H) degree to determine the sign. Applying the univariate construction twice gives eight orderings. To distinguish them, record whether four fixed polynomials

are positive (+) or negative (−):

first ring, then outer variable	inner	outer	$x-1$	$y-1$	$x-y$	$xy-1$
$(R[x])[y]$	L	L	−	−	+	−
$(R[x])[y]$	L	H	−	+	−	+
$(R[x])[y]$	H	L	+	−	+	−
$(R[x])[y]$	H	H	+	+	−	+
$(R[y])[x]$	L	L	−	−	−	−
$(R[y])[x]$	L	H	+	−	+	+
$(R[y])[x]$	H	L	−	+	−	−
$(R[y])[x]$	H	H	+	+	+	+

.

For example, with $(R[x])[y]$ and outer L, the deciding y^0 coefficient of $x-y$ is $x>0$, while that of $xy-1$ is $-1<0$. With outer H, their deciding coefficients are -1 and x, reversing those two signs. The other rows follow by interchanging x, y and choosing the specified inner signs for $x-1$ or $y-1$. No two sign rows agree. Thus no two of the eight orderings can be the same.

26 Homomorphisms and Factor Rings

Exercise 26.1: Homomorphisms of the product of integer rings

Plan. A homomorphism is determined by $u = \phi(1,0)$ and $v = \phi(0,1)$, since $\phi(m,n) = mu + nv$. The identities $(1,0)^2 = (1,0)$, $(0,1)^2 = (0,1)$ and $(1,0)(0,1) = 0$ force $u^2 = u$, $v^2 = v$, and $uv = 0$. An integer idempotent satisfies $t(t-1) = 0$, so each coordinate of u, v is 0 or 1. In each output coordinate, the pair of coefficients is therefore $(0,0)$, $(1,0)$, or $(0,1)$. Consequently each of the two coordinates of $\phi(m,n)$ can independently be 0, m, or n: there are $3^2 = 9$ maps. Conversely these nine maps preserve coordinatewise sums and products, so the list is exhaustive. Homomorphisms here are not required to preserve the unity of the codomain.

Exercise 26.2: A two-element subring

A subring isomorphic to $\mathbb{Z}_2$ must be $\{0, e\}$ with $e \neq 0$, $2e = 0$, and $e^2 = e$ modulo n. The additive cyclic group $\mathbb{Z}_n$ has an element of order two exactly when $n = 2m$, and then its only such element is $e = m$. Now $m^2 \equiv m \pmod{2m}$ is equivalent to $2 \mid (m-1)$. Thus m must be odd.

Conclusion. Exactly $n \equiv 2 \pmod 4$ work, with subring $\{0, n/2\}$. Its nonzero element acts as the identity within that subring.

Exercise 26.3: Ideals and quotients of the twelve-element ring

Every additive subgroup of $\mathbb{Z}_{12}$ is $\langle d \rangle$ for a divisor d of 12, and it is an ideal since multiplying a multiple of d by any residue still gives a multiple of d. Reduction modulo d has kernel $\langle d \rangle$, so

N	$\mathbb{Z}_{12}/N$
$\langle 1 \rangle = \mathbb{Z}_{12}$	$\{0\}$
$\langle 2 \rangle = \{0,2,4,6,8,10\}$	$\mathbb{Z}_2$
$\langle 3 \rangle = \{0,3,6,9\}$	$\mathbb{Z}_3$
$\langle 4 \rangle = \{0,4,8\}$	$\mathbb{Z}_4$
$\langle 6 \rangle = \{0,6\}$	$\mathbb{Z}_6$
$\langle 12 \rangle = \{0\}$	$\mathbb{Z}_{12}$

These are all the ideals because an ideal must first be an additive subgroup.

Exercise 26.4: Tables for an even-integer quotient

Write $A = 8\mathbb{Z}$, $B = 2 + 8\mathbb{Z}$, $C = 4 + 8\mathbb{Z}$, and $D = 6 + 8\mathbb{Z}$. Add or multiply representatives and then reduce modulo 8:

$+$	A	B	C	D
A	A	B	C	D
B	B	C	D	A
C	C	D	A	B
D	D	A	B	C

$\cdot$	A	B	C	D
A	A	A	A	A
B	A	C	A	C
C	A	A	A	A
D	A	C	A	C

For example $BD = 12 + 8\mathbb{Z} = C$. Every element squares to A or C, and no element is a multiplicative identity: $uC = A \neq C$ for all u. Since $\mathbb{Z}_4$ has an identity, these rings are not isomorphic, although their additive groups are isomorphic.

Exercise 26.5: Correcting isomorphism

The condition $\operatorname{Ker}\phi = \{0\}$ gives injectivity, but not surjectivity onto the declared codomain. An isomorphism $R \to R'$ is a *bijective* ring homomorphism. For instance inclusion $\mathbb{Z} \hookrightarrow \mathbb{Q}$ has zero kernel and is not an isomorphism onto $\mathbb{Q}$.

Exercise 26.6: The ideal definition

The definition is correct: an ideal is an additive subgroup N with $rn, nr \in N$ for every $r \in R, n \in N$. Both absorption conditions are needed in a noncommutative ring; they coincide when multiplication is commutative.

Exercise 26.7: Correcting the kernel

The kernel consists of *inputs*, not equations or outputs:

$$\operatorname{Ker}\phi = \{r \in R : \phi(r) = 0_{R'}\}.$$

For example the kernel of reduction $\mathbb{Z} \to \mathbb{Z}_5$ is $5\mathbb{Z}$, a subset of $\mathbb{Z}$.

Exercise 26.8: Differentiation

Differentiation preserves addition, but the product rule is $\delta(fg) = f'g + fg'$, rather than $f'g'$. Take $f = g = x$: $\delta(x^2) = 2x$ and $\delta(x)\delta(x) = 1$. Thus δ is not a ring homomorphism. Its kernel is the subring of constant functions (a differentiable function with derivative zero is constant). Example 26.12 shows this subring is not an ideal; this is another obstruction, since kernels of ring homomorphisms must be ideals.

Exercise 26.9: An image identity need not be the codomain identity

Let $\phi : \mathbb{Z} \to \mathbb{Z} \times \mathbb{Z}$ be $\phi(n) = (n, 0)$. Coordinatewise calculation gives $\phi(m + n) = \phi(m) + \phi(n)$ and $\phi(mn) = \phi(m)\phi(n)$. Yet $\phi(1) = (1, 0)$ is nonzero and differs from the codomain identity $(1, 1)$. It is the identity only on the image.

Exercise 26.10: True and false statements

(a) True. A homomorphism induces $R/\operatorname{Ker}\phi \cong \phi[R]$.

(b) False. Inclusion $\mathbb{Z} \hookrightarrow \mathbb{Q}$ sends the ideal $\mathbb{Z}$ to $\mathbb{Z}$, which is not an ideal of $\mathbb{Q}$, since $(1/2)1 \notin \mathbb{Z}$.

(c) True. $\phi(r) = \phi(s)$ exactly when $r - s \in \operatorname{Ker}\phi$.

(d) False. $1 \in \mathbb{Q}$ but $\sqrt{2}\,1 \notin \mathbb{Q}$, so absorption in $\mathbb{R}$ fails.

(e) True. Additive closure is required, and ideal absorption includes closure under multiplication of its own elements.

(f) False. $\mathbb{Z} \subset \mathbb{Q}$ is the counterexample from (b).

(g) True. $(r+N)(s+N) = rs+N = sr+N$ if R is commutative.

(h) True. Reduction modulo four induces $\mathbb{Z}/4\mathbb{Z} \cong \mathbb{Z}_4$.

(i) True. If $1 \in N$, every $r = r1 \in N$; conversely $N = R$ contains 1.

(j) True. These are precisely the respective substructures allowing well-defined quotient operations.

Exercise 26.11: The extreme quotients

The ring R/R has just the one coset R and is the zero ring. The map $r + \{0\} \mapsto r$ identifies $R/\{0\}$ with R. Thus the two extreme quotients contain no new structure, although both are useful boundary cases in general theorems.

Exercise 26.12: A domain with a field quotient

Take the integral domain $\mathbb{Z}$ and its ideal $2\mathbb{Z}$. Reduction modulo two gives $\mathbb{Z}/2\mathbb{Z} \cong \mathbb{Z}_2$, which is a field because its only nonzero element is its own inverse.

Exercise 26.13: A domain with zero divisors in a quotient

Again start with $\mathbb{Z}$, but quotient by $6\mathbb{Z}$. In $\mathbb{Z}/6\mathbb{Z}$, the cosets $2+6\mathbb{Z}$ and $3+6\mathbb{Z}$ are nonzero and their product is $6+6\mathbb{Z} = 0+6\mathbb{Z}$. Thus an integral domain can have a quotient with zero divisors.

Exercise 26.14: Removing zero divisors by a quotient

The ring $R = \mathbb{Z} \times \mathbb{Z}$ has $(1,0)(0,1) = 0$ with both factors nonzero. Projection $(a,b) \mapsto a$ has ideal kernel $N = \{0\} \times \mathbb{Z}$ and gives $R/N \cong \mathbb{Z}$, an integral domain.

Exercise 26.15: A subring that is not an ideal

The diagonal $D = \{(n,n) : n \in \mathbb{Z}\}$ is closed under differences and products and contains zero. But $(1,0)(1,1) = (1,0) \notin D$, so it is not an ideal of $\mathbb{Z} \times \mathbb{Z}$.

Exercise 26.16: Representatives and quotient elements

(a) The student has used r, s for elements of R/N and then written $rs - sr \in N$, which should concern elements of R. Quotient elements are cosets, so the types of objects have been confused. **(b)** A correct start is: for all $r, s \in R$, commutativity gives $(r+N)(s+N) = (s+N)(r+N)$. **(c)** That equality is equivalent to $rs+N = sr+N$, which is equivalent to $rs - sr \in N$. Reading these equivalences in either direction proves both implications. Notice that $rs = sr$ in R itself is not required.

Exercise 26.17: A quadratic integer ring and matrices

Plan. Match the coefficients of 1 and $\sqrt{2}$ with matrix entries. If $u = a + b\sqrt{2}$ and $v = c + d\sqrt{2}$, then

$$u - v = (a-c) + (b-d)\sqrt{2}, \qquad uv = (ac+2bd) + (ad+bc)\sqrt{2}.$$

These formulas show R is a nonempty subring of $\mathbb{R}$. Set $M(a,b) = \begin{pmatrix} a & 2b \\ b & a \end{pmatrix}$. Matrix subtraction and multiplication give

$$M(a,b) - M(c,d) = M(a-c, b-d), \quad M(a,b)M(c,d) = M(ac+2bd, ad+bc),$$

so R' is a subring of $M_2(\mathbb{Z})$ and $\phi(a+b\sqrt{2}) = M(a,b)$ preserves the operations. Coefficients in $a+b\sqrt{2}$ are unique: otherwise irrationality of $\sqrt{2}$ would fail. Hence the map is well defined; its kernel is zero by its entries, and every displayed matrix is an image. Thus it is a ring isomorphism.

Exercise 26.18: Homomorphisms from a field

Let $\phi : F \to R$ have a nonzero kernel element a. For any $x \in F$, write $x = (xa^{-1})a$. Then $\phi(x) = \phi(xa^{-1})\phi(a) = 0$, so ϕ is the zero map. If no such a exists, the kernel is zero, and ϕ is injective. These are the only possibilities.

Exercise 26.19: Composing ring homomorphisms

For $a, b \in R$, apply the defining identities twice:

$$\psi\phi(a+b) = \psi(\phi(a) + \phi(b)) = \psi\phi(a) + \psi\phi(b),$$
$$\psi\phi(ab) = \psi(\phi(a)\phi(b)) = \psi\phi(a)\,\psi\phi(b).$$

Thus $\psi \circ \phi : R \to R''$ preserves both operations and is a ring homomorphism.

Exercise 26.20: Frobenius

In a commutative ring the binomial theorem applies. For prime p, each $\binom{p}{k}$ with $0 < k < p$ is divisible by p, so characteristic p gives

$$(a+b)^p = a^p + b^p.$$

Commutativity also gives $(ab)^p = a^p b^p$. Therefore $a \mapsto a^p$ preserves addition and multiplication and is a ring homomorphism. Commutativity is essential to these expansions.

Exercise 26.21: The image of the identity in a ring without zero divisors

Put $e = \phi(1)$. It satisfies $e^2 = e$. It is nonzero, because otherwise $\phi(r) = \phi(1r) = e\phi(r) = 0$ for every r, contrary to the nonzero-image assumption. For any $y \in R'$, we have $e(ey - y) = 0$. Since $e \neq 0$ and R' has no zero divisors, $ey = y$. Likewise $(ye - y)e = 0$ gives $ye = y$. Thus e is a two-sided identity for all of R', even when ϕ is not onto.

Exercise 26.22: Images and inverse images of ideals

(a) The image $\phi[N]$ is an additive subgroup. For $\phi(r) \in \phi[R]$ and $\phi(n) \in \phi[N]$, both products equal images of rn or nr in N, so absorption holds inside $\phi[R]$. **(b)** Inclusion $\mathbb{Z} \hookrightarrow \mathbb{Q}$ with $N = \mathbb{Z}$ fails absorption by $1/2$, giving the required counterexample. **(c)** Let $I = \phi^{-1}[N']$. Then $0 \in I$, and for $a, b \in I$, $\phi(a-b) = \phi(a) - \phi(b) \in N'$. For $r \in R$, both $\phi(ra) = \phi(r)\phi(a)$ and $\phi(ar) = \phi(a)\phi(r)$ lie in N'. Thus I is an ideal. This uses only absorption by elements in $\phi[R]$, so either stated choice of ambient ring for N' is sufficient.

Exercise 26.23: Polynomials vanishing on a set

Let N_S be the polynomials vanishing at each point of $S \subset F^n$. The zero polynomial belongs to N_S. If $f, g \in N_S$, then $(f-g)(a) = f(a) - g(a) = 0$ for every $a \in S$. If h is any polynomial, $(hf)(a) = h(a)f(a) = 0$ and similarly for fh. Thus the subring test and absorption show that N_S is an ideal. If S is empty, the condition is vacuous and $N_S = F[x_1, \ldots, x_n]$, still an ideal.

Exercise 26.24: Quotients of a field

An ideal N of a field F is either zero or all of F: if $0 \neq a \in N$, then $1 = a^{-1}a \in N$, and hence every $x = x1 \in N$. In the two cases, F/N is respectively isomorphic to F or to the one-element zero ring.

Exercise 26.25: Unity in a proper quotient

The coset $1 + N$ multiplies any $r + N$ to $r + N$ on either side. It is nonzero because $1 \in N$ would force $N = R$, contrary to the assumption. Thus a proper quotient of a ring with unity again has a nonzero unity.

Exercise 26.26: An annihilator ideal

Let $I_a = \{x : ax = 0\}$. It contains zero. If $x, y \in I_a$, then $a(x-y) = 0 - 0 = 0$, so it is an additive subgroup. For $r \in R$, commutativity gives $a(rx) = r(ax) = 0$, hence $rx = xr \in I_a$. Thus I_a is an ideal. Without commutativity, a one-sided statement is automatic but both sides need not follow.

Exercise 26.27: Intersections of ideals

For a family (I_j) of ideals, zero belongs to every I_j. If a, b belong to their intersection, $a - b \in I_j$ for every j, hence lies in the intersection. Likewise ra and ar belong to every I_j for every $r \in R$. This proves the ideal axioms. The intersection of an empty family is understood as R, also an ideal.

Exercise 26.28: Induced quotient homomorphism

Define $\overline{\phi}(r + N) = \phi(r) + N'$. If $r + N = s + N$, then $r - s \in N$ and $\phi(r) - \phi(s) \in \phi[N] \subset N'$, so the output cosets agree. Thus the formula is well defined. Moreover

$$\overline{\phi}((r + N) + (s + N)) = \phi(r + s) + N' = \overline{\phi}(r + N) + \overline{\phi}(s + N),$$

and the identical calculation using products proves multiplicativity. No surjectivity assumption is needed.

Exercise 26.29: Images of units under a surjection

Because ϕ is onto, $\phi(1)$ acts as a two-sided identity on every element of R': write it as $\phi(r)$ and use $1r = r1 = r$. Also $\phi(1) \neq 0$ since $R' \neq \{0\}$. If $uv = vu = 1$ in R, applying ϕ gives $\phi(u)\phi(v) = \phi(v)\phi(u) = 1_{R'}$. Thus $\phi(u)$ is a unit with inverse $\phi(v)$.

Exercise 26.30: The nilradical

Let N consist of nilpotent elements in a commutative ring. It contains zero. If $a^m = 0$ and $b^n = 0$, expand $(a-b)^{m+n-1}$. Each term contains either at least m factors a or at least n factors b, so every term is zero. Thus $a - b \in N$. For $r \in R$, $(ra)^m = r^m a^m = 0$, so $ra = ar \in N$. These facts prove that N is an ideal. Commutativity is used in both power calculations.

Exercise 26.31: Three nilradicals

In $\mathbb{Z}_{12}$ a nilpotent residue must be divisible by both primes $2, 3$, since reducing a vanishing power modulo either prime gives a vanishing element in a field. Conversely $6^2 \equiv 0 \pmod{12}$. Thus the nilradical is $\{0, 6\} = \langle 6 \rangle$. In $\mathbb{Z}$, a vanishing integer power forces the integer to be zero, so the nilradical is $\{0\}$. In $\mathbb{Z}_{32}$, precisely the even residues are nilpotent: $(2k)^5 \equiv 0$, while an odd residue is a unit and cannot have a zero power. Thus the nilradical is $\langle 2 \rangle$.

Exercise 26.32: Quotienting by all nilpotents

Suppose $(a + N)^m = N$ in R/N. Then $a^m \in N$, and because N is the nilradical, $(a^m)^n = 0$ for some n. Hence $a^{mn} = 0$, so $a \in N$ and $a + N = N$. Thus the quotient has no nonzero nilpotent elements.

Exercise 26.33: Lifting nilpotence

For any $a \in R$, the hypothesis that every coset is nilpotent gives $(a + N)^m = N$, hence $a^m \in N$. Every element of N is nilpotent, so $(a^m)^n = 0$ for some n. Thus every $a \in R$ is nilpotent, which says the nilradical is all of R. The exponents may depend on a.

Exercise 26.34: The radical of an ideal

Let $\sqrt{N} = \{a : a^m \in N \text{ for some } m \geq 1\}$. It contains zero. If $a^m, b^n \in N$, expand $(a-b)^{m+n-1}$; each term has a factor a^m or b^n, so each term belongs to N. This puts $a - b$ in $\sqrt{N}$. For any $r \in R$, $(ra)^m = r^m a^m \in N$, so $ra = ar \in \sqrt{N}$. Therefore $\sqrt{N}$ is an ideal. This proof does not require all elements of N themselves to be nilpotent.

Exercise 26.35: A radical can grow or remain the same

(a) In $\mathbb{Z}$, take $N = 4\mathbb{Z}$. Then $2^2 \in 4\mathbb{Z}$ but $2 \notin 4\mathbb{Z}$, so $\sqrt{N} \neq N$ (in fact $\sqrt{N} = 2\mathbb{Z}$). **(b)** Take $N = 2\mathbb{Z}$. If a^m is even then a is even, so $\sqrt{2\mathbb{Z}} = 2\mathbb{Z}$. Both examples use proper ideals.

Exercise 26.36: Radical versus quotient nilradical

An element of the nilradical of R/N is a *coset* $a + N$ such that $(a + N)^m = N$, equivalently $a^m \in N$. Hence

$$\mathrm{Nil}(R/N) = \{a + N : a \in \sqrt{N}\} = \sqrt{N}/N.$$

It is not literally $\sqrt{N}$, whose elements lie in R rather than R/N.

Exercise 26.37: Complex numbers as real matrices

Set $M(a,b) = \begin{pmatrix} a & b \\ -b & a \end{pmatrix}$. Direct multiplication gives

$$M(a,b)M(c,d) = M(ac - bd, ad + bc),$$

which matches $(a + bi)(c + di) = (ac - bd) + (ad + bc)i$. Addition also matches entry by entry. The map is injective because $M(a,b) = 0$ forces $a = b = 0$, and it is onto its image by definition. Therefore $\mathbb{C} \cong \phi[\mathbb{C}] \subset M_2(\mathbb{R})$. The sign convention corresponds to multiplication on coordinates $(x, -y)$ instead of (x, y) and is consistent.

Exercise 26.38: A ring version of the regular representation

(a) Left multiplication λ_a is additive because $a(x + y) = ax + ay$, so it is an endomorphism of $(R, +)$.
(b) Endomorphisms are added pointwise and multiplied by composition. Thus

$$\lambda_a - \lambda_b = \lambda_{a-b}, \qquad \lambda_a \circ \lambda_b = \lambda_{ab}.$$

Together with $\lambda_0 = 0$, these identities show that $R' = \{\lambda_a : a \in R\}$ is a subring. **(c)** The map $a \mapsto \lambda_a$ is a surjective ring homomorphism onto R'. If $\lambda_a = \lambda_b$, evaluate at 1 to obtain $a = b$. Thus it is injective and $R \cong R'$. Unity is what makes this final injectivity argument work.

27 Prime and Maximal Ideals

Exercise 27.1: Prime and maximal ideals in a six-element ring

Every ideal is $\langle d\rangle$ for $d \mid 6$, and $\mathbb{Z}_6/\langle d\rangle \cong \mathbb{Z}_d$. A proper ideal is prime exactly when its quotient is an integral domain, and maximal exactly when its quotient is a field. Among the divisors $1, 2, 3, 6$, exactly $2, 3$ are prime. Thus both lists are $\langle 2\rangle = \{0, 2, 4\}$ and $\langle 3\rangle = \{0, 3\}$.

Exercise 27.2: Prime and maximal ideals in a twelve-element ring

For $d \mid 12$, reduction modulo d gives $\mathbb{Z}_{12}/\langle d\rangle \cong \mathbb{Z}_d$. The quotient is a domain, equivalently a field in this finite case, exactly when $d = 2$ or 3. Hence the prime ideals and the maximal ideals are both $\langle 2\rangle = \{0, 2, 4, 6, 8, 10\}$ and $\langle 3\rangle = \{0, 3, 6, 9\}$.

Exercise 27.3: A product of two two-element fields

In a product $A \times B$ with identities, every ideal is $I \times J$: multiply (a, b) by $(1, 0)$ and $(0, 1)$ to separate the coordinates. A domain quotient cannot retain both nonzero factors, since $(1, 0)(0, 1) = 0$. Thus the prime and maximal ideals of $\mathbb{Z}_2 \times \mathbb{Z}_2$ are exactly $\{0\} \times \mathbb{Z}_2$ and $\mathbb{Z}_2 \times \{0\}$. Each quotient is $\mathbb{Z}_2$.

Exercise 27.4: A product involving a four-element ring

The same product argument shows that a prime ideal must be all of one factor times a prime ideal in the other. In $\mathbb{Z}_2$ the only prime ideal is zero, and in $\mathbb{Z}_4$ it is $\langle 2\rangle = \{0, 2\}$. Therefore the only prime ideals are

$$\{0\} \times \mathbb{Z}_4, \qquad \mathbb{Z}_2 \times \{0, 2\}.$$

Both quotients are $\mathbb{Z}_2$, so these are also exactly the maximal ideals.

Exercise 27.5: Quadratic quotients over the three-element field

A quotient by a nonconstant polynomial is a field exactly when that polynomial is irreducible. A quadratic over a field is irreducible exactly when it has no root. In $\mathbb{Z}_3$, the possible squares are $0, 1$. The equation $x^2 = -c$ has no solution exactly when $-c = 2$, or $c = 1$. Thus only $c = 1$ works.

Exercise 27.6: A cubic family with fixed quadratic term

A cubic over a field is reducible exactly when it has a root. At $x = 0, 1, 2$, the values of $x^3 + x^2$ modulo three are $0, 2, 0$. Thus $x^3 + x^2 + c$ has a root if $c = 0$ or 1, but for $c = 2$ its values are $2, 1, 2$, all nonzero. The required value is $c = 2$.

Exercise 27.7: A cubic family with a variable coefficient

Evaluate $x^3 + cx^2 + 1$ at $0, 1, 2$. For $c = 0$ the value at 2 is zero; for $c = 1$ the value at 1 is zero; for $c = 2$ the three values are $1, 1, 2$, all nonzero. Thus exactly $c = 2$ makes the cubic irreducible and the quotient a field.

Exercise 27.8: Quadratics with a linear term over five elements

Since the characteristic is not two, $x^2 + x + c$ has a root exactly when the discriminant $1 - 4c$ is a square. The squares in $\mathbb{Z}_5$ are $0, 1, 4$. For $c = 0, 1, 2, 3, 4$, the discriminants are respectively $1, 2, 3, 4, 0$. Thus the nonsquare cases are $c = 1, 2$, and precisely those quotients are fields.

Exercise 27.9: Quadratics with constant term one

For $x^2 + cx + 1$, the discriminant is $c^2 - 4$. As c runs through $0, 1, 2, 3, 4$ in $\mathbb{Z}_5$, the discriminants are $1, 2, 0, 0, 2$. Nonsquares occur for $c = 1, 4$, so exactly those values give fields.

Exercise 27.10: Correcting maximal ideal

A maximal ideal M is a *proper* ideal of R such that the only ideals containing it are M and R. Merely saying that no other ideal contains it would make R the only candidate, because R contains every ideal.

Exercise 27.11: Correcting prime ideal

In a commutative ring, a prime ideal P is a proper ideal such that $ab \in P$ implies $a \in P$ or $b \in P$ for all $a, b \in R$. It need not be generated by a prime number; for example $(x) \subset \mathbb{Q}[x]$ is prime because its quotient is $\mathbb{Q}$.

Exercise 27.12: Prime field

The definition is correct: a prime field has no proper subfields. Equivalently it equals the subfield generated by its unity. Up to isomorphism these fields are $\mathbb{Q}$ and $\mathbb{Z}_p$ for primes p.

Exercise 27.13: Principal ideal

The definition is correct. In a commutative ring with unity, the smallest ideal containing a is $(a) = aR = \{ar : r \in R\}$. It contains $a = a1$, is closed under differences and absorption, and every ideal containing a contains all its multiples.

Exercise 27.14: True and false statements

(a) False. $(0) \subset \mathbb{Z}$ is prime but not maximal, since $\mathbb{Z}$ is a domain and not a field.

(b) True. The quotient by a maximal ideal is a field and hence a domain.

(c) True. Every subfield of $\mathbb{Q}$ contains 1, all integers, and all nonzero integer inverses, hence all rationals.

(d) False. The prime subfield of $\mathbb{C}$ is $\mathbb{Q}$.

(e) True. The unity generates a copy of $\mathbb{Q}$ in characteristic zero or of $\mathbb{Z}_p$ in characteristic p.

(f) True. $\mathbb{Z}_2 \times \mathbb{Z}_2$ has zero divisors and contains the diagonal field $\{(0,0),(1,1)\}$.

(g) True. Send m/n to $(m1)(n1)^{-1}$; characteristic zero ensures no nonzero integer maps to zero.

(h) False. $(x^2) \subset F[x]$ is not prime: x^2 belongs but x does not.

(i) True. Choose a nonzero element of smallest degree and use the division algorithm; include the zero ideal separately.

(j) False. (x^2) is principal but its quotient has a nonzero nilpotent element $x + (x^2)$, so is not a field.

Exercise 27.15: A maximal ideal of an integer product

Let $M = 2\mathbb{Z} \times \mathbb{Z}$. Projection onto the first coordinate followed by reduction modulo two is onto $\mathbb{Z}_2$ and has kernel M. Thus $(\mathbb{Z} \times \mathbb{Z})/M \cong \mathbb{Z}_2$, a field, and M is maximal.

Exercise 27.16: A prime ideal that is not maximal

Let $P = \{0\} \times \mathbb{Z}$. Projection onto the first coordinate gives quotient $\mathbb{Z}$, a domain but not a field. Thus P is prime and not maximal. Explicitly $P \subsetneq 2\mathbb{Z} \times \mathbb{Z} \subsetneq \mathbb{Z} \times \mathbb{Z}$.

Exercise 27.17: An ideal that is not prime

Let $I = 4\mathbb{Z} \times \mathbb{Z}$. It is proper and nonzero. Neither factor $(2,0)$ of $(2,0)(2,0) = (4,0)$ lies in I, but their product does. Hence I is not prime.

Exercise 27.18: A reducible rational quadratic quotient

Since $x^2 - 5x + 6 = (x-2)(x-3)$, the defining polynomial is reducible over $\mathbb{Q}$. In the quotient, the two linear-factor cosets are nonzero (their degrees are too small to be multiples of the quadratic), and their product is zero. Thus the quotient is not a field.

Exercise 27.19: An irreducible rational quadratic quotient

The discriminant of $x^2 - 6x + 6$ is $36 - 24 = 12$, whose square root $2\sqrt{3}$ is irrational. Thus the quadratic has no rational root and is irreducible over $\mathbb{Q}$. Its generated ideal is maximal, so the quotient is a field, isomorphic to $\mathbb{Q}(\sqrt{3})$ by evaluating at $3 + \sqrt{3}$.

Exercise 27.20: Synopsis: maximality produces a field quotient

If M is maximal and $a \notin M$, the ideal $M + (a)$ must be R, so $1 = m + ra$ for some $m \in M, r \in R$. Passing to cosets gives $(r + M)(a + M) = 1 + M$, so every nonzero coset has an inverse and R/M is a field.

Exercise 27.21: Synopsis: a field quotient forces maximality

If R/M is a field and I is an ideal properly containing M, choose $a \in I \setminus M$; an inverse of $a + M$ gives $ra - 1 \in M$ for some $r \in R$, so $1 \in I$ and $I = R$. Thus M has no proper intermediate ideal and is maximal.

Exercise 27.22: Synopsis: ideals in a polynomial ring

For a nonzero ideal $I \subset F[x]$, choose $g \in I$ of smallest degree. Dividing any $f \in I$ by g gives a remainder in I of smaller degree, hence zero; therefore $I = (g)$. The zero ideal is (0).

Exercise 27.23: Synopsis: maximal polynomial ideals are irreducible

If (p) is maximal but $p = fg$ with both factors of positive degree, then $(p) \subsetneq (f) \subsetneq F[x]$, a contradiction. Hence a nonconstant generator of a maximal ideal must be irreducible.

Exercise 27.24: Finite prime quotients

If P is prime, R/P is an integral domain and is finite because R is finite. For any nonzero a in a finite domain, the multiplication map $x \mapsto ax$ is injective by cancellation, hence surjective. It therefore takes some x to 1, giving an inverse for a. Thus R/P is a field, and P is maximal.

Exercise 27.25: Different cyclic-ring subrings

It is possible when the subrings need not share the ambient unity. In $R = \mathbb{Z}_2 \times \mathbb{Z}_3$, the subrings $\mathbb{Z}_2 \times \{0\}$ and $\{0\} \times \mathbb{Z}_3$ are isomorphic to $\mathbb{Z}_2$ and $\mathbb{Z}_3$, respectively. Their identities are $(1, 0)$ and $(0, 1)$, not the ambient $(1, 1)$. This distinction avoids incorrectly equating their characteristics with that of R.

Exercise 27.26: Two different prime-field subrings

The same example $\mathbb{Z}_p \times \mathbb{Z}_q$, for distinct primes p, q, contains the field subrings $\mathbb{Z}_p \times \{0\}$ and $\{0\} \times \mathbb{Z}_q$. Each is a field under the inherited operations, with its own identity. Thus the answer is yes under the book's subring convention.

Exercise 27.27: The domain case

In an integral domain, a nonzero idempotent e satisfies $e(e - 1) = 0$, so $e = 1$. Therefore the identity of any field subring is the ambient identity. A copy of $\mathbb{Z}_p$ forces the ambient characteristic to be p, and a copy of $\mathbb{Z}_q$ forces it to be q. Since $p \neq q$ is impossible, no such domain exists. Equivalently distinct nonzero orthogonal identities would create zero divisors.

Exercise 27.28: Direct proof that maximal ideals are prime

Suppose $ab \in M$ and $a \notin M$. The ideal $M + (a)$ properly contains M, so maximality makes it R. Hence $1 = ra + m$ for some $r \in R, m \in M$. Multiply by b:

$$b = rab + mb \in M,$$

since both terms belong to M. This proves $a \in M$ or $b \in M$, and M is proper by definition, so it is prime.

Exercise 27.29: Maximal ideals and simple quotient rings

Ideals J containing N correspond to ideals J/N of R/N: inverse image under the quotient map reverses this assignment. To see the image direction, multiplication by $r + N$ sends $j + N$ to $rj + N \in J/N$; the inverse-image direction follows from the ideal test. Therefore the only ideals of R/N are zero and the whole quotient exactly when the only ideals containing N are N and R. Requiring a nontrivial quotient is equivalent to $N \neq R$. These statements say precisely that N is maximal if and only if R/N is simple.

Exercise 27.30: Nonzero prime ideals of a one-variable polynomial ring

Write the proper nonzero prime ideal as (p), using the principal-ideal theorem for $F[x]$. If $p = fg$ with $\deg f, \deg g > 0$, primality puts f or g in (p), impossible by degree. Thus p is irreducible. The quotient by an irreducible polynomial over a field is a field, so (p) is maximal.

Exercise 27.31: Divisibility and membership

By definition, $(f) = \{hf : h \in F[x]\}$. Thus $g \in (f)$ means exactly that $g = hf$ for some polynomial h, which is the statement $f \mid g$. This includes $f = 0$: membership and divisibility both then require $g = 0$.

Exercise 27.32: The ideal generated by two polynomials

The set $N = (f, g)$ contains zero and is closed under subtraction:

$$(rf + sg) - (r'f + s'g) = (r - r')f + (s - s')g.$$

Multiplying by any $h \in F[x]$ gives $(hr)f + (hs)g$, again in N, so it is an ideal. If both f, g were irreducible and had distinct degrees, their only common divisors would be constants: a nonconstant common divisor of an irreducible is an associate of it, forcing equal degrees. The Euclidean algorithm then gives $uf + vg = 1$, which would make $N = F[x]$. This contradicts the stated hypothesis; thus they cannot both be irreducible.

Exercise 27.33: A one-variable Nullstellensatz

Plan. Replace the finitely generated ideal by a single greatest common divisor. Assume the Fundamental Theorem of Algebra. Write $(f_1, \ldots, f_r) = (h)$ using the principal-ideal theorem. If

h is a nonzero constant, the ideal is the whole ring and the assertion is immediate. If $h = 0$, all f_i vanish everywhere; then g vanishes everywhere, so it is the zero polynomial and belongs to the ideal. Otherwise factor $h = c\prod_j (x - a_j)^{e_j}$. Its zeros are exactly the common zeros of the f_i, since h divides every f_i and is their polynomial linear combination. Each a_j is therefore a zero of g. Taking $m \geq \max_j e_j$ makes $h \mid g^m$, so $g^m \in (f_1, \ldots, f_r)$. Conversely assume the stated Nullstellensatz. If a nonconstant f had no complex zero, the condition on $g = 1$ would be vacuously true. It would follow that $1 = 1^m \in (f)$, impossible since a nonconstant polynomial cannot divide 1. Thus every nonconstant complex polynomial has a zero.

Exercise 27.34: The sum of two ideals

(a) Zero is $0 + 0 \in A + B$. Subtracting $a + b$ and $a' + b'$ yields $(a - a') + (b - b') \in A + B$. For $r \in R$, both $r(a + b) = ra + rb$ and $(a + b)r = ar + br$ lie in $A + B$. Thus this is an ideal even if R is noncommutative. **(b)** Since $0 \in A \cap B$, write $a = a + 0$ and $b = 0 + b$. This proves $A \subset A + B$ and $B \subset A + B$.

Exercise 27.35: The product of two ideals

(a) The zero sum can be represented as $0 \cdot 0$. Sums of finite sums of products remain such sums; a negative is obtained by replacing a_i by $-a_i$. Also

$$r\sum_i a_i b_i = \sum_i (ra_i)b_i, \qquad (\sum_i a_i b_i)r = \sum_i a_i(b_i r),$$

and the changed factors stay in the appropriate ideals. Hence AB is an ideal. **(b)** Each $a_i b_i$ lies in A by right absorption and in B by left absorption. Both ideals are closed under finite sums, so $AB \subset A \cap B$.

Exercise 27.36: The ideal quotient

Let $I = A : B = \{r : rb \in A \text{ for every } b \in B\}$. Zero belongs to I. If $r, s \in I$, then $(r-s)b = rb-sb \in A$ for every $b \in B$, so $r - s \in I$. If $t \in R$, then $(tr)b = t(rb) \in A$. Thus $tr \in I$, and commutativity gives the other side too. Therefore $A : B$ is an ideal.

Exercise 27.37: A one-sided matrix ideal

Let S be the matrices with second row zero. Differences and products of such matrices still have second row zero, so S is a subring. For any $B \in M_2(F)$, the second row of AB is zero whenever $A \in S$, so S is a right ideal. Using matrix units, $E_{11} \in S$ but $E_{21}E_{11} = E_{21} \notin S$. Therefore multiplication on the left by an arbitrary ambient matrix need not stay in S, and S is not a left ideal.

Exercise 27.38: Simplicity of a matrix ring

Let I be a nonzero two-sided ideal in $M_2(\mathbb{Z}_2)$, and choose $A \in I$ with nonzero entry $a_{ij} = 1$. For any $r, s \in \{1, 2\}$,

$$E_{ri}AE_{js} = a_{ij}E_{rs} = E_{rs} \in I.$$

Thus I contains all four matrix units. Every binary matrix is the sum of the matrix units corresponding to its nonzero entries, so $I = M_2(\mathbb{Z}_2)$. Hence the only two-sided ideals are zero and the whole ring, proving simplicity.

28 Gröbner Bases for Ideals

Exercise 28.1: Decreasing lexicographic order, 1

Plan. With $z < y < x$, compare the exponent triples $(\deg_x, \deg_y, \deg_z)$ lexicographically: the first unequal coordinate decides, regardless of total degree. The required arrangement is

$$-3x^3 + 7x^2y^2z - 5x^2yz^3 + 2xy^3z^5.$$

No coefficient or monomial changes; only the order of writing the terms changes.

Exercise 28.2: Decreasing lexicographic order, 2

Plan. With $z < y < x$, compare the exponent triples $(\deg_x, \deg_y, \deg_z)$ lexicographically: the first unequal coordinate decides, regardless of total degree. The required arrangement is

$$-4x + 5y^3z^3 + 3y^2z^5 - 8z^7.$$

No coefficient or monomial changes; only the order of writing the terms changes.

Exercise 28.3: Decreasing lexicographic order, 3

Plan. With $z < y < x$, compare the exponent triples $(\deg_x, \deg_y, \deg_z)$ lexicographically: the first unequal coordinate decides, regardless of total degree. The required arrangement is

$$2x^2yz^2 - 2xy^2z^2 - 7x + 3y + 10z^3.$$

No coefficient or monomial changes; only the order of writing the terms changes.

Exercise 28.4: Decreasing lexicographic order, 4

Plan. With $z < y < x$, compare the exponent triples $(\deg_x, \deg_y, \deg_z)$ lexicographically: the first unequal coordinate decides, regardless of total degree. The required arrangement is

$$-8xy - 4xz + 3yz^3 + 2yz + 38.$$

No coefficient or monomial changes; only the order of writing the terms changes.

Exercise 28.5: Reversed variable priority, 1

Now $x < y < z$, so compare $(\deg_z, \deg_y, \deg_x)$, starting with the exponent of z. For equal z exponents, the y exponent decides before the x exponent. Thus the polynomial from Exercise 1 becomes

$$2xy^3z^5 - 5x^2yz^3 + 7x^2y^2z - 3x^3.$$

Exercise 28.6: Reversed variable priority, 2

Now $x < y < z$, so compare $(\deg_z, \deg_y, \deg_x)$, starting with the exponent of z. For equal z exponents, the y exponent decides before the x exponent. Thus the polynomial from Exercise 2 becomes

$$-8z^7 + 3y^2z^5 + 5y^3z^3 - 4x.$$

Exercise 28.7: Reversed variable priority, 3

Now $x < y < z$, so compare $(\deg_z, \deg_y, \deg_x)$, starting with the exponent of z. For equal z exponents, the y exponent decides before the x exponent. Thus the polynomial from Exercise 3 becomes

$$10z^3 - 2xy^2z^2 + 2x^2yz^2 + 3y - 7x.$$

Exercise 28.8: Reversed variable priority, 4

Now $x < y < z$, so compare $(\deg_z, \deg_y, \deg_x)$, starting with the exponent of z. For equal z exponents, the y exponent decides before the x exponent. Thus the polynomial from Exercise 4 becomes

$$3yz^3 + 2yz - 4xz - 8xy + 38.$$

Exercise 28.9: The first twenty degree-lexicographic monomials

First sort by total degree, then by lexicographic order with $z < y < x$. There are $1, 3, 6, 10$ monomials of degrees $0, 1, 2, 3$, respectively, so these are exactly the first twenty:

$$1;$$
$$z,\ y,\ x;$$
$$z^2,\ yz,\ y^2,\ xz,\ xy,\ x^2;$$
$$z^3,\ yz^2,\ y^2z,\ y^3,\ xz^2,\ xyz,\ xy^2,\ x^2z,\ x^2y,\ x^3.$$

For instance $y^3 < xz^2$ despite the larger y exponent, because the latter has a positive x exponent.

Exercise 28.10: Decreasing degree-lexicographic order, 1

Compare total degrees first, putting larger degree first. Break ties by the exponent triple $(\deg_x, \deg_y, \deg_z)$ in decreasing lexicographic order. This gives

$$2xy^3z^5 - 5x^2yz^3 + 7x^2y^2z - 3x^3.$$

This differs from pure lexicographic ordering whenever a term of lower total degree has a larger x exponent.

Exercise 28.11: Decreasing degree-lexicographic order, 2

Compare total degrees first, putting larger degree first. Break ties by the exponent triple $(\deg_x, \deg_y, \deg_z)$ in decreasing lexicographic order. This gives

$$3y^2z^5 - 8z^7 + 5y^3z^3 - 4x.$$

This differs from pure lexicographic ordering whenever a term of lower total degree has a larger x exponent.

Exercise 28.12: Decreasing degree-lexicographic order, 3

Compare total degrees first, putting larger degree first. Break ties by the exponent triple $(\deg_x, \deg_y, \deg_z)$ in decreasing lexicographic order. This gives

$$2x^2yz^2 - 2xy^2z^2 + 10z^3 - 7x + 3y.$$

This differs from pure lexicographic ordering whenever a term of lower total degree has a larger x exponent.

Exercise 28.13: Decreasing degree-lexicographic order, 4

Compare total degrees first, putting larger degree first. Break ties by the exponent triple $(\deg_x, \deg_y, \deg_z)$ in decreasing lexicographic order. This gives

$$3yz^3 - 8xy - 4xz + 2yz + 38.$$

This differs from pure lexicographic ordering whenever a term of lower total degree has a larger x exponent.

Exercise 28.14: A one-step reduction of the largest leading monomial

The largest leading monomial is x^2y, in $f_2 = x^2y + 4xy$. It is divisible by the leading monomial xy of $f_3 = xy - y^2$. Replace

$$f_2 \longmapsto f_2 - xf_3 = xy^2 + 4xy.$$

The new basis is $\{xy^2 - 2x,\ xy^2 + 4xy,\ xy - y^2\}$. Its maximum monomial is xy^2, smaller than x^2y. The ideal is unchanged because the replacement can be reversed by adding xf_3.

Exercise 28.15: Reducing a mixed cubic basis

The first polynomial has leading monomial xy; the third, $x - y^4$, has leading monomial x. Replace the first by

$$(xy + y^3) - y(x - y^4) = y^5 + y^3.$$

The resulting basis $\{y^5 + y^3,\ y^3 + z,\ x - y^4\}$ generates the same ideal. Its largest monomial is now x, whereas the previous largest was xy. In pure lex order a monomial involving x exceeds every monomial involving only y, z, regardless of degree.

Exercise 28.16: When the maximum cannot be reduced in one step

The leading monomials are xyz, x^3, and x^2yz^3. The largest is x^3. Neither xyz nor x^2yz^3 divides x^3, because each contains positive y, z exponents. Thus no division step using another basis element can cancel that largest monomial. One can reduce the third polynomial using the first, but the unchanged x^3 still remains the maximum. Consequently a single-step reduction that strictly decreases the maximum monomial of the entire basis is impossible.

Exercise 28.17: A reduction involving only two variables

The largest leading monomial is y^3z^2, divisible by y^2z^2, the leading monomial of the third polynomial. Replace

$$(y^3z^2 - 2z) - y(y^2z^2 + 3) = -3y - 2z.$$

Thus $\{y^2z^3 + 3,\ -3y - 2z,\ y^2z^2 + 3\}$ generates the same ideal. The maximum has dropped from y^3z^2 to y^2z^3.

Exercise 28.18: A linear Groebner basis

Name the three displayed generators f_1, f_2, f_3 in their source order. Ordinary elimination produces

$$\begin{aligned} g_1 &= 3f_1 - f_3 = 2w + 11z - 4, \\ g_2 &= -7f_1 + 2f_2 + 3f_3 = 4x - 29z + 14, \\ g_3 &= -5f_1 + 2f_2 + f_3 = 4y - 23z + 18. \end{aligned}$$

The coefficient transformation has nonzero determinant -8, so it is invertible over $\mathbb{R}$: both triples generate the same ideal. The leading monomials are w, x, y, pairwise relatively prime. Their S-polynomials therefore reduce to zero, so $\{g_1, g_2, g_3\}$ is a Groebner basis. Equivalently this is row reduction of a linear system, with z free.

Exercise 28.19: An inconsistent linear ideal

Let the source generators be f_1, f_2, f_3. Subtract twice the first from the second to obtain $6x - 5y + 1$. Subtract the first from the third to obtain $-6x + 5y - 3$. Adding gives

$$f_2 + f_3 - 3f_1 = -2.$$

Thus the ideal contains the nonzero constant -2, hence also 1. It is the whole polynomial ring, and $\{1\}$ is a Groebner basis. Its algebraic variety is empty.

Exercise 28.20: A one-variable common divisor

Let $f = x^4 + x^3 - 3x^2 - 4x - 4$ and $g = x^3 + x^2 - 4x - 4$. The first division gives

$$f - xg = x^2 - 4, \qquad g = (x + 1)(x^2 - 4).$$

Therefore $(f, g) = (x^2 - 4)$. A single generator of a one-variable ideal is a Groebner basis: the leading term of every multiple is divisible by the generator's leading term. The answer is $\{x^2 - 4\}$.

Exercise 28.21: Three one-variable generators

For the first two generators f_1, f_2, division gives

$$f_1 - (x - 3)f_2 = 6(x^2 - 3x + 2), \qquad f_2 = (x + 2)(x^2 - 3x + 2).$$

Their gcd is $h = (x - 1)(x - 2)$. The third generator is $f_3 = (x - 1)^2(x + 2)$, and

$$f_3 - (x + 3)h = 4(x - 1).$$

All three original polynomials are divisible by $x - 1$, while the displayed steps put $x - 1$ in their ideal. Hence the ideal is $(x - 1)$ and a Groebner basis is $\{x - 1\}$.

Exercise 28.22: Euclidean algorithm giving the unit ideal

Write the two source generators as f, g. The successive remainder equations are

$$\begin{aligned} f &= (x^2+x)g + r_1, & r_1 &= x^2+3x-5,\\ g &= (x-4)r_1 + r_2, & r_2 &= 18x-21,\\ r_1 &= \left(\frac{x}{18}+\frac{25}{108}\right) r_2 - \frac{5}{36}. \end{aligned}$$

Each remainder lies in the ideal of the preceding pair, so the original ideal contains the nonzero constant $-5/36$, and therefore 1. The required Groebner basis is $\{1\}$.

Exercise 28.23: A two-point variety

Put $f = x^2y - x - 2$ and $g = xy + 2y - 9$. Then

$$f - (x-2)g = 4(2x + y - 5).$$

Thus $h = 2x+y-5$ lies in the ideal; furthermore $g-(y/2)h = -(y^2-9y+18)/2$. Set $k = y^2-9y+18$. These steps are reversible, so $(f, g) = (h, k)$. The leading monomials x and y^2 are relatively prime; the pair is a Groebner basis. Since $k = (y-3)(y-6)$, solve $y = 3$ or 6 and $x = (5-y)/2$. The real variety is $\{(1, 3), (-1/2, 6)\}$.

Exercise 28.24: A variety consisting of the origin

Let $f = x^2y + x$ and $g = xy^2 - y$. Their leading-term cancellation gives $yf - xg = 2xy$, so $h = xy$ lies in the ideal. Then $f - xh = x$ and $g - yh = -y$. Hence (f, g) contains both x, y. Conversely both f, g belong to (x, y), so the ideals are equal. Thus $\{x, y\}$ is a Groebner basis, and its common zero set is exactly $\{(0, 0)\}$.

Exercise 28.25: A cubic curve intersection reduced to one variable

For $f = x^2y + x + 1$ and $g = xy^2 + y - 1$, compute $yf - xg = x + y = h$. Dividing by h gives

$$g = y^2h - k, \qquad f = (xy - y^2 + 1)h + k, \qquad k = y^3 - y + 1.$$

These identities show $(f, g) = (h, k)$. The leading monomials x and y^3 are relatively prime, so $\{x + y, y^3 - y + 1\}$ is a Groebner basis. The real variety is $\{(-t, t) : t^3 - t + 1 = 0\}$. This cubic has exactly one real zero: its local maximum and minimum, at $t = \mp 1/\sqrt{3}$, are both positive, while its value tends to $-\infty$ as $t \to -\infty$. Thus the description gives a single real point without requiring a radical expression.

Exercise 28.26: A line and an isolated point

Let $f = x^2y + xy^2$ and $g = xy - x$. Set $h = f - (x + y + 1)g = x^2 + x$. Thus $(f, g) = (h, g)$. For lex order, their leading monomials are x^2 and xy. The only S-polynomial is

$$yh - xg = x^2 + xy = h + g,$$

which reduces to zero. Hence $\{x^2 + x, xy - x\}$ is a Groebner basis. The equations are $x(x + 1) = 0$ and $x(y - 1) = 0$. For $x = 0$, any y works; for $x = -1$, we must have $y = 1$. The variety is the entire y-axis together with the point $(-1, 1)$.

Exercise 28.27: True and false statements

(a) True. Here the book uses $F[\mathbf{x}] = F[x_1, \ldots, x_n]$. The Hilbert Basis Theorem gives a finite generating set for every ideal. This is not a claim that a multivariable ideal is principal.

(b) False. For example $\mathbb{R}^2 \setminus \{(0, 0)\}$ is not the zero set of polynomials: zero sets are closed, whereas this set is not.

(c) True. It is the zero set of the constant polynomial 1.

(d) True. A point (a, b) is the zero set of $(x - a)^2 + (y - b)^2$; a finite union of points is the zero set of the product of these polynomials.

(e) True. A line is the zero set of $ax + by + c$ with $(a, b) \neq (0, 0)$.

(f) True. A finite union of lines is the zero set of the product of their defining linear polynomials.

(g) True. The Euclidean algorithm preserves common divisors at each remainder step and terminates as degrees decrease.

(h) This depends on the reader's experience. If one has used the Euclidean algorithm in one variable or row reduction for linear polynomials, one has already computed special cases of Groebner bases; it is not a universal mathematical assertion about every reader.

(i) False. For example $\{x\}$ and $\{2x\}$ are distinct Groebner bases of (x). A *reduced monic* Groebner basis for a fixed monomial order is unique.

(j) False. $x \in (x, y)$ but $x \notin (x^2, y^2)$, since the latter ideal has no terms of degree one. Equal zero sets need not give equal ideals.

Exercise 28.28: Order is not divisibility

With $y < x$ in lex order, $y < x$ because the x exponent of y is zero and that of x is one. Yet y does not divide x: a product of y with a monomial must still have positive y exponent. This is the required counterexample.

Exercise 28.29: The ideal generated by a finite list

Source correction. The displayed coefficient condition $c_i \in I$ is circular; the intended condition is $c_i \in R$. With that correction define $I = \{\sum_{i=1}^{r} c_i f_i : c_i \in R\}$. Zero is obtained by all coefficients zero. A difference is obtained by subtracting the coefficients, so I is an additive subgroup. For $a \in R$,

$$a \sum_i c_i f_i = \sum_i (a c_i) f_i \in I.$$

Commutativity gives absorption on the right as well, proving the ideal claim. Unity ensures each f_i belongs to I, so this is exactly the smallest ideal containing the list.

Exercise 28.30: Common divisors survive a remainder step

If $d \mid f$ and $d \mid g$, then $d \mid (f - gq) = r$. Conversely if $d \mid g$ and $d \mid r$, then $d \mid (gq + r) = f$. Thus the common divisors of f, g are exactly those of g, r. This elementary equivalence is the reason the Euclidean algorithm computes a gcd.

Exercise 28.31: Checking a simple Groebner basis

For $f = xy$ and $g = y^2 - y$, the leading monomials under lex order $y < x$ are xy and y^2. Their least common multiple is xy^2, and

$$S(f, g) = yf - xg = xy = f.$$

It reduces to zero by subtracting f. Buchberger's criterion therefore proves that $\{xy, y^2 - y\}$ is a Groebner basis of the ideal it generates.

Exercise 28.32: The vanishing ideal of a set

The zero polynomial vanishes on S. If f, g vanish on S, then $(f - g)(s) = 0$ for every $s \in S$. If h is any polynomial, $(hf)(s) = h(s)f(s) = 0$ for every $s \in S$. These are precisely closure under differences and absorption, so $I(S)$ is an ideal in $F[x_1, \ldots, x_n]$.

Exercise 28.33: A set lies in its double construction

Take $s \in S$. By definition every $f \in I(S)$ vanishes at s. Hence s is a common zero of all polynomials in $I(S)$, which says $s \in V(I(S))$. Since s was arbitrary, $S \subset V(I(S))$.

Exercise 28.34: The double construction can enlarge a set

Take $S = \{(t, 0) : t \in \mathbb{R},\ t \neq 0\}$. If f vanishes on S, the one-variable polynomial $f(x, 0)$ has infinitely many roots and is therefore zero. Thus f is divisible by y, so $I(S) = (y)$. Consequently $V(I(S))$ is the full x-axis and contains $(0, 0)$, which is absent from S.

Exercise 28.35: An ideal lies in its double construction

If $f \in N$ and $s \in V(N)$, the definition of $V(N)$ gives $f(s) = 0$. Therefore f vanishes on every point of $V(N)$ and lies in $I(V(N))$. Thus $N \subset I(V(N))$. If $V(N)$ is empty, every polynomial vacuously vanishes on it and the inclusion still holds.

Exercise 28.36: The double construction can enlarge an ideal

Choose $N = (x^2, y^2) \subset \mathbb{R}[x, y]$. Its zero set is $\{(0, 0)\}$, so x vanishes on $V(N)$ and belongs to $I(V(N))$. But $x \notin N$: every term of every polynomial $ax^2 + by^2$ has total degree at least two. Hence $I(V(N)) \neq N$. In fact $I(V(N)) = (x, y)$.

29 Introduction to Extension Fields

Exercise 29.1: A shifted square root is algebraic

For $\alpha = 1 + \sqrt{2}$, we have $(\alpha - 1)^2 = 2$. Expanding gives $\alpha^2 - 2\alpha - 1 = 0$. Thus the nonzero rational polynomial $f(x) = x^2 - 2x - 1$ witnesses algebraicity over $\mathbb{Q}$.

Exercise 29.2: A sum of two square roots

For $\alpha = \sqrt{2} + \sqrt{3}$, square to obtain $\alpha^2 = 5 + 2\sqrt{6}$. Isolate the radical and square again:

$$(\alpha^2 - 5)^2 = 24.$$

Hence $\alpha^4 - 10\alpha^2 + 1 = 0$, so $f(x) = x^4 - 10x^2 + 1 \in \mathbb{Q}[x]$ proves algebraicity. For this exercise an annihilating polynomial suffices; minimality is not required.

Exercise 29.3: A shifted imaginary unit

For $\alpha = 1 + i$, the equation $(\alpha - 1)^2 = -1$ gives $\alpha^2 - 2\alpha + 2 = 0$. Therefore $x^2 - 2x + 2$ is a nonzero rational polynomial vanishing at α.

Exercise 29.4: A nested square and cube root

Let $\alpha = \sqrt{1 + \sqrt[3]{2}}$. Then $\alpha^2 - 1 = \sqrt[3]{2}$, so $(\alpha^2 - 1)^3 = 2$. Thus

$$f(x) = (x^2 - 1)^3 - 2 = x^6 - 3x^4 + 3x^2 - 3$$

is a rational annihilating polynomial. Every displayed operation is applied to the actual given number, so no converse claim about extraneous roots is needed.

Exercise 29.5: A complex nested radical

For $\alpha = \sqrt{\sqrt[3]{2} - i}$, we have $(\alpha^2 + i)^3 = 2$. Set $A(x) = (x^2 + i)^3 - 2$. Multiplying by its coefficient-conjugate gives a polynomial in $\mathbb{Q}[x]$:

$$A(x)\overline{A}(x) = x^{12} + 3x^8 - 4x^6 + 3x^4 + 12x^2 + 5.$$

The first factor vanishes at α, so the product does too. For checking the multiplication, write $A(x) = (x^6 - 3x^2 - 2) + i(3x^4 - 1)$; the product is the sum of squares of these two real polynomials.

Exercise 29.6: Minimal polynomial of a nested quadratic radical

Let $\alpha = \sqrt{3 - \sqrt{6}}$. Then $(\alpha^2 - 3)^2 = 6$, giving $p(x) = x^4 - 6x^2 + 3$. To prove minimality, first $\sqrt{6} = 3 - \alpha^2 \in \mathbb{Q}(\alpha)$. In $K = \mathbb{Q}(\sqrt{6})$, the element $3 - \sqrt{6}$ is not a square: the norm of a square $(u + v\sqrt{6})^2$ is $(u^2 - 6v^2)^2$, a rational square, but the norm of $3 - \sqrt{6}$ is $9 - 6 = 3$, not a rational square. Thus $\alpha \notin K$ and $[K(\alpha) : K] = 2$. Since $[K : \mathbb{Q}] = 2$, α has degree four. The monic annihilating quartic must therefore be $\mathrm{irr}(\alpha, \mathbb{Q}) = p(x)$.

Exercise 29.7: A rational shift inside a nested radical

For $\alpha = \sqrt{1/3 + \sqrt{7}}$, the equation $(\alpha^2 - 1/3)^2 = 7$ gives

$$p(x) = x^4 - \frac{2}{3}x^2 - \frac{62}{9}.$$

The field $\mathbb{Q}(\alpha)$ contains $K = \mathbb{Q}(\sqrt{7})$ because $\sqrt{7} = \alpha^2 - 1/3$. The norm of $1/3 + \sqrt{7}$ down to $\mathbb{Q}$ is $1/9 - 7 = -62/9$. It cannot be the norm of a square in K, since such a norm is a square of a rational number and hence nonnegative. Therefore adjoining α to K has degree two, and the full degree is four. Thus $\mathrm{irr}(\alpha, \mathbb{Q}) = p(x)$ and $\deg(\alpha, \mathbb{Q}) = 4$. Multiplying by nine gives an integer polynomial but the irreducible polynomial is conventionally monic.

Exercise 29.8: Minimal polynomial of a real-imaginary sum

Let $\alpha = \sqrt{2} + i$. Its equation $(\alpha^2 - 1)^2 = -8$ gives $p(x) = x^4 - 2x^2 + 9$. For minimality, $3/\alpha = \sqrt{2} - i$ because $(\sqrt{2} + i)(\sqrt{2} - i) = 3$. Hence

$$\sqrt{2} = \frac{\alpha + 3/\alpha}{2}, \qquad i = \frac{\alpha - 3/\alpha}{2}.$$

Thus $\mathbb{Q}(\alpha) = \mathbb{Q}(\sqrt{2}, i)$. The real field $\mathbb{Q}(\sqrt{2})$ does not contain i, so adjoining i doubles its degree two, giving degree four. Therefore p is the monic minimal polynomial and $\deg(\alpha, \mathbb{Q}) = 4$.

Exercise 29.9: The imaginary unit over the rationals

The number i satisfies $x^2 + 1 = 0$. This quadratic has no rational root (indeed no real root), so it is irreducible over $\mathbb{Q}$. Thus i is algebraic of degree two over $\mathbb{Q}$.

Exercise 29.10: A nonreal number over the reals

The number $1 + i$ satisfies $x^2 - 2x + 2 = 0$. Its discriminant is $-4 < 0$, so the polynomial is irreducible over $\mathbb{R}$. Thus $1 + i$ is algebraic of degree two over $\mathbb{R}$.

Exercise 29.11: A square root of a transcendental number

The number $\sqrt{\pi}$ is transcendental over $\mathbb{Q}$. If it were algebraic, its square would be algebraic because algebraic elements form a field (or because powers of an algebraic element lie in its finite simple extension). This would make π algebraic over $\mathbb{Q}$, contradicting the known transcendence of π.

Exercise 29.12: The same number over the reals

Since $\sqrt{\pi} \in \mathbb{R}$, the polynomial $x - \sqrt{\pi} \in \mathbb{R}[x]$ vanishes at it. Its minimal polynomial over $\mathbb{R}$ is linear and its degree is one. Algebraicity always depends on the base field.

Exercise 29.13: A square root over a rational-function field

Put $t = \sqrt{\pi}$. Since $\pi = t^2$, the base field is $\mathbb{Q}(t^2)$. The polynomial $x^2 - \pi$ shows degree at most two. It has no zero in $\mathbb{Q}(\pi)$: if a rational function $P(\pi)/Q(\pi)$ squared to π, transcendence would give the polynomial identity $P(T)^2 = TQ(T)^2$. The exponent of T on the left is even, and on the right odd, a contradiction after removing any common powers. Thus the degree is two.

Exercise 29.14: A square of pi over the rationals

The number π^2 is transcendental over $\mathbb{Q}$. If a nonzero $f \in \mathbb{Q}[x]$ satisfied $f(\pi^2) = 0$, the nonzero polynomial $f(x^2)$ would vanish at π, contradicting transcendence. Therefore no algebraic degree is assigned.

Exercise 29.15: An element already in the base field

Here $\pi^2 \in \mathbb{Q}(\pi)$, so its minimal polynomial over that field is $x - \pi^2$ and its degree is one. The coefficient $-\pi^2$ is allowed because it lies in the specified base field.

Exercise 29.16: Two different powers of a transcendental number

Set $T = \pi^3$. The element π^2 satisfies $x^3 - T^2 = 0$ over $\mathbb{Q}(T)$. A reducible cubic over a field has a root in that field. But a rational root $P(T)/Q(T)$ would imply $P(T)^3 = T^2 Q(T)^3$. Comparing exponents of the irreducible factor T gives a multiple of three on the left and an integer congruent to two modulo three on the right. This is impossible. Hence the cubic is irreducible, and π^2 has degree three over $\mathbb{Q}(\pi^3)$.

Exercise 29.17: Splitting the four-element-field quadratic

In characteristic two, $-\alpha = \alpha$ and $\alpha^2 = \alpha + 1$. Polynomial division gives

$$x^2 + x + 1 = (x + \alpha)(x + \alpha + 1),$$

since the sum of the roots is 1 and their product is $\alpha(\alpha + 1) = \alpha^2 + \alpha = 1$. Thus the second root is $\alpha + 1$.

Exercise 29.18: Constructing the nine-element field

(a) The values of $x^2 + 1$ at $0, 1, 2$ in $\mathbb{Z}_3$ are $1, 2, 2$, so it has no root and is irreducible. **(b)** With $\alpha^2 = -1 = 2$, each element is uniquely $a + b\alpha$ for $a, b \in \mathbb{Z}_3$. Use

$$(a + b\alpha) + (c + d\alpha) = (a + c) + (b + d)\alpha,$$

$$(a + b\alpha)(c + d\alpha) = (ac - bd) + (ad + bc)\alpha,$$

reducing coefficients modulo three. For example $(1+2\alpha)(2+\alpha) = 2+5\alpha+2\alpha^2 = 2\alpha$. In the requested order, the full tables are

$+$	0	1	2	α	2α	$1+\alpha$	$1+2\alpha$	$2+\alpha$	$2+2\alpha$
0	0	1	2	α	2α	$1+\alpha$	$1+2\alpha$	$2+\alpha$	$2+2\alpha$
1	1	2	0	$1+\alpha$	$1+2\alpha$	$2+\alpha$	$2+2\alpha$	α	2α
2	2	0	1	$2+\alpha$	$2+2\alpha$	α	2α	$1+\alpha$	$1+2\alpha$
α	α	$1+\alpha$	$2+\alpha$	2α	0	$1+2\alpha$	1	$2+2\alpha$	2
2α	2α	$1+2\alpha$	$2+2\alpha$	0	α	1	$1+\alpha$	2	$2+\alpha$
$1+\alpha$	$1+\alpha$	$2+\alpha$	α	$1+2\alpha$	1	$2+2\alpha$	2	2α	0
$1+2\alpha$	$1+2\alpha$	$2+2\alpha$	2α	1	$1+\alpha$	2	$2+\alpha$	0	α
$2+\alpha$	$2+\alpha$	α	$1+\alpha$	$2+2\alpha$	2	2α	0	$1+2\alpha$	1
$2+2\alpha$	$2+2\alpha$	2α	$1+2\alpha$	2	$2+\alpha$	0	α	1	$1+\alpha$

$\cdot$	0	1	2	α	2α	$1+\alpha$	$1+2\alpha$	$2+\alpha$	$2+2\alpha$
0	0	0	0	0	0	0	0	0	0
1	0	1	2	α	2α	$1+\alpha$	$1+2\alpha$	$2+\alpha$	$2+2\alpha$
2	0	2	1	2α	α	$2+2\alpha$	$2+\alpha$	$1+2\alpha$	$1+\alpha$
α	0	α	2α	2	1	$2+\alpha$	$1+\alpha$	$2+2\alpha$	$1+2\alpha$
2α	0	2α	α	1	2	$1+2\alpha$	$2+2\alpha$	$1+\alpha$	$2+\alpha$
$1+\alpha$	0	$1+\alpha$	$2+2\alpha$	$2+\alpha$	$1+2\alpha$	2α	2	1	α
$1+2\alpha$	0	$1+2\alpha$	$2+\alpha$	$1+\alpha$	$2+2\alpha$	2	α	2α	1
$2+\alpha$	0	$2+\alpha$	$1+2\alpha$	$2+2\alpha$	$1+\alpha$	1	2α	α	2
$2+2\alpha$	0	$2+2\alpha$	$1+\alpha$	$1+2\alpha$	$2+\alpha$	α	1	2	2α

Irreducibility ensures this quotient is a field, so each nonzero row of the multiplication table contains 1.

Exercise 29.19: Correcting algebraic element

An element α of an extension field E/F is algebraic over F exactly when it is a zero of a *nonzero polynomial in* $F[x]$. Both qualifications matter: the zero polynomial vanishes everywhere, and a polynomial with arbitrary coefficients in E could always be $x - \alpha$.

Exercise 29.20: Correcting transcendental element

An element $\beta \in E$ is transcendental over F exactly when no *nonzero* polynomial in $F[x]$ vanishes at β. The zero polynomial is excluded, since it vanishes at every element.

Exercise 29.21: Correcting monic polynomial

A nonzero polynomial is monic when its *leading coefficient* is 1; its other coefficients may be arbitrary. For example $x^2 - 3x + 7$ is monic although not all its coefficients are one.

Exercise 29.22: Correcting simple extension

The extension E/F is simple if $E = F(\alpha)$ for some $\alpha \in E$. Equivalently no proper subfield of E that contains *both* F *and* α exists. Omitting the condition that the subfield contain F changes the

meaning.

Exercise 29.23: True and false statements

(a) True. The transcendence of π over $\mathbb{Q}$ is a known theorem used here.

(b) True. $\mathbb{C} = \mathbb{R}(i)$.

(c) True. For $a \in F$, the nonzero polynomial $x - a \in F[x]$ vanishes at a.

(d) True. The rational field is a subfield of $\mathbb{R}$.

(e) False. A subfield has the same identity and characteristic; $\mathbb{Q}$ has characteristic zero and $\mathbb{Z}_2$ characteristic two.

(f) True. The minimal polynomial divides every rational annihilating polynomial, so their degrees are at least n.

(g) False. Take $\alpha = \sqrt{2}$, of degree two over $\mathbb{Q}$, and $f = x - \sqrt{2} \in \mathbb{R}[x]$, of degree one.

(h) True. Kronecker's construction adjoins a zero using a quotient by an irreducible factor.

(i) False. $x^2 + 1 \in \mathbb{R}[x]$ has no zero in the extension $\mathbb{R}$ itself.

(j) True. Evaluation $\mathbb{Q}[x] \to \mathbb{Q}[\pi]$ is onto and is injective by transcendence of π.

Exercise 29.24: Making transcendental numbers algebraic over larger fields

(a) Take $F = \mathbb{Q}(\pi^3)$. The polynomial $x^3 - \pi^3$ vanishes at π, and has no root in F: treating $T = \pi^3$ as an indeterminate, $P(T)^3 = TQ(T)^3$ contradicts exponents of T. A cubic with no root is irreducible. Hence the degree is three. **(b)** Take $E = \mathbb{Q}(e^{10})$. The element $u = e^2$ is transcendental over $\mathbb{Q}$, since otherwise e would be algebraic over $\mathbb{Q}$. For any transcendental u, put $L = \mathbb{Q}(u^m)$ and let A be the L-span of $1, u, \ldots, u^{m-1}$. These powers are independent: after clearing denominators in a proposed relation, powers occurring in different terms have different exponent residues modulo m, so no nonzero polynomial relation can vanish at the transcendental u. Thus $\dim_L A = m$. Reducing powers using $u^m \in L$ shows that A is closed under multiplication and contains 1. For $0 \neq a \in A$, multiplication by a is an injective linear map $A \to A$, because A lies in the field $\mathbb{Q}(u)$. Finite dimension makes it surjective, so $ab = 1$ for some $b \in A$. Hence A is a field. It contains L and u and is contained in $\mathbb{Q}(u)$, so $A = \mathbb{Q}(u)$ and $[\mathbb{Q}(u) : \mathbb{Q}(u^m)] = m$. Taking $m = 5$ proves e^2 has degree five over E.

Exercise 29.25: Splitting an irreducible binary cubic

(a) The cubic $p = x^3 + x^2 + 1$ takes value 1 at both elements $0, 1$ of $\mathbb{Z}_2$, so has no linear factor and is irreducible. **(b)** If $p(\alpha) = 0$, then $\alpha^3 = \alpha^2 + 1$. Squaring an equation in characteristic two shows that α^2 and α^4 are roots too. They are distinct: α has multiplicative order seven in the field of eight elements, so its first, second, and fourth powers differ. Reducing powers gives $\alpha^4 = \alpha^2 + \alpha + 1$. Thus

$$p(x) = (x + \alpha)(x + \alpha^2)(x + \alpha^2 + \alpha + 1).$$

The first long division gives $p(x) = (x + \alpha)(x^2 + (1 + \alpha)x + \alpha^2 + \alpha)$; the other two displayed roots factor that quadratic. This supplies the actual linear factors, not just their existence.

Exercise 29.26: The additive and multiplicative groups of eight elements

Degree three over $\mathbb{Z}_2$ gives the unique representation $a + b\alpha + c\alpha^2$ with binary coefficients. Addition is coordinatewise, so the additive group is $\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2$. There are seven nonzero elements. Their multiplicative group has prime order seven, so any nonidentity element generates it by Lagrange's theorem. Hence it is $\mathbb{Z}_7$. The additive group is not cyclic: every nonzero element has additive order two.

Exercise 29.27: Why minimal polynomial is an appropriate name

Division by the monic irreducible polynomial $p = \operatorname{irr}(\alpha, F)$ shows that any $f \in F[x]$ with $f(\alpha) = 0$ is a multiple of p. Indeed its remainder would have smaller degree and still vanish at α, so must be zero. Thus p has the smallest possible degree among nonzero annihilating polynomials, and monicity makes it unique.

Exercise 29.28: Synopsis of Kronecker construction

Choose an irreducible factor p of the given nonconstant $f \in F[x]$ and form the field $E = F[x]/(p)$. The constant cosets embed F in E, and the coset $\alpha = x + (p)$ satisfies $p(\alpha) = 0$, hence $f(\alpha) = 0$ because p divides f.

Exercise 29.29: Exchanging a transcendental generator

Since α is algebraic over $F(\beta)$, clear denominators in an annihilating equation to obtain

$$H(\alpha, \beta) = 0, \qquad H(X, Y) \in F[X, Y] \setminus \{0\}.$$

Here clearing denominators can be done without making every coefficient zero when evaluated at β; start with actual nonzero coefficients in $F(\beta)$ and choose polynomial representatives. Regard $H(\alpha, Y)$ as a polynomial over $F(\alpha)$. It is not the zero polynomial: some coefficient is a nonzero polynomial in α, and transcendence of α prevents its vanishing. Its value at $Y = \beta$ is zero. Its degree in Y cannot be zero either, because that would give a nonzero element equal to zero. Thus β is algebraic over $F(\alpha)$.

Exercise 29.30: Counting a finite simple extension

If $\operatorname{irr}(\alpha, F)$ has degree n, every element of $F(\alpha)$ has a unique expression

$$a_0 + a_1\alpha + \cdots + a_{n-1}\alpha^{n-1}, \qquad a_i \in F.$$

Spanning follows from division by the minimal polynomial, and uniqueness from its minimal degree. Each of the n coefficients has q choices, independently. Thus there are exactly q^n elements.

Exercise 29.31: A twenty-seven-element field

(a) The polynomial $x^3 + 2x + 1$ takes value 1 at each of $0, 1, 2$ in $\mathbb{Z}_3$, so has no root and is irreducible.
(b) Therefore $\mathbb{Z}_3[x]/(x^3 + 2x + 1)$ is a field. Each coset has a unique representative $a + bx + cx^2$ with $a, b, c \in \mathbb{Z}_3$, giving $3^3 = 27$ elements.

Exercise 29.32: Quadratic extensions in every prime characteristic

(a) For odd p, the nonzero residues pair as $\{a, -a\}$, with two distinct members per pair. Their squares agree, and equality of squares implies this same pairing because $(a-b)(a+b) = 0$ in the field. Thus there are only $(p-1)/2$ nonzero squares, fewer than $p-1$, so a nonsquare exists. **(b)** Choose a nonsquare $d \in \mathbb{Z}_p$. Then $x^2 - d$ has no root and is irreducible, and its quotient field has p^2 elements. For the omitted prime $p = 2$, use $x^2 + x + 1$, which has no root in $\mathbb{Z}_2$ and produces four elements. This covers every prime.

Exercise 29.33: Nonconstant rational functions stay transcendental

Let $\beta = f(\alpha)/g(\alpha) \notin F$, where $f, g \in F[x]$ and $g(\alpha) \neq 0$. The polynomial $f(x) - \beta g(x) \in F(\beta)[x]$ vanishes at α. It is nonzero, since otherwise any nonzero coefficient of g would show $\beta \in F$. Thus α is algebraic over $F(\beta)$. If β were algebraic over F, adjoining it gives a finite extension, and then adjoining α gives another finite extension. Their composite would make α algebraic over F, contradicting its transcendence. Hence β is transcendental over F.

Exercise 29.34: A cubic-radical field without checking every axiom

The polynomial $x^3 - 2$ is irreducible over $\mathbb{Q}$ by Eisenstein's criterion at 2. Therefore $\mathbb{Q}(\sqrt[3]{2})$ has basis $1, \sqrt[3]{2}, (\sqrt[3]{2})^2$ and consists exactly of the displayed expressions with rational coefficients. Since it is a simple algebraic extension, it is already a field. This identifies the set as a subfield of $\mathbb{R}$ without separately proving formulas for inverses.

Exercise 29.35: Fields of orders eight, sixteen, and twenty-five

For order eight use $\mathbb{Z}_2[x]/(x^3 + x + 1)$; the cubic has no binary root, so is irreducible. For order sixteen use $\mathbb{Z}_2[x]/(x^4 + x + 1)$. It has no binary root. A reducible quartic with no linear factor would be a product of irreducible quadratics, but the only monic irreducible binary quadratic is $x^2 + x + 1$. Modulo that quadratic, $x^3 = 1$ and $x^4 = x$, so $x^4 + x + 1$ has remainder 1 and is not divisible by it. Hence it is irreducible. For order twenty-five use $\mathbb{Z}_5[x]/(x^2 + 2)$; a root would have square 3, which is not among $0, 1, 4$. Unique remainders of degrees less than $3, 4, 2$ give respectively $2^3, 2^4, 5^2$ elements.

Exercise 29.36: Every element of a finite field is algebraic

Let $|F| = q$. Its nonzero elements form a group of order $q-1$, so Lagrange's theorem gives $a^{q-1} = 1$ for every $a \neq 0$. Thus all elements, including zero, satisfy $a^q - a = 0$. The nonzero polynomial $x^q - x$ has coefficients in the prime subfield $\mathbb{Z}_p$, so every element is algebraic over that subfield.

Exercise 29.37: Prime-power order

List the finitely many elements as $a_1, \ldots, a_r$ and form the chain $F_0 = \mathbb{Z}_p$, $F_i = F_{i-1}(a_i)$. By Exercise 36, each a_i is algebraic even over $\mathbb{Z}_p$, hence also over F_{i-1}. If its degree there is d_i, Exercise 30 gives $|F_i| = |F_{i-1}|^{d_i}$. Starting from $|F_0| = p$ and iterating yields $|F| = p^{d_1 \cdots d_r}$, a prime power. Elements already adjoined simply contribute $d_i = 1$.

30 Vector Spaces

Exercise 30.1: Three disjoint bases

Use the three sets $B_1 = \{(1,0),(0,1)\}$, $B_2 = \{(2,0),(0,2)\}$, and $B_3 = \{(3,0),(0,3)\}$. No vector appears in two sets. Each spans $\mathbb{R}^2$ because $(x,y) = (x/k)(k,0) + (y/k)(0,k)$ for $k = 1,2,3$. The same equation has unique coefficients, so each pair is independent and is a basis.

Exercise 30.2: Testing a symmetric triple

Put the vectors into the columns of

$$A = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix}.$$

Its determinant is $1(0-1) - 1(1-0) = -2 \neq 0$, so its columns are independent and form a basis of $\mathbb{R}^3$. More explicitly, solving $a+b=x$, $a+c=y$, $b+c=z$ gives uniquely $a = (x+y-z)/2$, $b = (x+z-y)/2$, $c = (y+z-x)/2$. This also directly verifies spanning and uniqueness.

Exercise 30.3: A dependent triple

Name the first two vectors $u = (-1,1,2)$ and $v = (2,-3,1)$. Then

$$-2u + 4v = (2,-2,-4) + (8,-12,4) = (10,-14,0),$$

the third vector. Therefore the triple is linearly dependent and is not a basis of $\mathbb{R}^3$. This conclusion does not require checking spanning separately.

Exercise 30.4: A quadratic rational extension as a vector space

A basis for $\mathbb{Q}(\sqrt{2})$ over $\mathbb{Q}$ is $\{1, \sqrt{2}\}$. The polynomial $x^2 - 2$ is irreducible by irrationality of $\sqrt{2}$, so reduction by it gives spanning by these two powers. If $a + b\sqrt{2} = 0$ with rational a, b and $b \neq 0$, it would make $\sqrt{2}$ rational; hence $a = b = 0$.

Exercise 30.5: A radical already in the base field

Since $\sqrt{2} \in \mathbb{R}$, adjoining it changes nothing: $\mathbb{R}(\sqrt{2}) = \mathbb{R}$. A basis over $\mathbb{R}$ is $\{1\}$, because every real number is a unique real multiple of 1.

Exercise 30.6: A cubic rational extension as a vector space

A basis is $\{1, \sqrt[3]{2}, (\sqrt[3]{2})^2\}$. Indeed $x^3 - 2$ is irreducible over $\mathbb{Q}$ by Eisenstein at 2, so the simple extension has degree three and the first three powers form a basis. Every higher power reduces using $(\sqrt[3]{2})^3 = 2$.

Exercise 30.7: Complex numbers over the reals

The set $\{1, i\}$ is a basis for $\mathbb{C}$ over $\mathbb{R}$. Every complex number is $a + bi$ with $a, b \in \mathbb{R}$, giving spanning, and equality $a + bi = 0$ forces both its real and imaginary parts to vanish, giving independence.

Exercise 30.8: Gaussian rationals as a vector space

The set $\{1, i\}$ is also a basis for $\mathbb{Q}(i)$ over $\mathbb{Q}$. The polynomial $x^2 + 1$ is irreducible over $\mathbb{Q}$, so all elements are uniquely $a + bi$ for rational a, b. The ambient field here is smaller than $\mathbb{C}$, but the basis argument is the same.

Exercise 30.9: A fourth-root extension

Let $t = \sqrt[4]{2}$. By Eisenstein at 2, $x^4 - 2$ is irreducible over $\mathbb{Q}$. Hence $\{1, t, t^2, t^3\}$ is a basis for $\mathbb{Q}(t)$ over $\mathbb{Q}$. The relation $t^4 = 2$ reduces higher powers, and a dependence of these four powers would give an annihilating polynomial of degree less than four, contradicting irreducibility.

Exercise 30.10: The other nontrivial element of the four-element field

Let $\beta = 1 + \alpha$, where $\alpha^2 + \alpha + 1 = 0$ in characteristic two. Then $\beta^2 = 1 + \alpha^2 = \alpha$, and

$$\beta^2 + \beta + 1 = \alpha + (1 + \alpha) + 1 = 0.$$

Neither 0 nor 1 is a root of $x^2 + x + 1$ in $\mathbb{Z}_2$, so the polynomial is irreducible. Thus $\operatorname{irr}(1 + \alpha, \mathbb{Z}_2) = x^2 + x + 1$.

Exercise 30.11: Correcting spanning

A subset S spans V if every vector of V can be expressed as a *finite* linear combination of elements of S. Uniqueness is not required. For example $\{(1, 0), (0, 1), (1, 1)\}$ spans $\mathbb{R}^2$ but permits multiple expressions for $(1, 1)$.

Exercise 30.12: Correcting linear independence

The vectors in S are linearly independent if the *only* finite linear combination of distinct vectors of S equal to zero has every coefficient zero. Zero always has the trivial expression with all coefficients zero; what independence excludes is a nontrivial one.

Exercise 30.13: Dimension as a minimum spanning size

The proposed definition is correct for a finite-dimensional vector space. A basis spans, so its size is an attainable spanning size. Conversely deleting redundant elements from any finite spanning set yields a basis, so no spanning set can have fewer elements than a basis. This minimum equals the usual dimension.

Exercise 30.14: Correcting basis

A basis is a subset that spans the vector space and is linearly *independent*. Replace "dependent" by "independent." The two requirements ensure respectively existence and uniqueness of coordinates.

Exercise 30.15: True and false statements

(a) True. Addition is a binary operation on V.

(b) False as a general assertion. Scalars add in F, and F need not be the vector set V; for example in $\mathbb{R}^2$ a real scalar is not an ordered pair. Particular realizations may identify some scalars with vectors.

(c) True. A field is closed under multiplication.

(d) True. Scalar multiplication is a map $F \times V \to V$.

(e) False. $F[x]$ has no finite spanning set: the degrees in finitely many polynomials are bounded.

(f) False. A basis is independent by definition.

(g) False. A set containing zero has the nontrivial relation $1 \cdot 0 = 0$.

(h) True. α^2 lies in the finite extension $F(\alpha)$, whose elements are algebraic over F.

(i) True. The same theorem applies to $\alpha + \alpha^2 \in F(\alpha)$.

(j) True under the usual axiom of choice. Zorn's lemma gives a maximal independent subset, which must span or another vector could be adjoined.

Exercise 30.16: Subspaces and their intersection

(a) A subspace is a subset $W \subset V$ that is itself a vector space over the same field under the inherited operations. Equivalently $0 \in W$, and $u + v$ and cu belong to W whenever $u, v \in W, c \in F$. **(b)** Every subspace contains zero, so their intersection does. A sum of two vectors in every subspace lies in every subspace, as does any scalar multiple. Hence the intersection is a subspace. An empty intersection is V, which also works.

Exercise 30.17: The span generated by a subset

(a) Define $\mathrm{Span}(S)$ as the intersection of all subspaces containing S. This family is nonempty because V itself belongs to it. The intersection is a subspace by Exercise 16 and is the smallest such subspace. **(b)** Let L be the finite linear combinations of elements of S, including the zero combination. It is closed under addition and scalar multiplication by combining or scaling coefficients, so it is a subspace containing S. Thus $\mathrm{Span}(S) \subset L$. Every subspace containing S contains all its finite linear combinations, so $L \subset \mathrm{Span}(S)$. Therefore equality holds.

Exercise 30.18: Finite direct sums of vector spaces

Define $V_1 \oplus \cdots \oplus V_n$ as the set of tuples $(v_1, \ldots, v_n)$ with $v_i \in V_i$. Add componentwise and set $c(v_1, \ldots, v_n) = (cv_1, \ldots, cv_n)$. The zero is $(0, \ldots, 0)$ and the inverse is $(-v_1, \ldots, -v_n)$. All vector-space identities hold coordinate by coordinate. For example $c(u+v)$ has coordinate $c(u_i+v_i) = cu_i+cv_i$, the corresponding coordinate of $cu + cv$. The two scalar associativity and distributive laws and the identity law are inherited in the same way. Hence this direct sum is a vector space over F.

Exercise 30.19: The coordinate vector space over an arbitrary field

For F^n use componentwise addition and scalar multiplication. Each vector-space axiom follows from the field laws in each coordinate. The standard vectors e_i with a 1 in coordinate i and zeros elsewhere are a basis:

$$(a_1, \ldots, a_n) = a_1e_1 + \cdots + a_ne_n.$$

This proves spanning, and the coordinates of a zero combination show each $a_i = 0$, proving independence. Thus the dimension is n.

Exercise 30.20: Isomorphism of vector spaces

An isomorphism $T : V \to V'$ over the same field F is a bijection satisfying $T(u + v) = T(u) + T(v)$ and $T(cu) = cT(u)$ for all $u, v \in V, c \in F$. Preserving only the additive group is not enough: the scalar multiplication must also be preserved.

Exercise 30.21: Bases and unique coordinates

Suppose first that $B = \{\beta_1, \ldots, \beta_n\}$ is a basis. Spanning gives an expression for every vector. If $\sum a_i\beta_i = \sum b_i\beta_i$, subtraction yields $\sum(a_i - b_i)\beta_i = 0$. Independence forces $a_i = b_i$ for all i, giving uniqueness. Conversely, if every vector has a unique expression, existence gives spanning. A relation $\sum a_i\beta_i = 0$ is another expression for zero besides the all-zero one, so uniqueness forces every $a_i = 0$. Thus the vectors are independent and form a basis.

Exercise 30.22: Linear systems as a vector-space problem

(a) Let $\alpha_j = (a_{1j}, \ldots, a_{mj})$ be column j. Coordinate i of $\sum_j x_j\alpha_j$ is $\sum_j a_{ij}x_j$. Thus all m scalar equations hold exactly when $\sum_j x_j\alpha_j = \beta = (b_1, \ldots, b_m)$. A solution exists precisely when β belongs to the span of the columns. **(b)** If $m = n$ and the columns form a basis of F^n, every β has a unique basis expansion by Exercise 21. Its coefficients x_j are therefore the unique solution of the system. This argument uses field operations only and is valid in finite characteristic as well.

Exercise 30.23: Coordinates give an isomorphism

Choose a basis $\beta_1, \ldots, \beta_n$ for V. Define $T(\sum a_i\beta_i) = (a_1, \ldots, a_n)$. Uniqueness of basis coordinates makes this well defined and injective; every tuple has the preimage $\sum a_i\beta_i$, so it is onto. Adding vectors or multiplying by a scalar adds or scales their coordinates, respectively. Hence T is an isomorphism of F-vector spaces $V \cong F^n$.

Exercise 30.24: Prescribing a linear transformation on a basis

(a) Every vector has a unique finite expansion $v = \sum_i a_i \beta_i$. Linearity forces $T(v) = \sum_i a_i T(\beta_i)$, so knowing the images of all basis vectors determines every output. **(b)** Conversely prescribe arbitrary $\beta_i' \in V'$ and define $T(\sum_i a_i \beta_i) = \sum_i a_i \beta_i'$. Unique finite expansions make the formula well defined, including for an infinite basis. Combining coefficients proves additivity and homogeneity. The identity from (a) proves no other linear map can have the prescribed basis values. The target vectors may be dependent or repeat.

Exercise 30.25: Kernel and isomorphism for a linear map

(a) A linear transformation is the vector-space analogue of a group or ring homomorphism: it preserves all designated operations. **(b)** Define $\operatorname{Ker} T = \{v : T(v) = 0\}$. It contains zero. For $u, v \in \operatorname{Ker} T$ and $c \in F$, $T(u+v) = 0$ and $T(cu) = 0$. Hence it is a subspace. **(c)** A linear map is an isomorphism onto V' exactly when it is onto and has zero kernel. Indeed $T(u) = T(v)$ is equivalent to $u - v \in \operatorname{Ker} T$, so zero kernel is exactly injectivity. Its inverse then preserves both operations by applying T to their desired identities.

Exercise 30.26: Quotient vector spaces

Use the additive cosets $V/S = \{v + S : v \in V\}$. Define

$$(u + S) + (v + S) = (u + v) + S, \qquad c(v + S) = cv + S.$$

If $u - u', v - v' \in S$, then $(u+v) - (u'+v') \in S$; also $cv - cv' = c(v - v') \in S$. Thus neither operation depends on representatives. The additive group axioms and scalar laws now descend from those of V; for instance $c((u + v) + S) = (cu + cv) + S$. The zero is S and the negative of $v + S$ is $-v + S$. Thus this is a vector space over F.

Exercise 30.27: Image and rank-nullity

(a) The image contains $T(0) = 0$. Its vectors satisfy $T(u)+T(v) = T(u+v)$ and $cT(u) = T(cu)$, so it is a subspace of V'. **(b)** Choose a basis $u_1, \ldots, u_k$ of $\operatorname{Ker} T$, and extend it to a basis $u_1, \ldots, u_k, v_1, \ldots, v_r$ of V. Applying T to basis expansions shows that $T(v_1), \ldots, T(v_r)$ span the image, since each $T(u_i) = 0$. If $\sum_j c_j T(v_j) = 0$, then $\sum_j c_j v_j$ lies in the kernel and can be expressed using the u_i. Independence of the full basis forces every $c_j = 0$. Thus the displayed image vectors form a basis and $\dim \operatorname{im} T = r$. Since $\dim V = k + r$, we obtain $\dim \operatorname{im} T = \dim V - \dim \operatorname{Ker} T$.

31 Algebraic Extensions

Exercise 31.1: A basic quadratic extension

The polynomial $x^2 - 2$ is irreducible over $\mathbb{Q}$, so the degree is two and a basis is $\{1, \sqrt{2}\}$. Spanning follows by reducing powers with $(\sqrt{2})^2 = 2$; independence is equivalent to irrationality of $\sqrt{2}$.

Exercise 31.2: A biquadratic extension

First $[\mathbb{Q}(\sqrt{2}) : \mathbb{Q}] = 2$. The element $\sqrt{3}$ is not in $\mathbb{Q}(\sqrt{2})$: squaring $a + b\sqrt{2} = \sqrt{3}$ gives $a^2 + 2b^2 = 3$, $2ab = 0$. If $b = 0$ this would make 3 a rational square; if $a = 0$ it would make $3/2$ a rational square. Both are impossible by prime factorization of a reduced rational square. Thus adjoining $\sqrt{3}$ has degree two. The tower law gives degree four and the product basis $\{1, \sqrt{2}, \sqrt{3}, \sqrt{6}\}$.

Exercise 31.3: A redundant radical

Since $\sqrt{18} = 3\sqrt{2}$, the extra displayed generator already belongs to $\mathbb{Q}(\sqrt{2}, \sqrt{3})$. Therefore the field is unchanged from Exercise 2: its degree is four and a basis is $\{1, \sqrt{2}, \sqrt{3}, \sqrt{6}\}$.

Exercise 31.4: A cubic radical and a square root

Put $t = \sqrt[3]{2}$. Eisenstein at 2 gives $[\mathbb{Q}(t) : \mathbb{Q}] = 3$. If $\sqrt{3}$ belonged to that field, the tower law would require its degree two to divide three, impossible. Thus $x^2 - 3$ remains irreducible over $\mathbb{Q}(t)$, and the total degree is six. Multiplying the bases gives

$$\{1, t, t^2, \sqrt{3}, t\sqrt{3}, t^2\sqrt{3}\}.$$

Exercise 31.5: Square and cube roots of the same integer

Set $u = \sqrt[6]{2}$. Then $u^3 = \sqrt{2}$ and $u^2 = \sqrt[3]{2}$, while $u = \sqrt{2}/\sqrt[3]{2}$. Hence the given field equals $\mathbb{Q}(u)$. The polynomial $x^6 - 2$ is Eisenstein at 2, so its degree is six and a basis is $\{1, u, u^2, u^3, u^4, u^5\}$.

Exercise 31.6: A sum generating a biquadratic field

Let $s = \sqrt{2} + \sqrt{3}$. Since $(\sqrt{3} - \sqrt{2})s = 1$, the field $\mathbb{Q}(s)$ contains $1/s = \sqrt{3} - \sqrt{2}$. Adding and subtracting gives

$$\sqrt{3} = \frac{s + 1/s}{2}, \qquad \sqrt{2} = \frac{s - 1/s}{2}.$$

Therefore $\mathbb{Q}(s) = \mathbb{Q}(\sqrt{2}, \sqrt{3})$, of degree four. A basis can be $\{1, \sqrt{2}, \sqrt{3}, \sqrt{6}\}$, or equivalently $\{1, s, s^2, s^3\}$ since s has degree four.

Exercise 31.7: A product of square roots

The single generator is $\sqrt{2}\sqrt{3} = \sqrt{6}$, so the field is $\mathbb{Q}(\sqrt{6})$. The polynomial $x^2 - 6$ has no rational root and is irreducible. Thus the degree is two and a basis is $\{1, \sqrt{6}\}$. A product of radicals should not be mistaken for adjoining each radical separately.

Exercise 31.8: Two relatively prime simple degrees

The degrees of $\sqrt{2}$ and $t = \sqrt[3]{5}$ over $\mathbb{Q}$ are two and three. Let $K = \mathbb{Q}(\sqrt{2}, t)$. The tower law says both 2 and 3 divide $[K : \mathbb{Q}]$. On the other hand adjoining t to $\mathbb{Q}(\sqrt{2})$ has degree at most three, so $[K : \mathbb{Q}] \leq 6$. Hence its degree must be six. A basis is $\{1, t, t^2, \sqrt{2}, t\sqrt{2}, t^2\sqrt{2}\}$.

Exercise 31.9: Three displayed cube roots

Put $t = \sqrt[3]{2}$, $u = \sqrt[3]{3}$. Then $\sqrt[3]{6} = tu$ and $\sqrt[3]{24} = 2u$, so the field in the question is exactly $\mathbb{Q}(t, u)$. The proof in Exercise 24 shows $x^3 - 3$ is irreducible over $\mathbb{Q}(t)$. Thus adjoining u has degree three and the tower law gives $[\mathbb{Q}(t, u) : \mathbb{Q}] = 9$. A basis is

$$\{t^i u^j : 0 \leq i \leq 2,\ 0 \leq j \leq 2\}.$$

This is a product of the bases at the two stages. Merely observing that each radical separately has degree three would not by itself establish degree nine.

Exercise 31.10: A quadratic extension over a quadratic base

The top field is $\mathbb{Q}(\sqrt{2}, \sqrt{6}) = \mathbb{Q}(\sqrt{2}, \sqrt{3})$ because $\sqrt{6}/\sqrt{2} = \sqrt{3}$. It has degree four over $\mathbb{Q}$, while the base $\mathbb{Q}(\sqrt{3})$ has degree two. The tower law therefore gives relative degree two. A basis over $\mathbb{Q}(\sqrt{3})$ is $\{1, \sqrt{2}\}$.

Exercise 31.11: A sum over a base containing one summand

Over $F = \mathbb{Q}(\sqrt{3})$, adjoining $\sqrt{2} + \sqrt{3}$ is the same as adjoining $\sqrt{2}$, because we can subtract $\sqrt{3} \in F$. The top field has rational degree four and F degree two. Thus the relative degree is two and a basis is $\{1, \sqrt{2}\}$.

Exercise 31.12: A generator already defining the whole field

Exercise 6 established $\mathbb{Q}(\sqrt{2} + \sqrt{3}) = \mathbb{Q}(\sqrt{2}, \sqrt{3})$. The top and bottom fields here are therefore equal. The relative degree is one, and a basis is $\{1\}$.

Exercise 31.13: A less immediate containment of fields

Write $s = \sqrt{3} + \sqrt{5}$ and $F = \mathbb{Q}(s)$. Since $(\sqrt{5} - \sqrt{3})s = 2$, we recover $\sqrt{5} = (s + 2/s)/2$ and $\sqrt{3} = (s - 2/s)/2$. Thus $F = \mathbb{Q}(\sqrt{3}, \sqrt{5})$, a biquadratic field of rational degree four. The top field is $E = \mathbb{Q}(\sqrt{2}, \sqrt{2}s)$. It contains $s = (\sqrt{2}s)/\sqrt{2}$, so $E = F(\sqrt{2})$. To see that $\sqrt{2} \notin F$, use the independent sign changes of $\sqrt{3}, \sqrt{5}$. If an element of F has square 2, each sign change sends it to itself or its negative. A simultaneous eigenvector for these two sign changes is a rational multiple of exactly one of $1, \sqrt{3}, \sqrt{5}, \sqrt{15}$, as follows by comparing coefficients in their basis. None can square to 2, since $2, 2/3, 2/5, 2/15$ are not rational squares. Thus the relative degree is two and $\{1, \sqrt{2}\}$ is a basis over F.

Exercise 31.14: Correcting algebraic extension

An extension E/F is algebraic when *every element* of E is a zero of a nonzero polynomial in $F[x]$. It need not be generated by finitely many algebraic elements; the field of all algebraic numbers over $\mathbb{Q}$ is a standard infinite-degree example. A finite list of algebraic generators would define a finite extension, which is a stronger condition.

Exercise 31.15: Correcting finite extension

A finite extension E/F means that E is finite dimensional as an F-vector space: $[E : F] < \infty$. Having finitely many field generators is not enough; for example $F(t)$ has one transcendental generator but contains the linearly independent set $1, t, t^2, \ldots$ and has infinite dimension over F.

Exercise 31.16: Algebraic closure inside an extension

The definition is correct: the algebraic closure of F in E is $\overline{F}_E = \{a \in E : a \text{ is algebraic over } F\}$. It is a field because finite sums, products, and inverses of algebraic elements are algebraic. This relative closure need not itself be algebraically closed in the absolute sense.

Exercise 31.17: Correcting algebraically closed

A field F is algebraically closed if every *nonconstant polynomial in* $F[x]$ has a zero in F. Constant nonzero polynomials have no zero in any field, so they must be excluded. By repeatedly factoring off roots, this is equivalent to every nonconstant polynomial splitting into linear factors over F.

Exercise 31.18: A relative closure need not be algebraically closed

Take $F = \mathbb{Q}$ and $E = \mathbb{Q}(\sqrt{2})$, a proper algebraic extension. Every element of E is algebraic over F, so $\overline{F}_E = E$. But E is not algebraically closed: it is a real field and $x^2 + 1$ has no zero in it. This separates the relative and absolute uses of closure.

Exercise 31.19: True and false statements

(a) False. $\mathbb{Q}(\sqrt{2})/\mathbb{Q}$ has degree two but infinitely many elements.

(b) True. In an n-dimensional extension, $1, a, \ldots, a^n$ are dependent for every a.

(c) False. The field of all algebraic numbers over $\mathbb{Q}$ has unbounded element degrees; Exercise 36 gives a proof.

(d) True. Repeated application of the tower law multiplies finitely many finite degrees.

(e) False. $\sqrt{2} \in \mathbb{R}$ is algebraic over $\mathbb{Q}$ and is not rational.

(f) True. Any element algebraic over the already algebraically closed $\mathbb{C}$ must be a root of a linear factor, hence belongs to $\mathbb{C}$.

(g) False. x has no square root in $\mathbb{C}(x)$: a putative rational-function square would give an even exponent of the factor x equal to an odd exponent.

(h) False. Every field has an algebraic closure, including the rational-function field $\mathbb{C}(x)$; containing $\mathbb{C}$ does not supply roots involving the new indeterminate.

(i) False. An algebraic closure of $\mathbb{Z}_p$ is algebraically closed and still has characteristic p.

(j) False. $\mathbb{C}$ is algebraically closed and extends $\mathbb{Q}$, but contains the transcendental number π.

Exercise 31.20: Synopsis of finite implies algebraic

If $[E : F] = n$, the $n + 1$ vectors $1, a, \ldots, a^n$ are linearly dependent over F, providing a nonzero polynomial in $F[x]$ that vanishes at the arbitrary element $a \in E$.

Exercise 31.21: Synopsis of the tower law

Multiply each vector in an F-basis of E by each vector in an E-basis of K. Expanding first over E and then over F proves that these products span K, and grouping a proposed relation by the E-basis and using independence at both stages proves they are independent. Thus their number is $[K : E][E : F]$.

Exercise 31.22: Every nonreal complex number generates over the reals

For $z = a + bi$ with real a, b and $b \neq 0$, the field $\mathbb{R}(z)$ contains $(z - a)/b = i$. It therefore contains $\mathbb{R}(i) = \mathbb{C}$. The reverse inclusion holds because $z \in \mathbb{C}$, so $\mathbb{R}(z) = \mathbb{C}$. Division by b is legitimate because it is a nonzero element of the base field.

Exercise 31.23: Prime-degree extensions are simple

Take $a \in E \setminus F$. The intermediate field $F(a)$ satisfies

$$[E : F] = [E : F(a)][F(a) : F].$$

The left side is prime and the second factor is greater than one because $a \notin F$. Hence $[F(a) : F] = [E : F]$ and $[E : F(a)] = 1$. Thus $E = F(a)$ for every such a, proving simplicity.

Exercise 31.24: A cubic stays irreducible over a pure cubic field

Put $t = \sqrt[3]{2}$ and $K = \mathbb{Q}(t)$. A reducible cubic $x^3 - 3$ over K would have a zero in K. Since $K \subset \mathbb{R}$, that zero must be $u = \sqrt[3]{3}$. Suppose $u = a + bt + ct^2$ with $a, b, c \in \mathbb{Q}$. The rational polynomial $H(X) = (a + bX + cX^2)^3 - 3$ vanishes at t, so its minimal polynomial $X^3 - 2$ divides $H(X)$. Let ω be a nonreal cube root of unity. All three roots $t, \omega t, \omega^2 t$ of $X^3 - 2$ therefore make H vanish. The three values $a + bt + ct^2$, $a + b\omega t + c\omega^2 t^2$, and $a + b\omega^2 t + c\omega t^2$ are consequently cube roots of three. They are distinct. Indeed u has degree three over $\mathbb{Q}$ by Eisenstein at three. Under the assumption $u \in K$, the tower law gives $\mathbb{Q}(u) = K$. Substituting any of the three roots for t defines a field embedding of K into $\mathbb{C}$, because $K \cong \mathbb{Q}[X]/(X^3 - 2)$. If two such embeddings had the same value at u, they would

agree on all of $\mathbb{Q}(u) = K$, including t, contradicting their distinct assigned values of t. Hence the three values are exactly $u, \omega u, \omega^2 u$. Their sum is zero, but summing the three substituted expressions gives $3a$, so $a = 0$. The sum of their squares is also zero; with $a = 0$, summing $(bt + ct^2)^2$ over the three substitutions leaves $6bc\,t^3 = 12bc$. Thus $bc = 0$. If $c = 0$, cubing gives $b^3 = 3/2$; if $b = 0$, cubing gives $c^3 = 3/4$. Neither rational number is a rational cube, by its prime exponents. Both cases are impossible. Thus $u \notin K$, and the cubic is irreducible.

Exercise 31.25: Degrees arising from repeated square roots

Adjoining a square root of a nonsquare has degree exactly two. A tower of r such steps therefore has total degree 2^r by the tower law. Any element in that tower has degree over the original field dividing 2^r, hence itself a power of two. The polynomial $x^{14} - 3x^2 + 12$ is Eisenstein at 3: every nonleading coefficient is divisible by 3 and the constant 12 is not divisible by 9. It is irreducible over $\mathbb{Q}$, so each of its zeros has degree fourteen. Since fourteen is not a power of two, no zero can belong to a finite tower of quadratic extensions. In particular no expression made from finitely many rational functions and nested square roots of rationals can give such a zero.

Exercise 31.26: An intermediate domain in a finite extension

Let $0 \neq d \in D$. It is algebraic over F because $D \subset E$ and E/F is finite. Its minimal polynomial has nonzero constant term: otherwise it would have a factor x, contradicting irreducibility for a nonzero root. Write

$$d^n + a_{n-1}d^{n-1} + \cdots + a_1 d + a_0 = 0, \qquad a_0 \neq 0.$$

Rearrange to obtain

$$d^{-1} = -a_0^{-1}(d^{n-1} + a_{n-1}d^{n-2} + \cdots + a_1) \in D.$$

The coefficients lie in $F \subset D$ and D is closed under its ring operations. Thus every nonzero element of D has its inverse in D, proving it is a field.

Exercise 31.27: Recovering two quadratic radicals from their sum

Let $s = \sqrt{3} + \sqrt{7}$. The inclusion $\mathbb{Q}(s) \subset \mathbb{Q}(\sqrt{3}, \sqrt{7})$ is immediate. Since $s \neq 0$ and $(\sqrt{7} - \sqrt{3})s = 4$, the first field contains $4/s = \sqrt{7} - \sqrt{3}$. Consequently

$$\sqrt{7} = \frac{s + 4/s}{2}, \qquad \sqrt{3} = \frac{s - 4/s}{2}.$$

It contains both generators of the second field, yielding the reverse inclusion and equality.

Exercise 31.28: The general sum-of-two-square-roots identity

Put $s = \sqrt{a} + \sqrt{b} \neq 0$. Then

$$\frac{a - b}{s} = \sqrt{a} - \sqrt{b}.$$

Thus

$$\sqrt{a} = \frac{1}{2}\left(s + \frac{a - b}{s}\right), \qquad \sqrt{b} = \frac{1}{2}\left(s - \frac{a - b}{s}\right).$$

Since $a, b \in \mathbb{Q}$ and s is nonzero, both expressions belong to $\mathbb{Q}(s)$. This proves $\mathbb{Q}(\sqrt{a}, \sqrt{b}) \subset \mathbb{Q}(s)$; the reverse inclusion is immediate. The formula remains valid when $a = b$ provided the stated nonzero-sum condition holds.

Exercise 31.29: A divisibility obstruction to roots

Suppose $a \in E$ were a root of the irreducible polynomial $p \in F[x]$. Its minimal polynomial is the monic associate of p, so $[F(a) : F] = \deg p$. The tower law would then give $[E : F] = [E : F(a)] \deg p$, making $\deg p$ a divisor of $[E : F]$. This contradicts the hypothesis. Therefore no zero lies in E.

Exercise 31.30: Squaring an odd-degree element

The element α^2 lies in the finite extension $F(\alpha)$, so is algebraic over F. Moreover α satisfies $x^2 - \alpha^2$ over $F(\alpha^2)$, so $[F(\alpha) : F(\alpha^2)]$ is one or two. The tower law makes this degree divide the odd integer $[F(\alpha) : F]$, forcing it to be one. Hence $F(\alpha) = F(\alpha^2)$, and α^2 has the same odd degree as α.

Exercise 31.31: Transitivity of arbitrary algebraic extensions

If K/F is algebraic, every element of $E \subset K$ is algebraic over F, and every annihilating polynomial over F is also one over E. Thus E/F and K/E are algebraic. Conversely suppose both intermediate extensions are algebraic, and take $a \in K$. A polynomial annihilating a over E has only finitely many coefficients $c_0, \ldots, c_n$. They are algebraic over F, so $L = F(c_0, \ldots, c_n)$ is a finite extension of F by repeatedly adjoining one algebraic element. The element a is algebraic over L, hence $L(a)/L$ is finite. The finite tower law shows $L(a)/F$ is finite, so a is algebraic over F. This proves transitivity without assuming the original extensions finite.

Exercise 31.32: Elements outside the relative algebraic closure

Let $A = \overline{F}_E$. This field is algebraic over F by its definition. If $a \in E$ were algebraic over A, transitivity from Exercise 31 would make a algebraic over F, and hence place it in A. Therefore every $a \in E \setminus A$ must be transcendental over A.

Exercise 31.33: Relative closure inside an algebraically closed field

Let $A = \overline{F}_E$, where E is algebraically closed. Take any nonconstant $p \in A[x]$. Since $p \in E[x]$, it has a zero $a \in E$. This a is algebraic over A, and A/F is algebraic, so transitivity makes a algebraic over F. Hence $a \in A$ by the definition of A. Thus every nonconstant polynomial over A has a zero there, so A is algebraically closed.

Exercise 31.34: Containing all roots of base-field polynomials

Work in a common algebraic closure containing the given algebraic extensions. Every element of an algebraic closure $\overline{F}$ is a zero of a nonzero polynomial in $F[x]$, so the hypothesis implies $\overline{F} \subset E$. Any $a \in E$ is algebraic over F, hence also over $\overline{F}$. Its polynomial over the algebraically closed field $\overline{F}$ factors into linear factors, so a must belong to $\overline{F}$. Therefore $E = \overline{F}$ and is algebraically closed.

Equivalently, the hypothesis that all algebraic roots have already been included leaves no possible further algebraic element.

Exercise 31.35: A finite field is not algebraically closed

Suppose $|F| = q$ and its characteristic is odd. Squaring pairs the nonzero elements as $\{a, -a\}$, with distinct members, and equal squares arise precisely from those pairs. There are $(q-1)/2$ nonzero squares, so choose a nonsquare $d \in F$. Then $x^2 - d$ has no zero in F, proving F is not algebraically closed. For completeness, in any finite field every a satisfies $a^q = a$, so $x^q - x - 1$ evaluates to $-1 \neq 0$ at every element. This proves the parenthetical characteristic-two claim as well.

Exercise 31.36: The algebraic numbers have infinite degree

For every positive integer n, the polynomial $x^n - 2$ is Eisenstein at 2, so a chosen root has degree n over $\mathbb{Q}$. All these roots lie in the algebraic closure of $\mathbb{Q}$ in $\mathbb{C}$. If that closure had finite degree d, every element degree would divide d and in particular be at most d. Taking $n > d$ contradicts this. Thus the algebraic closure is not a finite extension of $\mathbb{Q}$.

Exercise 31.37: Finite extensions of the reals

A real polynomial factors over $\mathbb{R}$ into linear and irreducible quadratic factors, by pairing conjugate roots in its complex factorization. Thus every irreducible polynomial over $\mathbb{R}$ has degree one or two. If a finite extension $E/\mathbb{R}$ has no element outside $\mathbb{R}$, it is $\mathbb{R}$. Otherwise choose $a \in E \setminus \mathbb{R}$. Its minimal polynomial is an irreducible quadratic $x^2 + bx + c$ with $c - b^2/4 > 0$. The element $j = (a + b/2)/\sqrt{c - b^2/4}$ satisfies $j^2 = -1$, so $\mathbb{R}(a) = \mathbb{R}(j) \cong \mathbb{C}$. This intermediate field is algebraically closed, while E is algebraic over it. An algebraically closed field has no proper algebraic extension: the minimal polynomial of any algebraic element must be linear. Hence $E = \mathbb{R}(a) \cong \mathbb{C}$.

Exercise 31.38: Maximal ideals from Zorn lemma

Let I be the given proper ideal, and partially order the proper ideals containing I by inclusion. The family is nonempty because it contains I. For any nonempty chain, let J be its union. Two elements of J lie together in one chain member, so their difference lies in J; absorption follows within the member containing the chosen element. Thus J is an ideal containing I. It is proper: if $1 \in J$, then 1 lies in one chain member, which would be the whole ring, contradicting its propriety. The empty chain has upper bound I. Zorn's lemma now yields a maximal member M. Any proper ideal properly containing M would still contain I, contrary to maximality in this ordered family. Hence M is a maximal ideal of R containing I.

32 Geometric Constructions

Exercise 32.1: Deriving the cosine triple-angle identity

Euler's formula gives $e^{3i\theta} = (\cos\theta + i\sin\theta)^3$. Its real part is

$$\cos 3\theta = \cos^3\theta - 3\cos\theta\sin^2\theta.$$

Substitute $\sin^2\theta = 1 - \cos^2\theta$ and collect terms:

$$\cos 3\theta = \cos^3\theta - 3\cos\theta + 3\cos^3\theta = 4\cos^3\theta - 3\cos\theta.$$

The imaginary terms in the cube do not contribute to the real part.

Exercise 32.2: True and false statements

(a) True for a nondegenerate cube. If its constructible edge is $a > 0$ and the doubled cube had constructible edge b, then $b/a = \sqrt[3]{2}$ would be constructible, contrary to its degree three.

(b) True. Since even a unit cube cannot be doubled, no procedure can double every constructible cube; in fact (a) gives the stronger individual impossibility.

(c) True for a nonzero radius. Squaring a circle of constructible radius r would construct a square edge $r\sqrt{\pi}$; division by r and squaring would make π constructible and therefore algebraic, a contradiction.

(d) False. A right angle can be trisected into 30° angles, which are constructible.

(e) True. A constructible number lies in a tower of quadratic extensions, so its degree divides a power of two and is itself a power of two.

(f) False. Only a necessary degree condition was proved; being of power-of-two degree alone does not establish membership in a tower of quadratic extensions.

(g) True. The final obstruction uses that a degree divisible by three cannot divide a power of two, a consequence of prime factorization.

(h) True as the intended methodological observation: counting dimensions rules out constructions without trying all diagrams.

(i) True. Each individual constructible number has a finite construction by definition.

(j) False. A finite sequence gives only finitely many constructed points or segments, while the constructible numbers already include infinitely many integers. There is no common finite construction listing all of them.

Exercise 32.3: Why a regular nonagon is not constructible

If a regular 9-gon were constructible, its central angle 40° would be constructible. Let $u = 2\cos 40^\circ$. Since $3 \cdot 40^\circ = 120^\circ$, the triple-angle identity yields

$$u^3 - 3u = 2\cos 120^\circ = -1, \qquad u^3 - 3u + 1 = 0.$$

The monic cubic has only possible rational roots ± 1, and neither is a root. Thus it is irreducible over $\mathbb{Q}$ and u has degree three. Every constructible number has degree a power of two, so u and the regular 9-gon are not constructible.

Exercise 32.4: Constructing thirty degrees algebraically

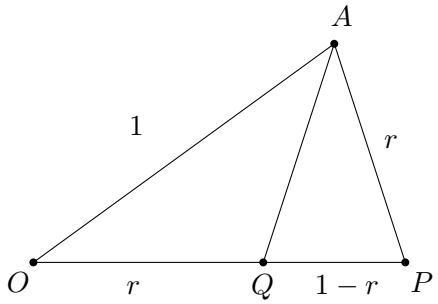
The numbers $\cos 30^\circ = \sqrt{3}/2$ and $\sin 30^\circ = 1/2$ are constructible: constructible numbers include rationals and are closed under positive square roots. Hence the unit-circle point $(\sqrt{3}/2, 1/2)$ is constructible, and the ray from the origin through it makes a 30° angle with the positive horizontal axis. This gives the requested algebraic justification.

Exercise 32.5: The golden-ratio chord of a decagon

In the source triangle, $OA = OP = 1$, $\angle O = 36^\circ$, and $AP = r$. Thus the base angles at A, P are 72°. The bisector AQ gives $\angle QAP = 36^\circ$, so $\triangle OAP \sim \triangle APQ$ with correspondence $O \leftrightarrow A$, $A \leftrightarrow P$, $P \leftrightarrow Q$. Also $AQ = AP = r$, and $OQ = AQ = r$, so $PQ = 1 - r$. The similar triangles give $1/r = r/(1-r)$, hence $r^2 + r - 1 = 0$. Positivity yields

$$r = \frac{\sqrt{5} - 1}{2},$$

a constructible length. A chord of this length in a unit circle subtends 36°; the law of cosines equivalently gives $\cos 36^\circ = 1 - r^2/2 = (1 + \sqrt{5})/4$. Thus one can construct the decagon by stepping this chord around the circle. Joining alternate vertices gives a regular pentagon.



Exercise 32.6: Constructing a regular twenty-gon

Exercise 5 constructs the central angle 36° of a decagon. Bisect it to obtain $18^\circ = 360^\circ/20$, using the standard compass-and-straightedge angle-bisector construction. Repeatedly mark this angle around a circle to obtain the twenty equally spaced vertices. Thus the regular 20-gon is constructible.

Exercise 32.7: Constructing a regular thirty-gon

A regular hexagon gives 60° and a decagon gives 36°. Subtracting constructible angles gives 24°, then bisecting gives $12^\circ = 360^\circ/30$. Both subtraction and bisection are standard straightedge-and-compass constructions. Marking this central angle around a circle constructs the regular 30-gon.

Exercise 32.8: Trisecting seventy-two degrees

The angles 60° and 36° are constructible, so their difference 24° is constructible. Since $72^\circ = 3 \cdot 24^\circ$, this gives a trisector. For the given 72° angle, copy 24° from one side twice inside it. Thus some angles that are readily constructible can also be trisected, despite the impossibility of a universal trisection procedure.

Exercise 32.9: Constructing a regular fifteen-gon

Subtract the decagon angle 36° from the hexagon angle 60° to construct 24°. This is $360^\circ/15$, so fifteen equal successive central angles locate the vertices of a regular 15-gon. Equivalently alternate vertices of the constructible regular 30-gon give it.

Exercise 32.10: A brief explanation for a geometry teacher

Every coordinate obtainable by compass and straightedge is built from rational numbers using arithmetic and repeated square roots. Consequently its algebraic degree must be a power of two. If 60° could be trisected, the constructible number $u = 2\cos 20^\circ$ would satisfy $u^3 - 3u - 1 = 0$, an irreducible cubic since neither possible rational root 1 nor -1 works. Its degree would be three, contradicting the power-of-two condition.

33 Finite Fields

Exercise 33.1: A power of two

The factorization is $4096 = 2^{12}$. For every prime p and positive integer n, there exists a finite field of order p^n. Thus a field with 4096 elements exists, namely $\mathrm{GF}(2^{12})$.

Exercise 33.2: An integer with two distinct prime factors

We have $3127 = 53 \cdot 59$. Both factors are prime and distinct, so this is not a prime power. Every finite field has prime-power order; therefore no field of 3127 elements exists.

Exercise 33.3: A cube of a prime

Compute $41^2 = 1681$ and $1681 \cdot 41 = 68{,}921$. Since 41 is prime, $68{,}921 = 41^3$ is a prime power, so $\mathrm{GF}(41^3)$ exists and has the required number of elements.

Exercise 33.4: Primitive eighth roots

The nonzero elements of GF(9) form a cyclic group of order 8. A primitive eighth root is exactly a generator of this group. A cyclic group of order eight has $\varphi(8) = 4$ generators, corresponding to powers with exponents $1, 3, 5, 7$ relatively prime to eight. Thus the answer is four.

Exercise 33.5: Primitive eighteenth roots

The multiplicative group of GF(19) is cyclic of order 18. Its elements of order 18 are its $\varphi(18) = 18(1 - 1/2)(1 - 1/3) = 6$ generators. Hence there are six primitive eighteenth roots of unity.

Exercise 33.6: Primitive fifteenth roots

The multiplicative group of GF(31) is cyclic of order 30. It has a unique subgroup of order 15, whose generators are exactly the elements of order 15 in the whole group. Their number is $\varphi(15) = 15(1 - 1/3)(1 - 1/5) = 8$. Thus the answer is eight.

Exercise 33.7: An order absent from the multiplicative group

The nonzero elements of GF(23) form a group of order 22. An element's order must divide 22, but 10 does not. Therefore there are no elements of order 10, and hence zero primitive tenth roots of unity. Merely computing $\varphi(10)$ without checking divisibility would give an incorrect answer.

Exercise 33.8: True and false statements

(a) True. The multiplicative group of a finite field is cyclic.

(b) False. The additive group of $\mathrm{GF}(p^n)$ is $(\mathbb{Z}_p)^n$; for $n > 1$ every nonzero element has order p, so it is not cyclic.

(c) True. The complex zeros of $x^{28} - 1$ are $e^{2\pi ik/28}$ and form the cyclic group generated by $e^{2\pi i/28}$.

(d) False. $60 = 2^2 \cdot 3 \cdot 5$ is not a prime power.

(e) True. $125 = 5^3$ is a prime power.

(f) False. $36 = 2^2 \cdot 3^2$ is not a prime power.

(g) True. The powers $i, i^2 = -1, i^3 = -i, i^4 = 1$ show that i has order four.

(h) True. Over every finite field there is an irreducible polynomial of each positive degree, including degree 58 over $\mathbb{Z}_2$.

(i) False. Any proposed rational generator has only finitely many primes in its numerator and denominator; its integer powers cannot produce a prime absent from that list. Thus $\mathbb{Q}^*$ is not cyclic.

(j) True. An isomorphic image has the same finite size as F, and within the fixed algebraic closure there is exactly one subfield of that size, the roots of $x^{|F|} - x$. The image is therefore F itself.

Exercise 33.9: Two cubic generators inside one algebraic closure

Both binary cubics $x^3 + x^2 + 1$ and $x^3 + x + 1$ have value 1 at $0, 1$, so both are irreducible. Thus $\mathbb{Z}_2(\alpha)$ and $\mathbb{Z}_2(\beta)$ each have eight elements. Every element of either field satisfies $x^8 - x = 0$. In the common algebraic closure this polynomial has at most eight roots, and each of these fields already consists of eight roots. They must therefore be the same set, proving equality of fields rather than just an abstract isomorphism.

Exercise 33.10: Irreducibles divide a Frobenius polynomial

Let $f \in \mathbb{Z}_p[x]$ be irreducible of degree n, replacing it by a monic associate if necessary. The quotient $E = \mathbb{Z}_p[x]/(f)$ is a field of p^n elements. For $\alpha = x + (f)$, Lagrange's theorem in E^* (and the zero case) gives $\alpha^{p^n} = \alpha$. The minimal polynomial of α is the monic associate of f, so it divides every polynomial vanishing at α, in particular $x^{p^n} - x$. Hence f divides that polynomial up to its harmless unit coefficient.

Exercise 33.11: A multiplicative generator has full degree

Every nonzero element of F is a power of α, so belongs to the subfield $\mathbb{Z}_p(\alpha)$. Zero also belongs, hence $\mathbb{Z}_p(\alpha) = F$. If $d = \deg(\alpha, \mathbb{Z}_p)$, the simple extension has p^d elements. Comparing with $|F| = p^n$ gives $d = n$.

Exercise 33.12: Subfields of a finite field

Let $E = \mathrm{GF}(p^n)$ and $m \mid n$. Set $K = \{a \in E : a^{p^m} = a\}$. Iterated Frobenius gives $(a+b)^{p^m} = a^{p^m} + b^{p^m}$ and $(ab)^{p^m} = a^{p^m} b^{p^m}$, so K is closed under addition, multiplication, and negatives; inverses of nonzero

fixed elements are fixed too. It contains $0, 1$, hence is a subfield. Because $p^m - 1$ divides $p^n - 1$, the cyclic group E^* has exactly $p^m - 1$ solutions of $a^{p^m-1} = 1$. Including zero gives $|K| = p^m$. Any subfield of size p^m consists of roots of $x^{p^m} - x$, so must equal this K by counting. This proves uniqueness. Conversely any subfield has size p^m, and the tower law makes m divide n.

Exercise 33.13: Factoring a Frobenius polynomial

The derivative of $x^{p^n} - x$ is -1, so it has no repeated irreducible factor. An irreducible monic polynomial f of degree d divides it exactly when its roots lie in $\mathrm{GF}(p^n)$. If such a root a lies there, $\mathbb{Z}_p(a)$ is a subfield of order p^d, so $d \mid n$ by the tower law. Conversely if $d \mid n$, the field $\mathbb{Z}_p(a)$ of p^d elements embeds as the unique subfield of $\mathrm{GF}(p^n)$ of that size; all its elements satisfy $x^{p^n} = x$, so f divides the polynomial. Thus its monic irreducible factors are exactly those whose degrees divide n, each appearing once. Since the polynomial is monic, the product has no additional nonunit constant.

Exercise 33.14: Euler criterion and a quadratic over seventeen elements

(a) Choose a generator g of $\mathbb{Z}_p^*$, which has even order $p - 1$. Write the nonzero residue $a = g^k$. It is a square exactly when k is even: the squares are the even powers of g. Also

$$a^{(p-1)/2} = 1 \quad \Longleftrightarrow \quad (p-1) \mid k(p-1)/2 \quad \Longleftrightarrow \quad 2 \mid k.$$

This proves the criterion, including both implications. **(b)** Modulo seventeen, $6^2 \equiv 2$, $6^4 \equiv 4$, and $6^8 \equiv 16 = -1 \neq 1$. Hence six is not a quadratic residue, so $x^2 - 6$ has no root in $\mathbb{Z}_{17}$ and is irreducible there.

34 Isomorphism Theorems

Exercise 34.1: A homomorphism from $\mathbb{Z}_{12}$ to $\mathbb{Z}_3$

Plan. An additive homomorphism is determined by the image of 1, so $\phi(x) = 2x \pmod 3$. **(a)** $2x \equiv 0 \pmod 3$ holds exactly when $3 \mid x$. Thus $K = \{0, 3, 6, 9\}$. **(b)** The distinct cosets are

$$K = \{0, 3, 6, 9\}, \quad 1 + K = \{1, 4, 7, 10\}, \quad 2 + K = \{2, 5, 8, 11\}.$$

(c) The first-isomorphism map is $x + K \mapsto \phi(x)$, hence $K \mapsto 0$, $1 + K \mapsto 2$, $2 + K \mapsto 1$. Changing x by an element of K does not change $\phi(x)$, so the map is well defined; it is a bijective homomorphism, not merely a matching of three sets.

Exercise 34.2: A homomorphism from $\mathbb{Z}_{18}$ to $\mathbb{Z}_{12}$

(a) Here $\phi(x) = 10x \pmod{12}$. Its kernel solves $12 \mid 10x$, equivalently $6 \mid x$, so $K = \{0, 6, 12\}$. **(b)** The six cosets are $r + K = \{r, r + 6, r + 12\}$ for $0 \le r \le 5$:

$$\{0, 6, 12\},\ \{1, 7, 13\},\ \{2, 8, 14\},\ \{3, 9, 15\},\ \{4, 10, 16\},\ \{5, 11, 17\}.$$

(c) Multiples of 10 modulo 12 give $\phi[\mathbb{Z}_{18}] = \{0, 2, 4, 6, 8, 10\}$. **(d)** Under $r + K \mapsto 10r \pmod{12}$, the cosets listed above map respectively to $0, 10, 8, 6, 4, 2$. Addition is preserved because $10(r + s) = 10r + 10s$; the kernel calculation makes the quotient map injective.

Exercise 34.3: The second isomorphism theorem in $\mathbb{Z}_{24}$

Plan. In an additive cyclic group, $H + N = \langle \gcd(4, 6) \rangle$ and $H \cap N = \langle \operatorname{lcm}(4, 6) \rangle$. **(a)** Thus $H + N = \{0, 2, 4, 6, 8, 10, 12, 14, 16, 18, 20, 22\}$ and $H \cap N = \{0, 12\}$. **(b)** The cosets in $(H + N)/N$ are

$0 + N$	$\{0, 6, 12, 18\}$
$2 + N$	$\{2, 8, 14, 20\}$
$4 + N$	$\{4, 10, 16, 22\}$

(c) Put $I = H \cap N$. The cosets in H/I are

$0 + I$	$\{0, 12\}$
$4 + I$	$\{4, 16\}$
$8 + I$	$\{8, 20\}$

(d) Use the representatives from H: $h + N \mapsto h + I$. Explicitly the correspondences are

$$0 + N \longmapsto 0 + I, \quad 4 + N \longmapsto 4 + I, \quad 2 + N \longmapsto 8 + I.$$

The representative from H matters: an arbitrary displayed integer representing a coset modulo N need not belong to H. If $h_1 + N = h_2 + N$ with $h_i \in H$, then $h_1 - h_2 \in H \cap N = I$, proving well-definedness. The same observation proves injectivity, and every $h + I$ is reached.

Exercise 34.4: The second isomorphism theorem in $\mathbb{Z}_{36}$

Plan. In an additive cyclic group, $H + N = \langle \gcd(6, 9) \rangle$ and $H \cap N = \langle \operatorname{lcm}(6, 9) \rangle$. **(a)** Thus $H + N = \{0, 3, 6, 9, 12, 15, 18, 21, 24, 27, 30, 33\}$ and $H \cap N = \{0, 18\}$. **(b)** The cosets in $(H + N)/N$ are

$0 + N$	$\{0, 9, 18, 27\}$
$3 + N$	$\{3, 12, 21, 30\}$
$6 + N$	$\{6, 15, 24, 33\}$

(c) Put $I = H \cap N$. The cosets in H/I are

$0 + I$	$\{0, 18\}$
$6 + I$	$\{6, 24\}$
$12 + I$	$\{12, 30\}$

(d) Use the representatives from H: $h + N \mapsto h + I$. Explicitly the correspondences are

$$0 + N \longmapsto 0 + I, \quad 6 + N \longmapsto 6 + I, \quad 3 + N \longmapsto 12 + I.$$

The representative from H matters: an arbitrary displayed integer representing a coset modulo N need not belong to H. If $h_1 + N = h_2 + N$ with $h_i \in H$, then $h_1 - h_2 \in H \cap N = I$, proving well-definedness. The same observation proves injectivity, and every $h + I$ is reached.

Exercise 34.5: The third isomorphism theorem in $\mathbb{Z}_{24}$

(a) $H = \langle 4 \rangle$, so the following table lists every coset of H in G:

coset	elements
$0 + \langle 4 \rangle$	$\{0, 4, 8, 12, 16, 20\}$
$1 + \langle 4 \rangle$	$\{1, 5, 9, 13, 17, 21\}$
$2 + \langle 4 \rangle$	$\{2, 6, 10, 14, 18, 22\}$
$3 + \langle 4 \rangle$	$\{3, 7, 11, 15, 19, 23\}$

(b) $K = \langle 8 \rangle$. Its cosets in G are

coset	elements
$0 + \langle 8 \rangle$	$\{0, 8, 16\}$
$1 + \langle 8 \rangle$	$\{1, 9, 17\}$
$2 + \langle 8 \rangle$	$\{2, 10, 18\}$
$3 + \langle 8 \rangle$	$\{3, 11, 19\}$
$4 + \langle 8 \rangle$	$\{4, 12, 20\}$
$5 + \langle 8 \rangle$	$\{5, 13, 21\}$
$6 + \langle 8 \rangle$	$\{6, 14, 22\}$
$7 + \langle 8 \rangle$	$\{7, 15, 23\}$

(c) In H/K there are two elements, namely K and $4 + K$; their underlying sets are

$$K = \{0, 8, 16\}, \qquad 4 + K = \{4, 12, 20\}.$$

(d) Write $C_r = r + K$ for the cosets in part (b). The cosets of H/K in G/K are precisely

$$\{C_r, C_{r+4}\} \qquad (0 \le r < 4).$$

This displays the elements of each outer coset: each is itself one of the two explicitly listed K-cosets.
(e) The required isomorphism is

$$r + H \longmapsto \{C_r, C_{r+4}\} = (r + K) + (H/K).$$

Adding representatives preserves addition of outer cosets. Its kernel before quotienting is H, which explains both well-definedness and injectivity.

Exercise 34.6: The third isomorphism theorem in $\mathbb{Z}_{36}$

(a) $H = \langle 9 \rangle$, so the following table lists every coset of H in G:

coset	elements
$0 + \langle 9 \rangle$	$\{0, 9, 18, 27\}$
$1 + \langle 9 \rangle$	$\{1, 10, 19, 28\}$
$2 + \langle 9 \rangle$	$\{2, 11, 20, 29\}$
$3 + \langle 9 \rangle$	$\{3, 12, 21, 30\}$
$4 + \langle 9 \rangle$	$\{4, 13, 22, 31\}$
$5 + \langle 9 \rangle$	$\{5, 14, 23, 32\}$
$6 + \langle 9 \rangle$	$\{6, 15, 24, 33\}$
$7 + \langle 9 \rangle$	$\{7, 16, 25, 34\}$
$8 + \langle 9 \rangle$	$\{8, 17, 26, 35\}$

(b) $K = \langle 18 \rangle$. Its cosets in G are

coset	elements
$0 + \langle 18 \rangle$	$\{0, 18\}$
$1 + \langle 18 \rangle$	$\{1, 19\}$
$2 + \langle 18 \rangle$	$\{2, 20\}$
$3 + \langle 18 \rangle$	$\{3, 21\}$
$4 + \langle 18 \rangle$	$\{4, 22\}$
$5 + \langle 18 \rangle$	$\{5, 23\}$
$6 + \langle 18 \rangle$	$\{6, 24\}$
$7 + \langle 18 \rangle$	$\{7, 25\}$
$8 + \langle 18 \rangle$	$\{8, 26\}$
$9 + \langle 18 \rangle$	$\{9, 27\}$
$10 + \langle 18 \rangle$	$\{10, 28\}$
$11 + \langle 18 \rangle$	$\{11, 29\}$
$12 + \langle 18 \rangle$	$\{12, 30\}$
$13 + \langle 18 \rangle$	$\{13, 31\}$
$14 + \langle 18 \rangle$	$\{14, 32\}$
$15 + \langle 18 \rangle$	$\{15, 33\}$
$16 + \langle 18 \rangle$	$\{16, 34\}$
$17 + \langle 18 \rangle$	$\{17, 35\}$

(c) In H/K there are two elements, namely K and $9 + K$; their underlying sets are

$$K = \{0, 18\}, \qquad 9 + K = \{9, 27\}.$$

(d) Write $C_r = r + K$ for the cosets in part (b). The cosets of H/K in G/K are precisely

$$\{C_r, C_{r+9}\} \qquad (0 \le r < 9).$$

This displays the elements of each outer coset: each is itself one of the two explicitly listed K-cosets. **(e)** The required isomorphism is

$$r + H \longmapsto \{C_r, C_{r+9}\} = (r + K) + (H/K).$$

Adding representatives preserves addition of outer cosets. Its kernel before quotienting is H, which explains both well-definedness and injectivity.

Exercise 34.7: Normality of an intersection

Let $h \in H$ and $x \in H \cap N$. Since H is a subgroup, $hxh^{-1} \in H$. Since $N \trianglelefteq G$ and $h \in G$, also $hxh^{-1} \in N$. Thus $h(H \cap N)h^{-1} \subseteq H \cap N$. Apply the same argument to h^{-1} to obtain the reverse inclusion. Hence $H \cap N \trianglelefteq H$, directly by the conjugation definition of normality.

Exercise 34.8: Successive quotient groups

(a) For $gH \in A$ and $kH \in B$, $(gH)(kH)(gH)^{-1} = (gkg^{-1})H \in B$ because $K \trianglelefteq G$. This proves $B \trianglelefteq A$; replacing K by L proves $C \trianglelefteq A$. Also $K \subseteq L$ implies $B \subseteq C$. If the original inclusion is strict, so is this one: $l \in L \setminus K$ gives $lH \in C \setminus B$. **(b)** The third isomorphism theorem gives $A/B \cong G/K$ and identifies C/B with L/K. Therefore

$$(A/B)/(C/B) \cong (G/K)/(L/K) \cong G/L.$$

Explicitly the final map sends the class represented successively by gH, then by $(gH)B$, to gL. Its kernel consists precisely of representatives in L.

Exercise 34.9: Two complementary normal subgroups

Because $K, L \trianglelefteq G$, the product KL is a subgroup containing both; hence $KL = K \vee L = G$. Restrict the quotient map $G \to G/K$ to L. Every $g = kl$ has $gK = lK$ (normality permits changing kl to lk'), so $l \mapsto lK$ is onto. Its kernel is $L \cap K = \{e\}$, so it is an isomorphism $L \cong G/K$. Interchanging K and L gives $K \cong G/L$. These conclusions require normality, not merely trivial intersection.

35 Series of Groups

Throughout this section $D_4 = \langle r, s\rangle$ denotes the order-8 square-symmetry group, with $r^4 = s^2 = e$ and $srs = r^{-1}$. We follow the textbook's definitions of principal series and solvability; qualifications are stated where they affect an answer.

Exercise 35.1: Refining the $10\mathbb{Z}$ and $25\mathbb{Z}$ series

Insert subgroups to obtain

$$\{0\} < 50\mathbb{Z} < 10\mathbb{Z} < 5\mathbb{Z} < \mathbb{Z},$$
$$\{0\} < 50\mathbb{Z} < 25\mathbb{Z} < 5\mathbb{Z} < \mathbb{Z}.$$

The first factor in each is $50\mathbb{Z} \cong \mathbb{Z}$. The remaining factors are, respectively, $\mathbb{Z}_5, \mathbb{Z}_2, \mathbb{Z}_5$ and $\mathbb{Z}_2, \mathbb{Z}_5, \mathbb{Z}_5$. Thus the factors match after reordering. Here $a\mathbb{Z}/b\mathbb{Z} \cong \mathbb{Z}_{b/a}$ when $a \mid b$; remember that a larger integer labels a smaller subgroup.

Exercise 35.2: Refining the $60\mathbb{Z}$ and $245\mathbb{Z}$ series

A common initial nonzero subgroup is $2940\mathbb{Z}$, since $\operatorname{lcm}(60, 245) = 2940$. Use

$$\begin{aligned}\{0\} < 2940\mathbb{Z} < 420\mathbb{Z} < 60\mathbb{Z} < 20\mathbb{Z}\\ < 10\mathbb{Z} < 5\mathbb{Z} < \mathbb{Z},\\ \{0\} < 2940\mathbb{Z} < 1470\mathbb{Z} < 735\mathbb{Z} < 245\mathbb{Z}\\ < 49\mathbb{Z} < 7\mathbb{Z} < \mathbb{Z}.\end{aligned}$$

Every original subgroup remains. In the first series the factors after $2940\mathbb{Z} \cong \mathbb{Z}$ have orders $7, 7, 3, 2, 2, 5$; in the second they have orders $2, 2, 3, 5, 7, 7$. Each factor is cyclic, so this is an isomorphism of series.

Exercise 35.3: Refinements in $\mathbb{Z}_{24}$

All subgroups here are cyclic and normal. The refinements

$$\{0\} < \langle 12\rangle < \langle 6\rangle < \langle 3\rangle < \mathbb{Z}_{24},$$
$$\{0\} < \langle 8\rangle < \langle 4\rangle < \langle 2\rangle < \mathbb{Z}_{24}$$

have factor orders $2, 2, 2, 3$ and $3, 2, 2, 2$. They retain respectively $\langle 3\rangle$ and $\langle 8\rangle$. Each factor is cyclic of the indicated prime order, giving the required matching.

Exercise 35.4: Refinements in $\mathbb{Z}_{72}$

Use

$$\{0\} < \langle 36\rangle < \langle 18\rangle < \langle 9\rangle < \langle 3\rangle < \mathbb{Z}_{72},$$
$$\{0\} < \langle 24\rangle < \langle 12\rangle < \langle 6\rangle < \langle 3\rangle < \mathbb{Z}_{72}.$$

The orders of these subgroups are respectively $1, 2, 4, 8, 24, 72$ and $1, 3, 6, 12, 24, 72$. Consecutive order ratios give cyclic factors of orders $2, 2, 2, 3, 3$ and $3, 2, 2, 2, 3$. These agree up to order and all original terms remain.

Exercise 35.5: Refining two series of $\mathbb{Z}^2$

Put $L = 60\mathbb{Z} \times 80\mathbb{Z}$ and $0 = \{(0,0)\}$. The refinements are

$$\begin{aligned} 0 < L < 60\mathbb{Z} \times 20\mathbb{Z} < 60\mathbb{Z} \times \mathbb{Z} \\ < 10\mathbb{Z} \times \mathbb{Z} < \mathbb{Z}^2, \\ 0 < L < 10\mathbb{Z} \times 80\mathbb{Z} < \mathbb{Z} \times 80\mathbb{Z} \\ < \mathbb{Z} \times 20\mathbb{Z} < \mathbb{Z}^2. \end{aligned}$$

The initial factor is $L \cong \mathbb{Z}^2$ in both. The other factors are $\mathbb{Z}_4, \mathbb{Z}_{20}, \mathbb{Z}_6, \mathbb{Z}_{10}$ in the first and $\mathbb{Z}_6, \mathbb{Z}_{10}, \mathbb{Z}_4, \mathbb{Z}_{20}$ in the second. This follows by quotienting the coordinate that changes at each step; the unchanged coordinate cancels.

Exercise 35.6: All composition series of $\mathbb{Z}_{60}$

Plan. A cyclic group has exactly one subgroup U_d of each order $d \mid 60$, namely $U_d = \langle 60/d \rangle$. A cyclic factor is simple exactly when its order is prime. Consequently *all* the series are

$$U_1 < U_{p_1} < U_{p_1 p_2} < U_{p_1 p_2 p_3} < U_{60},$$

where (p_1, p_2, p_3, p_4) runs over the distinct permutations of $(2, 2, 3, 5)$. This is an explicit parametrization with no duplication: the subgroup orders recover the sequence of primes. There are $4!/2! = 12$ such series. All have factors $\mathbb{Z}_2, \mathbb{Z}_2, \mathbb{Z}_3, \mathbb{Z}_5$, in some order, so all are isomorphic.

Exercise 35.7: All composition series of $\mathbb{Z}_{48}$

Let $U_d = \langle 48/d \rangle$ be the unique subgroup of order d. The possible successive subgroup orders are

$$\begin{gathered} 1, 3, 6, 12, 24, 48 \\ 1, 2, 6, 12, 24, 48 \\ 1, 2, 4, 12, 24, 48 \\ 1, 2, 4, 8, 24, 48 \\ 1, 2, 4, 8, 16, 48. \end{gathered}$$

Replacing every order d by U_d lists all five series. Indeed each step must multiply the order by a prime, and $48 = 2^4 \cdot 3$, so the only choice is the position of the factor 3 among five steps. Each series has four factors $\mathbb{Z}_2$ and one $\mathbb{Z}_3$.

Exercise 35.8: Composition series of $\mathbb{Z}_5^2$

Regard $\mathbb{Z}_5^2$ as a two-dimensional vector space over $\mathbb{Z}_5$. Its order-5 subgroups are exactly the six lines

$$L_t = \langle (1, t) \rangle \ (t = 0, 1, 2, 3, 4), \qquad L_\infty = \langle (0, 1) \rangle.$$

They exhaust the possibilities: a nonzero vector with nonzero first coordinate scales uniquely to $(1, t)$; otherwise it scales to $(0, 1)$. The six composition series are $0 < L_t < \mathbb{Z}_5^2$, including $t = \infty$. Both factors have order 5. No other composition series is possible, since simple factors of an abelian group have prime order.

Exercise 35.9: Composition series of $S_3 \times \mathbb{Z}_2$

Put $T = A_3 \times \{0\}$, $B = S_3 \times \{0\}$, $A = A_3 \times \mathbb{Z}_2$, $U = \{e\} \times \mathbb{Z}_2$, and $C = \{(\sigma, \epsilon) : \operatorname{sgn}(\sigma) = (-1)^\epsilon\}$. Then all four series are

$$1 < T < A < G, \qquad 1 < U < A < G,$$
$$1 < T < B < G, \qquad 1 < T < C < G.$$

Here $A \cong \mathbb{Z}_6$, while $B \cong C \cong S_3$. To check exhaustion, the commutator subgroup of G is T, and $G/T \cong \mathbb{Z}_2^2$. A maximal normal subgroup must therefore be one of the three inverse images of its lines, namely A, B, C (a simple quotient of this solvable group has prime order). The cyclic group A has the two indicated series, and each S_3 factor has only its A_3 series. The factor orders are $3, 2, 2$ or $2, 3, 2$, so all are isomorphic.

Exercise 35.10: Composition series of $\mathbb{Z}_2 \times \mathbb{Z}_5 \times \mathbb{Z}_7$

The product is cyclic of order 70 because 2, 5, and 7 are pairwise coprime. For each of the six permutations (p, q, r) of $(2, 5, 7)$, take

$$0 < U_p < U_{pq} < G,$$

where U_d is the unique subgroup of order d. In product notation U_p is the coordinate subgroup for prime p, and U_{pq} is the product of the p and q coordinate subgroups. These are all the series because simple cyclic factors have prime order; the successive factor orders must be a permutation of $2, 5, 7$.

Exercise 35.11: The center of $S_3 \times \mathbb{Z}_4$

An element (g, z) commutes with every (h, w) if and only if $gh = hg$ for every h; the second coordinate already commutes because $\mathbb{Z}_4$ is abelian. Since $Z(S_3) = \{e\}$,

$$Z(S_3 \times \mathbb{Z}_4) = \{e\} \times \mathbb{Z}_4.$$

For example a transposition fails to commute with a different transposition, and a nonidentity 3-cycle fails to commute with a transposition, verifying the stated center of S_3.

Exercise 35.12: The center of $S_3 \times D_4$

Write $D_4 = \langle r, s \rangle$ with $r^4 = s^2 = e$ and $sr = r^{-1}s$. Its center is $\{e, r^2\}$: r and r^3 fail to commute with s, and every reflection fails to commute with r. Commutation in a product is coordinatewise, so

$$Z(S_3 \times D_4) = \{e\} \times \{e, r^2\}.$$

In the textbook's notation $e = \rho_0$ and $r^2 = \rho_2$.

Exercise 35.13: Ascending central series of $S_3 \times \mathbb{Z}_4$

The first nontrivial term is $Z_1 = \{e\} \times \mathbb{Z}_4$. The quotient $G/Z_1 \cong S_3$ has trivial center. Taking the inverse image of that trivial center gives $Z_2 = Z_1$, and the same reasoning repeats. Thus the series, including its initial identity term, is

$$\{(e, 0)\} \leq \{e\} \times \mathbb{Z}_4 \leq \{e\} \times \mathbb{Z}_4 \leq \cdots .$$

It never reaches G. Repeated terms must be retained when describing the ascending central sequence.

Exercise 35.14: Ascending central series of $S_3 \times D_4$

The first term is $Z_1 = \{e\} \times \{e, r^2\}$. Now $G/Z_1 \cong S_3 \times \mathbb{Z}_2^2$, whose center is $\{e\} \times \mathbb{Z}_2^2$. Its inverse image is $Z_2 = \{e\} \times D_4$. Since $G/Z_2 \cong S_3$ has trivial center, every later term equals Z_2. Thus

$$1 \leq \{e\} \times \{e, r^2\} \leq \{e\} \times D_4 \leq \{e\} \times D_4 \leq \cdots .$$

Each term is normal in G, as required for the construction.

Exercise 35.15: Definition of a composition series

The proposed definition is correct, provided "maximal normal" means maximal among *proper* normal subgroups. The finite strict chain starts at $\{e\}$ and ends at G, and each $H_i \trianglelefteq H_{i+1}$ has no normal subgroup strictly between it and H_{i+1}. By the correspondence theorem this is equivalent to H_{i+1}/H_i being simple.

Exercise 35.16: Definition of solvability

Replace "a composition series of abelian groups" by "a composition series whose *factor groups* are abelian." The intermediate groups need not be abelian: $1 < A_3 < S_3$ is a solvable composition series even though S_3 is nonabelian. This uses the textbook's Definition 35.18. It restricts solvability to groups admitting such a finite composition series; the broader modern definition uses a finite subnormal series with abelian factors, allowing infinite groups such as $\mathbb{Z}$.

Exercise 35.17: True or false: group series

(a) True. Normality in G implies normality in every intermediate subgroup.

(b) False. In D_4, $1 < \langle s \rangle < \langle s, r^2 \rangle < D_4$ is subnormal, but $\langle s \rangle$ is not normal in D_4.

(c) True under Definition 35.12 in this text: a principal series is normal with simple factors, hence subnormal with simple factors.

(d) False. The composition series in (b) has order-2 factors but is not normal in D_4.

(e) False. $\mathbb{Z}_6$ has $0 < \langle 2 \rangle < \mathbb{Z}_6$ and $0 < \langle 3 \rangle < \mathbb{Z}_6$; moreover $\mathbb{Z}$ has no composition series.

(f) True. Repeatedly choose a maximal proper normal subgroup; orders strictly decrease, so the process terminates.

(g) False. Simple factors occur in every composition series by definition; S_5 has one but is not solvable, because A_5 is a nonabelian factor.

(h) False. $1 < A_7 < S_7$ has the nonabelian simple factor A_7.

(i) True. The simple factor groups, like primes in an integer factorization, are determined up to order and isomorphism, though the intermediate subgroups need not be unique.

(j) True. A group of prime order is cyclic, so its one factor is abelian and simple.

The textbook's definition of a principal/chief series here is stronger than the usual modern chief-series convention; the answers follow its stated convention.

Exercise 35.18: A composition series of S_3^2

Take

$$1 < A_3 \times 1 < S_3 \times 1 < S_3 \times A_3 < S_3 \times S_3.$$

Each term is normal in the next: this follows from $A_3 \trianglelefteq S_3$ and coordinatewise conjugation. The factors, from bottom to top, are $\mathbb{Z}_3, \mathbb{Z}_2, \mathbb{Z}_3, \mathbb{Z}_2$. They are simple and abelian, so the series is a composition series and $S_3 \times S_3$ is solvable.

Exercise 35.19: Solvability of the square-symmetry group

The chain $1 < \langle r^2 \rangle < \langle r \rangle < D_4$ has subgroup orders $1, 2, 4, 8$. The first subgroup is central, and the rotation subgroup has index 2, hence is normal. Every factor has order 2, so every factor is simple and abelian. Therefore D_4 is solvable. The definition does not require D_4 itself to commute.

Exercise 35.20: Schreier arrays in $\mathbb{Z}_{36}$

Plan. Retain repeated entries while forming the rectangular Schreier arrays. In $\mathbb{Z}_n$, $\langle a \rangle + \langle b \rangle = \langle \gcd(a,b) \rangle$ and $\langle a \rangle \cap \langle b \rangle = \langle \operatorname{lcm}(a,b) \rangle$ when $a, b \mid n$; write $\langle n \rangle = 0$. Use $H_{ij} = H_i + (H_{i+1} \cap K_j)$ and $K_{ji} = K_j + (K_{j+1} \cap H_i)$. The arrays for chains (3) and (4) are respectively

$$\begin{array}{ccccc} 0 & \leq & 0 & \leq & \langle 12 \rangle \\ \langle 12 \rangle & \leq & \langle 6 \rangle & \leq & \langle 3 \rangle \\ \langle 3 \rangle & \leq & \langle 3 \rangle & \leq & G \end{array} \qquad \begin{array}{ccccccc} 0 & \leq & 0 & \leq & \langle 18 \rangle & \leq & \langle 18 \rangle \\ \langle 18 \rangle & \leq & \langle 6 \rangle & \leq & \langle 3 \rangle & \leq & G \end{array}.$$

Read each row left to right, identifying its last term with the first term of the next row. The factor-order array for the first chain is

$$\begin{pmatrix} 1 & 3 \\ 2 & 2 \\ 1 & 3 \end{pmatrix}.$$

The second chain has the transpose array of factor orders: the factor in row i, column j of the first chain corresponds to row j, column i of the second. All factors are cyclic; order 1 denotes a repeated group and thus a trivial factor. Deleting repeated entries gives the requested isomorphic refinements. This also exhibits every factor correspondence, rather than only their unordered lists.

Exercise 35.21: Schreier arrays in $\mathbb{Z}_{24}$

Plan. Retain repeated entries while forming the rectangular Schreier arrays. In $\mathbb{Z}_n$, $\langle a \rangle + \langle b \rangle = \langle \gcd(a,b) \rangle$ and $\langle a \rangle \cap \langle b \rangle = \langle \operatorname{lcm}(a,b) \rangle$ when $a, b \mid n$; write $\langle n \rangle = 0$. Use $H_{ij} = H_i + (H_{i+1} \cap K_j)$ and $K_{ji} = K_j + (K_{j+1} \cap H_i)$. The arrays for chains (3) and (4) are respectively

$$\begin{array}{ccccccc} 0 & \leq & \langle 12 \rangle & \leq & \langle 12 \rangle & \leq & \langle 12 \rangle \\ \langle 12 \rangle & \leq & \langle 12 \rangle & \leq & \langle 12 \rangle & \leq & \langle 4 \rangle \\ \langle 4 \rangle & \leq & \langle 2 \rangle & \leq & G & \leq & G \end{array} \qquad \begin{array}{ccccccc} 0 & \leq & \langle 12 \rangle & \leq & \langle 12 \rangle & \leq & \langle 6 \rangle \\ \langle 6 \rangle & \leq & \langle 6 \rangle & \leq & \langle 6 \rangle & \leq & \langle 3 \rangle \\ \langle 3 \rangle & \leq & \langle 3 \rangle & \leq & G & \leq & G \end{array}.$$

Read each row left to right, identifying its last term with the first term of the next row. The factor-order array for the first chain is

$$\begin{pmatrix} 2 & 1 & 1 \\ 1 & 1 & 3 \\ 2 & 2 & 1 \end{pmatrix}.$$

The second chain has the transpose array of factor orders: the factor in row i, column j of the first chain corresponds to row j, column i of the second. All factors are cyclic; order 1 denotes a repeated group and thus a trivial factor. Deleting repeated entries gives the requested isomorphic refinements. This also exhibits every factor correspondence, rather than only their unordered lists.

Exercise 35.22: Normality in an intersection

Let $x \in H \cap K$ and $y \in H^* \cap K$. Since $x \in H$ and $H^* \trianglelefteq H$, the conjugate $xyx^{-1} \in H^*$. Since $x, y \in K$, it also lies in K. Thus conjugation by x preserves $H^* \cap K$; using x^{-1} yields equality. Therefore $H^* \cap K \trianglelefteq H \cap K$.

Exercise 35.23: Multiplying finite factor orders

Start with $|H_0| = 1$. If $|H_i|$ is finite and $[H_{i+1} : H_i] = s_{i+1}$, then H_{i+1} is the disjoint union of s_{i+1} cosets, each with $|H_i|$ elements. Hence $|H_{i+1}| = s_{i+1}|H_i|$. Induction gives $|H_j| = s_1 \cdots s_j$ for every j. At $j = n$ this proves $|G| = s_1 \cdots s_n$. Normality is needed to call the factors groups, although the coset-counting argument itself only uses subgroup indices.

Exercise 35.24: No infinite abelian composition series

A simple abelian group is cyclic of prime order: a nonidentity element generates a nontrivial normal subgroup, hence the whole group; an infinite cyclic group or a finite cyclic group of composite order has a proper nontrivial subgroup. If an abelian group had a composition series, all its factors would consequently be finite of prime order. Exercise 35.23 would make the group finite. This contradicts the hypothesis that it is infinite.

Exercise 35.25: Finite products of solvable groups

For two factors choose composition series $1 = H_0 < \cdots < H_r = H$ and $1 = K_0 < \cdots < K_s = K$ with abelian factors. Form

$$1 < H_1 \times 1 < \cdots < H \times 1 < H \times K_1 < \cdots < H \times K.$$

Normality holds coordinatewise. The factors in the first portion are H_i/H_{i-1}; those in the second are K_j/K_{j-1}. Thus every factor is simple and abelian, so this is a solvable composition series. Apply the two-factor construction repeatedly for any finite number of groups.

Exercise 35.26: Subgroups of solvable groups

Let $1 = H_0 < \cdots < H_n = G$ have abelian simple factors, hence cyclic factors of prime order. Set $K_i = K \cap H_i$. For $x \in K_i$, conjugation preserves $K \cap H_{i-1}$, so $K_{i-1} \trianglelefteq K_i$. The map $K_i \to H_i/H_{i-1}$, $x \mapsto xH_{i-1}$, has kernel K_{i-1}. Therefore K_i/K_{i-1} is isomorphic to a subgroup of a prime-order cyclic group. It is either trivial or that entire prime-order group. After deleting repetitions the intersections form a composition series of K with abelian factors. Thus K is solvable. The prime-order property is essential: intersections with an arbitrary composition series need not have simple factors.

Exercise 35.27: Inserting a simple normal subgroup

Because $N \trianglelefteq G$, each H_iN is a subgroup. Also $H_{i-1}N \trianglelefteq H_iN$: conjugation by H_i preserves both H_{i-1} and N, while conjugation by $n \in N$ fixes every coset modulo N and thus preserves $H_{i-1}N$. The composite map

$$H_i \longrightarrow H_iN/H_{i-1}N$$

is onto, with kernel $H_i \cap (H_{i-1}N)$ containing H_{i-1}. Hence its target is a quotient of the simple group H_i/H_{i-1}, and is either trivial or simple. Prepend 1 to $N = H_0N \leq H_1N \leq \cdots \leq G$. Its initial factor is the assumed simple group N; all later nontrivial factors are simple by the preceding argument. Delete repeated terms to obtain the required composition series.

Exercise 35.28: Composition series descend to quotients

Let $\gamma : G \to G/N$ be the quotient map. For $x \in H_i$ and $y \in H_{i-1}$, $\gamma(x)\gamma(y)\gamma(x)^{-1} = \gamma(xyx^{-1}) \in \gamma(H_{i-1})$. Thus the image terms form a subnormal chain. The onto map

$$H_i/H_{i-1} \longrightarrow \gamma(H_i)/\gamma(H_{i-1}), \qquad hH_{i-1} \longmapsto \gamma(h)\gamma(H_{i-1})$$

is well defined, since H_{i-1} maps into $\gamma(H_{i-1})$. A quotient of a simple group is either trivial or simple. The image chain starts at 1 and ends at G/N; deleting its repetitions therefore produces a composition series of G/N.

Exercise 35.29: Homomorphic images preserve solvability

Let $f : G \to Q$ be a homomorphism, and use a composition series of G with cyclic prime-order factors. Repeat the argument of Exercise 35.28 with the onto map $G \to f(G)$: each successive image factor is a quotient of a cyclic prime-order group, hence either trivial or cyclic of that prime order. Delete repetitions. The result is a composition series of $f(G)$ with abelian factors, proving that the image is solvable. No surjectivity onto the originally named codomain Q is needed; the assertion concerns $f(G)$.

36 Sylow Theorems

For a finite group of order $p^a m$, where $p \nmid m$, a Sylow p-subgroup has order p^a; its number n_p divides m and satisfies $n_p \equiv 1 \pmod{p}$. A unique Sylow subgroup is normal because conjugation preserves its order.

Exercise 36.1: The Sylow 3-order when $|G| = 12$

Factor $12 = 2^2 \cdot 3$. A Sylow 3-subgroup has the largest power of 3 dividing $|G|$, namely $3^1 = 3$, as its order. This is the subgroup's order, not the number of such subgroups.

Exercise 36.2: The Sylow 3-order when $|G| = 54$

Since $54 = 2 \cdot 3^3$, the largest power of 3 dividing 54 is 27. Hence every Sylow 3-subgroup has order 27.

Exercise 36.3: Counting Sylow 2-subgroups for order 24

Sylow's counting theorem gives $n_2 \mid 24/8 = 3$ and $n_2 \equiv 1 \pmod{2}$. Both divisors 1 and 3 satisfy these conditions, so the blanks are 1 and 3. The theorem gives possible values, without asserting that both occur in a particular group.

Exercise 36.4: Counting Sylow subgroups for order 255

For $p = 3$, $n_3 \mid 85$ and $n_3 \equiv 1 \pmod{3}$. Of $1, 5, 17, 85$, only $1, 85$ qualify. For $p = 5$, $n_5 \mid 51$ and $n_5 \equiv 1 \pmod{5}$; of $1, 3, 17, 51$, only $1, 51$ qualify. Thus the requested alternatives are $n_3 = 1$ or 85, and $n_5 = 1$ or 51. Only the counting theorem is being used here, as requested.

Exercise 36.5: Sylow 3-subgroups of S_4

A Sylow 3-subgroup has order 3, since $24 = 3 \cdot 8$. Every nonidentity element in one is a 3-cycle; each 3-cycle and its inverse generate the same subgroup. The full list is

$$\langle (123) \rangle, \quad \langle (124) \rangle, \quad \langle (134) \rangle, \quad \langle (234) \rangle.$$

There are eight 3-cycles, two per subgroup, so nothing is missing. Conjugation relabels a cycle: $\sigma(ijk)\sigma^{-1} = (\sigma(i)\,\sigma(j)\,\sigma(k))$. Starting from $\langle (123) \rangle$, conjugate by (34), (234), and (1234) to get the other three subgroups respectively. Thus all four are conjugate explicitly.

Exercise 36.6: Two conjugate Sylow 2-subgroups of S_4

Take $P = \langle (1234), (13) \rangle$. With $r = (1234)$, $s = (13)$, we have $r^4 = s^2 = e$ and $srs = r^{-1}$. Its eight distinct elements are

$$e, (1234), (13)(24), (1432), (13), (24), (14)(23), (12)(34).$$

Since 8 is the largest power of 2 dividing 24, P is Sylow. Let $\tau = (34)$ and set $Q = \tau P \tau^{-1} = \langle (1243), (14) \rangle$. It again has eight elements and is Sylow. It differs from P because $(1243) \notin P$. The displayed equation itself demonstrates that P and Q are conjugate, rather than merely appealing to the general theorem.

Exercise 36.7: Definition of a p-group

An element need not have order exactly p. The correct definition is: G is a p-group if every element has order a power p^k with $k \geq 0$ (the identity has order $p^0 = 1$). For finite G this is equivalent to $|G|$ being a power of p. For example $\mathbb{Z}_{p^2}$ is a p-group and has an element of order p^2.

Exercise 36.8: Definition of the normalizer

The normalizer is a set of *elements of* G, not a set of automorphisms:

$$N_G(H) = \{g \in G : gHg^{-1} = H\}.$$

Equivalently it consists of those elements whose induced inner automorphism preserves H. Distinct group elements can induce the same inner automorphism, so replacing elements by automorphisms changes the set and potentially its order.

Exercise 36.9: Definition of a Sylow subgroup

For a finite group with $|G| = p^a m$ and $p \nmid m$, a Sylow p-subgroup is a subgroup of order p^a. Equivalently it is a maximal p-subgroup under inclusion. Say *a* Sylow subgroup, since there may be several. "The largest" suggests a unique subgroup containing all others, which usually does not exist.

Exercise 36.10: True or false: Sylow theory

(a) True: this is Sylow conjugacy.

(b) True: $n_5 \mid 3$ and $n_5 \equiv 1 \pmod 5$ force $n_5 = 1$.

(c) True by the definition of a Sylow p-subgroup.

(d) False: $\{0, 2\} < \mathbb{Z}_4$ is a 2-subgroup but is not Sylow; the entire group has order 4.

(e) True: conjugation is trivial in an abelian group, and Sylow subgroups are conjugate.

(f) False: in S_3, the normalizer of $\langle (12) \rangle$ equals $\langle (12) \rangle$, which is not normal in S_3.

(g) True: every element of the normalizer conjugates H onto itself.

(h) True: uniqueness implies conjugation invariance, and normality combined with Sylow conjugacy implies uniqueness.

(i) False: in an abelian group $N_G(H) = G$, which can strictly contain H.

(j) False: the whole group is its Sylow p-subgroup.

Exercise 36.11: The normalizer is a subgroup

Certainly $eHe^{-1} = H$. If $aHa^{-1} = H$ and $bHb^{-1} = H$, then $(ab)H(ab)^{-1} = a(bHb^{-1})a^{-1} = H$. Also $aHa^{-1} = H$ implies $a^{-1}Ha = H$, by conjugating the equality with a^{-1}. Hence the normalizer contains the identity and is closed under products and inverses, so it is a subgroup.

Exercise 36.12: A unique proper Sylow subgroup is normal

Let P be the only Sylow p-subgroup. For each $g \in G$, conjugation is an isomorphism, so gPg^{-1} has the same order as P and is again Sylow. Uniqueness gives $gPg^{-1} = P$. Thus $P \trianglelefteq G$. It is nontrivial because $p \mid |G|$, and proper because a distinct prime q divides $|G|$. This is a nontrivial proper normal subgroup, so G is not simple.

Exercise 36.13: A normal order-9 subgroup for order 45

Since $45 = 3^2 \cdot 5$, Sylow 3-subgroups have order 9. Their number divides 5 and is 1 modulo 3. Of the divisors 1 and 5, only 1 is 1 modulo 3. Thus there is a unique Sylow 3-subgroup; conjugation preserves its order and hence preserves it. This is the required normal subgroup of order 9.

Exercise 36.14: Finite p-groups and prime-power orders

If $|G| = p^a$, Lagrange's theorem makes every element order divide p^a, so each order is a power of p. Conversely suppose every element order is a power of p. If a different prime q divided $|G|$, Cauchy's theorem would produce an element of order q, contradicting the assumption. Thus no prime other than p divides $|G|$, so $|G| = p^a$. The trivial group corresponds to $a = 0$.

Exercise 36.15: A Sylow normalizer is self-normalizing

Put $M = N_G(P)$. Then $P \trianglelefteq M$, and P remains a Sylow subgroup of M because $P \subseteq M \subseteq G$ leaves no room for a larger power of p. Hence P is the unique Sylow p-subgroup of M. If $g \in N_G(M)$, conjugation by g is an automorphism of M and carries P to another Sylow p-subgroup of M. Uniqueness forces $gPg^{-1} = P$, so $g \in M$. This proves $N_G(M) \subseteq M$. Every subgroup normalizes itself, giving the reverse inclusion. Therefore $N_G(N_G(P)) = N_G(P)$.

Exercise 36.16: Conjugating a p-subgroup into a fixed Sylow subgroup

Let the p-group H act by left multiplication on the left-coset set G/P. Its size $[G : P]$ is not divisible by p. The orbit congruence therefore implies that the number of cosets fixed by all of H is nonzero modulo p, hence positive. Choose a fixed coset xP. For every $h \in H$, $hxP = xP$, so $x^{-1}hx \in P$. Thus $x^{-1}Hx \subseteq P$. Taking $g = x^{-1}$ yields the requested $gHg^{-1} \subseteq P$.

Exercise 36.17: A normal order-125 subgroup

The group order is $(35)^3 = 5^3 \cdot 7^3$. A Sylow 5-subgroup has order 125. Its number divides 343, whose divisors are $1, 7, 49, 343$; their residues modulo 5 are $1, 2, 4, 3$. Only 1 is allowed by Sylow's congruence.

Therefore the order-125 subgroup is unique and hence normal.

Exercise 36.18: Nonsimplicity at order 255

A Sylow 17-subgroup has order 17. Its number divides 15 and is 1 modulo 17. The divisors $1, 3, 5, 15$ include only one such number, namely 1. Thus the subgroup is unique, normal, nontrivial, and proper. Consequently no group of order 255 is simple.

Exercise 36.19: The small-cofactor criterion and its prime exception

Qualification of the printed statement. As written it includes $m = 1, r = 1$, so it is false: $\mathbb{Z}_p$ is simple and has order $p^1 \cdot 1$ with $1 < p$. For the intended composite-order assertion, let $|G| = p^r m$ with $1 < m < p$. Since $p \nmid m$, a Sylow p-subgroup has order p^r. Its number divides m and is congruent to 1 modulo p. But every positive divisor of m lies between 1 and $m < p$, so only 1 is possible. The unique Sylow subgroup is nontrivial, normal, and proper. If $m = 1$ and $r > 1$, the group is a p-group. Its center is nontrivial by Exercise 36.20. If the center is proper it is the required normal subgroup; if the group is abelian, any order-p subgroup is normal and proper. Thus only the prime-order case must be excluded.

Exercise 36.20: Conjugation fixed points and the center

(a) A point $x \in G$ is fixed by every conjugation precisely when $gxg^{-1} = x$ for every g, equivalently $gx = xg$ for every g. Thus the fixed-point set is $Z(G)$. **(b)** When $|G| = p^n$ with $n \geq 1$, the orbit congruence for this action gives $|Z(G)| \equiv |G| \equiv 0 \pmod{p}$. The center contains e, so its order is a positive multiple of p and in particular at least p. Hence it is nontrivial.

Exercise 36.21: Normal subgroups of every prime-power order

We prove the stronger assertion of a nested normal chain by induction on n. For $n = 0$, take $H_0 = G = 1$. For $n \geq 1$, the center is nontrivial, so Cauchy's theorem gives a central subgroup C of order p. Centrality makes $C \trianglelefteq G$. The quotient G/C has order p^{n-1}. By induction it has a nested chain of subgroups $\overline{H}_j \trianglelefteq G/C$ of orders p^j for $0 \leq j \leq n-1$. If $\pi : G \to G/C$, set $H_0 = 1$ and $H_{j+1} = \pi^{-1}(\overline{H}_j)$. Preimages of normal subgroups are normal, and each preimage has order $|C|\,|\overline{H}_j| = p^{j+1}$. They are nested because the quotient terms are nested. This gives all required H_i, with $H_n = G$.

Exercise 36.22: Normal p-subgroups lie in every Sylow subgroup

Let S be any Sylow p-subgroup. Since $P \trianglelefteq G$, PS is a subgroup. Its order is

$$|PS| = \frac{|P||S|}{|P \cap S|},$$

a power of p, since all three orders are powers of p. It contains S, which already has the largest possible p-power order in G. Thus $PS = S$, and hence $P \subseteq S$. Since S was arbitrary, this holds for every Sylow p-subgroup.

37 Applications of the Sylow Theory

Exercise 37.1: Conjugacy classes of the square-symmetry group

(a) Write $D_4 = \{e, r, r^2, r^3, s, rs, r^2s, r^3s\}$ with $srs = r^{-1}$. Conjugating a rotation by s reverses it; conjugating a reflection by r advances its exponent by 2. The classes are therefore

$$\{e\}, \quad \{r^2\}, \quad \{r, r^3\}, \quad \{s, r^2s\}, \quad \{rs, r^3s\}.$$

These are disjoint and account for all eight elements. In the textbook's labels the last two classes are the two axis reflections and the two diagonal reflections, while $r = \rho_1$. **(b)** Their sizes give the class equation $8 = 1 + 1 + 2 + 2 + 2$. The two singleton classes are exactly the center $\{e, r^2\}$.

Exercise 37.2: Composite orders below 60

Plan. For completeness, organize the finite list instead of making an unsupported assertion for each order. If a Sylow number is 1, its subgroup is normal. A nontrivial p-group has nontrivial center and, if its order is composite, is not simple. The prime-power composite orders are

$$4, 8, 9, 16, 25, 27, 32, 49.$$

For the following remaining orders, the indicated prime has a unique Sylow subgroup, directly from $n_p \mid |G|/p^a$ and $n_p \equiv 1 \pmod p$:

p	$\lvert G\rvert$
3	$6, 18, 54$
5	$10, 15, 20, 35, 40, 45, 50$
7	$14, 21, 28, 42$
11	$22, 33, 44$
13	$26, 39, 52$
17	$34, 51$
19	$38, 57$
23	46
29	58
11	55

(The rows may use any prime giving uniqueness.) This leaves $12, 24, 30, 36, 48, 56$.

- For 24 and 48, n_2 is 1 or 3. If it is 3 and G were simple, its nontrivial conjugation action on the three Sylow subgroups would embed G into S_3, impossible by orders.
- For 12 and 36, n_3 is 1 or 4. In the latter case simplicity would embed G into S_4. The sign map must then be trivial (a simple group of order greater than 2 has no nontrivial map to $\mathbb{Z}_2$), so G embeds into A_4. Order 36 is too large; at order 12 this would give $G \cong A_4$, which has its normal order-4 subgroup of double transpositions.
- For 30, unless a Sylow 3- or 5-subgroup is normal, there are ten order-3 subgroups and six order-5 subgroups. Their nonidentity elements are disjoint, requiring $10 \cdot 2 + 6 \cdot 4 = 44 > 29$ nonidentity elements.

- For 56, n_7 is 1 or 8. In the latter case the order-7 subgroups already occupy 48 nonidentity elements. Only seven nonidentity elements remain. Every Sylow 2-subgroup has exactly seven nonidentity elements, all among these remaining ones; hence all such subgroups coincide and the Sylow 2-subgroup is normal.

This exhausts every composite order below 60. The conjugation actions used above are nontrivial and, under the assumption of simplicity, have trivial kernel, justifying the embeddings.

Exercise 37.3: True or false: applications of Sylow theory

(a) True. $159 = 3 \cdot 53$, and $53 \not\equiv 1 \pmod 3$; the two-prime criterion gives a cyclic group. Explicitly $n_{53} = 1$ and $n_3 \mid 53$, $n_3 \equiv 1 \pmod 3$ force $n_3 = 1$, so the normal factors commute and their product is cyclic.

(b) True. $102 = 2 \cdot 3 \cdot 17$, and $n_{17} \mid 6$, $n_{17} \equiv 1 \pmod{17}$ force a normal order-17 subgroup.

(c) False. $\mathbb{Z}_6$ is solvable but 6 is not a prime power.

(d) True. The normal chain of Exercise 36.21 has prime-order factors, so a finite p-group is solvable.

(e) This is a judgment about the method rather than a formal theorem; "true" describes the intended observation. Individual Sylow counts and class equations for many distinct orders become lengthy, and some cases require additional arguments.

(f) True. For order 21, $n_7 \mid 3$ and $n_7 \equiv 1 \pmod 7$ give $n_7 = 1$.

(g) True. Its center has order a positive multiple of 5 by the class equation for 5-groups, so it contains at least five elements.

(h) True. For order 42, $n_7 \mid 6$ and $n_7 \equiv 1 \pmod 7$ give $n_7 = 1$.

(i) False. A subgroup order must divide 42, but $8 \nmid 42$.

(j) False. There are other simple groups, for example $\mathrm{GL}(3, \mathbb{F}_2)$ of order $(8-1)(8-2)(8-4) = 168$, which is not of prime order or the order of an alternating group. One way to check simplicity is to count its conjugacy classes: their sizes are $1, 21, 56, 42, 24, 24$ (the nonidentity element orders are respectively $2, 3, 4, 7, 7$). No union of these classes containing the identity has a proper nontrivial size dividing 168. A normal subgroup is a union of conjugacy classes and its order divides 168, so none exists. The class sizes follow from centralizer orders $8, 3, 4, 7, 7$, respectively, by counting invertible matrices commuting with representatives.

Exercise 37.4: Groups of order $5 \cdot 7 \cdot 47$

Apply Sylow to each prime. The possible counts satisfy

$$n_{47} \mid 35, \quad n_{47} \equiv 1 \pmod{47}; \qquad n_7 \mid 235, \quad n_7 \equiv 1 \pmod 7.$$

The divisors of 35 are all below 47, so $n_{47} = 1$. The divisors of $235 = 5 \cdot 47$ are $1, 5, 47, 235$, with residues $1, 5, 5, 4$ modulo 7, so $n_7 = 1$. Also $n_5 \mid 329$; its possible divisors $1, 7, 47, 329$ have residues $1, 2, 2, 4$ modulo 5, forcing $n_5 = 1$. Thus all three Sylow subgroups are normal, cyclic, and of pairwise coprime orders. Two distinct ones commute: a commutator of their elements belongs to both normal

subgroups, whose intersection is trivial by Lagrange's theorem. Their product has all $5 \cdot 7 \cdot 47$ elements and is the direct product of three cyclic groups of pairwise coprime orders. Hence it is cyclic and therefore abelian.

Exercise 37.5: No simple group of order 96

A Sylow 2-subgroup has order 32, and $n_2 \mid 3$, so $n_2 = 1$ or 3. If $n_2 = 1$, it is a proper nontrivial normal subgroup. If $n_2 = 3$, conjugation on the three Sylow subgroups defines a nontrivial homomorphism $G \to S_3$. Were G simple, the kernel, a normal subgroup, would be trivial because the action is nontrivial. This would embed 96 elements into the six-element group S_3, impossible. Thus the kernel is itself a nontrivial proper normal subgroup, proving nonsimplicity in this case too.

Exercise 37.6: No simple group of order 160

Since $160 = 2^5 \cdot 5$, the Sylow 2-number divides 5 and is odd, so it is 1 or 5. The first case gives a normal subgroup of order 32. In the second case the conjugation action gives a nontrivial homomorphism $G \to S_5$. Its kernel cannot be trivial, because an injective map would require $160 \le |S_5| = 120$; and it is not all of G, since the action is nontrivial. Thus it is a nontrivial proper normal subgroup. Both cases exclude simplicity.

Exercise 37.7: An order-15 subgroup in every order-30 group

The counting argument of Example 37.12 shows that an order-30 group has a normal subgroup N of order 3 or of order 5: otherwise ten order-3 subgroups and six order-5 subgroups would require 44 nonidentity elements. If $|N| = 3$, the quotient G/N has order 10, and Cauchy's theorem gives a subgroup Q of order 5. Its preimage under the quotient map has order $3 \cdot 5 = 15$. If $|N| = 5$, the quotient has order 6 and has an order-3 subgroup; its preimage has order $5 \cdot 3 = 15$. Preimages of subgroups are subgroups, so either alternative proves the assertion. The order computation counts $|N|$ elements in each coset of the preimage.

Exercise 37.8: Cycle type and partitions

(a) The permutation $\tau\sigma\tau^{-1}$ sends $\tau(a_i)$ to $\tau(a_{i+1})$, with the last sent to the first, and fixes points outside this cycle. Thus it is $(\tau(a_1)\ \ldots\ \tau(a_m))$. **(b)** For two cycles of the same length, define τ to send their entries in cyclic order to one another and extend this bijection arbitrarily to the other points. Part (a) gives the desired conjugation. **(c)** For products of disjoint cycles with the same lengths, do the same on each disjoint support and combine these bijections. Conjugation respects products, so it sends the first product to the second. **(d)** Conversely conjugation only relabels entries and therefore preserves all cycle lengths, including fixed points as length-1 cycles. Thus conjugacy classes correspond exactly to unordered lists of positive lengths summing to n, namely partitions of n. **(e)** The numbers for $n = 1, 2, 3, 4, 5, 6, 7$ are

$$1, 2, 3, 5, 7, 11, 15.$$

For example a systematic way to compute them is to let $P(n, k)$ count partitions with parts at most k and use $P(n, k) = P(n, k-1) + P(n-k, k)$, with $P(0, k) = 1$ and negative arguments zero. The two terms distinguish partitions having no part k from those obtained by removing one part k.

Exercise 37.9: Conjugacy classes and class equation of S_4

By Exercise 37.8 there is one class for each cycle type. A class with m_k cycles of length k has size

$$\frac{n!}{\prod_k k^{m_k} m_k!}.$$

To see the denominator, rotate the entries within each length-k cycle in k ways and permute equal-length cycles in $m_k!$ ways; these do not change the permutation. For S_4 the classes are

type	representative	size
1^4	e	1
$2\,1^2$	(12)	6
2^2	$(12)(34)$	3
$3\,1$	(123)	8
4	(1234)	6

Here each row means all permutations of that type. Hence the class equation is $24 = 1 + 6 + 3 + 8 + 6$.

Exercise 37.10: Class equations of S_5 and S_6

Use the cycle-type formula $n!/\prod_k k^{m_k} m_k!$ from Exercise 37.9. For S_5 the complete table is

type	1^5	$2\,1^3$	$2^2\,1$	$3\,1^2$	$3\,2$	$4\,1$	5
size	1	10	15	20	20	30	24

Thus $120 = 1 + 10 + 15 + 20 + 20 + 30 + 24$. For S_6, list the eleven partitions as follows:

type	size	type	size
1^6	1	$2\,1^4$	15
$2^2\,1^2$	45	2^3	15
$3\,1^3$	40	$3\,2\,1$	120
3^2	40	$4\,1^2$	90
$4\,2$	90	$5\,1$	144
6	120		

Consequently

$$720 = 1 + 15 + 45 + 15 + 40 + 120 + 40 + 90 + 90 + 144 + 120.$$

The displayed sums equal the group orders, checking that all types and counts have been included.

Exercise 37.11: Conjugacy classes and abelian p-groups

The structure theorem writes an abelian group of order p^n uniquely up to isomorphism as

$$\mathbb{Z}_{p^{r_1}} \times \cdots \times \mathbb{Z}_{p^{r_t}}, \quad r_i \geq 1, \quad r_1 \leq \cdots \leq r_t, \quad \sum r_i = n.$$

The unordered exponent list is a partition of n, and different lists give nonisomorphic groups by uniqueness of the prime-power decomposition. Thus there are $p(n)$ isomorphism types of abelian groups of order p^n. Exercise 37.8 gives exactly $p(n)$ conjugacy classes in S_n, since cycle lengths provide the same partitions. The shared count comes from the common partition parameter, not from an asserted group isomorphism.

Exercise 37.12: The center of S_n for $n > 2$

Let $\sigma \neq e$, choose i with $\sigma(i) = j \neq i$, and choose k different from both i, j; this is possible because $n > 2$. Conjugation gives

$$\sigma(i\ k)\sigma^{-1} = (j\ \sigma(k)).$$

This transposition cannot equal $(i\ k)$, because its support contains j, which is neither i nor k. Therefore σ does not commute with $(i\ k)$ and is not central. Every nonidentity permutation fails centrality, while the identity commutes with everything. Hence $Z(S_n) = \{e\}$.

38 Free Abelian Groups

Exercise 38.1: A basis of $\mathbb{Z}^3$ with no zero coordinates

Take $u = (1, 1, 1)$, $v = (1, 2, 2)$, $w = (1, 2, 3)$. For any $(e, f, g) \in \mathbb{Z}^3$, solving $xu + yv + zw = (e, f, g)$ by subtracting coordinates gives

$$z = g - f, \qquad y = 2f - e - g, \qquad x = 2e - f.$$

These are integers and substitution recovers all three coordinates. The formulas also give a unique coefficient triple, so the vectors generate $\mathbb{Z}^3$ and are independent over $\mathbb{Z}$. Thus they form a basis, with every coordinate nonzero.

Exercise 38.2: The vectors $(2, 1)$ and $(3, 1)$

Yes. For an arbitrary (e, f), the equations $2x + 3y = e$, $x + y = f$ give

$$y = e - 2f, \qquad x = 3f - e.$$

Both coefficients are integers, so every integer pair is generated. They are uniquely determined, so the only representation of zero is $x = y = 0$. These are precisely the generating and independence requirements for a basis of a free abelian group.

Exercise 38.3: The vectors $(2, 1)$ and $(4, 1)$

No. Any integer combination $x(2, 1) + y(4, 1) = (2x + 4y, x + y)$ has even first coordinate. Therefore $(1, 0)$ is not generated. The two vectors are linearly independent over $\mathbb{R}$ (their determinant is -2), but that does not make them a $\mathbb{Z}$-basis: integer coefficients, rather than arbitrary real coefficients, are required.

Exercise 38.4: The determinant criterion for an integer basis

Put $D = ad - bc$, the determinant of the matrix with columns (a, b) and (c, d). If $D = \pm 1$, solving $x(a, b) + y(c, d) = (e, f)$ gives

$$x = \frac{de - cf}{D}, \qquad y = \frac{af - be}{D}.$$

These coefficients are integers and unique for all $e, f \in \mathbb{Z}$, proving the basis property. Conversely, if the two vectors are a basis, there is an integer matrix B whose columns express $(1, 0), (0, 1)$ in that basis. For $A = \begin{pmatrix} a & c \\ b & d \end{pmatrix}$ this gives $AB = I$. Taking determinants yields $D \det B = 1$. Both factors are integers, so $D = \pm 1$. Hence the necessary and sufficient condition is $ad - bc = \pm 1$.

Exercise 38.5: Definition of rank

Replace “a generating set” by “a basis.” The rank is the cardinality of any basis of the free abelian group. Generating sets can contain redundant elements: $\{1\}$ and $\{1, 2\}$ both generate $\mathbb{Z}$ but have different sizes. Bases have invariant cardinality.

Exercise 38.6: Definition of an abelian basis

The proposed definition is correct. The zero relation with distinct basis elements forces every integer coefficient to vanish, and the set generates the group. The reverse implication in the independence condition is automatic, since a sum with all coefficients zero is zero. A group having such a basis is free abelian.

Exercise 38.7: A proper subgroup of the same rank

For any positive integer r, take $G = \mathbb{Z}^r$ and $K = 2\mathbb{Z} \times \mathbb{Z}^{r-1}$. It is proper since $(1, 0, \ldots, 0) \notin K$. The r vectors $2e_1, e_2, \ldots, e_r$ generate K and are independent over $\mathbb{Z}$, so K has rank r, just as G does. In particular already $2\mathbb{Z} < \mathbb{Z}$ gives a rank-1 example.

Exercise 38.8: True or false: free abelian groups

(a) True. If a multiple of an element is zero, uniqueness of basis coefficients makes every coefficient zero.

(b) True by the structure theorem: the torsion cyclic factors vanish, leaving $\mathbb{Z}^r$.

(c) True: $\mathbb{Z}^r$ has the standard r-element basis.

(d) True. If G has Betti number r and is generated by r elements, write $G \cong \mathbb{Z}^r \oplus T$. A prime $p \mid |T|$ would give $G/pG \cong \mathbb{F}_p^r \oplus T/pT$ of dimension greater than r, impossible with r generators. Thus $T = 0$.

(e) True. The larger set still contains every generator in X, so still generates all of G.

(f) False. $\{1\}$ is a basis of $\mathbb{Z}$, but $\{1, 2\}$ is dependent: $2 \cdot 1 - 1 \cdot 2 = 0$.

(g) False. $\mathbb{Z}$ has only two bases, $\{1\}$ and $\{-1\}$.

(h) True. If $x_1, x_2, \ldots$ is a basis, replacing x_1 by $x_1 + kx_2$ gives a different basis for each $k \in \mathbb{Z}$, since the old x_1 is recovered by subtracting kx_2.

(i) True: a subgroup of $\mathbb{Z}^r$ is free abelian, as proved in this section.

(j) False. For $G = \mathbb{Z}$, $K = 2\mathbb{Z}$, the quotient is $\mathbb{Z}_2$, which has nonzero torsion and is not free abelian.

Exercise 38.9: The coordinate isomorphism

For a basis $x_1, \ldots, x_r$, define $\phi(\sum n_i x_i) = (n_1, \ldots, n_r)$. Uniqueness of a basis expansion makes this well defined, including the zero vector. If $a = \sum n_i x_i$ and $b = \sum m_i x_i$, then $a + b = \sum (n_i + m_i) x_i$, so $\phi(a + b) = \phi(a) + \phi(b)$. If $\phi(a) = 0$, all coefficients are zero, hence $a = 0$; thus the map is injective. Given any integer tuple $(n_1, \ldots, n_r)$, its preimage is $\sum n_i x_i$, so the map is onto. This completes the claimed isomorphism $G \cong \mathbb{Z}^r$.

Exercise 38.10: A free abelian group has no nonzero torsion

Let $a = \sum_{i=1}^{r} n_i x_i$ in a basis, with finitely many coefficients involved, and suppose $ma = 0$ for an integer $m \neq 0$. Then $\sum_i (mn_i)x_i = 0$. Independence forces $mn_i = 0$ in $\mathbb{Z}$ for each i. Since $m \neq 0$, every $n_i = 0$, so $a = 0$. Thus a nonzero element cannot have finite order; the argument works even for an infinite basis because each expansion is finite.

Exercise 38.11: Products of free abelian groups

If X is a basis of G and Y a basis of G', use

$$\{(x, 0) : x \in X\} \ \cup \ \{(0, y) : y \in Y\}.$$

It generates $G \times G'$ by expanding each coordinate separately. A zero integer combination has zero first coordinate, forcing all X-coefficients to be zero, and zero second coordinate, forcing all Y-coefficients to be zero. Thus the displayed set is a basis, and the product is free abelian. This is a finite product of groups; even when the bases are infinite, each individual expansion remains finite.

Exercise 38.12: Finitely generated torsion-free groups

A free abelian group of finite rank has a finite basis and hence is finitely generated; Exercise 38.10 proves it is torsion free. Conversely the structure theorem writes a finitely generated abelian group as $\mathbb{Z}^r \oplus T$, where T is finite. If there is no nonzero element of finite order, T must be trivial. Thus the group is $\mathbb{Z}^r$, which has its standard basis. The zero group is included by the usual empty-basis/rank-zero convention; if "free abelian" is restricted to nonzero groups as in the section's initial definition, state the equivalence for nonzero groups.

Exercise 38.13: Why $\mathbb{Q}$ is not free abelian

Two nonzero rational numbers $x = a/b$ and $y = c/d$ satisfy $(bc)x - (ad)y = 0$, with both coefficients nonzero when $a, c \neq 0$. Thus no independent set over $\mathbb{Z}$ can contain two distinct nonzero rational numbers; zero cannot be a basis element either. A basis for the nonzero group $\mathbb{Q}$ would therefore contain just one nonzero rational q. But its generated subgroup consists of integer multiples of q, and $q/2$ is not one of them. Hence one element cannot generate $\mathbb{Q}$, so $\mathbb{Q}$ is not free abelian.

Exercise 38.14: The p-primary torsion subgroup

The set T_p contains zero. If $p^a x = 0$ and $p^b y = 0$, let $c = \max(a, b)$; then $p^c(x - y) = 0$. Hence $x - y$ has order a power of p (or is zero), and lies in T_p. The one-step subgroup criterion proves $T_p \leq T$. The commutativity of T is used when distributing multiplication by p^c over a difference.

Exercise 38.15: Locating the p-primary factors

In a prime-power product decomposition of T, an element is killed by a power of p exactly when all its coordinates in q-power factors with $q \neq p$ vanish. Indeed, multiplication by p is invertible on a cyclic q-group, because $\gcd(p, q) = 1$. Its repeated powers therefore cannot kill a nonzero coordinate.

Conversely the remaining p-power coordinates are all killed by some common power of p. Restricting the product isomorphism to T_p thus identifies it exactly with the product of the cyclic p-power factors.

Exercise 38.16: Elements killed by an integer

The multiplication map $\mu_n : G \to G$, $x \mapsto nx$, is a homomorphism because G is abelian. Therefore $G[n] = \ker \mu_n$ is a subgroup. Directly, $n0 = 0$, and if $nx = ny = 0$, then $n(x - y) = nx - ny = 0$, proving the same fact without invoking kernels. For a general nonabelian group the power map need not be a homomorphism; the abelian assumption is essential.

Exercise 38.17: The p-torsion of a cyclic p-group

In $\mathbb{Z}_{p^r}$ the equation $px = 0$ means $p^r \mid px$, equivalently $p^{r-1} \mid x$. Thus

$$\mathbb{Z}_{p^r}[p] = \{0, p^{r-1}, 2p^{r-1}, \ldots, (p-1)p^{r-1}\}.$$

It is cyclic generated by p^{r-1} and has order p, so the map $\mathbb{Z}_p \to \mathbb{Z}_{p^r}[p]$ sending k to kp^{r-1} is a well-defined bijective homomorphism.

Exercise 38.18: Counting p-torsion in a product

An element of a direct product is killed by p exactly when every coordinate is killed by p. Hence

$$\left(\prod_{i=1}^{m} \mathbb{Z}_{p^{r_i}}\right)[p] = \prod_{i=1}^{m} (\mathbb{Z}_{p^{r_i}}[p]) \cong \mathbb{Z}_p^m$$

by Exercise 38.17. In particular its size is p^m; this size is intrinsic to the group, independent of a chosen decomposition.

Exercise 38.19: Uniqueness of the p-power exponents

(a) Apply the intrinsic construction $T_p[p]$ to the two decompositions. Their orders are p^m and p^n, so $m = n$. **(b)** Arrange exponents increasingly. Suppose that $r_i = s_i$ for $i < j$ and, seeking a contradiction, $r_j < s_j$. The group $p^{r_j}T_p$ has from the first decomposition one nonzero cyclic factor for each $r_i > r_j$, at most $m - j$ factors. From the second decomposition every index $i \geq j$ has $s_i \geq s_j > r_j$, giving at least $m - j + 1$ nonzero factors; the indices before j have the same behavior in both decompositions. Part (a), applied to this intrinsic subgroup, forbids different numbers of nonzero p-power cyclic factors. Thus $r_j < s_j$ is impossible. Interchange the decompositions to rule out $s_j < r_j$. Start at $j = 1$ and induct to obtain $r_i = s_i$ for all i.

Exercise 38.20: From invariant factors to prime powers

Factor each torsion coefficient m_i into pairwise coprime prime powers. The Chinese remainder isomorphism gives

$$\mathbb{Z}_{m_i} \cong \prod_{p \mid m_i} \mathbb{Z}_{p^{v_p(m_i)}}.$$

Taking the product over i and regrouping by prime produces the prime-power decomposition of T. Its uniqueness follows from Exercises 38.14–19. Factors of order 1 are omitted, since they contribute only a trivial coordinate.

Exercise 38.21: Recovering the largest torsion coefficient

Because $m_i \mid m_{i+1}$, the valuations $v_p(m_i)$ are nondecreasing for each prime p. The largest exponent for p in the entire prime-power decomposition is consequently $v_p(m_r)$. The same intrinsic largest exponent is $v_p(n_s)$ in the other decomposition. Therefore, writing it as h_p,

$$m_r = \prod_{p \mid\mid T\mid} p^{h_p} = n_s.$$

All primes dividing T occur in this product, so equality of the prime valuations proves equality of the integers.

Exercise 38.22: Recovering all torsion coefficients

For each prime, list its exponents in the unique prime-power decomposition in increasing order. Pad each prime's list on the *left* with zeros to a common length. The last torsion coefficient is the product of the prime powers in the last column; the preceding coefficient is the product in the next-to-last column, and so on. This reconstructs m_{r-1} and n_{s-1} identically: for each prime take its second-largest exponent, or zero if it appears only once. Continue using third-largest exponents and so forth. Divisibility forces precisely this alignment of the lists, because valuations are nondecreasing. The number of columns is the largest number of nonzero exponents in any prime's list: otherwise the first coefficient would be 1, contrary to $m_1, n_1 > 1$. Thus the common length is both r and s, and every coefficient agrees. This proves $r = s$ and $m_i = n_i$ for all i.

39 Free Groups

Exercise 39.1: Reduced words and their inverses

Plan. Combine only *adjacent* powers of the same letter; for an inverse, reverse the syllables and negate every exponent. **(a)** In $a^2b^{-1}b^3a^3c^{-1}c^4b^{-2}$ the adjacent b powers combine to b^2, and the adjacent c powers to c^3. Thus the reduced word and its inverse are

$$a^2b^2a^3c^3b^{-2}, \qquad b^2c^{-3}a^{-3}b^{-2}a^{-2}.$$

(b) In $a^2a^{-3}b^3a^4c^4c^2a^{-1}$, combine the first two a powers and the two c powers. The answers are

$$a^{-1}b^3a^4c^6a^{-1}, \qquad ac^{-6}a^{-4}b^{-3}a.$$

Multiplying either displayed word by its displayed inverse cancels from the middle outward to the empty word, checking the results.

Exercise 39.2: The same products in a free abelian group

Now commuting letters may be regrouped, so add *all* exponents of each letter. **(a)** The exponent totals are $a : 2 + 3 = 5$, $b : -1 + 3 - 2 = 0$, $c : -1 + 4 = 3$. The product is a^5c^3 and its inverse is $a^{-5}c^{-3}$. **(b)** The totals are $a : 2 - 3 + 4 - 1 = 2$, $b : 3$, $c : 4 + 2 = 6$. The product is $a^2b^3c^6$ and its inverse is $a^{-2}b^{-3}c^{-6}$. Unlike Exercise 39.1, rearrangement is allowed because this group is abelian.

Exercise 39.3: Homomorphisms from a rank-two free group

Let its basis be x, y. Any choice of the two images defines one and only one homomorphism, by the universal property of a free group. No commuting condition is imposed. Thus the number of maps to a finite group H is $|H|^2$. The answers are **(a)** $4^2 = 16$, **(b)** $6^2 = 36$, and **(c)** $|S_3|^2 = 36$.

Exercise 39.4: Surjections from a rank-two free group

The images of the two basis elements must generate the codomain, and every ordered generating pair gives exactly one surjection. **(a)** In $\mathbb{Z}_4$, a pair generates if at least one entry is odd. Of 16 pairs, exactly four have both entries even, leaving 12. **(b)** In $\mathbb{Z}_6$, the reductions must generate modulo both 2 and 3. The Chinese remainder theorem makes the choices independent: modulo 2 there are $2^2 - 1 = 3$ nonzero pairs, and modulo 3 there are $3^2 - 1 = 8$. Thus there are $3 \cdot 8 = 24$. **(c)** In S_3, an ordered pair generates exactly when it consists of two distinct transpositions ($3 \cdot 2 = 6$ choices), or one transposition and one 3-cycle (two positions, three transpositions, two cycles: 12 choices). The other pairs lie in a proper cyclic subgroup. Thus there are $6 + 12 = 18$ surjections.

Exercise 39.5: Homomorphisms from a rank-two free abelian group

A map from $\mathbb{Z}^2$ is determined by an ordered *commuting* pair of images. Every pair commutes in the cyclic codomains, so **(a)** and **(b)** again give 16 and 36. For **(c)**, count commuting pairs in S_3. The

identity commutes with all six elements. Each of three transpositions has a centralizer of size 2, and each of two 3-cycles has a centralizer of size 3. Thus the answer is

$$6 + 3 \cdot 2 + 2 \cdot 3 = 18.$$

For any commuting pair (g, h) the homomorphism is $(m, n) \mapsto g^m h^n$, verifying existence as well as the count.

Exercise 39.6: Surjections from a rank-two free abelian group

For the abelian codomains the generating-pair counts are unchanged from Exercise 39.4: **(a)** 12 maps onto $\mathbb{Z}_4$ and **(b)** 24 onto $\mathbb{Z}_6$. **(c)** There are 0 maps onto S_3, since the image of an abelian group is abelian: $f(x)f(y) = f(x + y) = f(y + x) = f(y)f(x)$. A nonabelian group therefore cannot be the entire image.

Exercise 39.7: Definition of a reduced word

The definition is correct. Adjacent syllables involving the same letter can be combined by adding their exponents, and a syllable of exponent zero can be deleted. A word is reduced exactly when neither elementary contraction is possible. The empty word is reduced as well.

Exercise 39.8: Definition of free-group rank

Replace "a set of generators" by "a basis," meaning a free generating set. The rank of a free group is the cardinality of a basis. An arbitrary generating set can contain redundant elements and therefore need not have the invariant size of a basis.

Exercise 39.9: Reflecting on an apparently obvious assertion

Consider the assertion that the final reduced word does not depend on the order of contractions. It is plausible from examples, but examples alone do not prove it. One systematic justification expands syllables into individual letters $a^{\pm 1}$ and scans left to right with a stack: cancel the next letter when it is inverse to the stack's top; otherwise append it. The stack after a prefix is the reduced representative of that prefix. Deleting an adjacent inverse pair anywhere leaves the final stack unchanged, so any sequence of such deletions gives the same result. Combining adjacent equal-letter powers is equivalent to these cancellations together with collecting identical adjacent signs. Thus the apparent obviousness can be replaced by an invariant algorithm. It is reasonable to accept the assertion while learning the main construction, while recognizing what a complete proof needs.

Exercise 39.10: True or false: free groups

(a) True: the subgroup theorem for free groups (Nielsen–Schreier), stated in the section, includes proper subgroups.

(b) False. $2\mathbb{Z} \times \mathbb{Z}$ is a proper subgroup of the free abelian group $\mathbb{Z}^2$, but is abelian of rank 2 and hence is not a free group.

(c) False. $\mathbb{Z} \twoheadrightarrow \mathbb{Z}_2$ is a homomorphism from a free group, but a nontrivial finite group is not free.

(d) True: this is the defining property of free abelian groups.

(e) False. The finitely generated abelian group $\mathbb{Z}_2$ is not free abelian, because it has nonzero torsion.

(f) False. A free group is, by definition, free on a basis; $\mathbb{Z}$ is a basic example.

(g) False. $\mathbb{Z}$ is both free abelian and free as a group on one generator.

(h) True. A free group on at least two basis elements is nonabelian (the reduced words xy and yx differ), whereas a free abelian group is abelian. A rank-greater-than-one free abelian group is neither trivial nor cyclic, so it cannot instead be a free group of rank 0 or 1.

(i) False. $\mathbb{Z}$ and the free group on two letters have different abelian/nonabelian behavior.

(j) True. A bijection between bases extends to an isomorphism by matching finite integer combinations.

Exercise 39.11: A basis allowing torsion

(a) In $\mathbb{Z}_4$ the relation $1 \cdot 2 + 2 \cdot 3 = 0$ holds, but the individual summands are both 2, not zero. Thus $\{2, 3\}$ fails this exercise's independence condition. The singleton $\{1\}$ is a basis: it generates, and a one-term relation already has its one term zero. **(b)** The same singleton $\{1\}$ is a basis of $\mathbb{Z}_6$. Also $\{2, 3\}$ generates because $3 - 2 = 1$. If $2m + 3n = 0 \pmod 6$, reducing modulo 3 gives $m = 0 \pmod 3$, and reducing modulo 2 gives $n = 0 \pmod 2$. Consequently $2m = 0$ and $3n = 0$ individually. This proves the required independence. **(c)** Yes. A Section 38 basis forces each integer coefficient to be zero in a zero sum, which in particular forces each summand to be zero. It also generates the group. **(d)** Write a finite abelian group as $\mathbb{Z}_{d_1} \times \cdots \times \mathbb{Z}_{d_n}$ with $d_i \mid d_{i+1}$, using its invariant-factor decomposition. Let b_i be the inverse image of the standard generator in coordinate i. They generate, have orders d_i, and a zero sum has zero in each coordinate, forcing each summand $m_i b_i$ to be zero. Thus they form a basis in the present torsion-allowing sense.

Exercise 39.12: The universal abelian quotient

(a) If $\phi_i : G \twoheadrightarrow G_i^*$ have the stated property, there are maps $\theta_1 : G_2^* \to G_1^*$ and $\theta_2 : G_1^* \to G_2^*$ with $\phi_1 = \theta_1\phi_2$ and $\phi_2 = \theta_2\phi_1$. Then $(\theta_1\theta_2)\phi_1 = \phi_1$. Since ϕ_1 is onto, $\theta_1\theta_2$ is the identity on G_1^*. Similarly $\theta_2\theta_1$ is the identity on G_2^*, so the groups are isomorphic. **(b)** Let $G^* = G/[G, G]$ and let ϕ be the quotient map. For a homomorphism $\psi : G \to A$ with A abelian, every commutator maps to the identity, so $[G, G] \subseteq \ker \psi$. Therefore $\theta(g[G, G]) = \psi(g)$ is well defined, is a homomorphism, and satisfies $\psi = \theta\phi$. **(c)** This is the abelianization of G, its largest universal abelian quotient. The factor map is unique because ϕ is onto.

Exercise 39.13: The universal free group

(a) Let $u : G_1 \to G_2$ and $v : G_2 \to G_1$ be the unique homomorphisms with $ug_1 = g_2$ and $vg_2 = g_1$. Then $vug_1 = g_1$, and both vu and the identity are homomorphisms extending g_1. Uniqueness forces $vu = \mathrm{id}_{G_1}$. Similarly $uv = \mathrm{id}_{G_2}$, giving an isomorphism. For clarity the universal property also implies that g_i is injective: distinguish two points of S by a function to $\mathbb{Z}_2$ and factor it. It implies that $g_i(S)$

generates: for $H = \langle g_i(S) \rangle$, factor $S \to H$ through a homomorphism $G_i \to H$; its composite with the inclusion into G_i is the identity by uniqueness, hence $H = G_i$. **(b)** Construct $F[S]$ as the group of reduced words on the alphabet S, with $g(s)$ the one-letter word. Any function $f : S \to G'$ extends by replacing letters with their images and using integer powers. Cancellation preserves the result; this is the unique extending homomorphism. **(c)** The "blop group" is exactly the free group on S.

Exercise 39.14: A universal property for free abelian groups

For a set S, require an abelian group A and a map $i : S \to A$ such that for every abelian group B and every function $f : S \to B$, there is a unique homomorphism $\overline{f} : A \to B$ with $\overline{f}\, i = f$. Existence follows by taking $A = \bigoplus_{s \in S} \mathbb{Z}$, the finitely supported integer functions on S, and $i(s)$ the function that is 1 at s and zero elsewhere. Define

$$\overline{f}\left(\sum_s n_s i(s)\right) = \sum_s n_s f(s).$$

The sums are finite, and commutativity in B makes this an additive map. Every element has a unique such expansion, so the extension exists and is unique. Conversely the universal property characterizes this group up to a unique isomorphism carrying the specified generators to one another, by the two-composites argument of Exercise 39.13(a). Thus it characterizes the free abelian group with basis S.

40 Group Presentations

Exercise 40.1: Presentations of $\mathbb{Z}_4$

One, two, and three generators can be used as follows:

$$\langle a \mid a^4 = 1\rangle, \qquad \langle a, b \mid a^4 = 1,\ b = a^2\rangle,$$

$$\langle a, b, c \mid a^4 = 1,\ b = a^2,\ c = a^3\rangle.$$

In each case eliminate the additional generators using their defining relations. Every element is a power of a, with exponent reduced modulo 4, so there are at most four elements. Sending a to $1 \in \mathbb{Z}_4$, b to 2 and c to 3 (where present) satisfies the relations and maps onto $\mathbb{Z}_4$, proving there are exactly four elements.

Exercise 40.2: A three-generator presentation of S_3

Use

$$\langle a, b, c \mid a^3 = 1,\ b^2 = 1,\ bab = a^{-1},\ c = ab\rangle.$$

The relation for c lets us eliminate it. The conjugation relation gives $ba = a^{-1}b$, so every word becomes $a^i b^j$ with $0 \le i < 3$, $0 \le j < 2$. Thus there are at most six elements. Map $a \mapsto (123)$, $b \mapsto (12)$, $c \mapsto (123)(12)$. These permutations satisfy the relations and generate S_3, so the presented group has at least six elements. It is therefore isomorphic to S_3.

Exercise 40.3: Tables for the octic and quaternion groups

Plan. Write every element as $a^i b^j$, with $0 \le i < 4$, $j \in \{0, 1\}$. The rule $ba = a^{-1}b$ moves b past a power of a. If $b^2 = a^c$, then

$$(a^i b^j)(a^u b^v) = a^{\,i+(-1)^j u + cjv} b^{\,j+v \bmod 2},$$

where the exponent of a is reduced modulo 4. Use $c = 0$ for the octic group and $c = 2$ for the quaternion group. The octic table, in the requested order, is

$\cdot$	1	a	a^2	a^3	b	ab	a^2b	a^3b
1	1	a	a^2	a^3	b	ab	a^2b	a^3b
a	a	a^2	a^3	1	ab	a^2b	a^3b	b
a^2	a^2	a^3	1	a	a^2b	a^3b	b	ab
a^3	a^3	1	a	a^2	a^3b	b	ab	a^2b
b	b	a^3b	a^2b	ab	1	a^3	a^2	a
ab	ab	b	a^3b	a^2b	a	1	a^3	a^2
a^2b	a^2b	ab	b	a^3b	a^2	a	1	a^3
a^3b	a^3b	a^2b	ab	b	a^3	a^2	a	1

The quaternion table is

$\cdot$	1	a	a^2	a^3	b	ab	a^2b	a^3b
1	1	a	a^2	a^3	b	ab	a^2b	a^3b
a	a	a^2	a^3	1	ab	a^2b	a^3b	b
a^2	a^2	a^3	1	a	a^2b	a^3b	b	ab
a^3	a^3	1	a	a^2	a^3b	b	ab	a^2b
b	b	a^3b	a^2b	ab	a^2	a	1	a^3
ab	ab	b	a^3b	a^2b	a^3	a^2	a	1
a^2b	a^2b	ab	b	a^3b	1	a^3	a^2	a
a^3b	a^3b	a^2b	ab	b	a	1	a^3	a^2

For example, the product $(a^2b)(a^3b)$ is $a^{2-3} = a^3$ in the first group and $a^{2-3+2} = a$ in the second. This checks the effect of the different square of b.

Exercise 40.4: All groups of order 14

Sylow gives a unique order-7 subgroup $\langle a\rangle$, so it is normal. Cauchy's theorem gives b of order 2 outside it. The elements a^ib^j exhaust the group, since the subgroup generated by a, b has order divisible by both 7 and 2. Conjugation has $bab^{-1} = a^r$ with r nonzero modulo 7. Since $b^2 = 1$, $r^2 \equiv 1 \pmod 7$, so $(r-1)(r+1) = 0$ in the field $\mathbb{Z}_7$ and $r = 1$ or 6. The first case gives $\mathbb{Z}_{14}$. The second gives $\langle a, b \mid a^7 = b^2 = 1,\ ba = a^6b\rangle$, the dihedral group D_7. Exercise 40.13 proves it has order 14; it is nonabelian because inversion acts nontrivially on a. These are the only two possibilities, and their different commutativity makes them nonisomorphic.

Exercise 40.5: All groups of order 21

The order-7 Sylow subgroup $\langle a\rangle$ is normal. Choose b of order 3 by Cauchy's theorem; it is outside $\langle a\rangle$, and together the elements generate all 21 elements. The action $bab^{-1} = a^r$ must satisfy $r^3 \equiv 1 \pmod 7$. Checking the six nonzero residues gives $r = 1, 2, 4$. For $r = 1$ the group is $\mathbb{Z}_{21}$. For $r = 2$ we obtain

$$\langle a, b \mid a^7 = b^3 = 1,\ ba = a^2b\rangle,$$

a nonabelian group of order 21 by Exercise 40.13. The choice $r = 4$ gives the same isomorphism type: replacing b by b^2 changes the action exponent from 2 to $2^2 = 4$, while b^2 still generates the order-3 subgroup. Thus there is exactly one abelian and one nonabelian isomorphism type.

Exercise 40.6: Consequences of relators

Include conjugates, which are missing from the proposed definition. A consequence is an element of the normal closure of the relators in the free group, equivalently a finite product

$$\prod_{j=1}^{k} w_j r_{i_j}^{\epsilon_j} w_j^{-1}, \qquad \epsilon_j \in \{1, -1\}.$$

Allowing arbitrary integer powers gives the same normal closure. The empty product is the identity. Products merely of powers of the original relators need not form a normal subgroup.

Exercise 40.7: Isomorphic presentations

Two presentations are isomorphic when the groups they present are isomorphic. A bijective renaming of generators and relators is sufficient but not necessary. For example $\langle a \mid a^4 = 1\rangle$ and $\langle a, b \mid a^4 = 1, b = a^2\rangle$ both present $\mathbb{Z}_4$, as proved in Exercise 40.1, but their generator sets have different sizes, so no bijective renaming exists. The printed proposed definition is therefore too restrictive.

Exercise 40.8: True or false: presentations

(a) True. Take a free group mapping onto the group and use all kernel elements as relators.

(b) True. Introduce arbitrarily many new generators together with relations expressing each as an old word.

(c) False. By the definition used here, all presentations of the same group are isomorphic presentations.

(d) False. A finite presentation gives a finitely generated, hence countable, group. The uncountable additive group $\mathbb{R}$ cannot have one.

(e) False. $\langle a \mid \rangle$ presents the infinite cyclic group with a finite generator set and no relators.

(f) True. A cyclic group has $\langle a \mid a^n = 1\rangle$ if its order is n, or $\langle a \mid \rangle$ if it is infinite. For the trivial group use $n = 1$.

(g) True. Consequences form the normal closure, which contains every conjugate of each relator.

(h) False. $\langle a \mid a^2 = 1\rangle$ and $\langle a \mid a^3 = 1\rangle$ have the same number of generators but give nonisomorphic groups of orders 2 and 3.

(i) True. If F/R is abelian, all commutators in F map to the identity, so $[F, F] \subseteq R$.

(j) False. $\langle a, b \mid b = 1\rangle$ presents the free group $\mathbb{Z}$ but has the nonidentity free word b as a relator.

Exercise 40.9: Groups of order 15 are abelian

Sylow gives a normal subgroup $\langle a\rangle$ of order 5. Cauchy's theorem gives b of order 3, and a, b generate the group. If $bab^{-1} = a^r$, conjugating three times gives $r^3 \equiv 1 \pmod 5$. Among the nonzero residues modulo 5 only $r = 1$ satisfies this (or use that their multiplicative group has order 4, coprime to 3). Thus a and b commute. The group is generated by commuting elements, hence abelian; indeed ab has order 15, so it is cyclic. There is no nonabelian group of order 15.

Exercise 40.10: An order-six nonabelian presentation

Apply Exercise 40.13 with $m = 3, n = 2, r = 2$. Since $2^2 \equiv 1 \pmod 3$, its normal-form model has exactly $3 \cdot 2 = 6$ elements. It realizes a with order 3 and $ba = a^2b$. If a, b commuted, cancellation would give $a = a^2$, hence $a = 1$, contradicting its order 3. Thus the group is nonabelian.

Exercise 40.11: Uniqueness of the nonabelian group of order six

For any group of order 6, the Sylow 3-subgroup $\langle a\rangle$ is unique and normal. An element b of order 2 exists by Cauchy's theorem and lies outside it, so a, b generate. Conjugation sends a to a or a^2, the two generators of $\langle a\rangle$. The first case makes the group abelian and cyclic of order 6. The second gives exactly the presentation in Exercise 40.10, which has order 6 and thus maps isomorphically onto the given group. Since S_3 is nonabelian of order 6, it realizes the second case. Concretely one may send a to (123) and b to (12).

Exercise 40.12: No subgroup of order six in A_4

An order-six subgroup would, by Exercise 40.11, be cyclic of order 6 or isomorphic to S_3. The cyclic possibility is impossible because elements of A_4 have orders only $1, 2, 3$: they are the identity, double transpositions, and 3-cycles. The S_3 possibility is also impossible. Its three involutions do not commute, and the product of two distinct involutions has order 3. In A_4, the only involutions are the three double transpositions; they form a Klein four group, commute, and multiply in pairs to another involution. An isomorphism preserves these products and orders, giving a contradiction.

Exercise 40.13: The normal-form multiplication criterion

(a) Write a formal element as (s, t) with $s \in \mathbb{Z}_m$ and $0 \le t < n$, and define

$$(s,t)(u,v) = (s + r^t u \bmod m,\ t + v \bmod n).$$

Assume $r^n \equiv 1 \pmod m$. Then r is invertible modulo m, and replacing $t+v$ by its residue modulo n leaves r^{t+v} unchanged modulo m. Thus the first coordinates of both associations of a triple product equal $s + r^t u + r^{t+v} w$, and the second equal $t + v + z \pmod n$. The operation is associative. The identity is $(0, 0)$. For (s, t) take $v = -t \pmod n$ and $u = -r^{-t} s \pmod m$; the resulting (u, v) is a two-sided inverse. This proves a group of order mn. Conversely, for $n > 1$ suppose the operation is a group. Then $a = (1, 0)$ has order m, $b = (0, 1)$ has order n, and $bab^{-1} = a^r$. Repeating conjugation n times gives $a = a^{r^n}$, hence $r^n \equiv 1 \pmod m$. *Degenerate case:* if $n = 1$, the displayed operation is always just addition in $\mathbb{Z}_m$, independent of r. Thus the literal "if and only if" in part (a) needs $n > 1$; its sufficiency remains valid for $n = 1$. **(b)** If the congruence holds, the group constructed above satisfies the presentation relations. Conversely those relations put every word in one of the mn forms $a^i b^j$, so the presented group maps onto this order-mn model and has at most mn elements; it therefore has exactly mn. If the presentation has order mn, its mn normal forms must all be distinct. In particular a has order m. The relations $b^n = 1$ and $bab^{-1} = a^r$ imply $a = a^{r^n}$ and hence the congruence. This also proves part (b) for $n = 1$, where $b = 1$ forces $a = a^r$.

Exercise 40.14: A unique nonabelian group of order pq

Let $p < q$ be primes with $p \mid q - 1$. For any order-pq group, the Sylow q-subgroup is unique because its number divides $p < q$ and is 1 modulo q. Write it $\langle a\rangle$, and choose b of order p by Cauchy's theorem. The two subgroups intersect trivially and generate the group, so it has a presentation

$$\langle a, b \mid a^q = b^p = 1,\ ba = a^r b\rangle, \qquad r^p \equiv 1 \pmod q.$$

The unit group $\mathbb{Z}_q^*$ is cyclic of order $q - 1$, so the pth roots of 1 form its unique order-p subgroup. Its nonidentity elements are $r, r^2, \ldots, r^{p-1}$ for any one generator r. The root 1 gives the abelian

cyclic group. Every other root gives a nonabelian order-pq group by Exercise 40.13. Replacing b by b^j, $1 \leq j < p$, changes the action exponent to r^j without changing its generated order-p subgroup. Therefore all nontrivial roots give isomorphic presentations, proving existence and uniqueness of the nonabelian isomorphism type.

41 Simplicial Complexes and Homology Groups

We use integer coefficients and ordinary, unreduced homology. Brackets list the vertices of an oriented simplex; exchanging two vertices reverses its sign. The zero group is denoted by 0.

Exercise 41.1: Boundary of the specified 2-chain

Use $[ijk] = P_iP_jP_k$ with the order indicating orientation. Reversing two vertices changes the sign, so the chain in the question becomes

$$c = 2[134] - 4[346] - 3[234] - [146].$$

(a) Apply $\partial[ijk] = [jk] - [ik] + [ij]$ term by term:

$$\begin{aligned}\partial c &= 2([34] - [14] + [13]) - 4([46] - [36] + [34]) \\ &\quad - 3([34] - [24] + [23]) - ([46] - [16] + [14]) \\ &= 2[13] - 3[14] + [16] - 3[23] + 3[24] \\ &\quad - 5[34] + 4[36] - 5[46].\end{aligned}$$

(b) No: distinct oriented edges form a basis of the chain group, and for instance the coefficient of [46] is $-5 \neq 0$. Thus $\partial c \neq 0$. **(c)** Yes: $\partial(\partial c) = 0$ because the boundary of a boundary is zero. This can also be checked by replacing each $[ij]$ by $P_j - P_i$; every vertex coefficient cancels.

Exercise 41.2: The boundary of a tetrahedron boundary

First

$$\partial_3[1234] = [234] - [134] + [124] - [123].$$

Applying ∂_2 gives

$$\begin{aligned}&([34] - [24] + [23]) - ([34] - [14] + [13]) \\ &\quad + ([24] - [14] + [12]) - ([23] - [13] + [12]) = 0.\end{aligned}$$

Every edge occurs twice with opposite sign. Thus $\partial_2\partial_3[1234] = 0$, as required. The signs, rather than an absence of geometric edges, are what cause the cancellation.

Exercise 41.3: The chain groups of one point

There is exactly one vertex and no positive-dimensional simplex. Hence $C_0(P) = \mathbb{Z}P$ and $C_i(P) = 0$ for $i > 0$. Since ∂_0 maps to zero, $Z_0(P) = C_0(P)$. Since $C_1(P) = 0$, $B_0(P) = 0$. Therefore $H_0(P) \cong \mathbb{Z}$. For $i > 0$, all of C_i, Z_i, B_i, H_i are zero. These are ordinary homology groups; reduced H_0 would instead vanish.

Exercise 41.4: Two isolated points

Here $C_0(X) = \mathbb{Z}P \oplus \mathbb{Z}P'$ and $C_i(X) = 0$ for $i > 0$, because no connecting edge is present. Thus $Z_0 = C_0$, $B_0 = 0$, and $H_0 \cong \mathbb{Z}^2$. All positive-dimensional chain, cycle, boundary, and homology groups are zero. The two independent generators of H_0 record the two connected components; the geometric line segment in the surrounding Euclidean space contributes nothing because it is not part of X.

Exercise 41.5: A closed line segment

Let $e = [12]$. Then $C_1 = \mathbb{Z}e$, $C_0 = \mathbb{Z}P_1 \oplus \mathbb{Z}P_2$, and all C_i for $i \geq 2$ vanish. The boundary satisfies $\partial_1(ne) = n(P_2 - P_1)$. Since the vertices are independent, the boundary is zero only if $n = 0$. Thus $Z_1 = 0$; also $B_1 = 0$, so $H_1 = 0$. In dimension zero, $Z_0 = C_0$ and $B_0 = \mathbb{Z}(P_2 - P_1)$. The sum-of-coefficients map $aP_1 + bP_2 \mapsto a + b$ is onto $\mathbb{Z}$ and has kernel B_0, so $H_0 \cong \mathbb{Z}$. In higher dimensions all groups are zero. This describes each of C_i, Z_i, B_i, H_i explicitly.

Exercise 41.6: True or false: chains, cycles, and boundaries

(a) True, because $\partial^2 = 0$.

(b) False. A once-around cycle of a circle cannot bound a 2-chain in that circle, since the circle has no 2-simplexes.

(c) True. One orientation of each n-simplex is a free abelian basis for $C_n(X)$.

(d) True. $B_n(X)$ is a subgroup of the free abelian group $C_n(X)$, so the subgroup theorem for free abelian groups applies.

(e) True for the same reason, with $Z_n(X) \leq C_n(X)$.

(f) True. $H_n = Z_n/B_n$ is a quotient of an abelian group.

(g) False. The boundary of a 3-simplex is a signed sum of four 2-simplexes, not a single 2-simplex.

(h) True. The boundary of a 2-simplex is a signed sum of edges, hence a 1-chain.

(i) True. A 3-cycle has boundary zero, and zero is a valid 2-chain.

(j) True. Quotienting a group by itself gives the one-element group.

Exercise 41.7: Oriented simplexes in arbitrary dimension

(a) An n-simplex is the convex hull of $n+1$ affinely independent vertices $P_0, \ldots, P_n$. An orientation is a choice of ordering modulo even permutations; an odd permutation reverses the orientation (for $n = 0$ there is just its single vertex). **(b)** Its oriented boundary is

$$\partial_n[P_0 \cdots P_n] = \sum_{i=0}^{n} (-1)^i [P_0 \cdots \widehat{P_i} \cdots P_n],$$

where a hat means omission. For example the formula gives $\partial[P_0P_1] = P_1 - P_0$. **(c)** A codimension-one face is obtained by omitting one vertex, with the induced sign $(-1)^i$ in this boundary formula. More generally a k-face is spanned by any $k+1$ of the vertices. In the textbook's oriented-face terminology, choose its orientation so that it appears with positive sign when it is the specified boundary summand.

Exercise 41.8: Chain, cycle, and boundary groups in general

Use the same definitions with an arbitrary integer subscript n: $C_n(X)$ is the free abelian group on a choice of orientation for every n-simplex; opposite orientations have opposite signs. Define

$\partial_n : C_n \to C_{n-1}$ on each basis simplex by the alternating face formula in Exercise 41.7, and extend by integer linearity. Put $C_n = 0$ for $n < 0$ and $\partial_0 = 0$. Then

$$Z_n(X) = \ker \partial_n, \qquad B_n(X) = \operatorname{im} \partial_{n+1}.$$

Thus a cycle is a chain whose boundary is zero, and a boundary is the boundary of a chain one dimension higher. This gives a uniform answer regardless of how large the dimension is.

Exercise 41.9: Why the boundary squared is zero

By linearity it suffices to start with one oriented n-simplex $[P_0 \cdots P_n]$. Every codimension-two face omits two vertices P_i, P_j with $i < j$. There are exactly two ways for this face to occur in the double boundary. Omitting P_i first moves P_j into position $j-1$, giving sign $(-1)^i(-1)^{j-1}$. Omitting P_j first leaves P_i in position i, giving sign $(-1)^j(-1)^i$. These signs are opposites, so the two copies cancel. Every summand is paired this way, proving $\partial_{n-1}\partial_n = 0$ for $n \geq 2$. For $n = 1$ the conclusion follows from $\partial_0 = 0$.

Exercise 41.10: Coboundaries in the solid tetrahedron

Fix increasing vertex order as the positive orientation. The coefficient of a simplex τ in $\delta\sigma$ is the coefficient of σ in $\partial\tau$; this retains the orientation signs in the question. **(a)** Since $\partial[ij] = P_j - P_i$, we have

$$\delta^{(0)}P_1 = -[12] - [13] - [14], \qquad \delta^{(0)}P_4 = [14] + [24] + [34].$$

(b) The oriented edge $P_3P_2 = -[23]$ occurs with coefficient -1 in the boundaries of $[123]$ and $[234]$, and not in the other two faces. Thus

$$\delta^{(1)}(P_3P_2) = -[123] - [234].$$

(c) Since $P_3P_2P_4 = -[234]$ and $[234]$ has coefficient $+1$ in $\partial[1234]$,

$$\delta^{(2)}(P_3P_2P_4) = -[1234].$$

Each negative sign describes the opposite orientation, not an omitted geometric face.

Exercise 41.11: Defining the coboundary and checking its square

(a) For the finite complexes used here identify $C^{(n)}(X)$ with integer coefficient vectors on the oriented n-simplexes. If D_{n+1} is the boundary matrix, define $\delta^{(n)}$ to have matrix D_{n+1}^{T}. Equivalently, sum the oriented cofaces with their incidence signs and extend linearly. **(b)** Since $D_{n+1}D_{n+2} = 0$, transposing gives

$$D_{n+2}^{\mathsf{T}} D_{n+1}^{\mathsf{T}} = 0.$$

This is precisely $\delta^{(n+1)}\delta^{(n)} = 0$. *Scope qualification.* For an arbitrary infinite complex, cochains should be defined as $\operatorname{Hom}(C_n(X), \mathbb{Z})$, allowing arbitrary coefficient functions; they need not equal the finite-support chain group. Define $(\delta\alpha)(c) = \alpha(\partial c)$ there. The same square-zero proof then works without a finiteness assumption. The textbook's identification of chains and cochains is valid for its finite examples.

Exercise 41.12: Cocycles and coboundaries

Define

$$Z^{(n)}(X) = \ker \delta^{(n)}, \qquad B^{(n)}(X) = \operatorname{im} \delta^{(n-1)}.$$

For $n = 0$ set $B^{(0)} = 0$. Kernels and images of homomorphisms are subgroups. If $b = \delta^{(n-1)}c$, then $\delta^{(n)}b = 0$ because $\delta^2 = 0$. Therefore every coboundary is a cocycle and $B^{(n)}(X) \leq Z^{(n)}(X)$, making the next quotient definition possible.

Exercise 41.13: Cohomology of the tetrahedron surface

Define $H^{(n)}(X) = Z^{(n)}(X)/B^{(n)}(X)$. For the surface S there are four vertices, six edges, four faces, and no 3-simplex. In dimension zero $\delta f = 0$ says that f takes the same value at both ends of every edge, so f is constant. Hence $H^{(0)}(S) \cong \mathbb{Z}$. For dimension one, a cocycle has edge values u_{ij} satisfying $u_{jk} - u_{ik} + u_{ij} = 0$ on every triangular face. Put $t_1 = 0$, $t_j = u_{1j}$; the face $[1ij]$ gives $u_{ij} = t_j - t_i$. Thus every 1-cocycle is the coboundary of a vertex function and $H^{(1)}(S) = 0$. In dimension two every cochain is a cocycle, since there are no 3-simplexes. Write e_{ijk} for the cochain dual to a face. Coboundaries of edges give, among others, the relations

$$e_{123} + e_{124} = 0, \quad -e_{123} + e_{134} = 0, \quad e_{123} + e_{234} = 0.$$

The other edge relations follow from these. Thus all four face classes are signed multiples of one unrestricted generator; there is no torsion relation. Consequently

$$H^{(2)}(S) \cong \mathbb{Z}, \qquad H^{(n)}(S) = 0 \ (n > 2).$$

These are cohomology computations from the coboundary maps, rather than an assumption that cohomology always equals homology.

42 Computations of Homology Groups

All homology groups have integer coefficients. Each answer includes H_0: connected spaces contribute one copy of $\mathbb{Z}$ there, whereas disjoint unions contribute one copy per component. All spaces in the exercises are at most two-dimensional, so groups above dimension two vanish.

Exercise 42.1: A figure eight

The two circles meet at exactly one point, so the space is connected and $H_0 \cong \mathbb{Z}$. Triangulate each circle with the shared point as a vertex. A 1-cycle can go independently around either circle, giving two independent cycle generators. There are no 2-simplexes and hence no nonzero boundaries in dimension one. Thus $H_1 \cong \mathbb{Z}^2$, while $H_n = 0$ for $n \geq 2$. This is the wedge of two circles, not the disjoint union of two circles.

Exercise 42.2: Two tangent 2-spheres

The touching point connects the two spheres, so $H_0 \cong \mathbb{Z}$. Each sphere has an independent fundamental 2-cycle, and no 3-chains impose a relation between them. Thus $H_2 \cong \mathbb{Z}^2$. A 1-cycle on either sphere bounds, and passing through the common point does not introduce a new circular passage, so $H_1 = 0$. All H_n for $n \geq 3$ vanish. Algebraically this is the wedge of two 2-spheres, whose positive-dimensional homology is the direct sum of that of the two pieces.

Exercise 42.3: A sphere and a disjoint annulus

There are two components, giving $H_0 \cong \mathbb{Z}^2$. Chains split uniquely between the disjoint components, so each positive-dimensional homology group is the direct sum of those for the pieces. The annulus retracts onto a circle and contributes $\mathbb{Z}$ in dimension one; the sphere contributes $\mathbb{Z}$ in dimension two. Therefore

$$H_1 \cong \mathbb{Z}, \qquad H_2 \cong \mathbb{Z}, \qquad H_n = 0 \ (n \geq 3).$$

Disjointness affects H_0 even though the higher-dimensional contributions can simply be added.

Exercise 42.4: An annulus attached along an equator

Move each point of the annulus radially toward its inner boundary, keeping that boundary and the entire sphere fixed. This deformation stays within the space and retracts the attached annulus onto its inner circle, which already lies on the sphere. Hence the whole space has the homology of the sphere:

$$H_0 \cong \mathbb{Z}, \quad H_1 = 0, \quad H_2 \cong \mathbb{Z}, \quad H_n = 0 \ (n \geq 3).$$

There is no additional dimension-one generator: the annulus's circular generator is the sphere's equator and bounds a hemisphere in the sphere.

Exercise 42.5: A circle touching a sphere

At the one touching point the two components become connected, giving $H_0 \cong \mathbb{Z}$. The circle supplies one 1-cycle that cannot bound in the union, while the sphere supplies one fundamental 2-cycle. Thus

$$H_1 \cong \mathbb{Z}, \qquad H_2 \cong \mathbb{Z}, \qquad H_n = 0 \ (n \geq 3).$$

This is a wedge $S^1 \vee S^2$; the independent positive-dimensional cycles lie in different pieces.

Exercise 42.6: A sphere with one handle

A sphere with one handle is a torus. Cut along a meridian and a longitude to obtain the square model with opposite edges identified. These two loops generate H_1, and the boundary of the square is $a + b - a - b = 0$, so it imposes no nonzero relation in homology. The coherently oriented whole surface is one independent 2-cycle. Consequently

$$H_0 \cong \mathbb{Z}, \quad H_1 \cong \mathbb{Z}^2, \quad H_2 \cong \mathbb{Z}, \quad H_n = 0 \ (n \geq 3).$$

Exercise 42.7: True or false: homology invariance

(a) True: homeomorphism invariance is one of the stated basic properties.

(b) False. A point and a closed interval both have $H_0 = \mathbb{Z}$ and all higher groups zero, but are not homeomorphic (they even have different cardinalities).

(c) False. S^n and E^n have different top-dimensional homology for $n > 0$; for $n = 0$, S^0 has two points and E^0 one.

(d) False. The connected circle has $H_1 \cong \mathbb{Z}$.

(e) True. A contractible space has the homology of a point, so its positive-dimensional groups vanish.

(f) False. For $n > 0$, $H_n(S^n) \cong \mathbb{Z}$.

(g) True. E^n is contractible.

(h) False. A torus has $H_1 \cong \mathbb{Z}^2$, but $H_1(S^2) = 0$.

(i) False. A torus has nonzero H_1, H_2, while the disk E^2 is contractible.

(j) True. A sphere with one handle is a standard geometric model of the torus.

Exercise 42.8: Two disjoint torus surfaces

A torus has $H_0 = \mathbb{Z}$, $H_1 = \mathbb{Z}^2$, and $H_2 = \mathbb{Z}$. For a disjoint union the chain complex, and thus its homology, is the direct sum of the two chain complexes. Therefore

$$H_0 \cong \mathbb{Z}^2, \qquad H_1 \cong \mathbb{Z}^4, \qquad H_2 \cong \mathbb{Z}^2, \quad H_n = 0 \ (n \geq 3).$$

There are two independent 2-cycles, because the two oriented torus surfaces are separate.

Exercise 42.9: Two tori stacked along a common circle

The stacked inner-tube surfaces meet along one circle, essential in each torus. The union is connected. Label their common circle b, and their separate meridian cycles a_1, a_2. Cutting and reducing cycles as for a torus leaves three independent 1-generators a_1, a_2, b: the two copies of b are identified. Each whole torus still supplies a separate 2-cycle. No extra 2-cycle arises, because a nonzero multiple of their common circle does not bound a chain on either torus. Hence

$$H_0 \cong \mathbb{Z}, \qquad H_1 \cong \mathbb{Z}^3, \qquad H_2 \cong \mathbb{Z}^2, \quad H_n = 0 \ (n \geq 3).$$

For an algebraic check, the circle-to-pieces map on H_1 sends 1 to $(b, -b)$ in $\mathbb{Z}^2 \oplus \mathbb{Z}^2$. It is primitive and injective, so identifying it removes exactly one rank from dimension one and adds none in dimension two. This uses the intended stacked configuration, whose surfaces touch along a circle.

Exercise 42.10: A balloon wearing an inner tube

The intersection circle is essential in the torus but bounds a hemisphere in the sphere. Thus gluing the sphere to the torus kills the torus's corresponding 1-generator b, leaving its other generator a. There are two independent 2-cycles: the whole sphere and the whole torus. There is no additional one, because the common circle maps nontrivially into the torus's H_1 (its map into the sphere's H_1 is zero). Therefore

$$H_0 \cong \mathbb{Z}, \qquad H_1 \cong \mathbb{Z}, \qquad H_2 \cong \mathbb{Z}^2, \quad H_n = 0 \ (n \geq 3).$$

The gluing calculation can be checked by the map $\mathbb{Z} \to \mathbb{Z}^2 \oplus 0$ sending 1 to the primitive common-circle class: it is injective, and its cokernel is $\mathbb{Z}$.

Exercise 42.11: A sphere with two handles

Cutting along two meridian–longitude pairs gives four 1-generators a_1, b_1, a_2, b_2. The boundary of the remaining polygon traverses each generator once in each direction, so its homological boundary is zero and imposes no relation on these four generators. The sum of the coherently oriented triangles of the closed surface gives one 2-generator, and any 2-cycle must use equal coefficients across adjacent triangles. Hence

$$H_0 \cong \mathbb{Z}, \quad H_1 \cong \mathbb{Z}^4, \quad H_2 \cong \mathbb{Z}, \quad H_n = 0 \ (n \geq 3).$$

This surface is connected; it is different from a disjoint pair of tori.

Exercise 42.12: A sphere with n handles

Each handle supplies a meridian and longitude. Cutting along all $2n$ loops gives one polygon whose attaching boundary is

$$a_1 b_1 a_1^{-1} b_1^{-1} \cdots a_n b_n a_n^{-1} b_n^{-1}.$$

In the abelian chain calculation this boundary is zero, since each edge occurs once with each sign. After the standard elimination of interior triangulation edges there is one vertex, $2n$ cycle generators, and one coherently oriented surface generator, with zero boundary maps. Thus for $n \geq 0$,

$$H_0 \cong \mathbb{Z}, \quad H_1 \cong \mathbb{Z}^{2n}, \quad H_2 \cong \mathbb{Z}, \quad H_k = 0 \ (k \geq 3).$$

For $n = 0$ this recovers the sphere; for $n = 1$ it recovers the torus. The surface is orientable, so its top-dimensional generator survives over $\mathbb{Z}$.

43 More Homology Computations and Applications

Exercise 43.1: Two triangulations of a square

In the first triangulation there are four corner vertices, four boundary edges and one diagonal, and two triangles. Thus

$$\chi(X) = V - E + F = 4 - 5 + 2 = 1.$$

In the second, the crossing of the diagonals is a vertex, and the horizontal line also introduces a vertex at each midpoint of the left and right sides. Thus there are $4 + 1 + 2 = 7$ vertices. There are six boundary edges and six interior edges (four half-diagonals and two horizontal half-segments), making twelve edges, and six triangles. Hence

$$\chi(X) = 7 - 12 + 6 = 1.$$

Counting intersections and subdivided sides correctly is essential; the two answers agree despite different simplex counts.

Exercise 43.2: Euler characteristic and Betti numbers

The theorem says $\chi = b_0 - b_1 + b_2$ for these two-dimensional spaces, where b_i is the free rank of H_i; torsion contributes no rank. **(a)** The annulus triangulation has five inner and five outer vertices, ten boundary edges and ten other edges, and ten triangles. Thus $\chi = 10 - 20 + 10 = 0$. Its homology has ranks $(1, 1, 0)$, giving $1 - 1 + 0 = 0$. **(b)** In the torus's 3×3 square grid with each small square cut diagonally, opposite sides are identified. This leaves $V = 9$, $E = 27$ (nine horizontal, nine vertical, nine diagonal), and $F = 18$. Thus $\chi = 9 - 27 + 18 = 0$. Its ranks $(1, 2, 1)$ give $1 - 2 + 1 = 0$. **(c)** The same grid with the Klein-bottle edge identifications again gives $V = 9, E = 27, F = 18$, hence $\chi = 0$. Its groups are $H_0 = \mathbb{Z}$, $H_1 = \mathbb{Z} \oplus \mathbb{Z}_2$, $H_2 = 0$, so its ranks $(1, 1, 0)$ give $1 - 1 + 0 = 0$. The $\mathbb{Z}_2$ factor does not change the Euler characteristic.

Exercise 43.3: Fixed points on a square and on two disks

For a square the answer is yes: it is homeomorphic to the closed disk E^2, and conjugating a continuous self-map by such a homeomorphism reduces the question to Brouwer's fixed-point theorem. A fixed point of the conjugate map pulls back to one of the original map. For two disjoint disks the answer is no. Choose matching coordinates on the disks and define a map that sends every point of the first to the corresponding point of the second, and conversely. It is continuous on each component, hence on their disjoint union, but has no fixed point because every point changes components. The connected disk hypothesis cannot simply be dropped.

Exercise 43.4: A sphere touching a Klein bottle

The union is connected and is the wedge of a sphere and a Klein bottle at one point. Its positive-dimensional homology is the direct sum of that of the two pieces: cycles on either piece have no new identification except at a vertex. The Klein bottle contributes $\mathbb{Z} \oplus \mathbb{Z}_2$ in dimension one and nothing in dimension two; the sphere contributes $\mathbb{Z}$ in dimension two. Thus

$$H_0 \cong \mathbb{Z}, \quad H_1 \cong \mathbb{Z} \oplus \mathbb{Z}_2, \quad H_2 \cong \mathbb{Z}, \quad H_n = 0 \ (n \geq 3).$$

Exercise 43.5: Two disjoint Klein bottles

A disjoint union splits its chain complex into the two component chain complexes. Taking direct sums of $H_0 = \mathbb{Z}$, $H_1 = \mathbb{Z} \oplus \mathbb{Z}_2$, $H_2 = 0$ for a Klein bottle gives

$$H_0 \cong \mathbb{Z}^2, \qquad H_1 \cong \mathbb{Z}^2 \oplus \mathbb{Z}_2^2, \qquad H_n = 0 \ (n \geq 2).$$

There are two torsion generators, one from each component, and two H_0 generators because the components are disjoint.

Exercise 43.6: True or false: general homology properties

(a) False for ordinary homology: a nonempty contractible space has $H_0 \cong \mathbb{Z}$. Its positive-dimensional groups vanish.

(b) True. A continuous map induces homomorphisms in each dimension; on simplicial chains the construction sends cycles to cycles and boundaries to boundaries.

(c) True. Each is a quotient of abelian groups.

(d) False. A Klein bottle has a $\mathbb{Z}_2$ torsion summand in H_1, which cannot occur in a free abelian group.

(e) True. H_0 is free abelian on the connected components (for a finite triangulation, one copy of $\mathbb{Z}$ per component).

(f) True. If there are no $(n+1)$-simplexes, $B_n = 0$ and $H_n = Z_n \leq C_n$. A subgroup of a free abelian group is free abelian.

(g) True by the definition of the boundary map.

(h) True because applying one more boundary gives zero.

(i) True because $\partial^2 = 0$.

(j) False. A group with torsion, such as H_1 of the Klein bottle, cannot be a subgroup of the torsion-free group C_1. In general homology classes are cosets, not distinguished chains.

Exercise 43.7: Euler characteristic of a surface with handles

For the sphere with n handles, Exercise 42.12 gives ranks $b_0 = 1$, $b_1 = 2n$, $b_2 = 1$, and all others zero. The Euler–Poincaré formula therefore gives

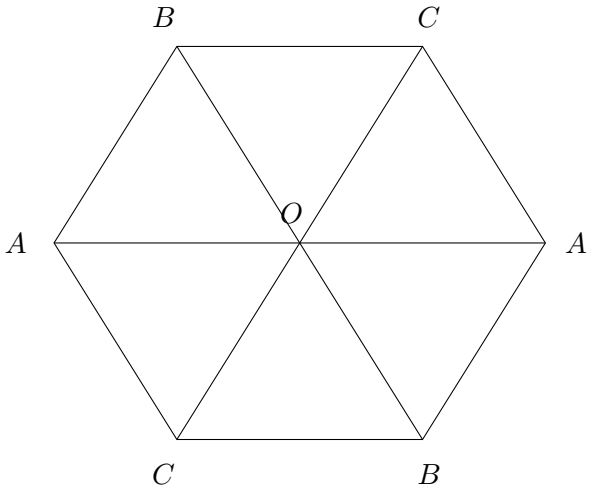
$$\chi = 1 - 2n + 1 = 2 - 2n.$$

This equals 2 for the sphere, 0 for the torus, and decreases by two for every added handle. A polygon cell model also gives the same count: one vertex, $2n$ edges, and one face.

Exercise 43.8: Triangulating and computing the projective plane

Build a genuine triangulation from the disk model.

Replace the circular boundary by a hexagon, keeping the antipodal identification. The paired boundary vertices have labels A, B, C, A, B, C in order; join all six to an interior vertex O.



Before identifying the boundary, barycentrically subdivide each of the six triangles: place a midpoint on each edge, a center in each triangle, and draw the six small triangles corresponding to vertex–edge–triangle flags. Now identify opposite boundary vertices and edge pieces. The subdivision prevents different triangles from having the same vertex set after gluing, so this gives an actual simplicial complex rather than a polygon with folded edges. It has 19 vertices, 54 edges, and 36 triangular faces, with Euler characteristic 1.

Compute the cycles and relations.

Eliminate interior edges using boundaries of the small triangles, just as in the torus calculation. All 1-cycles reduce to multiples of one loop a along the identified semicircles. Traversing the disk boundary goes around a twice in the same direction, so the remaining face boundary is $2a$. Equivalently, the reduced chain calculation has one vertex, one edge a, and one face F, with

$$\partial_1 a = 0, \qquad \partial_2 F = 2a.$$

The elimination consists of choosing and removing interior tree edges and cancelling adjacent face–edge pairs, so it leaves homology unchanged. Thus no nonzero multiple of F is a 2-cycle, while the dimension-one relation is exactly $2a = 0$. The space is connected. Therefore

$$H_0(\mathbb{RP}^2) \cong \mathbb{Z}, \quad H_1(\mathbb{RP}^2) \cong \mathbb{Z}_2, \quad H_n(\mathbb{RP}^2) = 0 \ (n \geq 2).$$

The factor 2 comes from the reversed edge gluing, not from a count of two independent loops.

Exercise 43.9: A sphere with q cross caps

For $q \geq 1$ the polygon model has one vertex, loops $a_1, \ldots, a_q$, and one face whose boundary word is $a_1^2 \cdots a_q^2$. Triangulate the polygon and eliminate interior edges as in Exercise 43.8. The resulting

chain maps are

$$\partial_1 = 0, \qquad \partial_2(1) = 2(a_1 + \cdots + a_q).$$

This map $\mathbb{Z} \to \mathbb{Z}^q$ is injective, so $H_2 = 0$. The vector $u = a_1 + \cdots + a_q$, together with $a_2, \ldots, a_q$, is an integer basis: solve $a_1 = u - a_2 - \cdots - a_q$. In this basis the single relation is $2u = 0$. Hence

$$H_0 \cong \mathbb{Z}, \qquad H_1 \cong \mathbb{Z}^{q-1} \oplus \mathbb{Z}_2, \qquad H_n = 0 \ (n \geq 2).$$

For $q = 1$ this recovers the projective plane; for $q = 2$ it recovers the Klein bottle. With no cross caps, $q = 0$, the space is the sphere instead, with $H_2 = \mathbb{Z}$ and $H_1 = 0$.

Exercise 43.10: Induced maps of torus coverings

In the figure, varying ϕ gives the meridian a, while varying θ gives the longitude b. Use $([a], [b])$ as the ordered basis of $H_1 \cong \mathbb{Z}^2$ and the oriented whole torus $[T]$ as the generator of $H_2 \cong \mathbb{Z}$. In every case $H_0 \cong \mathbb{Z}$ is sent by the identity, since a point maps to a point in a connected space.

(a) For $(\theta, \phi) \mapsto (2\theta, \phi)$, the meridian wraps once and the longitude twice: $[a] \mapsto [a]$, $[b] \mapsto 2[b]$. Thus the H_1 matrix is $\mathrm{diag}(1, 2)$. The whole oriented surface wraps twice, so $[T] \mapsto 2[T]$.

(b) For $(\theta, \phi) \mapsto (\theta, 2\phi)$, $[a] \mapsto 2[a]$, $[b] \mapsto [b]$. The H_1 matrix is $\mathrm{diag}(2, 1)$ and $[T] \mapsto 2[T]$.

(c) For doubling both angles, $[a] \mapsto 2[a]$, $[b] \mapsto 2[b]$. The H_1 matrix is $2I$, and the surface wraps twice in each direction, giving $[T] \mapsto 4[T]$.

The dimension-two multipliers are the determinants of the dimension-one matrices; the positive signs agree with preservation of orientation in each angular direction.

Exercise 43.11: Maps from the torus onto its longitude

The target circle b has $H_0(b) = \mathbb{Z}$, $H_1(b) = \mathbb{Z}[b]$, and $H_2(b) = 0$. Each map sends the H_0 point generator to the point generator, so its H_0 map is the identity. **(a)** For $f(\theta, \phi) = (\theta, 0)$, a meridian becomes a point and the longitude goes around once. Hence $f_*[a] = 0$, $f_*[b] = [b]$, or $(r, s) \mapsto s$ on H_1. **(b)** For $f(\theta, \phi) = (2\theta, 0)$, the same meridian collapses and the longitude goes around twice. Thus $(r, s) \mapsto 2s$ on H_1. In both cases the H_2 map is zero because its target group is zero. Geometrically no two-dimensional surface class can survive a map whose image is a circle.

Exercise 43.12: The same maps viewed as torus self-maps

Now the target has $H_1 = \mathbb{Z}[a] \oplus \mathbb{Z}[b]$ and $H_2 = \mathbb{Z}[T]$, but the maps still factor through the circle b. For part **(a)**, $[a] \mapsto 0$ and $[b] \mapsto [b]$, so the H_1 matrix is $\begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$. For part **(b)**, $[a] \mapsto 0$ and $[b] \mapsto 2[b]$, so it is $\begin{pmatrix} 0 & 0 \\ 0 & 2 \end{pmatrix}$. Both maps induce the identity on H_0. Both induce zero on H_2, even though the target H_2 is nonzero: factorization through b gives a factor through $H_2(b) = 0$. The determinant-zero H_1 matrices provide a consistency check.

Exercise 43.13: Projecting the Klein bottle onto b

The Klein bottle has $H_0 = \mathbb{Z}$, $H_1 = \mathbb{Z}_2[a] \oplus \mathbb{Z}[b]$, and $H_2 = 0$ in the conventions of Fig. 43.2. The projection to the vertical circle b collapses each horizontal a loop to a point and carries b onto itself once. Therefore the induced maps are

$$H_0 :\ m \mapsto m, \qquad H_1 :\ (\overline{r}, s) \mapsto s, \qquad H_2 :\ 0 \longrightarrow 0.$$

The dimension-one formula is well defined on the order-2 class of a, since that class maps to zero. It also follows algebraically that any homomorphism from its $\mathbb{Z}_2$ summand to the torsion-free group $\mathbb{Z}$ must vanish.

44 Homological Algebra

An exact sequence has image equal to the next kernel at every term. We write 0 for the trivial additive group and use ordinary homology with integer coefficients. A relative group $H_k(X, Y)$ is computed from chains modulo the chains in Y; when Y is nonempty, comparison with X/Y uses reduced homology in dimension zero.

Exercise 44.1: An exact sequence with two nonzero terms

Exactness at A says $\ker f = \operatorname{im}(0 \to A) = 0$, so f is injective. Exactness at B says $\operatorname{im} f = \ker(B \to 0) = B$, so f is surjective. An injective and surjective homomorphism is an isomorphism. Thus the given map itself supplies $A \cong B$.

Exercise 44.2: A short exact sequence

(a) Exactness at C gives $\operatorname{im} j = \ker(C \to 0) = C$, so j is onto. **(b)** Exactness at A gives $\ker i = \operatorname{im}(0 \to A) = 0$, so i is injective and is an isomorphism from A to its image in B. **(c)** Exactness at B gives $\ker j = i[A]$. The first isomorphism theorem, together with surjectivity of j, gives $B/i[A] \cong C$, via $b + i[A] \mapsto j(b)$. Equality of the kernel with $i[A]$ proves that this coset formula is well defined and injective.

Exercise 44.3: Three equivalent conditions in an exact sequence

For the exact sequence $A \xrightarrow{i} B \xrightarrow{j} C \xrightarrow{k} D$, exactness says $\operatorname{im} i = \ker j$ and $\operatorname{im} j = \ker k$. Hence

$$\begin{aligned} i \text{ onto} &\iff \operatorname{im} i = B \iff \ker j = B \\ &\iff j = 0 \iff \operatorname{im} j = 0 \iff \ker k = 0 \iff k \text{ injective.} \end{aligned}$$

The middle condition $j = 0$ means it maps every element of B to the zero element of C. This proves all three requested conditions equivalent.

Exercise 44.4: When the middle map is an isomorphism

For $A \xrightarrow{g} B \xrightarrow{h} C \xrightarrow{i} D \xrightarrow{j} E \xrightarrow{k} F$, exactness yields

$$\operatorname{im} h = \ker i, \quad \operatorname{im} i = \ker j, \quad \operatorname{im} g = \ker h, \quad \operatorname{im} j = \ker k.$$

Thus $h = 0$ is equivalent to injectivity of i, and $j = 0$ is equivalent to surjectivity of i. Together these say exactly that i is an isomorphism $C \to D$, proving (a) $\iff$ (b). Also g is onto exactly when $\ker h = B$, namely when $h = 0$; and k is injective exactly when $\operatorname{im} j = 0$, namely when $j = 0$. This proves (a) $\iff$ (c), completing the equivalence.

Exercise 44.5: The induced-quotient-map principle

Exercise 14.39 gives the criterion: a homomorphism $f : G \to G'$ induces $G/H \to G'/H'$ by $gH \mapsto f(g)H'$ if $f(H) \subseteq H'$. For Theorem 44.4, the chain-map equation first shows $f_k(Z_k(A)) \subseteq Z_k(A')$, since

$\partial' f_k z = f_{k-1}\partial z = 0$. It also shows $f_k(B_k(A)) \subseteq B_k(A')$, since $f_k(\partial c) = \partial'(f_{k+1}c)$. Apply the quotient-map criterion to Z_k/B_k to obtain the induced homology map. For Theorem 44.7, $\partial_k(A'_k) \subseteq A'_{k-1}$ by the subcomplex assumption. Apply the same criterion with $G = A_k$, $H = A'_k$, $G' = A_{k-1}$, $H' = A'_{k-1}$, and $f = \partial_k$. This gives the well-defined quotient boundary $c + A'_k \mapsto \partial c + A'_{k-1}$. Its square is zero because $\partial^2 = 0$. Thus both theorems use precisely the same representative-independence mechanism as Exercise 14.39.

Exercise 44.6: Relative homology of a torus modulo a

After eliminating interior triangulation edges, use the torus chain model with one vertex, edges a, b, and one face F. Its boundaries are $\partial_1 = 0$ and $\partial_2 F = a + b - a - b = 0$. The circle subcomplex a contains the vertex and edge a. In the relative chain complex these are set to zero, leaving

$$C_2(X, a) = \mathbb{Z}F, \qquad C_1(X, a) = \mathbb{Z}b, \qquad C_0(X, a) = 0,$$

with zero boundary maps. Hence

$$H_2(X, a) \cong \mathbb{Z}, \quad H_1(X, a) \cong \mathbb{Z}, \quad H_0(X, a) = 0,$$

and higher groups vanish. Collapsing a gives a pinched torus with the positive-dimensional homology of $S^2 \vee S^1$. There is a necessary qualification to the printed parenthesis: relative homology corresponds to *reduced* homology of the quotient. The quotient's ordinary H_0 is $\mathbb{Z}$, whereas the relative H_0 computed here is zero.

Exercise 44.7: The exact sequence of the torus pair

Use $H_1(X) = \mathbb{Z}[a] \oplus \mathbb{Z}[b]$ and choose matching orientations for the whole-surface generators. The nonzero part of the exact sequence is

$$0 \longrightarrow \mathbb{Z} \xrightarrow{1} \mathbb{Z} \xrightarrow{0} \mathbb{Z} \xrightarrow{u} \mathbb{Z}^2$$
$$\xrightarrow{v} \mathbb{Z} \xrightarrow{0} \mathbb{Z} \xrightarrow{1} \mathbb{Z} \longrightarrow 0,$$

where the groups in order are $H_2(X)$, $H_2(X, a)$, $H_1(a)$, $H_1(X)$, $H_1(X, a)$, $H_0(a)$, $H_0(X)$. Here $u(t) = (t, 0)$ and $v(r, s) = s$. The first map carries the whole torus to the same relative 2-cycle. The first connecting map is zero because that 2-cycle has zero boundary. The second connecting map is zero because b already has zero boundary. To check exactness directly: the identity map has full image, equal to the next zero map's kernel; the zero image equals $\ker u = 0$; $\operatorname{im} u = \mathbb{Z} \times 0 = \ker v$; v is onto, matching the next zero map's full kernel; and the final identity has zero kernel. This checks every position.

Exercise 44.8: Relative homology of a Klein bottle modulo a

For the Klein-bottle polygon use one vertex, edges a, b, and a face F with $\partial_2 F = 2a$ and $\partial_1 = 0$. Set the vertex and the edge a to zero in the relative complex. Then $\partial_2 F$ also becomes zero, leaving one free generator in dimensions two and one and no dimension-zero generator. Thus

$$H_2(X, a) \cong \mathbb{Z}, \qquad H_1(X, a) \cong \mathbb{Z}, \qquad H_0(X, a) = 0,$$

with higher groups zero. This explains why collapsing the torsion loop creates a top-dimensional cycle: its original boundary was twice the collapsed edge. As in Exercise 44.6, the connected pinched quotient has ordinary $H_0 = \mathbb{Z}$; its *reduced* homology agrees with these relative groups.

Exercise 44.9: The exact sequence of the Klein-bottle pair

Here $H_2(X) = 0$, $H_1(X) = \mathbb{Z}_2[a] \oplus \mathbb{Z}[b]$, and $H_0(X) = \mathbb{Z}$. The relevant sequence is

$$0 \longrightarrow \mathbb{Z} \xrightarrow{2} \mathbb{Z} \xrightarrow{u} \mathbb{Z}_2 \oplus \mathbb{Z}$$
$$\xrightarrow{v} \mathbb{Z} \xrightarrow{0} \mathbb{Z} \xrightarrow{1} \mathbb{Z} \longrightarrow 0.$$

The first $\mathbb{Z}$ is $H_2(X, a)$; the next is $H_1(a)$; the later three $\mathbb{Z}$ groups are $H_1(X, a)$, $H_0(a)$, $H_0(X)$. The connecting map sends F to its boundary $2a$, so it is multiplication by 2. Also $u(t) = (\overline{t}, 0)$ and $v(\overline{r}, s) = s$. Check exactness: multiplication by 2 is injective, $\ker u = 2\mathbb{Z}$, $\operatorname{im} u = \mathbb{Z}_2 \oplus 0 = \ker v$, v is onto, the following map is zero, and the final identity has zero kernel. These identities verify every term, including the torsion contribution.

Exercise 44.10: An annulus relative to both boundary circles

Let the boundary circles have the same angular orientation, and let Y be their union. A coherently oriented sum of the annulus's triangles has boundary equal to the difference of those two circles, which becomes zero modulo Y. It gives one relative 2-cycle F. A radial path e from one boundary component to the other has boundary in Y, so it gives one relative 1-cycle. Cutting the annulus along this path produces a disk; eliminating interior triangle edges shows that F and e exhaust the independent relative cycles. In a reduced relative chain model, there is one generator in degrees two and one, no generator in degree zero, and zero boundary maps. Hence

$$H_2(X, Y) \cong \mathbb{Z}, \qquad H_1(X, Y) \cong \mathbb{Z}, \qquad H_0(X, Y) = 0,$$

and higher groups vanish. The relative 1-generator is a radial path, not the circular loop, since that loop is homologous to a boundary circle and is zero in the relative group.

Exercise 44.11: The exact sequence for the annulus pair

Use the same angular orientation on the two boundary circles. Then inclusion maps $H_1(Y) = \mathbb{Z}^2 \to H_1(X) = \mathbb{Z}$ by $s(r, t) = r + t$. The analogous H_0 map is also s. Choose the orientation of the relative fundamental 2-chain and radial 1-chain so that their connecting maps are $d(k) = (k, -k)$. The nonzero part of the sequence is

$$0 \longrightarrow \mathbb{Z} \xrightarrow{d} \mathbb{Z}^2 \xrightarrow{s} \mathbb{Z} \xrightarrow{0} \mathbb{Z} \xrightarrow{d} \mathbb{Z}^2 \xrightarrow{s} \mathbb{Z} \longrightarrow 0.$$

The groups are respectively $H_2(X, Y)$, $H_1(Y)$, $H_1(X)$, $H_1(X, Y)$, $H_0(Y)$, $H_0(X)$; $H_2(X)$ and $H_0(X, Y)$ are zero. Directly, d is injective, $\operatorname{im} d = \{(k, -k)\} = \ker s$, and s is onto. These facts check exactness at both pairs of $\mathbb{Z}^2$ and $\mathbb{Z}$ terms. At the middle relative $\mathbb{Z}$, the incoming zero map has image zero, equal to the next d's kernel; at the preceding $\mathbb{Z}$, the onto map s fills the kernel of the zero map.

Exercise 44.12: The quotient projection is a chain map

Let $j_k(c) = c + A'_k$. By the definition of the quotient boundary,

$$\overline{\partial}_k(j_k(c)) = \overline{\partial}_k(c + A'_k) = \partial_k c + A'_{k-1} = j_{k-1}(\partial_k c).$$

Thus $\overline{\partial}_k j_k = j_{k-1}\partial_k$ for every k. This proves the squares commute, so the quotient projections constitute a chain map.

Exercise 44.13: Well-definedness of the connecting map

A relative homology class is represented by $c \in A_k$ with $\partial c \in A'_{k-1}$. Since $\partial^2 c = 0$, this boundary is a cycle of A', so $[\partial c] \in H_{k-1}(A')$ is meaningful. Two choices representing the same relative homology class differ by

$$c' - c = \partial d + a', \qquad d \in A_{k+1}, \quad a' \in A'_k.$$

Indeed equality modulo relative boundaries is precisely this condition, including the possible change of coset representative in A_k/A'_k. Taking boundaries gives $\partial c' - \partial c = \partial a'$, a boundary *inside* A'. Thus $[\partial c'] = [\partial c]$ there, proving independence of both choices. Finally $\partial(c_1 + c_2) = \partial c_1 + \partial c_2$, so the rule $[c] \mapsto [\partial c]$ preserves addition. It is therefore a well-defined homomorphism.

Exercise 44.14: Consecutive homology-sequence maps compose to zero

There are three kinds of consecutive composites to check. First $j_* i_* [z'] = [z' + A'_k] = 0$, since a chain in the subcomplex is zero in the quotient. Second $\partial_* j_* [z] = [\partial z] = 0$, because a homology representative z in A is a cycle. Third $i_* \partial_* [c] = [\partial c] = 0$ in $H_{k-1}(A)$, because ∂c is a boundary in A (even though it need not be a boundary in A'). Thus every consecutive composite is zero. All groups are abelian and all maps are homomorphisms, so the long sequence is indeed a chain complex.

Exercise 44.15: Exactness of the long homology sequence

Use $[z]$ for an absolute homology class, and use $[c]_{\mathrm{rel}}$ for a relative class, with $\partial c \in A'$.

(a) If a class comes from $H_k(A')$, its chain is zero in A_k/A'_k. Thus $\operatorname{im} i_* \subseteq \ker j_*$.

(b) If $j_*[z] = 0$, the cycle $z \in A_k$ satisfies $z = \partial d + a'$ with $d \in A_{k+1}$ and $a' \in A'_k$. Taking boundaries gives $\partial a' = 0$. Hence $[z] = [a'] = i_*[a']$ in $H_k(A)$, so $\ker j_* \subseteq \operatorname{im} i_*$.

(c) For an absolute cycle z, $\partial_* j_* [z] = [\partial z] = 0$. Thus $\operatorname{im} j_* \subseteq \ker \partial_*$.

(d) If $\partial_* [c]_{\mathrm{rel}} = 0$, there is $a' \in A'_k$ with $\partial c = \partial a'$. Then $z = c - a'$ is a cycle in A, and $j_*[z] = [c]_{\mathrm{rel}}$ since a' vanishes in the quotient. Therefore $\ker \partial_* \subseteq \operatorname{im} j_*$.

(e) The class $\partial_* [c]_{\mathrm{rel}} = [\partial c]$ maps to zero in $H_{k-1}(A)$ because it bounds c there. Thus $\operatorname{im} \partial_* \subseteq \ker i_*$.

(f) If $[a'] \in H_{k-1}(A')$ maps to zero in $H_{k-1}(A)$, then $a' = \partial c$ for some $c \in A_k$. Its boundary is in A', so it represents a relative cycle, and $\partial_* [c]_{\mathrm{rel}} = [a']$. Thus $\ker i_* \subseteq \operatorname{im} \partial_*$.

The six inclusions pair into equality at all three types of term, in every dimension. This proves exactness of the entire sequence.

Exercise 44.16: Chain homotopies induce equal homology maps

Let $z \in Z_k(A)$ represent a homology class. Because $\partial z = 0$, the given homotopy identity reduces to

$$f_k(z) - g_k(z) = \partial'_{k+1}(D_k z) + D_{k-1}(0) = \partial'_{k+1}(D_k z).$$

Thus the two image cycles differ by a boundary in A', so they represent the same class in $H_k(A')$. Since every homology class has a cycle representative, $f_{*k} = g_{*k}$ for every k. The chain-map hypotheses

ensure both induced maps are defined; the homotopy supplies the boundary witnessing equality on each class.

45 Unique Factorization Domains

Exercise 45.1: An integer prime

A nonzero nonunit is irreducible when every factorization has a unit factor. If $5 = ab$ in $\mathbb{Z}$, then $|a||b| = 5$. Since 5 is prime, one of $|a|, |b|$ is 1. Thus one factor is a unit, and 5 is irreducible.

Exercise 45.2: A negative prime

The units of $\mathbb{Z}$ are $1, -1$. If $-17 = ab$, then $17 = |a||b|$, so one factor has absolute value 1. Thus -17 is irreducible. Equivalently, it is an associate of the irreducible 17.

Exercise 45.3: A composite integer

The factorization $14 = 2 \cdot 7$ uses two nonunits of $\mathbb{Z}$. Therefore 14 is reducible.

Exercise 45.4: A primitive linear polynomial

Suppose $2x - 3 = gh$ in $\mathbb{Z}[x]$. Degrees add, so one factor, say g, is a nonzero integer c. It divides both 2 and -3, hence $c = \pm 1$. It is a unit. The polynomial is nonconstant, so is itself a nonunit; consequently it is irreducible.

Exercise 45.5: A nonprimitive linear polynomial

In $\mathbb{Z}[x]$, $2x - 10 = 2(x - 5)$. The constant 2 is not a unit of $\mathbb{Z}[x]$, and the nonconstant polynomial $x - 5$ is not a unit either. Therefore the polynomial is reducible.

Exercise 45.6: A linear polynomial over the rationals

Every nonzero constant in $\mathbb{Q}[x]$ is a unit. A factorization of a degree-one polynomial has a degree-zero factor, so that factor is a unit. Thus $2x - 3$ is irreducible in $\mathbb{Q}[x]$.

Exercise 45.7: The same expression in a different domain

Although $2x - 10 = 2(x - 5)$, the factor 2 is a unit in $\mathbb{Q}[x]$. Every factorization has a constant factor because the degree is 1. Therefore $2x - 10$ is irreducible in $\mathbb{Q}[x]$. Irreducibility depends on the coefficient domain.

Exercise 45.8: A linear polynomial modulo eleven

The coefficient 2 is nonzero in the field $\mathbb{Z}_{11}$. Thus $2x - 10$ has degree 1 in $\mathbb{Z}_{11}[x]$ and is irreducible by degree addition. Its constant term is congruent to 1, but this does not change its degree.

Exercise 45.9: Associates in three polynomial rings

Associates are unit multiples. In $\mathbb{Z}[x]$ the only units are ± 1, so the only associates are $2x-7$ and $-2x+7$; four different ones do not exist. In $\mathbb{Q}[x]$, four examples are $2x-7, 4x-14, 6x-21, 8x-28$. In $\mathbb{Z}_{11}[x]$ the same four unit multiples are distinct after reducing coefficients: $2x+4, 4x+8, 6x+1, 8x+5$. Distinctness follows also by comparing their leading coefficients.

Exercise 45.10: Factorization over three domains

First extract the integer content:

$$4x^2 - 4x + 8 = 2 \cdot 2\,(x^2 - x + 2).$$

The discriminant of $x^2 - x + 2$ is $1 - 8 = -7$, so it has no rational root. A quadratic over a field is irreducible exactly when it has no root. Gauss's lemma therefore makes this primitive polynomial irreducible in $\mathbb{Z}[x]$ as well. This is the factorization into irreducibles over $\mathbb{Z}$. Over $\mathbb{Q}$, 4 is a unit, and $4(x^2 - x + 2)$ is an irreducible polynomial times a unit. Modulo 11, the roots of $x^2 - x + 2$ are 5 and 7, because $5 + 7 = 1$ and $5 \cdot 7 = 2$ modulo 11. Thus

$$4x^2 - 4x + 8 = 4(x - 5)(x - 7) \quad \text{in } \mathbb{Z}_{11}[x].$$

The two linear factors are irreducible and 4 is a unit.

Exercise 45.11: All gcds of three integers

Euclidean division gives $3250 = 13(234) + 208$, $234 = 208 + 26$, and $208 = 8(26)$. Also $1690 = 65(26)$. Thus the positive gcd is 26. Since the only units of $\mathbb{Z}$ are ± 1, all gcds are 26 and -26.

Exercise 45.12: All gcds with a negative entry

We may replace -1960 by 1960 without changing divisibility. The calculations $1960 = 2(784) + 392$, $784 = 2(392)$, $448 = 392 + 56$, and $392 = 7(56)$ give positive gcd 56. All gcds are $56, -56$.

Exercise 45.13: All gcds of four integers

The divisions $2178 = 5(396) + 198$ and $396 = 2(198)$ give $\gcd(2178, 396) = 198$. Since $792 = 4(198)$ and $594 = 3(198)$, the remaining entries do not change it. The gcds are $198, -198$.

Exercise 45.14: Integer content

The positive gcd of $18, -12, 48$ is 6. Consequently $18x^2 - 12x + 48 = 6(3x^2 - 2x + 8)$. The coefficients $3, -2, 8$ have gcd 1, so the second factor is primitive. Changing the sign of both factors gives the equivalent associate choice.

Exercise 45.15: Content over a field

In $\mathbb{Q}$ every nonzero element is a unit and is a gcd of these coefficients. One valid choice is content 1 and primitive factor $18x^2 - 12x + 48$ itself. For comparison, the choice $6(3x^2 - 2x + 8)$ is also valid. Over a field every nonzero polynomial is primitive: every common nonzero divisor of its coefficients is already a unit.

Exercise 45.16: A polynomial already primitive

The coefficients $2, -3, 6$ have gcd 1, since any common divisor of 2 and 3 is a unit. Thus $2x^2 - 3x + 6 = 1(2x^2 - 3x + 6)$ is the required decomposition in $\mathbb{Z}[x]$.

Exercise 45.17: Content modulo seven

The coefficient ring is the field $\mathbb{Z}_7$. Take content 1 and primitive factor $2x^2 + 4x + 6$. Alternatively choose content 2 and write $2(x^2 + 2x + 3)$, since $2^{-1} = 4$ modulo 7. All nonzero constants are units, so these choices are equivalent.

Exercise 45.18: Definition of associates

The version using a/b is not suitable when $b = 0$. The correct definition for all $a, b \in D$ is: a and b are associates if $a = ub$ for some unit $u \in D$. This includes 0 being an associate of itself. If $b \neq 0$, the quotient criterion in the fraction field is equivalent, provided that quotient is a unit of D.

Exercise 45.19: Definition of irreducible

An irreducible is a nonzero nonunit $p \in D$ such that whenever $p = ab$ with $a, b \in D$, at least one of a, b is a unit. One cannot prohibit every factorization, because $p = 1 \cdot p$ is always available.

Exercise 45.20: Definition of prime

A prime is a nonzero nonunit p such that $p \mid ab$ implies $p \mid a$ or $p \mid b$, for every $a, b \in D$. The expression "smaller" has no intrinsic meaning in an arbitrary integral domain.

Exercise 45.21: Ten statements about factorization

(a) True. In a field there are no nonzero nonunits, so the factorization requirement is vacuous.

(b) True. The only ideals of a field are (0) and (1).

(c) True. The PID factorization theorem proves existence and uniqueness.

(d) False. $\mathbb{Q}[x, y]$ is a UFD, but (x, y) is not principal: a common divisor of x, y is a unit, which would make that ideal the whole ring.

(e) True, by Gauss's theorem applied to the UFD $\mathbb{Z}$.

(f) False. The irreducibles 2 and 3 of $\mathbb{Z}$ are not associates.

(g) False. $\mathbb{Z}$ is a PID but $(2, x)$ is not principal in $\mathbb{Z}[x]$.

(h) True, by the polynomial UFD theorem.

(i) False literally. Associates may replace p: $2 \mid 6$, but $6 = (-2)(-3)$ has no factor literally equal to 2. The correct claim is that an associate of p appears.

(j) True. Being an integral domain is part of the definition of UFD.

Exercise 45.22: Irreducibles in a polynomial UFD

There are two types. Constant irreducibles are precisely the irreducibles of D. A nonconstant $f \in D[x]$ is irreducible precisely when it is primitive and irreducible in $F[x]$. Indeed, a nonunit content would itself give a factorization; after removing content, Gauss's lemma transfers reducibility between $D[x]$ and $F[x]$. Constant nonzero elements become units in $F[x]$, so they must be treated separately.

Exercise 45.23: Irreducible after extending coefficients

Take $D = \mathbb{Z}$, $F = \mathbb{Q}$, and $g(x) = 2x + 2$. It is degree 1, hence irreducible in $\mathbb{Q}[x]$. In $\mathbb{Z}[x]$ the factorization $g = 2(x + 1)$ has two nonunit factors, so it is reducible. Its nonunit content is precisely what disappears on passing to F.

Exercise 45.24: Why zero divisors obstruct factorization

The element $(1, 0)$ is a nonzero nonunit of $\mathbb{Z} \times \mathbb{Z}$, but $(1, 0) = (1, 0)(1, 0)$, a product of two nonunits. More strongly, it cannot be a product of irreducibles. Suppose $(1, 0) = \prod_j (a_j, b_j)$. All a_j are ± 1 and at least one $b_j = 0$. That factor $(\pm 1, 0)$ is reducible because $(\pm 1, 0) = (1, 0)(\pm 1, 0)$. Hence there is no irreducible factorization of $(1, 0)$. The domain hypothesis prevents precisely this kind of idempotent behavior.

Exercise 45.25: A prime is irreducible

Let p be prime and write $p = ab$. Since $p \mid ab$, assume, after exchanging factors if necessary, that $p \mid a$. Write $a = pc$. Then $p = pcb$; cancel the nonzero p in the domain to obtain $1 = cb$. Thus b is a unit. Since a prime is by definition a nonzero nonunit, this proves irreducibility.

Exercise 45.26: An irreducible in a UFD is prime

Suppose $p \mid ab$, so $ab = pc$. If a or b is zero, p divides that factor. Otherwise factor a, b, c into irreducibles, allowing a unit factor when necessary. The concatenated factorizations of a, b and the factorization p followed by that of c represent the same nonzero element. Uniqueness forces an associate of p to occur in the factorization of a or b. It therefore divides that factor, as does p.

Exercise 45.27: Associates form an equivalence relation

Reflexivity follows from $a = 1a$. If $a = ub$ with u a unit, then $b = u^{-1}a$, proving symmetry. If additionally $b = vc$ with v a unit, then $a = (uv)c$, and uv is a unit, proving transitivity. These arguments need no cancellation and work even in a commutative ring with zero divisors.

Exercise 45.28: Products of nonzero nonunits

If a, b are nonzero nonunits, the domain condition gives $ab \neq 0$. If ab were a unit with inverse c, then $a(bc) = 1$ and $b(ac) = 1$, making both factors units, a contradiction. Thus the set is closed. It is not a group: an identity on this nonempty set would satisfy $ea = a$ for nonzero a, hence $e = 1$ by cancellation, but 1 is excluded. If the domain is a field, the set is empty and is still not a group.

Exercise 45.29: A divisor of a primitive polynomial

Write the primitive polynomial as $f = gh$ in $D[x]$. If a nonunit irreducible p divided every coefficient of g, then it would divide every coefficient of $gh = f$. That contradicts the content of f being a unit. Since every nonunit of the UFD has an irreducible divisor, the coefficients of g have no common nonunit divisor; hence g is primitive. The same argument applies to any divisor, constant or nonconstant.

Exercise 45.30: Containment in a maximal ideal

The statement needs the word *proper*: the ideal D itself is not contained in a maximal ideal. For a proper ideal I in a PID, suppose no maximal ideal contains it. Then I is not maximal, so there is a proper ideal I_1 strictly containing I. Repeating produces $I \subsetneq I_1 \subsetneq I_2 \subsetneq \cdots$. The ascending chain condition for a PID forbids this. Thus the process terminates at a maximal ideal containing I.

Exercise 45.31: Factoring a difference of cubes

We have $x^3 - y^3 = (x - y)(x^2 + xy + y^2)$. Regard these as polynomials in x over $\mathbb{Q}[y]$. The first is primitive and degree 1, so is irreducible over $\mathbb{Q}(y)$ and therefore over $\mathbb{Q}[y]$. The second is primitive and has discriminant $-3y^2$. This is not a square in $\mathbb{Q}(y)$: otherwise dividing a putative square root by y would give a rational function whose square is -3, impossible in the ordered field $\mathbb{Q}(y)$ (or by comparing coprime numerator and denominator). Thus this quadratic is irreducible over $\mathbb{Q}(y)$; Gauss's lemma gives irreducibility in $\mathbb{Q}[x, y]$.

Exercise 45.32: Three equivalent finiteness conditions

Plan. Prove a cycle of implications for ideals ordered by inclusion.

ACC implies the maximum condition.

If a nonempty collection has no maximal member, choose an ideal in it and then a strictly larger member repeatedly. This contradicts ACC.

The maximum condition implies finite generation.

For an ideal I, consider all ideals generated by finite subsets of I (including the empty generating set). A maximal such ideal J exists. If $a \in I \setminus J$, adjoining a gives a strictly larger finitely generated ideal inside I, a contradiction. Thus $J = I$.

Finite generation implies ACC.

Given $I_1 \subseteq I_2 \subseteq \cdots$, its union I is an ideal: any two elements lie together in a sufficiently late term, and multiplication by ring elements remains in that term. A finite generating set of I lies in some I_N, so $I = I_N$ and all later terms coincide with it. This is ACC.

Exercise 45.33: Descending chains and minimal ideals

As usual the minimum condition must concern *nonempty* collections; an empty collection cannot contain a minimal member. If DCC holds and a nonempty collection had no minimal member, successively choosing a smaller member would produce an infinite strictly descending chain. Conversely, apply the minimum condition to the set of terms of a descending chain. If I_N is minimal among those terms, then for $n \geq N$, $I_n \subseteq I_N$ forces $I_n = I_N$. Thus the chain stabilizes and DCC holds.

Exercise 45.34: ACC without DCC

Use $\mathbb{Z}$. Every ideal is $n\mathbb{Z}$, hence finitely generated, so ACC holds by Exercise 45.32. But

$$2\mathbb{Z} \supsetneq 4\mathbb{Z} \supsetneq 8\mathbb{Z} \supsetneq \cdots$$

is an infinite strictly descending chain: 2^k belongs to the kth ideal but not the next. Therefore DCC fails.

46 Euclidean Domains

Exercise 46.1: The square of absolute value

Yes. This map has nonnegative integer values. Ordinary integer division gives $a = bq + r$ with $0 \le r < |b|$, hence $r = 0$ or $r^2 < b^2$. For nonzero integers a, b, $a^2 \le a^2b^2$ because $b^2 \ge 1$. Both norm axioms hold.

Exercise 46.2: Degree over the integers

No. Divide $a = 1$ by $b = 2$ in $\mathbb{Z}[x]$. A nonzero remainder would have to satisfy $\deg r < \deg 2 = 0$, impossible. A zero remainder would say $1 = 2q$ in $\mathbb{Z}[x]$, also impossible. Degree division works over a field, not over arbitrary integer coefficients.

Exercise 46.3: Leading-coefficient size

No. Take dividend x and divisor 1. Every nonzero polynomial has norm at least 1, so a remainder with norm smaller than that of 1 must be zero; here zero remainder is possible, so this example alone is no obstruction. Instead divide x by x^2; zero remainder is impossible by degrees, while any nonzero remainder has leading-coefficient absolute value at least $1 = v(x^2)$. Thus the required division is impossible.

Exercise 46.4: A rational-valued proposal

No under the stated definition, since a Euclidean norm must take values in the nonnegative *integers*; $v(1/2) = 1/4$ does not. The monotonicity axiom also fails: $v(1) = 1 > v(1 \cdot 1/2) = 1/4$.

Exercise 46.5: A constant norm on a field

Yes. For $b \neq 0$ in $\mathbb{Q}$, set $q = a/b$ and $r = 0$. This gives the division axiom without needing a smaller norm. For nonzero a, b, $v(a) = 50 = v(ab)$ gives monotonicity.

Exercise 46.6: An explicit integer Bezout identity

Back substitution along the Euclidean algorithm gives

$$\begin{aligned} 22471 &= 6(3266) + 2875, & 3266 &= 2875 + 391, \\ 2875 &= 7(391) + 138, & 391 &= 2(138) + 115, \\ 138 &= 115 + 23, & 115 &= 5(23). \end{aligned}$$

Hence $23 = 138 - 115 = 3(138) - 391 = 3(2875) - 22(391) = 25(2875) - 22(3266) = 25(22471) - 172(3266)$. Direct multiplication verifies $561775 - 561752 = 23$, so $\lambda = 25, \mu = -172$.

Exercise 46.7: A larger integer gcd

Successive divisions give

$$\begin{aligned} 49349 &= 3(15555) + 2684, & 15555 &= 5(2684) + 2135,\\ 2684 &= 2135 + 549, & 2135 &= 3(549) + 488,\\ 549 &= 488 + 61, & 488 &= 8(61). \end{aligned}$$

The last nonzero remainder is 61, so a gcd is 61 (and -61 is an associate choice).

Exercise 46.8: Back substitution for the larger gcd

Using the remainders just found,

$$\begin{aligned} 61 &= 549 - 488 = 4(549) - 2135\\ &= 4(2684) - 5(2135) = 29(2684) - 5(15555)\\ &= 29(49349) - 92(15555). \end{aligned}$$

Thus $\lambda = 29, \mu = -92$. Checking the final equality gives $1431121 - 1431060 = 61$.

Exercise 46.9: A polynomial gcd

Let f, g denote the displayed degree-ten and degree-six polynomials. Exact polynomial division gives

$$\begin{aligned} f &= (x^4 - 2x)g + r,\\ r &= -x^4 - 3x^3 - 2x^2 - 5x + 3\\ &= (-x - 3)(x^3 + 2x - 1),\\ g &= (x^3 - 3x^2 + x - 2)(x^3 + 2x - 1). \end{aligned}$$

The cofactors $-x - 3$ and $x^3 - 3x^2 + x - 2$ are relatively prime: substituting $x = -3$ into the latter gives $-59 \neq 0$, so its linear possible common factor does not divide it. Consequently a monic gcd is $x^3 + 2x - 1$. The first division and the displayed factorizations can be checked by multiplying out; they also show directly why no further factor is common.

Exercise 46.10: A gcd of many elements

Compute $d_2 = \gcd(a_1, a_2)$, then $d_3 = \gcd(d_2, a_3)$, and continue until d_n. At each step, an element divides d_k exactly when it divides all $a_1, \ldots, a_k$. This follows by induction from the defining common-divisor property of a gcd. Therefore d_n is a gcd of the entire list. Ignore zero entries, or use $\gcd(a, 0) = a$ up to associates.

Exercise 46.11: Four integers by successive gcds

First $2178 = 5(396) + 198$ and $396 = 2(198)$, so the first gcd is 198. Since $792 = 4(198)$, it remains 198 after the third entry. Finally $726 = 3(198) + 132$, $198 = 132 + 66$, and $132 = 2(66)$. Thus the positive gcd is 66.

Exercise 46.12: The ideal of even constant terms

(a) $\mathbb{Z}[x]$ is a UFD because $\mathbb{Z}$ is a UFD and Gauss's theorem preserves this property on adjoining x.

(b) Let I consist of polynomials with even constant term. Differences preserve this property; multiplying such a polynomial by any integer polynomial multiplies its constant term by an integer, so it remains even. Thus I is an ideal, equal to $(2, x)$.

(c) If $I = (h)$, then h divides both 2 and x. Dividing 2 forces h to be a constant integer divisor of 2; dividing x then forces $h = \pm 1$. This would put 1 in I, contrary to its odd constant term. Hence I is not principal.

(d) A Euclidean domain is a PID. Since this ring is not a PID, it is not Euclidean.

Exercise 46.13: Ten statements about Euclidean domains

(a) True: use a nonzero element of smallest norm in an ideal and divide by it.

(b) False: for example $\mathbb{Z}[(1 + \sqrt{-19})/2]$ is a PID but is not Euclidean. The implication has no converse in general.

(c) True, since every PID is a UFD.

(d) False: $\mathbb{Z}[x]$ is a UFD but not a PID, hence not Euclidean.

(e) True: in the field $\mathbb{Q}$, $1/2$ divides every rational, and every common divisor divides it.

(f) True: successive nonnegative integer norms strictly decrease until the remainder is zero.

(g) True: apply $v(a) \leq v(ab)$ with $a = 1$.

(h) False: in $\mathbb{Z}$, $v(-1) = v(1)$ for $v(n) = |n|$, although $-1 \neq 1$.

(i) True: equality $v(a) = v(1)$ would make a a unit by the norm characterization, proved in Exercise 46.16.

(j) True, with the degree norm and polynomial division over F.

Exercise 46.14: A norm measures, rather than changes, arithmetic

No. Addition, multiplication, units, divisibility, ideals, and gcds are defined by the domain operations. A chosen norm helps find quotients and remainders, but does not alter these operations. Different permissible norms can change which division choices the algorithm makes; the final gcds still differ only by units.

Exercise 46.15: Equal norms for associates

If $a = bu$ with u a unit, monotonicity gives $v(b) \leq v(bu) = v(a)$. Also $b = au^{-1}$, so $v(a) \leq v(b)$. Thus $v(a) = v(b)$. The norms are defined here for nonzero associates.

Exercise 46.16: Strict growth detects a nonunit factor

If b is a unit, a and ab are associates, so their norms are equal. Conversely, suppose $v(a) = v(ab)$. Divide a by ab: write $a = abq + r$, with $r = 0$ or $v(r) < v(ab) = v(a)$. But $r = a(1 - bq)$, so if $r \neq 0$, monotonicity gives $v(a) \leq v(r)$, a contradiction. Thus $r = 0$, and cancellation of $a \neq 0$ yields $1 = bq$. Therefore b is a unit. As $v(a) \leq v(ab)$ always, nonunit b is exactly the strict-inequality case.

Exercise 46.17: The nonunits need not form an ideal

The proposed set is the nonunits together with zero, by the unit characterization. In $\mathbb{Z}$ with absolute-value norm, both 2 and 3 lie in it, but $3 - 2 = 1$ does not. It is therefore not an additive subgroup and is not an ideal in general.

Exercise 46.18: Every field is Euclidean

Define $v(a) = 0$ for every nonzero field element. Divide a by nonzero b by taking $q = ab^{-1}$ and $r = 0$. This satisfies the division axiom, and $v(a) = v(ab) = 0$ satisfies monotonicity. Thus every field is Euclidean.

Exercise 46.19: Changing the scale of a norm

(a) Since $v(a) \geq v(1)$, the assumption $s + v(1) \geq 0$ ensures $\eta(a) \geq 0$. Adding the same integer to two norms preserves both their strict and weak inequalities, so it preserves the two axioms.

(b) Multiplication by a positive integer t preserves nonnegative integer values and all inequalities, so $\lambda(a) = tv(a)$ is a Euclidean norm.

(c) Set $\mu(a) = 1 + 100(v(a) - v(1))$. This is a positive integer norm by (a),(b); it equals 1 on units. A nonunit has integer norm strictly larger than $v(1)$, so $\mu(a) \geq 101 > 100$.

Exercise 46.20: Least common multiples

Let $I = (a) \cap (b)$, the ideal of common multiples. It is nonzero because $ab \in I$ and $ab \neq 0$. A Euclidean domain is a PID, so $I = (c)$ for some nonzero c. Membership of c in both ideals says $a \mid c, b \mid c$. Every common multiple belongs to (c), hence is divisible by c. These are exactly the required lcm properties. Multiplying c by a unit gives another lcm.

Exercise 46.21: Generating the additive integers

The subgroup generated by r, s is $\{ur + vs : u, v \in \mathbb{Z}\}$. Bezout's theorem identifies this with $d\mathbb{Z}$, where $d = \gcd(r, s) > 0$: every combination is divisible by d, and a Bezout combination equals d. It is all $\mathbb{Z}$ precisely when $d = 1$.

Exercise 46.22: A solvable congruence when the coefficient is coprime

If $\gcd(a, n) = 1$, choose integers u, v with $ua + vn = 1$. Multiplying by b gives $a(ub) + n(vb) = b$. Thus $x = ub$ satisfies $ax \equiv b \pmod{n}$. In $\mathbb{Z}_n$ this says the nonzero residue of a is invertible when it is coprime to n.

Exercise 46.23: The general congruence criterion

Put $d = \gcd(a, n) > 0$. If $ax \equiv b \pmod{n}$, then $b = ax - kn$ for some integer k, and $d \mid b$. Conversely, if $b = dc$, multiply a Bezout identity $ua + vn = d$ by c. It gives $a(uc) \equiv b \pmod{n}$. In the residue ring $\mathbb{Z}_n$, the principal ideal generated by $\bar{a}$ consists exactly of residues represented by multiples of d; solvability asks whether $\bar{b}$ lies in this ideal.

Exercise 46.24: Constructing a solution

Run the extended Euclidean algorithm to obtain $d = ua + vn$. If $d \nmid b$, no solution exists. Otherwise $x_0 = u(b/d)$ is a solution; all solutions are $x = x_0 + k(n/d)$. For $22x \equiv 18 \pmod{42}$, $d = 2$ and $2 = 2(22) - 42$. Thus $x_0 = 2(18/2) = 18$. Check: $22(18) - 18 = 378 = 9(42)$. All integer solutions are $x = 18 + 21k$; modulo 42 these are 18 and 39.

47 Gaussian Integers and Multiplicative Norms

Exercise 47.1: Splitting five

In $\mathbb{Z}[i]$, $5 = (2+i)(2-i)$. Both factors have norm 5, a rational prime, so a factorization of either would force one factor to have norm 1 and hence be a unit. Both are irreducible.

Exercise 47.2: Seven stays irreducible

The norm of 7 is 49. If $7 = \alpha\beta$ with both factors nonunits, their norms must both be 7. But $a^2 + b^2 = 7$ has no integer solutions: the possible squares at most 7 are 0,1,4 and none sum to 7. Thus such a factorization is impossible, and 7 is irreducible in $\mathbb{Z}[i]$.

Exercise 47.3: Factoring a norm twenty-five element

A direct multiplication gives $4 + 3i = i(2-i)^2$. Indeed $(2-i)^2 = 3 - 4i$ and multiplying by i gives $4 + 3i$. The factor i is a unit; $N(2-i) = 5$ is prime, so $2 - i$ is irreducible.

Exercise 47.4: Factoring a norm eighty-five element

We have $6 - 7i = (2+i)(1-4i)$. Expanding gives $2 - 8i + i - 4i^2 = 6 - 7i$. The norms are 5 and 17, respectively, both rational primes, so both factors are irreducible.

Exercise 47.5: Failure of unique factorization

Put $s = \sqrt{-5}$. Then $6 = 2 \cdot 3 = (1+s)(1-s)$. The norm $N(a+bs) = a^2 + 5b^2$ is multiplicative, and its units have norm 1. There are no elements of norm 2 or 3. Thus 2 (norm 4), 3 (norm 9), and $1 \pm s$ (norm 6) cannot split into two nonunits: their proper possible norm factors would require norm 2 or 3. All four displayed factors are irreducible. Their norms $4, 9, 6, 6$ show neither 2 nor 3 is associate to either $1+s$ or $1-s$. Hence the two factorizations are different up to associates.

Exercise 47.6: One Gaussian division

Compute

$$\frac{7+2i}{3-4i} = \frac{(7+2i)(3+4i)}{25} = \frac{13+34i}{25}.$$

Round both coordinates to nearest integers to choose $\sigma = 1 + i$. Then $(3-4i)(1+i) = 7 - i$, so $\rho = 3i$. We have $7 + 2i = (3-4i)(1+i) + 3i$, and $N(3i) = 9 < 25 = N(3-4i)$.

Exercise 47.7: A Gaussian gcd

The divisions

$$8 + 6i = i(5 - 15i) + (-7 + i), \qquad 5 - 15i = (-1 + 2i)(-7 + i)$$

have first remainder norm 50, smaller than 250. The next remainder is zero. Thus the last nonzero remainder $-7 + i$ is a gcd. Its associates, for example $7 - i$, are equally valid.

Exercise 47.8: Ten statements about norms

(a) True. The Gaussian integer norm makes the ring Euclidean, hence a PID.

(b) True. Round real and imaginary coordinates in Gaussian division.

(c) True: $n = n + 0i$.

(d) False: $1/2$ has no integer real coordinate.

(e) True, since the nonnegative remainder norms decrease.

(f) True. Norm divisibility restricts possible factor norms, as in Exercises 47.1–47.4.

(g) True. From $N(u)N(u^{-1}) = N(1) = 1$, integer norms give $|N(u)| = 1$.

(h) False. Degrees add rather than multiply; also the norm of 1 would be 0 rather than 1.

(i) True. For nonzero polynomials, degree addition gives $2^{\deg(fg)} = 2^{\deg f}2^{\deg g}$; declaring $N(0) = 0$ handles zero products and $N(1) = 1$.

(j) True. It is a subring of $\mathbb{C}$, hence a domain; Exercise 47.5 gives nonunique factorization.

Exercise 47.9: Smallest nonunit norm

Suppose $\pi = ab$ with both factors nonunits. Since $\pi \neq 0$ and the ring is a domain, $a, b \neq 0$. By the hypothesis relating norm and units, both $|N(a)|, |N(b)|$ are integers greater than 1. Minimality gives $|N(a)| \geq |N(\pi)|$. Multiplicativity then gives $|N(\pi)| = |N(a)||N(b)| > |N(a)| \geq |N(\pi)|$, a contradiction. Thus every factorization has a unit factor, proving irreducibility.

Exercise 47.10: Which integer primes split

(a) $2 = -i(1 + i)^2$, and $1 + i$ is irreducible because its norm is the prime 2.

(b) If $p \equiv 1 \pmod 4$, the sum-of-two-squares theorem gives $p = a^2 + b^2 = (a + bi)(a - bi)$ with both factors nonunits. If an odd prime $p \equiv 3 \pmod 4$ were reducible, both nonunit factor norms would equal p. But an integer square is 0 or 1 modulo 4, so $a^2 + b^2$ cannot be 3 modulo 4. This excludes norm p and proves irreducibility. Every odd prime is 1 or 3 modulo 4, completing both directions.

Exercise 47.11: The Gaussian norm identity

For $\alpha = a + bi$, $\beta = c + di$, multiplication gives $\alpha\beta = (ac - bd) + (ad + bc)i$. Therefore

$$\begin{aligned} N(\alpha\beta) &= (ac - bd)^2 + (ad + bc)^2 \\ &= a^2c^2 + b^2d^2 + a^2d^2 + b^2c^2 \\ &= (a^2 + b^2)(c^2 + d^2) = N(\alpha)N(\beta). \end{aligned}$$

The cross terms cancel. Since $N(\alpha) = a^2 + b^2$ is a sum of squares, it is nonnegative. Also $N(\alpha) = 0$ exactly for $a = b = 0$, and $N(1) = 1$, giving the stated norm properties.

Exercise 47.12: A multiplicative norm with square root minus five

Write $\alpha = a + b\sqrt{-5}$ and $\beta = c + d\sqrt{-5}$. Their product is $(ac - 5bd) + (ad + bc)\sqrt{-5}$. Expanding gives

$$(ac - 5bd)^2 + 5(ad + bc)^2 = (a^2 + 5b^2)(c^2 + 5d^2).$$

Thus $N(\alpha\beta) = N(\alpha)N(\beta)$. The values are nonnegative integers, $N(0) = 0$, and $N(1) = 1$.

Exercise 47.13: Existence of irreducible factorizations from a norm

Use strong induction on the positive integer $m = |N(a)| > 1$. If a is irreducible we are done. Otherwise $a = bc$ with b, c nonunits, necessarily nonzero. Their absolute norms are integers greater than 1 and multiply to m, so both are strictly smaller than m. By induction both factors are products of irreducibles; concatenate these factorizations. The smallest possible nonunit norm supplies the base case, since it cannot split as a product of two smaller norms greater than 1.

Exercise 47.14: Another Gaussian gcd

Gaussian division yields

$$16 + 7i = (1 + i)(10 - 5i) + (1 + 2i), \qquad 10 - 5i = (-5i)(1 + 2i).$$

The first remainder has norm 5, below $N(10 - 5i) = 125$, and the second remainder is zero. Thus $1 + 2i$ is a gcd, up to multiplication by $1, -1, i, -i$.

Exercise 47.15: Finite Gaussian residue rings

(a) Division by nonzero α gives each residue class a representative r with $N(r) < N(\alpha)$. There are finitely many integer lattice points in this bounded disk, so there are finitely many residue classes. More exactly the quotient has $N(\alpha)$ elements: multiplication by $a+bi$ has integer lattice matrix $\left[\begin{smallmatrix} a & -b \\ b & a \end{smallmatrix}\right]$ of determinant $a^2 + b^2$.

(b) An irreducible in this PID is prime, so its principal ideal is maximal. Its quotient is therefore a field.

(c) For (3), there are $N(3) = 9$ residues and characteristic 3. For $(1 + i)$, there are 2 residues and characteristic 2 because $2 = -i(1 + i)^2$. For $(1 + 2i)$, there are 5 residues and characteristic 5, since $5 = (1+2i)(1-2i)$. These are fields because the indicated divisors are Gaussian irreducibles (7-like reasoning for 3; prime norm for the other two).

Exercise 47.16: Imaginary quadratic norms

Let $s = \sqrt{-n}$, with n positive and square free.

(a) Multiplication and expansion give $N((a + bs)(c + ds)) = (ac - nbd)^2 + n(ad + bc)^2 = (a^2 + nb^2)(c^2 + nd^2)$. Values are integers, with $N(1) = 1$ and $N(\alpha) = 0$ only for $\alpha = 0$.

(b) If α is a unit, multiplicativity forces $N(\alpha) = 1$. Conversely, if $N(\alpha) = 1$, then $\alpha\bar{\alpha} = 1$, and $\bar{\alpha} = a - bs$ is in the same ring, giving an inverse.

(c) Every nonzero nonunit has integer norm greater than 1. Apply the strong-induction proof of Exercise 47.13 to obtain a factorization into irreducibles. This proves existence, not uniqueness.

Exercise 47.17: Real quadratic norms

For the intended real quadratic case assume the square-free integer $n > 1$. The representation $a+b\sqrt{n}$ is then unique because $\sqrt{n}$ is irrational. Define $N(a + b\sqrt{n}) = a^2 - nb^2$, allowing negative values.

(a) Direct expansion gives $(ac+nbd)^2-n(ad+bc)^2 = (a^2-nb^2)(c^2-nd^2)$, so the norm is multiplicative. A nonzero element has nonzero norm by irrationality of $\sqrt{n}$.

(b) The appropriate statement is $|N(\alpha)| = 1$ if and only if α is a unit. A unit has integer norm dividing 1. Conversely, for norm ±1, $\alpha^{-1} = \bar{\alpha}/N(\alpha)$ lies in the ring. Requiring $N = 1$ literally would be false, e.g. $1 + \sqrt{2}$ has norm -1 and inverse $\sqrt{2} - 1$.

(c) The absolute norm is a positive integer, greater than 1 on nonzero nonunits. The induction in Exercise 47.13 therefore proves existence of an irreducible factorization.

The original reference includes square-free $n = 1$; here the expression $a^2 - b^2$ is not well defined on $a + b\sqrt{1}$ (1 has representations $(1, 0)$ and $(0, 1)$). Thus that degenerate case must be excluded for this norm construction.

Exercise 47.18: Euclidean division with square root minus two

For nonzero β, write $\alpha/\beta = u + v\sqrt{-2}$ with $u, v \in \mathbb{Q}$. Choose integers m, k with $|u - m| \le 1/2$ and $|v - k| \le 1/2$. Put $q = m + k\sqrt{-2}$ and $r = \alpha - \beta q$. Then

$$N(r) = N(\beta)\big((u - m)^2 + 2(v - k)^2\big) \le \tfrac{3}{4}N(\beta) < N(\beta).$$

If $r = 0$ the required alternative holds as well. All nonzero norms are positive integers and multiplication multiplies them by an integer at least 1, so the second Euclidean axiom holds too. Hence $\mathbb{Z}[\sqrt{-2}]$ is Euclidean.

48 Automorphisms of Fields

Exercise 48.1: Conjugates of a square root

The minimal polynomial over $\mathbb{Q}$ is $x^2 - 2$, irreducible by Eisenstein at 2. Its two zeros are $\sqrt{2}, -\sqrt{2}$, so these are exactly the conjugates.

Exercise 48.2: Changing the base field

Since $\sqrt{2} \in \mathbb{R}$, its minimal polynomial over $\mathbb{R}$ is $x - \sqrt{2}$. Its only conjugate over $\mathbb{R}$ is itself. Conjugacy always depends on the base field.

Exercise 48.3: A translated square root

Subtracting 3 gives $(x-3)^2 - 2 = x^2 - 6x + 7$. This is irreducible over $\mathbb{Q}$ because a rational zero would make $\sqrt{2}$ rational. Its roots, hence all conjugates, are $3 + \sqrt{2}$ and $3 - \sqrt{2}$.

Exercise 48.4: A difference of square roots

Independent changes of signs of $\sqrt{2}$ and $\sqrt{3}$ give $\pm\sqrt{2} \pm \sqrt{3}$. They are four distinct conjugates of $\sqrt{2} - \sqrt{3}$. Squaring twice gives $x^4 - 10x^2 + 1 = 0$. To see the degree is 4, put $\alpha = \sqrt{2} - \sqrt{3}$. Since $\alpha^{-1} = -(\sqrt{2} + \sqrt{3})$, both square roots lie in $\mathbb{Q}(\alpha)$. Thus this field is the degree-four field $\mathbb{Q}(\sqrt{2}, \sqrt{3})$, and the list is complete.

Exercise 48.5: A mixed real and imaginary generator

The four sign choices $\pm\sqrt{2} \pm i$ are the images under the four automorphisms of $\mathbb{Q}(\sqrt{2}, i)$. They are distinct. The field has degree 4 because i is not in the real field $\mathbb{Q}(\sqrt{2})$. Thus the minimal polynomial is $x^4 - 2x^2 + 9$, obtained by multiplying the four linear factors, and the four listed numbers are all conjugates.

Exercise 48.6: Conjugacy over the real field

Over $\mathbb{R}$, the minimal polynomial is $(x - \sqrt{2})^2 + 1$, which has no real zero. The conjugates are therefore $\sqrt{2} + i$ and $\sqrt{2} - i$. The base field fixes $\sqrt{2}$, so its sign cannot change.

Exercise 48.7: A nested square root over the rationals

Let $\alpha = \sqrt{1 + \sqrt{2}}$. Its square gives $\sqrt{2} = \alpha^2 - 1$, so $\mathbb{Q}(\sqrt{2}) \subseteq \mathbb{Q}(\alpha)$. The element $1 + \sqrt{2}$ is not a square in $\mathbb{Q}(\sqrt{2})$: its norm to $\mathbb{Q}$ is -1, whereas the norm of a square is a rational square. Hence $[\mathbb{Q}(\alpha) : \mathbb{Q}] = 4$. Eliminating the inner root gives $x^4 - 2x^2 - 1$. Its four zeros are $\pm\sqrt{1 + \sqrt{2}}$ and $\pm\sqrt{1 - \sqrt{2}}$, with the latter square root taken in $\mathbb{C}$. These are all conjugates.

Exercise 48.8: A nested square root over a quadratic base

Over $\mathbb{Q}(\sqrt{2})$, the polynomial is $x^2 - (1 + \sqrt{2})$. The norm argument in Exercise 48.7 shows it is irreducible. Thus only the two outer signs can change: the conjugates are $\sqrt{1 + \sqrt{2}}$ and $-\sqrt{1 + \sqrt{2}}$.

Exercise 48.9: A radical fixed by a sign change

The automorphism τ_2 changes only the sign of $\sqrt{2}$. Therefore $\tau_2(\sqrt{3}) = \sqrt{3}$.

Exercise 48.10: Applying a sign change to a sum

An automorphism preserves addition and fixes the other generating radicals. Thus $\tau_2(\sqrt{2} + \sqrt{5}) = -\sqrt{2} + \sqrt{5}$.

Exercise 48.11: Composing two sign changes

Apply the rightmost map first. The map τ_2 sends $\sqrt{2} + 3\sqrt{5}$ to $-\sqrt{2} + 3\sqrt{5}$; τ_3 fixes both radicals appearing there. Hence $(\tau_3\tau_2)(\sqrt{2} + 3\sqrt{5}) = -\sqrt{2} + 3\sqrt{5}$.

Exercise 48.12: An automorphism of a quotient

The map $\tau_5\tau_3$ flips $\sqrt{5}$ and $\sqrt{3}$ and fixes $\sqrt{2}$. Because field automorphisms preserve division,

$$(\tau_5\tau_3)\left(\frac{\sqrt{2} - 3\sqrt{5}}{2\sqrt{3} - \sqrt{2}}\right) = \frac{\sqrt{2} + 3\sqrt{5}}{-2\sqrt{3} - \sqrt{2}}.$$

The denominator remains nonzero because an injective map cannot send a nonzero element to zero.

Exercise 48.13: Simplifying a power of an automorphism

Since τ_5^2 is the identity and $\sqrt{45} = 3\sqrt{5}$, the remaining $\tau_3\tau_2$ flips only the $\sqrt{2}$ term. The result is $-\sqrt{2} + \sqrt{45}$.

Exercise 48.14: A nested composition

First $(\tau_2\tau_5)(\sqrt{30}) = \sqrt{30}$ because two radical signs change. Applying τ_5 to the bracket gives $\sqrt{2} - \sqrt{3} - \sqrt{30}$. Finally τ_3 changes the signs of the last two radical terms. The answer is $\sqrt{2} + \sqrt{3} + \sqrt{30}$.

Exercise 48.15: Fixed fields in a biquadratic extension

Write an element uniquely as $a + b\sqrt{2} + c\sqrt{3} + d\sqrt{6}$ with rational coefficients. In the source notation σ_1 flips $\sqrt{2}$, σ_2 flips $\sqrt{3}$, and σ_3 flips both.

(a) Fixing both σ_1 and σ_3 forces $b = c = d = 0$, so the fixed field is $\mathbb{Q}$.

(b) Fixing σ_3 forces $b = c = 0$, leaving $\mathbb{Q}(\sqrt{6})$.

(c) Fixing both σ_2 and σ_3 again forces $b = c = d = 0$, giving $\mathbb{Q}$.

Exercise 48.16: The fixed field of one radical flip

Use the basis $1, \sqrt{2}, \sqrt{3}, \sqrt{5}, \sqrt{6}, \sqrt{10}, \sqrt{15}, \sqrt{30}$ of E. The map τ_3 negates exactly those basis terms containing $\sqrt{3}$. Their coefficients must vanish in a fixed element. The surviving terms span $\mathbb{Q}(\sqrt{2}, \sqrt{5})$, which is the fixed field.

Exercise 48.17: The square of a radical flip

The map τ_3^2 is the identity, so every element is fixed. Its fixed field is $E = \mathbb{Q}(\sqrt{2}, \sqrt{3}, \sqrt{5})$.

Exercise 48.18: Fixing two separate sign changes

An element fixed by both τ_2 and τ_3 has no basis term involving $\sqrt{2}$ or $\sqrt{3}$. Only $1, \sqrt{5}$ survive, so the fixed field is $\mathbb{Q}(\sqrt{5})$.

Exercise 48.19: Fixing two simultaneous sign changes

The map $\tau_5\tau_2$ negates $\sqrt{2}, \sqrt{5}$ and fixes $\sqrt{3}$. A basis monomial survives exactly when it contains both or neither of $\sqrt{2}, \sqrt{5}$. The surviving basis is $1, \sqrt{3}, \sqrt{10}, \sqrt{30}$, so the fixed field is $\mathbb{Q}(\sqrt{3}, \sqrt{10})$.

Exercise 48.20: Fixing all three simultaneous signs

A basis monomial is fixed by $\tau_5\tau_3\tau_2$ exactly when it contains an even number of the three radicals. Thus the fixed field has basis $1, \sqrt{6}, \sqrt{10}, \sqrt{15}$ and equals $\mathbb{Q}(\sqrt{6}, \sqrt{10})$, since $\sqrt{6}\sqrt{10} = 2\sqrt{15}$.

Exercise 48.21: Fixing each of the three generators

Every nonconstant basis monomial contains at least one generating radical and is negated by its sign flip. Requiring invariance under all three flips leaves just rational multiples of 1. The fixed field is $\mathbb{Q}$.

Exercise 48.22: The eight-element sign-change group

(a) Each τ_j is nonidentity, since it negates a nonzero radical, and its square fixes all generators. It therefore has order 2.

(b) Put $a = \tau_2, b = \tau_3, c = \tau_5$. They commute because independent sign changes commute on each

generator. The elements are $1, a, b, c, ab, ac, bc, abc$, with table

	1	a	b	c	ab	ac	bc	abc
1	1	a	b	c	ab	ac	bc	abc
a	a	1	ab	ac	b	c	abc	bc
b	b	ab	1	bc	a	abc	c	ac
c	c	ac	bc	1	abc	a	b	ab
ab	ab	b	a	abc	1	bc	ac	c
ac	ac	c	abc	a	bc	1	ab	b
bc	bc	abc	c	b	ac	ab	1	a
abc	abc	bc	ac	ab	c	b	a	1

The rule is to cancel repeated letters, since $a^2 = b^2 = c^2 = 1$.

(c) Any rational automorphism sends each $\sqrt{j}$ to $\pm\sqrt{j}$ because it preserves its minimal polynomial. There are at most $2^3 = 8$ choices. The eight maps above realize every choice, so they form the full group, isomorphic to $\mathbb{Z}_2^3$.

Exercise 48.23: Correcting the conjugate criterion

Replace "the same polynomial" by "the same irreducible polynomial over F." Equivalently, α, β have the same monic minimal polynomial over F. Merely being roots of one polynomial is insufficient: 0 and 1 are both roots of $x(x-1)$, but have different minimal polynomials.

Exercise 48.24: The kernel criterion for conjugacy

The statement is correct. The evaluation kernel at an algebraic element is the principal ideal generated by its monic minimal polynomial. Equality of the kernels is equivalent to equality of their monic generators, hence to conjugacy over F.

Exercise 48.25: A shifted generator gives the same automorphism

(a) The other conjugate of $\alpha = 3 + \sqrt{2}$ is $\beta = 3 - \sqrt{2}$, since both satisfy $x^2 - 6x + 7$.

(b) Both conjugation automorphisms fix $\mathbb{Q}$ and send $\sqrt{2} = \alpha - 3$ to $-\sqrt{2} = \beta - 3$. Since $\sqrt{2}$ generates the field, they are exactly the same map; only the chosen generator used to describe it differs.

Exercise 48.26: Frobenius on four elements

Write the field as $\mathbb{Z}_2(\alpha)$ with $\alpha^2 + \alpha + 1 = 0$, so $\alpha^2 = \alpha + 1$. Squaring gives

$$0 \mapsto 0, \quad 1 \mapsto 1, \quad \alpha \mapsto \alpha + 1, \quad \alpha + 1 \mapsto \alpha.$$

Thus precisely 0 and 1 are fixed; the fixed field is $\mathbb{Z}_2$.

Exercise 48.27: Frobenius on nine elements

Here $\alpha^2 = -1 = 2$ in characteristic 3. For $a, b \in \mathbb{Z}_3$, cubing gives $(a + b\alpha)^3 = a - b\alpha$. Thus 0,1,2 are fixed; the other six elements are exchanged in the pairs

$$\alpha \leftrightarrow 2\alpha, \quad 1 + \alpha \leftrightarrow 1 + 2\alpha, \quad 2 + \alpha \leftrightarrow 2 + 2\alpha.$$

The fixed field is $\mathbb{Z}_3$.

Exercise 48.28: Frobenius on an infinite field

Let $F = \mathbb{Z}_p(t)$ with t transcendental. The image of $r(t) \mapsto r(t)^p$ is $\mathbb{Z}_p(t^p)$. It does not contain t: if $(P(t)/Q(t))^p = t$ with P, Q nonzero polynomials, comparing degrees in $P(t)^p = tQ(t)^p$ gives $p \deg P = 1 + p \deg Q$, impossible modulo p. Frobenius remains injective, but need not be onto.

Exercise 48.29: Ten statements about field automorphisms

(a) False. An automorphism fixes 0 and 1 and cannot map one to the other.

(b) False. $\mathbb{Q}(0) = \mathbb{Q}$ and $\mathbb{Q}(\sqrt{2})$ are not isomorphic: any isomorphism fixes $\mathbb{Q}$, and their rational degrees differ.

(c) True, by the conjugation isomorphism theorem.

(d) True. It fixes 1, hence integers and their quotients, giving the prime field.

(e) False. The identity automorphism of a finite field fixes only finitely many elements.

(f) True: it fixes the distinct elements 0 and 1.

(g) True: it fixes the infinite prime subfield $\mathbb{Q}$.

(h) True: composition, identity, and inverse preserve field operations.

(i) True: fixed elements are closed under differences, products and inverses of nonzero elements, and include 0,1.

(j) True. Fixing the larger field E implies fixing F, and these maps form a subgroup.

Exercise 48.30: Synopsis of the existence direction

When α and β have the same minimal polynomial p, identifying both $F(\alpha)$ and $F(\beta)$ with $F[x]/(p)$ gives the unique isomorphism fixing F and sending α to β. This works because evaluation at either root has exactly the kernel (p).

Exercise 48.31: Synopsis of the necessity direction

An isomorphism fixing the coefficients in F carries the equation $p(\alpha) = 0$ to $p(\beta) = 0$, so the minimal polynomials of α and β coincide. Applying the inverse gives the reverse divisibility if one spells this out via minimal polynomials.

Exercise 48.32: Bounding the number of embeddings

An embedding fixing F must send α to a root of its degree-n minimal polynomial. There are at most n such distinct roots in an algebraic closure. Its image on α determines the embedding on every polynomial in α, hence on all $F(\alpha)$. Therefore there are at most n embeddings.

Exercise 48.33: Images of generators determine a map

Every element of $F(\alpha_1, \ldots, \alpha_n)$ is a rational expression in those generators with coefficients in F. A homomorphism fixing F evaluates that expression at $\sigma(\alpha_1), \ldots, \sigma(\alpha_n)$. Its denominator stays nonzero by injectivity. Thus two such automorphisms agreeing on the generators agree on every field element.

Exercise 48.34: Permuting the conjugates present in a field

Let p be the minimal polynomial of α. If $\beta \in E$ satisfies $p(\beta) = 0$, then $p(\sigma(\beta)) = \sigma(p(\beta)) = 0$, because σ fixes every coefficient of p. So σ carries the set of such roots into itself. Its inverse has the same property, proving the restriction is a bijection, hence a permutation.

Exercise 48.35: A generating set has the same fixed field as its group

Since $S \subseteq H$, any element fixed by H is fixed by S. Conversely, if x is fixed by every $s \in S$, it is fixed by s^{-1} too: apply s^{-1} to $s(x) = x$. It is then fixed by any finite composition of generators and their inverses. Those compositions comprise H, so $x \in E_H$. Thus $E_S = E_H$.

Exercise 48.36: Prime cyclotomic fields

(a) A zero ζ of Φ_p is a nonidentity pth root of unity. Since p is prime, it has order p, and $\zeta, \ldots, \zeta^{p-1}$ are distinct, nonidentity roots. They exhaust the $p - 1$ zeros of Φ_p.

(b) Each choice $\sigma_a(\zeta) = \zeta^a$, $1 \leq a < p$, gives a rational embedding onto the same field because a is invertible modulo p. Composition satisfies $\sigma_a \sigma_b = \sigma_{ab}$, proving the group is abelian and has order $p - 1$.

(c) Multiplication by ζ carries the usual basis $1, \ldots, \zeta^{p-2}$ to the basis $\zeta, \ldots, \zeta^{p-1}$. The automorphisms permute its basis vectors transitively. If $\sum_{j=1}^{p-1} c_j \zeta^j$ is fixed by every map, uniqueness of basis coefficients forces all c_j equal, say to c. The sum is then $c(\zeta + \cdots + \zeta^{p-1}) = -c \in \mathbb{Q}$. Conversely rationals are fixed, so the fixed field is $\mathbb{Q}$.

Exercise 48.37: Transcendental generators

Yes. Define $P(\alpha)/Q(\alpha) \mapsto P(\beta)/Q(\beta)$ for $P, Q \in F[x]$, $Q \neq 0$. Transcendence ensures neither evaluation denominator vanishes, and equality of rational expressions is exactly a polynomial identity, so the map is well defined. It preserves both operations and has inverse obtained by exchanging α, β. Thus it is an isomorphism fixing F.

Exercise 48.38: All automorphisms of a rational function field

They are precisely

$$x \longmapsto \frac{ax+b}{cx+d}, \qquad a,b,c,d \in F, \quad ad-bc \neq 0.$$

Such a substitution has inverse $x \mapsto (dx-b)/(-cx+a)$, so it is an automorphism. Multiplying all four coefficients by the same nonzero scalar gives the same map. For completeness, if an automorphism sends x to $r = P(x)/Q(x)$ in lowest terms, put $m = \max(\deg P, \deg Q)$. The standard degree computation gives $[F(x) : F(r)] = m$: $P(T) - rQ(T)$ is the minimal polynomial of x over $F(r)$. Indeed, with r replaced by an independent indeterminate u, the polynomial $P(T) - uQ(T)$ is irreducible in $F[T,u]$, since a factorization of this degree-one polynomial in u would require a nonconstant common divisor of P, Q; Gauss's lemma transfers this to $F(u)[T]$. An automorphism has $F(r) = F(x)$, so $m = 1$. The numerator and denominator are therefore linear or constant, and coprimality/nonconstancy is exactly $ad - bc \neq 0$.

Exercise 48.39: Rigidity of real-field automorphisms

(a) If $u = v^2$, then $\sigma(u) = \sigma(v)^2$. Applying the inverse shows squares map onto squares.

(b) Positive real numbers are exactly the nonzero squares. An automorphism preserves nonzero elements, so it maps positive reals to positive reals.

(c) If $a < b$, then $b - a > 0$, hence $\sigma(b) - \sigma(a) = \sigma(b-a) > 0$. Thus it preserves strict order.

(d) Every automorphism fixes $\mathbb{Q}$. If $\sigma(x) > x$, choose a rational q with $x < q < \sigma(x)$. Order preservation gives $\sigma(x) < \sigma(q) = q$, contradiction. If $\sigma(x) < x$, a rational between them gives the reverse contradiction. Hence $\sigma(x) = x$ for every real x.

49 The Isomorphism Extension Theorem

Exercise 49.1: Extending an identity on two radicals

An embedding of E is a sign choice on $\sqrt{2}, \sqrt{3}, \sqrt{5}$. Fixing $\sqrt{2}$ forces its sign to be positive; fixing $\sqrt{15} = \sqrt{3}\sqrt{5}$ forces the other two signs to agree. Thus the two triples of images are $(\sqrt{2}, \sqrt{3}, \sqrt{5})$ and $(\sqrt{2}, -\sqrt{3}, -\sqrt{5})$. Each is realized by a sign-change automorphism, and there are no others.

Exercise 49.2: Extending a prescribed negative product

Here $\sqrt{2}$ must stay fixed and the product $\sqrt{3}\sqrt{5}$ must change sign. Thus the two triples of images are $(\sqrt{2}, -\sqrt{3}, \sqrt{5})$ and $(\sqrt{2}, \sqrt{3}, -\sqrt{5})$. They both respect the prescribed map, and exhaust its possible extensions.

Exercise 49.3: Extending conjugation on a triple product

The product of the three radical signs must be negative. The four triples are

$$(-\sqrt{2}, \sqrt{3}, \sqrt{5}),\ (\sqrt{2}, -\sqrt{3}, \sqrt{5}),\ (\sqrt{2}, \sqrt{3}, -\sqrt{5}),\ (-\sqrt{2}, -\sqrt{3}, -\sqrt{5}).$$

Every triple defines a sign-change automorphism of E, and exactly the odd numbers of sign changes negate $\sqrt{30}$.

Exercise 49.4: Embeddings of a real cubic field

Put $r = \sqrt[3]{2}$ and $\omega = (-1 + i\sqrt{3})/2$. The polynomial $x^3 - 2$ is irreducible by Eisenstein and has distinct roots $r, r\omega, r\omega^2$. For each $j = 0, 1, 2$, there is one rational embedding sending r to $r\omega^j$. Its value on $a + br + cr^2$ is $a + br\omega^j + cr^2\omega^{2j}$. These are all three embeddings into $\overline{\mathbb{Q}}$.

Exercise 49.5: Adding an independent square root

Let r, ω be as in Exercise 49.4. The field $\mathbb{Q}(r)$ has degree 3 and cannot contain the quadratic field $\mathbb{Q}(\sqrt{3})$, by the tower law. Thus adjoining $\sqrt{3}$ doubles the degree. For each of the three choices $r \mapsto r\omega^j$, the extension theorem allows both roots $\sqrt{3} \mapsto \pm\sqrt{3}$ of its transported quadratic minimal polynomial. These six independent pairs describe all embeddings; a field map is determined by these generators.

Exercise 49.6: Extending a prescribed quadratic conjugation

The source field is $\mathbb{Q}(i, \sqrt{3}, r)$ and the map must send $\sqrt{3}$ to $-\sqrt{3}$. We can independently send i to $\pm i$ and r to $r\omega^j$ for $j = 0, 1, 2$. To justify existence, $\mathbb{Q}(i, \sqrt{3})$ is quadratic over $\mathbb{Q}(\sqrt{3})$, and $x^3 - 2$ remains irreducible over it: if a cubic root lay in this degree-four field, its degree-three rational subfield would violate the tower law. Thus there are exactly $2 \cdot 3 = 6$ extensions. For each, values on the three generators determine the entire map.

Exercise 49.7: A transcendental sign change

(a) The fixed field is $\mathbb{Q}(\pi^2)$. Indeed, $\mathbb{Q}(\pi)$ has basis $1, \pi$ over $\mathbb{Q}(\pi^2)$ because π is transcendental and $x^2 - \pi^2$ gives degree 2. The sign change fixes $A + B\pi$ precisely when $B = 0$.

(b) The element $\sqrt{\pi}$ satisfies $x^2 - \pi$ over $\mathbb{Q}(\pi)$. Transporting coefficients under $\pi \mapsto -\pi$ gives $x^2 + \pi$, whose roots in the algebraic closure are $\pm i\sqrt{\pi}$. The two extensions therefore send $\sqrt{\pi}$ to these two values. They exist by the extension theorem and are determined by this image.

Exercise 49.8: Ten statements about extensions of isomorphisms

(a) False. Take $F = \mathbb{Q}$ and α transcendental; no embedding of $\mathbb{Q}(\alpha)$ into the algebraic closure $\overline{\mathbb{Q}}$ can send a transcendental element to an algebraic one.

(b) True, by the isomorphism extension theorem for algebraic extensions.

(c) False. The identity on $\mathbb{Q}$ has one extension to $\mathbb{Q}$ but two to $\mathbb{Q}(\sqrt{2})$.

(d) True. An isomorphism extends to an isomorphism of algebraic closures.

(e) False. $\mathbb{Q}$ and $\mathbb{Q}(\sqrt{2})$ are not isomorphic, but share an algebraic closure up to isomorphism.

(f) True. Both closures are algebraic closures of $\mathbb{Q}$.

(g) True. The number of embeddings is at most the finite degree.

(h) True. Count extensions of each lower embedding through a tower; each has the same number of extensions as the relative index.

(i) True. Successively extending across finitely many algebraic generators constructs the map; the infinite case needs the maximality argument.

(j) False. $\mathbb{C}$ is uncountable, whereas $\overline{\mathbb{Q}}$ is countable, so they cannot be isomorphic.

Exercise 49.9: An algebraic self-embedding of a closed field

The image $\sigma[K]$ is algebraically closed because it is isomorphic to the algebraically closed field K. By assumption every element of K is algebraic over this image. An algebraically closed field has no proper algebraic extension: each minimal polynomial is linear. Therefore every element of K already lies in $\sigma[K]$. Thus the embedding is onto and is an automorphism.

Exercise 49.10: Extending to an automorphism of the closure

Let $\sigma : E \to \overline{F}$ fix F. The extension theorem extends it to an embedding $\tau : \overline{F} \to \overline{F}$. Its image contains F, and $\overline{F}$ is algebraic over F, hence algebraic over the larger image $\tau[\overline{F}]$. Exercise 49.9 now makes τ onto. Its restriction to E is the desired original map.

Exercise 49.11: Closures of algebraically related fields

An algebraic closure $\overline{E}$ of the algebraic extension E/F is algebraic over F by transitivity of algebraicity. It is also algebraically closed. Thus it is an algebraic closure of F. Uniqueness of algebraic closures up to an isomorphism fixing F gives $\overline{E} \cong \overline{F}$.

Exercise 49.12: Closures of rational function fields

Since π is transcendental, $\sqrt{\pi}$ is transcendental too, and substitution $\sqrt{\pi} \mapsto x$ gives $\mathbb{Q}(\sqrt{\pi}) \cong \mathbb{Q}(x)$. Their algebraic closures are consequently isomorphic. The extension $\overline{\mathbb{Q}}(x)/\mathbb{Q}(x)$ is algebraic, so an algebraic closure of either is an algebraic closure of the other by Exercise 49.11. Finally the algebraic closure of $\mathbb{Q}(\sqrt{\pi})$ inside $\mathbb{C}$ is indeed algebraically closed: a root in $\mathbb{C}$ of a polynomial over it is algebraic over that field, hence algebraic over $\mathbb{Q}(\sqrt{\pi})$. These observations prove the requested isomorphism.

Exercise 49.13: Index is bounded by degree

For a simple algebraic extension $F(\alpha)/F$, the index is the number of distinct roots of the minimal polynomial, hence is at most its degree. Write a finite extension as a finite tower of simple extensions. Index and degree are both multiplicative along the tower, so multiplying the inequalities at each step gives $\{E : F\} \leq [E : F]$. Here braces denote the number of embeddings into a fixed algebraic closure, not a quotient field.

50 Splitting Fields

Exercise 50.1: Splitting a quadratic

The roots of x^2+3 are $\pm\sqrt{-3}$. Thus the splitting field is $\mathbb{Q}(\sqrt{-3})$ of degree 2, since the polynomial has no rational root.

Exercise 50.2: Fourth roots of unity

The roots of x^4-1 are $1,-1,i,-i$. Adjoining i suffices and is necessary. The splitting field $\mathbb{Q}(i)$ has degree 2 over $\mathbb{Q}$.

Exercise 50.3: Two distinct quadratic radicals

The splitting field is $\mathbb{Q}(\sqrt{2},\sqrt{3})$. It has degree 4: if $\sqrt{3}=a+b\sqrt{2}$ with rational a,b, squaring would give $3=a^2+2b^2+2ab\sqrt{2}$, hence $ab=0$. The alternatives would require either a rational square root of 3 or a rational square root of $3/2$, both impossible. Thus the second quadratic doubles the first degree.

Exercise 50.4: Splitting an irreducible cubic

Let $r=\sqrt[3]{3}$ and ω be a primitive cube root of unity. The splitting field is $\mathbb{Q}(r,\omega)$. Eisenstein at 3 gives $[\mathbb{Q}(r):\mathbb{Q}]=3$, and ω is nonreal while $\mathbb{Q}(r)\subset\mathbb{R}$. Its quadratic polynomial therefore doubles the degree, giving 6.

Exercise 50.5: Third roots of unity

Factor $x^3-1=(x-1)(x^2+x+1)$. The quadratic has discriminant -3, so its splitting field is $\mathbb{Q}(\sqrt{-3})=\mathbb{Q}(\omega)$, of degree 2.

Exercise 50.6: Combining square and cube roots

The roots require $\sqrt{2}, r=\sqrt[3]{2}$, and ω. Put $s=2^{1/6}$; then $s=\sqrt{2}/r$, while $s^2=r$ and $s^3=\sqrt{2}$. Thus the splitting field is $\mathbb{Q}(s,\omega)$. Eisenstein makes x^6-2 irreducible, so $[\mathbb{Q}(s):\mathbb{Q}]=6$. This real field does not contain the nonreal ω, giving total degree 12.

Exercise 50.7: Automorphisms of the real cubic field

The order is 1. An automorphism sends the real $r=\sqrt[3]{2}$ to a root of x^3-2 inside the real field $\mathbb{Q}(r)$. Its only real root is r, so every automorphism fixes this generator and is the identity.

Exercise 50.8: Automorphisms of its splitting field

The splitting field $E = \mathbb{Q}(r, i\sqrt{3})$ has degree 6 over $\mathbb{Q}$. Characteristic zero gives six embeddings, all of which are automorphisms because E is a splitting field. Thus the group order is 6.

Exercise 50.9: Fixing the real cubic generator

The extension $E/\mathbb{Q}(r)$ has degree 2, obtained by adjoining $i\sqrt{3}$. Its two automorphisms fix r and send $i\sqrt{3}$ to itself or its negative. Hence the relative group has order 2.

Exercise 50.10: A cubic splits in an eight-element field

The polynomial $f = x^3 + x^2 + 1$ has no root in $\mathbb{Z}_2$, so is irreducible. Thus $\mathbb{Z}_2(\alpha)$ has eight elements. Frobenius preserves roots of polynomials over $\mathbb{Z}_2$, so $\alpha, \alpha^2, \alpha^4$ are roots. They are distinct: a nonzero nonidentity element of the group of seven nonzero elements has order 7, and the exponents 1,2,4 are distinct modulo 7. Hence these exhaust its three roots and f splits there. From $\alpha^3 = \alpha^2 + 1$, the last root is $\alpha^4 = \alpha^2 + \alpha + 1$.

Exercise 50.11: Correcting the splitting-field criterion

In the stated condition, "every polynomial" must be replaced by "every irreducible polynomial." Otherwise even the splitting field $E = F = \mathbb{Q}$ fails the proposed condition for $(x - 1)(x^2 - 2)$. An algebraic extension E/F is a splitting field of a family of polynomials precisely when each irreducible polynomial over F with one zero in E has all its zeros in E.

Exercise 50.12: Correcting the meaning of splitting

A polynomial splits over E when it is a nonzero constant times a product of *linear* polynomials in $E[x]$. Merely factoring into lower-degree factors is insufficient: $(x^2 - 2)(x^2 - 3)$ factors over $\mathbb{Q}$ but does not split there.

Exercise 50.13: Bounds for a splitting-field degree

For a nonconstant degree-n polynomial, $1 \leq [E : F] \leq n!$. Adjoin a first root of degree at most n, factor out its linear factor, then adjoin a root of the remaining factor of degree at most $n - 1$, and continue. The tower law bounds the product by $n(n - 1) \cdots 1$. No irreducibility assumption gives a stronger positive lower bound: a polynomial already split over F has degree-one splitting field. For a nonzero constant polynomial the splitting field is F as well.

Exercise 50.14: Ten statements about splitting fields

(a) True. A conjugation isomorphism extends to the closure, and normality preserves E; the converse follows by applying an automorphism to a minimal polynomial.

(b) False. $\mathbb{R}/\mathbb{Q}$ is not algebraic, whereas a splitting field is algebraic.

(c) True: the trivial extension is the splitting field of x.

(d) True: $\mathbb{C}$ splits x^2+1 over $\mathbb{R}$ and is generated by its roots.

(e) True: it splits x^2+1 over $\mathbb{Q}$.

(f) True: $\mathbb{Q}(\pi)$ splits $x^2-\pi^2$ over $\mathbb{Q}(\pi^2)$.

(g) False. An isomorphism need not fix F or even have its image in the same field; send the transcendental generator of $E=F=\mathbb{Q}(t)$ to an independent indeterminate.

(h) False. For $E=F=\mathbb{Q}(t)$, the embedding $t\mapsto t^2$ has image $\mathbb{Q}(t^2)\subsetneq E$ and lies inside an algebraic closure of F.

(i) True. This is the defining embedding property of a normal algebraic extension.

(j) True. An algebraic closure is generated by all roots of all polynomials over F.

Exercise 50.15: Why irreducibility is essential

Take $F=E=\mathbb{Q}$, a splitting field of x over $\mathbb{Q}$. The reducible polynomial $(x-1)(x^2-2)$ has the zero 1 in E but its other two zeros are $\pm\sqrt{2}$, outside E. Thus the corollary about an irreducible polynomial having one root in a splitting field does not extend to reducible polynomials.

Exercise 50.16: Group orders in a tower

The intended multiplicativity formula has $|G(K/F)|$ on the left; the printed formula repeats $G(E/F)$ there.

(a) No. For $F=\mathbb{Q}$, $E=\mathbb{Q}(\sqrt[3]{2})$ and $K=\mathbb{Q}(\sqrt[3]{2},i\sqrt{3})$, the group orders are 6,1,2 respectively. Thus $6\neq 2\cdot 1$.

(b) Yes when the relevant fields are splitting fields over the bottom field. Their group orders equal their embedding indices, including for K/E, since a normal extension over F is normal over every intermediate field. The multiplicativity of indices then gives $|G(K/F)|=|G(K/E)||G(E/F)|$. This argument does not require separability.

Exercise 50.17: One polynomial suffices for a finite normal extension

Choose finitely many generators $\alpha_1,\ldots,\alpha_m$ for E/F and let p_j be their minimal polynomials over F. Since E is normal, each p_j splits in E. Let $f=\prod_j p_j$. Its splitting field inside E contains every α_j, so contains E; it is also contained in E, since all its roots are there. Hence it equals E.

Exercise 50.18: A quadratic extension is a splitting field

Choose $\alpha\in E\setminus F$. The tower law gives $E=F(\alpha)$ and a degree-two minimal polynomial x^2+bx+c. Its other root, counted with multiplicity, is $-b-\alpha\in E$. Thus the polynomial splits in E, and one root generates E, proving the claim even in characteristic 2 (where the roots may coincide).

Exercise 50.19: Containing all conjugates

If E/F is a splitting field, every irreducible polynomial with a root in E splits there, so all conjugates of every element belong to E. Conversely, for each $\alpha \in E$, take its minimal polynomial. The hypothesis places all its roots in E. The splitting field of this family is contained in E and contains every α, hence is exactly E.

Exercise 50.20: No nontrivial automorphism of the real cubic

Every automorphism of $\mathbb{Q}(r)$ fixes $\mathbb{Q}$, and the image of $r = \sqrt[3]{2}$ must satisfy $x^3 - 2 = 0$. The other two roots $r\omega, r\omega^2$ are nonreal, whereas the field is real. Thus r is fixed. Since it generates the field, the automorphism is the identity.

Exercise 50.21: The cyclic subgroup fixing the quadratic radical

Put $r = \sqrt[3]{2}$ and $\omega = (-1 + i\sqrt{3})/2$. The field $\mathbb{Q}(i\sqrt{3}) = \mathbb{Q}(\omega)$ has degree 2 over $\mathbb{Q}$, so its relative degree in the degree-six splitting field is 3. The map $\rho(r) = r\omega$, $\rho(\omega) = \omega$ is an automorphism, and its powers send r successively to $r\omega^2$ and r. It has order 3 and accounts for all three relative embeddings. Thus the group is cyclic, isomorphic to $(\mathbb{Z}_3, +)$.

Exercise 50.22: The permutation representation on roots

(a) For an automorphism *fixing* F, $f(\sigma(\alpha)) = \sigma(f(\alpha)) = 0$. Its inverse shows that this gives a permutation of the roots. The base-field qualification is necessary.

(b) The roots generate the splitting field over F. Agreement on all roots and on F implies agreement on every rational expression in them, hence on the whole field.

(c) Restriction to the roots is therefore an injective homomorphism $G(E/F) \hookrightarrow S_\Omega$, where Ω is the set of distinct roots. Its image is the subgroup of permutations that preserve all algebraic relations among the roots over F.

Exercise 50.23: The full cubic permutation group

(a) The field $E = \mathbb{Q}(r, \omega)$ has degree 6 and is separable in characteristic zero. All six rational embeddings are automorphisms, so $|G(E/\mathbb{Q})| = 6$.

(b) Its action on the three roots gives an injection into S_3. As both groups have six elements, the injection is onto, proving $G(E/\mathbb{Q}) \cong S_3$.

Exercise 50.24: A prime-root-of-unity extension

For a prime p, every nonidentity pth root ζ is primitive, and its powers give all roots of $x^p - 1$. Thus its splitting field is $\mathbb{Q}(\zeta)$. The minimal polynomial $1 + x + \cdots + x^{p-1}$ is irreducible over $\mathbb{Q}$ by the prime cyclotomic theorem, so the extension degree is $p - 1$ (including degree 1 for $p = 2$).

Exercise 50.25: Uniqueness of splitting fields in different closures

There is an isomorphism $\tau : \overline{F} \to \overline{F}'$ fixing F. It maps roots of f to roots of f because the coefficients are fixed. Its inverse gives the reverse inclusion, so it bijects the two full root sets. Since each splitting field is generated by the respective roots, $\tau(E) = E'$. The restriction is the desired F-isomorphism.

51 Separable Extensions

Exercise 51.1: Combining square and cube roots

Choose $\alpha = \sqrt{2}/\sqrt[3]{2} = 2^{1/6}$. This quotient is in the given field. Direct exponent calculations give $\alpha^3 = \sqrt{2}$ and $\alpha^2 = \sqrt[3]{2}$, which are polynomials in α with rational coefficients. Thus each original generator lies in $\mathbb{Q}(\alpha)$, while α lies in their field, proving equality.

Exercise 51.2: Combining fourth and sixth roots

Choose $\alpha = \sqrt[4]{2}/\sqrt[6]{2} = 2^{1/12}$, an element of the given field. Then $\sqrt[4]{2} = \alpha^3$ and $\sqrt[6]{2} = \alpha^2$. These explicit polynomial expressions prove $\mathbb{Q}(\sqrt[4]{2}, \sqrt[6]{2}) = \mathbb{Q}(\alpha)$.

Exercise 51.3: A primitive biquadratic generator

Let $\alpha = \sqrt{2} + \sqrt{3}$, which belongs to the given field. Cubing and simplifying gives $\alpha^3 = 11\sqrt{2} + 9\sqrt{3}$. Subtracting suitable multiples of α yields

$$\sqrt{2} = \frac{\alpha^3 - 9\alpha}{2}, \qquad \sqrt{3} = \frac{11\alpha - \alpha^3}{2}.$$

Thus both generators are rational polynomials in α, giving the desired equality.

Exercise 51.4: A primitive generator for a complex sextic field

Put $r = \sqrt[3]{2}$ and choose $\alpha = r + i$. From $(\alpha - i)^3 = 2$ and $i^2 = -1$,

$$i = \frac{\alpha^3 - 3\alpha - 2}{3\alpha^2 - 1}.$$

The denominator is nonzero: α has imaginary part 1 and positive real part, so its square has nonzero imaginary part. This already proves that $i, r \in \mathbb{Q}(\alpha)$. To meet the requested *polynomial* form, exact reduction gives

$$i = \frac{-12\alpha^5 + 9\alpha^4 - 40\alpha^3 + 78\alpha^2 - 78\alpha - 91}{22},$$
$$r = \frac{12\alpha^5 - 9\alpha^4 + 40\alpha^3 - 78\alpha^2 + 100\alpha + 91}{22}.$$

Here is a way to check these expressions without assuming the answer. Squaring the first displayed numerator equation for i gives

$$m(\alpha) = 0, \quad m(x) = x^6 + 3x^4 - 4x^3 + 3x^2 + 12x + 5.$$

Polynomial division verifies that $(3x^2 - 1)$ times the displayed polynomial for i is congruent to $x^3 - 3x - 2$ modulo $m(x)$. Evaluation at α proves the formula; subtracting it from α gives the formula for r. Hence the field is $\mathbb{Q}(\alpha)$.

Exercise 51.5: Where multiplicity is measured

The factorization must be in $\overline{F}[x]$, not necessarily in $F[x]$, since α may lie outside F. For a *nonzero* polynomial f, multiplicity $v \geq 1$ means $(x-\alpha)^v \mid f$ but $(x-\alpha)^{v+1} \nmid f$ in $\overline{F}[x]$. The zero polynomial has no finite root multiplicity under this definition.

Exercise 51.6: The definition of separable element

The proposed definition is correct: an algebraic element is separable over F when it has multiplicity 1 as a root of its minimal polynomial over F. All roots of an irreducible polynomial have the same multiplicity, so this also says the minimal polynomial has distinct roots in an algebraic closure.

Exercise 51.7: A repeated polynomial over a perfect field

Take $f(x) = (x^2+1)^2$. It has no rational root, and its complex roots $i, -i$ each have multiplicity 2. This does not contradict perfection of $\mathbb{Q}$, which concerns *irreducible* polynomials. The minimal polynomial x^2+1 has distinct roots; the repetitions were introduced by squaring it.

Exercise 51.8: Ten statements about separability

(a) False. Over $\mathbb{Z}_p(t)$, $x^p - t$ is irreducible and has just one distinct root, so adjoining that root gives an inseparable finite extension.

(b) True. Every finite field is perfect.

(c) True. An irreducible polynomial in characteristic zero cannot share a factor with its smaller-degree nonzero derivative.

(d) False. $(x-1)^2$ has a repeated root over any field.

(e) False for the same example, even over $\mathbb{Q}$.

(f) True. This is the defining polynomial characterization of a perfect field.

(g) True. Every irreducible polynomial over an algebraically closed field is linear.

(h) True. An algebraic closure is algebraic over F and is perfect.

(i) True. Separability makes the embedding index equal the degree, and splitting makes every embedding an automorphism.

(j) True. The index of a finite extension divides its degree (the quotient is its inseparable degree), and in a splitting field the index equals the group order.

Exercise 51.9: Separable elements are closed under field operations

The minimal polynomial of β over $F(\alpha)$ divides its separable minimal polynomial over F, so it is separable. Therefore $F(\alpha,\beta)/F(\alpha)$ and $F(\alpha)/F$ are finite separable extensions; the tower theorem makes $F(\alpha,\beta)/F$ separable. Every element of this field is then separable over F, including $\alpha+\beta$, $\alpha\beta$, and α/β when $\beta \neq 0$.

Exercise 51.10: A basis for a Frobenius extension

Let $K = \mathbb{Z}_p(y^p)$. The relation $y^p \in K$ makes $K[y]$ spanned by $1, y, \ldots, y^{p-1}$. Being a finite-dimensional domain over K, it is a field: multiplication by any nonzero element is an injective linear map, hence surjective, giving an inverse. Thus $K[y] = \mathbb{Z}_p(y)$. For independence, a relation $\sum_{j=0}^{p-1} r_j(y^p)y^j = 0$ can be multiplied by a common denominator to give $\sum_j P_j(y^p)y^j = 0$. Exponents coming from different j lie in different residue classes modulo p, so no cancellation between the groups is possible; all P_j vanish. Thus the displayed spanning set is a basis, and the degree is p. Setting $t = y^p$, the polynomial $x^p - t$ has y as a root and the same degree as its minimal polynomial. Hence it is irreducible over $\mathbb{Z}_p(t)$.

Exercise 51.11: Algebraic extensions of a perfect field

Let α be algebraic over E. Since E/F is algebraic, α is algebraic over F as well. The minimal polynomial of α over the perfect field F is separable. Its minimal polynomial over E divides it in $E[x]$, so also has no repeated roots. Thus every element algebraic over E is separable over E, proving E perfect.

Exercise 51.12: Transitivity for infinite separable extensions

Take $\alpha \in K$. Its separable minimal polynomial over E uses finitely many coefficients $c_1, \ldots, c_m \in E$. Set $E_0 = F(c_1, \ldots, c_m)$; this is a finite separable extension of F. The same polynomial is irreducible over E_0, because a factorization there would be one over E, and it remains separable. Thus $E_0(\alpha)/E_0$ is finite separable. The finite tower theorem gives $E_0(\alpha)/F$ separable, so α is separable over F. This holds for every $\alpha \in K$, proving transitivity without assuming the entire tower finite.

Exercise 51.13: The separable closure inside an algebraic extension

Let S be the elements of E separable over F. It contains F, including 0 and 1. Exercise 51.9 shows it is closed under sums, products, and quotients by nonzero elements; additive inverses are included by multiplying by $-1 \in F$. Hence S is a subfield. It consists exactly of the separable elements, so is the largest intermediate extension separable over F, called the separable closure of F in E.

Exercise 51.14: Frobenius generates a finite-field Galois group

(a) In a field of p^n elements, $a^{p^n} = a$ for every a, so $\sigma_p^n = 1$. If $\sigma_p^d = 1$ for $0 < d < n$, all p^n elements would be roots of $x^{p^d} - x$, whose degree p^d is smaller than p^n, impossible. Thus its order is n.

(b) The extension has degree n, is separable because finite fields are perfect, and is a splitting extension since it splits $x^{p^n} - x$. Its Galois group therefore has order n. The n distinct powers of Frobenius already account for that many elements, so the full group is cyclic with this generator.

Exercise 51.15: The additive derivative homomorphism

(a) If $f = \sum a_j x^j$ and $g = \sum b_j x^j$, differentiating coefficient by coefficient gives $(f+g)' = \sum j(a_j + b_j)x^{j-1} = f' + g'$. This proves it is a homomorphism of additive groups; it is generally not a

ring homomorphism.

(b) In characteristic zero, $ja_j = 0$ for $j > 0$ implies $a_j = 0$. Thus the kernel is the constant polynomials F.

(c) In characteristic p, j vanishes as a field scalar precisely when $p \mid j$. Hence the kernel is $F[x^p]$, the polynomials all of whose nonzero exponents are multiples of p. Their coefficients need not themselves be pth powers.

Exercise 51.16: Formal derivative rules

(a) For $c \in F$, coefficientwise differentiation gives $(cf)' = \sum_j jca_j x^{j-1} = cf'$.

(b) First check $(x^i x^j)' = (i+j)x^{i+j-1} = x^i(jx^{j-1}) + (ix^{i-1})x^j$. Expand arbitrary polynomials into monomials and use additivity and (a) to obtain $(fg)' = fg' + f'g$.

(c) Induct on m. For $m = 1$ the assertion is immediate. If $(f^m)' = mf^{m-1}f'$, the product rule gives $(f^{m+1})' = f^m f' + (f^m)'f = (m+1)f^m f'$. The integers are interpreted as elements of F, so this proof also works in positive characteristic.

Exercise 51.17: Multiplicity and vanishing derivative

In $\overline{F}[x]$, write $f = (x-\alpha)^v g$ with $g(\alpha) \neq 0$. If $v = 1$, the product rule gives $f'(\alpha) = g(\alpha) \neq 0$. If $v > 1$, both terms of

$$f' = v(x-\alpha)^{v-1}g + (x-\alpha)^v g'$$

vanish at α, including when the scalar v is zero in the field. Thus $f'(\alpha) = 0$ exactly when the root has multiplicity greater than 1.

Exercise 51.18: Characteristic zero forces irreducible separability

Let $p \in F[x]$ be nonconstant and irreducible in characteristic zero. Its derivative is nonzero and has smaller degree. If a root α were repeated, it would be a root of p' too, so its minimal polynomial p would divide p'. The degree inequality rules this out. Therefore all roots of p are simple, proving separability.

Exercise 51.19: The derivative test for an inseparable irreducible

For an irreducible q, a repeated root implies $q \mid q'$ by minimality. Since $\deg q' < \deg q$ unless $q' = 0$, this is equivalent to $q' = 0$. In characteristic p, coefficientwise differentiation gives $q' = 0$ exactly when each exponent with nonzero coefficient is divisible by p. Conversely a nonconstant polynomial with zero derivative cannot have a simple root by Exercise 51.17. Thus the criterion holds in both directions.

Exercise 51.20: Common factors over the algebraic closure

By Exercise 51.17 a repeated root is exactly a common root of f and f'. Over the algebraically closed field $\overline{F}$, a common root α supplies the common linear factor $x - \alpha$. Conversely a nonconstant

common factor has a root in the closure, which is a common root of f, f', hence repeated in f. This proves the requested equivalence for nonzero f.

Exercise 51.21: Bringing the gcd test back to the coefficient field

If f, f' have no nonconstant common divisor in $F[x]$, their gcd is 1, so Bezout's identity gives $Af + Bf' = 1$ with $A, B \in F[x]$. This same identity holds in $\overline{F}[x]$, excluding any nonconstant common factor there. Conversely a nonconstant common factor in $F[x]$ remains one over the closure. Combining with Exercise 51.20 proves the equivalence over F itself. The argument includes nonzero constant f, whose gcd with its zero derivative is a unit.

Exercise 51.22: A root-free computational test

For a nonzero polynomial, form its formal derivative and run the Euclidean algorithm in $F[x]$ on f, f'. Normalize the final nonzero remainder to a monic gcd d. A repeated root exists exactly when $\deg d > 0$. In particular, if f is nonconstant and $f' = 0$, the gcd is the monic associate of f, so there is repetition. If f is a nonzero constant, the gcd has degree 0 and there are no roots. No numerical root approximation is required.

52 Totally Inseparable Extensions

For degree checks involving two independent indeterminates, one useful elementary rule is

$$[\mathbb{F}_3(y,z) : \mathbb{F}_3(y^a z^b, y^c z^d)] = |ad - bc|$$

when the two exponent vectors are independent. Here is why the determinant appears. The subgroup of $\mathbb{Z}^2$ generated by $(a,b),(c,d)$ has $|ad-bc|$ cosets. Choose one monomial for each exponent coset. Every Laurent monomial is a coefficient-field monomial times one chosen representative, so these representatives span; their span is a finite-dimensional domain over the coefficient field, hence is a field and includes rational functions too. They are independent because clearing coefficient denominators leaves terms in distinct exponent cosets, which cannot cancel. Thus they form a basis. This supplies the degree certificates below; separability and the power tests establish the actual closure fields.

Exercise 52.1: A separable closure with two independent exponents

Let $K = \mathbb{Z}_3(y^{12}, z^{18})$ and $L = \mathbb{Z}_3(y,z)$. Set $S = \mathbb{Z}_3(y^3, z^9)$. The generators y^3 and z^9 satisfy respectively $X^4 - y^{12}$ and $X^2 - z^{18}$ over K, whose nonzero roots are simple in characteristic 3. Thus S/K is separable. Also $y^3, z^9 \in S$, so every element of L has some 3-power in S; in fact its ninth power is in S. Hence L/S is purely inseparable. An element separable over K is separable over S too, since its new minimal polynomial divides its old separable one; if it also lies in the purely inseparable extension L/S, it must be in S. Therefore S is exactly the separable closure. The monomial degree counts are $[L:K] = 12 \cdot 18 = 216$, $[L:S] = 3 \cdot 9 = 27$, and $[S:K] = 8$.

Exercise 52.2: A separable closure with a mixed monomial

Now $K = \mathbb{Z}_3(y^{12}, y^2 z^{18})$. Put $S = \mathbb{Z}_3(y^3, yz^9)$. The first generator satisfies the separable equation $X^4 - y^{12} = 0$ and the second the separable equation $X^2 - y^2 z^{18} = 0$, so S/K is separable. Moreover $y^3 \in S$ and $z^{27} = (yz^9)^3/y^3 \in S$. Thus $L = \mathbb{Z}_3(y,z)$ is purely inseparable over S. The separable/purely-inseparable intersection argument from Exercise 52.1 makes S the full separable closure. For a degree check, the exponent lattice of K has determinant $12 \cdot 18 = 216$, while that of S, generated by $(3,0),(1,9)$, has determinant 27; hence $[S:K] = 8$.

Exercise 52.3: A totally inseparable closure in the independent case

The field is $P = \mathbb{Z}_3(y^4, z^2)$. Its generators have a third and a ninth power, respectively, in $K = \mathbb{Z}_3(y^{12}, z^{18})$, so P/K is purely inseparable. The extension L/P is separable, because y satisfies $X^4 - y^4$ and z satisfies $X^2 - z^2$, both separable polynomials. If $a \in L$ has a 3-power in K, it is purely inseparable over P but also separable over P, so $a \in P$. This proves maximality. Thus the totally inseparable closure is P; its degree over K is 27.

Exercise 52.4: A totally inseparable closure in the mixed case

Take $P = \mathbb{Z}_3(y^4, y^2 z^2)$. It contains $K = \mathbb{Z}_3(y^{12}, y^2 z^{18})$, since $y^{12} = (y^4)^3$ and $y^2 z^{18} = (y^2 z^2)^9/(y^4)^4$. Its generators have 3-powers in K:

$$(y^4)^3 = y^{12}, \qquad (y^2 z^2)^{27} = (y^{12})^4 (y^2 z^{18})^3.$$

Hence P/K is purely inseparable. Over P, first adjoining y is separable via its fourth-power equation, and then adjoining z is separable via $z^2 = (y^2z^2)/y^2$. Thus L/P is separable. The same intersection argument as in Exercise 52.3 shows that every element of L purely inseparable over K lies in P. Its exponent lattice, generated by $(4,0),(2,2)$, has determinant 8, so $[P:K] = 216/8 = 27$.

Exercise 52.5: Ten statements about totally inseparable extensions

In this section Fraleigh reserves "totally inseparable" for a *proper* extension with one embedding. The more common "purely inseparable" convention also permits the trivial extension.

(a) False. $\mathbb{Z}_3(t)(\sqrt{t})/\mathbb{Z}_3(t)$ is a proper separable extension, since $X^2 - t$ is irreducible and has nonzero derivative.

(b) True. A totally inseparable algebraic element has some p^t-power in F.

(c) False. $\mathbb{Z}_5(y)/\mathbb{Z}_5(y^5)$ has degree 5 and only one embedding; its minimal polynomial is $X^5 - y^5$.

(d) False. $\mathbb{Z}_5(y)/\mathbb{Z}_5(y^{10})$ has inseparable degree 5 and separable degree 2, since $X^{10} - y^{10}$ has two distinct roots, each repeated five times.

(e) False. The same mixed extension has two embeddings, so is not totally inseparable either.

(f) False. The element y over $\mathbb{Z}_5(y^{10})$ in (d) is neither separable nor totally inseparable.

(g) True. The separable elements form a subfield by Exercise 51.13.

(h) False with the book's strict convention: if E/F is separable, its separable closure is E, and E/E is not a proper extension. The corrected claim allows E to equal that closure, or uses the inclusive "purely inseparable" convention.

(i) True. If E/F is not separable, its separable closure is proper, and every element outside it is totally inseparable over it.

(j) True, by the minimal-polynomial characterization of total inseparability.

Exercise 52.6: Purely inseparable elements form a field

Include the elements of F as requested. For a, b in the proposed set, choose a common power $q = p^m$ with $a^q, b^q \in F$. Then $(a+b)^q = a^q + b^q \in F$, $(ab)^q = a^q b^q \in F$, and if $b \neq 0$, $(a/b)^q = a^q/b^q \in F$. Additive inverses work in the same way. Thus it is closed under field operations and contains 0,1. Every element outside F with such a power is totally inseparable, by the power criterion, so these closure calculations prove the assertion. In characteristic zero no proper totally inseparable elements exist, and the set is simply F.

Exercise 52.7: Perfection and surjective Frobenius

If F is perfect and $a \in F$, choose a root b of $X^p - a$ in a closure. That polynomial equals $(X-b)^p$, so b has only one conjugate. Separability forces its degree to be 1; hence $b \in F$ and $a = b^p$. Thus $F^p = F$. Conversely suppose $F^p = F$. An inseparable irreducible polynomial would have derivative zero, so have form $\sum a_j X^{pj}$. Write $a_j = b_j^p$. Then it equals $(\sum b_j X^j)^p$, contradicting irreducibility. Therefore all irreducibles are separable, and F is perfect.

Exercise 52.8: Perfection through a finite extension

Let $n = [E : F]$. Frobenius is an isomorphism from E onto E^p carrying F onto F^p, so $[E^p : F^p] = n$. If $F^p = F$, then $F \subseteq E^p \subseteq E$ and both $[E^p : F]$ and $[E : F]$ equal n, forcing $E^p = E$. Conversely, if $E^p = E$, the equality above reads $[E : F^p] = n$. In the tower $F^p \subseteq F \subseteq E$, it follows that $n = n[F : F^p]$, so $[F : F^p] = 1$ and $F^p = F$. Finiteness of n is essential for this degree argument.

53 Galois Theory

Exercise 53.1: The embedding index of a multiquadratic field

The characteristic is zero, so the extension is separable. Thus its index, the number of embeddings fixing $\mathbb{Q}$, equals its given degree: $\{K : \mathbb{Q}\} = 8$.

Exercise 53.2: The group order of a multiquadratic field

All independent sign changes of $\sqrt{2}, \sqrt{3}, \sqrt{5}$ are automorphisms. They give $2^3 = 8$ maps, exhausting the degree bound. Thus $|G(K/\mathbb{Q})| = 8$.

Exercise 53.3: The subgroup fixing the bottom field

By definition $\lambda(\mathbb{Q}) = G(K/\mathbb{Q})$, so its order is 8. The whole rational Galois group fixes the bottom field.

Exercise 53.4: Fixing two specified radicals

The field $\mathbb{Q}(\sqrt{2}, \sqrt{3})$ has degree 4 over $\mathbb{Q}$. The subgroup fixing it has order $[K : \mathbb{Q}(\sqrt{2}, \sqrt{3})] = 8/4 = 2$, consisting of the identity and the sign change of $\sqrt{5}$.

Exercise 53.5: Fixing a product radical

The field $\mathbb{Q}(\sqrt{6})$ has degree 2 over $\mathbb{Q}$, so its fixing subgroup has order $8/2 = 4$. Equivalently the signs of $\sqrt{2}$ and $\sqrt{3}$ must agree, while the sign of $\sqrt{5}$ is arbitrary.

Exercise 53.6: Fixing a triple-product radical

The quadratic field $\mathbb{Q}(\sqrt{30})$ has degree 2 over $\mathbb{Q}$, so its fixing subgroup has order 4. Its sign changes are exactly the choices with a positive product of the three radical signs.

Exercise 53.7: Fixing a sum of two radicals

Put $\alpha = \sqrt{2} + \sqrt{6}$. Its square is $8 + 4\sqrt{3}$, so $\sqrt{3} = (\alpha^2 - 8)/4 \in \mathbb{Q}(\alpha)$. Also $\sqrt{2} = \alpha/(1 + \sqrt{3})$ lies in this field. Thus $\mathbb{Q}(\alpha) = \mathbb{Q}(\sqrt{2}, \sqrt{3})$ has degree 4; its fixing subgroup has order $8/4 = 2$.

Exercise 53.8: Fixing the entire top field

An automorphism fixing every element of K is the identity. Therefore $\lambda(K)$ is the trivial subgroup, of order 1.

Exercise 53.9: The group of fourth roots of unity

The splitting field of $x^4 - 1$ is $\mathbb{Q}(i)$. The two automorphisms send i to i and $-i$. Their composition table is that of the cyclic group $\mathbb{Z}_2$; the nonidentity map is complex conjugation.

Exercise 53.10: A relative finite-field Frobenius

Since $729 = 9^3$, the degree of GF(729) over GF(9) is 3. Its Galois group is cyclic of order 3, generated by $a \mapsto a^9$. The third power is $a \mapsto a^{729} = a$; a smaller positive power cannot fix all 729 elements by the polynomial root bound.

Exercise 53.11: The cubic splitting-field correspondence

Let $r = \sqrt[3]{2}$, $\omega = (-1 + i\sqrt{3})/2$, $K = \mathbb{Q}(r, \omega)$, $\rho(r) = r\omega$, $\rho(\omega) = \omega$, and let κ be complex conjugation.

(a) The six maps and their images are

σ	$\sigma(r)$	$\sigma(i\sqrt{3})$
1	r	$i\sqrt{3}$
ρ	$r\omega$	$i\sqrt{3}$
ρ^2	$r\omega^2$	$i\sqrt{3}$
κ	r	$-i\sqrt{3}$
$\rho\kappa$	$r\omega$	$-i\sqrt{3}$
$\rho^2\kappa$	$r\omega^2$	$-i\sqrt{3}$

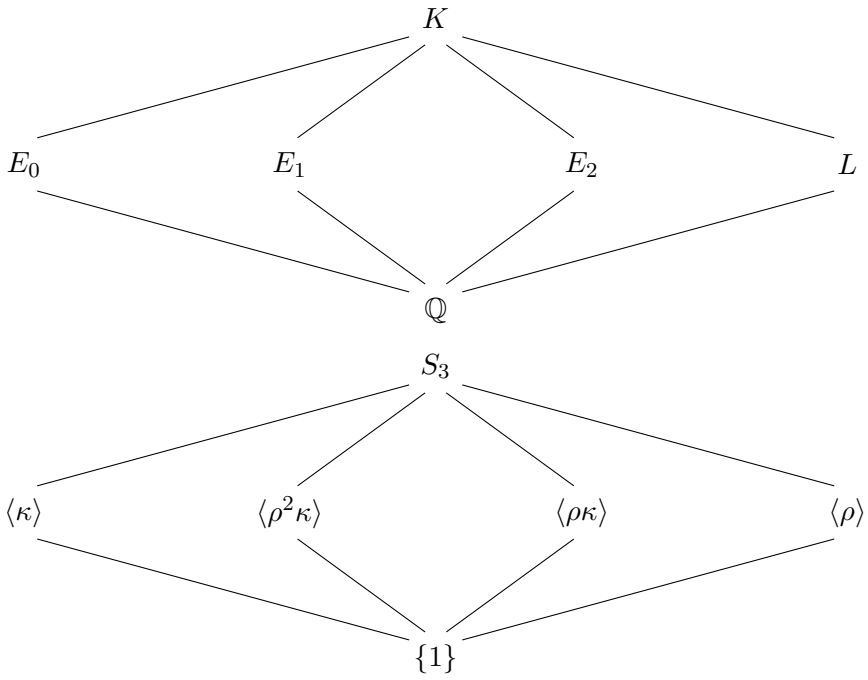
These account for all six rational embeddings of the degree-six splitting field.

(b) Their faithful action on the three cubic roots identifies the group with S_3. In particular $\rho^3 = \kappa^2 = 1$ and $\kappa\rho\kappa = \rho^{-1}$.

(c) The correspondence is

subgroup	fixed field
S_3	$\mathbb{Q}$
$\langle\rho\rangle$	$L = \mathbb{Q}(\omega)$
$\langle\kappa\rangle$	$E_0 = \mathbb{Q}(r)$
$\langle\rho\kappa\rangle$	$E_2 = \mathbb{Q}(r\omega^2)$
$\langle\rho^2\kappa\rangle$	$E_1 = \mathbb{Q}(r\omega)$
$\{1\}$	K

For instance $\rho\kappa(r\omega^2) = r\omega \cdot \omega^{-2} = r\omega^2$. The field degrees (3 for the E_j, 2 for L) and the Galois correspondence prove these are the full fixed fields. The diagrams below have only the displayed covering relations; invert one to obtain the other.



Exercise 53.12: The group of two quadratic factors

Since $x^4 - 5x^2 + 6 = (x^2 - 2)(x^2 - 3)$, its splitting field is $\mathbb{Q}(\sqrt{2}, \sqrt{3})$ of degree 4. The independent sign flips of the two radicals commute and each has order 2. Thus its group is $\mathbb{Z}_2 \times \mathbb{Z}_2$, the Klein four-group.

Exercise 53.13: The group of third roots of unity

The splitting field is $\mathbb{Q}(\omega)$ with $\omega^2 + \omega + 1 = 0$. Its two maps send ω to ω and ω^2, so the group is cyclic of order 2.

Exercise 53.14: An abstract group does not determine its field

Take $K_1 = \mathbb{Q}(\sqrt{2})$ and $K_2 = \mathbb{Q}(\sqrt{3})$ over $F = \mathbb{Q}$. Both have Galois group $\mathbb{Z}_2$. They are not isomorphic as fields: any isomorphism fixes $\mathbb{Q}$ and would send $\sqrt{2}$ to a square root of 2 in K_2. Writing $(a + b\sqrt{3})^2 = 2$ forces $ab = 0$, leading to $a^2 = 2$ or $b^2 = 2/3$, neither with rational solutions. Thus such an image does not exist.

Exercise 53.15: Ten statements about the Galois correspondence

Here "finite normal" has Fraleigh's definition: a *separable splitting extension*, often called a finite Galois extension.

(a) False. The subgroup-to-fixed-field correspondence is one-to-one.

(b) False. Inclusion reverses: $E \subseteq L$ implies $\lambda(L) \subseteq \lambda(E)$.

(c) True. The same splitting polynomials work over E, and separability descends along the tower.

(d) True. Each degree equals the order of its Galois group.

(e) True. A normal subgroup corresponds exactly to a normal intermediate extension of F.

(f) False. $\mathbb{Q}(\zeta_5)/\mathbb{Q}$ is simple but has cyclic group of order 4, which has a proper subgroup of order 2.

(g) False. A quadratic separable extension has simple Galois group $\mathbb{Z}_2$.

(h) True. Finite-field Galois groups are cyclic.

(i) False in characteristic 2 with this definition. $\mathbb{Z}_2(t^{1/2})/\mathbb{Z}_2(t)$ has degree 2 but is inseparable, hence is not "finite normal" as Fraleigh defines it (it is normal in the modern splitting-only sense).

(j) True. A quadratic in characteristic different from 2 is separable and splits in its root field.

Exercise 53.16: Abelian Galois extensions in a tower

The group $G(K/E)$ is a subgroup of the abelian group $G(K/F)$, hence is abelian. If E/F is normal, restriction gives $G(E/F) \cong G(K/F)/G(K/E)$. A quotient of an abelian group is abelian. Thus K/E and E/F are both abelian extensions in the requested sense. In fact every subgroup of an abelian group is normal, so the normality of intermediate E/F follows automatically here.

Exercise 53.17: Norm and trace lie in the base field

For any $\tau \in G(K/F)$, left multiplication $\sigma \mapsto \tau\sigma$ permutes the group. Hence

$$\tau\left(\prod_{\sigma}\sigma(\alpha)\right) = \prod_{\sigma}(\tau\sigma)(\alpha) = \prod_{\sigma}\sigma(\alpha),$$

and the analogous equality holds for the sum. Thus norm and trace are fixed by the entire group. Its fixed field is exactly F by the Galois correspondence, so both belong to F.

Exercise 53.18: Eight norm and trace calculations

The four embeddings independently change the signs of $\sqrt{2}, \sqrt{3}$.

(a) $N(\sqrt{2}) = (\sqrt{2})^2(-\sqrt{2})^2 = 4$.

(b) $N(\sqrt{2}+\sqrt{3}) = (\sqrt{2}+\sqrt{3})(\sqrt{2}-\sqrt{3})(-\sqrt{2}+\sqrt{3})(-\sqrt{2}-\sqrt{3}) = 1$, since the first pair is -1 and the second pair is -1.

(c) $N(\sqrt{6}) = (\sqrt{6})^2(-\sqrt{6})^2 = 36$.

(d) Every embedding fixes 2, so $N(2) = 2^4 = 16$.

(e) The two copies each of $\sqrt{2}, -\sqrt{2}$ sum to 0.

(f) The four sign choices of $\sqrt{2}+\sqrt{3}$ sum to 0.

(g) The two copies each of $\sqrt{6}, -\sqrt{6}$ sum to 0.

(h) The four copies of 2 sum to 8.

Exercise 53.19: Norm and trace from minimal-polynomial coefficients

Because $K = F(\alpha)$ is finite normal, its n automorphisms give exactly the n distinct roots of the minimal polynomial of α. Thus

$$x^n + a_{n-1}x^{n-1} + \cdots + a_0 = \prod_{\sigma \in G(K/F)} (x - \sigma(\alpha)).$$

Comparing constant terms gives $a_0 = (-1)^n \prod_\sigma \sigma(\alpha)$, so (a) $N_{K/F}(\alpha) = (-1)^n a_0$. Comparing the coefficient of x^{n-1} gives (b) $T_{K/F}(\alpha) = -a_{n-1}$.

Exercise 53.20: The group order divides the factorial

Let Ω be the set of distinct roots of f in its splitting field K, and let $m = |\Omega| \leq n$. Because all irreducible factors are separable, K/F is finite normal in the book's sense. The faithful action of $G(K/F)$ on its generating roots embeds it into S_m. Lagrange's theorem gives $|G(K/F)| \mid m!$, and $m! \mid n!$. Hence the order divides $n!$. Repeated irreducible factors in f cause no difficulty when one uses distinct roots.

Exercise 53.21: The natural permutation group of a polynomial

For $\sigma \in G(K/F)$, its values on the roots define a permutation, since applying σ to $f(\alpha) = 0$ gives $f(\sigma\alpha) = 0$. Restriction respects composition. Its kernel fixes every root, hence fixes the field they generate over F, namely K, so its kernel is trivial. Thus the Galois group is naturally isomorphic to its image, a subgroup of the permutations of the distinct roots.

Exercise 53.22: Cyclotomic extensions over a general field

(a) The powers of a primitive nth root ζ give all roots of $x^n - 1$. Its derivative nx^{n-1} has no common root with it under the characteristic hypothesis. Thus $F(\zeta)$ is a finite separable splitting extension, hence finite normal.

(b) Any automorphism sends ζ to another primitive root ζ^a, where a is invertible modulo n. The map assigning a to that automorphism is injective, and composition multiplies exponents modulo n. It embeds the group into the abelian unit group $(\mathbb{Z}/n\mathbb{Z})^\times$. Therefore the Galois group is abelian, even when not every invertible exponent occurs.

Exercise 53.23: Cyclic extensions and their intermediate fields

(a) A subgroup of a cyclic group is cyclic, so $G(K/E)$ is cyclic. If E/F is normal, its group is the quotient $G(K/F)/G(K/E)$, also cyclic. In fact the normality follows automatically because every subgroup of a cyclic group is normal.

(b) If $|G(K/F)| = n$, a cyclic group has exactly one subgroup of each order dividing n. An intermediate field of degree d over F corresponds to a subgroup of index d, or order n/d. Hence for each divisor d of n there is exactly one such intermediate field.

Exercise 53.24: The polynomial formed from every automorphism

(a) Each $\tau \in G(K/F)$ permutes the factors of $f(x) = \prod_\sigma (x - \sigma\alpha)$, since $\sigma \mapsto \tau\sigma$ permutes the group. Thus every coefficient is fixed by the group, and belongs to F.

(b) All distinct conjugates of α occur, by the extension theorem, and each occurs the same number of times: the maps taking α to a specified conjugate form a coset of its stabilizer $G(K/F(\alpha))$. Thus

$$f(x) = \operatorname{irr}(\alpha, F)^{[K:F(\alpha)]}.$$

Since the minimal polynomial is nonconstant, this equals the minimal polynomial exactly when the exponent is 1, namely when $K = F(\alpha)$.

Exercise 53.25: A compositum corresponds to an intersection

An automorphism fixes the join $E \vee L$ if and only if it fixes both E and L: one direction follows from containment, and the other because it then fixes every rational expression in elements of the two fields. Therefore

$$G(K/(E \vee L)) = G(K/E) \cap G(K/L).$$

This is the reversal of taking the smallest field containing two given fields.

Exercise 53.26: An intersection corresponds to a generated subgroup

Put $H = \langle G(K/E), G(K/L) \rangle$. An element is fixed by H exactly when it is fixed by both generating subgroups, so

$$K_H = K_{G(K/E)} \cap K_{G(K/L)} = E \cap L.$$

The bijective Galois correspondence now gives

$$G(K/(E \cap L)) = \langle G(K/E), G(K/L) \rangle.$$

One must generate a subgroup, rather than merely take the union, which need not be a group.

54 Illustrations of Galois Theory

Exercise 54.1: Irreducibility of the eighth cyclotomic polynomial

Make the invertible substitution $x = y + 1$. Then

$$(y+1)^4 + 1 = y^4 + 4y^3 + 6y^2 + 4y + 2.$$

Eisenstein's criterion at 2 applies: every coefficient except the leading one is divisible by 2, and the constant 2 is not divisible by 4. Thus the substituted polynomial is irreducible over $\mathbb{Q}$. A factorization of $x^4 + 1$ would yield a factorization after substitution, so $x^4 + 1$ is irreducible too.

Exercise 54.2: Checking the entire octic fixed-field diagram

Put $\alpha = \sqrt[4]{2}$ and $K = \mathbb{Q}(\alpha, i)$. We have $[K : \mathbb{Q}] = 8$, since $x^4 - 2$ is Eisenstein and $\mathbb{Q}(\alpha)$ is real. To avoid relying on a picture alone, write $r(\alpha) = i\alpha, r(i) = i$ and $s(\alpha) = \alpha, s(i) = -i$. Then $r^4 = s^2 = 1$ and $srs = r^{-1}$. In the source notation $r = \rho_1$, $r^2 = \rho_2$, $s = \mu_1$, $r^2 s = \mu_2$, $rs = \delta_1$, and $r^3 s = \delta_2$. The proposed fields can all be verified by their fixed generators and degrees:

subgroup	proposed fixed field	degree over $\mathbb{Q}$
$\langle r^2, s\rangle = H_1$	$\mathbb{Q}(\alpha^2) = \mathbb{Q}(\sqrt{2})$	2
$\langle r\rangle = H_2$	$\mathbb{Q}(i)$	2
$\langle r^2, rs\rangle = H_3$	$\mathbb{Q}(i\alpha^2) = \mathbb{Q}(\sqrt{-2})$	2
$\langle s\rangle = H_4$	$\mathbb{Q}(\alpha)$	4
$\langle r^2 s\rangle = H_5$	$\mathbb{Q}(i\alpha)$	4
$\langle r^2\rangle = H_6$	$\mathbb{Q}(\alpha^2, i)$	4
$\langle rs\rangle = H_7$	$\mathbb{Q}((1+i)\alpha)$	4
$\langle r^3 s\rangle = H_8$	$\mathbb{Q}((1-i)\alpha)$	4

For each row, substitution of r, s checks that the indicated generator is fixed by the subgroup. For H_1, H_2, H_3, the generator is a nonrational quadratic radical, giving degree 2. For H_4, H_5, the fourth power is 2 and $x^4 - 2$ is irreducible. For H_6, the real quadratic field gains the nonreal i, giving degree 4. For H_7, H_8, if $\gamma = (1 \pm i)\alpha$, then $\gamma^2 = \pm 2i\sqrt{2}$ and $\gamma^4 = -8$. Its square generates $\mathbb{Q}(\sqrt{-2})$, and $\pm 2i\sqrt{2}$ is not a square there: writing $(a + b\sqrt{-2})^2 = \pm 2\sqrt{-2}$ would force $a^2 - 2b^2 = 0$ and $ab = \pm 1$, impossible with rational a, b. Thus these fields have degree 4 too. In every row this degree equals the subgroup index, proving equality with the full fixed field. The trivial subgroup fixes K; the full group fixes $\mathbb{Q}$. This verifies all ten fields and, by reversing subgroup containment, all inclusions in the diagram.

Exercise 54.3: Primitive generators and all their minimal polynomials

Use the notation of Exercise 54.2. A complete list, including top and bottom, is

field	primitive element	minimal polynomial
$\mathbb{Q}$	0	x
$\mathbb{Q}(\sqrt{2})$	$\sqrt{2}$	x^2-2
$\mathbb{Q}(i)$	i	x^2+1
$\mathbb{Q}(\sqrt{-2})$	$i\sqrt{2}$	x^2+2
$\mathbb{Q}(\alpha)$	α	x^4-2
$\mathbb{Q}(i\alpha)$	$i\alpha$	x^4-2
$\mathbb{Q}(\sqrt{2},i)$	$\sqrt{2}+i$	x^4-2x^2+9
$\mathbb{Q}((1+i)\alpha)$	$(1+i)\alpha$	x^4+8
$\mathbb{Q}((1-i)\alpha)$	$(1-i)\alpha$	x^4+8
K	$\alpha+i$	$x^8+4x^6+2x^4+28x^2+1$

The degrees and irreducibility in the first nine rows follow from Exercise 54.2 and the four distinct sign conjugates of $\sqrt{2}+i$. For the last row, the eight images of $\alpha+i$ are $i^j\alpha+\varepsilon i$ for $0\le j<4$ and $\varepsilon=\pm1$. They are all distinct: an equality with different j would express α as an element of $\mathbb{Q}(i)$, impossible because its rational degree is 4; with equal j the signs must agree. Thus this element has degree 8 and generates K. Multiplying

$$((x-i)^4-2)((x+i)^4-2)$$

gives the displayed monic polynomial, proving it is its minimal polynomial. The identical quartics in several rows do not imply identical subfields: an abstract isomorphism can send one primitive root to another.

Exercise 54.4: The fifth-root-of-unity correspondence

Let ζ be a primitive fifth root.

(a) Its powers $1,\zeta,\zeta^2,\zeta^3,\zeta^4$ are all roots of x^5-1, so $K=\mathbb{Q}(\zeta)$ is exactly its splitting field.

(b) An automorphism preserves order and the equation $\zeta^5=1$, so sends ζ to ζ^a for $a=1,2,3,4$.

(c) Φ_5 is irreducible of degree 4. Thus every listed image gives an automorphism, $\sigma_a(\zeta)=\zeta^a$. Composition multiplies exponents modulo 5. Since 2 has order 4 modulo 5, the group is $\langle\sigma_2\rangle\cong\mathbb{Z}_4$.

(d) Its only proper nontrivial subgroup is $\langle\sigma_4\rangle$, generated by complex conjugation. Its fixed field is $M=\mathbb{Q}(t)$ where $t=\zeta+\zeta^{-1}$. Dividing $1+\zeta+\cdots+\zeta^4=0$ by ζ^2 gives $t^2+t-1=0$, so $M=\mathbb{Q}(\sqrt{5})$ has degree 2 and is the full fixed field. The two diagrams are the chains

$$\begin{array}{cc} K & \langle\sigma_2\rangle \\ | & | \\ M & \langle\sigma_4\rangle \\ | & | \\ \mathbb{Q} & \{1\} \end{array}$$

with correspondence $K\leftrightarrow\{1\}$, $M\leftrightarrow\langle\sigma_4\rangle$, and $\mathbb{Q}\leftrightarrow\langle\sigma_2\rangle$. Each step has index or field degree 2.

Exercise 54.5: A fifth radical over a cyclotomic base

Put $F = \mathbb{Q}(\zeta_5)$ and $r = \sqrt[5]{2}$. Since $[F : \mathbb{Q}] = 4$ and $[\mathbb{Q}(r) : \mathbb{Q}] = 5$, the tower law applied to $F(r)$ shows $5 \mid 4[F(r) : F]$. But adjoining a root of $x^5 - 2$ gives degree at most 5, so $[F(r) : F] = 5$. The presence of ζ_5 makes all roots $r\zeta_5^j$ lie in $F(r)$, so this is the splitting field. The map $r \mapsto r\zeta_5$, fixing F, has order 5 and generates all five automorphisms. Thus the group is $\mathbb{Z}_5$.

Exercise 54.6: The seventh-root-of-unity correspondence

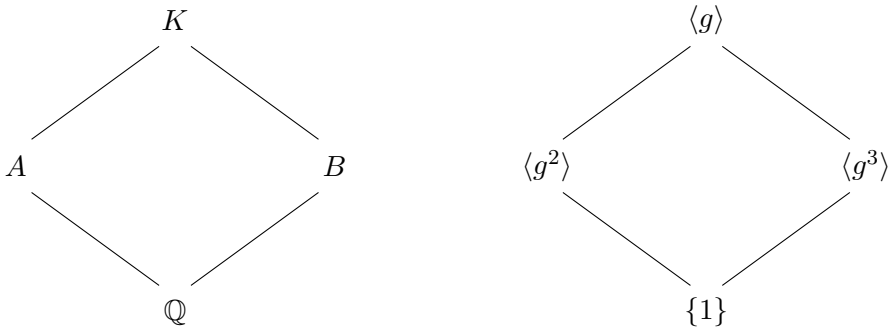
Let ζ have order 7 and $K = \mathbb{Q}(\zeta)$.

(a) Its seven powers are precisely the roots of $x^7 - 1$, so it generates the splitting field.

(b) Every automorphism sends ζ to one of ζ^a, $1 \le a \le 6$.

(c) Irreducibility of Φ_7 gives degree 6, so all six exponent maps occur. Their group is $(\mathbb{Z}/7\mathbb{Z})^\times$, cyclic of order 6 generated by 3. Write $g = \sigma_3$.

(d) The two proper nontrivial subgroups are $\langle g^2\rangle = \{\sigma_1, \sigma_2, \sigma_4\}$ of order 3 and $\langle g^3\rangle = \{\sigma_1, \sigma_6\}$ of order 2. Their fixed fields are respectively $A = \mathbb{Q}(\sqrt{-7})$ and $B = \mathbb{Q}(\zeta + \zeta^{-1})$. For the first, put $\eta = \zeta + \zeta^2 + \zeta^4$. The other three powers sum to $\bar\eta$, and direct multiplication gives $\eta + \bar\eta = -1$, $\eta\bar\eta = 2$. Thus η satisfies $x^2 + x + 2$, has degree 2, and generates A. The subgroup fixes this sum, so it fixes exactly A. For B, complex conjugation fixes $t = \zeta + \zeta^{-1}$ and ζ satisfies $X^2 - tX + 1$; since B is real and ζ nonreal, $[K : B] = 2$ and $[B : \mathbb{Q}] = 3$. For a check, $t^3 + t^2 - 2t - 1 = 0$ follows by dividing $\Phi_7(\zeta) = 0$ by ζ^3. The field and group diagrams are diamonds:



The correspondence reverses order: $A \leftrightarrow \langle g^2\rangle$, $B \leftrightarrow \langle g^3\rangle$, with top and bottom paired oppositely. There are no other intermediate fields because a cyclic group of order 6 has no other subgroups.

Exercise 54.7: Eighth roots without recomputing a table

The splitting field is $\mathbb{Q}(\zeta_8) = \mathbb{Q}(\sqrt{2}, i)$. Its group is $(\mathbb{Z}/8\mathbb{Z})^\times = \{1, 3, 5, 7\}$ under multiplication modulo 8. Each of its three nonidentity elements squares to 1, so it is the Klein four-group $\mathbb{Z}_2 \times \mathbb{Z}_2$.

Exercise 54.8: A complete quartic splitting-field correspondence

Find the field and its degree.

Put $a = \sqrt{2+\sqrt{5}}$, $b = i/a$, and $K = \mathbb{Q}(a, i)$. Then $b^2 = 2 - \sqrt{5}$ and $ab = i$, so the roots are $a, -a, b, -b$ and K is their splitting field. The norm of $2 + \sqrt{5}$ from $\mathbb{Q}(\sqrt{5})$ to $\mathbb{Q}$ is -1, so it is not a square there. Thus $[\mathbb{Q}(a) : \mathbb{Q}] = 4$; this field is real and gains degree 2 on adjoining i, so $[K : \mathbb{Q}] = 8$.

Describe the group.

Define

$$r(a) = b, \quad r(i) = -i, \qquad s(a) = a, \quad s(i) = -i.$$

To justify existence, conjugacy gives an embedding $\mathbb{Q}(a) \to \mathbb{Q}(b)$ sending a to b. The element i is outside $\mathbb{Q}(b)$: otherwise $a = i/b$ would also belong to that degree-four field, contradicting $[K : \mathbb{Q}] = 8$. Thus $x^2 + 1$ is still irreducible over $\mathbb{Q}(b)$, and the embedding extends with either choice $i \mapsto \pm i$. The map s is ordinary complex conjugation. This constructs the asserted automorphisms, rather than merely proposing images. They satisfy $r^4 = s^2 = 1$, $srs = r^{-1}$, and yield eight distinct maps $1, r, r^2, r^3, s, rs, r^2s, r^3s$. Thus the group is the dihedral group of order 8.

List every subgroup and its field.

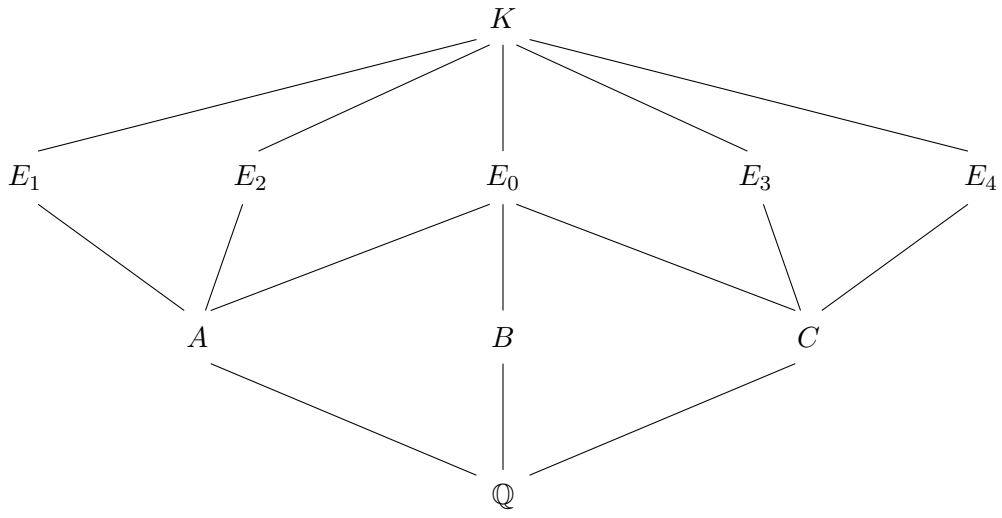
Define the field labels by this table:

subgroup	fixed field	[field : $\mathbb{Q}$]
G	$\mathbb{Q}$	1
$\langle r \rangle$	$B = \mathbb{Q}(i\sqrt{5})$	2
$\langle r^2, s \rangle$	$A = \mathbb{Q}(\sqrt{5})$	2
$\langle r^2, rs \rangle$	$C = \mathbb{Q}(i)$	2
$\langle r^2 \rangle$	$E_0 = \mathbb{Q}(\sqrt{5}, i)$	4
$\langle s \rangle$	$E_1 = \mathbb{Q}(a)$	4
$\langle r^2 s \rangle$	$E_2 = \mathbb{Q}(ia)$	4
$\langle rs \rangle$	$E_3 = \mathbb{Q}(a + b)$	4
$\langle r^3 s \rangle$	$E_4 = \mathbb{Q}(a - b)$	4
$\{1\}$	K	8

For verification, r negates both i and $\sqrt{5} = a^2 - 2$, hence fixes $i\sqrt{5}$. The subgroup $\langle r^2, s \rangle$ fixes a^2, and $\langle r^2, rs \rangle$ fixes i. The individual involutions fix their displayed generators: r^2 fixes a^2, i; s fixes a; r^2s fixes ia; rs interchanges a, b; r^3s interchanges $a, -b$. The quadratic degrees are immediate. The field $E_1 = \mathbb{Q}(a)$ has degree 4 as proved above. For E_2, the element ia satisfies $X^4 + 4X^2 - 1 = 0$. Its images under the already constructed automorphisms include $ia, -ia, ib, -ib$, four distinct numbers: $\pm ia$ are nonzero imaginary and $\pm ib = \mp 1/a$ are nonzero real. Its minimal polynomial therefore has at least four distinct roots and degree at most 4, so its degree is exactly 4. For E_0, i doubles the real quadratic degree. Finally $(a \pm b)^2 = 4 \pm 2i$, which is not a square in $\mathbb{Q}(i)$: the norm of a putative square would be the rational square 20, impossible. Hence E_3, E_4 have degree 4, with minimal polynomial $x^4 - 8x^2 + 20$. All proposed fields therefore have the required index and equal the full fixed fields.

Give all inclusions.

The following is the full field diagram. Reverse its edges and replace each field by its subgroup from the table to obtain the full subgroup diagram. In particular E_0 contains all three quadratic fields; E_1, E_2 contain only A, and E_3, E_4 contain only C.



To make the subgroup diagram explicit as well, its covering relations are

upper subgroup	immediately lower subgroups
G	$\langle r\rangle,\ \langle r^2, s\rangle,\ \langle r^2, rs\rangle$
$\langle r\rangle$	$\langle r^2\rangle$
$\langle r^2, s\rangle$	$\langle r^2\rangle, \langle s\rangle, \langle r^2 s\rangle$
$\langle r^2, rs\rangle$	$\langle r^2\rangle, \langle rs\rangle, \langle r^3 s\rangle$
each order-two subgroup	$\{1\}$

These relations and the field diagram exhibit the entire correspondence, not just its normal subgroups.

Exercise 54.9: Two symmetric-function identities

Write $s_1 = y_1 + y_2 + y_3$, $s_2 = y_1y_2 + y_1y_3 + y_2y_3$, $s_3 = y_1y_2y_3$.

(a) Expanding s_1^2 gives $y_1^2 + y_2^2 + y_3^2 + 2s_2$, so the answer is $s_1^2 - 2s_2$.

(b) Put the six displayed ratios over the denominator s_3. Their numerator is $\sum_{i\neq j} y_i^2 y_j$. On expanding s_1s_2, this sum appears together with $3y_1y_2y_3$. Thus the answer is $(s_1s_2 - 3s_3)/s_3$. The equality is an identity of rational functions, and numerical evaluation requires each original denominator nonzero.

Exercise 54.10: Polynomials of a sum and of squares

Vieta's identities for $x^3 - 4x^2 + 6x - 2$ give $s_1 = 4, s_2 = 6, s_3 = 2$ for its three roots.

(a) Their sum is 4, so the monic polynomial with that sole zero is $x - 4$.

(b) The sum of the squared roots is $s_1^2 - 2s_2 = 4$. Their pairwise products sum to $s_2^2 - 2s_1s_3 = 36 - 16 = 20$, and their product is $s_3^2 = 4$. Hence the required polynomial is $x^3 - 4x^2 + 20x - 4$. These squared roots are distinct: an opposite pair of original roots would force the third root to be 4, but the original polynomial does not vanish at 4.

Exercise 54.11: Every finite group occurs over a suitable field

Let H be finite. By Cayley's theorem embed it into S_n for some n. Take independent indeterminates $y_1, \dots, y_n$ over $\mathbb{Q}$, and put $L = \mathbb{Q}(y_1, \dots, y_n)$ and $B = \mathbb{Q}(s_1, \dots, s_n)$, using their elementary symmetric functions. The symmetric-function theorem gives $G(L/B) \cong S_n$. Regard H as a subgroup and let $F = L_H$. The Galois correspondence gives $G(L/F) = H$, and L/F is a finite normal extension. This proves existence over a suitable field F; it does not assert that every finite group occurs over $\mathbb{Q}$.

Exercise 54.12: Discriminants and even permutations

Let $\Delta = \prod_{i<j}(\alpha_i - \alpha_j)$, with roots repeated according to the multiplicities in f.

(a) $\Delta = 0$ exactly when two entries in this root list coincide, meaning f has a repeated root. Since each irreducible factor is separable, repetitions occur exactly when some irreducible factor appears at least twice. This is the stated square-divisibility criterion.

(b) An automorphism permutes the root list and sends Δ to $\pm\Delta$. Squaring removes the sign, so Δ^2 is fixed by the full Galois group and belongs to F.

(c) The equivalence needs two additional hypotheses: f must have distinct roots and $\operatorname{char} F \neq 2$. Under those hypotheses the root action embeds $G(K/F)$ into S_n, and

$$\sigma(\Delta) = \operatorname{sgn}(\sigma)\Delta.$$

This sign rule follows because interchanging two roots negates the alternating product and transpositions generate S_n. Since $\Delta \neq 0$ and $1 \neq -1$, it is fixed exactly by even permutations. Thus $G(K/F) \subseteq A_n$ if and only if every automorphism fixes Δ, equivalently $\Delta \in F$. As printed, the unrestricted assertion is false. In characteristic 2, x^2+x+1 over $\mathbb{Z}_2$ has splitting field of order 4 and group S_2, but $\Delta = \alpha - \alpha^2 = 1 \in \mathbb{Z}_2$. With repeated roots, $\Delta = 0 \in F$ regardless of the group, and the set of distinct roots may have fewer than n elements, so the stated identification with permutations of n distinct roots is not available. The preceding corrected version is the usual discriminant test.

Exercise 54.13: Algebraic integers form a ring

Let α, β satisfy monic integer polynomials of degrees m, n. Higher powers of either can be reduced using its monic relation with integer coefficients. Thus the additive group

$$M = \sum_{0 \le j < m,\ 0 \le k < n} \mathbb{Z}\alpha^j \beta^k$$

is finitely generated, contains 1, and is closed under multiplication by α and β, hence by $\alpha + \beta$, $\alpha\beta$, and $-\alpha$. It is torsion free as a subgroup of the additive group of $\mathbb{C}$, so it is a free abelian group of finite rank. For each γ among these three elements, multiplication by γ on M is represented in a $\mathbb{Z}$-basis by an integer matrix A. Its characteristic polynomial is monic with integer coefficients, and Cayley–Hamilton gives $p(A) = 0$. Applying that identity to $1 \in M$ yields $p(\gamma) = 0$ in $\mathbb{C}$. Thus $\alpha + \beta$, $\alpha\beta$, and $-\alpha$ are algebraic integers. Finally 0 and 1 satisfy x and $x - 1$. These facts verify all subring conditions.

55 Cyclotomic Extensions

Exercise 55.1: Multiplying the primitive eighth-root factors

Let ζ have order 8, so $\zeta^4 = -1$ and $\zeta^8 = 1$. The primitive roots are $\zeta, \zeta^3, \zeta^5 = -\zeta, \zeta^7 = -\zeta^3$. Pairing opposite roots gives

$$\begin{aligned}\Phi_8(x) &= (x^2 - \zeta^2)(x^2 - \zeta^6) \\ &= x^4 - (\zeta^2 + \zeta^6)x^2 + \zeta^8 = x^4 + 1,\end{aligned}$$

because $\zeta^6 = -\zeta^2$. This computes all coefficients rather than just identifying the zeros.

Exercise 55.2: The twentieth-root group as an abelian group

The group is $(\mathbb{Z}/20\mathbb{Z})^\times$. Since $20 = 4 \cdot 5$ with coprime factors, the Chinese remainder theorem gives

$$(\mathbb{Z}/20\mathbb{Z})^\times \cong (\mathbb{Z}/4\mathbb{Z})^\times \times (\mathbb{Z}/5\mathbb{Z})^\times \cong \mathbb{Z}_2 \times \mathbb{Z}_4.$$

The factors have orders 2 and 4, so the group has order 8. This is its invariant-factor classification because $2 \mid 4$.

Exercise 55.3: Three Euler totients

For $n = \prod p_j^{e_j}$, $\varphi(n) = \prod p_j^{e_j - 1}(p_j - 1)$, since a residue must not be divisible by any prime divisor of n.

(a) $60 = 2^2 \cdot 3 \cdot 5$, so $\varphi(60) = 2 \cdot 2 \cdot 4 = 16$.

(b) $1000 = 2^3 \cdot 5^3$, so $\varphi(1000) = 4 \cdot 100 = 400$.

(c) $8100 = 2^2 \cdot 3^4 \cdot 5^2$, so $\varphi(8100) = 2 \cdot 54 \cdot 20 = 2160$.

Exercise 55.4: The first thirty constructible regular polygons

The criterion is $n = 2^a$ times a product of distinct Fermat primes. Enumerating these numbers increasingly, starting at $n \geq 3$, gives

$$\begin{gathered}3, 4, 5, 6, 8, 10, 12, 15, 16, 17, \\ 20, 24, 30, 32, 34, 40, 48, 51, 60, 64, \\ 68, 80, 85, 96, 102, 120, 128, 136, 160, 170.\end{gathered}$$

For completeness of the enumeration up to 170, its only possible odd prime factors are 3,5,17; the next Fermat prime is 257. No square of an odd prime is allowed. Listing $2^a d \leq 170$ for $d \in \{1, 3, 5, 15, 17, 51, 85\}$, omitting 1 and 2 and sorting, gives exactly the displayed thirty numbers.

Exercise 55.5: The smallest constructible integral-degree angle

The answer is 3°. A regular 120-gon is constructible because $120 = 2^3 \cdot 3 \cdot 5$, so its central angle $360^\circ/120 = 3^\circ$ is constructible. A 1° angle would construct a regular 360-gon, but $360 = 2^3 \cdot 3^2 \cdot 5$ contains a square of the odd prime 3. A 2° angle would similarly construct a regular 180-gon, and $180 = 2^2 \cdot 3^2 \cdot 5$ fails the same criterion. Thus neither smaller positive integral angle is constructible.

Exercise 55.6: Twelfth roots and a group of exponent two

(a) The splitting field is $K = \mathbb{Q}(\zeta_{12})$, of degree $\varphi(12) = 4$ over $\mathbb{Q}$.

(b) Its automorphisms are the exponent maps for $1, 5, 7, 11$ modulo 12. Each exponent squares to 1 modulo 12, so every automorphism squares to the identity. A group of order 4 with no element of order 4 is $\mathbb{Z}_2 \times \mathbb{Z}_2$, giving the requested classification.

Exercise 55.7: Two cyclotomic polynomials in finite characteristic

Over $\mathbb{Z}_2$, the primitive third roots are the nonidentity roots of $x^3 - 1$, so $\Phi_3(x) = x^2 + x + 1$. Over $\mathbb{Z}_3$, the same opposite-root computation as Exercise 55.1 gives $\Phi_8(x) = x^4 + 1$. It factors there as $(x^2 + x + 2)(x^2 - x + 2)$; each quadratic has no root in $\mathbb{Z}_3$. This illustrates that a cyclotomic polynomial need not be irreducible over a finite field.

Exercise 55.8: A repeated polynomial already split over the prime field

In characteristic 3,

$$x^6 - 1 = (x^2 - 1)^3 = (x - 1)^3(x + 1)^3.$$

Its only distinct roots are $1, -1$, already in $\mathbb{Z}_3$. Therefore the splitting field is $\mathbb{Z}_3$ itself, with 3 elements. Counting the formal degree as the number of distinct roots would give the wrong answer.

Exercise 55.9: Ten statements about cyclotomic fields

(a) False. Over $\mathbb{C}$, for instance, every cyclotomic polynomial of degree greater than 1 splits into linear factors. Irreducibility here is over $\mathbb{Q}$.

(b) True. These are exactly its defining zeros.

(c) False. The group has order $\varphi(n)$; for example $n = 3$ gives order 2, not 3.

(d) True. It is isomorphic to the multiplicative group of units modulo n.

(e) True. $[\mathbb{Q}(\zeta_n) : \mathbb{Q}] = \varphi(n)$ and the extension is Galois.

(f) False. $25 = 5^2$ has a squared odd prime factor; also $\varphi(25) = 20$ is not a power of 2.

(g) True. $17 = 2^{2^2} + 1$ is a Fermat prime.

(h) True for an ordinary polygon, so prime $p \geq 3$: the criterion says an odd prime p must be Fermat. If one includes the degenerate $p = 2$ case, it must be treated separately; 2 is not a Fermat prime.

(i) False. $2^{2^5}+1 = 4294967297 = 641 \cdot 6700417$ is composite.

(j) True by definition: a Fermat prime is a prime among the numbers $2^{2^k}+1$.

Exercise 55.10: Partitioning roots by their orders

Assume the characteristic does not divide n. The derivative of $x^n - 1$ is nx^{n-1}, so all its roots in a closure are distinct. Each root has a unique multiplicative order d dividing n, and is then a primitive dth root. The root set is therefore the disjoint union of the primitive-root sets for all $d \mid n$. Multiplying their monic linear-factor products gives

$$x^n - 1 = \prod_{d|n} \Phi_d(x).$$

All the cyclotomic factors have coefficients in the prime field, hence in F, so this is an identity in $F[x]$.

Exercise 55.11: The first six cyclotomic polynomials

Use $x^n - 1 = \prod_{d|n} \Phi_d(x)$ and divide by factors from proper divisors. This gives

n	$\Phi_n(x)$
1	$x-1$
2	$x+1$
3	x^2+x+1
4	x^2+1
5	$x^4+x^3+x^2+x+1$
6	x^2-x+1

For example, $x^6 - 1 = (x-1)(x+1)(x^2+x+1)(x^2-x+1)$, so removing the factors for divisors 1,2,3 leaves the factor for 6.

Exercise 55.12: The twelfth cyclotomic polynomial

The divisors of 12 are $1, 2, 3, 4, 6, 12$. An efficient cancellation is

$$x^{12} - 1 = (x^6 - 1)(x^6 + 1), \qquad x^6 + 1 = (x^2 + 1)(x^4 - x^2 + 1).$$

The first factor already contains $\Phi_1, \Phi_2, \Phi_3, \Phi_6$, and $x^2 + 1 = \Phi_4$. Thus the remaining factor is $\Phi_{12}(x) = x^4 - x^2 + 1$.

Exercise 55.13: Negating roots of odd order

If $n > 1$ is odd and ζ has order n, then $-\zeta$ has order $2n$: its nth power is -1, and if $(-\zeta)^k = 1$, then squaring gives $n \mid k$ and the odd multiple $k = n$ does not work. Conversely, negating a primitive $2n$th root gives a primitive nth root. Thus the zeros of $\Phi_n(-x)$ are exactly those of $\Phi_{2n}(x)$. For odd $n > 1$, $\varphi(n)$ is even, since invertible residues pair as $a, -a$, so $\Phi_n(-x)$ is monic. The two monic polynomials with the same distinct roots coincide.

Exercise 55.14: The compositum for coprime root orders

Let ζ have order nm, and set $\xi = \zeta^m$, $\eta = \zeta^n$, which have orders n, m. Their fields are contained in $\mathbb{Q}(\zeta)$. Conversely choose integers u, v with $um + vn = 1$. Then $\zeta = \xi^u \eta^v \in \mathbb{Q}(\xi, \eta)$. Thus

$$\mathbb{Q}(\zeta_{nm}) = \mathbb{Q}(\zeta_n, \zeta_m),$$

which is exactly the splitting field of $(x^n - 1)(x^m - 1)$.

Exercise 55.15: The product decomposition for coprime orders

Identify the three cyclotomic Galois groups with unit groups modulo nm, n, m. Restriction to the two smaller splitting fields is the map

$$(\mathbb{Z}/nm\mathbb{Z})^\times \longrightarrow (\mathbb{Z}/n\mathbb{Z})^\times \times (\mathbb{Z}/m\mathbb{Z})^\times, \quad a \longmapsto (a \bmod n, a \bmod m).$$

It is injective because equality modulo both coprime moduli implies equality modulo their product. The Chinese remainder theorem gives a unique residue modulo nm for each pair of unit residues, and that residue is also a unit, proving surjectivity. Multiplication is preserved. Therefore the group of $x^{nm} - 1$ is the direct product of the two smaller groups, as claimed.

56 Insolvability of the Quintic

Exercise 56.1: A radical extension in characteristic two

The polynomial $x^2 + x + 1$ is irreducible over $\mathbb{Z}_2$ and its splitting field is $K = \mathrm{GF}(4)$. A square root of an element of $\mathbb{Z}_2$ cannot generate it: the only elements are 0,1, whose square roots in any extension are uniquely 0,1 in characteristic 2. Nevertheless K is a radical extension. A root α of the polynomial satisfies $\alpha^3 = 1$, and $K = \mathbb{Z}_2(\alpha)$ is obtained by adjoining this cube root of the base-field element 1. A radical need not be a square root or the real positive choice of a root.

Exercise 56.2: An even polynomial of degree eight

Yes in characteristic zero. Substitute $y = x^2$ to get the quartic $ay^4 + by^3 + cy^2 + dy + e$. Every quartic is solvable by radicals, so all four y-roots lie in a radical extension. Adjoining square roots of each of them, one after another, gives a further radical extension containing both square roots of every y-root. These are exactly the roots of the original polynomial. Thus its splitting field is contained in a radical extension.

Exercise 56.3: Ten statements about radicals and solvability

(a) True. This is the definition of solvability by radicals.

(b) True in characteristic zero, by the solvable-Galois-group criterion.

(c) True. The roots of $x^{17} - 5$ are products of a seventeenth root of 5 and powers of a seventeenth root of unity; adjoining these radicals supplies a radical extension. Its group is therefore solvable.

(d) False. Although both are transcendental, they satisfy the polynomial relation $Y^2 - X = 0$ over $\mathbb{Q}$, so they are algebraically dependent.

(e) True. The group is cyclic, hence abelian and solvable.

(f) False. $x^5 - 2$ is solved by adjoining a fifth root of 2 and a fifth root of unity.

(g) True. The quartic formulas, or solvability of subgroups of S_4, prove it.

(h) False. A real cube root of 2 has degree 3 over $\mathbb{Q}$ and cannot belong to a tower of quadratic extensions, whose degree is a power of 2.

(i) False. The cubic $x^3 - x$ has rational roots $-1, 0, 1$, requiring no extension at all.

(j) True. A subnormal series with abelian quotients characterizes solvable groups, linking the group criterion to radical towers.

Exercise 56.4: The quadratic formula over a field

Multiplying $ax^2 + bx + c = 0$ by $4a$ and completing the square gives $(2ax + b)^2 = b^2 - 4ac$. Since $a \neq 0$ and characteristic is not 2, $2a$ is invertible. With $\delta^2 = b^2 - 4ac$, the roots are $(-b + \delta)/(2a)$ and $(-b - \delta)/(2a)$, so they lie in $F(\delta)$. Conversely either root determines $\delta = 2a\alpha + b$, up to sign, so

the field generated by the roots contains δ. Thus the splitting field is $F(\sqrt{b^2-4ac})$, including the trivial-extension cases where the discriminant is already a square or zero.

Exercise 56.5: A biquadratic solved by successive radicals

Put $y = x^2$. First solve $ay^2 + by + c = 0$ by adjoining $\delta = \sqrt{b^2-4ac}$, giving $y_\pm = (-b \pm \delta)/(2a)$. Next adjoin $\sqrt{y_+}$ and $\sqrt{y_-}$. The resulting tower is radical, and contains all four roots $\pm\sqrt{y_+}, \pm\sqrt{y_-}$, with coincidences allowed. Hence the splitting field is contained in this radical extension. The characteristic assumption permits division by $2a$.

Exercise 56.6: Refining a series with abelian factors

Consider one original step $N \trianglelefteq M$ with M/N abelian, and a refined chain $N = H_0 \trianglelefteq H_1 \trianglelefteq \cdots \trianglelefteq H_k = M$. For each j, H_j/N is a subgroup of the abelian group M/N, hence is abelian. By the third isomorphism theorem,

$$H_j/H_{j-1} \cong (H_j/N)/(H_{j-1}/N),$$

which is a quotient of an abelian group and is therefore abelian. Every new step lies within some original step, so this proves the assertion for the whole refinement.

Exercise 56.7: Refining solvable factors to abelian composition factors

For a step $N \trianglelefteq M$ whose quotient M/N is solvable, choose a subnormal series of M/N with abelian factors. Lift its subgroups under the quotient map $M \to M/N$; the correspondence and isomorphism theorems give a subnormal refinement from N to M with those same abelian quotients. Do this for every original step. Because the group is finite, refine further until every nontrivial quotient is simple, obtaining a composition series. Exercise 56.6 ensures the refined quotients remain abelian. Thus all composition factors are simple abelian finite groups, hence cyclic groups of prime order.

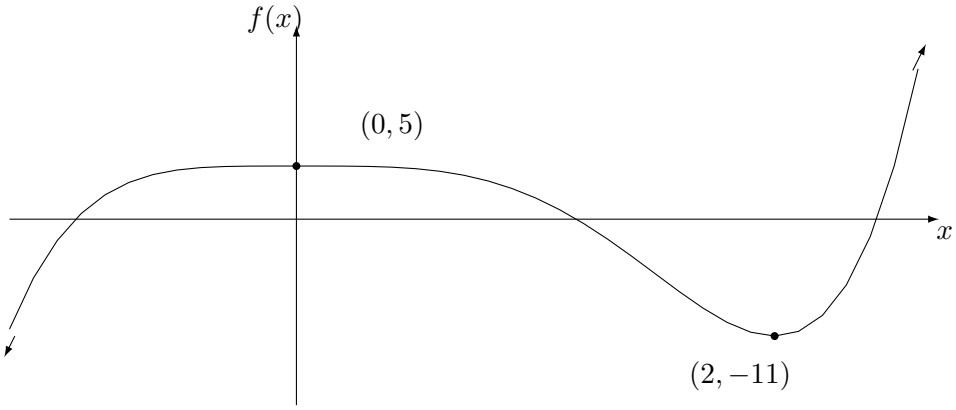
Exercise 56.8: An explicit quintic with no radical solution

(a) Label the letters by $\mathbb{Z}_5$ so the five-cycle is $c : j \mapsto j+1$. If the transposition is (u, v), its conjugates $c^k(u, v)c^{-k} = (u+k, v+k)$ all lie in H. Regard these as edges joining vertices differing by the nonzero residue $v - u$. Such edges make a connected graph, since every nonzero residue generates the additive group $\mathbb{Z}_5$. Edge transpositions of a connected graph generate all transpositions: along a path, conjugating adjacent edge swaps gives a swap of the path endpoints, by induction on its length. Hence H contains every transposition and therefore $H = S_5$.

(b) Irreducibility of the quintic makes the Galois action on its five roots transitive, so the orbit-stabilizer theorem gives $5 \mid |G|$. Cauchy's theorem gives an element of order 5, which in S_5 is a five-cycle. Complex conjugation exchanges exactly the two nonreal roots and fixes the three real ones, so its permutation is a transposition, not merely an arbitrary element of order 2. Part (a) now gives $G = S_5$.

(c) For $f(x) = 2x^5 - 5x^4 + 5$, Eisenstein at 5 proves irreducibility: 5 divides every nonleading coefficient, 25 does not divide the constant term, and 5 does not divide 2. The derivative is $f'(x) = 10x^3(x-2)$. Thus f increases on $(-\infty, 0)$, decreases on $(0, 2)$, and increases on $(2, \infty)$.

Its local maximum is $f(0) = 5 > 0$ and its local minimum is $f(2) = -11 < 0$. The odd degree with positive leading coefficient gives limits $-\infty$ and $+\infty$ at the two ends. Therefore continuity and strict monotonicity give exactly one real zero on each of these three intervals, and no others. Its remaining two roots are a nonreal conjugate pair. Here is a sketch, with the two extrema labeled and the curve drawn from actual sampled values; arrows indicate the tails.



By (b) its Galois group is S_5. This group is not solvable: its simple nonabelian normal subgroup A_5 obstructs an abelian-factor series. The solvability theorem therefore shows this quintic is not solvable by radicals over $\mathbb{Q}$.

A Appendix: Matrix Algebra

An $m \times n$ matrix has m rows and n columns. Addition and subtraction require the same dimensions and operate entry by entry. The product of an $m \times n$ matrix and an $n \times r$ matrix is an $m \times r$ matrix: its (i, j) entry is the dot product of row i with column j. Throughout the complex calculations, use $i^2 = -1$.

Exercise A.1: Adding two real matrices

Plan. Add entries occupying the same position; both matrices are 2×2.

$$\begin{bmatrix} -2 & 4 \\ 1 & 5 \end{bmatrix} + \begin{bmatrix} 4 & -3 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} -2+4 & 4-3 \\ 1+1 & 5+2 \end{bmatrix} = \boxed{\begin{bmatrix} 2 & 1 \\ 2 & 7 \end{bmatrix}}.$$

No row is multiplied by a column when adding matrices.

Exercise A.2: Adding two complex matrices

Both matrices are 2×3, so addition is defined. Add real parts to real parts and imaginary parts to imaginary parts:

$$\begin{aligned} &(1+i)+3 = 4+i, && -2+(i-1) = -3+i, && (3-i)+(-2+i) = 1, \\ &4+(3-i) = 7-i, && i+(1+i) = 1+2i, && (2-i)+0 = 2-i. \end{aligned}$$

Conclusion. The sum is $\begin{bmatrix} 4+i & -3+i & 1 \\ 7-i & 1+2i & 2-i \end{bmatrix}$.

Exercise A.3: Subtracting complex matrices

Subtraction is defined because both matrices are 3×2. Distribute each minus sign to the entire corresponding entry:

$$\begin{bmatrix} i & -1 \\ 4 & 1 \\ 3 & -2i \end{bmatrix} - \begin{bmatrix} 3-i & 4i \\ 2 & 1+i \\ 3 & -i \end{bmatrix} = \begin{bmatrix} i-(3-i) & -1-4i \\ 4-2 & 1-(1+i) \\ 3-3 & -2i-(-i) \end{bmatrix} = \boxed{\begin{bmatrix} -3+2i & -1-4i \\ 2 & -i \\ 0 & -i \end{bmatrix}}.$$

In particular, subtracting $-i$ adds i.

Exercise A.4: A product of two square matrices

Plan. The inner dimensions match. Compute four row–column dot products.

$$\begin{bmatrix} 1 & -1 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} 2 & 4 \\ -1 & 3 \end{bmatrix} = \begin{bmatrix} 1(2)+(-1)(-1) & 1(4)+(-1)(3) \\ 3(2)+1(-1) & 3(4)+1(3) \end{bmatrix} = \boxed{\begin{bmatrix} 3 & 1 \\ 5 & 15 \end{bmatrix}}.$$

The factors and the result are all 2×2.

Exercise A.5: A rectangular matrix product

A 2×2 matrix times a 2×3 matrix gives a 2×3 result:

$$\begin{bmatrix} 3 & 1 \\ -4 & 2 \end{bmatrix} \begin{bmatrix} 1 & 5 & -3 \\ 2 & 1 & 6 \end{bmatrix} = \begin{bmatrix} 3(1)+1(2) & 3(5)+1(1) & 3(-3)+1(6) \\ -4(1)+2(2) & -4(5)+2(1) & -4(-3)+2(6) \end{bmatrix} = \boxed{\begin{bmatrix} 5 & 16 & -3 \\ 0 & -18 & 24 \end{bmatrix}}.$$

Exercise A.6: Checking dimensions before multiplying

The first matrix is 2×2 and the second is 3×2. The number of columns of the first, namely 2, differs from the number of rows of the second, namely 3. For example, the proposed first entry would require a dot product between a two-entry row and a three-entry column.

Conclusion. The indicated matrix product is undefined.

Exercise A.7: A complex matrix product

Multiply rows by columns, remembering $i(3i) = 3i^2 = -3$:

$$\begin{bmatrix} i & 1 \\ -2 & 1 \end{bmatrix} \begin{bmatrix} 3i & 1 \\ 4 & -2i \end{bmatrix} = \begin{bmatrix} i(3i)+1(4) & i(1)+1(-2i) \\ (-2)(3i)+1(4) & (-2)(1)+1(-2i) \end{bmatrix}$$
$$= \boxed{\begin{bmatrix} 1 & -i \\ 4-6i & -2-2i \end{bmatrix}}.$$

Exercise A.8: The fourth power of a real matrix

Let $M = \begin{bmatrix} 1 & -1 \\ 1 & 0 \end{bmatrix}$. Matrix powers mean repeated matrix multiplication, not taking powers of the individual entries. First compute

$$M^2 = \begin{bmatrix} 1-1 & -1+0 \\ 1+0 & -1+0 \end{bmatrix} = \begin{bmatrix} 0 & -1 \\ 1 & -1 \end{bmatrix}.$$

Now square that result:

$$M^4 = (M^2)^2 = \begin{bmatrix} 0 & -1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 0 & -1 \\ 1 & -1 \end{bmatrix} = \boxed{\begin{bmatrix} -1 & 1 \\ -1 & 0 \end{bmatrix}}.$$

As a check, multiplying M^2 by M gives $M^3 = -I_2$, so $M^4 = -M$.

Exercise A.9: The fourth power of a complex matrix

Put $N = \begin{bmatrix} 1 & -i \\ i & 1 \end{bmatrix}$. Then

$$N^2 = \begin{bmatrix} 1+(-i)i & -i-i \\ i+i & i(-i)+1 \end{bmatrix} = \begin{bmatrix} 2 & -2i \\ 2i & 2 \end{bmatrix} = 2N.$$

Consequently $N^4 = (N^2)^2 = (2N)^2 = 4N^2 = 8N$, and therefore

$$\boxed{N^4 = \begin{bmatrix} 8 & -8i \\ 8i & 8 \end{bmatrix}}.$$

Only the scalar factors commute here; no interchange of matrix factors is needed.

Exercise A.10: An explicit failure of commutativity

Take integer matrices

$$P = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \qquad Q = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}.$$

Their products are

$$PQ = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \qquad QP = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}.$$

The upper-right entries differ, so $PQ \neq QP$. Both factors belong to $M_2(\mathbb{Z})$ and both products are defined, as required.

Exercise A.11: Inverting a signed permutation matrix

Let $J = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$. Multiplying gives $J^2 = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} = -I_2$. Thus $J(-J) = (-J)J = I_2$. An inverse must produce the identity on both sides, which proves

$$\boxed{J^{-1} = -J = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}}.$$

Exercise A.12: Inverting a diagonal matrix

For $D = \operatorname{diag}(2, 4, -1)$, choose $E = \operatorname{diag}(1/2, 1/4, -1)$. Products of diagonal matrices multiply their corresponding diagonal entries and have zero off-diagonal entries. Hence

$$DE = ED = \operatorname{diag}(2 \cdot \tfrac{1}{2}, 4 \cdot \tfrac{1}{4}, (-1)(-1)) = I_3.$$

This verifies directly that

$$\boxed{D^{-1} = \begin{bmatrix} 1/2 & 0 & 0 \\ 0 & 1/4 & 0 \\ 0 & 0 & -1 \end{bmatrix}}.$$

The inverse is taken over $\mathbb{R}$ or $\mathbb{C}$; it need not have integer entries.

Exercise A.13: The determinant of a triangular matrix

The given matrix is lower triangular: all entries above its main diagonal are zero. Its determinant is therefore the product of its diagonal entries:

$$\det \begin{bmatrix} 3 & 0 & 0 \\ 10 & -2 & 0 \\ 4 & 17 & 8 \end{bmatrix} = 3(-2)(8) = \boxed{-48}.$$

Alternatively, expanding along the first row gives $3 \det \begin{bmatrix} -2 & 0 \\ 17 & 8 \end{bmatrix} = 3(-16)$, the same answer.

Exercise A.14: Products of invertible matrices

What we must show. Given invertible $A, B \in M_n(\mathbb{C})$, exhibit two-sided inverses for AB and BA.

Plan. Reverse the order of the inverses. Matrix multiplication is associative, which allows regrouping without changing the order of any factors.

Check the inverse of AB.

$$(AB)(B^{-1}A^{-1}) = A(BB^{-1})A^{-1} = AI_nA^{-1} = I_n,$$
$$(B^{-1}A^{-1})(AB) = B^{-1}(A^{-1}A)B = B^{-1}I_nB = I_n.$$

Thus $(AB)^{-1} = B^{-1}A^{-1}$.

Check the inverse of BA.

Similarly,

$$(BA)(A^{-1}B^{-1}) = B(AA^{-1})B^{-1} = I_n, \qquad (A^{-1}B^{-1})(BA) = A^{-1}(B^{-1}B)A = I_n.$$

Conclusion. Both products are invertible, with $(AB)^{-1} = B^{-1}A^{-1}$ and $(BA)^{-1} = A^{-1}B^{-1}$.